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Financial Mathematics, Derivatives and Structured Products

Second Edition

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Dedicated to our loved ones,

Raymond: to Katharine and Jason

Yves: to Véronique, Antoine, Céline, and Sophie

Spike: to Carie and Josh

Xun: to Yun

Preface

This book originates from various courses taught by the authors, including:

- “Financial Mathematics” (undergraduate course) at The Chinese University of Hong Kong by the first author;
- “Mathematical Financial Modelling” (master’s course) at The Chinese University of Hong Kong by the second and third authors;
- “Structured Products: Analysis and Pricing” (master’s course) at The Hong Kong University of Science and Technology, “Stochastic Models” and “Derivatives Market” (master’s courses) at The Chinese University of Hong Kong (SZ), and “Derivatives Market and Financial Engineering” (FMBA) at Tsinghua University by the second author;
- “Financial Mathematics I” (undergraduate course) at the National University of Singapore and Hong Kong Polytechnic University by the fourth author.

Many examples and analysis from an industry perspective are featured in this book, provided by the second author who has over 20 years of banking experience in financial engineering for his work at a large French bank and two other top American investment banks. The book targets to bridge the gap between theory and practice, which makes it suitable to both students and practitioners with or without a background in science or finance. The book can also be used as a textbook for courses such as:

- Financial Mathematics (undergraduate level)
- Stochastic Modelling in Finance (postgraduate level)
- Financial Markets and Derivatives (undergraduate level)
- Structured Products and Solutions (undergraduate/postgraduate level)

This book aims to equip readers with the necessary knowledge of financial markets for working as product structurers, traders, sales, risk managers, or quantitative analysts. Both financial practitioners and academics may be interested in this book because it intends to unify derivatives modelling and financial engineering practice in the market.

Since the first edition of this book was published, the authors have received numerous constructive comments from the book readers and reviewers. Together with the new materials accumulated in recent years and typos spotted in the first edition, a second edition of the book became a viable option. The major new materials and enhanced contents include, but not limited to, the role of central counterparties for derivatives transactions, the reference rates to replace LIBOR, risk-neutral modelling for futures and forward, discussions and analysis on risk-neutral framework and numéraires, discrete dividend modelling, variance reduction techniques for Monte Carlo method, finite difference method analysis, tree method, FX modelling, multi-name credit derivatives modelling, local volatility model, forward variance model, and local-stochastic volatility model to reflect market practice.

With the publication of the second edition, the authors aim to optimize the reading experience by balancing the writing style and level of detail, making it both efficient and effective. Due to the comprehensive scope of the book, it is advisable for readers to jump directly to the interested topics, with complementary back-and-forth reading of related concepts, as the book is mostly self-contained. Part I and Part IV provide an overview of financial market instruments and structured products that do not require a mathematical background. However, beginners are advised to skip the chapters and sections marked with an asterisk (*) during the first reading of the book to avoid being overwhelmed by too many detailed or advanced concepts.

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Central, Hong Kong
Sha Tin, Hong Kong
Hung Hom, Hong Kong
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Acronyms

ABS	Asset-Backed Security
ADI	Alternating Direction Implicit
ADTV	Average Daily Trading Volume
AKI	American Knock-In
ALM	Asset Liability Management
APT	Arbitrage Pricing Theory
ASB	Accelerated Shares Buy-Back
ATM	At-the-Money
ATMF	At-the-Money Forward
ATMS	At-the-Money Spot
AUM	Assets Under Management
AXKI	At Expiry Knock-In
BBM	Buy Below the Market
BCBS	Basel Committee on Banking Supervision
BEN	Bonus Enhanced Note
BGM	Brace–Gątarek–Musiela
BPV	Basis Point Value
BSM	Black–Scholes–Merton
CAL	Capital Allocation Line
CAPM	Capital Asset Pricing Model
CB	Convertible Bond
CBBC	Callable Bull/Bear Contract
CBOE	Chicago Board Options Exchange
CCP	Central Counterparty
CCS	Cross-Currency Swap
CD	Certificate of Deposit
CDO	Collateralized Debt Obligation
CDS	Credit Default Swap
CET	Central European Time
CEV	Constant Elasticity of Variance
CIR	Cox–Ingersoll–Ross

CLN	Credit-Linked Note
CME	Chicago Mercantile Exchange
CMS	Constant Maturity Swap
CP	Commercial Paper
CPPI	Constant Proportion Portfolio Insurance
CSA	Credit Support Annex
CTA	Commodity Trading Advisor
CTD	Cheapest-to-Deliver
CTM	Collateralized-to-Market
CVA	Credit Valuation Adjustment
DAC	Daily Range Accrual Callable Note
DCI	Dual Currency Investment
DCM	Debt Capital Market
DVA	Debt (or Debit) Valuation Adjustment
DvP	Delivery versus Payment
EAD	Exposure at Default
ECM	Equity Capital Market
EFFR	Effective Federal Funds Rate
ELN	Equity-Linked Note
EPE	Expected Positive Exposure
ERR	Expiry Reference Rate
ETF	Exchange-Traded Fund
EONIA	Euro Overnight Index Average
EPS	Earnings-per-Share
EURIBOR	Euro Interbank Offered Rate
€STR	Euro Short-term Rate
FCN	Fixed Coupon Callable Note
FMP	Fixed Maturity Plan
FoP	Free of Payment
FRA	Forward Rate Agreement
FRN	Floating Rate Note
FTD	First-to-Default
FX	Foreign Exchange
GMRA	Global Master Repurchase Agreement
GMSLA	Global Master Securities Lending Agreement
HJM	Heath–Jarrow–Morton
HNW	High Net Worth
HWM	High Water Mark
IA	Independent Amount
IBOR	Interbank Offered Rate
ICE	Intercontinental Exchange
ICMA	International Capital Market Association
IIC	International Index Company
IOSCO	Board of the International Organization of Securities Commissions
IPO	Initial Public Offering

IRB	Internal Ratings-Based
IRR	Internal Rate of Return
IRS	Interest Rate Swap
ISDA	International Swaps and Derivatives Association
ITM	In-the-Money
KMV	Kealhofer, McQuown, and Vašíček
KO	Knock-Out
LGD	Loss Given Default
LGM	Linear Gaussian Model
LIBOR	London Interbank Offered Rate
LMM	Libor Market Model
LTV	Loan-to-Value
MAC	Market Agreed Coupon
MD	Modified Duration
MLF	Medium-term Lending Facility
MPT	Modern Portfolio Theory
MTA	Minimum Transfer Amount
MtM	Mark-to-Market
MVO	Mean-Variance Optimization
NAFMII	National Association of Financial Market Institutional Investors
NAV	Net Asset Value
NDF	Non-Deliverable Forward
NDS	Non-Deliverable Swap
NPV	Net Present Value
OBPI	Option-Based Portfolio Insurance
OIS	Overnight Indexed Swap
O/N	Overnight
OTC	Over-the-Counter
OTM	Out-of-the-Money
PCA	Principal Component Analysis
PD	Participating Dealer
PD	Probability of Default
PDE	Partial Differential Equation
PFE	Potential Future Exposure
PIDE	Partial Integro-Differential Equation
P/L	Profit & Loss
PnL	Profit & Loss
PR	Participation Rate
PR	Prime Rate
PV	Present Value
QFI	Qualified Foreign Institution
Repo	Repurchase Agreement
RFR	Risk-Free Rate
RICI	Rogers International Commodity Index
RWA	Risk-Weighted Asset

SAC	Securities Association of China
SAM	Sell above the Market
SARON	Swiss Average Rate Overnight
SBL	Securities Borrowing and Lending
SDE	Stochastic Differential Equation
SEF	Swap Execution Facility
SHIBOR	Shanghai Interbank Offered Rate
S/N	Spot/Next
SOFR	Secured Overnight Financing Rate
SONIA	Sterling Overnight Index Average
SPV	Special Purpose Vehicle
STIR	Short-Term Interest Rate
STM	Settlement-to-Market
T-Bill	Treasury Bill
TER	Total Expense Ratio
TIBOR	Tokyo Interbank Offered Rate
TIPS	Treasury Inflation-Protected Securities
TIPP	Time Invariant Portfolio Protection
T/N	Tomorrow/Next
TONA	Tokyo Overnight Average Rate
TRF	Target Redemption Forward
TRS	Total Return Swap
TWAP	Time-Weighted Average Price
WTI	West Texas Intermediate
VaR	Value at Risk
VIX	CBOE Volatility Index
VS	Variance Swap
VWAP	Volume-Weighted Average Price
ZC(B)	Zero Coupon (Bond)

Notation

W	week, e.g., 1W means a 1-week tenor
M	month, e.g., 3M means a 3-month tenor
Y	year, e.g., 1Y means a 1-year tenor
k	thousand, e.g., USD50k
m/mio	million, e.g., USD10m
a.s.	almost surely
bn	billion, e.g., USD1bn
bps	basis point, which is $0.01\% = 0.0001$
c.d.f.	cumulative distribution function
i.i.d.	independent and identically distributed
LHS	left-hand side of an equation
RHS	right-hand side of an equation
r.v.	random variable
s.t.	such that
p.a.	per annum
p.d.f.	probability density function
w.r.t.	with respect to
$a \wedge b$	$\min\{a, b\}$
$a \vee b$	$\max\{a, b\}$
$(X)^+$	$\max\{0, X\}$
$\cdot$	vector inner product
$e, \exp$	exponential function
$\ln, \log$	natural logarithm
$\partial_x f$	$\frac{\partial f}{\partial x}$
$\partial_{xy} f$	$\frac{\partial^2 f}{\partial x \partial y}$
$\mathbb{E}[X \mathcal{G}]$	conditional expectation given σ -algebra $\mathcal{G}$
$\mathbb{E}_t[X]$	$\mathbb{E}[X \mathcal{F}_t]$
T, t, s, u	$\begin{cases} \text{day count fraction, if it is used in calculation;} \\ \text{index for time, otherwise (e.g. as a subscript).} \end{cases}$

Part I

Financial Markets

Chapter 1

Introduction to Financial Markets



There are a few objectives in this book. The first is to present the main properties of basic derivatives such as forward/futures, swaps, vanilla and exotic options. The second is to provide a modelling framework within which those derivatives can be priced and hedged. The third is to introduce structured products, which are pre-packaged and sometimes tailor-made financial instruments designed to take advantage of specific market views.

1.1 Investable Assets and Financial Instruments

The assets for investors in the financial market are mainly liquid investable assets such as bonds, stocks, currencies, and commodities. The liquidity of an asset is measured by the ease and cost of converting it into cash. Not all assets are tradable (e.g., a bank deposit is not tradable).

A *security* is a tradable financial asset, often identified by a code given by a well-recognized institution.¹ The basic types of securities are *debts* and *equities*, which are directly linked to economic activities. Debt securities take the form of bonds, bills, etc. Equity securities issued by corporations are called *stocks* or *shares*, representing claims of ownership in the corporation.

A *financial instrument* is a general term for a monetary contract between parties. It can be either a *cash instrument*, whose value is directly determined by the market (e.g., stocks, bonds), or a *derivative instrument* (or simply, *derivative*),

¹ For example, the BNP Paribas SA stock, has a RIC (Reuters Instrument Code) of “bnpp.pa”, an ISIN (International Securities Identification Number) code of “FR0000131104” and a BBGID (Bloomberg Global Identifier) of “BNP FP Equity”. Terminals such as Bloomberg, Reuters, FactSet provide financial market data and analysis, news and other services, such as risk analysis, pricing and trading.

whose payoff is derived from the price or level of the underlying asset or reference (called *underlier* or *underlying* for simplicity) such as equity security, fixed-income security, foreign currency, commodity, or interest rate.²

The *long position* of an instrument realizes a profit if the price of the instrument goes up, and suffers a loss if the price drops. Holding a security is a long position of that security. Conversely, the *short position* of an instrument realizes a profit if the instrument's price drops, and suffers a loss if the price goes up. A short position of a security can be built by borrowing (not buying) the security at time- t_1 and selling it at market price³ S_{t_1} . To close the short position at a later time- t_2 , the security must be bought back at market price S_{t_2} and returned to the security lender. If $S_{t_2} < S_{t_1}$, the short position generates a profit.

The most common derivatives are *forward contracts*, *futures*, *swaps* and *options*. Generally, we denote S_t as the time- t price or level of the underlier, N as the notional amount and T as the expiration date of the derivative instrument.

A *forward contract* is an agreement between two parties in which the buyer will purchase a certain asset from the seller⁴ at a pre-agreed price, called strike price or delivery price, at expiration date T . Mathematically, the payoff for the buyer is $N \times (S_T - K)$, where K is the strike price. The strike price that makes a forward contract value equal to zero at inception is called the *forward price* of the underlier, which is often denoted by F .

A *futures contract* is similar to a forward contract, but it is listed on an exchange with standardized terms. Instead of trading directly between themselves, the long party and the short party of a futures contract trade with the exchange serving as the central counterparty who settles daily P/L (Profit and Loss) with the long and short parties, based on the futures' market price.

A *swap* is a derivative contract in which two parties agree to exchange cash flows or securities based on the notional amount over a pre-defined period. A basic interest rate swap (IRS) is a typical example in which one party agrees to pay cash flows at a fixed rate and receive cash flows based on a floating rate.

Roughly speaking, an *option* is a financial instrument that offers a return (called payoff or payout) that is asymmetric to the upside and downside performance of the underlier(s). The most basic types of options are call options and put options, which are also known as vanilla options. A *European call option* gives the owner the right, but not the obligation, to buy the underlier from the seller at expiry with a pre-agreed price K , known as strike price or exercise price. The payoff at expiration date T of a European call option is $N \times \max\{S_T - K, 0\}$. Similarly, a *European put option* gives the owner the right, but not the obligation, to sell the underlier at the strike price leading to the payoff of $N \times \max\{K - S_T, 0\}$ at expiration date T .

² Generally speaking, a derivative instrument can be linked to multiple underliers.

³ For market stability, most markets apply the *uptick rule* that requires short sale of a security to be conducted at a higher price than the previous trade price.

⁴ For an underlier that is not an asset, the wording "buying (selling) the asset" should be understood as "taking a long (short) position of the underlier". See forward rate agreement in Sect. 4.1.

Any option that is not a vanilla option is referred to as an *exotic* option, e.g., barrier option, Asian option.

A *structured product* is based on derivatives to offer a specific risk-return objective with an investment-made-easy packaging. The world has witnessed an extraordinary development for structured products since the late 1990s.

The *mark-to-market* (MtM) price of a financial instrument is the price at which the instrument can be traded in the market. The mark-to-market value of a financial instrument is the current fair value that is consistent with the market.

1.2 Investment Returns and Risks

The investment return of a non-dividend-paying asset (i.e., an asset that does not generate income such as dividends, coupons, security lending fees, etc.) for the period between time u and time t ($u < t$), is defined as

$$R = \frac{S_t - S_u}{S_u} \quad \text{or} \quad R = \frac{S_t}{S_u} - 1, \quad \text{where } S_t \text{ is the price of the asset at time } t.$$

A return rate is an annualized concept in general. There are various types of return rates used in finance. One such rate is *internal rate of return* (IRR), which represents the value of r that satisfies the following equality:

$$S_t = S_u(1 + r)^\tau, \quad \text{where } \tau \text{ is the number of years for the period } [u, t].$$

Generally, the return for a portfolio (a single asset is a special case) can be

a *price return*: it is measured by the portfolio's values at the beginning and the end of the period. The dividends, coupons or other payments during the period are ignored;

a *total return*: it is obtained by re-investing all the incomes generated by each asset (e.g., dividends, coupons) back into the asset itself (see Sect. 5.2.6);

an *excess return*: it is defined as the portfolio's total return minus the funding cost, representing the return of a self-financing portfolio.

Example 1.1 Consider two assets, A and B. Asset B pays a \$1 dividend per unit at time t . The financing cost is 1% on the borrowed amount.

	S_u	S_t	Price return	Total return	Excess return
A	19	20	5.26%	5.26%	4.26%
B	90	99	10%	11.11%	10.11%

The total return for B is $\frac{99+1}{90} - 1$, assuming that the \$1 dividend is reinvested. $\square$

For financial analysis or modelling, it is often convenient to manufacture a *total return asset* (also known as *non-dividend paying asset*) that provides total return prices by the method described in Sect. 5.2.5.

The most important risks for a financial product include

Market Risk: it is the risk on the value and return of the product caused by the changes of the market price of the related financial instruments and other financial factors;

Counterparty Risk: it arises when one of the parties fails to fulfill its obligation towards the other party;

Liquidity Risk: the uncertainty about the condition and ability to make, unwind or hedge a transaction.

The term *arbitrage* refers to transactions leading to a risk-free profit (see Example 6.1 for illustration).

1.3 Investment Performance Measures

There are many indicators used to measure the performance of an investment strategy or an investment fund besides its total return R . Denoting r as the risk-free rate⁵ or a relevant reference rate, the most popular performance measures can be introduced as follows:

Sharpe ratio $= (R - r)/\sigma$, where σ is the standard deviation of $(R - r)$. It is the most used measure for risk-adjusted excess return;

Information ratio $= (R - R_m)/\sigma$, where R_m is the return of the relevant benchmark (e.g., an equity index, if the investment portfolio is composed of stocks) and σ the standard deviation of $(R - R_m)$;

Sortino ratio $= (R - r)/\sigma^-$, where σ^- is the “downside deviation”, calculated with negative returns only.⁶ This measure is preferred by people who consider that the Sharpe ratio punishes the “good risk” as positive returns are also used in its standard deviation calculation;

Maximum drawdown: the worst peak-to-trough decline in percentage during the considered period for an investment, i.e., $\min_i \{m_i/M_i - 1\}$ where M_i is the i -th peak value in the period and m_i is the lowest value thereafter before achieving a new peak (i.e., with higher value than M_i).

There are two other well-known performance indicators, *Jensen's Alpha* and *Treynor ratio* that are based on the Capital Asset Pricing Model (CAPM). These will be presented in Sect. 32.2.1.

⁵ The *risk-free rate* is the return of a risk-free investment, which does not exist strictly speaking. It is approximated by the return of a collateralized debt or a government debt.

⁶ $\sigma^- = \sqrt{\frac{252}{n} \sum_{i=1}^n \min\{R_i, 0\}^2}$ where R_i is the return for the i -th business day, n is the total number of business days for the period, and 252 is the number of business days in a year.

1.4 Financial Markets

A financial market is a marketplace where financial securities (e.g., stocks and bonds), currencies, commodities and derivatives are traded at prices that reflect supply and demand. The participants include financial institutions (such as banks, insurance companies, pension funds, mutual funds, and hedge funds), individual investors and corporations. Financial markets can be classified by the type of service they offer:

capital market provides financing through the issuance of equity and debt securities, and enables the subsequent trading thereof. It includes:

- *stock market* allows the participants to buy and sell stocks of listed companies. The value of a stock reflects the view about its expected dividend payments, future earnings, and the resources that the company will control;
- *bond market* refers to the market for long-term (over 1 year) debt instruments such as government, corporate and other bonds;

money market refers to the market for trading short-term (up to 1 year) debt instruments such as interbank loan/deposit, treasury bill, certificate of deposit, commercial paper, etc.;

fixed-income market refers to the market for trading interest rate and credit related cash or derivative instruments in general, including bond market and money market;

equity market is the market for trading stock and equity index related cash or derivative instruments in general, including stock market;

foreign exchange market is the market for trading foreign exchange related instruments;

commodity market is the market for trading commodities instruments, such as precious metals, industrial metals, energy products and agricultural products. Futures contracts are the most convenient instruments for commodities trading activities. A futures contract may be sold out before the commodity is due to be delivered.

Financial markets can also be classified using other criteria, such as

primary market: where new issues are sold through IPOs (Initial Public Offerings). The primary market businesses for debt and stocks are covered respectively by the DCM (debt capital market) and ECM (equity capital market) divisions of the *corporate finance* entity in investment banks;

secondary market: whereby for all subsequent trading after IPO between market participants takes place. It provides the support for financial products used in investment and risk management.

When classified by market organization, there are

listed market: public auction market with a central counterparty (CCP) for trading and clearing, such as stock exchanges (e.g., Euronext, New York Exchange, Nasdaq, Hong Kong Exchange). The exchanges are usually private companies and may be listed on an exchange themselves. A CCP is the buyer to every seller and the seller to every buyer of a specified set of contracts;

OTC (over-the-counter) market: where buy/sell contracts are made between the two counterparties. OTC market is often animated by brokers as intermediary agents.

1.5 Central Counterparty (CCP)

To support financial market activities, there are service providers (e.g., exchanges, depository institutions, clearing houses) for trading infrastructure, securities custody, clearing and settlement, etc.

In particular, a CCP (*central counterparty*) is a financial institution with good creditworthiness (e.g., Eurex Clearing, Hong Kong Exchanges and Clearing) that takes on counterparty credit risk between the parties to a financial transaction and provides clearing and settlement services. When a CCP is involved in a transaction agreed by two parties, either directly or anonymously, the transaction is actually carried out by two legal contracts, one between the buyer and the CCP whilst the other one between the seller and the CCP, and there is no direct transaction between the two engaged parties. The CCP mitigates the counterparty risk of the parties by collateral management (and daily P/L settlement for exchange traded products). CCPs are highly regulated and should operate efficiently to maintain their creditworthiness for the stability of financial system.

1.6 Securities Lending and Repo

Securities lending and Repo are secured transactions (by collaterals), which play an important role in the financial system. Repo facilitates secured lending of cash, while securities lending supports short position implementation of securities and other corporate action related activities. A Repo transaction is motivated by the need to borrow cash with securities as collateral, while a securities borrow transaction is mostly driven by the need for security with a collateral that is often in cash. The borrowed amount (i.e., cash for repo, securities for SBL) remains unchanged during the life of the trade, but the collateral amount is subject to adjustment.

A **securities borrowing & lending (SBL)** transaction allows the lender to lend securities to the borrower on either an *open* (i.e., callable at anytime) or *term* (fixed tenor) basis. The lender loses the legal ownership of the securities to the borrower but enjoys the economic benefits such as dividends that are “manufactured” by the

borrower. Upon the trade termination, the securities will be returned to the lender.⁷ The borrower posts collateral with daily adjustment and pays fees to the lender. The fee rate depends on lending supply/demand for the underlying security. The eligible collaterals can be cash or other securities as agreed by the parties. The cash collateral level is usually $\sim 105\%$ of the latest closing price of the security. The International Securities Lending Association (ISLA) has developed a standard **master agreement** (see Sect. 2.4.1) called *Global Master Securities Lending Agreement (GMSLA)*, which is used by most of the institutions. The motivation for a security borrower includes short position recovering, hedging of derivatives, corporate action arbitrage (such as scrip dividend, see Sect. 5.2.1), etc. While securities lending is mostly carried out as an OTC transaction, it may be facilitated by a central counterparty in some markets (e.g., China Securities Finance Corporation).

In a **sale and repurchase agreement (Repo)**, one party (repo party or collateral poster) borrows money while providing collaterals that are typically fixed-income assets (see Fig. 1.1). The repo party gains access to funds at a lower interest rate than are available elsewhere, due to the collateral. The eligibility and haircuts⁸ for collateral are negotiable between the repo counterparties. The standard master agreement for repo is *Global Master Repurchase Agreement (GMRA)*, published by the International Capital Market Association (ICMA). A reverse repo is the transaction seen by the other party (cash lender) of a repo trade. While repo and reverse repo are actively used by financial institutions, central banks also use them for open market operations to influence the money supply in the financial system.

If the collateral is held at a third party, usually a custodian bank, clearing institution or a central securities depository, the transaction is called a *Tri-Party Repo*. The third party often provides services in addition to custody, such as valuation and adjustment of the collateral. The residual risk in a tri-party repo is

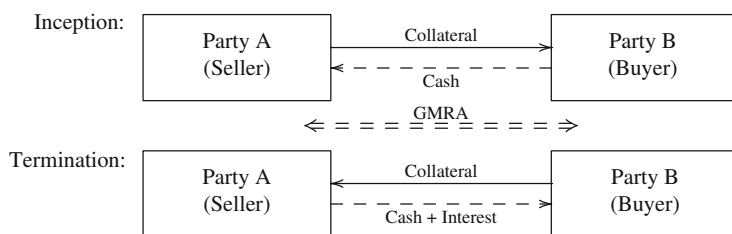


Fig. 1.1 Illustration of a Repo transaction

⁷ In case of failure-to-deliver by a party, stock exchanges may provide the “buying-in” service that consists of buying the securities in the market to be delivered to receiving party. The original delivering party has to pay the incurred cost. A similar practice, termed “selling-out,” prevails when a purchaser fails to take up his securities.

⁸ The haircut on an asset refers to the lower-than-market value of the asset, which reflects the perceived risk of the asset’s price drop in a sale or liquidation. Instead of applying hair-cut, a repo trade can be equivalently implemented with an initial margin concept as described in [55].

the correlation between the default risk of the collateral poster and the value of the collateral in custody.

Besides OTC repo market, a CCP may also provide a repo market (e.g., Eurex) where the CCP is the central counterparty and determines the collateral pool and haircuts,⁹ standardizes some of the contract features such as minimum hair-cut, and facilitates clearing and pledge of collateral.

1.7 Derivatives Activities

Derivative instruments are used by end-users for various purposes, including but not limited to

Gaining exposure to a specific view: derivatives can be used for gaining exposure to a specific view on future price changes. The simplest views are “bullish” (i.e., rising trend) and “bearish” (i.e., falling trend) views on the price of the considered financial instrument. However, there could be more complex views that derivatives may take advantage of.

For example, if one believes that the price of an equity index will move significantly in either direction at a certain time horizon, then buying both a call and a put at the same time (eventually with out-of-the-money strikes to reduce cost, see Sect. 10.1.2) can be a viable trading strategy for this view;

Hedging: which means offsetting or reducing the impact of market factors or events on the value of the considered asset or product.

For instance, a company is expected to convert a cash flow in Euro into US dollars in 3 months. To mitigate the currency exchange rate risk, the company enters into a 3-month EUR/USD forward transaction to sell the cash flow in EUR into USD at a fixed exchange rate. So, at any time until the expiry of the forward contract, the change in USD value of the future cash flow in Euro will be offset by the value change of the forward contract.

Leverage: derivatives allow leveraged exposure to the underlier, e.g., futures and options require much less fund (respectively, margins and option premium) than the notional of the exposure.

⁹ Such a collateral pool is called GC (General Collateral) basket. The securities in a GC basket are defined by the security types (not name specific) and can be substituted for one another without changing the repo rate. Such a repo is known as a *GC repo*.

Remark 1.1 (Determination of Hedging Ratio q_t) Generally speaking, to hedge a portfolio of value V_t at time- t against the impact of a certain market factor x_t (e.g., stock price or volatility) by a small value change ϵ with an appropriate self-financing derivative or cash instrument position of value D_t (e.g., stock, futures, swap, option), one can build a position with q_t instruments such that:

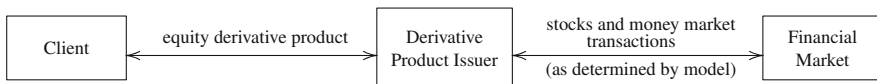
$$\Delta V_t(x_t) + q_t \Delta D_t(x_t) = 0,$$

where $\Delta V_t(x_t) = V_t(x_t + \epsilon) - V_t(x_t)$ and $\Delta D_t(x_t) = D_t(x_t + \epsilon) - D_t(x_t)$. Positive (negative) sign for q_t means long (short) position. q_t can be static (for portfolio with linear risk to x_t) or dynamically adjusted (for options). $\square$

Derivatives are a double-edged sword: they can help hedge out risks, but they may also allow the user to increase risk exposure. The risk of some derivatives may be complex and difficult to understand for non-sophisticated investors. Regulators for the financial industry, as well as financial institutions, keep implementing measures to prevent selling the products that are not matching the investors' investment/risk profiles.

Professional investors and traders usually use liquid and relatively simple derivative instruments such as forwards, futures, swaps and vanilla options for turnaround flexibility and lower transaction costs. Less sophisticated or passive investors usually make buy-and-hold investments and are more attracted by structured products that provide easy transaction management and flexibility to take advantage of specific and sometime customized market views.

Investment banks, as well as some other financial institutions (i.e., product issuers), offer derivatives and structured products as a service to clients (end-users) and earn commissions for the service. To offset the incurred profit and loss with clients, the issuer *replicates* or *hedges* the product payoff by simple transactions on the underlying securities or other instruments in the market, with the timing and quantities as determined by a mathematical model. It is a common misunderstanding that the derivatives issuer and the client are playing a zero-sum game where the client makes a profit when the issuer suffers a loss and vice versa. The following chart illustrates the replication of a derivative product linked to the performance of stocks:



The price of a hedgeable derivative product is based on its replication cost, while the price of a basic asset such as stock, currency, commodity is driven by its long-term fundamentals (see [90]) and supply/demand. The pricing and hedging of derivatives are based on the same techniques, which are analyzed in this book.

Exercises

1. An order book contains a list of buy (known as bid) orders with corresponding buy sizes and sell (known as ask) orders with corresponding sell sizes raised by market participants. The following is an order book for security A.

Bid volume	Bid price	Ask price	Ask volume
500	100	101	500
1000	99	102	500
1500	98	103	1000
$\vdots$	$\vdots$	$\vdots$	$\vdots$

An investor wants to buy 2000 shares of A using a *market order*, i.e., regardless of price level. Calculate the average price of the total positions in security A.

2. Let S_t be the price of a non-dividend paying asset at time t . Find out under what condition the *log-return* defined as $\ln(S_t/S_u)$ for $u < t$ is a close approximate of the standard return $(S_t - S_u)/S_u$.
3. Explain the difference between repo and securities transactions in terms of the purpose, collateral management and documentation.
4. Explain the risk for the investor in a tri-party repo based product.
5. Consider a new private fund with an initial NAV (net asset value) of 100. The investment started on 2 Jan 2017 and is fully invested in Stock A. On 15 April 2017, the fund switched all the investment into Stock B. On 3 July 2017, the fund paid a cash dividend of 3% of the NAV. The interest rate is 3% applicable with the formula $3\% \times \text{number of days}/365$ to calculate the interest. To simplify the exercise, we make the assumptions of zero transaction cost/management fee and that we can trade a fraction of a stock. The stock prices are given as follows

Date	Price of stock A	Price of stock B
2 Jan 2017	20	180
15 Apr 2017	22	160
3 Jul 2017	21.5	190
15 Sep 2017	23	198

Calculate the following values at 15 Sep 2017: (1) Fund NAV; (2) Total Return; (3) Price Return; (4) Excess Return.

6. Suppose the daily returns of an investment strategy are given by

$$\mathbf{r} = [2.5986\%, -0.4661\%, -1.1829\%, -1.3517\%, 0.7311\%]$$

for a certain week. Calculate the maximum drawdown for this period.

Chapter 2

Financial Transactions and Counterparty Risk Management*



In this chapter, we introduce the basic concepts related to the life cycle of derivatives transactions. Counterparty risk mitigation is a key target for all the transaction management approaches that will be presented in the following sections.

2.1 Concepts in the Life Cycle of a Financial Transaction

The buy/sell transaction life cycle of a cash security such as bond or stock is characterized by order placing, execution and clearing/settlement. For a derivative trade, in addition to the operations for cash securities, there are valuation, cash flow/securities transfer, and corporate actions to handle until the trade termination. The following concepts are related to the life cycle of a derivative trade.

order type: the condition to buy/sell securities in the market. The usual ones include limit order, market order, VWAP (volume weighted average price), TWAP (time weighted average price), etc.;

trade size: the amount, either in currency units or in securities units, used to determine the payment amount or the quantity of securities to deliver. Several terms are used without a clear distinction criteria: *face value* or *denomination*, often used for securities; *notional* or *nominal* (amount), usually used in derivatives contracts without payment of the amount; *principal* (amount), when there is payment of the amount;

trade date: the calendar date on which an order is executed, or a financial contract is concluded. The initial reference and strike prices are determined upon order execution.

tenor: the life time of a trade, e.g., 3 months, 2 years, etc. In money market, there are one business day tenor transactions including O/N (overnight), T/N (tom/next, i.e., from next business day to the following one), S/N (spot/next, i.e., from spot

- date to the following business day). The spot date for most of the currencies is $T + 2$ whereby T stands for trade date;
- value date*: the date on which the currency or security units involved in a transaction take effect or change ownership;
- expiration date* or *termination date*: the last day to exercise the right in a derivative instrument. It is often used as the last valuation date;
- maturity date*: on which the validity of a financial instrument ends;
- payoff* or *payout*: the realized cash flows or delivered securities for the long party of a derivative instrument during the whole period after its inception until the termination date;
- fixing date*: the date on which the price, rate or other reference level to be used in financial calculation is determined;
- clearing*: it is the process of establishing final positions for settlement, including confirming instructions, reconciliation, netting, etc. It is usually conducted by a clearing house;
- netting*: payment netting refers to the process of combining several payments of cash or securities between two counterparties to achieve an aggregated net payment (which will result in a reduced amount if there are offsetting payments). Close-out netting determines the net obligations of a defaulting counterparty and allows the non-defaulting party to deduct legitimate amounts from its obligations to the defaulting party;
- settlement*: the delivery of securities or fund units; *DvP* (delivery versus payment) method ¹ has the least settlement risk; *FoP* (free of payment) as well as other settlement methods² are used in some circumstances. The settlement date lag (see [110]) usually does not constitute a constraint to active trading, as investors can sell the purchased but not yet settled securities on the same day; ³
- custody*: safekeeping of securities by a custodian bank or a depository institution; A depository institution often offers the services of custody, clearing and settlement. The custody of securities can be done at several levels, through global custody, local custody, or sub-custody;
- collateral*: assets (cash, securities, etc.) serving as guarantee for a liability;
- corporate action*: usually initiated by the issuer of a security (a stock or a bond), a corporate action normally has ownership or financial impact. Examples include dividend, rights issue, merger, stock split, and shares buy-back;
- valuation*: the process of determining the mark-to-market price for accounting, termination or unwinding payments, counterparty risk exposure, etc. For an OTC transaction, a calculation agent must be specified to perform the valuation. For structured products, the product provider is usually the calculation agent;
- valuation date*: on which a valuation is carried out.

¹ CLS is an international organization providing PvP (payment vs. payment) service for currencies.

² such as PAD-payment after delivery and DAP-delivery after payment, ...

³ There are exceptions. At the time of writing this book, stocks cannot be sold if they are not yet settled in the Chinese stock market by regulation. So, investors cannot buy and sell the same stock on the same day.

Example 2.1 In the description of a structured note, we may see terms such as the note issuer, note denomination, payments description, calculation agent, and the following dates in its life cycle:

- trade date: on which the note investor’s buy order is executed with the initial reference price determined and the initial delta hedging (see Sect. 10.3.5) done by the product’s hedging party. The transaction starts to exist from this date;
- issue date: (e.g., trade date + 2 weeks) on which the initial payment is made by the investor and the note is issued with a security code such as ISIN;
- observation dates or fixing dates: on which the underlier price references for the note payments calculation are determined. The last observation date is often called *final valuation date*;
- maturity date: (e.g., final valuation date + 1 week) the final settlement date where the final payment or delivery is done, which marks the termination of the note.

□

2.2 Margining Methods for Mitigating Counterparty Risk

An OTC derivative involve potential payments between the parties in the future. To mitigate the *counterparty credit risk* that one party does not pay as obligated, there are two types of margins that are required for derivatives transactions between the counterparties (the counterparty with no future liability, such as an option holder, will not post margin.):

initial margin: the collateral taken at the inception of each trade for mitigating close-out risk in the event of default. It is based on the analysis of adverse price movements during the close-out period, typically set to be 10 business days as market practice (see [56] for the initial margin approach in ISDA SIMM™);

variation margin: the amount received or paid daily, reflecting the market-to-market value change of the derivatives and collaterals. There are two models for variation margin:

- *settlement-to-market* (STM): under this model, the MtM exposure is settled with effective payment on daily basis. It is generally used by the exchanges;
- *collateralized-to-market* (CTM): it consists of posting collateral to the positive MtM party. Any income generated by the collateral belongs to the collateral poster. Upon close-out, the non-defaulting counterparty may set off collateral it has received from the defaulting party against the owed amount. The contracts under ISDA/CSA generally follow the CTM model currently.

These two variation margining models lead to not only different ways to operate and to document but also different quantitative modelling approaches (see Sect. 16.9) and potential adjustment to align product economics (see PAI concept in Sect. 4.3.6).

2.3 Exchange-Traded Derivatives*

Exchange-traded derivatives include futures and option contracts on various asset classes. For all contracts listed at an exchange, a central counterparty (CCP) is in the middle of each transaction anonymously agreed by a buyer and a seller (in other terms, between the long and short parties). See Fig. 2.1.

To mitigate counterparty credit risk, the CCP applies a margining mechanism that has proven to be resilient even during severe financial crisis in the past. To illustrate the margin management, we take the case of futures contracts:

- The buyer and seller should have their “margin account” in place before trading futures;
- At the inception of each trade, both parties will deposit an “exchange initial margin”, which is typically a cash amount determined by the CCP according to the type of the futures;
- At the end of a trading day, each party will have its margin account debited or credited for the daily P/L.⁴ Such daily settled amount is a real P/L rather than a collateral;
- A minimum margin level called “maintenance margin”⁵ may be set for every margin account. If the account value is below this level, a “margin call” will be issued to bring back the account to the level of “exchange initial margin” which provides a cushion to avoid frequent margin transfers.

In general, exchange listed derivatives have the following advantages over OTC derivatives:

- low counterparty risk
- price transparency
- low transaction cost
- high liquidity
- leveraged exposure
- for financial institutions, more favorable treatment on risk capital requirement

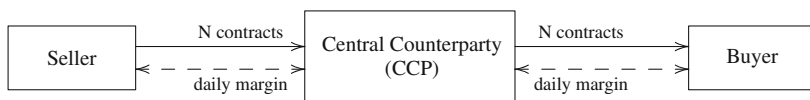


Fig. 2.1 Illustration of margining system

⁴ The daily P/L corresponds to *variation margin* concept.

⁵ Actually, “maintenance margin” concept is closer to *initial margin* presented in the previous section. The “exchange initial margin” here is equivalent to *initial margin* plus a buffer for *variation margin* (buffer = “exchange initial margin” – “maintenance margin”).

2.4 OTC Derivatives*

OTC derivatives products offer flexibility and customization to satisfy the financial needs of investors and other market participants.

2.4.1 OTC Derivatives Documentation*

Financial institutions offering OTC derivatives products and services monitor and control their counterparty risks. Some products such as structured notes imply one-way counterparty risk where the end investor takes the counterparty risk of the financial institution, but the financial institution has no risk from the end investor. However, in some other products such as forward and swap, both parties are taking counterparty risks.

The growth of OTC derivatives business prompted the standardization of contract terms and counterparty credit risk management to be used by institutions and corporates active in the financial market. In general, two parties set up their trading relationship for OTC derivatives by negotiating the applicability and necessary adjustment to the standard terms and conditions of a *master agreement* template. The mutually signed master agreement will normally govern all the bilateral transactions, and the confirmation of each derivative transaction only requires a short-form document known as a *trade confirmation*, *term sheet* or *transaction/pricing supplement* that contains only the specific terms and conditions of the transaction.

The *ISDA (International Swaps and Derivatives Association) Master Agreement*, initially developed in the 1980s to cover the IRS and currency swaps, has been progressively updated to include credit derivatives as well as other derivatives such as forward, swap and option linked to equities, commodities and funds. It sets out standard terms applied to all trades between the two parties.

The *CSA (Credit Support Annex)* outlines the terms for credit support (i.e., collateral) in OTC derivatives transactions, including acceptable collateral and haircuts. The “independent amount (IA)” of a trade is the *initial margin* (collateral) required by one party from the other,⁶ which is returned upon the trade’s termination. The IA level depends on trade volatility and counterparty creditworthiness. *Variation margin* is exchanged for consolidated exposure of all bilateral trades. The “threshold” and MTA (minimum transfer amount) for variation margin are specified

⁶ Usually, between dealers, both parties post initial margins while, between the dealer and the end user, it is common that only the end user posts initial margin.

in the CSA.⁷ A *margin call* is triggered for one party if $(\text{positive net exposure} - \text{threshold} - \text{collateral held}) > \text{MTA}$.

Normally, the credit agreement between two financial institutions is a two-way CSA in which both parties may post margins for their OTC trades. Between a financial institution and a corporate, the credit agreement may be a one-way CSA, meaning that only the corporate posts margins to the financial institution.

Example 2.2 Given a two-way ISDA/CSA agreement between party A and party B with Threshold = USD1m, MTA = USD500k, if Net Exposure = USD1.6m and Collateral Held = 0 for party A, a margin call will be triggered, and the margin amount will be USD600k to be paid by party B. □

Individual investors, small corporates and other non-active entities involved in financial market normally do not have master agreement with financial institutions and trade with a bespoke agreement or a long form confirmation that contains all terms and conditions for each trade.

The master agreement set ISDA/CSA is widely used internationally while some jurisdictions have their own documentation for OTC transactions, e.g., in China, NAFMII and SAC⁸ have established master agreement templates for transactions in China interbank market and exchange market respectively.

2.4.2 Centralized Clearing*

The 2008 financial crisis revealed weakness in risk management and transparency issues for OTC derivatives. In response, the Group of Twenty (G20) initiated a reform programme in 2009 (see [39]) that comprised the below

- all standard OTC derivatives should be traded on exchanges or electronic platforms and cleared through CCPs;
- OTC derivatives should be reported to trade repositories;
- non-centrally cleared derivatives should be subject to higher capital requirements.

For centrally cleared derivatives, a CCP acts as principal for each transaction facing its clearing members representing the long and short parties of the transaction. The clearing members of a CCP provide the transaction, clearing and settlement services to their customers. In general, a non-clearing member can only participate

⁷ Post 2008 financial crisis, most bilateral counterparties amended the threshold to be zero, which simplifies the counterparty credit risk management, in particular the CVA (credit value adjustment) calculation and hedging.

⁸ NAFMII (National Association of Financial Market Institutional Investors) is the self-regulatory organization for China Interbank Market; SAC (Securities Association of China) is the self-regulatory organization for Chinese securities market.

in the market through a clearing member. But if an OTC trade to be centrally cleared is agreed by the parties, it must be immediately brought to a CCP (through clearing members). The OTC trade is then split into two transactions with each party facing the CCP only.⁹ This process is known as *novation*.

A CCP mitigates its counterparty credit risk by requiring collaterals from clearing members, i.e., *initial margin* at each trade inception and regular *variation margins* to cover the exposure of mark-to-market value changes. Further measures for risk mitigation include trades netting and compression,¹⁰ limit management, etc., accompanied by the appropriate documentation for the contracts. Defaults are made less likely to happen and less severe if they ever happen. The implementation by each CCP may vary slightly in the details. Generally, a default shortfall beyond the margins is first drawn from the defaulting clearing member's contribution in the *default fund* (also known as clearing deposit or guaranty fund) in which, each clearing member should make its contribution. For further shortfall, CCP will follow a waterfall principal to draw from its dedicated fund, then the default funds of non-defaulting members, and ultimately the CCP's capital.

2.4.3 Non-Centrally Cleared Derivatives*

In 2011, G20 called upon BCBS (Basel Committee on Banking Supervision) and IOSCO (Board of the International Organization of Securities Commissions) to formulate the standards for the margin requirement for non-centrally cleared OTC derivatives, which led to the report: *Margin requirements for non-centrally cleared derivatives, April 2020*. The report recommends that all financial firms and systemically important non-financial entities that engage in non-centrally cleared derivatives must exchange initial and variation margin to mitigate counterparty risks. An initial margin is to be posted by each party to its counterparty (netting set) for each new trade with an amount corresponding to its potential future exposure (PFE)¹¹ increment, and it is subject to adjustment overtime for significant PFE changes. Then, on a regular basis, typically daily, a variation margin (mostly in cash) is exchanged with the necessary amount to fully collateralize the mark-to-market exposure already incurred by the parties. In general, the revenue from initial margin and variation margin, either cash or securities, belongs to the margin poster.

⁹ If the parties are using two different CCPs with an interoperability arrangement, the OTC trade will be split into three transactions with a trade between the two CCPs in addition.

¹⁰ Trade compression refers to the process of eliminating or reducing the size of OTC derivatives positions by terminating offsetting trades or replacing them with a reduced size or number of trades.

¹¹ BCBS/IOSCO recommends 99%, 10 day VaR (Value at Risk) as PFE, which is used in ISDA SIMM™.

2.5 Risk Management for Investment Financing*

Financial institutions often offer financing to their clients for investment through various implementations for counterparty credit risk management. We present here two practical examples.

A securities company offers margin trading service whereby, a client can buy securities on margin with the financing from the securities company. For the risk management of a client portfolio of securities and cash, the securities company monitors the collateral ratio R (= mark-to-market value of all the assets in the portfolio / total liability) with a few levels: *alert level* | *margin call level* | *liquidation level* (e.g., 150% | 140% | 130%, meaning that if $R < 150\% | 140\% | 130\%$, there will be actions for, respectively, *alert* | *margin call* | *liquidation*). A margin call requires the client to bring back the ratio R above alert ratio by posting more collateral or reducing the exposure.

A private bank allows their clients to make investment with leverage (e.g., investment in bonds, stocks, derivatives or structured products). The financing is applied through a *credit facility* for the client account based on a signed *deed of charge and assignment*. Each investment in the account is given three ratios: *Financing Ratio* (FR) | *Margin Call Ratio* (MCR) | *Liquidation Ratio* (LR) (e.g., 70% | 75% | 80%).¹² Let V_i be the mark-to-market value to the client for the i -th item in the account that can be an asset, a derivative transaction or a loan. V_i can be positive (e.g., a stock) or negative (e.g., a loan). Denote V_i^+ as the value of V_i if it is positive, and V_i^- as the absolute value of V_i if it is negative¹³. $\sum V_i^-$ is the client's liability. Define $L = \sum(V_i^- + \text{add-on}_i)$ where add-on_i is a conservative buffer to contain the volatility of V_i^- . Then, the client may get financing from the bank up to $\sum(V_i^+ \times \text{FR}_i) - L$. The bank may be willing to increase the financing for hedged or partially hedged positions (e.g., a single stock + long put option on the stock), and by considering the risk diversification effect if the account contains a large number of assets. A margin call is triggered if $L > \sum(V_i^+ \times \text{MCR}_i)$ and liquidation is triggered if $L > \sum(V_i^+ \times \text{LR}_i)$.

Exercises

1. Explain the role of a central counterparty in transactions and highlight the advantage of centrally cleared transactions over non-centrally cleared transactions.
2. What are the purposes of ISDA and CSA documentation? How can counterparties make a transaction without ISDA and CSA agreements?

¹² The bank may apply 100% FR to the cash in the account's currency and 0% FR to the investments that the bank is not willing to finance.

¹³ Mathematically, we have $V_i^+ = \max(0, V_i)$ and $V_i^- = -\min(0, V_i)$

Chapter 3

Interest Rate Instruments: I



A fixed-income instrument, such as a debt security, generally provides income on a fixed schedule, although the income amount can vary. The money market instruments such as treasury bill, commercial paper (CP) and interbank deposit are the basic short-term fixed-income instruments. The basic long-term fixed-income instruments include bonds, notes, and asset-backed securities (ABS). In this section, we will present the main concepts related to fixed-income instruments.

3.1 Interest Rate Conventions

3.1.1 Day Count Convention

A day count convention determines the time period *in years*, known as *day count fraction*, to be used to calculate the interest for an investment in financial instruments. It is also involved in calculating the present value of financial assets through discounting. Generally, the first day ($d_1/m_1/y_1$) of the considered period is included and the last day ($d_2/m_2/y_2$) is excluded.

- **30/360 methods** are a class of calculation methods with the following general formula

$$\text{Day Count Fraction} = \frac{360 \times (y_2 - y_1) + 30 \times (m_2 - m_1) + (d_2 - d_1)}{360},$$

where y , m , and d stand for calendar year, month, and day respectively. A non-exhaustive list includes 30/360 US (bond basis), 30E/360 (Eurobond basis), in which different treatments of d_1 and d_2 are involved.¹

¹ e.g., in 30/360, if d_1 is 31, then change d_1 to 30. If d_2 is 31 and d_1 is 30 or 31, then change d_2 to 30. But under 30E/360, if d_1 is 31, then change d_1 to 30. If d_2 is 31, then change d_2 to 30.

- **ACT methods**, where ACT stands for *actual*, utilize the number of days between two dates, denoted by $d_2 - d_1$. For instance,

ACT/360 (money market basis): $(d_2 - d_1)/360$

ACT/365 (English money market basis): $(d_2 - d_1)/365$

ACT/ACT ISDA: $(\text{days in a non-leap year})/365 + (\text{days in a leap year})/366$.

3.1.2 Business Day Convention

A business day convention specifies the adjustment when a date used in calculation is not a business day. The most common ones are

- *Following*: the following business day;
- *Preceding*: the preceding business day;
- *Modified Following*: the following business day, unless the day is in the next calendar month, in which case the adjusted date is the preceding business day. It can avoid the cash flow to be put into the next fiscal year;
- *End-of-Month*²: if the start date of a period is on the final business day of a particular calendar month, then the end date is on the final business day of the end month.

3.2 Interest Rate Types and Zero Coupon Bond

An interest rate represents the time value of money, measured as the value change per annum (e.g., 2.10% p.a.). It is the cost of borrowing money or the return for lending money. There are several types of interest rates. Some of them are used in financial contracts to determine cash flows, and some others, e.g., the continuous interest rate, are not directly used in real financial contracts and have been introduced for comparison and calculation purposes.

In the follows, we denote M as the initial amount at time-0 and the M_T for the terminal amount at time- T . We note that the symbol T in the calculation formulas represents the day count fraction for the period $[0, T]$.

² e.g., if the start date is 29-Apr-2011 that is the last business day of the month, then the end date for 1-month period will be 31-May-2011.

3.2.1 Simple Rate

Simple interest rate (r_m), known as *money market rate*, is applied to determine the terminal amount M_T according to the following formula:

$$M_T = M(1 + r_m T)$$

The short-term interest-bearing instruments such as loan/deposits and certificates of deposit use this type of interest rate.

On the other hand, a non-interest-bearing instrument in money market such as a treasury bill or a commercial paper is quoted in price for receiving a fixed amount at maturity. The implied rate measuring the discount is called *discount rate* (r_d), which satisfies: $M = M_T(1 - r_d T)$.

3.2.2 Compound Rate

A *compound* rate assumes re-investment of the interests from an investment. Suppose you deposit M in a bank at the interest rate of r with annual payments. At the end of year one, the bank account value will become $M(1 + r)$. After 2 years, you will have $M(1 + r) \cdot (1 + r) = M(1 + r)^2$. After T years, your account value will be $M_T = M(1 + r)^T$. If it is compounded m times per annum (i.e., the interests are paid m times a year), the terminal value of the investment after T years will be

$$M_T = M \left(1 + \frac{r}{m}\right)^{m \times T}. \quad (3.1)$$

3.2.3 Continuous Rate

The pricing of financial derivatives often involves continuous-time modelling, which uses ACT/365 day count convention in general. The interest rate used in continuous-time modelling is a compound rate applied to infinitesimal time intervals. In formula (3.1), if we apply compounding with very small time intervals of Δt per annum and denote $n = T/\Delta t$, we obtain the continuous interest rate r as the limiting case:

$$M_T = \lim_{\Delta t \rightarrow 0} M (1 + r \Delta t)^{T/\Delta t} = \lim_{n \rightarrow +\infty} M \left(1 + \frac{rT}{n}\right)^n = M e^{rT}$$

Alternatively, we can assume the differential equation $dM_t = r M_t dt$, which is the limiting expression of $\Delta M_t = r M_{t-\Delta t} \Delta t$ with $\Delta t \rightarrow 0$. Its resolution gives the same result: $M_T = M e^{rT}$.

More generally, if the continuous rate is a function of time, the term *instantaneous interest rate* will be used with the notation r_t . The resolution of $dM_t = r_t M_t dt$ over the time period $[t, T]$ leads to

$$M_T = M_t e^{\int_t^T r_s ds}.$$

In particular, the constant rate $r_t = r$ can be considered as a special case:

$$M_T = M_t e^{r(T-t)}.$$

where $T - t$ represents the year fraction for the period $[t, T]$.

3.2.4 Conversion of Interest Rates

One can easily transform the interest rates of different types and day count conventions through the equivalence in cash flow value. For instance, the continuous rate r that is not directly observable in the market-traded instruments can be derived from the money market rate r_m by the below no-arbitrage relationship

$$1 + r_m T_m = e^{rT}, \text{ or equivalently } r = \ln(1 + r_m T_m)/T,$$

where T_m is the day count fraction under the day count convention of the money market instrument while T usually follows ACT/365.

Due to the interchangeability of interest rate types, we are indifferent regarding the choice of rate type and convention for financial calculation.

3.2.5 Zero Coupon Bond

A *zero coupon (ZC) bond* (also called *discount bond*), denoted by $P_t(T)$, represents the time- t value of an instrument paying out \$1 value at maturity T . To lighten the notation, we will use $P(T)$ for $P_0(T)$ (time-0 usually refers to the spot date). Its value can be used as a *discount factor* to calculate the present value of a future cash flow. Zero coupon bonds constitute the building blocks for financial calculation.

Zero coupon bonds for different maturities can be extracted from market-traded instruments (see Sect. 4.4). Conversely, the interest rate for a specific instrument can be represented by zero coupon bonds. Since the interest rates in the instruments may use different rate types and day count conventions, it is convenient to use zero coupon bonds for financial calculation that are independent of any conventions.

The current market price (or fair value) of a future cash flow M_T is called the *present value (PV)* of M_T . It can be expressed as

$$PV = P(T) \times M_T.$$

Remark 3.1 (ZC bond v.s. Loan/Deposit) As the zero coupon bonds issued by real entities have limited supply, a long/short position of a ZC bond can be implemented with a money market loan/deposit by the relation $P(T) = \frac{1}{1+rT}$: a loan (deposit) of \$1 is equivalent to a short (long) ZC bond position of $1 + rT$ face value. Conversely, a short (long) ZC bond position of \$1 face value can be implemented with a loan (deposit) of the amount of $\frac{1}{1+rT}$. $\square$

3.3 Money Market Instruments

Money market instruments are short-term fixed-income instruments with tenor up to 1 year, typically including

- Interbank Deposit: issued by a deposit-taking institution such as a bank or a credit union; Deposit is not a tradable instrument.
- Treasury Bills (T-Bills): short-term government debt;
- Certificate of Deposit (CD): short-term debt issued by a bank;
- Commercial Paper (CP): short-term debt issued by a corporate;
- Repo: a collateralized loan that is normally not tradable.

3.4 Reference (Floating) Rates in Financial Market

There is a need for public references (i.e., *floating interest rates*) for the interest rate levels at a given time based on market-quoted instruments (e.g., interbank deposits) applied to homogeneous groups in terms of creditworthiness and profiles (e.g., retail, corporates, financial institutions). The daily time series of a reference rate forms an index that is published by the index administrator.

These reference rates, in particular the ones derived from interbank loan/deposit market,³ are then used in financial instruments and derivatives to determine the cash flows to be paid or received. The banks and mortgage lenders usually apply their own reference rate called PR (prime rate) to their creditworthy clients for lending.⁴ A PR may be set as a fixed value or as $R + \text{spread}$ where R is another industry loan reference.⁵ In the sequel, we introduce some of the most used reference rates.

³ It is a market for short-term loans/deposits between financial institutions (i.e., banks, insurance companies, investment funds, pension funds) for various purposes including financing, regulatory reserve and liquid management.

⁴ The lending rate to a particular client is PR plus (or minus) a specific spread to that client.

⁵ e.g., the LPR (Loan Prime Rate) in China uses the reference rate MLF (medium-term lending facility), which is the lending rate from the central bank to commercial banks.

3.4.1 *Front-Fixed Reference Rates*

The level of a front-fixed interest rate is determined and published *at the beginning* of the loan period. A front-fixed reference rate is a forward-looking rate derived from sources such as estimations submitted by contributor banks, market transactions or quotations, as determined by its methodology. There are mainly two types of front-fixed reference rates in the interbank market.

IBOR (Interbank Offered Rate)

IBOR is a generic name for a reference rate representing the average rate at which contributor banks (known as panel banks) can obtain unsecured funding (i.e., without collateral/guaranty) in a certain interbank market for a given period and currency. Such a reference rate reflects the averaged credit spread of contributor banks. Examples include LIBOR (London Interbank Offered Rate), EURIBOR, SHIBOR (Shanghai), TIBOR (Tokyo). The value of an IBOR is based on the estimated rates from the contributor banks. The most popular IBOR used to be LIBOR, published at 11:55am London time by ICE (Intercontinental Exchange),⁶ which has served as the main reference for hundreds of trillions of dollars of financial transactions. Some major currencies (namely, USD, EUR, GBP, JPY, CHF) are covered for tenors up to 1 year.⁷ The original fixing mechanism of LIBOR has shown some weaknesses such as lack of representativeness (not enough real transactions to support panel banks' estimation) and risk of manipulation by the panel banks, which has contributed to the decision of progressive cessation of LIBORs by 2023. The replacement of LIBORs are rear-fixed rates based on real transactions.⁸

We note that some IBORs such as EURIBOR will continue to be published with improved fixing mechanism using a “waterfall” methodology that prioritizes the implied rates from real transactions.

Repo Rate

In China, there are forward-looking reference rates based on interbank repo transactions made between 9:00am–11:30am and published daily after 11:30am. Currently, the covered tenors are 1 day, up to 7 days and up to 14 days (FDR001, FDR007, FDR014). See [88] for interest rate references in China.

⁶ LIBOR has been published daily since 1986. Its administration was transferred from BBA (British Banker's Association) to ICE in 2014. The improved methodology can be found in [54].

⁷ e.g., on 24 Jan 2022, ICE published USD LIBOR for the tenors: O/N, 1M, 3M, 6M and 12M.

⁸ As LIBOR is an unsecured rate while its replacement rate is nearly risk-free, it should be replaced by new rate + *spread* when novating existing contracts.

3.4.2 Rear-Fixed Reference Rates

For a rear-fixed reference rate,⁹ its level is determined and published *at the end* of the loan period. It is a backward-looking reference rate that is mainly derived from real transactions.

Overnight Reference Rates

An overnight rate is the interest rate for a borrowing/lending transaction with one business day tenor ($T - 1$ to T). Such transactions are mainly made between banks to manage their surplus/shortage of funds with respect to bank reserves¹⁰ at the end of each business day. An overnight rate is qualified as *secured* if it is a repo rate (due to collateralization) and *unsecured* if it is a 1-day loan/deposit rate. The overnight rate of a currency is monitored by the central bank, which may use reverse repo or repo to influence the interest rates according to its monetary policy.

The overnight reference rates based on real transactions are serving as the backbone for the financial market. In particular, the ones in the below table have been endorsed by regulators as replacement for LIBOR rates and are considered as RFRs (*risk-free rates*) as the risks are low:

Overnight rate	Description	Publication	Administrator
€STR (EUR) (Euro Short-Term Rate)	Reflects the wholesale euro unsecured overnight borrowing costs of banks located in the eurozone	9am CET since 2019	European Central Bank
SOFR (USD) (Secured Overnight Financing Rate)	Based upon U.S. Treasury repo transactions	8am Eastern Time since 2018	Federal Reserve Bank of New York
SONIA (GBP) (Sterling Overnight Index Average)	Reflects the average of the interest rates that banks pay to borrow GBP overnight from other financial institutions and other institutional investors	9am CET since 1997 and reformed in 2018	Bank of England
TONA (JPY) (Tokyo Overnight Average Rate)	Measures the cost of borrowing in JPY unsecured overnight money market	10am since 1992	Bank of Japan
SARON (CHF) (Swiss Average Rate Overnight)	Represents the overnight interest rate of the secured funding market for CHF. It is based on transactions and binding quotes posted in the Swiss repo market	Every 10 minutes from 8:30am CET since 2009	SIX Swiss Exchange

⁹ Rear-fixed rates were used for French francs before the introduction of euro, e.g., TMP (taux moyen pondéré—overnight rate), T4M (taux moyen mensuel du marché monétaire—1M rate) and TAM (taux annuel monétaire—12M rate).

¹⁰ Each bank must meet the minimum reserve requirement determined by the reserve ratio set by the central bank and applied to the bank's liabilities such as customer deposits. Each bank's reserve at the central bank includes the required minimum reserve and voluntary reserve.

For USD, another important reference rate is EFFR (Effective Federal Funds Rate), which is the weighted average overnight rate at which depository banks in the US lend to each other. Historically, especially after the 2008 financial crisis, it has been used as USD risk-free rate for various purposes including floating rate for OIS (overnight index swap), cash collateral interest rate to the poster, etc. But going forward, SOFR will play a more important role in financial products. It is worth noting that compared to EFFR and USD overnight LIBOR, SOFR may show higher volatility around monetary policy meeting dates and bank regulatory reporting dates where repo is used by banks to obtain liquidity.

Rear-Fixed Term Reference Rates

Rear-fixed term rates, such as 1-month, 3-month, 6-month rates, etc., can be constructed from the reference overnight rates. The usual method is to compound¹¹ daily reference overnight rates as shown in the following example with ACT/360 day count convention:

$$R = \left[\prod_{i=1}^{N_{\text{fix}}} \left(1 + \frac{r_i \times n_i}{360} \right) - 1 \right] \frac{360}{N},$$

where

R : reference rate for the considered period

N_{fix} : total number of fixings of overnight rate for the period

r_i : published reference overnight rate for day i

n_i : number of calendar days covered by r_i

N : total number of calendar days for the period.

Remark 3.2 Forward-looking references for rear-fixed rates start to exist, e.g., Intercontinental Exchange CE launched “ICE Term SOFR” in March 2022 offering 1, 3, 6 and 12 months forward-looking reference rates of SOFR that are based on OIS traded in the market.

3.5 Bonds

3.5.1 Bond Features and Types

A bond is a long-term debt security. It has a *face value* or *par value* (also called *principal amount*) representing the payment promised by the bond issuer to the bondholder when the bond matures.

¹¹ We note that CME applies compounding for its 3-month SOFR futures contracts but not for its 1-month SOFR futures.

Bonds may be classified according to the coupon types:

coupon bearing bond that makes periodic payments known as *coupons* over the lifetime of the bond, plus the repayment of the face value (usually at maturity). Typically, the coupon payment frequency is every 6 months or every year. A *fixed rate bond* pays coupons with a fixed rate. It is the most common bond type. A *floating rate note (FRN)* pays coupons linked to a reference rate, e.g., 6M SOFR+spread.

zero coupon bond that does not pay any coupon and only pays the bondholder the face value at maturity. Its sensitivity to interest rate is higher than a coupon bearing bond of the same maturity. Zero coupon bonds can be stripped from a coupon-bearing bond.

Bonds can also be classified according to other criteria such as issuer category and special features:

issuer: *Treasury bonds (aka rate bonds)*—issued by governments, often considered as credit risk free instruments; *Corporate bonds (aka credit bonds)*— usually with higher yield than Treasury bonds; *Municipal bonds*—often give tax benefits to interest income; *Foreign bonds*—issued by foreign corporates, e.g., samurai bonds (issued in Japan, denominated in JPY), dim-sum bonds (issued outside of mainland China, in CNH), panda bonds (issued in China, in CNY), masala bonds (issued outside of India, in INR), etc.;

seniority: *Senior bonds* have higher priority than *Subordinated bonds* for repayment and claim on the assets in case of issuer's liquidation;

others: A *convertible bond (CB)* can be converted into a predetermined amount of the issuing company's equity; *Inflation index bond* has its coupons and principal indexed on an inflation index, e.g., TIPS (treasury inflation protected securities). Some bonds embed the *callable feature* giving the issuer the right to buy back the bond at a predefined price on the “call dates”, or the *puttable feature* that gives the bondholder the right to sell back the bond at a predefined price on the “put dates”.

3.5.2 Bond Quotation and Yield to Maturity

The price for a bond at any time represents the sum of the present values of all the expected future cash flows.¹² In the market, bonds are quoted in either of the below units:

¹² The expectation is calculated with the “risk-neutral probability” that will be presented in the modelling part of the book.

dirty price (also called **full price** or **invoice price**), that we denote by B , is the price that the buyer will pay to the seller. For a fixed coupon bond, it is simply

$$B = \sum_{i=1}^n CF_i \times P(t_i),$$

where CF_i is the future cash flow of the bond payable at time t_i , $i = 1, \dots, n$. CF_{t_n} is usually composed of the last coupon and the face value;

clean price, which is defined as

$$\text{Clean Price} = B - \text{accrued coupon}$$

where *accrued coupon* is the interest that has been earned but not yet paid. The coupon calculation method is described in the bond prospectus, which usually follows the money market convention.

We note that

- transaction settlement is always done at dirty price;
- for quotation, dirty price is commonly used in the European bond market, whilst clean price is mostly used in the US bond market;
- clean price is more stable over time than dirty price that changes its value due to accrued coupon, in addition to economic reasons;

Yield to maturity (which is the IRR—internal rate of return) of a fixed coupon bond is the unique yield y that satisfies the following equation:

$$B = \sum_{i=1}^n \frac{CF_i}{(1 + y/q)^{t_i}},$$

where

q : number of coupon payments per year, e.g., 2 for half-yearly coupon payments;
 t_i : the number of periods from settlement date to the i -th coupon payment date.

For instance, with ACT/ACT convention, $t_1 = \tau/m_1$ and $t_i = i - 1 + \tau/m_1$ for $i > 1$ where τ is the number of calendar days from the settlement date to the coming coupon payment date and m_1 is the total number of days from the previous coupon payment (or the bond settlement date at issuance if the coming coupon is the first coupon of the bond after issuance) date to the coming coupon payment date.

If a bond pays a constant yearly fixed coupon rate C with final redemption at par, the yield to maturity satisfies

$$B = \sum_{i=1}^{n-1} \frac{C}{(1+y)^{t_i}} + \frac{1+C}{(1+y)^{t_n}},$$

where n is the number of coupon payments.

There are several other measures for the return of a bond investment, e.g., current yield, simple yield (also called Japanese yield),¹³ etc.

Example 3.1 Consider a fixed coupon bond in secondary market:

Face value: \$1000	Maturity date: 15 Nov 2017
Coupon rate: 3% p.a.	Day count: ACT/ACT
Coupon frequency: semi-annual	

An investor bought the bond at 99.583% of the face value (or \$995.83) with settlement date at 05 Jan 2017. The future coupon dates are on 15 May and 15 Nov. There are 130 days between the settlement date and the next coupon date (15 May 2017). The total number of days between the previous coupon date (15 Nov 2016) and the next coupon date is 181. The *yield to maturity* is 4% because $\frac{3\%/2}{(1+4\%/2)^{\frac{130}{181}}} + \frac{1+3\%/2}{(1+4\%/2)^{1+\frac{130}{181}}} = 99.583\% = \text{Price of the bond}$. There are 51 days from the previous coupon date to the settlement date, 05 Jan 2017. Hence, Accrued Coupon = $\frac{3\%}{2} \times \frac{51}{181} = 0.4227\%$ and Clean Price = $99.583\% - 0.4227\% = 99.160\%$. $\square$

3.5.3 Duration, Modified Duration, BPV, DV01 and Convexity

Duration, also known as **Macaulay Duration**, was introduced by the Canadian economist F. Macaulay in 1938. It is defined as the weighted average time to receive the discounted future cash flows from a bond.

$$\text{Duration} = \sum_{i=1}^n t_i \times \frac{CF_i}{(1+y)^{t_i}} / B;$$

¹³ Current Yield = Coupon Rate/Clean Price, which does not reflect the price impact on disposal of the bond. Simple Yield = $(C + \text{Adj})/\text{Clean Price}$, where Adj = (Final Redemption Value – Clean Price)/Life to Maturity.

Modified Duration (MD) measures the percentage change of bond price w.r.t. yield change, which is defined as

$$MD = -\frac{dB}{dy}/B.$$

Direct calculation of the derivative leads to

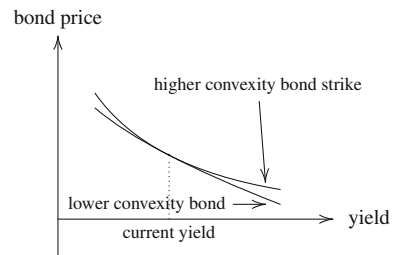
$$MD = \text{Duration}/(1 + y).$$

Modified duration and duration are measures for market risk. For a fixed coupon bond, the lower the coupon rate, the longer the duration. The zero-coupon bond has the longest duration (largest market risk). An FRN with the coming coupon rate yet to be fixed (e.g., rear-fixed floating rate or forward starting FRN) has zero modified duration and duration (see the explanation for formula (4.1));

BPV (basis point value, also known as **DV01**) measures the valuation change of a financial instrument with respect to a 0.01% parallel movement in the yield curve. It is widely used for risk management of trading positions;

Convexity of a bond measures the change of duration value with respect to the yield change of the bond. It is defined as the second-order derivative of bond price w.r.t. yield, i.e., $\frac{d^2B}{dy^2}/B$. For bonds without call/put features, the convexity is always positive for bondholders.¹⁴ For a given maturity, the lower the coupon rate, the higher the convexity. For bonds with uncertain maturities, such as callable bonds or mortgage-backed securities, the convexity can be negative. It is also possible that the convexity changes sign during the life of the product.

For two bonds with the same duration but different convexity values, the higher convexity bond outperforms the lower convexity bond when the yield moves up or down, as shown in the graphics. Hence, the higher convexity bond should trade with a lower yield in the market. It is intuitive that the convexity impact on bond price should be a positive function of yield volatility.



¹⁴ It can be easily checked that $\text{Convexity} = \frac{1}{(1+y)^2} \sum_{i=1}^n \frac{CF_i}{(1+y)^{t_i}} (t_i^2 + t_i)/B > 0$.

3.6 Credit Rating

A credit rating is an assessment of the creditworthiness of a debt security that is provided by a credit rating agency. The three major agencies are Standard & Poor's, Moody's and Fitch. For instance, the decreasing credit rating levels from S&P are AAA, AA, A, BBB, BB, B, CCC, CC, C.¹⁵ Each category between AA and CCC has three notches, e.g., A+, A, A-. The bonds with ratings at or above BBB- are called *investment grade (IG) bonds*.

3.7 Main Risks for a Debt Security

The main risks for a bond are

- *credit risk*, which is linked to the probability of default of the issuer before the maturity of the bond; The higher the default risk, the lower the price for the bond;
- *market risk*, which is linked to the interest rate changes. Higher interest rate will result in lower bond price because the future cash flows will be less valuable;
- *liquidity risk*, which is linked to the transaction bid/ask spread. A bond with smaller issuance size tends to have higher liquidity risk.

Exercises

1. Measure the year fraction from 2 Jan 2017 to 1 Mar 2017 with each of the following day count conventions: 30/360, ACT/365 and ACT/ACT ISDA. Modified Following is applied for business day convention.
2. For the period from 2 Jan 2014 to 31 Dec 2014, the money market interest rate for a certain currency is 2% under the day count convention ACT/360. What is the equivalent continuous rate level under ACT/365 day count convention?
3. Let A and B be two treasury bonds in the same currency with similar duration but higher convexity for A . Which bond will be quoted normally at a higher price?

¹⁵ For AAA bonds, the averaged default probability in 5 years is $\sim 0.10\%$. For BBB bonds, the default probability in 5 years is $\sim 1.9\%$.

Chapter 4

Interest Rate Instruments: II



In this chapter, we will cover the well-known fixed-income derivatives and a construction method for yield curves that can be used to price and hedge financial instruments.

4.1 Forward Rate Agreement and Single-Period Swap

A *forward rate agreement (FRA)* provides the P/L of a loan engagement with a pre-agreed fixed rate K , also known as strike, which will start at a time in the future. There are 2 dates in an FRA contract ($T < T_m$):

T : start date for the underlying loan, known as *effective date* (or *expiration date* for front-fixed reference rates such as EURIBOR);

T_m : termination date for the underlying loan. It is also the normal termination date for rear-fixed reference rates.

Example 4.1 The following is a 1×4 FRA trade.

Notional amount: EUR10m	Reference rate: 3M EURIBOR
Effective date T : in 1 month	Fixed rate: 1%
Termination date T_m : in 4 months	

The P/L for a long position (the opposite for a short position) to be settled in an FRA may be either of the following amounts:

$$\begin{cases} (1) \ N \times \frac{(r-K)\tau}{1+r\tau}, & \text{for settlement at } T \\ (2) \ N \times (r - K)\tau, & \text{for settlement at } T_m \end{cases}$$

where

N : notional amount;

r : effective reference rate for the period $[T, T_m]$, which is published at T for front-fixed rate or at T_m for rear-fixed rate;

K : fixed rate in the FRA contract, which is also known as *strike*;

τ : day count fraction for the period $[T, T_m]$;

For an FRA on a front-fixed reference rate, the settlement at T is a common market practice. While the natural settlement time for an FRA on a rear-fixed reference rate is T_m , it is nevertheless possible to settle the FRA at T , provided there exists a forward-looking fixing for the reference rate based on market instruments (see Remark 3.2). An FRA with settlement at the end of the forward period, i.e. T_m , is equivalent to a *single-period interest rate swap*, which is a particular case of IRS (see Sect. 4.3).

Typically, an FRA is provided by a bank as a financial service to a client who uses it to get the exposure to the reference interest rate. The bank, however, has the objective to achieve zero P/L (after deducting the servicing fee) from the FRA exposure by implementing a hedging strategy with financial market instruments. The business model of the banks is to earn servicing fee only from client trades.

Proposition 4.1 *Let F be the fixed rate (or strike) such that the value of an FRA is zero. Then, we have*

$$F = \frac{1}{\tau} \left(\frac{P(T)}{P(T_m)} - 1 \right),$$

and F is known as the **forward rate** for the FRA.

Proof The proof is based on the self-financing replication (i.e., hedging) of the FRA with transactions of two ZC bonds: bond A maturing at T_m and bond B maturing at T (see Remark 3.1). We first assume that the hedger is the short forward rate party of the FRA with \$1 notional. The below strategy allows the hedger to obtain the P/L for the long forward rate party.

T_0 : make the below transactions at inception time T_0

- borrow for the period $[T_0, T_m]$ by short selling m ($m = \frac{P(T)}{P(T_m)}$) units of bond A ; ¹
- lend for the period $[T_0, T]$ by buying 1 unit of bond B ;
- PV of the position is $-mP(T_m) + P(T) = 0$ (i.e., self-financing).

¹ At T_0 , investing in m ($= \frac{P(T)}{P(T_m)}$) units of bond A with PV $P(T_m)$ is equivalent to investing in 1 bond B with PV $P(T)$ that matures at T with value \$1, and simultaneously lending \$1 for the period $[T, T_m]$. Therefore, shorting m units of bond A is equivalent to shorting 1 bond B , and borrowing \$1 for $[T, T_m]$.

T : receive \$1 as B matures at T , and

- (1) for T -settled FRA, buy back A at its market price, which should be $m \frac{1}{1+r\tau}$ in the absence of arbitrage;
- (2) for T_m -settled FRA, lend \$1 in money market to receive $1 + r\tau$ at T_m and let A mature to receive m .

From the above self-financing strategy,

- for (1), P/L at T is $1 - m \frac{1}{1+r\tau} = \frac{(r-F)\tau}{1+r\tau}$;
- for (2), P/L at T_m is $(1 + r\tau) - m = (r - F)\tau$.

Hence, the hedger has replicated the P/L of the FRA with fixed rate F at zero cost, which implies that the initial value of the FRA with fixed rate F is zero.

If the hedger is the long forward rate party, the replication can be carried out by inverting all the transactions above (i.e., changing long to short, and vice versa). $\square$

If there is an FRA showing $K \neq F$, one can make a profit of $N|K - F|\tau$ by the following arbitrage transactions:

- if $K > F$, lend at K with the FRA, otherwise borrow at K with the FRA;
- apply the self-financing strategy as described in the proof.

The valuation of an FRA after its inception as well as an off-market FRA with $K \neq F$ follow the general approach in Remark 5.2. Specifically, the mark-to-market value of FRA is $N \times P(T_m)(F - K)\tau$ where τ is the remaining tenor and F is the forward rate for the FRA of tenor τ . The valuation change of an FRA is not linear to the underlying interest rate change, and it shows *convexity*.

4.2 Interest Rate Futures

An *interest rate futures* contract is an exchange-listed product with standardized features providing exposure to the underlying interest rate for a future period. Its price reflects market expectation for the underlying interest rate. Interest rate futures are preferred instruments for market participants to get exposure to their views on interest rates. They are also actively used to hedge interest rate risks and constitute a building block of the interest rate market.

4.2.1 Short-Term Interest Rate (STIR) Futures

STIR futures are standardized contracts listed on a exchange that provide similar exposures to interest rates as FRAs. The quotation of a futures contract with expiry at T is

$$p = 100 \times (1 - r),$$

where r is the underlying interest rate, which is normally the current market expected value of the interest rate to be observed at T . A futures contract settles daily P/L proportional to the difference between today's price and the previous day's price.

The following example is a Euribor futures contract listed at ICE. It is similar to the well-known Eurodollar² futures launched by the Chicago Mercantile Exchange (CME) in 1981. The daily P/L per contract is determined by the purchase and sale price difference in basis points times the BPV of the contract.

Contract notional	EUR 1 million
Quotation	$100 \times (1 - 3M \text{ EURIBOR})$
Minimum price fluctuation	0.5bp (i.e., 0.005%)
Contract months	Mar, Jun, Sep, Dec, and four serial months, such that 28 delivery months are available for trading
Last trading day	2 business days prior to the 3rd Wednesday of the delivery month
Daily P/L settlement	Cash settlement of the amount equal to daily price change $\times$ BPV
Basis point value (BPV)	$\text{€}1,000,000 \times \left(\frac{90}{360}\right) \times 0.01\% = \text{€}25$

In accordance with the introduction of rear-fixed rates to replace IBORs as reference floating rates, futures contracts on rear-fixed rates are being listed, e.g., 3-month SOFR futures that is similar to the above-mentioned Euribor futures.³

Remark 4.1 Unlike an FRA, a futures contract always has zero convexity with respect to the underlying interest rate as its BPV is a constant (e.g., €25). The convexity difference leads to the fact that when the interest rate decreases, the FRA's value increases more than the futures and when the rate increases, the FRA's value decreases less than the futures. Therefore, the implied rate from a futures price is traded higher than its corresponding FRA rate to neutralize the convexity impact on position carry. This strike rate difference is known as *futures/forward convexity adjustment*, which is positively correlated to the volatility of the interest rate and the tenor of the contract (see Sect. 16.9 for further analysis). $\square$

² Eurodollar is a term for US dollars held outside the United States. The term was coined for US dollars held in European banks. The “Euro-” prefix is now used to indicate any currency held in another country where it is not the official currency, e.g., Euroyen, Euroeuro.

³ The 3-month SOFR futures contract is referenced by the month that is 3 months earlier than its final settlement date. This makes it consistent with Eurodollar futures for the interest rate exposure period. For example, the “June” contracts for 3-month SOFR and Eurodollar contract both relate to the period between June to September.

4.2.2 Treasury Bond Futures*

The underlier of a *Treasury bond futures* contract is a virtual (or “notional”) security with standardized features for coupon rate $R\%$, maturity and size. The futures contract applies physical delivery at expiry. There is a set of deliverable bonds associated with each contract, which is determined by the exchange. For example,⁴ an active futures in Europe is of 10Y maturity with EUR 100,000 size and $R\% = 6\% p.a.$. The set of deliverable bonds is published by EUREX with remaining maturities of $8.5Y \rightarrow 10.5Y$.

For each deliverable bond, a **conversion factor** (cf) is published by the exchange, which is the clean price of the bond computed with the flat yield $R\%$ (we note that the price of the futures contract is at par of the face value with this flat yield curve).⁵ cf can be understood as the ratio between the bond price and the futures’ price with flat yield $R\%$ and it is applied to simplistically estimate the bond’s clean price from the futures’ market price F as illustrated by the below table:

Market data assumption	Futures price	Estimated clean price of the bond
Flat yield at $R\%$	100%	cf
Real market parameters	F	$F \times cf$

For a futures contract with settlement price F , if the i -th bond with accrued coupon acc^i is chosen for delivery at time T , then, by the rules of the exchange, the short party will

- receive the cash amount corresponding to “invoice price” $\mathcal{F}$, defined by $\mathcal{F} = F \times cf + acc^i$;
- deliver the i -th bond with the face value equal to the notional amount of the futures contract.

The exchange gives the right to the short party to choose the bond to deliver from the pool of deliverable ones. In practice, the bond to be delivered is the one that is the most in favor of the short party, and it is known as the **cheapest-to-deliver** (CTD) bond. At any time t before the delivery time T , the CTD bond is the one with lowest

⁴ More examples: in the US, there are 2Y, 3Y, 5Y, 10Y Treasury note futures, Treasury bond futures (deliverable bonds of $15Y \rightarrow 25Y$) and Ultra T-Bond futures (deliverable bonds of $>25Y$); in China, there are 2Y, 5Y, 10Y and 30Y maturities treasury bond futures with $R\% = 3\% p.a.$.

⁵ Consider a US treasury bond futures with $R\% = 6\% p.a.$. The cf for a bond with semi-annual coupons $5\% p.a.$ and 15 years and 3 months remaining maturity is

$$cf = \left[\sum_{i=0}^{30} \frac{2.5\%}{(1+3\%)^i} + \frac{100\%}{(1+3\%)^{30}} \right] / (1+3\%)^{\frac{3}{6}} - accrued, \text{ where } accrued = 2.5\% \times \frac{3}{6}.$$

net basis, which is $C_t - \mathcal{F}_t$ where C_t is the cost to be engaged by the short party from t until T for bond delivery⁶ and $\mathcal{F}_t$ is the invoice price ($= F_t \times cf + acc$ where F_t is the futures price at time- t). The CTD bond may change over time, although it is relatively stable in normal times.

The price of a futures contract F_t is related to C_t , the cost for the CTD bond delivery.⁷ Moreover, when futures are used in hedging, the BPVs of the CTD bonds should be used to determine the quantity of futures contracts q to hedge the BPV of a portfolio V (q is determined such that $BPV(V) + q \times BPV(\text{CTD bond})/cf = 0$).

4.3 Interest Rate Swap (IRS)

An interest rate swap (IRS) is a contract that entails periodic payments between two counterparties during the term of the trade. The swap market is very large and liquid. Swaps are actively used to hedge long-term interest rate risks. The market standard IRS (aka plain vanilla IRS) is a “fixed-floating” trade with two legs:

fixed leg: one party will pay, at the *end* of each covered period, the cash flows calculated with a fixed interest rate as predetermined in the contract;

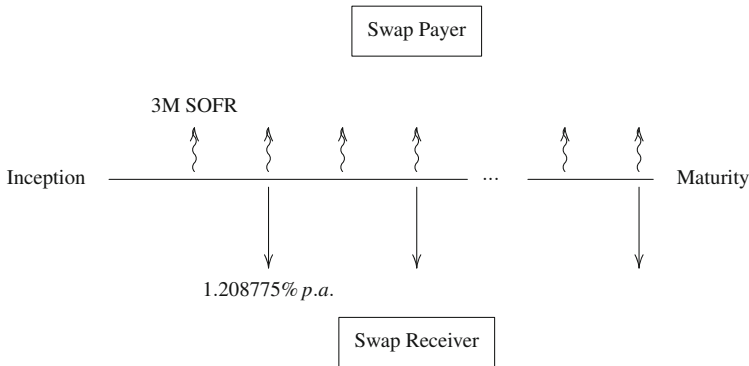
floating leg: the other party will pay, at the *end* of each covered period, the cash flows calculated with a reference rate + spread.

By market vocabulary, fixed rate payer (receiver) is referred to as the “swap payer” (“swap receiver”). The swap payment dates are usually determined backward from the maturity date. If the fixed and floating legs have the same payment dates, the swap becomes a strip of FRAs. An illustration example of a swap is as follows:

Maturity	5 years
Notional amount	USD 10m
Fixed leg	1.208775%, payable at the end of each semi-annual period
Floating leg	3M SOFR, payable at each quarter end

⁶ For a given bond, we have $C_t = B_t + f_t - I_t$, where B_t , f_t and I_t are respectively the bond's full price at t , financing cost for the long bond position for the period (t, T) and the bond's coupon income in the period (t, T) accrued to T .

⁷ Given the CTD bond, $F_t \approx C_t/cf = (B_t + f_t - I_t)/cf = (P_t + f_t - I_t + acc_t)/cf$ where B_t and P_t represent the full price and the clean price of the CTD bond. Among other factors impacting the futures' price, there may be the options for the short party found in some futures arising from their specific delivery process rules, e.g., *wild card option* in CME futures (see [29]).



Other combinations such as “floating-floating” with different floating rates and “fixed-fixed” with different payment schedules may be composed by “fixed-floating” swap trades.

4.3.1 Asset Swap as an IRS Application

An asset swap refers to the package of a fixed-coupon credit bond combined with a swap that turns the fixed coupons into floating coupons. It provides the exposure of a synthetic FRN in effect, which mitigates the interest rate risk.

For example, a financial institution having access to the interbank market for funding is interested in a 5-year, fixed-coupon bond with 3% coupon. The financial institution can have short-term funding at SOFR flat, and the prevailing market price of the swap with fixed rate of 3% is $\text{SOFR} + 1.5\%$. If the bond is trading at par (to simplify the illustration), the financial institution will have a net return of 1.5%, which corresponds mainly to the reward for credit risk taking.

4.3.2 IRS Valuation

The valuation of an interest rate swap is based on the cost of replication. The fixed coupons can be replicated by zero-coupon bonds, and the floating coupons at market forward rate can be replicated with zero-coupon bonds as well, according to the proof of Proposition 4.1. The PV of a receiver swap⁸ at settlement date t_0 is simply the PV of the cash flows of the fixed leg minus the PV of the cash flows of the floating leg:

$$\text{Swap } PV = PV^{\text{fixed}} - PV^{\text{floating}} = \sum_{j=1}^m P(t_j) s \tau_j - \sum_{i=1}^n P(t_i) F(t_{i-1}, t_i) \tau_i,$$

⁸ Actually, the PV calculation formula used here is consistent with the risk-neutral modelling.

where

- m : number of remaining cash flows for the fixed leg;
- n : number of remaining cash flows for the floating leg;
- s : rate of the fixed leg;
- τ_i : day count fraction for i -th period (t_{i-1}, t_i) ;
- $P(t)$: price of the zero-coupon bond with maturity t and settlement date t_0 ;
- $F(t_{i-1}, t_i)$: market forward rate for i -th period (t_{i-1}, t_i) .

For standard (vanilla) IRS, the valuation can be further simplified. By Proposition 4.1, the market quote for the floating rate covering the period (t_{i-1}, t_i) is $F(t_{i-1}, t_i) = \frac{1}{\tau_i} \left(\frac{P(t_{i-1})}{P(t_i)} - 1 \right)$. Replacing $F(t_{i-1}, t_i)$ in the PV of the floating leg of a new swap to price:

$$\begin{aligned} PV^{\text{floating}} &= \sum_{i=1}^n P(t_i) F(t_{i-1}, t_i) \tau_i = [P(t_0) - P(t_1)] + \cdots + [P(t_{n-1}) - P(t_n)] \\ &= P(t_0) - P(t_n) \end{aligned}$$

In particular, if t_0 is today, we have $PV^{\text{floating}} = P(t_0) - P(t_n) = 1 - P(t_n)$. This can also be understood by the fact that the PV of an investment of \$1 receiving regular interests based on floating rate and the \$1 back at the end should just be \$1, i.e., $1 = \sum_{i=1}^n P(t_i) F(t_{i-1}, t_i) \tau_i + P(t_n) = PV^{\text{floating}} + P(t_n)$.

A market swap is quoted with the fixed rate level that makes the swap valuation equal to zero, i.e., $PV^{\text{fixed}} = PV^{\text{floating}}$. For standard swaps, we have

$$\sum_{j=1}^m P(t_j) s \tau_j = P(t_0) - P(t_n). \quad (4.1)$$

Hence, the rate is given as

$$s = \frac{P(t_0) - P(t_n)}{A(t_m)}. \quad (4.2)$$

where $A(t_m) = \sum_{j=1}^m P(t_j) \tau_j$ is known as *annuity factor*.

In the above analysis, the same set of zero-coupon bonds is used to project future cash flows and to carry out discounting. This was the market practice before the 2008 financial crisis, which neglected the counterparty and funding risks. The market IBOR swaps are now valued with two curves: index curve for projecting futures cash flows and RFR curve for discounting (see Sect. 4.5).

4.3.3 *Overnight Indexed Swap (OIS)*

An overnight indexed swap (OIS) is an interest rate swap where the cash-flow of the floating leg is based on a floating rate calculated from the overnight reference rates, such as €STR and SOFR, for each period. It is the preferred swap instrument in the post-LIBOR era. The curve constructed with market OIS represents the near-RFRs, which is used to discount future cash flows in financial products.

OIS instruments in the market cover a wide spectrum of maturities, from short term to long term. For short-term OIS with maturity within 1 year, there is typically only a single cash flow at maturity. Now, the standard quotation of IBOR swaps are using RFR curve for discounting.

4.3.4 *Other Interest Rate Swaps**

There are other types of interest rate swaps (called exotic swaps in general). Their valuation methods usually involve interest rate term structure modelling. We will mention only two examples: constant maturity swap and callable swap.

A *constant maturity swap* (CMS) is a swap in which the reference floating rate used for each settlement is the prevailing market rate of the standard swap of a specified fixed tenor. For example, in a 2-year maturity CMS swap linked to USD10Y swap rate, the cash flow for the floating leg is based on the market rate of the USD10Y swap rate observed at each settlement.

It is worth highlighting that the *CMS rate* in a structured product refers to the market swap rate for a specified tenor (e.g., USD20Y swap rate).

A *callable swap* is a swap in which one of the parties has the right to early terminate the swap at some specified dates prior to the maturity. The early termination settlement value is specified in the product.

4.3.5 *Swap Clearing**

As explained in Sect. 1.5, standard derivatives should be cleared through CCPs and non-centrally cleared swaps are subject to higher regulatory capital requirement.

Nowadays, the majority of swaps are centrally cleared through CCPs. Once a swap is transacted OTC or on a SEF (swap execution facility, a trading platform), it is immediately “given up” by novation to a CCP that becomes the central counterparty to both swap counterparties.

4.3.6 Swap Futures*

Post 2008 financial crisis, the regulations reward standardization, especially the exchange traded products in terms of regulatory capital and initial margin. This favors the development of swap futures, e.g., MAC (market agreed coupon) USD swap futures and Eris[®] USD swap futures, both quoted as 100% + swap NPV with daily P/L settlement, listed quarterly at CME.

MAC (USD) swap futures is like a forward-starting standardized IRS that is deliverable at contract expiry (physical delivery). Upon the swap futures expiry, the long (receiving fixed rate) and short (paying fixed rate) parties will each be facing the exchange for the deliverable IRS and keep anonymity. The final settlement price is published by the exchange, which reflects the mark-to-market value of the underlying swap with agreed methodology used by the major swap clearers. At the listing of the contract, the fixed rate of the deliverable IRS is established at a level close to the forward rate for an IRS with the corresponding maturity.

Eris[®] (USD) swap futures has been designed to replicate the economics of a vanilla OTC swap while retaining all the operational, capital and cost efficiency of traditional futures contracts. Unlike the MAC swap futures, an Eris[®] futures contract is a spot starting swap with an expiration date set at the termination of the underlying swap. The contract's P/L on date t is based on the contract settlement value S_t that is defined as $S_t = 100\% + A_t + B_t - C_t$, where

A_t : NPV of future fixed and floating coupons discounted at SOFR;

B_t : accumulated historical fixed and floating payments to date;

C_t : SOFR interest on previous day's NPV, known as PAI (price alignment interest), mimicking the interest on the collateral (corresponding to the accumulated daily settlements of the Eris[®] futures contract) in an analogous non-centrally cleared OTC IRS. More precisely, $C_t = \text{NPV}_{t-1} r_{t-1} \Delta t + C_{t-1}$ where Δt is the day count fraction for the period $[t-1, t]$.

Actually, we have $S_t - S_{t-1} = (\text{NPV}_t - \text{NPV}_{t-1}) + (B_t - B_{t-1}) - \text{NPV}_{t-1} r_{t-1} \Delta t$.

4.4 Yield Curve Construction

4.4.1 Yield Curve

A *yield curve* is a set of interest rates with consistent credit risks across different tenors (also called pillars) in the same currency. The relationship between the interest rates of different tenors is called *interest rate term structure*. In this section, we illustrate a construction method for the yield curve to be used for pricing and risk management of financial instruments.

As an interest rate in any basis (e.g., continuous, money market, etc.) can be implied from ZC bonds (i.e., discount factors), the yield curve can be represented by

the so-called *zero-coupon curve* (or simply *zero curve*), which is computed with the interest rate instruments to be used for hedging. The pillars of the curve correspond to the maturity dates of these instruments. For an IBOR curve related to unsecured rates, it is customary to use interbank loan/deposit for the shortest end, followed by the liquid STIR futures (if applicable) and IRS. For a secured rate curve, OIS⁹ can be used for both short-term and long-term parts. The futures on secured reference rates, such as SOFR, may also be used as they are actively used in hedging and trading.⁹

4.4.2 Interpolation Method for Yield Curve

For any date t that is not on the pillars of the curve, the zero-coupon bond $P(t)$ can be calculated by interpolation¹⁰ from the zero-coupon bonds for its neighbouring dates t_1 and t_2 where $t_1 < t < t_2$. One commonly used interpolation method consists of assuming a constant forward rate r between t_1 and t_2 , such that $P(t_2) = P(t_1) \times e^{-r(t_2-t_1)/365}$, i.e., $r = \ln \frac{P(t_1)}{P(t_2)} \times \frac{365}{t_2-t_1}$. The interpolation formula for the zero-coupon bond is then

$$P(t) = P(t_1) \times e^{-r(t-t_1)/365}.$$

Actually, there are methods that take more than two known points for producing smoother interpolated values, but at the expense of losing stability as the value change on any point may change the value of the interpolated one.

4.4.3 Bootstrapping Method

Bootstrapping consists of calculating the ZC bonds implied by the instruments step-by-step, from the shortest tenor to the longest. The ZC bond for expiry date t is calculated from the already known ZC bonds for the dates prior to t and the rates implied from market instruments covering the tenor until t . The calculation depends on the type of instrument. For instance, if we know $P(s)$ and the money market rate r applied from s to t ($s < t$), then the bootstrapping procedure gives $P(t) = \frac{P(s)}{1+r\tau}$, where τ is the day count fraction for $t - s$.

We note that one can also solve a system of equations instead of using bootstrapping iterations.

⁹ As STIR futures has zero convexity to interest rate (see Remark 4.1), a convexity adjustment can be applied to the implied forward rate from futures price for building the zero curve, if the impact is significant.

¹⁰ For financial data, it is relatively safe to do interpolation. However, if we need a data point beyond the range of the available data, practitioners often use the value of the nearest available point (i.e., the crystallized end-point value) instead of doing extrapolation.

Table 4.1 Instruments for Euro rates—27 Nov 2014

Instrument	Tenor	Expiry date	Begin	End	Rate or price
Deposit	O/N		27 Nov 14	28 Nov 14	0.0000%
Deposit	T/N		28 Nov 14	01 Dec 14	0.1500%
Deposit	1M		01 Dec 14	02 Jan 15	0.0400%
Deposit	2M		01 Dec 14	02 Feb 15	0.0400%
Deposit	3M		01 Dec 14	02 Mar 15	0.1400%
Futures		15-Dec-14	17 Dec 14	17 Mar 15	99.910
Futures		16-Mar-15	18 Mar 15	18 Jun 15	99.915
Futures		15-Jun-15	17 Jun 15	17 Sep 15	99.930
Futures		14-Sep-15	16 Sep 15	16 Dec 15	99.925
Futures		14-Dec-15	16 Dec 15	16 Mar 16	99.910
Futures		14-Mar-16	16 Mar 16	16 Jun 16	99.890
Futures		13-Jun-16	15 Jun 16	15 Sep 16	99.870
Futures		19-Sep-16	21 Sep 16	21 Dec 16	99.840
Swap	3Y		01 Dec 14	01 Dec 17	0.2570%
Swap	4Y		01 Dec 14	03 Dec 18	0.3208%
Swap	5Y		01 Dec 14	02 Dec 19	0.3960%
Swap	6Y		01 Dec 14	01 Dec 20	0.4840%
Swap	7Y		01 Dec 14	01 Dec 21	0.5830%

4.4.4 Illustration Example for Yield Curve Construction

We illustrate the methodology through an example of IBOR zero curve with the instruments in Table 4.1, which uses three types of instruments. Although the trend is to use secured rate curve, the calculation methodology remains the same.

Interbank Loan/Deposit Instruments

Bootstrapping starts from O/N (overnight) deposit. Next, we move on to T/N (tomorrow/next, i.e., $T + 1$ to $T + 2$) deposit. Note that 29 Nov 2014 was a Saturday, leading to 3 calendar days in the calculation. Then, we consider 1-month deposit, ...

	O/N deposit	T/N deposit	1-month deposit
Day count convention	ACT/360	ACT/360	ACT/360
Rate	$r = 0.0000\%$	$r = 0.1500\%$	$r = 0.0400\%$
Day count	$s = 27 \text{ Nov } 14$	$s = 28 \text{ Nov } 14$	$s = 01 \text{ Dec } 14$
	$t = 28 \text{ Nov } 14$	$t = 01 \text{ Dec } 14$	$t = 02 \text{ Jan } 15$
	$\tau = 1/360$	$\tau = 3/360$	$\tau = 32/360$
$P(s)$	$P(s) = 1$	$P(s) = 1$	$P(s) = 0.9999875$
$P(t)$	$P(t) = 1$	$P(t) = 0.9999875$	$P(t) = 0.9999519$

Short-Term Interest Rate Futures

Interest rate futures are quoted in *price* at the exchange, i.e., $p = 1 - r$ where r is 3M EURIBOR observed at the expiration date of the contract. We use the term “Termination Date (TD)” for the date that is 3 months after the expiry date of the contract. Strictly speaking, a convexity adjustment should be applied to the implied EURIBOR rates from futures prices for the consistency with deposit rates and swap rates. But due to its insignificance, the convexity adjustment is not applied here, which is satisfactory for non fixed-income trading activities, such as equity derivatives, with main risks on securities prices, volatilities, etc. The application of convexity adjustment is however straightforward and is only about the input parameters.

For the first futures contract, we need to find the ZC bond at the beginning date of the contract, i.e., 17 Dec 14, which is not yet available. It has to be interpolated from the nearby $P(T/N)$ and $P(1M)$.

<i>First futures contract</i>		
Convention	ACT/360	
Price	p	$p = 99.91\%$
Implied rate	$r = 1 - p$	$r = 1 - 99.91\% = 0.09\%$
Day count	s : TD of the preceding contract	$s = 17 \text{ Dec } 14$
	t : TD of the current contract	$t = 17 \text{ Mar } 15$
	$\tau = (t - s)/360$	$\tau = 90/360 = 0.25$
$P(s)$	Interpolated	$P(s) = 0.9999797$
$P(t)$	$P(s)/(1 + r \times \tau)$	$P(t) = 0.9997448$

From the second futures contract onward, the calculation will be $P(t_2) = \frac{P(t_1)}{1+r \times \tau}$, where t_2 and t_1 are the termination dates of the current contract and the preceding contract, r being the implied rate for the current contract.

If the periods covered by two successive futures overlap or have a hole in between, any smooth treatment may be applied for this joining period. If the impact is negligible, one may choose to simply adjust the period length of one of the contracts by keeping the implied rate unchanged.

Interest Rate Swaps

Features of standard EUR swaps:

Fixed leg	Floating leg
Day count: 30/360	Day count: ACT/360
Frequency: annual	Frequency: semi-annual

For a swap with settlement date t_0 (two business days from today in our case), the PV of the floating leg is $P(t_0) - P(t_n)$. The bootstrapping of swap contracts is based on the below no-arbitrage relationship:

$$P(t_0) - P(t_n) = s \sum_{i=1}^n \tau_i P(t_i).$$

We take the ZC bond for 3Y pillar as an example where we have already know the following numbers through bootstrapping of instruments with shorter tenors:

$$P(t_0) = 0.9999875, P(1Y) = 0.9991986, P(2Y) = 0.9979836, s = 0.2570\%.$$

We can solve for $P(3Y)$ that makes the following equality hold:

$$P(t_0) - P(3Y) = s \times [\tau_1 \times P(1Y) + \tau_2 \times P(2Y) + \tau_3 \times P(3Y)].$$

Applying the numerical numbers, we obtain $P(3Y) = 0.9923045$.

4.5 Multiple Zero-Coupon Curves*

There are different types of zero curves for different purposes. The commonly used ones include:

- *RFR curve*: the zero curve used for discounting in financial calculations (also known as OIS discounting or CSA discounting);
- *Index curve*: constructed with market price/quotation of interest rate instruments, an index curve is used to determine the cash flows of the current and future periods in financial calculation;
- *Cross currency curve*: a zero curve constructed with the interest rates of a foreign currency that are implied from CCS and FX swaps. These rates are normally different from the interest rates of that currency observed in its domestic market due to supply/demand of the currency (see Sect. 6.4); For a currency that is not used for its accounting, an entity should use the cross currency curve for discounting;

Before the 2008 financial crisis, a single zero curve was normally used in financial market for projecting the future cash flows of financial instruments and for discounting, whereby counterparty and funding risks were not fully considered in financial calculation. Since the crisis, the market realized that these risks can be quite significant and started to apply adjustments (with numerous complex XVAs: CVA—credit value adjustment, FVA—funding value adjustment, etc.) to the valuation made with a single curve (see Sect. 16.8). To improve the valuation of financial instruments, two curves are used at the same time: index curve for projecting the

cash flows to come and RFR curve for discounting. The methodology presented in Sect. 4.4 is applicable to construct the different curves.

Now, active market participants, guided by regulators, have implemented measures to reduce counterparty risks, such as centralized clearing, zero threshold for collateral posting in the bilateral CSA between counterparties. Under such improved market conditions, a single RFR curve may be used to value fully collateralized financial instruments linked to RFR references.

Exercises

1. From the data in the yield curve example presented in the last section, calculate
 - (1) 4M deposit rate (for the period 2 Dec 2014–2 Apr 2015)
 - (2) zero-coupon bond values for 13 Feb 2014 and 2 May 2016 using the interpolation method presented in the last section
2. Consider the following 3×6 FRA contract issued on Tuesday, 8 May 2018:

Notional amount	EUR 30 million
Expiry (i.e., Effective date)	3 months (8 August 2018, Wed)
Termination date	6 months (8 November 2018, Thu)
3M EURIBOR	2.36%
6M EURIBOR	2.52%
Day count convention	ACT/360

- (1) Calculate the market forward rate for the FRA contract (i.e., the rate at which the contract value is zero at inception);
 - (2) At expiry, the 3M EURIBOR on the settlement date is 2.60%, what is the P/L for the long party?
3. An institution has obtained a loan of EUR10mio for 3 years at the interest rate of 6M EURIBOR + 1.5% with semi-annual interest payments. With this loan, the institution decided to invest in a bond paying 5% p.a. with semi-annual coupon payments.
 - (a) Describe the risks for the institution;
 - (b) The institution decided to enter into a 3Y standard IRS transaction with a bank in which it receives 6M EURIBOR and pays 1.70% fixed rate at zero cost. The payment dates coincide for the bond, the loan, and the IRS.
 - (1) Has the IRS reduced the interest rate risk for the institution?
 - (2) Who is the payer of the swap transaction? (institution or the bank)?
 - (3) Calculate the net cash flow the institution will receive every 6 months.
 - (4) For each 6-month period, when is the floating rate of the IRS fixed (i.e., at the beginning or the end)? and when is the floating coupon paid?

Chapter 5

Equities and Equity Indices



5.1 Equity

Equity is the claim of ownership of a company. Equity securities issued by corporations are called *stocks* or *shares*.¹ In this book, we only consider the *publicly traded stocks*, which are listed on stock exchange. The main equity securities are

Common stock (ordinary stock): a common share owner has voting right and is entitled to dividends. In case of liquidation of the company, the shareholder of a common share has the lowest priority for the liquidation assets.

Preferred stock: the holder of a preferred stock has no voting right but has the priority for dividend payments and liquidation assets over the common share holder. Many preferred shares pay dividends in the form of fixed coupons, like a perpetual bond. There are several types of preferred shares: a convertible preferred share gives its holder the right to convert the preferred share into a common share; a cumulative preferred share allows dividends omitted in the past (e.g. due to profitability issue) to be paid later.

A company's *free float* refers to the number of shares that are immediately tradable in the market. *Restricted share*, usually held by the insiders (such as company's executives and some other employees) or strategic investors, are not transferable until certain conditions are met. We have

$$\text{Free Float} = \text{Outstanding Shares} - \text{Restricted Shares}.$$

The *market capitalization* of a company is defined as the market price multiplied by the number of outstanding shares.

¹ The word *stock* consists of all the shares of a company, but may also be used to refer to a single *share*. In the context of this book, the words *stock* and *share* are used interchangeably.

5.2 Corporate Actions

5.2.1 Stock Dividend

A *stock dividend* is a payment (e.g. yearly or quarterly) made by a corporation to its shareholders for profit distribution. The dividend amount is proposed by the company's board to be approved at a general shareholders' meeting. There are different factors impacting dividend policy.² For a detailed discussion on dividend policy and practice, we refer to [99].

When a dividend is declared, an *ex-dividend date* is specified. It is the date on which shares bought no longer come attached with the right to be paid for the declared dividend. Before the ex-dividend date, the stock is said to be trading *cum-dividend*. Other factors remaining equal, the share price will drop on the ex-dividend date t by the amount equivalent to the after-tax dividend amount. Otherwise, one can make a risk-free profit by buying at S_{t-} to receive the dividend and then sell it at S_t where S_t denotes the stock price at time t and S_{t-} the price just before t .

A stock dividend can be either a *cash dividend* in monetary form, or a *stock dividend* (also known as *proportional dividends* in modelling) that involves the company issuing more shares to its existing shareholders. Unlike cash dividend, a stock dividend does not change the company's market capitalization. However, the share value is diluted to $S_t = \frac{M}{M+N} S_{t-}$ where M is the total number of shares before the ex-dividend date and N is the total number of new shares.

In some jurisdictions (e.g. UK), companies can issue the so-called *scrip dividend* that gives investors the option to receive additional newly issued shares instead of cash dividend. Usually, scrip dividends provide tax benefits and may help the company to keep more cash for business. There are trading activities on scrip dividends that involve stock borrowing/lending.

Suppose that a stock pays a dividend value of D with ex-dividend date t . The quantity $d = D/S_{t-}$ is called *dividend yield* or *dividend rate*. For example, if shareholders receive one new share for every twenty already owned shares, then $D = \frac{1}{20} S_t$ with $S_t = \frac{1}{1+1/20} S_{t-}$, and the dividend yield is $d = D/S_{t-} = 4.762\%$.

Generally, for a stock paying, at time- t , both proportional dividend at rate d and cash dividend amount a , we have

$$S_t = S_{t-} (1 - d) - a.$$

In continuous-time modelling, it is sometimes convenient to represent divided yield by a value η such that $e^{-\eta} = 1 - d$, which leads to

$$S_t = S_{t-} e^{-\eta} - a.$$

² For instance, for protecting equity investors' interest, a listed company in China should have distributed, in cash dividend, the cumulated sum of $\geq 30\%$ of the average yearly distributable profits for the past three years to be eligible for issuing new shares.

5.2.2 *Stock Split, Reverse Stock Split, Rights Issue**

A *stock split* consists of dividing each existing share of a company into multiple shares. It increases the number of outstanding shares while keeping the capitalization unchanged for the company, which results in the dilution of the share price by the split ratio. If a stock's price has reached such a high level that it becomes unaffordable to small investors, the stock split helps to increase the stock's liquidity.

A *reverse stock split* is the opposite of *stock split*. A company whose stock price has dropped to a very low level such that one *tic* (i.e., the minimum quotation unit) for its price becomes quite significant may consider undertaking a reverse stock split to bring back the price to a normal unit.

A company can raise money through a *rights issue* that is offered to existing shareholders for purchasing new shares in the company at a subscription price at a discount to the prevailing market price. The right can be either *renounceable*, allowing the shareholder to trade it in the market, or *non-renounceable* that is not tradable.

5.2.3 *Impact of Corporate Actions**

Some corporate actions, such as dividend, stock split and rights issue, systematically cause share value change. Denote

- t as the effective date of a corporate action that causes the share value change (e.g. ex-dividend date),
- S_{t-} as the share valuation just before the corporate action (e.g. cum-dividend share price),
- $b_t = (S_t - S_{t-})/S_{t-}$ as the stock value change ratio caused by the corporate action(s) at time- t .

We have the below summary:

Corporate action	Description (per existing share)	New price S_t	Change ratio b_t
Cash dividend	a in cash amount	$S_{t-} - a$	a/S_{t-}
Stock dividend	z new shares	$S_{t-}/(1+z)$	$1 - \frac{1}{1+z}$
Rights issue	n new shares issued at price P	$\frac{S_{t-} + n \times P}{1+n}$	$1 - \frac{1+nP/S_{t-}}{1+n}$
Split/Reverse split	m new shares	S_{t-}/m	$1 - \frac{1}{m}$

It is convenient to combine all the above corporate actions except the last one:

$$S_{t-} + nP = S_t(1+z+n) + a, \quad (5.1)$$

$$b_t = 1 - \frac{1 + n \frac{P}{S_{t-}} - \frac{a}{S_{t-}}}{1+z+n} \quad \text{and} \quad S_t = S_{t-}(1 - b_t).$$

5.2.4 No-Arbitrage Condition for Derivatives Price and Contract Terms Adjustment*

For predictable corporate actions that may create a jump in the underlying security's price, the derivatives price continuity at the event should be satisfied to avoid arbitrage. This *no-arbitrage condition* is taken into account in option pricing (see equations (23.14) and (23.15)). However, if the effective event details (e.g. dividend amount) deviate from the initial assumptions, the parties may have a valuation P/L.³ To avoid such risks, typically for the below situations,

- exceptional corporate actions, i.e. not scheduled at the inception of the contract
- large size or illiquid trades (e.g. strategic equity transactions, convertible bonds),

the parties of a trade can choose to perform adjustment to some of the contract terms such as strike price, notional amount (see Sect. 16.7.4) with a pre-agreed methodology. It is a common practice to price large derivative contracts with the assumption of no corporate action and to adjust the contract terms upon the occurrence of each corporate action, scheduled or not (e.g. regular and exceptional dividends).

5.2.5 Total Return Asset*

It is often necessary and convenient to work on total return assets that do not provide income (e.g. dividend, stock lending fee, coupon...). The **total return price process** S_t^* for an asset (stocks, bonds, funds, ...) can be constructed by re-investing all the income.

Firstly, we assume that all the income is re-invested into the asset itself. This is the usual understanding of total return concept.

- asset subject to discrete events (e.g. stock dividends, split):

$$S_t^* = S_t \prod_{0 < t_i \leq t} \frac{1}{1 - b_{t_i}} \quad (b_{t_i} \text{ is defined by formula (5.1)}) \quad (5.2)$$

where $\prod_{0 < t_i \leq t} \frac{1}{1 - b_{t_i}}$ represents the adjusted number of asset units.

- asset with continuous income (e.g. interests, securities lending fees):

$$S_t^* = S_t e^{\int_0^t q_u du} \quad (q_u \text{ is the income rate}) \quad (5.3)$$

³ For example, some financial institutions unexpectedly cancelled dividends in late 2019/2020 at the request of regulators due to Covid-19 pandemic. Consequently, several large equity derivatives trading books suffered significant loss on dividend risk exposure.

where $e^{\int_0^t q_u du}$ represents the accrued number of asset units starting with 1.

For the general case, especially for modelling (see Sects. 5.4 and 16.7), some of discrete incomes (with income rate a_t) are carried over in money market account and all the other incomes are re-invested into the asset itself. Then, the price process of the total return asset is

$$S_t^* = S_t \beta(t) + \sum_{t_k \leq t} a_k \beta(t_k) e^{\int_{t_k}^t r_u du} \quad (5.4)$$

where the term $\beta(t) = e^{\int_0^t q_u du} \prod_{0 < t_i \leq t} \frac{1}{1-b_{t_i}}$ represents the number of accrued asset units at time t (from 1 unit at inception). So, $a_k \beta(t_k)$ is the discrete income received at time t_k . The formula (5.4) can be equivalently written as

$$S_t^* = \beta(t)[S_t + \alpha(t)] \quad (5.5)$$

where $\alpha(t) = \sum_{0 < t_i < t} (a_i e^{\int_{t_i}^t (r_u - q_u) du} \prod_{t_i < t_k \leq t} (1 - b_{t_k}))$.

Remark 5.1 (Backward Quantity) For delivering one unit of an asset at time t , one can hold $1/\beta(u)$ amount of the asset at time u ($u < t$). $\square$

5.2.6 Historical Price Adjustment due to Corporate Actions*

It is convenient to adjust the time series of a stock price so that the real performance (total return) can be simply determined by the end date price and start date price for any considered period $[t_1, t_n]$. One can choose to keep either the last price unadjusted (backward adjustment),⁴ or the initial price unadjusted (forward adjustment). By formula (5.2) and Remark 5.1, we have $S_t^* = S_t \prod_{t_n \geq t_k > t} (1 - b_{t_k})$

($S_t^* = S_t \prod_{t_1 < t_k \leq t} \frac{1}{1-b_{t_k}}$) for backward (forward) adjusted price time series, if we assume that all the incomes are re-invested into the asset itself.

Example 5.1 backward and forward adjustment with a share dividend of 3% on ex-dividend date t_5 (numbers in *italic* are adjusted ones);

Type	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
Unadjusted	100	103.33	102	100	98.3	99.6	102.5	103.8
Backward adj	97	<i>100.23</i>	<i>98.94</i>	97	98.3	99.6	102.5	103.8
Forward adj	100	103.33	102	100	<i>101.34</i>	<i>102.68</i>	<i>105.67</i>	<i>107.01</i>

$\square$

⁴ For instance, *Yahoo! Finance*TM provides backward adjusted prices

5.3 Equity Index

An equity index tracks the price performance of the selected stocks for a particular target, e.g., to reflect the performance of a global stock market, a certain sector or the stocks with a specific feature. It is computed from the prices of selected stocks. An index is a mathematical construct that does not allow direct investment. People can trade an index through index futures, mutual funds or exchange-traded funds (ETFs) tracking the index.

An equity index with n stocks may be categorized according to its calculation method. Let I_t be the index value at date t and S_t^i be the closing price of the i -th stock for date t (converted) in the index currency. The two main classes of equity indices are as follows:

- *Capitalization Weighted Index:*

The majority of equity indices belong to this type, e.g. HSI, S&P 500 or Euro Stoxx 50. The calculation methodology for each index is a completion of the below formula:⁵

$$I_t = I_{t-1} \times \frac{\sum_{i=1}^n K_t^i}{\sum_{i=1}^n K_{t-1}^i} = I_{t-1} \times \frac{\sum_{i=1}^n S_t^i n_t^i \phi_t^i}{\sum_{i=1}^n S_{t-1}^i n_{t-1}^i \phi_{t-1}^i}$$

where

K_t^i : ($= S_t^i n_t^i \phi_t^i$) represents the adjusted capitalization for stock i at date t ;

n_t^i : number of shares for stock i at date t ;

ϕ_t^i : adjustment factor, which is typically the product of the free-float factor $f(i, t)$, cap factor $c(i, t)$ and the factor $p(t)$ serving to cancel the impact from extraordinary corporate actions or index constituent changes.

To illustrate the weighting by capitalization, we can rewrite the formula as $I_t = I_{t-1} \times \sum_{i=1}^n \frac{S_t^i}{S_{t-1}^i} w_t^i \gamma_t^i$ where $w_t^i = \frac{K_{t-1}^i}{\sum_{i=1}^n K_{t-1}^i}$ and $\gamma_t^i = n_t^i \phi_t^i / (n_{t-1}^i \phi_{t-1}^i)$ is the adjustment term for date t ($\gamma_t^i = 1$ implies no effective adjustment);

- *Price Weighted Index:*

There are only a few equity indices of this type. Dow Jones Industrial Average (DJIA) and Nikkei Index are the most well-known ones. The general calculation formula is

$$I_t = \frac{1}{D_t} \sum_{i=1}^n S_t^i \theta_t^i$$

⁵ It actually follows Paasche's index method. The index formula can be equivalently expressed, using a certain D_t , as $I_t = \frac{1}{D_t} \sum_{i=1}^n S_t^i n_t^i \phi_t^i$, which is used in many index descriptions, including S&P 500 or Euro Stoxx 50.

where

θ_t^i : weighting factor for i -th constituent stock (note: $\theta_t^i = 1$ for DJIA index);
 D_t : adjustment factor, which is used to normalize the index level and to keep its continuity upon extraordinary corporate actions or index constituent changes.

Obviously, relating the weighting to stock price level is not a rational index construction approach (note that $\sum_{i=1}^n S_t^i \theta_t^i = \sum_{i=1}^n \frac{S_t^i}{S_{t-1}^i} w_t^i$ where $w_t^i = S_{t-1}^i \theta_t^i$).

5.4 Equity Forward and Cash and Carry Strategy

An *equity forward* on a single stock is a contract between two parties where the long party is engaged to buy from the short party a notional quantity N of the stock at a pre-agreed delivery price or strike price K upon the expiry of the contract. The payoff for the long party is

$$\text{Payoff} = N \times (S_T - K), \text{ where } S_T \text{ is the stock price at expiry } T.$$

The *forward price*, denoted as F , is the value of K such that the forward contract value is zero at inception.

Let S_t be the stock price at time- t and $\tau = T - t$. In its general form, a stock is assumed to pay both cash and proportional dividends. Stock borrow/lending fee rate is called *repo rate*⁶ in modelling for equities. The time- t forward price F can be derived by replication.

Forward Price Formula Under Simplifying Assumptions

The simplifying assumptions are as follows

r : constant interest rate

q : constant repo rate

b : yield for the proportional dividend payable before T

η : a value such that $e^{-\eta} = 1 - b$

a : cash dividend payable at T .

We choose to re-invest all the repo return into the stock. Therefore, one stock at time- t will grow to $e^{q\tau+\eta}$ stocks at time T , or equivalently, $e^{-q\tau-\eta}$ will become one stock at T . Then, we have

Proposition 5.1 *Under the above assumptions, the forward price F is*

$$F = S_t e^{(r-q)\tau} (1 - b) - a \text{ or equivalently, } F = S_t e^{(r-q)\tau-\eta} - a \quad (5.6)$$

⁶ It is an abused vocabulary here as repo normally refers to a collateralized loan for cash.

Proof The short party can hedge the forward contract by a *Cash & Carry Strategy* that consists of buying $e^{-q\tau-\eta}$ stock at time- t financed by a loan with interest rate r and lending the stock until expiry. The table shows that, by setting $F = S_t e^{(r-q)\tau-\eta} - a$, the total P/L of the short forward position and the long stock position with a loan is zero, which means perfect hedging.

Time	Value of the forward	Stock position	Monetary position	Net position
t	0	$S_t e^{-q\tau-\eta}$	$-S_t e^{-q\tau-\eta}$	0
T	$-(S_T - F)$	S_T	$-S_t e^{(r-q)\tau-\eta} + a$	0

Similarly, the long position of a forward can be hedged with a *Reverse Cash & Carry Strategy*, which consists of a short position of the stock at time- t market price (e.g. by selling borrowed stocks or other available stocks). The cash from the stock short-selling is invested in money market to earn interests. To take into account the repo and proportional dividends, the long party needs to short $e^{-q\tau-\eta}$ stock at time- t , which will grow to 1 stock at T . $\square$

General Forward Price Formula*

Now, let's consider the forward price in its most general form that is used in practice. Denote

r_t : instantaneous interest rate

q_t : instantaneous repo rate⁷

b_i : yield for the proportional dividend applied at time t_i^-

a_i : cash dividend applied at time t_i^- .

We choose to re-invest all the repo return into the stock and the cash dividend is re-invested into money market.

Proposition 5.2 The forward price⁸ F for maturity T is

$$F = S_t n(0) e^{\int_t^T r_t dt} - \sum_{t_k \leq T} a_k n(k) e^{\int_{t_k}^T r_t dt} \quad (5.7)$$

where $n(k) = 1/\beta(k) = e^{-\int_{t_k}^T q_t dt} \prod_{t_k < t_i \leq T} (1 - b_i)$ represents the fraction of the stock at time t_k that will become one stock at T (see Remark 5.1).

Proof To hedge the forward on one stock, the short party needs to buy $n(0)$ stocks at time- t . The cash dividend received at t_k is $a_k n(k)$. Setting F by the formula 5.7 in the below table achieves the zero P/L of the hedging position.

⁷ If continuous dividend is used, q_t is the sum of repo rate and continuous dividend rate (see Sect. 16.7 for continuous dividend modelling). Normally, $q_t \geq 0$. But under some special market environment, q_t can be negative.

⁸ For further details including default risk on cash dividends, we refer to [86], Chapter 1.

Time	Value of the forward	Stock position	Monetary position	Net position
t	0	$S_t n(0)$	$-S_t n(0)$	0
T	$-(S_T - F)$	S_T	$-S_t n(0) e^{\int_t^T r_t dt} + \sum_{t_k \leq T} a_k n(k) e^{\int_{t_k}^T r_t dt}$	0

The long position of a forward can be hedged with *Reverse Cash & Carry Strategy*. The long party borrows $n(0)$ stock at time- t and returns 1 stock at T . $\square$

Remark 5.2 (Forward Valuation and Off-Market Forward Contract Pricing)

Generally, the pricing of a forward contract $N(S_T - K)$ with $K \neq F$ can be done by splitting the payoff into two parts: $N(S_T - K) = N(S_T - F) + N(F - K)$. As the valuation of the first term is zero, the forward can be valued as $N \times P(T)(F - K)$ where $P(T)$ denotes the discount factor. $\square$

5.5 Equity Index Futures

An *equity index futures* contract is similar to an equity forward linked to an equity index with standardized terms. It is listed on an exchange, with daily P/L settlement between the exchange and the investor. The daily settlement amount per contract is based on the Index Point Value representing the P/L per unit change of the futures price, i.e. $(F_t - F_{t-1}) \times \text{Index Point Value}$ where F_t is the futures settlement price at time- t . At expiration date, the futures settlement price coincides with the index price.

Example 5.2 E-mini S&P500 futures contracts have an index point value of \$50. An investor started a long position of 100 contracts at the price of 2010.50. If the futures settlement price for the next day is 2,001.00, then the investor's daily P/L will be $100 \times (2001.00 - 2010.50) \times \$50 = -\$47,500.00$, which means a loss for the investor. $\square$

The quotation for futures is usually subject to a minimum value called the *tick* (e.g., \$12.5 for E-mini S&P 500 futures). The observed difference (futures price – current index price) is called *basis* of futures contract.

5.6 Equity Swap*

An *equity swap* is an OTC contract between two parties where one party will receive cash flows linked to the performance of the underlying equity securities and pay the funding in cash form, usually linked to a fixed or floating rate (e.g., 3M SOFR + spread). It is normally a *total return swap (TRS)* where the dividends as

well as any other income/capital gain from the underlying securities are paid to the receiver of the equity performance.

An equity swap can be used to obtain leveraged exposure to the underlier if the long party does not fully collateralize the trade. Swaps with regular settlements known as *resettable swaps* have lower counterparty risk and require lower IA (independent amount, i.e. initial margin), which implies higher leverage.

Remark 5.3 Sometimes, the counterparties may have an ISDA agreement without a CSA. In such a case, to mitigate the counterparty risk taken by the payer of contingent cash flows, a *fully-funded* swap can be used whereby the payer of contingent cash flows receives the full notional amount at the inception instead of the funding payments during the life of the swap. The economic effect of a fully-funded swap is similar to a note. So, it may also be used when notes cannot be issued or cannot be distributed to investors (e.g., in some jurisdictions, offshore notes cannot be sold to onshore investors by regulation). $\square$

Exercises

1. Consider the stock CCB listed on the Hong Kong stock market (ticker: 0939.HK). It announced a cash dividend of \$0.319504 in HKD, with an ex-dividend date of 22 June 2017. The closing price on 21 June 2017 was 6.42. The market information on 22 June 2017 was

Date	Open	High	Low	Close
22 June 2017	6.11	6.21	6.1	6.12

- (1) Explain why the opening price on 22 June 2017 was not $6.42 - 0.319504$?
 - (2) Calculate the daily total return for 22 June 2017.
2. Given a non-dividend paying stock with its current price at \$100. The interest rate in the market is 2% p.a. (ACT/365).
 - (1) Calculate the 1-year forward price, stock lending rate is 2% p.a. (ACT/365).
 - (2) An investor is long a 6-month forward at the strike price of \$101 for 10,000 stocks. The stock price is \$115 at expiry. What is the P/L?
 - (3) Describe how to hedge a long position of an equity forward contract.
3. Explain the sentence “Unlike cash dividend, a stock dividend does not change the company’s market capitalization.”

Chapter 6

Foreign Exchange Instruments



The FX (foreign exchange) market, also called Forex or currency market, facilitates the trading of currencies. It starts on Monday morning in Wellington (New Zealand) and closes on Friday evening in New York (USA). A *non-convertible currency* is restricted for trading in offshore markets.

6.1 Quotation Conventions

A currency is represented by a three-letter ISO code, with the first two letters for the country and the last letter for the currency name in most of the cases (e.g., USD).¹

A currency exchange rate is quoted as the units of a “quote currency” (also called “counter currency” or “domestic” currency) per “base currency” (also called “foreign” currency).² In the market, the default order for being the base currency is EUR, GBP, AUD, NZD, USD, CAD, CHF, JPY, etc. (for instance, EUR/USD, USD/JPY). However, the default order is not mandatory and the precise definition of the FX notation used in a contract is necessary to avoid confusion.

In the market, the unit *pip* (price interest point) is used in quotations for the last meaningful digit: 1 pip for EUR/USD and most of the other pairs is 0.0001; 1 pip for USD/JPY is 0.01. 100 pips is one *big figure*. For instance, if EUR/USD exchange rate moved from 1.0425 to 1.0415, it is a decrease of 10 pips.

The exchange rate between any two non-USD currencies is called *cross-rate* that can be implied from the quotes against USD for each currency.

¹ Financial market practitioners sometimes use informal vocabularies for currency names: “Cable” or Sterling for GBP, “Kiwi” for New Zealand Dollar, etc. [27].

² The terms “domestic” and “foreign” currencies originally came from the holder’s location. In financial market, these terms are used to indicate one of the currencies in a FX pair but not associated with a location.

Example 6.1 $\text{USD/JPY} = 113.65$ and $\text{EUR/USD} = 1.0922$ imply $\text{EUR/JPY} = 1.0922 \times 113.65 = 124.13$. Otherwise, there will be *arbitrage*: if $\text{EUR/JPY} = 124.23$ is observed, one can make a profit of 0.10 JPY by selling EUR/buying JPY at 124.23, and simultaneously buying EUR/USD and buying USD/JPY.

6.2 FX Spot, Forward, Swap, Non-deliverable Forward (NDF)

A *spot* transaction is an immediate purchase or sales of one currency for another currency to be delivered in usually two working days, denoted as T+2 (T+1 for some pairs, e.g., USD/CAD).

A *forward* (also called *forward outright*) is an OTC contract between two parties to exchange currencies at a future date with a pre-set rate (called strike rate). The P/L for the long party can be calculated mathematically by an (implicit) payoff in domestic currency as

$$\text{Payoff} = N \times (X_T - K),$$

where

N : notional amount in *foreign* currency;

K : pre-agreed rate of the contract, i.e., strike rate;

X_T : foreign exchange rate at expiration date (which is also known as *value date*).

The *forward rate* is the strike rate F at which, the valuation of the forward contract is zero. In the market, the forward is quoted by the so-called “swap points” equal to (forward rate F – spot rate X_0), where X_0 can be anchored off spot mid-rate, i.e., the average of bid/ask spot rates.

An *NDF* (non-deliverable forward) is an outright forward for a non-convertible currency against a convertible currency (typically USD). At maturity, the counter-parties settle, in the convertible currency, the P/L from the difference between the contracted NDF rate and the prevailing spot rate on the agreed notional amount.

An *FX swap* is a contract between two parties in which one party agrees to buy (or sell) a certain amount of foreign currency with domestic currency at the current spot rate, and to sell (or buy) the same amount of foreign currency at a future time with a pre-agreed FX swap rate. It is composed of two transactions:

- (1) a spot transaction (buy or sell) at spot rate (e.g. spot mid-rate);
- (2) a forward transaction (sell or buy) at the pre-agreed FX swap rate (the FX swap rate that makes the value of the FX swap at zero is also the forward rate).

One can obtain a *synthetic loan* in domestic currency by combining a loan in foreign currency with an FX swap. For various reasons, there could be a difference between the cost of a normal loan in domestic currency and the cost of the synthetic loan for a company (see exercise 3 for the all-in-cost calculation).

Sometimes, there is a need for an FX forward or swap with a value date (i.e. expiration date) shorter than the spot date (which is typically $T + 2$). The formulas presented in the next section can be applied to determine the rate from the market observed spot rate and the interest rates.

An *FX futures contract* is similar to a standardized forward listed at an exchange with standardized terms and daily P/L settlement (e.g., CME's Euro FX futures).

6.3 Interest Rate Parity for FX Forward

For financial modelling, it is often convenient to consider the foreign currency as a stock, and to consider the interest rate of the foreign currency as a repo or continuous dividend yield on the stock.

Proposition 6.1 *According to covered interest rate parity (CIP), the market forward rate F should be*

$$F = X_0 \frac{P^f(T)}{P^d(T)},$$

where

X_0 : the spot foreign exchange rate;

$P^f(T)$, $P^d(T)$: zero coupon bond in foreign and domestic currency respectively.

Equivalently, the forward rate should be

- with money market rates R^d (domestic currency) and R^f (foreign currency):

$$F = X_0 \frac{1 + R^d T_d}{1 + R^f T_f},$$

where T_d and T_f are day count fractions for $[0, T]$ calculated with the conventions for domestic and foreign currencies respectively;

- with continuous rates (r^d and r^f for domestic and foreign currencies respectively, and T for the day count fraction for $[0, T]$ using ACT/365):

$$F = X_0 e^{(r^d - r^f)T}.$$

Proof If $K > F$, one can build the following arbitrage position at inception that is similar to the cash & carry strategy for stocks:

- sell one unit of forward at K ;
- buy $P^f(T)$ foreign currency, funded by a loan of $X_0 P^f(T)$ in domestic currency.

At maturity, we have

- short 1 unit of foreign currency, long K units of domestic currency;
- long 1 unit of foreign currency, short $X_0 \frac{P^f(T)}{P^d(T)}$ in domestic currency

$\Rightarrow$ The total P/L will be $K - X_0 \frac{P^f(T)}{P^d(T)} > 0$, which is an arbitrage.

Similarly, we can prove the case for $K < F$. □

6.4 Cross Currency Swap and Non-deliverable Swap (NDS)

A CCS (Cross Currency Swap) consists of exchanging the principals at the beginning using the spot rate, exchanging the interests during the tenor, and exchanging the principals back at maturity.

Example 6.2 Example of a USD/JPY cross currency swap assuming USD/JPY spot rate = 110.03:

At Inception:

- party A pays party B USD10m
- party B pays party A $\text{USD10m} \times 110.03 = \text{JPY1,100.3m}$

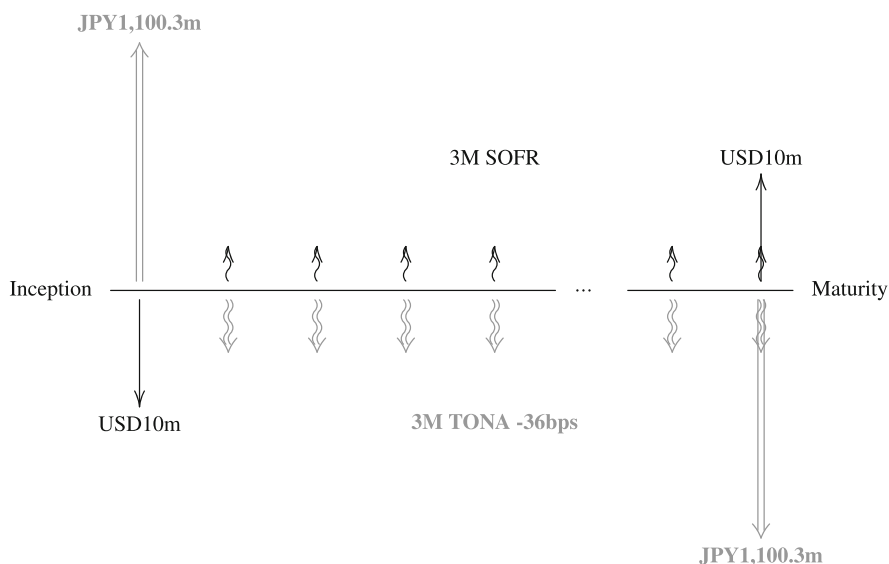
During the Trade Life:

- party A pays party B TONA -36 bps (if it is negative, party B will pay party A) on JPY notional amount
- party B pays party A interest of SOFR on USD notional amount

At Maturity:

- party B pays party A USD10m
- party A pays party B JPY1,100.3m

Graphical Illustration of CCS



A CCS can be understood as an instrument composed of two loans mutually offered in each currency. For each counterparty, one loan (short) in one currency is collateralized by the other loan (long) in the other currency. Hence, the counterparty credit risk is almost offset naturally and nicely.

The market quotation convention consists of applying a flat USD reference rate to the USD leg and a reference rate + *spread* on the non-USD currency when USD is involved, where *spread* represents the relative funding supply/demand for the currencies. A negative *spread* on a currency implies less funding need for that currency.

Between swap dealers, *Mark-to-Market CCS* is often a preferred CCS variation. It consists of adjusting the principal amount on one side (usually the USD leg, if applicable) periodically to bring the trade in line with the prevailing FX spot rate to minimize counterparty credit risk.³

CCS is a very useful instrument. It can be used to exchange cash flows in one currency for cash flows in another currency (combining with an IRS, if fixed cash flows are involved). See Sect. 30.7 for an example of its application in funding management.

For non-convertible currencies (e.g., KRW, CNY), there is a non-deliverable instrument called *NDS* (non-deliverable swap). It is a cross currency swap in which

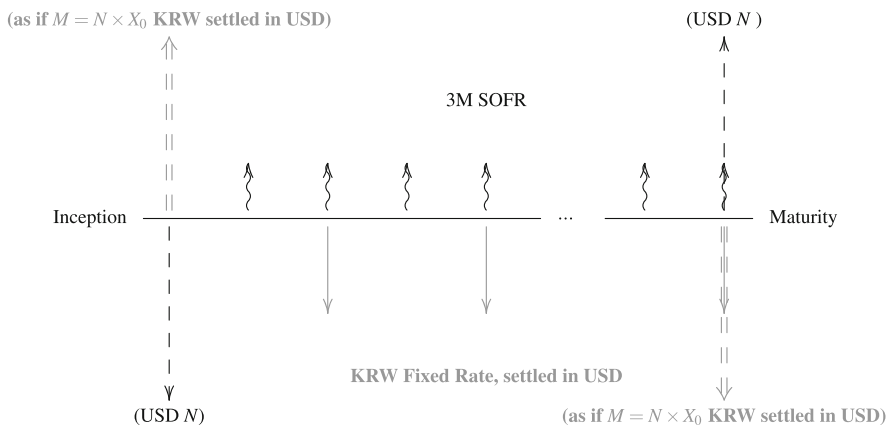
³ Here is a numerical illustration: initial EUR/USD spot 1.1600, exchange USD116m/EUR100m; at interest payment date, interest payments are based on USD116/EUR100m. Then, principal is adjusted to USD118m/EUR100m according to the prevailing EUR/USD spot 1.1800 and interest payments for the next period will be based on USD118/EUR100m; and so on.

one currency is non-convertible and the other one is convertible (typically USD). The special feature is that all cash flows are settled in the convertible currency by applying the prevailing exchange rate at the time of payment. This implies that

- at inception: no initial notional exchange (due to netting)
- at maturity: settlement in the convertible currency for (notional $\times$ difference between FX Rate at maturity and FX rate at inception).

The interest rate on the non-convertible currency is normally a fixed rate as there is no floating rate in offshore market.

Graphical Illustration of NDS on USD/KRW



Exercises

1. A student wants his HKD100,000 converted into CNY at a money exchange where the bid/ask for CNY/HKD are shown as 1.2 and 1.25 respectively. How much CNY will he/she receive?
2. Assume that the spot price of USD/CNH is 6.6741 (meaning 6.6741 CNH per USD). The 3-month money market rates are 4.2% for CNH (ACT/360) and 1.17% for USD (ACT/360).
 - (1) What is the base currency for the pair USD/CNH?
 - (2) If the spot price has increased by 20 pips, what is the new spot price?
 - (3) If the forward price for 3M tenor is 6.6751, what is the swap point?
 - (4) What is the 3-month forward price of USD/CNH, assuming the period has 90 calendar days?
 - (5) Someone agrees to make a 3-month USD/CNH forward transaction (either long or short) at the rate of 6.6800. How can I build an arbitrage position of USD20m notional to get a profit? And what is the P/L?

3. In a synthetic loan implemented by a company, the borrow cost in foreign currency is r^f and the FX swap rate is F . Prove that the equivalent all-in-cost of borrow in domestic currency r^d for the period T is

$$r^d = r^f \frac{\text{spot rate}}{\text{FX swap rate}} + \frac{\text{swap points}}{\text{FX swap rate}} / T \quad (\text{same day count convention is assumed})$$

Hint: (1) observe the cash flows in domestic currency at the spot date and the value date; (2) compare the above formula with $r^d = (\frac{F}{X_0}(1 + r^f T_f) - 1) / T_d$.

Chapter 7

Commodities



In this chapter, we outline the basic concepts and instruments related to commodities. For a more detailed introduction, we refer to [28, 101].

7.1 Commodities Overview

Commodities refer to raw or processed goods that are used in the production of other goods or for direct consumption. The major commodity categories are

- *precious metals*: gold, silver, platinum, etc.;
- *energy*: crude oil (e.g., WTI, Brent), natural gas, electricity;
- *environmental commodities*: carbon credits, carbon offsets, renewable energy certificates, water rights;
- *metals*: base (copper, zinc, nickel, aluminum), ferrous (iron, steel, iron ore);
- *agricultural*: grain/seeds (wheat, corn, soybean, rice, oats), soft commodities (coffee, sugar, cocoa, cotton, rubber), livestock (live cattle, lean hogs, pork bellies).

A *commodity index* represents the weighted average of the price (i.e., spot price or futures price) performance of a basket of commodities. Well-known commodity indices include S&P GSCI Index, Dow Jones-UBS Index, Thomson Reuters/Jefferies CRB Index, Rogers International Commodity Index (RICI). We note that the diversification is limited for most commodity indices, as energy often represents a heavy weight. Hence, such commodity indices are not the perfect tool for hedging inflation.

Investment exposure to commodity asset class is usually realized through the following instruments:

- commodity futures listed on an exchange;
- mutual fund making commodity related investments, or commodity ETF;
- swap, option with commodity underliers;
- structured product linked to commodity or commodity index;
- stocks of commodity companies.

7.2 Commodity Forward and Futures

The shortest-dated market commodity transaction is normally considered as the spot transaction. A commodity *forward* is an OTC contract between two parties. The long party is engaged to buy from the short party a certain quantity (notional) of the underlying commodity at a pre-agreed price (called delivery price or strike price) at the expiration date of the contract. Mathematically, the payoff of the long party is

$$\text{Payoff} = N \times (S_T - K),$$

where

N : notional amount of the commodity;

S_T : commodity price at expiry;

K : delivery price or strike price.

The final settlement of the forward may follow either of the below methods

- physical delivery: only real commodity users may choose physical settlement
- cash settlement: this is the most common for financial market participants

The forward price F is the strike price such that the valuation of the forward is zero.

Commodity futures contracts are similar to forward contracts with standardized terms that are transacted at an exchange, normally with daily P/L settlement and physical delivery at maturity. Holders of the contracts need to sell their holding to the market before expiry if they do not want the delivery of physical commodity.

Example 7.1 The Coffee C contract¹ is the world benchmark for Arabica coffee. The contract prices physical delivery of exchange-grade green beans, from one of 20 countries of origin in a licensed warehouse to a port in the U.S., Europe, etc.

¹ Source: extracted from Intercontinental Exchange (ICE) Website.

Contract size	37,500 pounds
Price quotation	Cents and hundredths of a cent up to two decimal places
Contract listing	March, May, July, September, December
Settlement	Physical delivery
Min. price movement	5/100 cent/lb., equivalent to \$18.75 per contract
Grade/standards	A Notice of Certification is issued based on testing the grade of the beans and the flavor
Deliverable growths	Mexico, Salvador, Guatemala, Costa Rica, Nicaragua, Kenya, New Guinea, ...
Deliverable points	Exchange licensed warehouses in the Port of New York District and the Port of Virginia (each at par), the Port of Antwerp, ...
...	

Spot-Forward Relationship

Except for a few commodities such as precious metals, it is difficult to implement cash-and-carry strategies for hedging forward contracts. Actually, forward/futures prices depend on several factors, including demand/offer, geopolitical and environmental changes.

In modelling, the below concepts are used to establish the relationship between the spot price S_0 and the forward price F :

- *interest rate* (denoted by r): the financing rate
- *storage cost* (denoted by c): actually $r + c$ represents the carry cost
- *convenience yield* (denoted by y): it represents the benefit of holding physical commodities instead of holding of forward/futures contracts. Its value can be positive or negative.

A forward price can be modelled by the below relation

$$F = S_0 e^{(r+c-y)T} \quad \text{where } T \text{ is the day count fraction for the period } [0, T].$$

As there are no market quotations for c and y , practitioners often use one parameter $\hat{y} = y - c$ so that $F = S_0 e^{(r-\hat{y})T}$. This formula is not useful for pricing forward/futures prices that are mostly driven by demand/offer. The parameter $\hat{y}$ is implied from market futures/forward prices, but not the other way around. However, the formula and the parameters are useful for other derivatives such as options.

Commodity Price Term Structure

In commodities market, people often observe two typical shapes for a futures/forward price term structure:

backwardation: the downward sloping curve is called *backwardation*.

contango: the upward sloping curve is called *contango*.

Futures price eventually converges to spot price by construction. In a contango market, futures contracts are trading at a premium to spot price. It may imply that the carry cost increases with time. Backwardation is the situation where future prices

are less expensive than spot price. The reason for backwardation may be shortages, political events or weather concerns.

Seasonality*

Some commodities, e.g., natural gas, electricity, show strong patterns of seasonality due to consumption seasonality and storage difficulties.

Roll-Over of Futures Contracts*

A futures contract has a fixed expiration date on which the price of the futures contract converges to the spot price of the commodity. Investors may choose to “roll” a futures contract before its expiry to a futures contract in a further-out month, i.e., close the contract about to expire and enter into a new contract with a later expiration date.

It is customary to roll the front-month contract to the second-nearest contract because the near contracts are more liquid. If the spot price of the commodity stays stable, the “roll yield” (i.e., P/L by rolling the contract) for the investor will be

- *positive*: in a backwardation futures market
- *negative*: in a contango futures market

Instead of rolling the front-month futures to the second-nearest futures, there are numerous sophisticated rolling strategies used by traders that are based on the dynamic term structure of the commodity.

7.3 A Special Commodity: Gold

Gold is the most popular precious metal for jewellery and investment. It is considered as

- a “currency”: relatively independent of interest rate policies. In most of the investment banks, the trading of gold is carried out by the FX team;
- a safe-haven asset: throughout history, gold has been a safe-haven asset, providing shelter in periods of financial crisis or turmoil;
- an inflation hedging asset: gold price is intimately linked to the real inflation rate of the economy.

The gold price reference largely used in the financial market is the *London Gold Fixing* by LBMA (London Bullion Market Association). It is the gold price from an auction that takes place twice daily at 10:30 am and 3:00 pm London time, with the price set in US dollars per fine troy ounce. A *spot* transaction is settled at T+2.

Gold is virtually an indestructible metal, as it does not corrode. This helps to develop a special type of transaction called a *gold lease* through which the lender of gold earns an income that can be paid in cash or in gold. For instance, a central bank may have an interest in lending out physical gold to a jewellery manufacturer.

Forward price for gold is $F = S_0 e^{(r - r_{gl})T}$ where r_{gl} is gold lease rate. A gold swap consists of buying physical gold at spot price and selling back at forward

price from the gold borrower's perspective. For physically settled transactions, *Loco* represents the place where gold with a certain grade is physically held. Examples are loco London, loco Zurich, etc.

Exercises

1. Explain why commodities are known to bring diversification benefits for investors.
2. Name the differences between a stock market index and a commodity index.
3. A forward curve can be represented by a graph that displays futures prices against futures maturities.
 - (1) Plot the forward curve examples for backwardation and contango structures;
 - (2) From a graphical perspective, explain which term structure tends to benefit the futures roll-over and how the slope of the forward curve comes into play?
4. The current price of a commodity is \$50, the continuous interest rate is 1.50% flat, the storage cost is 0.32% p.a. (continuous rate), and convenience yield is 0.60% p.a. (continuous rate).
 - (1) What is the forward price for a 2-year maturity contract with zero cost? Is it backwardation or contango?
 - (2) An investor is long 50,000 commodity forward with the forward price at \$52. Assuming the commodity price at expiry is \$52, what is the investor's payoff?

Chapter 8

Credit Derivatives



8.1 Credit Default Swap (CDS)

First introduced by JP Morgan in 1997, *CDS (Credit Default Swap)*¹ remains the main instrument for credit exposure and hedging. Centralized clearing was put in place following the risks evidenced by the 2008 financial crisis.

In a CDS, the *protection buyer*

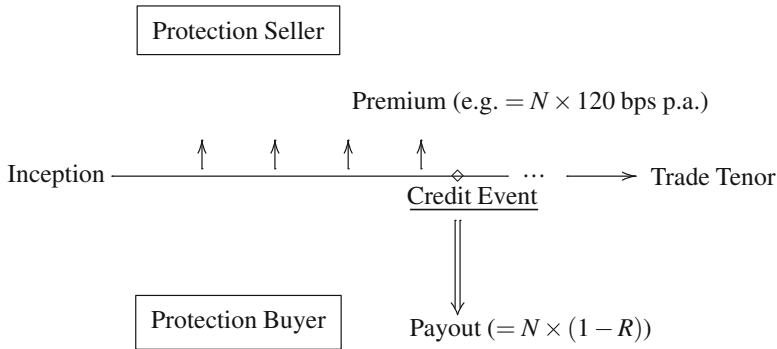
- makes periodic payments (called CDS spread or premium) to the *protection seller* until either a credit event or the trade tenor if no credit event occurred;
- will be made whole for the exposure to the reference entity (also known as the *name* for simplification).

Several standard credit events are referred to in the ISDA definitions for credit derivative transactions: “bankruptcy”, “failure to pay”, “restructuring”, “repudiation/moratorium”, “obligation acceleration”, “obligation default”, etc. There are two types of settlement in case of a credit event:

physical settlement: the protection buyer has to deliver a bond of seniority not lower than that of the reference obligation plus the accrued CDS interest to the protection seller for receiving the face value of the bonds (notional amount) in return;

cash settlement: the protection buyer receives “notional $\times$ (1 – recovery rate)”. The recovery rate is determined by the Credit Event Auction.

¹ In China Interbank market, there are similar instruments known as CRMA (Credit Risk Mitigation Agreement) and CRMW (Credit Risk Mitigation Warrant). CRMA is similar to CDS, but their details may differ slightly (e.g., default events definition). CRMW is a securitized instrument that requires the payment of the total protection premium at inception so that the issuer (protection seller) is not taking counterparty risk from the protection buyers and hence the security can be traded in secondary market (under some conditions by regulation though).



N : notional amount

R : recovery rate. For senior unsecured debt, R is usually assumed to be around 40% in pricing.

At CDS inception, the spread s is set at a level such that the fair value of the premium payments (known as the *premium leg*) is equal to the fair value of the protection leg (also known as the *default leg*). Actually, the general relationship between the default intensity λ and the spread s is $s \approx (1 - R)\lambda$, which is known as the *credit triangle*.²

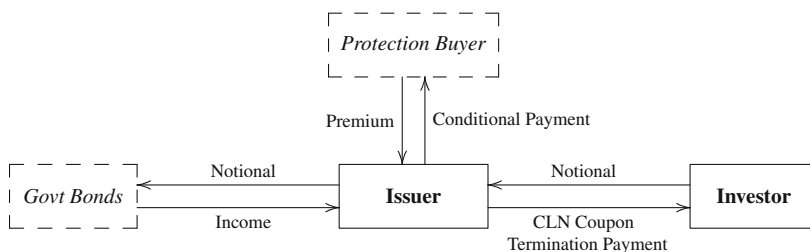
Remark 8.1 (Synthetic Exposure v.s. Bond) There is a link between a CDS and the credit spread implied from the credit bond price of the reference entity because the combination of “long *rate (treasury) bond* + short *protection through CDS*” produces a similar risk/return profile as the related credit bond. $\square$

8.2 CLN (Credit Linked Note)

A Credit Linked note (CLN) is a debt instrument in which the investor takes the credit risk of one or more reference entities in return for an enhanced yield. In case of a credit event, the investor will receive either a physical or cash settlement based on the recovery rate from credit auction, or a pre-agreed redemption level that is usually set to zero for boosting the note yield. A simple CLN is a funded version of a CDS issued by an SPV, whereby the investors are selling protection via CDS on the reference entity and buying “safe assets” such as treasury bonds serving as collateral for the CDS. The net income from the CDS and safe assets constitute the note yield. The CLN investor takes the risks of the reference entity, the “safe assets” as well as the correlation. CLN can also be issued by an investment bank, whereby the investor takes the risk of the bank in addition to the risk of the reference entity

² The proof will be given in Chap. 28.

while the CLN coupon will include both the funding spread of the bank and the CDS spread of the reference entity. A single-name CLN is similar to a corporate bond in risk profile. The credit exposure using CDS makes the product more attractive than the bond when there is a positive basis between CDS and bond spread, i.e., the CDS spread is higher than the spread for the bond.



CLN provides access and flexibility to the reference entities that may not have issued bonds or may not have liquid bonds for the maturity, size, and features required by investors. A popular CLN linked to a basket of reference entities (names) is the FTD (first-to-default) note. It offers a higher yield than a single-name CLN by taking the risk of the first credit event, which has a higher probability to occur. FTD notes and swaps were actively traded before the 2008 financial crisis for yield enhancement and correlation trading. However, in the post crisis era, these multi-name products tend to provide less yield enhancement to investors due to heightened correlations applied in pricing. While some market participants exited this activity, many structured credit desks in investment banks still trade them OTC.

8.3 Credit Index*

A credit default swap index is a standardized credit derivative that offers the credit exposure of a portfolio of reference entities. It is a basket of single-name CDS. Compared to a single-name CDS, it offers better liquidity and diversification. For example:

- CDX NA IG: 125 investment grade (IG) companies in North America (NA), administered by CDS Index Company and marketed by Markit Group Limited;
- iTraxx Europe: 125 investment grade companies in Europe, managed by the International Index Company (IIC), also owned by Markit.

These portfolios are updated (rolled over) on March 20th and September 20th every year.

Credit indices can be traded with licensed dealers providing liquidity. An index is traded with a fixed coupon (e.g., 1%), typically paid quarterly (20th of Mar, Jun, Sep, and Dec) by the protection buyer (who is taking a short position of the index).

Upfront payments are made at the initiation and at the closing of the trade to reflect the change in price of the index. The indices are quoted on a clean basis.

Example 8.1 (Illustration Example from [74]) An index is launched with the price of 100 on Sep 20th, fixed coupon of 60 bps, 5Y tenor.

- On Nov 30th, an investor buys USD10m protection on the index when the spread has moved to 90 and the corresponding price is 98.67 (the price is par minus the PV of the spread differences).

Upfront payment: $\text{USD}10\text{m} \times (100 - 98.67)/100 = \text{USD}133,000$

Receiving the accrual up to trade date: $71/360 \times \text{USD}10\text{m} \times 0.0060 = \text{USD}11,833.33$

Net Outflow = $\text{USD}133,000 - \text{USD}11,833.33 = \text{USD}121,166.67$;

- On Dec 20th, the investor pays the fixed coupon: $0.0060 \times \text{USD}10\text{m} \times 91/360 = \text{USD}15,166.67$;
- On Mar 13th, the investor closes the trade when the spread is 120 while the price is 97.44, and pays the accrued interest up to trade date and receives payment:
Inflow = $\text{USD}10\text{m} \times [(100 - 97.44)/100 - 0.0060 \times 84/360] = \text{USD}242,000$.

□

Upon credit event, protection buyer obtains the payout for the reference entity according to its weight w_i in the index:

$$N \times w_i \times (1 - \text{recovery rate}).$$

Normally, cash settlement is applied as a standard. Following a credit event of a constituent in the index, a new version of the index is published, which assigns a zero percent weight to the defaulted entity. The notional amount of the index trade is reduced by its weight in the index. For instance, consider a USD10m trade on an index with 100 names. After one default occurred, the new version of the index will contain 99 names and will have a revised notional of USD9.9m.

8.4 Collateralized Debt Obligation (CDO)*

Asset-Backed Securities (ABS) allow their issuers to raise cash and also to offload the risk of the underliers that are often illiquid and may not be easily sold on their own. Securitization consists of pooling assets together and creating tranching financial instruments with different risk/return profiles to be sold to investors. It offers credit enhancement to senior tranches with increased credit risk to the junior tranche.

An ABS with bonds or loans as underliers is known as a *cash* CDO (collateralized debt obligation). A CDO is commonly structured through an SPV (special purpose

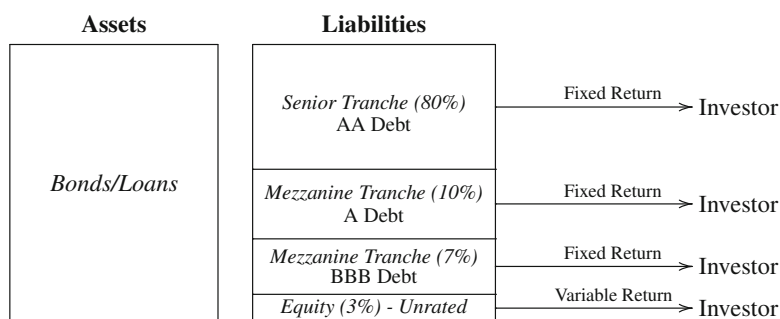
vehicle, often in the form of a company³) that buys bonds/loans as *assets* funded by the *debt* and *equity* of the SPV sold to investors. The net revenue (after all costs and losses due to defaults) from the assets is distributed to the debt and equity investors. Multiple tranches related to the debts and equity in the capital structure of the SPV are issued with different payment priorities. The tranches are categorized as *senior* debt, *mezzanine* debt and *equity* with increasing levels of risk compensated by increasing (expected) return rates. The highest senior tranche is also known as *super-senior* tranche. Unlike the debt tranches that usually have fixed rates, the return for equity tranche consists of the remaining distributable income after paying all the debt tranches. Each tranche is associated with two thresholds for principal loss:

- *attachment point*: the default ratio of the portfolio for triggering loss in this tranche;
- *detachment point*: the attachment point of the next tranche.

The losses are born by investors in a waterfall model, from junior to senior tranches, while loss recovery as well as prepayments or amortization normally reduce the most senior tranche first.

An illustration example of a CDO with 4 tranches is as follows:

Credit Tranche	Return rating	rate	Attachment	Detachment	Absorbing portfolio loss
Senior	AA	1%	20%	100%	from 20% until 100%
Mezzanine	A	1.6%	10%	20%	from 10% until 20%
Mezzanine	BBB	4%	3%	10%	from 3% until 10%
Equity	not rated	variable, e.g., 12%	0%	3%	from 0% until 3%



³ Other forms exist, e.g., *trusts* or *specific asset management plans* in China.

8.4.1 *Synthetic CDO**

By Remark 8.1, a CDO can be constructed by combinations of “long rate bond and short protection through CDS” instead of credit bonds, and it is known as a *synthetic CDO*. The premium payments from the component CDS’ constitute the revenue cash flows for the structure. Synthetic CDOs are easier, flexible and less costly to create than cash CDOs provided that there are liquid CDS supplies. However, as a synthetic CDO is built with CDS instead of bonds, its size is not limited by the available bonds and may be out of control. The most common trading instruments based on synthetic CDOs are:

- note on a single CDO tranche: the investor of a CDO note pays upfront the notional and receives periodic coupon payments (also called *premium leg*). The losses due to defaults (known as *default leg* or *protection leg*);
- swap on a single CDO tranche: it has a payoff leg and a premium leg. The protection buyer pays periodic premium to the seller. As the loss to the tranche increases, the premium notional of the tranche decreases.

The valuation of a CDO instrument depends on the valuation of the premium leg and the default leg. The initial pricing of a single tranche CDO swap consists of setting the premium leg equal to the default leg, which will be discussed in Sect. 28.4.3.

8.4.2 *Single Tranches on Credit Index**

Single tranches are traded on some credit indices. An investor can choose the exposure to a particular portion of an index loss distribution. For example, the tranches for CDX NA IG are 0–3%, 3–7%, 7–10%, 10–15%, 15–30% and 30–100%. The protection buyer of a tranche makes regular (e.g., quarterly) coupon payments to the protection seller and receives a payment in case of a credit event. Like a credit index trade, payments are made at the initiation and the closing of a single tranche trade to reflect the change in price. Coupon payments (e.g., 500 or 100 bps per annum) are made on the remaining notional until the tranche gets fully written down due to credit events or until maturity.

Following a credit event, equity tranche investors are the first to take the loss of $(100\% - \text{recovery rate})$. The principal of any mezzanine tranche is unaffected until the equity is wiped out and the most junior mezzanine becomes equity. The super-senior tranche may use a more complex definition, where the notional used to determine the coupon payment not only amortizes from the bottom when losses impact the capital structure, but may also be “amortized from the top” by the net recovery in case of default (because the recovered amount can no longer be lost), so that the notional on which the whole capital structure pays a coupon reduces similarly to the outstanding notional of the underlying basket of CDS. Readers can refer to [74] for detailed explanation with illustration examples.

Exercises

1. Consider an investment grade company and its corresponding 5Y CDS contract. Are the equity price and the CDS spread mostly moving in tandem or inversely related? Explain your rationale.
2. Assume a CDO of a total asset value of USD500m invested in high-yield bonds with a weighted average return of 5%. The total annual fee (management fees + other fees/charges) is 1% on the total asset value.

Tranches	Notional in M USD	Interest rate
AAA rated	N	2.5%
A rated	150	3.5%
BBB rated	100	R
Equity (unrated)	50	9.5%

Please determine the values of “ N ” and “ R ” in the above table after 1Y investment.

3. On 01 Dec 2017, the 5-year CDS spread on a corporate bond is 3.0%. The same corporate bond’s yield is 6% and the 5-year treasury bond yield is 2.58%. Calculate the basis and explain how you can arbitrage this situation.

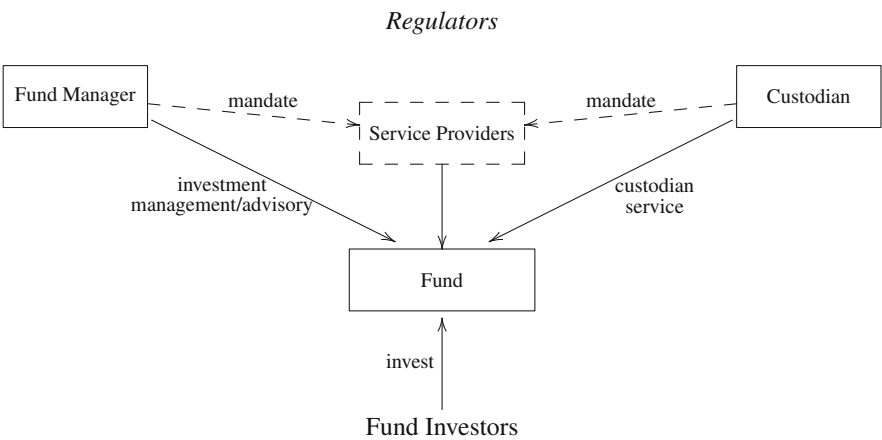
Chapter 9

Investment Funds



9.1 Funds

A fund is an investment scheme that pools money from investors. It is managed by investment advisors known as fund managers, usually from a management company that is legally and financially distinct from the fund and its assets. A fund has the objective to produce higher risk-adjusted returns than its benchmark (typically an index). Generally speaking, the investment universe for funds could include all investable assets and instruments. However, we only introduce here the funds investing in tradable securities and instruments such as bonds, listed stocks, etc. The parties involved in an investment fund activity may be illustrated by the below chart:



There are two main types of funds: *mutual funds* that are open to all investors, and *hedge funds* that are suitable for sophisticated and accredited investors.

9.1.1 Fund NAV and Fees

The value of a fund unit is represented by its *net asset value* (NAV), which is defined as

$$\text{NAV} = \frac{\text{value of the assets held by the fund} - \text{liabilities}}{\text{number of shares (or units)}}.$$

There are different charges linked to the investment in a fund:

- management fee: a fixed percentage applied to the NAV (e.g., 1.2% p.a.); the fee level depends on the required skills/involvement/added value of the fund manager;
- custody fee: e.g., 0.03% of the fund NAV;
- sales charges or loads: *front-end load* for subscription, *back-end load* charged upon redemption. Back-end load is often decreasing with the holding time to encourage long-term investment into the fund;
- other expenses, including distribution and marketing costs.

The *total expense ratio* (TER) of a fund measures the total cost to investors. It is defined as

$$\text{TER} = \frac{\text{management fee} + \text{all the other fund expenses}}{\text{total asset value}}.$$

9.1.2 Fund Organization*

A fund can be legally set up in several ways, including

- a unit trust: a fund that adopts a trust structure whereby investors are trust beneficiaries; e.g., investment funds in UK and Hong Kong;
- a corporation with variable capital: investors are shareholders; e.g., SICAV in France and Luxembourg, mutual funds in the US, OFC in Hong Kong and VCC in Singapore. A subscription (redemption) implies a capital increase (decrease) of the company;¹ A fund set up as a corporation is overseen by a board of directors and an annual report to fund investors is required;
- a contractual investment scheme: usually by a tri-party agreement between the investor, manager, and custodian of the fund, e.g., asset management funds in China, FCP (fonds commun de placement) in France;
- a limited partnership: the fund manager is the general partner and investors are limited partners. It is mostly used for hedge funds and private equity funds.

¹ SICAV—société d'investissement à capital variable; OFC—open-ended fund company; VCC—variable capital company.

There are different forms for a fund, including

open-end fund: all buy/sell trades of fund units are realized in primary market through subscription and redemption. Open-end funds constitute the majority of investment funds;

closed-end fund: a fixed number of non-redeemable shares are issued at initial offering. The shares are then traded in secondary market. The share price may deviate from the NAV of the fund because of demand/supply (often at a discount to the fund NAV). The maturity can be perpetual or fixed;

exchange-traded fund (ETF):² an index tracking fund listed on an exchange. ETF units are bought and sold between investors at the exchange like listed stocks. However, for each ETF, there are *participating dealers* (PDs) that are typically market makers of the ETF authorized to subscribe and redeem fund units with the ETF manager by an in-kind transfer of a basket of securities with the weights reflecting the ETF composition.³ This process prevents the trading price in secondary market at the exchange to be deviated significantly from the ETF NAV (by PDs' arbitrage);

fixed maturity plan (FMP): a fund having fixed maturity, which is close-ended in nature. It aims to invest in bonds with a similar maturity to the fund itself. So, the fund is similar to a principal-protected income product (subject to no default). We note that a fixed maturity fund can be used to offer formula-based returns linked to the underliers' performance. In this case, the fund is known as a *structured fund* and it is actually a structured product distributed as a fund.

9.1.3 Share Classes*

One may invest in a mutual fund through a certain *share class*, usually named by a single letter, which represents a type of fee/cost structure and the corresponding NAV. For instance, based on the same investment fund, “A” may be the share class with USD denomination while “B” is the share class with EUR denomination; “I” is often used for institutional share class, which has a lower management fee but a higher minimum subscription amount; A “distribution” share class pays regular dividends while an “accumulation” share class does not pay dividend (or equivalently, reinvests the dividend back into the fund).

² In China, there are *listed open-ended funds (LOF)* that are public open-end funds listed on an exchange. A LOF is traded at the fund's NAV and enjoys both the traditional distribution channel and the exchange channel for fund subscription/redemption.

³ In some markets, e.g., in China, investors of large size may also be authorized to subscribe/redeem fund units in-kind through participation dealers.

9.2 Mutual Fund

By *mutual funds* or *public funds*, we mean here the investment funds that are accessible to all types of investors, including individuals. They are highly regulated and supervised with a high standard of transparency and diversification:

- in the US, regulated by the Investment Company Act of 1940
- in Hong Kong, regulated by the SFC (Securities and Futures Commission)
- in Europe, the UCITS qualification offers the passport for selling funds across European countries

Mutual funds are managed by licensed investment advisors, i.e., fund managers.⁴ A mutual fund's assets are usually safeguarded by a custodian. For open-ended funds, there is usually daily liquidity for subscription/redemption. A prospectus (or explanatory memorandum) is required for each fund.

A mutual funds is generally classified by its main investment focus⁵ described in the prospectus:

- money market fund:⁶ investing in money market instruments and bonds with short expiration date;
- bond fund: e.g., emerging market bond fund, high-yield bond fund;
- equity fund: such as mid-cap fund, high dividend yield equity fund;
- hybrid fund: with its investment allocated to several asset classes.

The management fees for an actively managed mutual fund may range from 0.25 to 1.50% of the NAV. For passive funds such as ETFs and index funds, the management fee level is lower (e.g., 0.15%) because the fund management does not require securities research and investment skills.

9.3 Hedge Fund*

A hedge fund aims to achieve an absolute return by investing in a range of markets with a wide variety of styles and financial instruments that may include illiquid assets.⁷ It is restricted to sophisticated or accredited investors such as non-bank financial institutions and HNWI (high-net-worth individuals) with a minimum

⁴ e.g., in Hong Kong, the license Type 9 of SFC covers asset management.

⁵ The threshold of 80% is used in some jurisdictions to define the *main investment focus*.

⁶ It is a competing product to bank deposits. The NAV of a money market fund targets to be maintained at 1 dollar by paying and re-investing the accrued interest as dividend in fund units on a daily basis.

⁷ A special account known as *side-pocket* may be used to segregate illiquid assets that are hard to value. For a side-pocket created after some special situation, only existing investors have exposure to it usually.

investment size.⁸ Hedge fund managers often have significant stakes in the funds, which serves to align their own interests with those of investors.

The hedge funds and their managers have historically been less regulated compared to mutual funds. A hedge fund is normally not very transparent, offering limited liquidity for subscription/redemption (e.g., monthly, semi-annually).

The legal structure of a hedge fund varies depending on its location and investors. In the US, they are generally structured as limited partnerships so that investors are not subject to both entity-level and personal-level taxation. Many hedge funds are established in offshore financial centres to avoid adverse tax consequences for investors.

A hedge fund has one or more *prime brokers* providing the services including execution, custody, leverage, short-term financing, and securities lending. Prime brokers may even help fund managers meet with potential investors.

The fees charged by hedge fund management firms typically include:

- management fees: typically 2% on NAV;
- early redemption fee (or early withdrawal fee);
- performance fee: typically 20% on the out-performance above its HWM (*High Water Mark*) that is the highest NAV level (after fees) over a specified past period, or a *Hurdle* that is a benchmark rate (e.g., SOFR + spread). As the buying/selling time and NAV for investors may be different, the performance fee should be individualized.⁹

Examples of hedge fund strategies are:

- *long/short equity*: investment strategy with both long and short positions, e.g., 130/30 fund targets to have 130% in long position and 30% in short position;
- *global macro*: investment strategy in share, bond, or currency markets in anticipation of global macroeconomic events;
- *event-driven*: seeking investment opportunities in corporate events;
 - distressed securities: traded at a deep discount due to restructuring, bankruptcy, etc.;
 - risk arbitrage: typically to identify M&A deals, buy target companies and sell acquiring companies;
 - special situations: spin-off, shares buy-back, security issuance, asset sales, etc.;
- *relative value*: it is a market neutral strategy taking advantage of relative discrepancies in price between securities. The list of such strategy includes fixed-income arbitrage, equity market neutral, convertible bond arbitrage and capital structure arbitrage;

⁸ For instance, the minimum subscription size is USD50k for a single hedge fund in Hong Kong.

⁹ There are mainly two methods for individualized performance fee calculation: *equalisation*, which is based on one NAV for all investors, and *series accounting*, which consists of creating a new series with its own NAV for each dealing day.

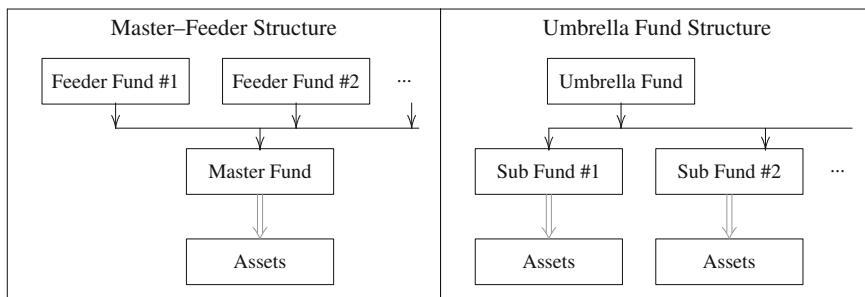
- *managed futures fund* (also known as CTA fund—commodity trading advisor fund): trades mainly futures on stock indices, commodities, forex, etc. The fund uses quantitative techniques, such as trend-following, technical pattern recognition, to take long and/or short positions.

An investment bank may offer structured products linked to the performance of a hedge fund by setting up a *managed account*, typically through the asset management arm of the bank. The hedge fund manager will be mandated as investment advisor for making the similar investment as the target hedge fund. However, the managed account is transparent for the constituent assets to the bank, which makes the delta hedging (see Sect. 10.3.5) for the structured product possible. To financial institutions, the regulatory capital risk weight for an investment in a managed account is often lower than that for an investment in the hedge fund itself.

9.4 Fund Structures*

Fund investment may be carried out with a stand-alone fund or through a more complex structure that offers flexibility and efficiency. Typically, there are

- *stand-alone fund*: it is the most common form for a fund;
- *master-feeder structure*: whereby multiple funds (feeder funds) using the same investment strategy can pool their investment and be managed as a bigger fund (master fund);
- *umbrella fund structure*: it allows a fund to create sub-funds and each sub-fund provides a different strategy or rights to investors. Investors can switch strategies without significant extra charges.



It is maybe worth mentioning the *tranchet fund*¹⁰ that allows the fund unit (called master unit) to be split into sub units providing different return/risk profiles that

¹⁰ In 2007, the first *tranchet fund* was set up in China. For various reasons, including the complexity and some reported risk sharing issues among the sub units investors, this type of fund is now highly regulated and restricted to private funds for accredited investors only.

can be traded separately (e.g., on an exchange). However, only the master unit can be redeemed so that the fund can be managed as a normal fund. A simple implementation consists of splitting the master unit into two sub units named as A and B whereby A provides funding to B who has the full P/L of leveraged exposure deducted by the funding cost paid to A. A tranching fund may also be used for more complex structures in which each sub unit is an option provided that the sum of all the sub units is just the fund itself.¹¹

9.5 Fund Derivatives*

Investors can get exposure to fund performance through derivative structures provided by investment banks. For hedging a derivative linked to a fund, the hedging party should have an agreement with the underlying fund for trading at NAV without subscription/redemption charges and also receiving a rebate of part of the management fee as a reward for increasing the fund's AUM (assets under management). The underlying funds are mostly mutual funds, as the lack of liquidity and transparency of hedge funds makes it difficult to dynamically hedge derivative structures. The most common structures are as follows:

- leveraged exposure: total return swap, leveraged note, etc.
- protected exposure:
 - option based structure e.g., an investor may purchase a call option (see Chap. 10) or a note with an embedded call option with a fund as underlier. The volatility target (see Sect. 32.1.1) is often used to avoid the option provider to hedge the fund volatility risk.
 - gap risk based structure, e.g., CPPI (see Sect. 32.1.3) that applies dynamic investment allocation between risky and safe assets to mitigate gap risk.

Exercises

1. Find out what are *target-date funds* and list their pros and cons.
2. An open-end fund has reported a net asset value of \$1 per share for all the historical trading days. Find out the type of this fund and explain such phenomenon.
3. Investors are becoming more aware of the Environmental, Social and Governance (ESG) criteria in an investment process. Explain how such criteria make an impact on index funds.

¹¹ Technically, a combination of options could be no more an option. For instance, a position of long call and short put with the same strike offers the same return as a forward contract.

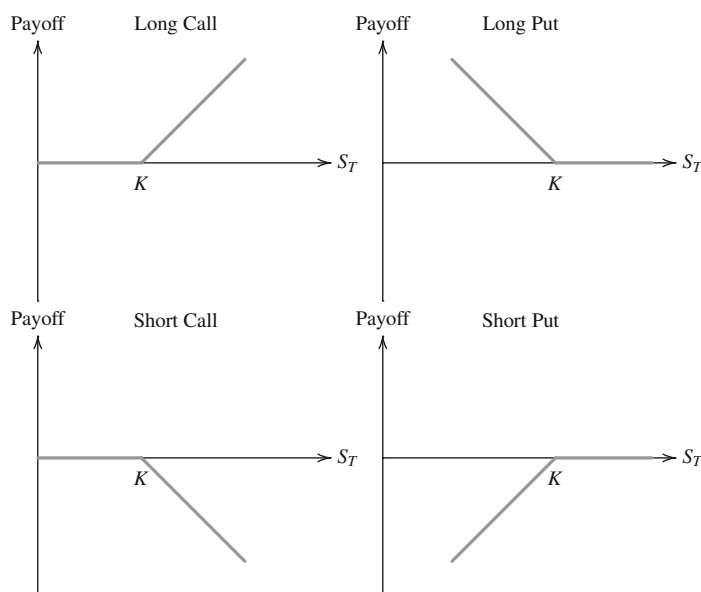
Chapter 10

Options



An option is a financial instrument that may offer an asymmetrical exposure to the performance of the underlier. There is a large variety of options, but call and put options are the most basic ones.

We recall that a *call option* gives the owner the right but not the obligation to buy the underlier from the seller at a pre-agreed price, known as strike price or exercise price. The payoff of a call option at expiry T is $N\max\{S_T - K, 0\}$ where S_T , K , N represent respectively the final underlier's price, the strike price and the notional amount of the option. Similarly, a *put option* gives the owner the right but not the obligation to sell the underlier at the strike price, and its payoff at expiry T is $N\max\{K - S_T, 0\}$. Graphically, we have



Options can be used for many purposes, including

- hedging: for instance, if an investor holding XYZ share is concerned about its price drop in the short term, he/she may buy a put option to hedge the long position of the stock;
- leverage: for instance, instead of investing directly N dollars into an asset, one can choose to buy a *call* offering the same upside exposure by paying the option premium that is a fraction of N ;
- directional exposure with protection: to speculate with a bearish view, one may build a short position with a forward or by short-selling the underlier, which will incur a loss if the underlier price rises instead of falling. But if he buys a *put* instead, the upside risk will be mitigated;
- specific view: we will see later that a combination of options allows one to take advantage of any directional market view for the considered horizon.

10.1 Option General Features

10.1.1 Option Style

There are mainly two styles related to options:

European: an option that may only be exercised at the expiry of the option

American: an option that may be exercised at any time before the expiry.

10.1.2 Moneyness of an Option

Denote S_0 as the initial price and S_t as the current price of the underlier of an option. The *option* is said to be

- in-the-money (ITM): if its intrinsic value is positive; It means $S_t > K$ for a call and $S_t < K$ for a put;
- at-the-money (ATM): if its intrinsic value is zero and that S_t is at the limit of being ITM; For a call or put option, it means $K = S_t$;
- out-of-the-money (OTM): if it is neither ATM nor ITM. It means $S_t < K$ for a call and $S_t > K$ for a put.

The *strike* K of the option is said to be

- at-the-money spot (ATMS) (or simply ATM): if $K = S_0$;
- at-the-money forward (ATMF): if K is the forward price of the underlier observed at inception;

- in-the-money (ITM): if the option's intrinsic value is positive at inception; It means $K < S_0$ for a call and $K > S_0$ for a put;
- out-of-the-money (OTM): if it is neither ATM nor ITM. It means $K > S_0$ for a call and $K < S_0$ for a put.

10.1.3 Notional Amount Definition

Each option contract has a notional amount used to calculate the payoff. There are two methods to define an option notional amount:

- number of units of the underlier (e.g., number of shares): For instance, the maturity payoff of an ATMS call option is

$$N \times \max\{S_T - S_0, 0\}, \text{ where } N \text{ represents the number of shares.}$$

- monetary amount (e.g., EUR amount): it is generally used in structured products. For instance, the maturity payoff of an ATMS call is written as

$$N \times \max\left\{\frac{S_T}{S_0} - 1, 0\right\}, \text{ where } N \text{ is in EUR.}$$

We often omit the notional amount N (i.e., $N = 1$) when it is not essential for the analysis.

10.2 Option Price, Intrinsic Value and Time Value

An option's mark-to-market price, or simply *option price*, is the price at which it can be bought or sold in the market before its expiry. In particular, the option price at inception is known as the *option premium*. Option price is made up of two components (see Fig. 10.1 for an illustration with a call option.):

- *intrinsic value*: the payoff that would be received if the underlier is at its current level S_t when the option expires. It is defined to be $N\max\{S_t - K, 0\}$ for a call and $N\max\{K - S_t, 0\}$ for a put;
- *time value*: the difference between the MtM price and the intrinsic value. It will be zero at expiry.

The change of option price with respect to the change of its underlier's price is called *delta*. The change of the option price with respect to the change of time-to-maturity is known as *theta* or *decay* of the option.

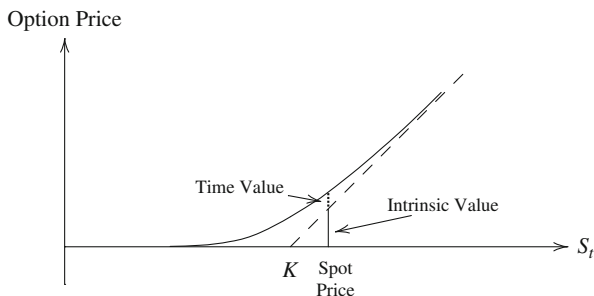


Fig. 10.1 Illustration of *Intrinsic value* and *Time value*

10.3 Vanilla Options

Vanilla options refer to the most liquid and basic options traded in the market that are call or put options. Any option that is not a vanilla option is an *exotic option*. In this section, we will introduce the vanilla options for the main tradable asset classes, namely, equities, interest rates, currencies, and commodities. Market quotations for vanilla options are based on the BSM (Black–Scholes–Merton) type of model, which will be introduced in Sect. 10.3.5.

10.3.1 Equity Options

Vanilla options linked to equities are call and put options of either European or American style on a single stock or an index. For the *payoff*, there are respectively call = $N \max\{S_T - K, 0\}$ or put = $N \max\{K - S_T, 0\}$, where S_T is the price of the underlying stock or equity index at maturity.

Equity index options are actively traded in the market. Many exchanges have listed equity index options that are traded with decent volumes. Although single stock options are listed at some exchanges (e.g., CBOE–Chicago Board Options Exchange since 1973), they are still mainly traded in the OTC market and their trading volume is far lower compared to index options. Single stock options listed at CBOE are of American style with physical settlement at expiry (i.e., with securities delivery), whereas index options apply cash settlement.

Example 10.1 The S&P500 option at CBOE is $\$100 \times \max\{\text{final index price} - \text{strike}, 0\}$. If the index value is 2100 at expiry and the strike price is 2000, the option will be exercised. The option settlement value will be $\$100 \times (2100 - 2000) = \$10,000$. $\square$

10.3.2 Foreign Exchange Options

The vanilla FX options are call/put options that give the right to buy/sell the base currency with the counter currency. Denoting X_t as the underlying exchange rate, the payoffs are call = $N \max\{0, X_T - K\}$ and put = $N \max\{0, K - X_T\}$.

One particular feature of an FX option (known as *FX Option Duality*, see Sect. 18.3) is that for a pair $ccy1/ccy2$, an N notional call option of strike K on $ccy1$ (the right to buy N $ccy1$ with $K \times N$ $ccy2$, with option premium in $ccy2$) is equivalent to a $K \times N$ notional put option of strike $1/K$ on $ccy2$ (the right to sell $K \times N$ $ccy2$ against N $ccy1$ with option premium in $ccy1$). That is why practitioners use the language like: “buying USD50m, 1-month at-the-money forward, dollar Call, yen Put”.

10.3.3 Commodity Options

Depending on the commodity, the vanilla option is linked to either spot price (e.g., gold price), or futures' price (e.g., WTI futures). For the *payoff*, there are call = $N \max\{0, S_T - K\}$ or put = $N \max\{0, K - S_T\}$, where S_T is the underlier price at maturity. At some exchanges (e.g., CME, ICE), both European and American options on futures contracts are listed.

10.3.4 Interest Rate Options*

There are two types of vanilla options on interest rates: cap/floor and swaption.

Cap/Floor

A European call option on a floating interest rate (e.g., 3M SOFR) for a considered period (T_{i-1}, T_i) is called a *caplet*, which takes the form

$$\text{payoff of an interest rate caplet} = N \times \tau_i \times \max\{R_i - K, 0\},$$

where N , τ_i , R_i represent respectively, the notional amount, the day count fraction and the reference interest rate covering (T_{i-1}, T_i) . The caplet buyer is entitled to receive a payment at the end of the period if the reference interest rate for the period exceeds the agreed strike rate K .

An *interest rate cap* is a sum of consecutive caplets in which the buyer receives payments at the end of each period where the interest rate exceeds the common strike rate for all caplets:

$$\text{payoff of an interest rate cap} = N \sum_{i=1}^m \tau_i \max\{R_i - K, 0\}.$$

Example 10.2 Someone obtained a USD1m, 10 year mortgage with monthly interest payments at 1-month SOFR +2% *p.a.*. Concerned by the risk of interest rate increase in the future, he/she decided to buy a USD1m, 10Y cap with the strike of 3%. Now, the effective interest will be limited by 5% *p.a.* because the impact from the 1-month SOFR above 3% will be paid by the short party of the cap. $\square$

Similarly, a European put option on interest rate is known as a *floorlet* when it is applied to a specific period. An *interest rate floor* is a sum of floorlets in which the buyer receives payments at the end of each period if the reference interest rate is below the strike K :

$$\text{payoff of an interest rate floor} = N \sum_{i=1}^m \tau_i \max\{K - R_i, 0\}.$$

Swaption

A *swaption* is an option that gives the option holder the right but not the obligation to enter into an IRS (interest rate swap) that starts at T and matures at T_m , at a pre-agreed swap rate K at the option expiration date T . The option's tenor is from today until the expiration date T . The swaption's final payoff at T is

$$\text{payoff} = N \times A(T, T_m) \max\{s(T, T_m) - K, 0\},$$

where

$s(T, T_m)$: market swap rate at T of the IRS;

$A(T, T_m)$: *annuity factor* for the IRS (see Sect. 4.3.2).

Example 10.3 A 1Y, 2.5% strike call *payer* swaption linked to 10Y USD swap gives its holder the right but not the obligation to enter into the swap at the expiration date in 1 year in which the option holder pays fixed rate of 2.5% to the short party. $\square$

10.3.5 Option Pricing, Hedging, and Execution*

Option Hedging/Pricing

The price of an option is the cost to replicate the payoff. In other words, option price is its hedging cost. As we will see in the modelling part of the book, the replication/hedging in *complete market*¹ is based on the sensitivity of the option price (V_t) to the change of the underlier's price (S_t) known as *delta* ($\Delta_t = \frac{dV_t}{dS_t}$). *Delta hedging* refers to the process of dynamically buying or selling underlier units according to Δ_t .

The *Black–Scholes–Merton* (BSM) model² is the first and the most popular model for option hedging and pricing. Under this model, the asset's price follows the stochastic differential equation: $dS_t/S_t = \mu dt + \sigma dW_t$, where μ is the drift, σ (known as *volatility*) represents the annualized standard deviation of the log return of the underlier price, and W_t is a *Brownian motion* that depicts the randomness in underlier prices.

Option Order Execution

The execution of an OTC option offered by an option provider (hedging party) to a client is directly based on the hedging of the option. The trade order is said to be “done” (or “filled”) when the initial delta is executed by the hedging party in the market. The execution of the delta means buying or selling of the underliers according to the quantity indicated by the *delta*. The price at which the delta is executed serves as the *initial price reference* or *initial spot reference* in the option contract. For placing a *limit order* for a trade with multiple underliers, the limit will be set for one underlier and the initial delta for other underliers will be executed *at market*. For an FX vanilla option, the order is usually given with fixed initial spot reference (instead of a limit order, etc.) and the execution is usually done live between the parties in order to “catch” the market spot level.

Vanilla Option Pricing

For vanilla options of European style, market participants mainly use the so-called B-S formula (Black&Scholes formula)³ derived under the BSM model:

$$c = S_0 N(d_1) - e^{-rT} K N(d_2), \quad p = e^{-rT} K N(-d_2) - S_0 N(-d_1) \quad (10.1)$$

¹ Roughly speaking, in a complete market, all contingent claims can be replicated with (dynamic) positions of underliers.

² In fact, the BSM model has been adapted to different asset classes with different names: the Garman–Kohlhagen model for currencies and the Black model for interest rates.

³ The B-S formula presented here is for non-dividend-paying stocks with zero repo rate. Similar formulas can be derived for the general case and for other asset classes.

where c and p are the call and put option premiums with maturity T and strike K , r is the continuous interest rate, S_0 is the spot price of the underlier with volatility σ ,

$$d_1 = d_2 + \sigma\sqrt{T}, \quad d_2 = \frac{\ln \frac{S_0}{K} + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad \text{and } N(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{x^2}{2}} dx.$$

Apart from option features such as tenor and strike price, other factors impacting option price in the BSM model are the volatility σ of the underlier and the interest rate r . σ is the most important parameter for an option trader or risk manager as the impact from r is relatively small. By the B-S formula, there is a one-to-one relationship between the option price and σ with the other parameters fixed.

Option Quotation

As a market practice, equity options are quoted by strike and tenor, with strike expressed in percentage of *initial price reference*.

In FX option market, people prefer to quote a vanilla FX option by its *delta* (calculated with the B-S formula) instead of its *strike*.⁴ The benchmark market quotes are ATMF, 25% Δ , and 10% Δ . Bisesti et al. [17] provides an overview of market practice for FX options.

For cap/floor, market quotes are available by strike and tenor. Swaptions are quoted in the market by strike, option tenor and the underlying swap tenor.

10.3.6 Implied Volatility*

An *implied volatility* is the volatility value derived (with other parameters fixed) from an option price quoted in the market. For a given underlier, although its volatility has a unique value in the BSM model, its implied volatility may be different for different tenors and option strikes. For a given tenor, the implied volatilities from the options with different strike prices show a pattern known as *volatility smile*.

For equity index options, the smile normally takes the shape shown in Fig. 10.2. There are various explanations on the volatility smile for equity underliers, including

- model imperfection: the BSM model implies a log normal distribution for the return of underliers. However, historical data confirm that the “real” distribution of an equity underlier shows fatter tails (kurtosis) and steeper asymmetry (skewness). Practitioners use adjusted volatility to produce the “correct” option price. As the adjusted volatility “corrects” rather the final distribution than the

⁴ Given the price of an option, there is a one-to-one relationship between *delta* and *strike* using the B-S formula.

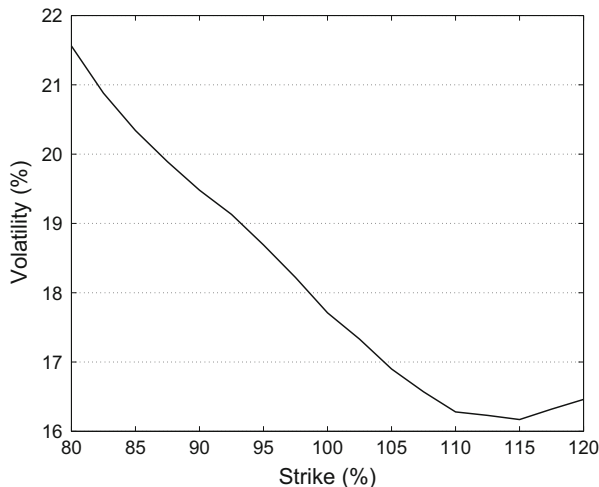


Fig. 10.2 Example of volatility smile for equity options

dynamic behaviour of the underlier, the BSM model with adjusted volatility is not suitable for pricing path dependent options;

- demand/offer: the demand/offer for options with different strike prices influences the option prices. For instance, institutional investors such as insurance companies and fund managers tend to buy OTM equity index put options to hedge their long exposure to equities. This leads to higher prices for OTM put options, i.e., higher implied volatility for lower strike options. While, for single stock options, especially warrants, the volatility smile may be less steep and sometimes inverted (i.e., implied volatility is higher on higher strike options).

The above analysis is also applicable to other underlier types.

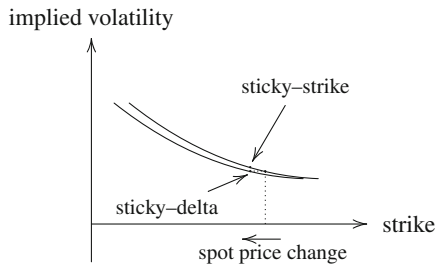
The implied volatilities for FX options are derived by the following market quotations for a given tenor:

- at-the-money-forward volatility (σ_{ATMF}): the volatility for the strike equal to the at-the-money forward;
- 25%-delta risk-reversal (25 Δ RR): $\sigma_{25\Delta call} - \sigma_{25\Delta put}$;
- 25%-delta strangle (also known as 25%-delta butterfly) 25 Δ BF: $\frac{1}{2}(\sigma_{25\Delta call} + \sigma_{25\Delta put}) - \sigma_{ATMF}$.

By σ_{ATMF} , 25 Δ RR and 25 Δ BF, the values of $\sigma_{25\Delta call}$ and $\sigma_{25\Delta put}$ can be solved using the above definitions. The implied volatilities for other delta points can be interpolated.⁵ Using the one-to-one correspondence between *strike* and *delta* given the option price, one can deduce the implied volatility by strike.

⁵ For strike levels beyond the range, people normally do not apply the simple extrapolation for the implied volatility. Instead, the far OTM and ITM options are quoted with a minimum time value.

Fig. 10.3 Sticky-strike and Sticky-delta



Regarding the dynamics of implied volatility smile, it could be a mixture of two types of behaviours when the underlier price moves over time while the market does not expect significant change in realized volatility (see Fig. 10.3):

- *sticky-strike*: the smile remains unchanged with strike prices. It is more applicable when the market is expected to be range-bound;
- *sticky-delta*: the smile remains unchanged with option's moneyness. it is more applicable when the markets are trending.

The volatility for a future period $[t_1, t_2]$, $0 = t_0 < t_1 < t_2$, can be derived with the additivity of variance: $\sigma(t_1, t_2)^2(t_2 - t_1) = \sigma_1^2 t_2 - \sigma_1^2 t_1$. It may be used for pricing forward-starting options. The volatility for the period $[t_0, t]$, $t_1 < t < t_2$ can be interpolated. One simple method consists of using $\sigma(t_1, t_2)$ for the period $[t_1, t]$, which leads to $\sigma(t)^2 t = \sigma_1^2 t_1 + \sigma(t_1, t_2)^2(t - t_1) = [1 - w(t)]\sigma_1^2 t_1 + w(t)\sigma_2^2 t_2$ where $w(t) = \frac{t-t_1}{t_2-t_1}$.

The sensitivity (the first-order mathematical derivative) of an option's price w.r.t. its underlier's volatility is called *vega* by financial market practitioners.

10.3.7 Put-Call Parity

Proposition 10.1 (Put-Call Parity) Let c_t , p_t and F_t be respectively the time- t prices for the European call, European put and forward contract with the same maturity T and strike K . Then, we have

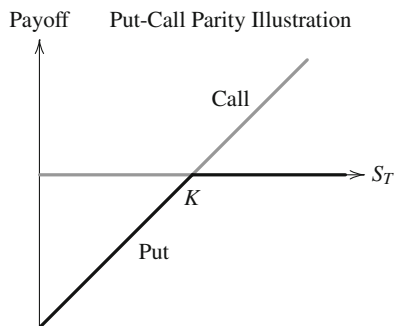
$$c_t(T, K) - p_t(T, K) = F_t(T, K), \quad \forall t \leq T.$$

In particular, we have (cap - floor = swap) for interest rate options.

Proof It is obvious that $\max\{S_T - K, 0\} - \max\{K - S_T, 0\} = S_T - K, \forall T > 0$. If *Put-Call Parity* is not verified, we then have an arbitrage opportunity by selling the overvalued leg and hedging with the other two legs with a lower price. $\square$

For a non-dividend-paying stock with zero repo rate, we have $F_t = S_t - e^{-r(T-t)}K$ leading to

$$c_t(T, K) - p_t(T, K) = S_t - e^{-r(T-t)}K.$$



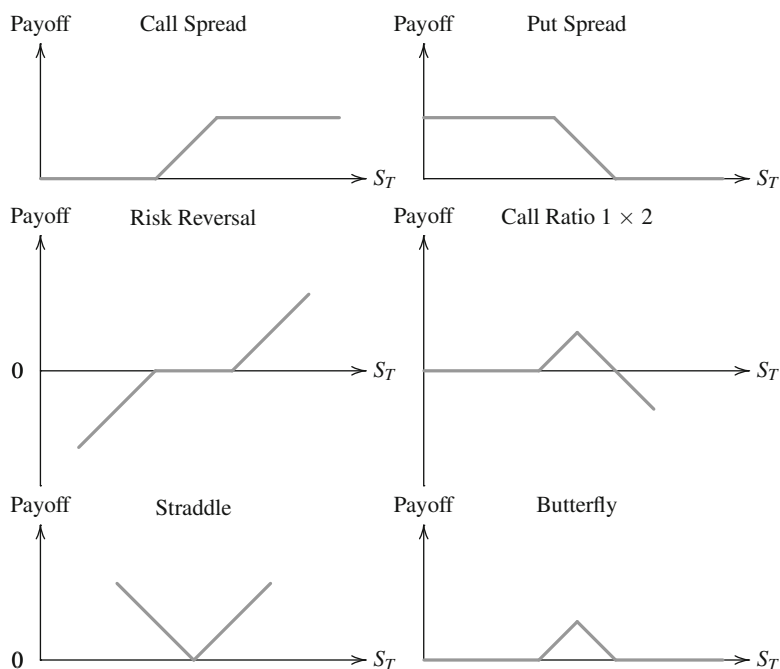
Remark 10.1

- Put-Call Parity, by arbitrage, forces the vanilla call and put options with the same strike level to have the same implied volatility value;
- The vega of a call option should be equal to the vega of the put option with the same strike. This is due to Put-Call Parity and the fact that the vega of a forward contract is zero;
- Put-Call Parity illustrates an interesting case where the combination of two options is no more an option. $\square$

10.3.8 Popular Strategies with European Options

European options can be used to take advantage of specific views on future prices of the underlier. It can be proven that any option payoff of European style can be represented as a combination of European call and put options (see Sect. 24) of the same maturity.⁶ Below is a list of well-known option strategies:

⁶ Call and put payoffs are half-lines with slope ± 1 (related to 45° angle). For the graphical representation based on a fixed notional, the slope will change if we change the exposure (effective notional) to the option. It is not difficult to imagine that any payoff at a given maturity date can be presented by a combination of call and put options with different levels of exposure.



Call spread (also called vertical bull spread): It is a position of long 1 call at K_1 , short 1 call at K_2 ($K_1 < K_2$). It is a cheaper way than a call to take advantage of a bullish view. The price of the whole package is sensitive to skew instead of volatility if K_1 and K_2 are close.

Put spread: It is a position of long 1 put at K_1 , short 1 put at K_2 ($K_1 > K_2$). It is a cheap way to take advantage of a bearish view.

Risk reversal: long 1 OTM call at K_{call} and short 1 OTM put at K_{put} ($K_{put} < K_{call}$) such that the option premium at inception is close to zero.

Butterfly: long call at strike K_1 , short 2 calls at strike K_2 and long call at strike K_3 where $K_1 < K_3$ and $K_2 = (K_1 + K_3)/2$. A butterfly spread leads to a profit if the stock price stays close to K_2 .

Collar: long share, long OTM put option and short OTM call option

Straddle: long call and long put with same strike (usually ATM) and expiry

Strangle: long OTM call and long OTM put with the same expiry

Call ratio: a popular strategy used by hedge funds is one-by-two (1×2) i.e., long one call at strike K_1 and short two calls at strike K_2 where $K_1 < K_2$

Covered call (also called *Buy/Write*): long position of the underlier and a short position of an OTM call on the underlier.⁷ It is a popular strategy to monetize volatility when the short-term view on the underlier is not strongly bullish.

⁷ In a covered call, the long underlier position may serve as collateral for the short call position to mitigate the counterparty credit risk for the long party (usually an investment bank).

Calendar spread: it is formed by options with different expiries. Since we usually use x -axis to denote time, it is also called *horizontal spread*. It can be used for forward volatilities arbitrage.

Hedge funds often trade slightly OTM call/put options to get directional exposure by paying less time value. Very often, the options are unwound or offset before expiry. That is why hedge funds tend to trade more liquid options instead of customized exotic options that are more difficult or costly to unwind.

10.3.9 American Options

An *American option* is one that may be exercised at *any time* prior to its expiration date by its holder. If the option can be exercised only at discrete time points (e.g., every month), it is called a *Bermudan option*. However, Bermudan options are sometimes called American options as well.

An American option holder must assess when it is the best time to exercise the option. As we will see in the modelling part, the optimal decision is when the exercise payoff is higher than the value of the remaining American option if the holder does not exercise.

Let c_t and p_t denote the time- t price of the European call option and put option with strike K and expiry T , while C_t and P_t denote that of the American call option and put option, respectively. Due to its exercise right at the discretion of option holder, an American option's price should be at least the price of its counterpart European option, i.e.,

Proposition 10.2 $C_t \geq c_t$ and $P_t \geq p_t$.

However, there is a particular case where the American call is worth the same as the European call.

Proposition 10.3 *The price of an American call is equal to the price of the corresponding European call under the following conditions:*

- *dividend and repo rate are all zero for the underlier during the tenor of the option, and*
- *the interest rate r is non-negative.*

Proof We first show that

$$c_t \geq \max\{0, S_t - e^{-r(T-t)} K\}, \quad \forall t < T.$$

Define two portfolios at time- t :

portfolio A: Long c_t + a bond with the value of $e^{-r(T-t)} K$

portfolio B: Long one share S_t

At maturity T , we have

portfolio A: $\max\{0, S_T - K\} + K = \max\{S_T, K\}$

portfolio B: S_T (no dividend payment)

Hence, we should have $A \geq B$, for any time until maturity, i.e., $c_t + e^{-r(T-t)}K \geq S_t$, or

$$c_t \geq \max\{0, S_t - e^{-r(T-t)}K\} \quad (\text{because } c_t \geq 0).$$

Consequently, we have the following inequalities

$$C_t \geq c_t \geq \max\{0, S_t - e^{-r(T-t)}K\} \geq \max\{0, S_t - K\} \quad (\text{as } r \geq 0).$$

Because the early exercise price is $\max\{0, S_t - K\}$, it is not optimal for the option holder to exercise the option before expiry (the option holder should sell the option instead of exercising it). Therefore, $\forall t < T$, we have

$$C_t = c_t.$$

□

Proposition 10.3 implies that one should not exercise an American call option on a non-dividend paying stock with zero repo rate before its expiry if interest rate is non-negative. If an option holder thinks that the stock is currently overpriced and wants to lock in the gain, it is better to sell the option rather than exercise it.⁸

For investors, with the same option premium, it is always more advantageous to hold the American option instead of the European one because, if one ever wants to terminate the option early, the American option has the certainty of receiving at least its intrinsic value. However, with the European option, the option provider usually charges an unwinding fee, which may lead to an unwind value lower than the intrinsic value.

10.4 Exotic Options

Any option that is not a vanilla option is an *exotic option*. Each exotic option offers specific exposure to the performance of the underlier. If a payoff is determined by more than one price of a single underlier over time, the option is called *path-dependent*. Actually, the proliferation of structured products and the advances in pricing techniques made it unnecessary to name each new product with an exotic and fancy name. However, well-studied exotic options named in the old days are

⁸ Nevertheless, it may be optimal to exercise an American put option on a non-dividend-paying stock before its maturity.

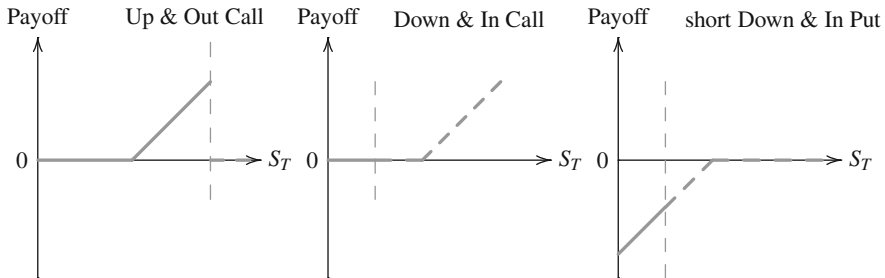
often used for comparing/understanding a financial product's features, risks, and pricing/hedging issues.

Let us introduce a few notations that will be used in the following context:

- 1_A is the *indicator function* that returns 1 if event A is true, otherwise 0;
- $\min_{t \in [t_1, t_2]} S_t$ and $\max_{t \in [t_1, t_2]} S_t$ denote respectively the minimum value and maximum value of S_t in the period $[t_1, t_2]$.

10.4.1 Barrier Option

A barrier option is a type of call or put option that either comes into existence (“knock-in”) or becomes worthless (“knock-out”) based on a certain condition linked to the performance of the underlier. Terms such as *down-and-in*, *down-and-out*, *up-and-in*, *up-and-out* are used to indicate the condition. For instance, the payoff of *down-and-in* put is $\max\{K - S_T, 0\} 1_{\min_{t \in [0, T]} S_t \leq L}$ and the payoff of *up-and-out* call is $\max\{S_T - K, 0\} 1_{\max_{t \in [0, T]} S_t < H}$.

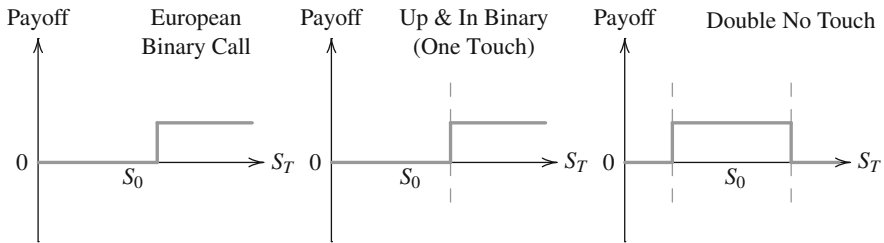


Due to the extra condition, either knock-in or knock-out, barrier options are cheaper than the underlying vanilla options, which explains their popularity in practice. Theoretical pricing formulas can be found in Appendix B. There are many applications of barrier options in financial products (e.g., Shark-Fin, CBBC and Twin-Win presented in Sect. 31.1.7, 31.1.16, and 31.1.11).

10.4.2 Binary (or Digital) Option

A binary option pays off a fixed amount (either cash or asset) if a certain condition (A) is met, otherwise 0. Mathematically, it takes the form of 1_A (*cash-or-nothing*) or $S_T 1_A$ (*asset-or-nothing*). There are two main types of binary options:

- European binary option: the condition depends on the final price of the underlier. For instance, the payoff of a binary call option is $N \mathbf{1}_{S_T \geq H}$. The payoff of a binary put is $N \mathbf{1}_{S_T \leq L}$.
- American binary option: the condition depends on the maximum or minimum price of the underlier observed over a period. It is also called a *touch* option. For instance, the payoff of a *one-touch* option is $\mathbf{1}_{\max_{t \in [0, T]} S_t \geq H}$, where $S_0 < H$. The payoff of a *double-no-touch* is $\mathbf{1}_{\min_{t \in [0, T]} S_t \geq L \cap \max_{t \in [0, T]} S_t \leq H}$, where $L < S_0 < H$. In the FX option market, touch options are actively traded and considered as vanilla options for major currency pairs.



An *up-and-out* call option with rebate c (paying out c when the call option is knocked out) can be seen as an *up-and-out* call (without rebate) plus an *up-and-in* binary option paying c (i.e., $\text{call} \cdot \mathbf{1}_{\max_{t \in [0, T]} S_t < H} + c \cdot \mathbf{1}_{\max_{t \in [0, T]} S_t \geq H}$).

10.4.3 In-Out Parity*

Similar to put-call parity for European options, there is an *in-out parity* for barrier options, which states that a vanilla option is equivalent to the sum of a *knock-in* option and a *knock-out* option. Actually, from $1 = \mathbf{1}_{\min_{t \in [0, T]} S_t \geq L} + \mathbf{1}_{\min_{t \in [0, T]} S_t < L}$, we obtain

$$\text{payoff}(T) = \text{payoff}(T) \cdot \mathbf{1}_{\min_{t \in [0, T]} S_t \geq L} + \text{payoff}(T) \cdot \mathbf{1}_{\min_{t \in [0, T]} S_t < L}.$$

For instance, a *down-and-in* put option is equivalent to a *vanilla* put option minus a *down-and-out* put option.

10.4.4 Asian Option or Average Option

An Asian option is similar to a vanilla option, except that the final underlier price and/or the strike price are replaced by the averaged value(s) of the underlier over a predefined period. For instance, the payoff of an Asian call option with averaging at

the end (we say *Asian tail* or *Asian out*) is $\max\{\frac{1}{m} \sum_{i=n-m+1}^n S_{t_i} - K, 0\}$. When the averaged price of the underlier is used as the strike, the option is called an *Asian strike* option, e.g., $\max\{S_{t_n} - \frac{1}{m} \sum_{i=0}^{m-1} S_{t_i}, 0\}$.

An Asian option offers a smoother payoff compared to a European vanilla option that depends entirely on the single price of the underlier at maturity. Another application of Asian option is for managing the risk of “striking” and “expiry” that will be explained further in Chap. 20.

Example 10.4 Warrants listed at HK Stock Exchange use the arithmetic average of the closing prices in the final 5 trading days as the final price reference for the payoff. $\square$

10.4.5 Lookback Option

In this type of option, the maximum or minimum asset price sampled during a certain period⁹ is used as either the performance (*lookback spot*) or as the initial reference (*lookback strike*) in the calculation of the option payoff. For instance, a call option paying off

$$\max\left\{\frac{S_T}{S_0^*} - 1, 0\right\}, \quad S_0^* = \min\{S_{t_1}, S_{t_2}, \dots, S_{t_m}\}, \quad t_m \leq T$$

is called a *lookback strike* call. A call option paying off

$$\max\left\{\frac{S_T^*}{S_0} - 1, 0\right\}, \quad S_T^* = \max\{S_{t_{n-m+1}}, \dots, S_{t_{n-1}}, S_{t_n}\}, \quad t_n = T$$

is called a *lookback spot* call.

10.4.6 Quanto Option and Composite Option

When the payoff currency is different from the underlier’s currency, there exist two well-known options called Quanto option and Composite option.

The term *quanto* (quantity-adjusted option) is used to indicate that the performance of the underlier is only measured in its domestic currency without taking into account any impact from the currency exchange rate. For instance, the payoff

⁹ The full period lookback is rarely used in practice.

for a USD5m, USD quanto call option linked to BNP Paribas stock price S_t^ϵ quoted in EUR is simply

$$\text{USD5m} \times \max \left\{ \frac{S_T^\epsilon}{S_0^\epsilon} - K, 0 \right\}, \quad \text{to be settled in USD.}$$

Quanto options are natural products for investors and widely used. They do not require investors to have a view on the FX rate changes.

A *composite option* (aka compo) is another type of option for the situation where the payoff currency is different from the underlier's currency. In this option, the performance of the underlier is derived from the underlier price converted into the payoff currency with the prevailing FX rate. For instance, the payoff for a USD5m, USD compo call option linked to BNP Paribas stock is

$$\text{USD5m} \times \max \left\{ \frac{S_T^\$}{S_0^\$} - K, 0 \right\}, \quad \text{to be settled in USD}$$

where $S_t^\$ = S_t^\epsilon X_t$ with S_t^ϵ being the price of BNP Paribas stock in EUR and X_t being the EUR/USD rate at time t . As only $S_t^\epsilon X_t$ is used in the payoff, it can be considered as the price of a “composite stock” denominated in the payoff currency. Hence, the option price is not sensitive to future changes in the interest rate of the stock's domestic currency. Composite options are not often used as it is harder for investors to have a view on $S_t^\epsilon X_t$ instead of S_t^ϵ alone. A composite option may be considered when the interest rate risk hedging for the underlier's currency is difficult or costly, which happens for some emerging markets.

10.4.7 Basket Option, Worst-of, and Rainbow Options

Denote S_t^i as the price of stock i at time t . A *basket call option* takes the form

$$\text{payoff} = \max \left\{ \frac{1}{n} \sum_{i=1}^n \frac{S_T^i}{S_0^i} - K, 0 \right\}.$$

Due to averaging, a basket option is generally smoother than a single underlier option. This option is (i.e., to the long party) long forward (i.e., the option's price will increase if the forward price¹⁰ of the underliers increases), long volatility, and long correlations between the underliers.

¹⁰ We recall that the forward price for a stock increases with interest rate but decreases with repo rate and dividend yield.

A *worst-of* option takes the minimum performance of a basket of underliers. The payoff of a worst-of call option is

$$\text{payoff} = \max \left\{ \min_i \left\{ \frac{S_T^i}{S_0^i} \right\} - K, 0 \right\}.$$

Due to their low prices, worst-of options are widely used in structured solutions that often have tight constraints on cost. A worst-of option is long forward and short correlation. So, the underlying stocks with low interest rate, high dividend/repo rate and low correlations tend to show a lower price. We note that the correlation of FX-stock as well as the FX volatility may impact the “forward” of foreign currency denominated stocks, which will be analyzed in the chapter on FX modelling.

A *best-of* option takes the maximum performance of the basket of the underliers. A best-of call option has the payoff like

$$\text{payoff} = \max \left\{ \max_i \left\{ \frac{S_T^i}{S_0^i} \right\} - K, 0 \right\}.$$

This option is long forward and long correlation. Due to their high premium, best-of options are not really used in practice.

A *rainbow basket* takes its performance as the weighted average of individual performances of underliers according to their performance ranking, e.g., 70% of the worst performance, 30% of the second worst and 0% for the rest of the underliers. Worst-of's and best-of's are special cases of *rainbow* options.

10.5 Derivatives Modelling Framework*

10.5.1 Purpose of Derivatives Modelling

In the normal course of business, a derivatives provider replicates the P/L of the derivatives based on pricing models. The replication (known as hedging) is a dynamic process over the entire life of the trade. For example, the seller of a call option linked to a stock will hold a dynamic quantity of stocks to produce the P/L of the option due to the changes in the stock price. The quantity of stocks to be held at any time is determined by a pricing model. We list the main purposes of models:

- pricing for new transactions (primary market), especially for the OTC derivatives that do not have liquid market quotations;
- pricing for size increase or unwinding of an existing trade (secondary market);
- mark-to-market valuation;
- determination of hedging-related quantities that are, as we will see, sensitivities (also called Greeks) of derivative products;
- risk reporting;
- regulatory capital calculation for financial institutions.

10.5.2 Model Input Parameters and Calibration*

Models require input parameters. General financial data are directly observable from market quotations of financial instruments. However, model-specific parameters are usually determined by a process known as *calibration* or *fitting*. Calibration targets to minimize the difference between the prices calculated by a model and the prices observed in the market for liquid derivatives linked to the considered underliers. Thus, calibration is an optimization problem if there is no direct calculation of model parameters from market-quoted instruments.¹¹ If its calculation is fast, calibration may be done “on-the-fly”, meaning that the model is re-calibrated just before each pricing.

Generally, liquid market-quoted derivatives are vanilla options of each asset class and other basic instruments such as forward/futures, swaps, and CDS. Vanilla options are normally priced with closed-form solutions in the market. For a typical financial institution offering and hedging derivatives, vanilla products traders are responsible for updating the market prices of the instruments used to calibrate the models. Exotic option traders, structurers, sales and risk controllers¹² are the main users of the calibrated models for pricing and risk management.

10.5.3 Historical Volatility and Correlation*

If the price process of an asset follows, approximately, a lognormal distribution, the annualized historical volatility of the asset price over a period with n daily observations, σ , can be determined as follows:

$$\sigma^2 = \frac{N}{n-1} \sum_{k=1}^n (R_k - \hat{R})^2,$$

where

- S_k : the price of the asset on day k ;
- $N = 252$: the number of trading days in a year;
- $R_k = \ln \frac{S_k}{S_{k-1}}$ or $\frac{S_k - S_{k-1}}{S_{k-1}}$: daily total return (see Sect. 5.2.6 for price adjustment);
- $\hat{R} = \frac{1}{n} \sum_{k=1}^n R_k$: average daily return. It is often assumed to be zero (e.g., in the historical variance formula in variance swap term sheets).

Historical volatilities can serve as a reference when implied volatilities are not easily available in the market.

¹¹ Levenberg–Marquardt algorithm is one of the most popular optimization method used in practice.

¹² Risk controllers are also responsible for supervising the correctness of the inputs from traders.

Similarly, for two asset price time series, S^i and S^j , the historical correlation ρ^{ij} over a period with n trading days can be calculated as¹³

$$\rho^{ij} = \frac{\sum_{k=1}^n (R_k^i - \hat{R}^i)(R_k^j - \hat{R}^j)}{\sqrt{\sum_{k=1}^n (R_k^i - \hat{R}^i)^2} \sqrt{\sum_{k=1}^n (R_k^j - \hat{R}^j)^2}}.$$

If we assume $\hat{R}^i = 0$ and $\hat{R}^j = 0$, we have

$$\rho^{ij} = \frac{\sum_{k=1}^n R_k^i R_k^j}{\sqrt{\sum_{k=1}^n (R_k^i)^2} \sqrt{\sum_{k=1}^n (R_k^j)^2}}.$$

In the above correlation calculation, weekly returns are often used for the assets that are listed in different time zones. Readers may find further details on historical correlation calculation and risk management in Sect. 20.2.2

Exercises

1. The implied volatilities for 30 days and 60 days (assuming same strike) are 15% and 12% respectively. Calculate the *forward volatility* between days 31 and 60.
2. An investor is long a call option on a stock that will expire in a month. The strike is \$100, the current stock price is \$110.
 - (1) what is the intrinsic value of the call?
 - (2) what is the term given to the difference between the option price and intrinsic value?
 - (3) Is the option premium the expected value of the payoff?
3. An investor has bought a 3-month ATM European call option for 10,000 stocks. The underlying stock does not pay any dividend and the stock lending rate is zero. Its current stock price is \$100. The interest rate in the market is 2% p.a. (continuous rate, ACT/365). The implied BSM volatility of the stock is 25%.
 - (1) What is the strike price of the option?
 - (2) Calculate the payout for the investor for the stock price at the expiration date at \$120 and \$70.
 - (3) Use the Black–Scholes formula for call options to calculate the initial call option price with stock price of \$100 and \$101. Deduce the approximate value of the initial delta.

¹³ For asset price processes driven by Brownian motions, pricing models require the correlation between the Brownian motions. The formula presented in this section calculates such correlation for log-normal distributed asset prices.

- (4) Based on the above result, how many shares (i.e., delta position) should the option seller hold for initial delta-hedging?
 - (5) If the share price moves up by 1%, what is the gain from the delta position? What is the gain if the option seller holds 10,000 stocks?
 - (6) Someone thinks that it is better for the option seller to hold 10,000 stocks instead of holding the delta amount for hedging as the gain will be higher if the stock price goes up. Do you agree with this hedging strategy? Why?
 - (7) Use Put-Call parity to deduce the European put option price with the same strike and tenor from the above result.
4. Describe the relationship between the price of a worst-of call option and volatilities of the underliers.

Chapter 11

Introduction to Structured Products



In this chapter, we provide an introduction to structured products. Detailed presentation on structured products and solutions, as well as their design techniques, are presented in Part IV. We note that most of the structured products involve option derivatives requiring advanced modelling for pricing and hedging, which will be covered by Part II and Part III.

11.1 Background and Purpose of Structured Products

A direct investment in an asset offers linear and symmetric exposure to the performance of the asset. However, an investor's view on the market may be more complex than a directional perspective, e.g., performance cycles, relative performance between assets, view on volatilities and correlations, etc. Risk appetite and tolerance are also different according to investors' profiles.

Structured products are pre-packaged investment strategies to achieve specified risk-return objectives linked to the performance of the *underliers* that are investable assets, instruments or financial references (e.g., equity index, reference rate). They are used as an alternative to direct investments for getting exposure to specific market views and risk profiles under the constraints of financial budget and regulations. Derivatives hedging (or replication) technology benchmarked by the Black–Scholes–Merton model has made it possible for banks to offer various structured products without warehousing significant market risk. Structured products have experienced a fantastic development since the late 1990s, when French banks pioneered the innovative equity-linked products in particular. Today, the major end investors of structured products are retail, high-net-worth individuals (HNWIs) and non-bank financial institutions (e.g., insurance or pension funds).

The investment targets of structured products may be classified as follows:

- *Participation*: exposure to the performance of the underlier, implemented with tracking/market access¹ structures;
- *Income*: enhanced yield with fixed, variable coupons or capped performance of the underlier(s);
- *Protection*: prudent investment with zero or limited potential loss;
- *Leverage*: leveraged exposure to the performance of the underlier.

A structured product is composed of three building blocks:

- *Wrapper or Instrument*: the legal form/carrier for the product. The most common wrappers include certificate/note, structured deposit and total return swap;
- *Underlier*: the specified investable asset, instrument, financial index or reference, which constitutes the performance engine;
- *Payoff*: the return/risk of the product based on the performance of the underlier.

The return or payoff of a structured product is linked to the performance of the underliers pre-defined in the contract. Price returns (see Sect. 1.2) measured from the product inception are used in the payoff calculation in general. Structured products offer a wide variety of payoffs to meet different purposes. Categorization of structured products can be made in different ways. One categorization criterion is by the level of principal protection.

11.2 Principal Protected Product Example

Principal protection is an important feature for risk averse investors. It may be achieved by two types of techniques:

- Option based structures, which is also known as OBPI (Option Based Portfolio Insurance). Most of the structured products belong to this category;
- Dynamic asset allocation based structures, i.e., CPPI (Constant Proportion Portfolio Insurance), that dynamically adjust the investment exposure based on gap risk (see Chap. 32).

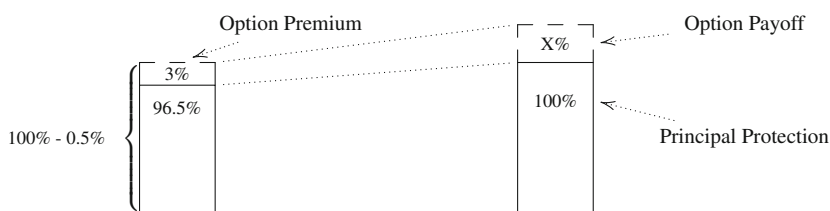
The simplest option-based principal protected structure consists of spending the available premium (which is part of the interest generated by the principal of the product) to buy an option on the underlier. Let's take the example of a 1-year, 100% principal protected note issued at face value N providing the maturity payoff

$$N + PR \times N \times \max \left\{ 0, \frac{S_T}{S_0} - 1 \right\} \quad (11.1)$$

¹ Some markets have cross-border investment restrictions. Offshore qualified foreign institutions (QFI) approved by the onshore regulator can issue *P-notes* (participatory notes), TRS, etc., which provide the performance of onshore securities to offshore investors.

The product may be structured (hedged) by the issuer as follows:

- invest into a 1-year zero coupon bond with redemption value N to provide the principal at maturity. Assume that the bond price is $(1 - 3.5\%)N$;
- if the total cost for note issuance and distribution is $0.5\%N$, the available premium to spend will be $(3.5 - 0.5\%)N = 3\%N$;
- invest the available premium of $3\%N$ to buy a call option on the underlier, say an equity index. If the option price is 8.1% of the notional, we can apply a *participation rate* (PR, also known as *gearing*) such that the option investment will be 3% : $PR = 37\% (= 3\%/8.1\%)$.



11.3 Non-principal Protected Product Example

In a non-principal protected product, the investor takes some risks linked to the underlier for a higher investment return. The most common risk-taking by bullish investors consists of selling a put option (implicitly) on the underlier. The premium of the put option is used to enhance the return of the structured product. Other types of risk taking include range view, bearish view, combination of views on different underliers, view on yield curve shape, etc. which will be presented in Part IV. The maximum loss for investors depends on the structure: e.g., 100% of invested amount for a note/certificate/long option, limited loss for a short put option but unlimited loss for a short call option or an FX swap, etc.

Equity Linked Note (ELN) is one of the most popular wealth management products, especially for private investors. It offers yield enhancement by taking downside risk on equities. An ELN is often a short-term product (typically, 1–6 months), issued at a discount to its face value. At maturity,

- if the underlying stock price has not dropped beyond a certain level (called strike, which is set at a level lower than its initial price at inception), the investor will receive 100% of the face value (i.e., denomination) of the note;
- Otherwise, the investor will receive a number of stocks at the strike price (physical settlement). Cash settlement of equivalent value can be applied alternatively.

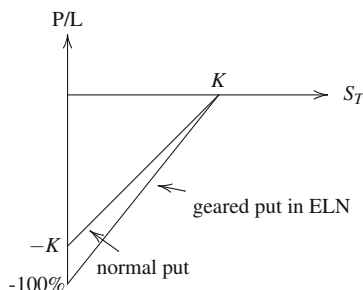
ELN is also known as Reverse Convertible Note as the conversion into stocks (physical settlement) occurs when the stock price has fallen, which is the opposite of a convertible bond. It is suitable for a stable or bullish view on the underlying stock for the tenor of the trade. The investor enjoys an attractive return if the stock does

Table 11.1 Illustration example of ELN (equity linked note)

Issuer	XYZ bank
Tenor	1 Month
Notional amount (N)	HKD 1m
Issue price	N (100%–Discount)
Discount	0.85%
Underlying stock	[•]
Initial stock price (S_0)	HKD 102.10
Strike (K)	93%
Maturity payoff	If final stock price (S_T) $> K \times S_0$, Investor will receive N ; Otherwise, Investor will receive $N/(K \times S_0)$ stocks

not underperform significantly. In case the investor will be delivered with stocks, the purchase price will be the strike price, which is lower than the initial stock price, implying a better return compared to direct investment for the same period at inception time. On the risk side, the investor will be obliged to buy the stocks at the strike price even though the stock price dropped below this level. See Table 11.1 for illustration.

In this product, the investor is selling a *geared* put (i.e., the effective notional amount for the put is not $100\% \times N$). This can be understood from the graphic comparing a normal put (with maximum loss of $N \times K$, where $K \leq 100\%$) and a geared put (with maximum loss of $N \times 100\%$). The put is leveraged by a factor of $1/K$.



Mathematically, we can show that the final redemption value is equivalent to²

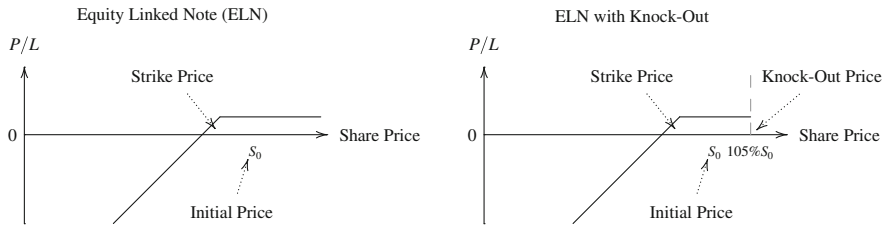
$$N - N \times \frac{1}{K} \max \left\{ 0, K - \frac{S_T}{S_0} \right\} \quad (11.2)$$

² Denoting $A = \{ \frac{S_T}{S_0} \geq K \}$, $A^c = \{ \frac{S_T}{S_0} < K \}$, and 1_A for the function that is equal to 1 if A occurred or 0 otherwise. Notice that $1_A + 1_{A^c} = 1$. Final redemption value is $N 1_A + S_T \frac{N}{K S_0} 1_{A^c} = N 1_A + N \left(1 + \frac{1}{K} \frac{S_T}{S_0} - 1 \right) 1_{A^c} = N \left[1 - \frac{1}{K} \left(K - \frac{S_T}{S_0} \right) 1_{A^c} \right] = N \left(1 - \frac{1}{K} \max \left\{ 0, K - \frac{S_T}{S_0} \right\} \right)$.

The term $\frac{1}{K}$ in front of the put option in the last line is known as *gearing*.

For ELNs as well as most of the bullish equity structures, the investor is short volatility and long forward (which implies long interest rate, short dividend and repo rate). So, the higher the volatility and the lower the forward, the lower the issue price of the product.

A simple variation of ELN, called KO ELN, consists of introducing an up-and-out barrier (e.g., 105% S_0) that is typically observed on a daily basis. Upon knock-out event, the structure will be early terminated at par (i.e., 100% of N).



11.4 Quanto Feature in Structured Products

Despite its historical labelling as an exotic type of option, quanto structure is a natural and common feature in structured solutions. In the principal protected product example, if the currency of the underlier is not the currency of the note, the embedded call option is then a quanto call option. Products with physical settlement involves foreign exchange for calculating the number of underlier units to be delivered. Let's take the example of an ELN, where the currency for the note is f and the currency for the stock is d . Denote the foreign exchange rate at time T by X_T (the number of currency d for 1 unit of currency f). The final payoff description becomes

Maturity payoff	If final stock price (S_T^d) > $K S_0^d$, Investor will receive N^f ; Otherwise, Investor will receive n stocks where $n = (N^f X_T)/(K S_0^d)$
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In the case of physical delivery, the value of the stocks received by the investor will be $n S_T^d$ in currency d , which is equivalent to $n S_T^d / X_T = N^f \left[1 - \frac{1}{K} \left(K - \frac{S_T^d}{S_0^d} \right) \right]$ in currency f . Combining the two cases of maturity payoff, we can conclude that the implied put option in the ELN is actually a quanto option in such a circumstance, i.e., $N^f \times \frac{1}{K} \max \left(0, K - \frac{S_T^d}{S_0^d} \right)$.

11.5 Risks of Structured Products to Investors

To investors, a structured product has the following risks in general:

- **Issuer and Counterparty Risk:** similar to any debt instrument, the credit-worthiness of the issuer is important to investors; For securitized instruments such as note/certificate, the issuer does not have any counterparty credit risk from the investors, but the investors are bearing the credit risk of the issuer. The investor's worst loss is the total amount spent on the security. For non-securitized and OTC transactions, we refer to Chap. 2 for counterparty risk management;
- **Liquidity and Early-exit Risk:** the risk of not being able to early redeem the product at a reasonable price; To terminate a structured product before its maturity, the investor needs to "unwind" the product at the bid price (also known as secondary market bid price) from the product issuer/counterparty. The unwind price includes transaction cost and is usually not transparent;
- **Market Risk:** it is the risk from all market factors (e.g., underlier's price or volatility) determining the final payoff as well as the mark-to-market value of the structured product.

Exercises

1. Consider the simplest option-based principal protected structure with a long call option. How will a low interest rate environment affect such a structure?
2. Implement a pricer for a standard ELN linked to a non-dividend paying stock and tweak the stock volatility to study the short volatility effect. Assume that the stock price follows the BSM model so that the B-S formula (10.1) can be applied.

Part II
Stochastic Calculus and Financial
Modelling

Chapter 12

Review of Basic Probability Concepts



In this chapter, we review the basic concepts and notations in probability that will be used or referenced to in the following chapters. We refer to [57] for further studies.

12.1 Probability Space, Measure, and Properties

A *sample space* Ω is a non-empty set of all the possible outcomes from a random experiment. For example, the coin toss of n times has $\Omega = \{\omega : \omega = \omega_1 \cdots \omega_n\}$ with 2^n elements and ω_i is H (head) or T (tail), $\forall i = 1, \dots, n$. A subset of Ω , $A \subseteq \Omega$, is called an *event*, say, first toss being a head.

A *partition* of Ω is defined as a collection of non-empty sets of Ω (known as *blocks* of the partition), $\{A_1, \dots, A_n\}$, which satisfies the following:

- (1) A_i are *disjoint*: $\forall i \neq j, A_i \cap A_j = \emptyset$
- (2) $A_1 \cup \dots \cup A_n = \Omega$.

Definition 12.1 (σ -Algebra) Let $\mathcal{F}$ be a collection of subsets of Ω , and it is called a σ -algebra if it satisfies:

- (1) $\mathcal{F}$ is closed under complements: $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$
- (2) $\mathcal{F}$ is closed under countable union: $\forall A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$
- (3) the empty set $\emptyset$ is in $\mathcal{F}$: $\emptyset \in \mathcal{F}$ (hence $\Omega \in \mathcal{F}$ as $\Omega = \emptyset^c$).

It can be shown that a σ -algebra is closed under intersections and set differences: $\forall A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$ and $A \setminus B \in \mathcal{F}$ ($A \setminus B$ is sometimes noted as $A - B$). In probability theory, σ -algebra is used to represent a set of events. The smallest σ -algebra generated by intervals in $\mathbb{R}$ is called **Borel** σ -algebra.

Definition 12.2 (Probability) Let $\mathcal{F}$ be a σ -algebra on Ω . A *probability measure* $\mathbb{P}$ is a function such that:

- (1) $0 \leq \mathbb{P}(A) \leq 1, \forall A \in \mathcal{F}$
- (2) $\mathbb{P}(\Omega) = 1$
- (3) Countable additivity: $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$, for disjoint sets $A_i \in \mathcal{F}, \forall i$.

The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is then called a *probability space*. If $\mathbb{P}(A) = 1$, then, we say that the event A occurs $\mathbb{P}$ -almost surely (or simply, *almost surely, a.s.*).

Example 12.1 Let $\Omega = \{1, 2, \dots\}$ be the set of all natural numbers, $A_1 \subset \Omega$ the set of all even numbers, $A_2 \subset \Omega$ the set of all odd numbers, and $\mathcal{F} = \{\emptyset, \Omega, A_1, A_2\}$ is a σ -algebra. Defining $\mathbb{P}(A_1) = \mathbb{P}(A_2) = \frac{1}{2}$, then, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. $\square$

Here are some properties for probability measure:

- (1) $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$
- (2) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
- (3) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- (4) if A and B are disjoint, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$
- (5) if $A \subseteq B$, then $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$
- (6) if $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Theorem 12.1 (Law of Total Probability) Let $\{B_1, \dots, B_n\}$ be a partition of Ω . Then, $\forall A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \cap B_i).$$

Example 12.2 Consider rolling a dice. Let $\{B_1, B_2\}$ be a partition of Ω with B_1 and B_2 being the set of odd numbers and even numbers respectively. Denote A the outcome being 1 or 6. Then by Law of Total Probability, $\mathbb{P}(A) = \sum_{i=1}^2 \mathbb{P}(A \cap B_i) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$. $\square$

12.2 Independence and Conditional Probability

Given $(\Omega, \mathcal{F}, \mathbb{P})$, the events $A \in \mathcal{F}$ and $B \in \mathcal{F}$ are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

A collection of events $\{A_1, \dots, A_n\}$ are independent if, for every sub-collection of at least two events $\{A_{i_1}, \dots, A_{i_k}\}, k = 2, \dots, n$,

$$\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_k}).$$

Example 12.3 Sophie has 3 coins in her pocket, 2 Euro coins and 1 Pound coin. She takes out randomly one coin, and then takes out another one *after* putting the first one back in. Let A be the event that the first coin is Euro and B be the event that the second coin is Euro. Then, $\mathbb{P}(A \cap B) = \frac{4}{9} = \frac{2}{3} \times \frac{2}{3} = \mathbb{P}(A)\mathbb{P}(B)$. $\square$

The *conditional probability* of event $A \in \mathcal{F}$ given event $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$ is defined as

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Properties of conditional probability are

- (1) $0 \leq \mathbb{P}(A|B) \leq 1$
- (2) if $B \subseteq A$, $\mathbb{P}(A|B) = 1$
- (3) $\mathbb{P}(A^c|B) = 1 - \mathbb{P}(A|B)$
- (4) $\mathbb{P}(A_1 \cup A_2|B) = \mathbb{P}(A_1|B) + \mathbb{P}(A_2|B) - \mathbb{P}(A_1 \cap A_2|B)$
- (5) if A_1 and A_2 are disjoint, then $\mathbb{P}(A_1 \cup A_2|B) = \mathbb{P}(A_1|B) + \mathbb{P}(A_2|B)$.

Example 12.4 From the previous example, suppose Sophie takes out the second coin *without* putting the first one back in. Let A be the event that the first coin is Euro and B the event that the second coin is Euro. Then, $\mathbb{P}(A) = \frac{2}{3}$ and $\mathbb{P}(A \cap B) = \frac{1}{3}$. The conditional probability for first getting Euro and then Euro again is $\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{1/3}{2/3} = \frac{1}{2}$. $\square$

Let $\{B_1, \dots, B_n\}$ be a partition of Ω and $\mathbb{P}(B_i) > 0$, $i = 1, \dots, n$. The following is a different version of Theorem 12.1.

Theorem 12.2 (Law of Total Probability) For all $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

Theorem 12.3 (Bayes' Rule) For all $A \in \mathcal{F}$ and $k = 1, \dots, n$,

$$\mathbb{P}(B_k|A) = \frac{\mathbb{P}(A|B_k)\mathbb{P}(B_k)}{\sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i)}.$$

Example 12.5 Revisit the last example. Let B_1 be the event that the second coin is Euro, and B_2 be the event that the second coin is Pound. Then, by Law of Total Probability, $\mathbb{P}(A) = \sum_{i=1}^2 \mathbb{P}(A|B_i)\mathbb{P}(B_i) = \frac{1}{2} \times \frac{2}{3} + 1 \times \frac{1}{3} = \frac{2}{3}$. The conditional probability for first getting Euro and then Euro again can be alternatively calculated with the Bayes' Rule, $\mathbb{P}(B_1|A) = \frac{\mathbb{P}(A|B_1)\mathbb{P}(B_1)}{\sum_{i=1}^2 \mathbb{P}(A|B_i)\mathbb{P}(B_i)} = \frac{1}{2} \times \frac{2/3}{2/3} = \frac{1}{2}$. $\square$

12.3 Random Variable and Distribution

Definition 12.3 (Random Variable) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and B be an interval¹ in $\mathbb{R}$. A (real) *random variable* is a function $X(\omega)$ (or, with simplified notation X): $\Omega \rightarrow \mathbb{R}$ such that

$$\{X(\omega) \in B\} \in \mathcal{F},$$

where $\{X(\omega) \in B\}$ is the notation for $\{\omega \in \Omega : X(\omega) \in B\}$, i.e., the collection of ω such that $X(\omega) \in B$.

It is customary to denote

- $\mathbb{P}\{X = x\}$, i.e., $\mathbb{P}\{\omega \in \Omega : X(\omega) = x\}$, by $\mathbb{P}(X = x)$ or $\mathbb{P}[X = x]$
- $\mathbb{P}\{X \in B\}$, i.e., $\mathbb{P}\{\omega \in \Omega : X(\omega) \in B\}$ by $\mathbb{P}(X \in B)$ or $\mathbb{P}[X \in B]$.

One particular random variable is the *indicator function* for $A \in \mathcal{F}$:

$$1_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ 0, & \text{if } \omega \notin A. \end{cases}$$

12.3.1 Distribution

The function $F(x) = \mathbb{P}\{X \leq x\}$ for random variable X and $x \in \mathbb{R}$ is called the *cumulative distribution function (c.d.f.)* of X . It has the following properties:

- (1) $F(x)$ is non-decreasing of x
- (2) $0 \leq F(x) \leq 1$, $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow \infty} F(x) = 1$
- (3) F is right continuous: $\lim_{y \downarrow x} F(y) = F(x)$.

For a continuous random variable, there is $\mathbb{P}\{a \leq X \leq b\} = F(b) - F(a)$, $\forall a \leq b$. For a discrete random variable, $\mathbb{P}\{X = x\} = F(x) - F(x_-)$.²

An integrable function f is called *probability density function (p.d.f.)* of X if

$$F(a) = \int_{-\infty}^a f(x)dx, \quad \forall a \in \mathbb{R}.$$

¹ More generally, a Borel set on the real line.

² x_- is the discrete random variable's value just below x if it exists. Otherwise, i.e., x is the smallest value, $F(x_-) = 0$.

We have $\mathbb{P}\{a \leq X \leq b\} = \int_a^b f(x)dx$. The derivative of *c.d.f.* gives back the *p.d.f.*, $F'(x) = f(x)$. Moreover, we have $\int_{-\infty}^{\infty} f(x)dx = \mathbb{P}\{X \in \mathbb{R}\} = 1$, which is the continuous case of probability adding up to 1.

Let $(X_1, X_2) \in \mathbb{R}^2$ be a vector of two random variables. The *joint distribution function* of (X_1, X_2) is defined as

$$F(a, b) = \mathbb{P}\{X_1 \leq a, X_2 \leq b\},$$

where $\{X_1 \leq a, X_2 \leq b\}$ represents the collection of the events for $\{X_1 \leq a\}$ and $\{X_2 \leq b\}$. If (X_1, X_2) is a continuous random vector, then, the *joint probability density function* is the function $f(x_1, x_2)$ such that

$$F(a, b) = \int_{-\infty}^b \int_{-\infty}^a f(x_1, x_2)dx_1dx_2.$$

For (X_1, X_2) with density $f(x_1, x_2)$, the *marginal cumulative distribution function* of X_1 is defined as

$$F_{X_1}(a) = \int_{-\infty}^a f_{X_1}(x_1)dx_1,$$

where $f_{X_1}(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2)dx_2$ is called the *marginal density*.

More generally, let $(X_1, \dots, X_n)$ be a vector of random variables. The *joint distribution function* of the vector is defined as

$$F(a_1, \dots, a_n) = \mathbb{P}\{X \leq a_1, \dots, X_n \leq a_n\}.$$

For a continuous random vector, then, the *joint probability density function* is the function $f(x_1, \dots, x_n)$ such that

$$F(a_1, \dots, a_n) = \int_{-\infty}^{a_n} \cdots \int_{-\infty}^{a_1} f(x_1, \dots, x_n)dx_1 \cdots dx_n.$$

Example 12.6 Consider a c.d.f. $F(x) = \max\{0, \min\{x, 1\}\}$ for X . $F(x)$ satisfies the properties of a *c.d.f.* The *p.d.f.* is $f(x) = F'(x) = \mathbf{1}_{\{x \in [0, 1]\}}$. X is said to follow the *standard uniform distribution* $\mathcal{U}[0, 1]$. For $X_1, X_2 \sim \mathcal{U}[0, 1]$, the joint probability density function $f(x_1, x_2)$ is $\mathbf{1}_{\{x_1 \in [0, 1]\}} \mathbf{1}_{\{x_2 \in [0, 1]\}}$. The marginal *c.d.f.* of X_1 is $F_{X_1}(a) = \int_0^a f_{X_1}(x_1)dx_1 = \int_0^a \left(\int_{-\infty}^{\infty} f(x_1, x_2)dx_2 \right) dx_1 = \int_0^a \mathbf{1}_{\{x_1 \in [0, 1]\}} dx_1$. Thus, $f_{X_1}(x_1) = \mathbf{1}_{\{x_1 \in [0, 1]\}}$. $\square$

12.3.2 Expectation

The *expectation* (also known as *expected value*, *mean*) of a random variable X is its probability weighted average, denoted as $\mathbb{E}[X]$:

- for a discrete random variable $X \in \{x_1, \dots, x_n\}$,

$$\mathbb{E}[X] = \sum_{i=1}^n x_i \mathbb{P}\{X = x_i\} \quad \text{or} \quad \mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega).$$

- for a continuous random variable in $\mathbb{R}$ with density $f(x)$,

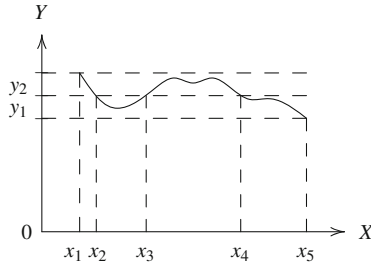
$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f(x) dx.$$

Example 12.7 The expectation of standard uniform distribution is $\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot 1_{\{x \in [0,1]\}} dx = \int_0^1 x dx = \frac{1}{2}$. $\square$

Expectation of a random variable can be denoted by *Lebesgue integral*. Readers may refer to [87, 100]. For a traditional *Riemann integral* $\int_a^b f(x) dx$, x is partitioned for interval $[a, b]$ with step size Δx_i and the integral's value is $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x_i$ where x_i is value of any point in the interval represented by Δx_i . The equivalent *Lebesgue integral* consists of partitioning the value of $f(x)$ ($\dots, f_i, f_{i+1}, \dots$) in the domain $D = \{f(x), \forall x \in [a, b]\}$. Lebesgue integral is defined as $\int_D f d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n f_i \mu\{f = f_i\}$ where $\mu\{f = f_i\}$ is the *measure* of $\{f = f_i\}$ that, in this case, is the sum of lengths of all the disjoint intervals $I_k^i, k = 1, 2, \dots$ such that $f_i \leq f(x) < f_{i+1}, \forall x \in I_k^i$. When both can be defined for the same problem, the two integrals will give the same value. In the literature, there are several equivalent notations for Lebesgue integration:

$$\int_D f d\mu = \int_D f(x) d\mu(x) = \int_D f(x) \mu(dx).$$

Example 12.8 Graphical illustration of a Lebesgue integral with measures constituted by intervals on x -axis:



$$\int_{\mathbb{R}} y d\mu \approx y_1 \times [(x_3 - x_2) + (x_5 - x_4)] + y_2 \times [(x_2 - x_1) + (x_4 - x_3)].$$

□

Lebesgue integral is more general than Riemann integral. When the measure is a probability measure, Lebesgue integral allows a unified notation of expectation for a random variable that can be either discrete or continuous:

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) d\mathbb{P}(\omega) \text{ or simply } \mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}.$$

In the above expression, $d\mathbb{P}(\omega)$ represents the probability measure for $X(\omega)$. For continuous case, $d\mathbb{P}(\omega) = f(x)dx$ where $f(x)$ is the probability density function. We have

$$\mathbb{E}[X \mathbf{1}_A] = \int_{\Omega} X \mathbf{1}_A d\mathbb{P} = \int_A X d\mathbb{P}.$$

In particular, the expectation of $\mathbf{1}_A$ can be reduced as the probability of A :

$$\mathbb{E}[\mathbf{1}_A] = \int_{\Omega} \mathbf{1}_A(\omega) d\mathbb{P}(\omega) = \int_A d\mathbb{P}(\omega) = \mathbb{P}(A).$$

Another way to show the result can be $\mathbb{E}[\mathbf{1}_A] = 1 \times \mathbb{P}(A) + 0 \times \mathbb{P}(A^c) = \mathbb{P}(A)$.

12.3.3 Variance and Covariance

The *variance* of a random variable X with $\mathbb{E}[X] = m$ is defined as

$$\text{Var}(X) = \mathbb{E}[(X - m)^2],$$

The *standard deviation* of X is defined as $\sigma = \sqrt{\text{Var}(X)}$. Let c be a constant, some properties of variance are

- (1) $\text{Var}(c) = 0$;
- (2) $\text{Var}(cX) = c^2 \text{Var}(X)$;
- (3) if X, Y are *independent*, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

The i -th *moment* of a random variable X is defined as $\mathbb{E}[X^i]$, $i = 1, 2, \dots$, if it exists. The *covariance* of two randoms X and Y with $\mathbb{E}[X] = m_X$ and $\mathbb{E}[Y] = m_Y$ is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - m_X)(Y - m_Y)].$$

The *correlation* between X and Y with variance σ_X^2 and σ_Y^2 is defined by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

and it can be proven that $-1 \leq \rho \leq 1$ (see Appendix Proposition A.3). We have

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

Let a and b be constants, the basic properties of covariance include

- (1) $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$
- (2) $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- (3) $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$
- (4) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

Let $\mathbf{x} \in \mathbf{R}^n$ be a random vector with mean $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^\top$ and standard deviation $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n)^\top$, $\forall \sigma_i > 0$. Correlation matrix $\mathbf{C}$ and covariance matrix $\boldsymbol{\Sigma}$ are defined by the symmetric matrices

$$\mathbf{C} = \begin{pmatrix} 1 & \rho_{21} & \dots & \rho_{1n} \\ \rho_{21} & 1 & \dots & \rho_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ \rho_{n1} & \rho_{n2} & \dots & 1 \end{pmatrix} \quad \boldsymbol{\Sigma} = \begin{pmatrix} 1 & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & 1 & \dots & \sigma_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & 1 \end{pmatrix} \quad (12.1)$$

where $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)]$ and $\boldsymbol{\Sigma} = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top]$ and it can be shown that $\boldsymbol{\Sigma}$ and $\mathbf{C}$ are *positive semi-definite* matrices (see Appendix Proposition A.3). We have the relation

$$\boldsymbol{\Sigma} = \mathbf{D}\mathbf{C}\mathbf{D}^\top \quad (12.2)$$

where $\mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$ is a $n \times n$ diagonal matrix, i.e., 0 for entries (i, j) , $\forall i \neq j$ and σ_i for entries (i, i) , $\forall i$.

12.3.4 Independent Random Variables

Random variables $X_1, \dots, X_n$ are said to be independent if for Borel sets B_i ,

$$\mathbb{P}\{X_1 \in B_1, \dots, X_n \in B_n\} = \mathbb{P}\{X_1 \in B_1\} \cdots \mathbb{P}\{X_n \in B_n\},$$

and the following properties hold:

- Joint cumulative distribution: $F(a_1, \dots, a_n) = F_{X_1}(a_1) \cdots F_{X_n}(a_n)$;
- Joint density: $f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$ (if the densities exist);
- Expectation: $\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n]$.

Given two independent random variables X and Y with distributions μ and ν , the probability distribution of $Z = X + Y$ is known as *convolution* of μ and ν , and it is denoted as $\mu * \nu$.

If X and Y are discrete random variables, we have

$$\mu * \nu(Z = z) = \sum_{k=-\infty}^{\infty} \mathbb{P}\{X = k\} \mathbb{P}\{Y = z - k\} = \sum_{k=-\infty}^{\infty} \mathbb{P}\{X = z - k\} \mathbb{P}\{Y = k\}.$$

If X and Y are continuous with densities f and g , the density of Z is

$$f * g(z) = \int_{-\infty}^{\infty} f(z - y)g(y)dy = \int_{-\infty}^{\infty} f(x)g(z - x)dx.$$

12.3.5 Conditional Probability Distribution

The *conditional cumulative distribution* of continuous random variable X given y , $F(a|y)$ is defined by

$$F(a|y) = \mathbb{P}\{X \leq a | Y = y\} = \lim_{|Y - y| \rightarrow 0} \mathbb{P}\{X \leq a | Y \in \Delta y\}.$$

Here $Y \in \Delta y$ means the event that Y is in a small interval including y . If $f(x|y)$ is a non-negative function and $F(a|y) = \int_{-\infty}^a f(x|y)dx$, then, $f(x|y)$ is called the *conditional probability density* of X given $Y = y$. We have

$$f(x, y) = f(x|y)f_Y(y),$$

where $f_Y(y)$ is the marginal distribution of Y .

12.3.6 Conditional Expectation Given Event

The *conditional expectation* of random variable X given event $B = \{Y = \bar{y}\}$ is defined as

$$\mathbb{E}[X|B] = \frac{1}{\mathbb{P}(B)}\mathbb{E}[X\mathbf{1}_B] \text{ and more precisely,}$$

- for discrete random variables:

$$\mathbb{E}[X|B] = \mathbb{E}[X|Y = \bar{y}] = \sum_i x_i \mathbb{P}\{X = x_i | Y = \bar{y}\};$$

- for continuous random variables:

$$\mathbb{E}[X|B] = \mathbb{E}[X|Y = \bar{y}] = \int_{\Omega} xf(x|\bar{y})dx.$$

If X and B are independent, then $\mathbb{E}[X\mathbf{1}_B] = \mathbb{E}[X]\mathbb{E}[\mathbf{1}_B]$ and $\mathbb{E}[X|B] = \mathbb{E}[X]$. When the joint probability density $f(x, y)$ exists for random variables X and Y , $\mathbb{E}[X|Y = \bar{y}]$ can be represented by Dirac delta function $\delta(x)$ (see Appendix A.6)

$$\mathbb{E}[X|Y = \bar{y}] = \frac{\mathbb{E}[X\delta(Y - \bar{y})]}{\mathbb{E}[\delta(Y - \bar{y})]}. \quad (12.3)$$

Let $f_Y(y)$ represent the marginal density of $f(x, y)$. Then, we can derive (12.3) by

$$\begin{aligned} \mathbb{E}[X|Y = \bar{y}] &= \int_{-\infty}^{\infty} xf(x|\bar{y})dx = \int_{-\infty}^{\infty} x \frac{f(x, \bar{y})}{f_Y(\bar{y})} dx \\ &= \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x\delta(y - \bar{y})f(x, y)dx dy}{\int_{-\infty}^{\infty} \delta(y - \bar{y})f_Y(y)dy} = \frac{\mathbb{E}[X\delta(Y - \bar{y})]}{\mathbb{E}[\delta(Y - \bar{y})]}. \end{aligned}$$

12.3.7 Characteristic and Moment Generating Functions

Let $F(x)$ be a distribution function for random variable X . The *characteristic function* (or *Fourier transform*) of X is the complex function of a real variable t :

$$\phi(t) = \mathbb{E}[e^{itX}], \quad t \in \mathbb{R}, \quad i^2 = -1.$$

There exist several inversion calculation formulas for $F(x)$ from $\phi(t)$ and we cite the following well-known ones (see Appendix A.9 for the details on derivation):

$$\begin{cases} F(x) = F(a) + \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-ita} - e^{-itx}}{it} \phi(t) dt, & \text{for } a < x; \\ F(x) = \frac{1}{2} + \frac{1}{2\pi} \int_0^{\infty} \frac{e^{itx} \phi(-t) - e^{-itx} \phi(t)}{it} dt; \\ F(x) = \frac{1}{2} - \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{e^{-itx} \phi(t)}{it} \right] dt. \end{cases}$$

The probability density function is

$$f(x) = F'(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi(t) dt.$$

The *moment generating function* of a random variable X is defined as below, if it exists,

$$\mathcal{M}_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

For a vector of random variables $(X_1, \dots, X_n)$, the characteristic function is defined below for $t_k \in \mathbb{R}$, $k = 1, \dots, n$,

$$\phi(t_1, \dots, t_n) = \mathbb{E}[e^{i \sum_{k=1}^n t_k X_k}],$$

and the moment generating function, if it exists, is

$$\mathcal{M}(t_1, \dots, t_n) = \mathbb{E}[e^{\sum_{k=1}^n t_k X_k}].$$

As $\frac{d^n}{dt^n} \mathbb{E}[e^{tX}] \Big|_{t=0} = \mathbb{E}[X^n e^{tX}] \Big|_{t=0} = \mathbb{E}[X^n]$, the moment generating function can be used to calculate the n -th moment.

Unlike the characteristic function, the moment generating function of a real-valued distribution does not always exist. A distribution function is uniquely determined by its characteristic function and moment generating function.

12.3.8 Normal Distribution

A *standard normal distribution* $X \sim \mathcal{N}(0, 1)$ satisfies

- probability density function: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$
- cumulative distribution function: $N(a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$

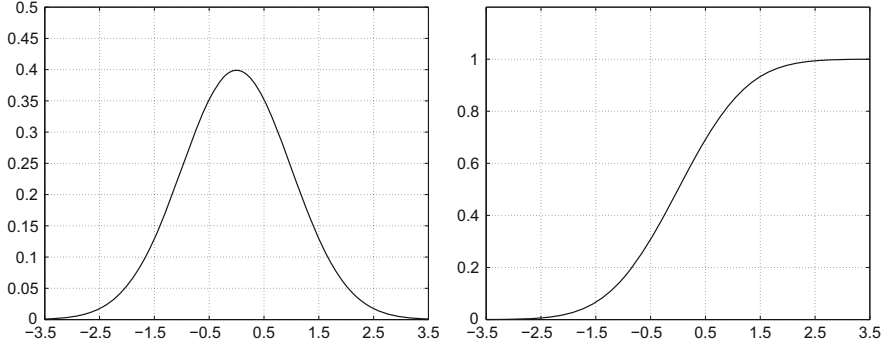


Fig. 12.1 The probability density function (left) and cumulative distribution function (right) for a standard normal random variable

The probability density function $f(x)$ is the famous “bell curve” (see Fig. 12.1). The mean or expected value of X is $\mathbb{E}[X] = 0$, and the variance of X is $\text{Var}(X) = 1$. We cite some useful properties: $N(a) + N(-a) = 1$, $N(-\infty) = 0$, $N(0) = \frac{1}{2}$, and $N(\infty) = 1$. See Appendix A.8 for its numerical approximation.

The moment generating function is

$$\mathcal{M}_X(t) = \mathbb{E}[e^{tX}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{x^2}{2}} dx = e^{\frac{t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-t)^2} dx = e^{\frac{t^2}{2}}.$$

It *generates* moments of X because its derivatives at 0 give

$$\begin{aligned} \mathcal{M}_X^{(1)} &= te^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X] = \mathcal{M}_X^{(1)}(0) = 0 \\ \mathcal{M}_X^{(2)} &= (1 + t^2)e^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X^2] = \mathcal{M}_X^{(2)}(0) = 1 \\ \mathcal{M}_X^{(3)} &= (3t + t^3)e^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X^3] = \mathcal{M}_X^{(3)}(0) = 0 \\ \mathcal{M}_X^{(4)} &= (3 + 6t^2 + t^4)e^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X^4] = \mathcal{M}_X^{(4)}(0) = 3 \end{aligned}$$

In general, the n -th moment can be directly calculated [87]

$$\mathbb{E}[X^n] = \begin{cases} 0, & \text{when } n \text{ is odd,} \\ 1 \cdot 3 \cdots (n-1), & \text{when } n \text{ is even.} \end{cases} \quad (12.4)$$

A general *normal distribution* with mean μ and variance σ^2 , denoted by $Y \sim \mathcal{N}(\mu, \sigma^2)$, satisfies

- probability density function: $f(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$, $-\infty < y < \infty$
- cumulative distribution function: $F(a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy$.

Note that

$$X \equiv \frac{Y - \mu}{\sigma} \sim \mathcal{N}(0, 1),$$

Then, the moment generating function can be directly obtained:

$$\mathbb{E}[e^{tY}] = \mathbb{E}[e^{t(\mu + \sigma X)}] = e^{\mu t} \mathbb{E}[e^{(t\sigma)X}] = e^{\mu t + \frac{1}{2}\sigma^2 t^2}.$$

Setting $t = 1$ in the moment generating function, we obtain a well-known result:

$$\mathbb{E}[e^Y] = e^{\mathbb{E}[Y] + \frac{1}{2}\text{Var}(Y)}.$$

A vector-valued random variable $(X_1, \dots, X_n)$ is said to follow a *multivariate normal distribution* if every linear combination of X_i follows a normal distribution. Probability density function of multivariate normal distribution is

$$f(x_1, \dots, x_n) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right),$$

where $\mathbf{x} = [x_1, \dots, x_n]^\top$ is a real n -dimensional column vector, $\boldsymbol{\mu}$ is the mean vector, and $|\Sigma|$ is the determinant of Σ , a $n \times n$ covariance matrix that is symmetric and positive definite.

12.3.9 Exponential Distribution

An exponential random variable τ ($\tau > 0$) follows the density below:

$$f(t) = \lambda e^{-\lambda t},$$

where $\lambda > 0$ and it is known as *intensity*. Properties of exponential distribution for $\tau > 0$ and $0 < s < t$ are

- Distribution: $\mathbb{P}\{\tau \leq t\} = \int_0^t f(s)ds = 1 - e^{-\lambda t}$ and $\mathbb{P}\{\tau > t\} = e^{-\lambda t}$
- Expectation: $\mathbb{E}[\tau] = \int_0^\infty t f(t)dt = \frac{1}{\lambda}$

- Variance: $\text{Var}(\tau) = \frac{1}{\lambda^2}$
- Conditional distribution: $\mathbb{P}\{\tau > s + t | \tau > s\} = \frac{\mathbb{P}\{\tau > s+t\}}{\mathbb{P}\{\tau > s\}} = e^{-\lambda t} = \mathbb{P}\{\tau > t\}$ (“Memorylessness property”).

Example 12.9 Consider a exponential random variables as waiting times. We set $t = 6$ and $s = 8$, then $\mathbb{P}(\tau > 14 | \tau > 8) = \mathbb{P}(\tau > 6)$. This means that the probability we have to wait 14 minutes or more from the start, given that we have already waited at least 8 minutes, is equal to the probability that we wait 6 minutes or more from the start.³ $\square$

12.3.10 Poisson Distribution

A Poisson random variable N is a discrete variable for counting the number of random events within a time interval. The arrival time for each random event follows the exponential distribution with intensity λ . Here are some properties of Poisson distribution for the event number, an integer $k \geq 0$ and an event rate $\lambda > 0$:

- Distribution: $\mathbb{P}\{N = k\} = \frac{\lambda^k}{k!} e^{-\lambda}$
- Expectation: $\mathbb{E}[N] = \lambda$
- Variance: $\text{Var}(N) = \lambda$.

12.4 Limit Theorems

12.4.1 Law of Large Numbers

Let $X_1, X_2, \dots$ be an infinite sequence of *i.i.d.* (independent and identically distributed) Lebesgue integrable random variables with expected value $\mathbb{E}[X_1] = \mathbb{E}[X_2] = \dots = \mu$. Then,

$$\lim_{n \rightarrow \infty} \frac{1}{n} (X_1 + X_2 + \dots + X_n) = \mu.$$

The below expressions are *unbiased estimators* for, respectively, μ and σ^2 with finite *i.i.d.* samples $X_1, \dots, X_n$ are

$$\frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad \frac{1}{n-1} \sum_{i=1}^n \left(X_i - \frac{1}{n} \sum_{j=1}^n X_j \right)^2$$

(see Appendix Proposition A.4).

³ Exponential random variables are the only continuous random variables with memorylessness property known so far.

12.4.2 Central Limit Theorem

Let $X_1, X_2, \dots$ be an infinite sequence of *i.i.d.* random variables with finite mean μ and variance σ^2 . Denote a new sequence of random variables as

$$S_n = \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mu}{\sigma/\sqrt{n}}.$$

Then, S_n converges to a standard normal random variable by

$$\lim_{n \rightarrow \infty} \mathbb{P}\{S_n \leq a\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^a e^{-\frac{x^2}{2}} dx.$$

12.4.3 Confidence Interval

A confidence interval is a range of values that the true value of a statistical parameter lies in with a certain probability known as confidence level (e.g., 95%). For Monte Carlo simulation (see Monte Carlo method in Chap. 21), confidence interval provides a measure for the accuracy of the estimated result.

Denote $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. By Central Limit Theorem, the confidence interval for the estimation of μ (i.e., $\mathbb{E}[X]$) is obtained from normal distribution:

$$\mathbb{P}\left\{-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right\} = 1 - \alpha,$$

or,

$$\mathbb{P}\left\{\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} < \mu < \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right\} = 1 - \alpha.$$

Given α , the value of $z_{\alpha/2}$ is known for normal distribution. For a large n , we have

$$\sigma \approx s, \text{ where } s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Example 12.10 For $1 - \alpha = 95\%$, we have $z_{\alpha/2} = 1.96$ from normal distribution. So, the 95% confidence interval for μ is

$$\left(\bar{X} - 1.96 \frac{s}{\sqrt{n}}, \bar{X} + 1.96 \frac{s}{\sqrt{n}}\right).$$

□

Exercises

1. Suppose X and Y are independent random variables, both following the standard uniform distribution $\mathcal{U}[0, 1]$. Find the conditional distribution and conditional expectation of Y given $X + Y > 1$.
2. Consider $X \sim \mathcal{N}(0, 2)$. Obtain the mean, variance, and probability density function of $Y = 3 + 2X$.
3. Assume $Z \sim \mathcal{N}(0, 1)$. Prove that Z and Z^2 are uncorrelated but dependent.
4. Assume $Z \sim \mathcal{N}(0, 1)$. Show that

$$\mathbb{E}[e^{-\frac{\sigma^2}{2} + \sigma Z} \mathbf{1}_{\{Z < x\}}] = N(x - \sigma),$$

where x and σ are real variables.

5. If $Y_1 \sim N(\mu_1, \sigma_1^2)$ and $Y_2 \sim N(\mu_2, \sigma_2^2)$ are independent, show that $Y_1 + Y_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.
6. Use RAND function to generate a number of samples in EXCEL to verify the mean of standard uniform distribution $\mathcal{U}[0, 1]$ according to the law of large numbers.

Chapter 13

Stochastic Calculus: I



13.1 Stochastic Process

The σ -algebra *generated* by a random variable X , denoted $\sigma(X)$, is the σ -algebra by the collection of $\{X \in B\}$ (i.e., $\{\omega : X(\omega) \in B\}$), where B is any interval in $\mathbb{R}$. Let $\mathcal{G}$ be a σ -algebra on Ω . Then X is said to be $\mathcal{G}$ -*measurable* if $\sigma(X) \subseteq \mathcal{G}$.

Definition 13.1 (Filtration) Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *filtration* $\mathcal{F}_t$ is an increasing collection of σ -algebras on Ω , such that for s, t in a totally ordered set $\mathcal{T}$ (e.g., “time”),

$$\mathcal{F}_s \subseteq \mathcal{F}_t, \quad s < t.$$

Definition 13.2 (Stochastic Process) A *stochastic process* is a collection of random variables indexed by $\mathcal{T}$:

$$\{X_t : t \in \mathcal{T}\}.$$

Here, X_t is a short notation for $X_t(\omega)$. For each t , X_t is reduced as a random variable. When there is no randomness, X_t is called *deterministic*.

A stochastic process X_t on $\mathcal{T}$ is said to be *adapted* to filtration $\mathcal{F}_t$ if, for every $t \in \mathcal{T}$, X_t is $\mathcal{F}_t$ -measurable. Given a stochastic process $X = \{X_t : t \in \mathcal{T}\}$, the **natural filtration** for this process is the filtration where $\mathcal{F}_t$ is generated by the events for all the possible values of $X_s, s \leq t$. A stochastic process is always adapted to its natural filtration.

Example 13.1 (Asset Price Example) Assume a simplistic multi-period asset price model where a price has only two possible movements for each period driven by a single economic factor with 2 status (“y”—yes or “n”—no):

- “y” drives the price up by a certain amount with probability p , or
- “n” drives the price down by a certain amount with probability q ($q = 1 - p$).

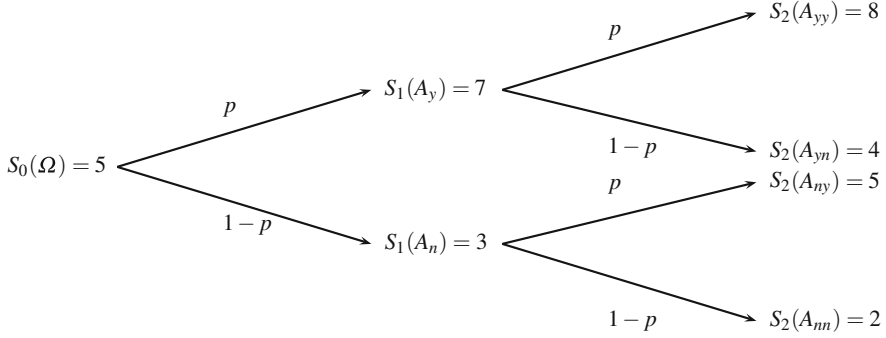


Fig. 13.1 A 2-period asset price model

Consider a 2-period asset price model S_t , $t = 0, 1, 2$, represented by the tree in Fig. 13.1, where $p = 0.5$ and $q = 1 - p = 0.5$.

As an example, the sample space of S_2 is $\Omega = \{yy, yn, ny, nn\}$. The “atom events” for S_2 are $A_{yy} = \{yy\}$, $A_{yn} = \{yn\}$, $A_{ny} = \{ny\}$, $A_{nn} = \{nn\}$. The other events can be generated by the atom events with union and complement operations: $A_y = A_{yy} \cup A_{yn}$, $A_n = A_{ny} \cup A_{nn}$, $\Omega = A_y \cup A_n$. Probabilities for the events are

$$\mathbb{P}(A_{yy}) = 0.25, \mathbb{P}(A_{yn}) = 0.25, \mathbb{P}(A_{ny}) = 0.25, \mathbb{P}(A_{nn}) = 0.25$$

$$\mathbb{P}(A_y) = 0.5, \mathbb{P}(A_n) = 0.5$$

$$\mathbb{P}(\Omega) = 1.$$

Filtrations for the periods are

$$\mathcal{F}_0 = \{\emptyset, \Omega\}$$

$$\mathcal{F}_1 = \{\emptyset, \Omega, A_y, A_n\}$$

$$\mathcal{F}_2 = \{\emptyset, \Omega, A_y, A_n, A_{yy}, A_{yn}, A_{ny}, A_{nn}, A_{yy}^c, A_{yn}^c, A_{ny}^c, A_{nn}^c, A_{yy} \cup A_{ny}, A_{yy} \cup A_{nn}, A_{yn} \cup A_{ny}, A_{yn} \cup A_{nn}\}.$$

□

13.2 Conditional Expectation Given σ -Algebra*

Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We review the definition of conditional expectation *given σ -algebra* that extends the concept of conditional expectation *given event* introduced in 12.3.6.

If $\mathcal{B}$ is the σ -algebra generated by some disjoint events of in $\mathcal{F}$: $\{B_1, B_2, \dots\}$, $\forall i, B_i \in \mathcal{F}$ and $B_i \cap B_j = \emptyset$ for $i \neq j$. The conditional expectation of random variable X given $\mathcal{B}$ is defined as

$$\mathbb{E}[X|\mathcal{B}] = \sum_i \mathbb{E}[X|B_i] \mathbf{1}_{B_i}, \quad \mathcal{B} \subseteq \mathcal{F}.^1$$

Generally, the conditional expectation given σ -algebra $\mathcal{G} \subseteq \mathcal{F}$ is the unique $\mathcal{G}$ -measurable random variable Z , denoted as $\mathbb{E}[X|\mathcal{G}]$, that satisfies²

$$\mathbb{E}[Z \mathbf{1}_B] = \mathbb{E}[X \mathbf{1}_B], \quad \forall B \in \mathcal{G}$$

(Note: $\mathbb{E}[Z \mathbf{1}_B] = \mathbb{E}[X \mathbf{1}_B]$ is equivalent to $\int_B \mathbb{E}[X|\mathcal{G}](\omega) d\mathbb{P}(\omega) = \int_B X(\omega) d\mathbb{P}(\omega)$ that is another usual notation for the definition of conditional expectation.) $\mathbb{E}[X|\mathcal{G}]$ is not a single value in general. It is a variable representing the “partial averaging” for each $B \in \mathcal{G}$. It is a generalization of the single valued conditional expectation with a single event. If the conditioning σ -algebra is a filtration $\mathcal{F}_t$, we can use indifferently the following two equivalent notations:

$$\mathbb{E}_t[X] \quad \text{or} \quad \mathbb{E}[X|\mathcal{F}_t].$$

We note that $\mathbb{E}[X] = \mathbb{E}[X|\mathcal{F}_0]$.

Example 13.2 (Conditional Expectation) In the 2-period asset price example, the partitions for $\mathcal{F}_2, \mathcal{F}_1, \mathcal{F}_0$ can be respectively $\{A_{yy}, A_{yn}, A_{ny}, A_{nn}\}, \{A_y, A_n\}, \{\Omega\}$. We have the following conditional expectation representations on $\mathcal{F}_2, \mathcal{F}_1, \mathcal{F}_0$:

$$\mathbb{E}[S_2|\mathcal{F}_2] = 8 \cdot \mathbf{1}_{A_{yy}} + 4 \cdot \mathbf{1}_{A_{yn}} + 5 \cdot \mathbf{1}_{A_{ny}} + 2 \cdot \mathbf{1}_{A_{nn}} = S_2$$

$$\begin{aligned} \mathbb{E}[S_2|\mathcal{F}_1] &= [8\mathbb{P}(A_{yy}|A_y) + 4\mathbb{P}(A_{yn}|A_y) + 5\mathbb{P}(A_{ny}|A_y) + 2\mathbb{P}(A_{nn}|A_y)] \mathbf{1}_{A_y} \\ &\quad + [8\mathbb{P}(A_{yy}|A_n) + 4\mathbb{P}(A_{yn}|A_n) + 5\mathbb{P}(A_{ny}|A_n) + 2\mathbb{P}(A_{nn}|A_n)] \mathbf{1}_{A_n} \\ &= [8\mathbb{P}(A_{yy}|A_y) + 4\mathbb{P}(A_{yn}|A_y)] \mathbf{1}_{A_y} + [5\mathbb{P}(A_{ny}|A_n) + 2\mathbb{P}(A_{nn}|A_n)] \mathbf{1}_{A_n} \\ &= 6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n} \end{aligned}$$

$$\begin{aligned} \mathbb{E}[S_2|\mathcal{F}_0] &= [8\mathbb{P}(A_{yy}|\Omega) + 4\mathbb{P}(A_{yn}|\Omega) + 5\mathbb{P}(A_{ny}|\Omega) + 2\mathbb{P}(A_{nn}|\Omega)] \mathbf{1}_\Omega \\ &= [8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn}) + 5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})] \mathbf{1}_\Omega \\ &= \mathbb{E}[S_2] \mathbf{1}_\Omega = \mathbb{E}[S_2] = 4.75. \end{aligned}$$

□

¹ We can show that $\mathbb{E}[X|\mathcal{B}]$ satisfies the general definition of conditional expectation given a σ -algebra, i.e., $\mathbb{E}[Z \mathbf{1}_B] = \mathbb{E}[X \mathbf{1}_B]$, $\forall B \in \mathcal{B}$. In fact, $B = B_{k_1} \cup B_{k_2} \cup \dots$ where $B_{k_j} \in \{B_1, B_2, \dots\}$ and $\mathbf{1}_B = \mathbf{1}_{B_{k_1}} + \dots$. Then, $\mathbb{E}[Z \mathbf{1}_B] = \mathbb{E}[\mathbb{E}[X|\mathcal{B}] \mathbf{1}_B] = \mathbb{E}[(\sum_i \mathbb{E}[X|B_i] \mathbf{1}_{B_i}) \mathbf{1}_B] = \mathbb{E}[\sum_j \mathbb{E}[X|B_{k_j}] \mathbf{1}_{B_{k_j}}] = \sum_j \mathbb{E}[\mathbb{E}[X|B_{k_j}] \mathbf{1}_{B_{k_j}}] = \sum_j \mathbb{E}[\mathbb{E}[X \mathbf{1}_{B_{k_j}}|B_{k_j}]] = \sum_j \mathbb{E}[X \mathbf{1}_{B_{k_j}}] = \mathbb{E}[X \sum_j \mathbf{1}_{B_{k_j}}] = \mathbb{E}[X \mathbf{1}_B]$.

² This general definition is related to the orthogonal projection in Hilbert space: $(X - Z) \cdot \mathbf{1}_B = 0$, or $X \cdot \mathbf{1}_B = Z \cdot \mathbf{1}_B$, where the inner product $a \cdot b$ is defined by $\mathbb{E}[ab]$ ($\mathbb{P}$ -a.s.).

Let $\mathcal{G}, \mathcal{H}$ be two σ -algebras and $\mathcal{G} \subset \mathcal{H}$. Then, we have the following properties:

- (1) **Independence:** If X is independent of $\mathcal{G}$, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$
- (2) **Law of Total Expectation:** $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[X]$
- (3) **Taking out Known Factors:** If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$
- (4) **Tower Property** (also known as **Iterated Conditioning**):
 - (a) $\mathbb{E}[\mathbb{E}[X|\mathcal{H}|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}]$
 - (b) $\mathbb{E}[\mathbb{E}[X|\mathcal{G}|\mathcal{H}] = \mathbb{E}[X|\mathcal{G}]$

Instead of providing pure mathematical proofs of the above properties (see [105] for instance), we prefer to illustrate with Asset Price Example in the below.

Illustration for (1) “Independence” Let $\mathcal{G}$ be a σ -algebra that is independent of S_2 . Assuming that $\{B_1, B_2\}$ is a partition of $\mathcal{G}$ (hence $\mathbf{1}_{B_1} + \mathbf{1}_{B_2} = 1$), then B_1, B_2 are independent of S_2 . Hence,

$$\begin{aligned}\mathbb{E}[S_2|\mathcal{G}] &= \mathbb{E}[S_2|B_1]\mathbf{1}_{B_1} + \mathbb{E}[S_2|B_2]\mathbf{1}_{B_2} \\ &= \mathbb{E}[S_2]\mathbf{1}_{B_1} + \mathbb{E}[S_2]\mathbf{1}_{B_2} = \mathbb{E}[S_2](\mathbf{1}_{B_1} + \mathbf{1}_{B_2}) = \mathbb{E}[S_2].\end{aligned}$$

Illustration for (2) “Law of Total Expectation”

$$\begin{aligned}\mathbb{E}[S_2|\mathcal{F}_1] &= [8\mathbb{P}(A_{yy}|A_y) + 4\mathbb{P}(A_{yn}|A_y)]\mathbf{1}_{A_y} + [5\mathbb{P}(A_{ny}|A_n) \\ &\quad + 2\mathbb{P}(A_{nn}|A_n)]\mathbf{1}_{A_n} \\ \mathbb{E}[\mathbb{E}[S_2|\mathcal{F}_1]] &= [8\mathbb{P}(A_{yy}|A_y) + 4\mathbb{P}(A_{yn}|A_y)]\mathbb{P}(A_y) + [5\mathbb{P}(A_{ny}|A_n) \\ &\quad + 2\mathbb{P}(A_{nn}|A_n)]\mathbb{P}(A_n) \\ &= 8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn}) + 5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn}) = \mathbb{E}[S_2].\end{aligned}$$

Illustration for (3) “Taking out Known Factors” Consider $\mathbb{E}[S_1 S_2|\mathcal{F}_1]$ where $S_1 = 7 \cdot \mathbf{1}_{A_y} + 3 \cdot \mathbf{1}_{A_n}$ is $\mathcal{F}_1$ -measurable.

$$\begin{aligned}\mathbb{E}[S_1 S_2|\mathcal{F}_1] &= [7 \cdot 8\mathbb{P}(A_{yy}|A_y) + 7 \cdot 4\mathbb{P}(A_{yn}|A_y)]\mathbf{1}_{A_y} + [3 \cdot 5\mathbb{P}(A_{ny}|A_n) \\ &\quad + 3 \cdot 2\mathbb{P}(A_{nn}|A_n)]\mathbf{1}_{A_n} \\ &= (7 \cdot \mathbf{1}_{A_y} + 3 \cdot \mathbf{1}_{A_n})\{[8\mathbb{P}(A_{yy}|A_y) + 4\mathbb{P}(A_{yn}|A_y)]\mathbf{1}_{A_y} \\ &\quad + [5\mathbb{P}(A_{ny}|A_n) + 2\mathbb{P}(A_{nn}|A_n)]\mathbf{1}_{A_n}\} \\ &= (7 \cdot \mathbf{1}_{A_y} + 3 \cdot \mathbf{1}_{A_n})\mathbb{E}[S_2|\mathcal{F}_1] = S_1 \mathbb{E}[S_2|\mathcal{F}_1].\end{aligned}$$

Illustration for (4.a) “Tower Property”

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[S_2|\mathcal{F}_2]|\mathcal{F}_1] &= \mathbb{E}[8 \cdot \mathbf{1}_{A_{yy}} + 4 \cdot \mathbf{1}_{A_{yn}} + 5 \cdot \mathbf{1}_{A_{ny}} + 2 \cdot \mathbf{1}_{A_{nn}}|\mathcal{F}_1] \\
&= 8\mathbb{E}[\mathbf{1}_{A_{yy}}|\mathcal{F}_1] + 4\mathbb{E}[\mathbf{1}_{A_{yn}}|\mathcal{F}_1] + 5\mathbb{E}[\mathbf{1}_{A_{ny}}|\mathcal{F}_1] \\
&\quad + 2\mathbb{E}[\mathbf{1}_{A_{nn}}|\mathcal{F}_1].
\end{aligned}$$

The first term is

$$8\mathbb{E}[\mathbf{1}_{A_{yy}}|\mathcal{F}_1] = 8\mathbb{P}(A_{yy}|A_y)\mathbf{1}_{A_y} + 8\mathbb{P}(A_{yy}|A_n)\mathbf{1}_{A_n} = 8\mathbb{P}(A_{yy}|A_y)\mathbf{1}_{A_y}.$$

We obtain similar results for the other terms. Hence,

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[S_2|\mathcal{F}_2]|\mathcal{F}_1] &= 8\mathbb{P}(A_{yy}|A_y)\mathbf{1}_{A_y} + 4\mathbb{P}(A_{yn}|A_y)\mathbf{1}_{A_y} \\
&\quad + 5\mathbb{P}(A_{ny}|A_n)\mathbf{1}_{A_n} + 2\mathbb{P}(A_{nn}|A_n)\mathbf{1}_{A_n} \\
&= 6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n} = \mathbb{E}[S_2|\mathcal{F}_1].
\end{aligned}$$

Illustration for (4.b) “Tower Property”

$$\mathbb{E}[\mathbb{E}[S_2|\mathcal{F}_1]|\mathcal{F}_2] = \mathbb{E}[6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n}|\mathcal{F}_2] = 6\mathbb{E}[\mathbf{1}_{A_y}|\mathcal{F}_2] + 3.5\mathbb{E}[\mathbf{1}_{A_n}|\mathcal{F}_2].$$

The first term is

$$\begin{aligned}
6\mathbb{E}[\mathbf{1}_{A_y}|\mathcal{F}_2] &= 6[\mathbb{P}(A_y|A_{yy})\mathbf{1}_{A_{yy}} + \mathbb{P}(A_y|A_{yn})\mathbf{1}_{A_{yn}} + \mathbb{P}(A_y|A_{ny})\mathbf{1}_{A_{ny}} \\
&\quad + \mathbb{P}(A_y|A_{nn})\mathbf{1}_{A_{nn}}] \\
&= 6(\mathbf{1}_{A_{yy}} + \mathbf{1}_{A_{yn}}) = 6 \cdot \mathbf{1}_{A_y}.
\end{aligned}$$

Similarly, we can obtain $3.5\mathbb{E}[\mathbf{1}_{A_n}|\mathcal{F}_2] = 3.5 \cdot \mathbf{1}_{A_n}$. Hence,

$$\mathbb{E}[\mathbb{E}[S_2|\mathcal{F}_1]|\mathcal{F}_2] = 6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n} = \mathbb{E}[S_2|\mathcal{F}_1].$$

Remark 13.1 (Conditional Expectation and Regression) Regression analysis is related to conditional expectation. To illustrate, let $\mathbf{y}$ and $\mathbf{x}$ be two centered random variables such that $\mathbf{y} = \mathbf{a}\mathbf{x} + \epsilon$ where ϵ is a white noise with $\mathbb{E}[\epsilon|\mathbf{x}] = 0$ (which implies $\mathbb{E}[\epsilon] = 0$). Then, we have $\mathbb{E}[\mathbf{y}|\mathbf{x}] = \mathbf{a}\mathbf{x}$ by taking conditional expectation on both sides of the equation. Hilbert orthogonal projection can be used to determine $\mathbf{a}$ whereby $\mathbf{a}\mathbf{x}$ is the projection of $\mathbf{y}$ in the space generated by all $\mathbf{c}\mathbf{x}$, $\forall \mathbf{c} \in \mathbb{R}$ with the inner product $X \cdot Y$ defined as $\mathbb{E}[XY]$ ($\mathbb{P}$ -a.s.). As $(\mathbf{y} - \mathbf{a}\mathbf{x})$ and $\mathbf{c}\mathbf{x}$ are orthogonal, we have $(\mathbf{y} - \mathbf{a}\mathbf{x}) \cdot \mathbf{x} = 0$, i.e., $\mathbb{E}[(\mathbf{y} - \mathbf{a}\mathbf{x})\mathbf{x}] = 0$, which leads to the linear regression formula $\mathbf{a} = \mathbb{E}[\mathbf{y}\mathbf{x}]/\mathbb{E}[\mathbf{x}^2]$. Appendix A.4 provides an equivalent method to determine $\mathbf{a}$ (least-squares regression method). $\square$

13.3 Martingale, Stopping Time

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with the filtration $\mathcal{F}_t$. Let $0 \leq s \leq t$. An $\mathcal{F}_t$ -adapted stochastic process M_t is said to be

- a *martingale*, if $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$
- a *supermartingale*, if $\mathbb{E}[M_t | \mathcal{F}_s] \leq M_s$
- a *submartingale*, if $\mathbb{E}[M_t | \mathcal{F}_s] \geq M_s$.

Supermartingales tend to fall and submartingales tend to rise. Processes that are both a supermartingale and a submartingale are martingales, thus they have no tendency to rise or fall. In particular, the expected value of a martingale M_t is its initial value M_0 , i.e., $\mathbb{E}[M_t] = M_0$.

Theorem 13.1 (Doob–Meyer Decomposition) *A supermartingale (submartingale) X_t admits a unique decomposition: $X_t = Y_t - A_t$ ($= Y_t + A_t$) where Y_t is a martingale and A_t is an increasing predictable³ process.*

Proof can be found in [94] [11].

A stopping time τ is a *random variable* taking values in $[0, \infty]$ and satisfying $\{\tau \leq t\} \equiv \{\omega \in \Omega; \tau(\omega) \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$.

Theorem 13.2 (Optional Sampling) *A martingale (supermartingale, submartingale) X_t stopped at a stopping time, i.e., $X_{t \wedge \tau}$ is a martingale (supermartingale, submartingale).*

We refer to [11] for the proof and further analysis.

13.4 Markov Property*

The Markov property ensures the memorylessness of a process, i.e., the past does not matter when predicting the future, only the present matters. An $\mathcal{F}_t$ -measurable stochastic process X_t is said to be a *Markov process* if, for any function⁴ f ,

$$\mathbb{E}[f(X_t) | \mathcal{F}_s] = \mathbb{E}[f(X_t) | X_s] = \varphi(s, X_s), \quad \forall t > s.$$

A continuous Markov process is called a *diffusion*.

Proposition 13.1 *The stochastic process $\varphi(s, X_s)$ is a martingale.*

³ Left continuous and $\mathcal{F}_t$ -adapted.

⁴ To be specific, any Borel measurable function.

Proof Let $0 \leq s \leq t \leq T$, then

$$\mathbb{E}[\varphi(t, X_t) | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[f(X_T) | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}[f(X_T) | \mathcal{F}_s] = \varphi(s, X_s).$$

□

13.5 Quadratic Variation

The *quadratic variation* of stochastic process X_t is defined as, if it exists,

$$\langle X, X \rangle_t = \lim_{\|P\| \rightarrow 0} \sum_i (X_{s_{i+1}} - X_{s_i})^2,$$

where $P = \{s_0, s_1, \dots, s_n\}$ represents a partition of interval $[0, t]$ and $\|P\|$ is the maximum step size. The quadratic variation is also denoted as $\langle X \rangle_t$.

Remark 13.2 If X_t has a continuous derivative X'_t and $\int_0^t |X'_s|^2 ds$ is finite, then $\langle X, X \rangle_t = 0$, which is illustrated in the below:

$$\begin{aligned} \langle X, X \rangle_t &= \lim_{\|P\| \rightarrow 0} \sum_i (X_{s_{i+1}} - X_{s_i})^2 \\ &= \lim_{\|P\| \rightarrow 0} \sum_i [X'_{s_i^*} (s_{i+1} - s_i)]^2, s_i^* \in [s_i, s_{i+1}] \quad (\text{by Mean Value Theorem}) \\ &\leq \lim_{\|P\| \rightarrow 0} \|P\| \sum_i |X'_{s_i^*}|^2 (s_{i+1} - s_i) = \lim_{\|P\| \rightarrow 0} \|P\| \cdot \int_0^t |X'_s|^2 ds = 0. \end{aligned}$$

The last equality holds, as $\int_0^t |X'_s|^2 ds$ is finite. □

The *cross variation* (or *quadratic covariation*) of two processes, X_t and Y_t , is defined as, if it exists,

$$\langle X, Y \rangle_t = \lim_{\|P\| \rightarrow 0} \sum_i (X_{s_{i+1}} - X_{s_i})(Y_{s_{i+1}} - Y_{s_i}).$$

An equivalent definition of *cross variation* is to use *polarization identity*: $\langle X, Y \rangle_t = \frac{1}{4}(\langle X + Y, X + Y \rangle_t - \langle X - Y, X - Y \rangle_t)$.

The incremental value change of $\langle X, Y \rangle_t$ corresponding to an infinitesimal change of t is called the *differential of $\langle X, Y \rangle_t$* that is denoted by $d\langle X, Y \rangle_t$ or $dX_t dY_t$ in this book. As t is a special case of stochastic processes, we also see the notation such as $dt dX$.

13.6 Brownian Motion

Definition 13.3 (Brownian Motion) A continuous process $W_t(\omega)$ is said to be a *standard Brownian Motion* if it satisfies

- (1) $W_0(\omega) = 0$ for all ω
- (2) for all ω , the map $t \mapsto W_t(\omega)$ is a continuous function for $t \geq 0$
- (3) W_t has independent increments, i.e., $W_u(\omega) - W_v(\omega)$ and $W_t(\omega) - W_s(\omega)$ are independent for all $0 \leq v \leq u \leq s \leq t$
- (4) W_t has stationary increments and every increment $W_t(\omega) - W_s(\omega)$, $0 \leq s < t$, follows the normal distribution $\mathcal{N}(0, t - s)$.

For simplicity, we will write $W_t(\omega)$ as W_t . We have

$$\mathbb{E}[W_t - W_s] = 0$$

$$\text{Var}(W_t - W_s) = t - s$$

$$W_t = \epsilon\sqrt{t} \text{ where } \epsilon \sim \mathcal{N}(0, 1) \text{ (because } W_t = W_t - W_0\text{)}.$$

We can see that W_t is not differentiable because

$$\lim_{\Delta t \rightarrow 0} \frac{W_{t+\Delta t} - W_t}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\epsilon\sqrt{\Delta t}}{\Delta t} = \frac{\epsilon}{\lim_{\Delta t \rightarrow 0} \sqrt{\Delta t}} \rightarrow \infty, \text{ a.s..}$$

See Fig. 13.2 for 50 sample paths of Brownian motion.

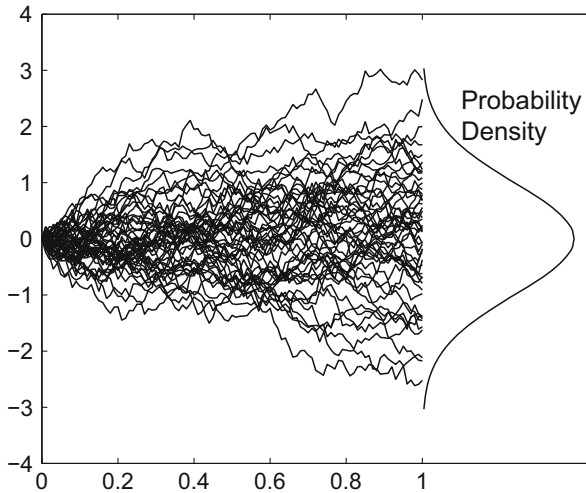


Fig. 13.2 Fifty Brownian paths with probability density at time 1

An n -dimensional Brownian motion is defined as

$$\mathbf{W}_t = (W_t^1, W_t^2, \dots, W_t^n),$$

where $W_t^i, i = 1, \dots, n$, is a standard Brownian motion and W_t^i and W_t^j ($\forall i \neq j$) are independent.

Properties of a Brownian Motion

Independence of Brownian increments implies that $W_t - W_s$ ($s < t$) is independent of $\mathcal{F}_s = \sigma(W_u, u \leq s)$. Hence, we have

$$\mathbb{E}[f(W_t - W_s) | \mathcal{F}_s] = \mathbb{E}[f(W_t - W_s)].$$

An $\mathcal{F}_t$ -adapted Brownian Motion is called $\mathcal{F}_t$ -Brownian Motion.

Theorem 13.3 (Properties of $\mathcal{F}_t$ -Brownian Motion)

- (1) W_t is an $\mathcal{F}_t$ -martingale
- (2) $W_t^2 - t$ is an $\mathcal{F}_t$ -martingale
- (3) $e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$ is an $\mathcal{F}_t$ -martingale
- (4) *time-change or scaling property*: $cW_t \xLeftrightarrow{(d)} W_{c^2 t}$ where the arrow with (d) means equivalence in terms of distribution.

Proof In the following, we assume that $u \geq t$.

- (1) $\mathbb{E}[W_u - W_t | \mathcal{F}_t] = \mathbb{E}[W_u - W_t] = 0$. Hence, $\mathbb{E}[W_u | \mathcal{F}_t] = \mathbb{E}[W_t | \mathcal{F}_t] = W_t$ (as W_t is $\mathcal{F}_t$ -measurable).
- (2) $\mathbb{E}[W_u^2 - W_t^2 | \mathcal{F}_t] = \mathbb{E}[(W_u - W_t)^2 + 2W_t(W_u - W_t) | \mathcal{F}_t] = \mathbb{E}[(W_u - W_t)^2 | \mathcal{F}_t] = u - t$. Hence, $\mathbb{E}[W_u^2 - u | \mathcal{F}_t] = \mathbb{E}[W_t^2 | \mathcal{F}_t] - t = W_t^2 - t$ (because W_t^2 is $\mathcal{F}_t$ -measurable).
- (3) $\mathbb{E}[e^{-\frac{1}{2}\sigma^2 u + \sigma W_u} | \mathcal{F}_t] = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t} \mathbb{E}[e^{\sigma(W_u - W_t)} | \mathcal{F}_t] = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t} \mathbb{E}[e^{\sigma(W_u - W_t)}]$. The expectation is given by the moment generating function of a normal random variable: $\mathbb{E}[e^{\sigma(W_u - W_t)}] = e^{\frac{1}{2}\sigma^2(u-t)}$. Hence, we obtain $\mathbb{E}[e^{-\frac{1}{2}\sigma^2 u + \sigma W_u} | \mathcal{F}_t] = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t} e^{\frac{1}{2}\sigma^2(u-t)} = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$.
- (4) can be checked by calculating the first and second order moments. □

Example 13.3 Let $X_t = W_t^2$. The process X_t is a submartingale from the Jensen's inequality [105]:

$$\mathbb{E}[X_t | \mathcal{F}_s] = \mathbb{E}[W_t^2 | \mathcal{F}_s] \geq (\mathbb{E}[W_t | \mathcal{F}_s])^2 = W_s^2 = X_s.$$

The Doob–Meyer decomposition for X_t is

$$X_t = Y_t + A_t = (W_t^2 - t) + t,$$

where $Y_t = W_t^2 - t$ is a martingale and $A_t = t$ is an increasing predictable (left continuous, $\mathcal{F}_t$ -adapted) process. □

Theorem 13.4 (Quadratic Variation of Brownian Motion) *Let W_t be a Brownian motion, then $\langle W, W \rangle_t = t$, almost surely. In differential form, we have the following calculation rules*

$$dW_t dW_t = dt, \quad dW_t dt = 0 \quad \text{and} \quad dt dt = 0.$$

Proof

- (1) We prove $dW_t dW_t = dt$ by showing that $\mathbb{E}[\langle W, W \rangle_t] = t$ and $\text{Var}(\langle W, W \rangle_t) = 0$. Let $P = \{s_0, s_1, \dots, s_n\}$ be a partition of $[0, t]$ with $\|P\| \rightarrow 0$.

$$\begin{aligned} \mathbb{E}[\langle W, W \rangle_t] &= \mathbb{E}\left[\lim_{\|P\| \rightarrow 0} \sum_i (W_{s_{i+1}} - W_{s_i})^2\right] \\ &= \lim_{\|P\| \rightarrow 0} \sum_i \text{Var}(\Delta W_{s_i}) = \lim_{\|P\| \rightarrow 0} \sum_i \Delta s_i = t, \end{aligned}$$

where $\Delta W_{s_i} = W_{s_{i+1}} - W_{s_i}$ and $\Delta s_i = s_{i+1} - s_i$. And,

$$\begin{aligned} \text{Var}\left(\sum_i (W_{s_{i+1}} - W_{s_i})^2\right) &= \text{Var}\left(\sum_i \Delta W_{s_i}^2\right) \\ &= \sum_i \text{Var}(\Delta W_{s_i}^2) \\ &= \sum_i \mathbb{E}[(\Delta W_{s_i}^2 - \mathbb{E}[\Delta W_{s_i}^2])^2] \\ &= \sum_i \mathbb{E}[(\Delta W_{s_i}^2 - \Delta s_i)^2] \\ &= \sum_i \mathbb{E}[\Delta W_{s_i}^4 - 2\mathbb{E}[\Delta W_{s_i}^2] \Delta s_i + \Delta s_i^2] \\ &= \sum_i 3\Delta s_i^2 - 2\Delta s_i^2 + \Delta s_i^2 = \sum_i 2\Delta s_i^2. \end{aligned}$$

Hence,

$$\begin{aligned} \text{Var}(\langle W, W \rangle_t) &= \text{Var}\left(\lim_{\|P\| \rightarrow 0} \sum_i (W_{s_{i+1}} - W_{s_i})^2\right) \\ &= \lim_{\|P\| \rightarrow 0} \text{Var}\left(\sum_i (W_{s_{i+1}} - W_{s_i})^2\right) \\ &= \lim_{\|P\| \rightarrow 0} \sum_i 2\Delta s_i^2 \leq \lim_{\|P\| \rightarrow 0} \|P\| \sum_i 2\Delta s_i \\ &= \lim_{\|P\| \rightarrow 0} 2\|P\|t = 0. \end{aligned}$$

In the above, $\mathbb{E}[\Delta W_{s_i}^4] = 3\Delta s_i^2$ is derived from the property of normal distribution that is proven in Sect. 12.3.8.

(2) We now prove that $dWdt = 0$:

$$\begin{aligned}\mathbb{E}[\langle W, s \rangle_t] &= \lim_{\|P\| \rightarrow 0} \sum_i \mathbb{E}[(W_{s_{i+1}} - W_{s_i})(s_{i+1} - s_i)] = 0, \\ \text{Var}(\langle W, s \rangle_t) &= \text{Var}\left(\lim_{\|P\| \rightarrow 0} \sum_i (W_{s_{i+1}} - W_{s_i})(s_{i+1} - s_i)\right) \\ &\leq \text{Var}\left(\lim_{\|P\| \rightarrow 0} \|P\| \sum_i (W_{s_{i+1}} - W_{s_i})\right) \\ &= \text{Var}\left(\lim_{\|P\| \rightarrow 0} \|P\| W_t\right) = \lim_{\|P\| \rightarrow 0} \|P\| t = 0.\end{aligned}$$

(3) Lastly, we prove $dt dt = 0$:

$$\lim_{\|P\| \rightarrow 0} \sum_i (s_{i+1} - s_i)^2 \leq \lim_{\|P\| \rightarrow 0} \|P\| \sum_i (s_{i+1} - s_i) = \lim_{\|P\| \rightarrow 0} \|P\| t = 0.$$

□

Remark 13.3 It is interesting to note that a Brownian motion is continuous everywhere but *not* of finite variation on any interval, i.e., $\sum_i |W_{s_{i+1}} - W_{s_i}| \rightarrow \infty$, which is different from continuous differentiable functions [94]. □

Correlated Brownian Motions

Two Brownian motions W_t and B_t are correlated by ρ if $\langle W, B \rangle_t = \rho t$. The parameters $\rho = 0$ and $\rho = 1$ give independence and perfect correlation respectively.

Now we show how to construct correlated Brownians. Consider

$$B_t = \rho W_t + \sqrt{1 - \rho^2} \widehat{W}_t, \quad -1 \leq \rho \leq 1,$$

where W_t and $\widehat{W}_t$ are *independent* standard Brownian motions. Lévy's Theorem shows that B_t is a Brownian motion: firstly B_t is a continuous martingale with $B_0 = 0$ and

$$d\langle B, B \rangle_t = dB_t dB_t = \rho^2 dW_t dW_t + (1 - \rho^2) d\widehat{W}_t d\widehat{W}_t = \rho^2 dt + (1 - \rho^2) dt = dt,$$

As $d\langle B, W \rangle_t = dB_t dW_t = \rho dt$, the two Brownian motions B_t and W_t are correlated with ρ . Note that B_t and $\widehat{W}_t$ are also correlated with $\sqrt{1 - \rho^2}$.

In this book, the correlation for Brownian-driven asset prices actually refers to the correlation between the Brownian motions.

13.7 Itô Integral and Itô Calculus

Definition 13.4 (Itô Integral) Let $P = \{t_0, t_1, \dots, t_n\}$ be a partition of interval $[0, T]$, f_t be a left-continuous $\mathcal{F}_t$ -adapted process (called *predictable process*) and $\mathbb{E} \left[\int_0^T f_t^2 dt \right] < +\infty$, the *Itô integral* is defined as

$$\int_0^T f_t dW_t = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} f_{t_i} (W_{t_{i+1}} - W_{t_i}).$$

Note that *the values of f_t are taken at the beginning of the sub intervals*. This is a crucial property for the application of Itô integral in finance. For instance, the integral can represent the cumulative trading P/L where f_t acts as a function of exposure size and is determined by a trading strategy before each time period. **We highlight that $\mathbb{E} \left[\int_0^T f_t^2 dt \right] < +\infty$ is assumed in the entire book**, which implies that the Itô integral is a martingale. As only Itô integrals are used in this book, we often use the simple term *stochastic integral* instead.

Properties of stochastic integral $I_t = \int_0^t f_s dW_s$ include

- (1) *Martingale*: I_t is an $\mathcal{F}_t$ -martingale
- (2) *Itô Isometry*: $\mathbb{E}[I_t^2] = \mathbb{E} \left[\int_0^t f_s^2 ds \right]$
- (3) *Quadratic Variation*: $\langle I, I \rangle_t = \int_0^t f_s^2 ds$.

Remark 13.4 (1) may be understood from the fact that, for $0 < t \leq T$,

$$\mathbb{E} \left[\int_0^T f_s dW_s \middle| \mathcal{F}_t \right] = \mathbb{E} \left[\int_0^t f_s dW_s \middle| \mathcal{F}_t \right] + \mathbb{E} \left[\int_t^T f_s dW_s \middle| \mathcal{F}_t \right].$$

The second term is actually zero because the expected value of any future Brownian increment is zero.

For (2), we notice that

$$\mathbb{E} \left[\left(\sum_{i=0}^{n-1} f_{t_i} (W_{t_{i+1}} - W_{t_i}) \right)^2 \right] = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \mathbb{E} [f_{t_i} f_{t_j} (W_{t_{i+1}} - W_{t_i})(W_{t_{j+1}} - W_{t_j})].$$

For each term of the form $i < j$:

$$\begin{aligned} \mathbb{E} [f_{t_i} f_{t_j} \Delta W_{t_i} \Delta W_{t_j}] &= \mathbb{E} [\mathbb{E} [f_{t_i} f_{t_j} \Delta W_{t_i} \Delta W_{t_j} \middle| \mathcal{F}_{t_j}]] \\ &= \mathbb{E} [f_{t_i} f_{t_j} \Delta W_{t_i} \mathbb{E} [\Delta W_{t_j} \middle| \mathcal{F}_{t_j}]] = 0 \text{ (as } \mathbb{E} [\Delta W_{t_j} \middle| \mathcal{F}_{t_j}] = 0). \end{aligned}$$

For terms with $i = j$:

$$\mathbb{E}[f_{t_i}^2 \Delta W_{t_i}^2] = \mathbb{E}[\mathbb{E}[f_{t_i}^2 \Delta W_{t_i}^2 | \mathcal{F}_{t_i}]] = \mathbb{E}[f_{t_i}^2 \mathbb{E}[\Delta W_{t_i}^2 | \mathcal{F}_{t_i}]] = \mathbb{E}[f_{t_i}^2 \Delta t_i].$$

Hence, $\mathbb{E}[(\int_0^T f_t dW_t)^2] = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} \mathbb{E}[f_{t_i}^2 \Delta W_{t_i}^2] = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} \mathbb{E}[f_{t_i}^2 \Delta t_i] = \mathbb{E}[\int_0^T f_t^2 dt]$.

Lastly, we have $\langle I, I \rangle_t = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} (I_{t_{i+1}} - I_{t_i})^2 = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} (f_{t_i} (W_{t_{i+1}} - W_{t_i}))^2$.

Then, following the proof for Theorem 13.4 for quadratic variation of Brownian motion, we obtain the result in property (3). $\square$

Definition 13.5 (One-dimensional Itô Process) A stochastic process X_t is called an Itô process if it follows:

$$X_t = X_0 + \int_0^t \theta_s ds + \int_0^t \phi_s dW_s,$$

where W_s is an $\mathcal{F}_t$ -Brownian motion and θ_s and ϕ_s are $\mathcal{F}_t$ -adapted processes.

In stochastic calculus, integration is a well-defined object of study. Differential of a stochastic process is understood as the incremental change corresponding to the infinitesimal change of t . It is often convenient to use the differential form of Itô process:

$$dX_t = \theta_t dt + \phi_t dW_t,$$

where θ_t is known as the *drift* of X_t . It is easily seen that X_t is a martingale if its drift is zero for all t . Given predictable processes f_t, g_t and Itô processes X_t, Y_t , we have the below calculation rule

$$d\langle f_t X_t, g_t Y_t \rangle = f_t g_t d\langle X_t, Y_t \rangle \quad (13.1)$$

The below theorem for functions of an Itô process is a key component of stochastic calculus.

Theorem 13.5 (Itô–Doebelin Formula) Let $f(t, x)$ be a function such that $\frac{\partial f}{\partial t}(t, x)$, $\frac{\partial f}{\partial x}(t, x)$ and $\frac{\partial^2 f}{\partial x^2}(t, x)$ are defined and continuous. Then, for $T > 0$,

$$\begin{cases} df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} d\langle X, X \rangle_t \\ df(t, X_t) = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} \theta_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \phi_t^2 \right) dt + \frac{\partial f}{\partial x} \phi_t dW_t. \end{cases}$$

Itô–Doebelin formula can be understood from the Taylor series of $f(t, X_t)$:

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} dX_t dX_t + [\text{other terms}].$$

Applying the calculation rules of Brownian Motion: $dW_t dW_t = dt$, $dW_t dt = 0$, $dt dt = 0$, the value of $[\text{other terms}]$ is zero. From $dX_t = \theta_t dt + \phi_t dW_t$, we obtain

$$dX_t dX_t = \phi_t^2 dW_t dW_t + \phi_t \theta_t dW_t dt + \theta_t \phi_t dt dW_t + \theta_t^2 dt dt = \phi_t^2 dt.$$

Hence, replacing dX_t and $dX_t dX_t$ in the Taylor series, we obtain

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} \theta_t dt + \frac{\partial f}{\partial x} \phi_t dW_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \phi_t^2 dt.$$

□

Readers may refer to [11] and [85] for a rigorous proof.

The integral forms are

$$\begin{cases} f(T, X_T) = f(0, X_0) + \int_0^T \frac{\partial f}{\partial t} dt + \int_0^T \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \int_0^T \frac{\partial^2 f}{\partial x^2} d\langle X, X \rangle_t \\ f(T, X_T) = f(0, X_0) + \int_0^T \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} \theta_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \phi_t^2 \right) dt + \int_0^T \frac{\partial f}{\partial x} \phi_t dW_t. \end{cases}$$

Similarly, we can obtain the below multi-dimensional Itô–Doebelin formula.

Theorem 13.6 (Multi-Dimensional Itô–Doebelin Formula) *Let $X_t^1, \dots, X_t^n$ be Itô processes and $f(t, x_1, \dots, x_n)$ be a function for which the required derivatives are defined and continuous. Then,*

$$df = \frac{\partial f}{\partial t} dt + \sum_{i=1}^n \frac{\partial f}{\partial x_i} dX_t^i + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j} d\langle X^i, X^j \rangle_t.$$

In particular, the two-dimensional Itô–Doebelin formula is

$$\begin{aligned} df(t, X_t, Y_t) &= \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{\partial f}{\partial y} dY_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} d\langle X, X \rangle_t \\ &\quad + \frac{\partial^2 f}{\partial x \partial y} d\langle X, Y \rangle_t + \frac{1}{2} \frac{\partial^2 f}{\partial y^2} d\langle Y, Y \rangle_t. \end{aligned}$$

Applying the two-dimensional Itô–Doebelin formula to $f(t, X_t, Y_t) = X_t Y_t$ we obtain the below very useful result.

Corollary 13.1 (Itô's Product Rule) *Let X_t and Y_t be Itô processes, then*

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d\langle X, Y \rangle_t.$$

Notice that $\langle X, Y \rangle_t$ is an additional term compared to Riemann calculus. The integral form of the Itô's product rule can also be regarded as the integration by parts formula [85].

Example 13.4 We calculate $d(e^{-\int_0^t r_s ds} X_t)$ with Itô's product rule, where r_t can be deterministic or stochastic and X_t is an Itô process. Then, $d\langle e^{-\int_0^t r_s ds}, X_t \rangle = de^{-\int_0^t r_s ds} dX_t = -r_t e^{-\int_0^t r_s ds} dt dX_t = 0$. Hence, we have

$$\begin{aligned} d(e^{-\int_0^t r_s ds} X_t) &= X_t d(e^{-\int_0^t r_s ds}) + e^{-\int_0^t r_s ds} dX_t + d\langle e^{-\int_0^t r_s ds}, X_t \rangle \\ &= -r_t e^{-\int_0^t r_s ds} X_t dt + e^{-\int_0^t r_s ds} dX_t \\ &= e^{-\int_0^t r_s ds} (-r_t X_t dt + dX_t). \end{aligned} \quad (13.2)$$

In particular, when r_t is a degenerate process $r_t = r$, we have

$$d(e^{-rt} X_t) = e^{-rt} (dX_t - r X_t dt). \quad (13.3)$$

□

The following theorem is useful to identify Brownian motions.

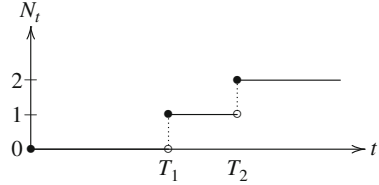
Theorem 13.7 (Lévy's Theorem) *Let M_t be a continuous martingale to its natural filtration $\mathcal{F}_t$, $M_0 = 0$, and $\langle M, M \rangle_t = t$. Then M_t is a Brownian motion.*

We can show that the moment-generating function of M_t is $\mathbb{E}[e^{uM_t}] = e^{\frac{1}{2}u^2 t}$ (i.e., $\mathbb{E}[e^{uM_t - \frac{1}{2}u^2 t}] = 1$), which is the one for normal distribution $\mathcal{N}(0, t)$. Applying Taylor expansion to $f(t, M_t) = e^{uM_t - \frac{1}{2}u^2 t}$ and using $d\langle M_t, M_t \rangle = dt$ and $d\langle M_t, t \rangle = 0$ (as its expected value and variance are both 0), we have $f = f_0 + \int_0^t (\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial M_t^2}) dt + \int_0^t \frac{\partial f}{\partial M_t} dM_t = 1 + \int_0^t \frac{\partial f}{\partial M_t} dM_t$. Taking expectation on both sides of the equation, we obtain $\mathbb{E}[e^{uM_t - \frac{1}{2}u^2 t}] = 1$. See [64] for a rigorous proof.

13.8 Poisson Process*

Let $\tau_k, k = 1, 2, \dots$ be *i.i.d.* random variables following an exponential distribution. Then, τ_k can be used to define an increasing sequence of random times by $T_n = \sum_{k=1}^n \tau_k$, which is known as the arrival time of the n -th event. The arrival times are known to follow the gamma density, see [105].

Fig. 13.3 A sample path for Poisson process N_t



A *Poisson process* N_t is a right continuous process that counts the number of events up to and including t (see Fig. 13.3). It is defined as

$$N_t = \sum_{n \geq 1} \mathbf{1}_{\{t \geq T_n\}}.$$

For a right continuous process, there exists a left continuous version and the value prior to the jump at time t is denoted by “ $t-$ ”. For instance, the first jump of Poisson process happens at $t = T_1$, and hence $N_{t-} = 0$, $N_t = 1$ with $\Delta N_t \equiv N_t - N_{t-} = 1$. Here are some properties of Poisson processes for integer $k \geq 0$ and $0 \leq s < t$ that can be proven with straightforward calculation where $e^x = \sum_{j=0}^{\infty} \frac{x^j}{j!}$ is sometimes used (see [105]):

- (1) Distribution: $\mathbb{P}\{N_t = k\} = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$
- (2) Mean: $\mathbb{E}[N_t - N_s] = \lambda(t - s)$
- (3) Variance: $\text{Var}(N_t - N_s) = \lambda(t - s)$
- (4) Martingale Property: $N_t - \lambda t = N_t - \mathbb{E}[N_t]$ is a martingale
- (5) Independent and stationary increments:

$$\mathbb{P}\{N_t - N_s = k\} = \frac{\lambda^k (t - s)^k}{k!} e^{-\lambda(t-s)} = \mathbb{P}\{N_{t-s} = k\}$$

- (6) Moment generating function: $\mathbb{E}[e^{uN_t}] = e^{\lambda t(e^u - 1)}$.

The parameter λ is the *intensity* in the exponential distribution for arrival time. The independence feature for increments of Poisson processes can be understood from the memorylessness of exponential distribution. The other properties can be obtained by direct calculation.

Note that

$$\mathbb{P}(\tau \leq t + \Delta t | \tau > t) = \frac{\mathbb{P}(t < \tau \leq t + \Delta t)}{\mathbb{P}(\tau > t)} = \frac{e^{-\lambda t} - e^{-\lambda(t+\Delta t)}}{e^{-\lambda t}} = 1 - e^{-\lambda \Delta t}.$$

Hence, $\lim_{\Delta t \rightarrow 0} \mathbb{P}(\tau \leq t + \Delta t | \tau > t) = \lambda dt$ using the Taylor expansion of $e^{-\lambda \Delta t}$. In financial modelling, λdt is often used as the probability of an event in $(t, t + dt)$, conditioned by no such event happened before t . It can be proven that there can be

at most one event arrival within an infinitesimally small interval $(t, t + dt)$ [94]. So, in differential notation, a Poisson process is often written as

$$dN_t = \begin{cases} 0, & \text{with probability } 1 - \lambda dt, \\ 1, & \text{with probability } \lambda dt. \end{cases}$$

Exercises

1. Let Ω be an infinite coin toss space

$$\Omega = \{\omega = \omega_1\omega_2\cdots; \omega_i = y \text{ or } n, i = 1, 2, \dots\}.$$

where y stands for a head and n stands for a tail with equal probability. Define the random variables

$$Y_i(\omega) = \begin{cases} 1, & \text{if } \omega_i = y, \\ -1, & \text{if } \omega_i = n. \end{cases}$$

Then, the random walk is defined as a discrete stochastic process $X_k = \sum_{i=1}^k Y_i$ with initial value $X_0 = 0$. Find $\sigma(X_2)$, the σ -algebra generated by the random walk at time 2.

2. Consider an Itô process $dX_t = W_t^2 dt + W_t^2 dW_t$, or

$$X_t = X_0 + \int_0^t W_s^2 ds + \int_0^t W_s^2 dW_s.$$

Derive $\mathbb{E}[X_t]$.

3. Let σ_t^1 and σ_t^2 be non-zero adapted processes, and we have two correlated Brownian motions given by $\langle W, \widehat{W} \rangle_t = \rho t$. Show that

$$B_t = \int_0^t \frac{\sigma_s^1 dW_s + \sigma_s^2 d\widehat{W}_s}{\sqrt{(\sigma_s^1)^2 + (\sigma_s^2)^2 + 2\rho\sigma_s^1\sigma_s^2}}$$

is a Brownian motion.

4. Suppose W_t is a one dimensional standard Brownian motion. Show that $\lim_{t \rightarrow \infty} \frac{W_t}{t} = 0$ (Hint: Let $X_i = W_i - W_{i-1}$ for $i = 1, 2, \dots$, write $W_n/n = (X_1 + \dots + X_n)/n$ and apply the Law of Large Numbers).

Chapter 14

Black–Scholes–Merton Model for Option Pricing



In this chapter, we start off the discussion of option pricing or derivatives modelling¹ with the pioneering work by Black, Scholes and Merton who proposed the first hedging (replication) framework in 1973. Their work laid the foundation for the rapid growth of derivative products. In recognition of their contributions, Scholes and Merton received the 1997 Nobel Prize in Economics.²

The option pricing and hedging framework of the *Black–Scholes–Merton Model* was originally derived with a PDE (Partial Differential Equation) approach. In later years, researchers have generalized the model and developed a rigorous and practical modelling framework with a probabilistic approach, which will be presented in Chap. 16 that readers may choose to read directly by skipping this chapter. After acquiring the basic knowledge of option modelling, it is helpful to revisit Sects. 14.4 and 14.5 for the discussions on option hedging.

14.1 The Black–Scholes–Merton Model

As indicated in Chap. 1, we often use the term *underlier* instead of underlying asset, security or underlying instrument for simplification. We also use *derivative* instead of derivative contract or product.

A derivative (e.g., option) price is sensitive to the change of the underlier's price. Take the example of a call option on a stock. If the stock price goes up, the price of the call option will increase, which implies a gain to option holder and a loss to

¹ Actually, the term *derivatives modelling* is more accurate than *option pricing* as the analysis and results are not limited to options but applicable to derivatives hedging and pricing in general. As *option pricing* had been widely used, we do not want to disturb readers by replacing it with *derivatives modelling*.

² Unfortunately, Black passed away in 1995.

option writer. But if the writer of the call has held a certain quantity of the stock, the increase in stock price will result in a gain that may mitigate or offset the loss on the short option position.

This intuitive analysis gives rise to the question of whether this strategy can perfectly hedge the option and what quantity of the stock the option writer needs to hold at a given point in time. The answer was given by Black, Scholes and Merton in the seminal work on option pricing published in 1973 [18, 78]. The authors demonstrated that investors can create a *riskless* (in terms of market risk) hedging portfolio where all market risks are eliminated under the model now known as the *BSM model* (*Black–Scholes–Merton Model*).

Under the Black–Scholes–Merton model, the price of a non-dividend paying underlier (i.e., total return asset) follows the so-called *geometric Brownian motion*:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (\sigma > 0), \quad (14.1)$$

where μ and σ are constants, representing the underlier's drift and volatility respectively.

Proposition 14.1 *Let S_t be the process defined in SDE (14.1). Then,*

$$S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma W_t} \text{ and } S_T = S_t e^{(\mu - \frac{\sigma^2}{2})(T-t) + \sigma(W_T - W_t)}. \quad (14.2)$$

Proof Define $f(t, x) = \ln x$. Then

$$\frac{\partial f}{\partial x} = \frac{1}{x}, \quad \frac{\partial^2 f}{\partial x^2} = -\frac{1}{x^2}, \quad \frac{\partial f}{\partial t} = 0.$$

By Itô–Doebelin formula, we obtain

$$\begin{aligned} d \ln(S_t) &= df(t, S_t) = 0 \cdot dt + \frac{\partial f}{\partial x}(t, S_t) dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, S_t) (dS_t)^2 \\ &= \frac{1}{S_t} dS_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) (dS_t)^2 = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t, \end{aligned}$$

where SDE (14.1) and $(dS_t)^2 = \sigma^2 S_t^2 dt$ are used. Integrating from time 0 to t , we obtain the below solution:

$$S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma W_t}.$$

Then, then second equation follows:

$$S_T = S_0 e^{(\mu - \frac{\sigma^2}{2})T + \sigma W_T} = S_t e^{(\mu - \frac{\sigma^2}{2})(T-t) + \sigma(W_T - W_t)}.$$

□

Corollary 14.1 *The logarithm of S_t is an arithmetic Brownian motion*

$$\ln S_t = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t \sim \mathcal{N}\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right).$$

Thus, the distribution of S_t is said to be log-normal. Furthermore,

$$\mathbb{E}[S_t] = S_0 e^{\mu t} \quad \text{and} \quad \text{Var}(S_t) = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1).$$

Proof The distribution of $\ln S_t$ follows from taking logarithm on both sides of Eq. (14.2). Note that $W_t \sim \mathcal{N}(0, t)$, then $W_t = \sqrt{t}\epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$. By the moment generating function for normal variables (see Sect. 12.3.8), we have

$$\mathbb{E}[S_t] = \mathbb{E}\left[S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma W_t}\right] = S_0 e^{(\mu - \frac{\sigma^2}{2})t} \mathbb{E}[e^{\sigma \sqrt{t}\epsilon}] = S_0 e^{(\mu - \frac{\sigma^2}{2})t} e^{\frac{\sigma^2}{2}t} = S_0 e^{\mu t}.$$

Similarly, we have

$$\mathbb{E}[S_t^2] = \mathbb{E}\left[S_0^2 e^{(2\mu - \sigma^2)t + 2\sigma W_t}\right] = S_0^2 e^{(2\mu - \sigma^2)t} \mathbb{E}[e^{2\sigma \sqrt{t}\epsilon}] = S_0^2 e^{(2\mu + \sigma^2)t}.$$

$$\text{Hence } \text{Var}(S_t) = \mathbb{E}[S_t^2] - \mathbb{E}[S_t]^2 = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1). \quad \square$$

14.2 Derivation of the Black–Scholes Equation

The BSM model assumes a “perfect market” in which

- there is no arbitrage opportunity;
- underlier can be bought or sold at any time for any quantity (including a fraction of the underlier);
- there is no short selling restriction and any security can be borrowed in the market;
- there is no transaction cost;
- there exists a risk-free interest rate r for borrowing and lending.³

Without loss of generality, we present the option modelling for a European option V linked to a single underlier whose total return price process is S_t . However, the results can be generalized to V being a portfolio of European options linked to a common underlier.

³ r can be generalized to be time-dependent (denoted as r_t).

Denote V_t as the option price process. By Itô–Doebelin formula, we obtain V_t as an Itô process

$$dV_t = \left(\mu S_t \frac{\partial V_t}{\partial S_t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S^2} + \frac{\partial V_t}{\partial t} \right) dt + \sigma_t S_t \frac{\partial V_t}{\partial S_t} dW_t. \quad (14.3)$$

As the below analysis remains valid beyond the assumption of constant volatility, we denote the volatility parameter by σ_t .

Now, we construct a portfolio Π_t consisting of a short position of the option and a position of Δ_t units⁴ of the underlier at time- t :

$$\Pi_t = -V_t + \Delta_t S_t. \quad (14.4)$$

The key idea is to find the value of Δ_t so that the portfolio becomes instantaneously deterministic, i.e., without the term dW_t representing the randomness of the underlier price. The change in the value of Π_t in an infinitesimal time step dt is

$$d\Pi_t = -dV_t + \Delta_t dS_t. \quad (14.5)$$

Applying Eqs. (14.1) and (14.3) to Eq. (14.5), we get

$$\begin{aligned} d\Pi_t = & - \left(\mu S_t \frac{\partial V}{\partial S_t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \frac{\partial V}{\partial t} - \mu \Delta_t S_t \right) dt \\ & + \sigma_t S_t \left(-\frac{\partial V}{\partial S_t} + \Delta_t \right) dW_t. \end{aligned} \quad (14.6)$$

The first term on the right-hand side is “deterministic” while the second term is “stochastic” and “risky” as it involves W_t . By setting

$$\Delta_t = \frac{\partial V_t}{\partial S_t}, \forall t, \quad (14.7)$$

the risk from the randomness of underlier price change is eliminated and (14.6) becomes

$$d\Pi_t = - \left(\frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2} + \frac{\partial V_t}{\partial t} \right) dt. \quad (14.8)$$

⁴ Here $\Delta_t < 0$ implies a short position of the underlier. For instance, if the option is a put, we may need to short sell a certain amount of underlier to offset the risk. In that case, $\Delta_t < 0$.

Under the no-arbitrage assumption, we should have the below equation satisfied if there is no risk-taking

$$d\Pi_t = r\Pi_t dt, \text{ or equivalently, } -dV_t + rV_t dt + \Delta_t dS_t - \Delta_t r dS_t dt = 0. \quad (14.9)$$

In fact, if $d\Pi_t$ were greater than $r\Pi_t dt$, an arbitrageur could make a riskless profit by borrowing an amount Π_t at rate r to invest in the portfolio. Conversely, if $d\Pi_t$ were less than $r\Pi_t dt$, then the arbitrageur would short the portfolio and invest Π_t at the risk-free rate r .

Combining Eqs. (14.8), (14.9) and (14.4), we get

$$\left(\frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2} + \frac{\partial V_t}{\partial t} \right) dt = r(V_t - \Delta_t S_t) dt \quad \left(\text{with } \Delta_t = \frac{\partial V_t}{\partial S_t} \right), \quad (14.10)$$

which leads to the famous **Black–Scholes equation**:

$$\frac{\partial V_t}{\partial t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2} = rV_t - rS_t \frac{\partial V_t}{\partial S_t}. \quad (14.11)$$

The price of any European derivative instrument whose payoff depends only on the underlier prices up to maturity date must satisfy the Black–Scholes equation under the no-arbitrage assumption. For a given derivative, the terminal and boundary conditions are specified so that (14.11) has a unique solution to V_t .

Note that the Black–Scholes equation (14.11) does not contain the drift parameter μ of the underlier. The price will depend on the risk-free interest rate r and the volatility σ_t however. A consequence of this is that two people may have quite different views on μ , yet still agree on the value of an option.

Delta-hedging process can be conceptually considered as a succession of the below operations for each infinitesimal period dt until the termination of the option:

- at the beginning of dt : short the call option at price of V_t and invest the proceeds at rate r ; long Δ_t units of the underlier funded at rate r ;
- at the end of dt : close the whole position to cash;⁵ The P/L Z_t is $-dV_t$ (by option price change) $+ rV_t dt$ (carry) $+ \Delta_t dS_t$ (by underlier price change) $- \Delta_t r dS_t dt$ (funding cost for underlier units). By the second equation in (14.9), Z_t should be zero.

In practical applications of the BSM model, the parameter σ_t is the “best” estimation by market consensus at time- t of the “constant” volatility to be realized from the valuation date until the maturity of the derivative (aka *implied volatility* as its value is reverse-calculated from the market price of vanilla options). Obviously,

⁵ In practical implementation, only the net position of cash and underlier units will be executed. But it is clearer to make the analysis as stated.

this market's view on σ_t changes over time. For avoiding arbitrage, σ_t should be (re)set dynamically at the implied volatility level so that the Black–Scholes equation (14.11) holds at any time.

To illustrate the PDE method for option pricing, let us focus on European call option $c(t, S)$, with exercise price K and expiry date T . The PDE method requires terminal and boundary conditions. For a European option, the terminal condition is its payoff:

$$c(T, S_T) = \max\{S_T - K, 0\}. \quad (14.12)$$

The ‘spatial’ boundary conditions are applied at $S_t = 0$ and at $S_t \rightarrow \infty$. We can see from (14.1) that if S_t ever touches zero, it will stay at zero forever, leading to 0 payoff for the call option. Hence,

$$c(t, 0) = 0 \quad \text{for all } t \geq 0. \quad (14.13)$$

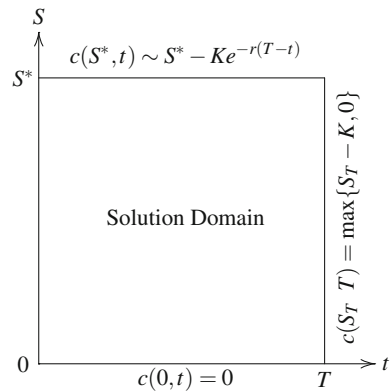
If $S_t \rightarrow \infty$, it becomes certain that the option will be exercised and the value of the option will be the underlier price minus the discounted exercise price. Hence, we have, for all $t > 0$,

$$c(t, S_t) = S_t - Ke^{-r(T-t)}, \quad \text{as } S_t \rightarrow \infty. \quad (14.14)$$

The terminal and boundary conditions (14.12)–(14.14) allow the Black–Scholes equation (14.11) to be solved for the option price V_t (i.e., $c(t, S_t)$ for the European call option case). Figure 14.1 illustrates the solution domain and the boundary conditions.

Similarly, for a vanilla European put option price $p(t, S_t)$, the terminal condition is $p(T, S_T) = \max\{K - S_T, 0\}$ while the boundary conditions are, for all $t \geq 0$, $p(t, 0) = Ke^{-rt}$ and $p(t, S_t) \rightarrow 0$ as $S_t \rightarrow \infty$.

Fig. 14.1 Solution domain of a European call option and the boundary conditions. Here S^* is a sufficiently large value



We note that, by Feynman-Kac theorem, the PDE problem from the BSM model can be resolved by computing the related expectation. This is consistent with the risk-neutral analysis in Chap. 16.

14.3 Black–Scholes Formulas for Vanilla Options

By solving the PDE, closed-form formulas are obtained for European call and put options [18]. In this section, we only verify that the formulas are solutions of the Black–Scholes equation.

Proposition 14.2 (Black–Scholes Formula) *The time- t value of a vanilla European call with strike price K and maturity T is given by*

$$c(t, S_t) = S_t N(d_1) - K e^{-r(T-t)} N(d_2), \quad (14.15)$$

where $N(x)$ is the cumulative distribution function for standard normal random variables,

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T-t}. \quad (14.16)$$

Proof Firstly, $c(t, S_t)$ in (14.15) satisfies the Black–Scholes equation (14.11). This can be checked by direct calculation of partial derivative terms $\frac{\partial c}{\partial t}$, $\frac{\partial c}{\partial S}$, $\frac{\partial^2 c}{\partial S^2}$ in Eq. (14.11). We note the below for calculating the partial derivatives

$$\frac{\partial N(d_i)}{\partial x} = \frac{\partial N(d_i)}{\partial d_i} \frac{\partial d_i}{\partial x} = \frac{e^{-\frac{d_i^2}{2}}}{\sqrt{2\pi}} \frac{\partial d_i}{\partial x}, \quad i = 1, 2.$$

For the boundary condition (14.13), we first note that by (14.16), $d_1, d_2 \rightarrow -\infty$ as $S_t \rightarrow 0$. Obviously, $N(-\infty) = 0$. Hence

$$c(t, 0) = 0 \times N(-\infty) - K e^{-r(T-t)} N(-\infty) = 0.$$

For the boundary condition (14.14), we note again that $d_1, d_2 \rightarrow \infty$ as $S_t \rightarrow \infty$ where $N(\infty) = 1$. Hence

$$c(t, S_t) \rightarrow S_t N(\infty) - K e^{-r(T-t)} N(\infty) \sim S_t - K e^{-r(T-t)}, \quad \text{as } S_t \rightarrow \infty.$$

Finally, we consider the final condition (14.12). At $t = T$, if $S_T > K$, then $d_1, d_2 \rightarrow \infty$. Hence $c(T, S_T) = S_T - K$. If $S_T < K$, then $d_1, d_2 \rightarrow -\infty$. Hence $c(T, S_T) = 0$. If $S_T = K$, by continuity, $c(T, S_T) = 0$. $\square$

Next we give the formula for European put options.

Proposition 14.3 *The value of a vanilla European put is given by*

$$p(t, S_t) = K e^{-r(T-t)} N(-d_2) - S_t N(-d_1), \quad (14.17)$$

where d_1 and d_2 are given in (14.16).

Proof We can derive the put option price formula (14.17) immediately by using the put-call parity formula (see Proposition 10.1) $c(t, S_t) - p(t, S_t) = S_t - K e^{-r(T-t)}$, Proposition 14.2 and the equality $N(d) + N(-d) \equiv 1$. $\square$

By straightforward derivation, we can obtain the deltas of vanilla calls and puts:

$$\Delta_t^c = N(d_1) \quad \text{and} \quad \Delta_t^p = N(d_1) - 1.$$

We remark that although (14.15) and (14.17) seem to be closed-form solutions for vanilla options, one still has to compute the integral $N(d_i)$ numerically by quadrature rules such as Simpson's rule or Gaussian rule [9].

A Useful Approximation Formula

For at-the-money forward (ATMF) call/put options, we have the below approximation formula for estimating option price without using a pricer:

$$c = p \approx 0.4 S_0 \sigma \sqrt{T}.$$

Proof The strike of an ATMF option is $K = e^{rT} S_0$, where S_0 is the spot level at time-0. In this case, the European call and put options have the same price according to put-call parity. Let us check the approximation formula for an ATMF call option. Firstly,

$$S_0 = e^{-rT} K, \quad d_1 = \frac{\sigma \sqrt{T}}{2} \quad \text{and} \quad d_2 = -\frac{\sigma \sqrt{T}}{2}.$$

From Taylor expansion, we have

$$N(x) = N(0) + N'(0)x + \frac{1}{2}N''(0)x^2 + O(x^3).$$

Using $N'(0) = \frac{1}{\sqrt{2\pi}} \approx 0.4$, we obtain

$$\begin{aligned} c &= S_0 N(d_1) - e^{-rT} K N(d_2) = S_0 \left[N\left(\frac{\sigma \sqrt{T}}{2}\right) - N\left(-\frac{\sigma \sqrt{T}}{2}\right) \right] \\ &\approx S_0 N'(0) \sigma \sqrt{T} \approx 0.4 S_0 \sigma \sqrt{T}. \end{aligned}$$

$\square$

14.4 Derivatives Price Sensitivities (Greeks)

Consider a portfolio Π composed of options linked to a single underlier S a quantity of the underlier itself,⁶ and units of money market account (i.e. cash). As a particular case, the portfolio can be composed of an option alone.

In the Black–Scholes–Merton model, the price of portfolio Π_t depends on the underlier price, time-to-maturity of the options, volatility, and risk-free interest rate: $\Pi_t = \Pi(T - t, S_t, \sigma_t, r)$. The sensitivities with respect to these parameters are denoted by Greek letters by financial market practitioners:⁷

- Delta $\Delta_t = \frac{\partial \Pi_t}{\partial S_t}$
- Theta $\Theta_t = \frac{\partial \Pi_t}{\partial t}$
- Rho $\rho_t = \frac{\partial \Pi_t}{\partial r}$
- Gamma $\Gamma_t = \frac{\partial^2 \Pi_t}{\partial S_t^2}$
- Vega $\mathcal{V}_t = \frac{\partial \Pi_t}{\partial \sigma_t}$

(Figs. 14.2 and 14.3 illustrate the delta, gamma, and vega for a call option.) The derivative values can be computed by finite difference methods (see 23.1.1).

From Itô–Doebelin formula, we observe that the price change of Π_t can be expressed by its Greeks:

$$d\Pi_t = \Theta_t dt + \Delta_t dS_t + \frac{1}{2} S_t^2 \sigma_t^2 \Gamma_t dt. \quad (14.18)$$

From the Black–Scholes equation $\frac{\partial \Pi_t}{\partial t} + r S_t \frac{\partial \Pi_t}{\partial S_t} + \frac{1}{2} \frac{\partial^2 \Pi_t}{\partial S_t^2} \sigma_t^2 S_t^2 = r \Pi_t$,

$$\Theta_t + r S_t \Delta_t + \frac{1}{2} \Gamma_t \sigma_t^2 S_t^2 = r \Pi_t.$$

In case that Π_t is delta-neutral (i.e., $\Delta_t = 0$, which is the case for a portfolio of delta-hedged options), the Black–Scholes equation becomes

$$\frac{1}{2} \Gamma_t \sigma_t^2 S_t^2 = -\Theta_t + r \Pi_t. \quad (14.19)$$

This is known as **Gamma-Theta relationship** under the continuous delta-hedging assumption. It shows that gamma and theta should have opposite signs if their values are significant.

As each Greek represents a certain risk exposure for the portfolio, the derivative product providers (i.e., hedging party) always apply limits in dollar amount on the major sensitivity indicators such as (residual) delta and vega.

⁶ For clarity and simplification, we assume a single underlier for the analysis. But the analysis is also valid for a portfolio with options linked to single or multiple underliers.

⁷ Actually, vega $\mathcal{V}_t$ is not a Greek letter.

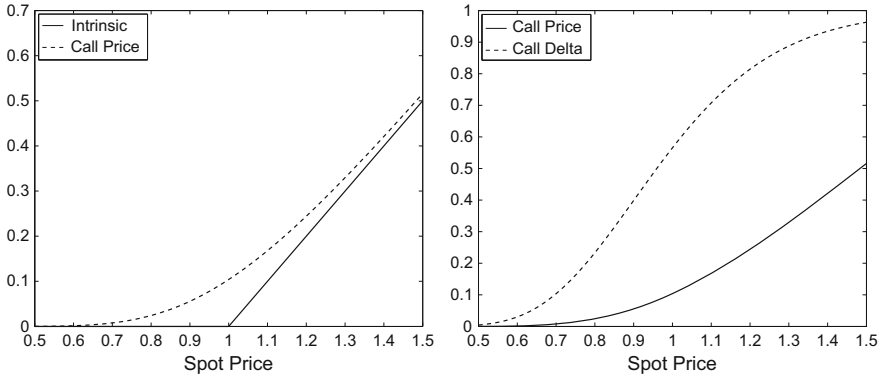


Fig. 14.2 Price and delta of a call option with $K = 1$, $r = 0.01$, $\sigma = 0.25$ and $T = 1$

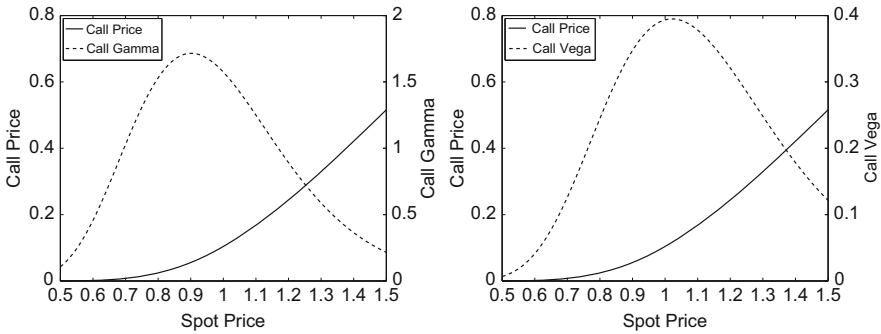


Fig. 14.3 Gamma and vega of a call option with $K = 1$, $r = 0.01$, $\sigma = 0.25$ and $T = 1$

Delta: Δ_t

The delta of a portfolio or an option measures the value change w.r.t. the price change of the underlier. Delta-hedging consists of holding (long or short depending on the sign of delta) Δ_t units of the underlier (called delta amount or, simply, delta) at any time t . As Δ_t is not a constant for option derivatives, the delta position needs to be adjusted frequently (continuously in theory, but usually once at the end of the day in practice). Any adjustment should be carried out with the self-financing principal, i.e., any underlier purchase is financed by a loan and any cash is lent out. The long position of underlier is lent to the market. The negative delta position is implemented by borrowing and selling units of the underlier.

As the underlier price changes, a P/L will be produced by the delta. According to the BSM model, the cumulative P/L from the continuous delta-hedging for a single option combined with the option premium will be exactly the option price for any time t . In particular, when t is the expiry time, the option price is the payoff amount.

Let's take the example of the hedging for a call sold by an option writer who borrowed $\Delta_0 S_0$ for the delta at inception. The initial portfolio is composed of the

call option (short), option premium (long), Δ_0 units of the underlier and a loan of $\Delta_0 S_0$. As the option premium offsets the value of the short option, we have $\Pi_0 = 0$. During the life of the trade, Δ_t units of the underlier should be maintained at any time t . The cash in the portfolio is lent out, and the underlier position is also lent to the market at repo rate (i.e., security lending rate).

Gamma: Γ_t

Gamma measures the change of delta w.r.t. the price change of underlier. If gamma is zero, the delta amount is constant throughout the entire considered tenor, i.e., static hedging. A non-zero gamma portfolio leads to dynamic delta hedging instead of static hedging as the delta changes over time. Gamma represents the convexity part that is not captured by delta-hedging when the derivative price change is not linear to underlier's price change. For a continuously delta-hedged portfolio, the gamma impact is offset by theta and the portfolio's carry, as indicated by (14.19). However, in practice, only discrete hedging is possible and the natural offsetting of gamma impact by theta does not hold anymore. Further discussion on gamma will be provided in the discrete hedging section.

Theta: Θ_t

Theta measures the value change of a portfolio or an option with time passing. It is also called *time decay*, which reflects the loss in value due to having less time for exercising the option. Taking the example of a long call option on a stock with zero dividend and repo, the option price curve will be approaching the intrinsic value line progressively from the above over time. For a delta-neutral portfolio, theta is compensated by the gamma term through Gamma-Theta relationship (14.19).

Vega: $\mathcal{V}_t$

In the BSM model, the volatility is a constant, which is a simplistic assumption. In practice, it is a parameter that is difficult to estimate, especially it is for the future period until the option's expiry.

Vega measures the impact of volatility to option price. A vega position can be hedged with other options having the opposite amount of vega. For the motivation of vega hedging, we can imagine a pure theoretical extreme case that the underlier price has never moved after the option's inception (i.e., realized volatility is zero), which implies that the time value is not compensated by gamma. This risk can be mitigated by vega hedging. In practice, the frequency of vega hedging is far less than delta hedging.

Under the BSM model, there is a **Vega-Gamma relationship** for European options: $\mathcal{V}_t = \sigma(T-t)S_t^2\Gamma_t$ (the proof will be provided in Chap. 16), which implies that if the gamma of a portfolio of European options is zero, its vega should also be zero, and vice versa.

Rho: ρ_t

The parameter rho measures the sensitivity of the portfolio's value w.r.t. the change of interest rate level. In practice, the risk-free rate is replaced by the borrow/lending

rate available for hedging transactions, such as the internal funding rate from the same institution.

14.5 Practical Issues in Hedging*

14.5.1 Hedging Instruments and Hedging Ratios

Let Π_t be a portfolio comprising one or more derivatives, which is sensitive to a variety of factors $x_t^1, \dots, x_t^n$ that can be securities' prices and their volatilities, interest rates, foreign exchange rates, etc.

The assumptions of BSM model are only partially satisfied in financial market and the delta-hedging alone is not perfect. As hedging is performed in discrete time, Taylor expansion should be used for the analysis, while Itô–Doebelin formula is applicable for continuous time analysis. If Π_t is at least twice differentiable with respect to the related factors, the portfolio's value change over time interval Δt can be represented by the Taylor expansion of Π_t :

$$\Delta \Pi_t = \frac{\partial \Pi_t}{\partial t} \Delta t + \sum_{i=1}^n \frac{\partial \Pi_t}{\partial x_t^i} \Delta x_t^i + \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 \Pi_t}{\partial x_t^i \partial x_t^j} \Delta x_t^i \Delta x_t^j + \varepsilon_t \quad (14.20)$$

where $\Delta \Pi_t = \Pi_{t+\Delta t} - \Pi_t$ and $\Delta x_t^i = x_{t+\Delta t}^i - x_t^i$ and ε_t represents the total value of the higher order terms.

Dynamic hedging consists of offsetting the impact of the value change of the factors to Π_t by dynamically including a self-financing position of liquid instruments that are also sensitive to these factors. In practice, not all the terms in (14.20) can be perfectly hedged with simple and liquid instruments (e.g., cross derivative terms, higher order terms).

In the below, we provide some examples w.r.t. a single factor x_t with a liquid instrument $f(x_t, \dots)$ which provides efficient sensitivity to x_t . The hedging quantity of $f(x_t, \dots)$ denoted by Q_t satisfies the following general relation for hedging the first or second order sensitivities:

$$\frac{\partial^k}{\partial x_t^k} \left(\Pi_t + Q_t (f_t - c) \right) = 0, \text{ or equivalently, } Q_t = -\frac{\partial^k \Pi_t}{\partial x_t^k} / \frac{\partial^k f_t}{\partial x_t^k}. \quad (14.21)$$

where $k = 1, 2$ and c represents the cost for including f_t in the portfolio at time- t , i.e. $c = f_t$, which implies that the portfolio's value remain unchanged at the time of hedging.

- for $k = 1$ and x_t being the underlier's price S_t , we can use $f(S_t) = S_t$ to obtain the **delta-hedging** amount $Q_t = -\frac{\partial \Pi_t}{\partial S_t}$;

- for $k = 1$ and x_t being the underlier's implied volatility σ_t , $f(S_t, \sigma_t)$ can be a delta-hedged option, which leads to the **vega-hedging** with $Q_t = -\frac{\partial \Pi_t}{\partial \sigma_t} / \frac{\partial f_t}{\partial \sigma_t}$;
- for $k = 2$ and x_t being the underlier's price S_t , $f(S_t, \sigma_t)$ can be a delta-hedged option, which leads to the **gamma-hedging** with $Q_t = -\frac{\partial^2 \Pi_t}{\partial S_t^2} / \frac{\partial^2 f_t}{\partial S_t^2}$.

As $\frac{\partial \Pi_t}{\partial x_t} \Delta x_t = \frac{\partial \Pi_t}{\partial x_t} \frac{\Delta x_t}{\Delta y_t} \Delta y_t$, y_t can be used instead of x_t for hedging when it is more convenient, e.g., index futures for delta-hedging an ETF option.

14.5.2 Discrete Hedging

If delta-hedging is continuous, the impact of gamma and theta are offset according to Gamma-Theta relationship (14.19). In practice, delta-hedging can only be carried out at discrete times (one or a few times a day at most) and the P/L impact from gamma and theta may not be offset according to (14.19) between two delta-hedging times, despite that the implied volatility is adjusted to make the price and theta “fair” for the projected market condition: if the underlier price has almost not changed, theta impact should be more significant than gamma impact that is near zero, while if the underlier price makes large moves, the gamma impact will be more significant than theta impact. A negative gamma position will make the portfolio under-hedged while a positive gamma position will make the portfolio over-hedged.

Example 14.1 Consider a delta-hedged long call position with ATM/F strike. Thus, we have $\Gamma > 0$. At inception, the hedging is done by shorting Δ_0 shares at spot price S_0 , where $\Delta_0 \approx 0.5$. Then, the next day,

Scenario	Hedging	Result
S_t drops and $\Delta_t \rightarrow 0.45$	Need to short less, buy back at cheaper price	Buy low
S_t rises and $\Delta_t \rightarrow 0.55$	Need to short more, sell at higher price	Sell high

□

Actually, in discrete re-hedging, the $\Gamma > 0$ position enjoys a “buy low / sell high” benefit while $\Gamma < 0$ position leads to a “buy high / sell low” situation. But the re-hedging P/L is counterbalanced by the theta impact ($\Theta < 0$ for positive gamma position and $\Theta > 0$ for negative gamma position).

The decision on the amount of theta and gamma for the portfolio is one of the major focuses for option traders, which depends on their day-to-day forecast on the scale of the price move to be compared with the theta from the position carry. In a market with liquid options, the gamma position of a portfolio (i.e., long, short or neutral) can be constructed by trading delta-hedged options having the opposite gamma amount as shown in Sect. 14.5.1.

14.5.3 Delta-Hedging P/L

The realized volatility during the life of an option trade may differ from the implied volatility used for pricing and hedging. Let us analyze the P/L impact of the realized volatility assuming a constant implied volatility for pricing and hedging. For an in-depth analysis of hedging P/L under various models, we refer to [15].

Let S_t be the underlier price that follows the below general process for pricing

$$dS_t = \mu S_t dt + \sigma_t S_t W_t,$$

where σ_t represents the realized volatility for the infinitesimal period dt .

Denote $\hat{\sigma}_t$ as the implied volatility used for hedging/pricing that satisfies the *Black–Scholes equation* (14.11) (otherwise, there is arbitrage according to the analysis in Sect. 16.3). Let r be the interest rate. Then, we have the following important result on option's delta-hedging.

Proposition 14.4 *The P/L for the short party of a delta-hedged contingent claim paying out V_T at maturity T is*

$$\frac{1}{2} \int_0^T e^{r(T-t)} \Gamma_t S_t^2 (\sigma_t^2 - \hat{\sigma}_t^2) dt. \quad (14.22)$$

Proof Note that V_t satisfies the *Black–Scholes equation* that is related to the hedging volatility $\hat{\sigma}_t$,

$$\frac{\partial V}{\partial t} = - \left(r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \hat{\sigma}_t^2 S_t^2 - r V_t \right).$$

Applying the Itô–Doebelin formula to V_t as a function of t , S_t and $\hat{\sigma}_t$,

$$dV_t = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma_t^2 S_t^2 dt.$$

Substituting $\frac{\partial V}{\partial t}$ in the formula for dV_t , we get

$$\begin{aligned} dV_t &= r V_t dt - r S_t \frac{\partial V}{\partial S} dt + \frac{\partial V}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} S_t^2 (\sigma_t^2 - \hat{\sigma}_t^2) dt \\ dV_t - r V_t dt &= \Delta_t (-r S_t dt + dS_t) + \frac{1}{2} \Gamma_t S_t^2 (\sigma_t^2 - \hat{\sigma}_t^2) dt \\ d(e^{-rt} V_t) &= \Delta_t d(e^{-rt} S_t) + \frac{1}{2} e^{-rt} \Gamma_t S_t^2 (\sigma_t^2 - \hat{\sigma}_t^2) dt \end{aligned}$$

For a portfolio composed of the short position of a derivative and its delta-hedging, the portfolio's delta is zero. Hence, the discounted hedging P/L for the dt period is

$$\frac{1}{2}e^{-rt}\Gamma_t S_t^2(\sigma_t^2 - \hat{\sigma}_t^2)dt$$

and the cumulative hedging P/L of the portfolio at T is (14.22). $\square$

Actually, the above result can be extended to more general models with non-constant interest rate, local volatility, etc.

14.5.4 Motivation for Volatility Modelling

To mitigate the delta-hedging P/L related to the realized gamma and variance, traders may choose to trade delta-hedged vanilla options to offset the gamma (i.e., gamma-hedging). As the derivative price V_t represents the hedging portfolio's value, it will be a function of S_t and option price that depends on the implied volatility for vanilla options $\hat{\sigma}_t$, i.e., $V_t(t, S_t, \hat{\sigma}_t)$. Generally, $\hat{\sigma}_t$ is random and correlated to S_t . If we apply Itô–Doebelin formula to $V_t(t, S_t, \hat{\sigma}_t)$, there would be second order terms $\frac{d^2 V_t}{d\hat{\sigma}_t^2}$ (known as *volga*) and $\frac{d^2 V_t}{dS_t d\hat{\sigma}_t}$ (known as *vanna* or *spot-vega*) that are not hedged out by delta/gamma hedging. Apply modelling to $\hat{\sigma}_t$ (not to the realized volatility σ_t) leads to stochastic volatility models that offer a consistent pricing and risk management framework with all the $\hat{\sigma}_t$ related sensitivities quantified and a theta offsetting all the second orders terms in particular. Stochastic volatility models will be introduced in Chap. 25.

Remark 14.1 In some products, such as snowball autocallable structures with KI-barrier (see Sect. 31.1.5), volga ($= \frac{\partial}{\partial \hat{\sigma}_t}$ vega) and vanna ($= \frac{\partial}{\partial S_t}$ vega) may become quite significant during the life of the product, which may result in large vega re-hedging risk under some market condition (i.e. a sharp increase in volatility may trigger the KI-barrier and makes trading to buy back volatility at a higher level). Currently, the liquid volatility instruments are vanilla options, which do not provide efficient hedging for these sensitivities. $\square$

14.5.5 Transaction Cost*

The framework of option pricing with transaction costs presented here was proposed by Leland [69]. Let k be the round trip transaction cost per dollar value of the underlier. A self-financing hedging portfolio with no transaction cost satisfies

$$\delta V \approx \Delta \delta S + r(V - \Delta S)\delta t,$$

where δt is a small discrete change in time. The cost for adjusting the Δ position during the time interval δt is

$$\delta K = \frac{k}{2} S |\delta \Delta| \approx \frac{k}{2} S |\Gamma \sigma S \delta W|. \quad \left(\Gamma = \frac{\delta \Delta}{\delta S}, \delta S \approx \sigma S \delta W \right)$$

In this analysis, $\mathbb{E}[|\delta W|]$ is applied instead of $|\delta W|$. By direct integration and the fact that $\delta W \sim \mathcal{N}(0, \delta t)$, we can obtain

$$\begin{aligned} \mathbb{E}[|\delta W|] &= \int_{-\infty}^{\infty} \frac{|x|}{\sqrt{2\pi\delta t}} e^{-\frac{x^2}{2\delta t}} dx = 2 \int_0^{\infty} \frac{x}{\sqrt{2\pi\delta t}} e^{-\frac{x^2}{2\delta t}} dx \\ &= \sqrt{\frac{2\delta t}{\pi}} \int_0^{\infty} e^{-\frac{x^2}{2\delta t}} d\left(\frac{x^2}{2\delta t}\right) = \sqrt{\frac{2\delta t}{\pi}} \int_0^{\infty} e^{-y} dy = \sqrt{\frac{2\delta t}{\pi}}. \end{aligned}$$

Defining Leland number, $\text{Le} = \sqrt{\frac{2}{\pi}} \left(\frac{k}{\sigma \sqrt{\delta t}} \right)$, we get $\delta K = \frac{1}{2} \sigma^2 S^2 |\Gamma| \text{Le} \delta t$. Adding in the cost for rebalancing the delta δK , we obtain

$$\delta V \approx \Delta \delta S + r(V - \Delta S) \delta t - \frac{1}{2} \sigma^2 S^2 |\Gamma| \text{Le} \delta t.$$

From the Itô–Doebelin formula, we have $dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} dt$. Therefore,

$$\frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} dt = \Delta dS + r(V - \Delta S) dt - \frac{1}{2} \sigma^2 S^2 |\Gamma| \text{Le} dt.$$

Arranging the terms, we obtain the modified BSM model with transaction cost:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \frac{1}{2} \sigma^2 S^2 \left| \frac{\partial^2 V}{\partial S^2} \right| \text{Le} + rS \frac{\partial V}{\partial S} - rV = 0.$$

If $1 + \text{sgn}(\Gamma) \text{Le} \geq 0$, we have

$$\frac{\partial V}{\partial t} + \frac{1}{2} \tilde{\sigma}^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

with adjusted volatility

$$\tilde{\sigma}^2 = \sigma^2 (1 + \text{sgn}(\Gamma) \text{Le}).$$

The impact on option price for small $(\tilde{\sigma} - \sigma)$ is roughly $\text{vega} \times (\tilde{\sigma} - \sigma)$ with $\tilde{\sigma} - \sigma \approx \frac{k}{\sqrt{2\pi\delta t}} \text{sgn}(\Gamma)$ ($\tilde{\sigma}^2 - \sigma^2 = \sigma^2 \text{sgn}(\Gamma) \text{Le}$, $\tilde{\sigma}^2 - \sigma^2 = (\tilde{\sigma} + \sigma)(\tilde{\sigma} - \sigma) \approx 2\sigma(\tilde{\sigma} - \sigma)$). We can see that the more frequent the rebalancing (smaller δt), the higher will be the transaction cost.

Exercises

1. Without using the put-call parity, verify that the Black–Scholes put option formula (14.17) satisfies the Black–Scholes equation (14.11) and the corresponding boundary conditions and terminal condition.
2. Let $N'(d)$ denote the derivative of the standard normal c.d.f.. Show that

$$S_t N'(d_1) = K e^{-r(T-t)} N'(d_2) \quad (14.23)$$

and use this equality to derive the delta of a vanilla call.

3. Transform the Black–Scholes equation with log price by making the variable change $x = \ln S$, and obtain the corresponding boundary conditions and terminal condition.
4. Find the value of Black–Scholes call price when volatility approaches ∞ . What does the corresponding delta imply?

Chapter 15

Stochastic Calculus: II



15.1 Change of Probability

Changing probability measure plays an important role in risk-neutral option pricing. We first review the following important theorem.

Theorem 15.1 (Radon–Nikodym) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. There exists a unique $Z > 0$, $\mathbb{E}[Z] = 1$, $\mathbb{P}$ -a.s., such that a new probability measure $\tilde{\mathbb{P}}$ can be defined*

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega), \quad \forall A \in \mathcal{F}.$$

Furthermore,

$$\tilde{\mathbb{E}}[X] = \mathbb{E}[ZX] \quad \text{and} \quad \tilde{\mathbb{E}}\left[\frac{X}{Z}\right] = \mathbb{E}[X],$$

Z is called the Radon–Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$. We often denote it as $Z = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$. Furthermore, $\tilde{\mathbb{P}}$ is equivalent to $\mathbb{P}$ (noted as $\tilde{\mathbb{P}} \sim \mathbb{P}$), i.e., they agree on the probability zero sets.

The proof can be found in real analysis or probability textbooks such as [100].

For example, $Z = 1.5 \times \mathbf{1}_A + 0.5 \times \mathbf{1}_{A^c}$ is a Radon–Nikodym derivative for the probability on a binary event A , where $\mathbb{P}(A) = \mathbb{P}(A^c) = 0.5$.

Next, we define the Radon–Nikodym derivative process from the Radon–Nikodym derivative.

Lemma 15.1 *Let Z be the Radon–Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$. If we define $Z_t = \mathbb{E}[Z|\mathcal{F}_t]$, then Z_t is a $\mathbb{P}$ -martingale and*

$$\tilde{\mathbb{E}}[Y] = \mathbb{E}[YZ_t],$$

where Y is an $\mathcal{F}_t$ -measurable random variable for any t .

Proof For any $s < t$, the following arguments show that Z_t is a martingale:

$$\mathbb{E}[Z_t|\mathcal{F}_s] = \mathbb{E}[\mathbb{E}[Z|\mathcal{F}_t]|\mathcal{F}_s] = \mathbb{E}[Z|\mathcal{F}_s] = Z_s.$$

For the expectation under $\tilde{\mathbb{P}}$, the Radon–Nikodym theorem leads to

$$\tilde{\mathbb{E}}[Y] = \mathbb{E}[YZ] = \mathbb{E}[\mathbb{E}[YZ|\mathcal{F}_t]] = \mathbb{E}[Y\mathbb{E}[Z|\mathcal{F}_t]] = \mathbb{E}[YZ_t].$$

□

Obviously, $\mathbb{E}[Z_t] = 1$, and Z_t is called a Radon–Nikodym derivative process and is denoted as $\left.\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}\right|_{\mathcal{F}_t}$. As $\mathbb{E}[Z_t|\mathcal{F}_s] = Z_s$ implies $\mathbb{E}\left[\frac{Z_t}{Z_s}|\mathcal{F}_s\right] = 1$, it is natural to question the equivalence between $\tilde{\mathbb{E}}[Y|\mathcal{F}_s]$ and $\mathbb{E}[Y\frac{Z_t}{Z_s}|\mathcal{F}_s]$ ($= \frac{1}{Z_s}\mathbb{E}[YZ_t|\mathcal{F}_s]$) that seems to be the “conditioned version” of Radon–Nikodym theorem. This can be confirmed by the following formula that is very useful for calculating the conditional expectation under measure $\tilde{\mathbb{P}}$.

Theorem 15.2 (Bayes’ Formula) *Let $0 \leq s \leq t$, Y be an $\mathcal{F}_t$ -measurable random variable, $Z_t = \left.\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}\right|_{\mathcal{F}_t}$ be a Radon–Nikodym derivative process, then*

$$\tilde{\mathbb{E}}[Y|\mathcal{F}_s] = \frac{1}{Z_s}\mathbb{E}[YZ_t|\mathcal{F}_s].$$

Proof We only need to check the partial averaging property of conditional expectation. For any $A \in \mathcal{F}_s$,

$$\begin{aligned} \tilde{\mathbb{E}}\left[\mathbf{1}_A\left(\frac{1}{Z_s}\mathbb{E}[YZ_t|\mathcal{F}_s]\right)\right] &= \mathbb{E}[\mathbf{1}_A\mathbb{E}[YZ_t|\mathcal{F}_s]] = \mathbb{E}[\mathbb{E}[\mathbf{1}_A YZ_t|\mathcal{F}_s]] \\ &= \mathbb{E}[\mathbf{1}_A YZ_t] = \tilde{\mathbb{E}}[\mathbf{1}_A Y]. \end{aligned}$$

□

For the first and last equalities, Lemma 15.1 had been used.

Now, we introduce a special type of martingale known as *exponential martingale*. It is an important tool for applying the upcoming Girsanov Theorem.

Proposition 15.1 (Exponential Martingale) *Let λ_t be an $\mathcal{F}_t$ -adapted process satisfying Novikov's condition $\mathbb{E}[\exp(\frac{1}{2} \int_0^t |\lambda_s|^2 ds)] < \infty$. The stochastic process*

$$Z_t = \exp \left\{ \int_0^t \lambda_s dW_s - \frac{1}{2} \int_0^t \lambda_s^2 ds \right\}$$

is an $\mathcal{F}_t$ -martingale and $\mathbb{E}[Z_t] = Z_0 = 1$.

Proof Let $f(x) = e^x$ and $X_t = \int_0^t \lambda_s dW_s - \frac{1}{2} \int_0^t \lambda_s^2 ds$, which implies

$$dX_t = \lambda_t dW_t - \frac{1}{2} \lambda_t^2 dt.$$

By Itô–Doebelin formula, we have

$$\begin{aligned} dZ_t &= df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) dX_t dX_t \\ &= e^{X_t} \left(\lambda_t dW_t - \frac{1}{2} \lambda_t^2 dt \right) + \frac{1}{2} e^{X_t} \lambda_t^2 dt = \lambda_t Z_t dW_t. \end{aligned}$$

Hence, Z_t is an $\mathcal{F}_t$ -martingale and $\mathbb{E}[Z_t] = Z_0 = 1$. □

Theorem 15.3 (Girsanov Theorem) *Let W_t be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and θ_t be an adapted process. Taking the exponential $\mathbb{P}$ -martingale $Z_t = \exp \left\{ -\frac{1}{2} \int_0^t \theta_s^2 ds - \int_0^t \theta_s dW_s \right\}$ (or, equivalently, $dZ_t = -\theta_t Z_t dW_t$ with $Z_0 = 1$) as Radon–Nikodym derivative process $\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}$, and define*

$$\tilde{W}_t = W_t + \int_0^t \theta_s ds,$$

then $\tilde{W}_t$ is a Brownian motion under the probability measure $\tilde{\mathbb{P}}$, assuming that $\mathbb{E}[\int_0^t \theta_s^2 Z_s^2 ds] < \infty$.

Under a similar condition but taking $dZ_t = -\sum_{i=1}^n \theta_t^i Z_t dW_t^i$ where W_t^i and W_t^j are independent $\forall i \neq j$ (or, equivalently, $Z_t = \exp \left\{ -\frac{1}{2} \int_0^t \|\theta_s\|^2 ds - \sum_{i=1}^n \int_0^t \theta_s^i dW_s^i \right\}$, $\|\theta_s\|^2 = \sum_{i=1}^n (\theta_s^i)^2$ as Radon–Nikodym derivative process, we have the **multi-dimensional Girsanov Theorem** by which $\tilde{W}_t^i = W_t^i + \int_0^t \theta_s^i ds$.

Proof Both one-dimensional or multi-dimensional Girsanov Theorems can be proven by a direct application of the upcoming Girsanov–Meyer Theorem 15.4 (with $dZ_t = -\theta_t Z_t dW_t$ for 1-D case). □

To get an intuition of Girsanov Theorem, let's take $\theta_t (= \theta)$, $W_t = \sqrt{t}\epsilon$ where $\epsilon \sim \mathcal{N}(0, 1)$, $\tilde{W}_t = W_t + \theta t$ and $\tilde{W}_t = \sqrt{t}\tilde{\epsilon}$ (i.e. $\tilde{\epsilon} = \epsilon + \theta\sqrt{t}$). Then, we can check that

$$\frac{1}{\sqrt{2\pi}}e^{-\frac{\epsilon^2}{2}} = \frac{1}{\sqrt{2\pi}}e^{-\frac{\tilde{\epsilon}^2}{2}} \left(e^{-\frac{1}{2}\theta^2 t - \theta W_t} \right) = \frac{1}{\sqrt{2\pi}}e^{-\frac{\tilde{\epsilon}^2}{2}} Z_t.$$

Due to the appearance of a drift term $\int_0^t \theta_s ds$, $\tilde{W}_t$ is not a martingale under the original probability measure $\mathbb{P}$ in general. However, the Girsanov Theorem states that $\tilde{W}_t$ becomes a martingale under $\tilde{\mathbb{P}}$. As a result, $\tilde{W}_t$ enjoys all the Brownian motion properties under $\tilde{\mathbb{P}}$, for instance, $\tilde{\mathbb{E}}[\tilde{W}_t] = 0$ (note that $\mathbb{E}[\tilde{W}_t] \neq 0$).

Next we introduce a more general form of Girsanov theorem known as Girsanov–Meyer Theorem.

Lemma 15.2 Suppose $\tilde{\mathbb{P}} \sim \mathbb{P}$ with Radon–Nikodym derivative process $Z_t = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}\Big|_{\mathcal{F}_t}$. Then, $\tilde{X}_t$ is a $\tilde{\mathbb{P}}$ -martingale if and only if $\tilde{X}_t Z_t$ is a $\mathbb{P}$ -martingale.

Proof Let $\tilde{X}_t$ be a $\tilde{\mathbb{P}}$ -martingale. From Bayes' formula, we have

$$\tilde{X}_s = \tilde{\mathbb{E}}[\tilde{X}_t | \mathcal{F}_s] = \frac{1}{Z_s} \mathbb{E}[\tilde{X}_t Z_t | \mathcal{F}_s], \quad \forall 0 < s < t,$$

i.e., $\tilde{X}_s Z_s = \mathbb{E}[\tilde{X}_t Z_t | \mathcal{F}_s]$. Conversely, let $\tilde{X}_t Z_t$ be a $\mathbb{P}$ -martingale, then we have $\tilde{X}_s Z_s = \mathbb{E}[\tilde{X}_t Z_t | \mathcal{F}_s]$, i.e.,

$$\tilde{X}_s = \frac{1}{Z_s} \mathbb{E}[\tilde{X}_t Z_t | \mathcal{F}_s] = \tilde{\mathbb{E}}[\tilde{X}_t | \mathcal{F}_s],$$

by Bayes' formula again. □

Theorem 15.4 (Girsanov–Meyer Theorem) Suppose $\tilde{\mathbb{P}} \sim \mathbb{P}$ with a continuous Radon–Nikodym derivative process $Z_t = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}\Big|_{\mathcal{F}_t}$. Moreover, X_t is a continuous $\mathbb{P}$ -martingale. Define $\tilde{X}_t$ by

$$d\tilde{X}_t = dX_t - \frac{1}{Z_t} d\langle X, Z \rangle_t.$$

Then $\tilde{X}_t$ is a continuous $\tilde{\mathbb{P}}$ -martingale.

Proof As X_t and Z_t are continuous $\mathbb{P}$ -martingales, they can be represented by Brownian motions by martingale representation theorem. Hence, $d\langle X, Z \rangle_t$ is of the form $[\cdots]dt$ and $d\langle X, Z \rangle_t = 0$. Therefore,

$$d\langle \tilde{X}, Z \rangle_t = \left(dX_t - \frac{1}{Z_t} d\langle X, Z \rangle_t \right) dZ_t = d\langle X, Z \rangle_t - \frac{1}{Z_t} d\langle \langle X, Z \rangle, Z \rangle_t = d\langle X, Z \rangle_t.$$

Now, according to the Lemma 15.2, it is enough to check that $\tilde{X}_t Z_t$ is $\mathbb{P}$ -martingale. Note that

$$\begin{aligned} d(\tilde{X}_t Z_t) &= Z_t d\tilde{X}_t + \tilde{X}_t dZ_t + d\langle \tilde{X}, Z \rangle_t = Z_t d\tilde{X}_t + \tilde{X}_t dZ_t + d\langle X, Z \rangle_t \\ &= Z_t dX_t - d\langle X, Z \rangle_t + \tilde{X}_t dZ_t + d\langle X, Z \rangle_t = Z_t dX_t + \tilde{X}_t dZ_t. \end{aligned}$$

Hence, $\tilde{X}_t Z_t$ is a $\mathbb{P}$ -martingale. $\square$

The above theorem is convenient to use when Z_t is correlated to the $\mathbb{P}$ -martingale X_t , e.g., Z_t and X_t are driven by different but correlated Brownian motions.

15.2 Predictable Martingale Representation

An $\mathcal{F}_t$ -adapted process is said to be *predictable* if it is left continuous. For a continuous martingale X_t , it is tempted to believe that we may find a process ϕ_t such that the variance of $\frac{dX_t}{\phi_t}$ is dt , $\forall t \geq 0$, and hence $\frac{X_t}{\phi_t}$ is a Brownian motion by Levy's Theorem (Theorem 13.7). In other terms, dX_t can be represented by $\phi_t dW_t$, which is confirmed by the below theorem.

Theorem 15.5 (Predictable Martingale Representation Theorem) *Let W_t be a Brownian with $\mathcal{F}_t$ as its natural filtration and X_t be an $\mathcal{F}_t$ -measurable continuous martingale, then, there is a unique predictable process ϕ_s , such that*

$$X_T = X_t + \int_t^T \phi_s dW_s, \quad \text{where } X_t = \mathbb{E}_t[X_T], \quad \forall T > t.$$

The proof can be found in [64] and [94]. Intuitively, ϕ_s is the parameter to make $\phi_s dW_s$ behave like dX_s . The following illustrations may help to understand the theorem:

- if $\langle X, X \rangle_t = t$ for all t , then, $X_T = X_0 + \int_0^T dW_s$
- if $\langle X, X \rangle_t = ct$ for all t , then, $X_T = X_0 + \int_0^T \sqrt{c} dW_s$
- if $\frac{d\langle X, X \rangle_t}{dt} = \lambda_t$ exists for all t , then, $X_T = X_0 + \int_0^T \sqrt{\lambda_s} dW_s$.

The above theorem is the one-dimensional case of the *multi-dimensional martingale representation* that is more general and beyond Brownian motions. Let $\mathcal{F}_t^{\mathbf{Y}}$ be the natural filtration of a martingale set $\mathbf{Y}$. The necessary and sufficient condition for X_T to admit the predictable representation with the integral of $\mathbf{Y}$ ($X_T = X_0 + \int_0^T \boldsymbol{\phi}_s \cdot d\mathbf{Y}_s$) is that there is a unique probability measure $\mathbb{P}$ under which X_t is an $\mathcal{F}_t^{\mathbf{Y}}$ -martingale [94]. It can be shown that $\mathbb{P}$ is unique if $\mathbf{Y}$ is a set of Brownians, i.e., multi-dimensional Brownian motion.

Martingale representation theorem plays a fundamental role in the hedging process of option derivatives. For any continuous $\mathbb{P}$ -martingale $X_t = X_0 + \int_0^t \phi_s dW_s$

with $\phi_s \neq 0$ ($\mathbb{P}$ -a.s., $\forall s \geq 0$), we have $dW_s = \frac{1}{\phi_s} dX_s$ ($\mathbb{P}$ -a.s.). Hence, any continuous $\mathbb{P}$ -martingale Z_t can be represented by X_t ($\mathbb{P}$ -a.s.). We will see that under $\widehat{\mathbb{P}}$, known as risk-neutral probability (obtained by a probability change), the discounted option price $e^{-rt} V_t$ can be represented by discounted underlier price $e^{-rt} S_t$, which describes a self-financing hedging process.

15.3 Stochastic Differential Equations

Consider Itô processes of the form:

$$dX_t = b(t, X_t)dt + \phi(t, X_t)dW_t,$$

where $b(t, x)$ is the drift and $\phi(t, x)$ the diffusion coefficient. The above equation is a stochastic differential equation (SDE).

Suppose $X_t = z$ is the initial condition at time $t \geq 0$, $x \in \mathbb{R}$. Then, the solution of the SDE is a stochastic process X_T , $T > t$, such that

$$\begin{cases} X_t = z, \\ X_T = X_t + \int_t^T b(s, X_s)ds + \int_t^T \phi(s, X_s)dW_s. \end{cases} \quad (15.1)$$

Theorem 15.6 (Existence and Uniqueness Theorem for One-Dimensional SDE)

Let b and ϕ be continuous functions such that, there exists a $K < +\infty$, and

- (1) $|b(t, x) - b(t, y)| + |\phi(t, x) - \phi(t, y)| \leq K|x - y|$
- (2) $|b(t, x)| + |\phi(t, x)| \leq K|1 + x|$

for all $t \leq T$, $x, y \in \mathbb{R}$. Then, (15.1) admits a unique solution.

Readers are referred to [85] for the proof.

The following processes are frequently used in financial modelling.

Arithmetic Brownian Motion

The following SDE is used in Bachelier model [10]

$$dS_t = \mu dt + \sigma dW_t,$$

which gives normally distributed $S_T = z + \mu(T - t) + \sigma(W_T - W_t)$ with $S_t = z$.

Geometric Brownian Motion

Consider the following SDE used in the Black–Scholes–Merton model,

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ and σ ($\sigma > 0$) are constants. Here $b(t, x) = \mu x$ and $\phi(x, t) = \sigma x$. With $S_t = z$, the unique solution is the lognormally distributed geometric Brownian motion

$$S_T = ze^{(\mu - \frac{1}{2}\sigma^2)(T-t) + \sigma(W_T - W_t)}.$$

Ornstein–Uhlenbeck Process

OU (Ornstein–Uhlenbeck) process is a widely applied in financial modelling. It is the unique solution of

$$dX_t = -\kappa X_t dt + \sigma dW_t, \quad X_0 = z, \kappa > 0,$$

which is a *mean reverting* Gaussian process drifting towards its mean function over time. κ is known as mean-reversion speed. Let $Y_t = X_t e^{\kappa t}$ and apply Itô's product rule to it, we get $dY_t = \sigma e^{\kappa t} dW_t$. Hence, we have

$$X_t = ze^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s,$$

where $e^{-\kappa(t-s)}$ is helpful to control the impact of the random term over time. Obviously, we have

$$\mathbb{E}[X_t] = ze^{-\kappa t}.$$

Using Itô isometry,

$$\begin{aligned} \text{Var}(X_t) &= \mathbb{E}[(X_t - \mathbb{E}[X_t])^2] = \sigma^2 e^{-2\kappa t} \mathbb{E}\left[\left(\int_0^t e^{\kappa s} dW_s\right)^2\right] \\ &= \sigma^2 e^{-2\kappa t} \mathbb{E}\left[\int_0^t e^{2\kappa s} ds\right] = \sigma^2 \frac{1 - e^{-2\kappa t}}{2\kappa}. \end{aligned}$$

A more general form of OU process is

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t \quad (\theta \text{ is a constant})$$

with $\mathbb{E}[X_t] = ze^{-\kappa t} + \theta(1 - ze^{-\kappa t})$.

Cox–Ingersoll–Ross Square Root Process

CIR (Cox–Ingersoll–Ross) process is the unique solution of

$$dX_t = (a - bX_t)dt + \sigma\sqrt{X_t}dW_t, \quad X_t \in [0, \infty), \quad X_0 = x > 0, a > 0, b > 0.$$

It is a positive process with mean reversion behavior, which means that X_t tends to rise when $X_t < \frac{a}{b}$ and tends to fall when $X_t > \frac{a}{b}$. The parameter b characterizes the speed of mean reversion.

Let $\tau_0 = \inf\{t \geq 0 | X_t = 0\}$ be the first time of hitting 0. The properties of CIR process are given below

- if $a \geq \frac{\sigma^2}{2}$, $\mathbb{P}\{\tau_0 = \infty\} = 1$
- if $0 \leq a < \frac{\sigma^2}{2}$, $\mathbb{P}\{\tau_0 < \infty\} = 1$.

It is also known that given the current value, the future values of a CIR process follow a non-central Chi-squared distribution extended to non-integer degrees of freedom that can be calculated in closed-form. The references for the proof are given in Appendix E where we show that the CIR process can be represented with a Squared Bessel process that is well studied for its properties. The relevant properties of Squared Bessel process are also introduced in Appendix E.

Proposition 15.2 (Markov Property of SDE Solutions) *Let X_s , $s \geq 0$ be a solution to an SDE with initial condition given at time 0. Then for any function h ,*

$$\mathbb{E}[h(X_T) | \mathcal{F}_t] = \mathbb{E}^{t, X_t}[h(X_T)],$$

where $\mathbb{E}^{t, X_t}[\cdot]$ denotes an expectation evaluated with initial value X_t at time t .

A proof can be found in [85] and we refer to [11] for further reading on Markov property analysis. Actually, from Itô–Doebelin Formula, we have

$$\begin{aligned} h(X_T) &= h(X_0) + \int_0^T [\cdots] ds + \int_0^T [\cdots] dW_s \\ &= \left(h(X_0) + \int_0^t [\cdots] ds + \int_0^t [\cdots] dW_s \right) + \int_t^T [\cdots] ds + \int_t^T [\cdots] dW_s \\ &= h(X_t) + \int_t^T [\cdots] ds + \int_t^T [\cdots] dW_s. \end{aligned}$$

We see that $h(X_T)$ depends on the initial value X_t , which is $\mathcal{F}_t$ -measurable, and the increments of X_s , $t \leq s \leq T$, which, in turn, are determined by increments of W_s . As increments of W_s are independent of $\mathcal{F}_t$, it seems natural to drop the conditioning of $\mathcal{F}_t$ in the expectation.

The below theorem links the expectation computation from an SDE related problem with a PDE problem.

Theorem 15.7 (Feynman–Kac Theorem) *Let X_t follow the SDE*

$$dX_t = b(t, X_t)dt + \phi(t, X_t)dW_t.$$

Assume that h satisfies $\mathbb{E}^{t,x}[|h(X_T)|] < \infty$. For any $x \in \mathbb{R}$ and $t < T$, let

$$g(t, x) = \mathbb{E}^{t,x}[h(X_T)].$$

Then, $g(t, x)$ satisfies the below PDE

$$\frac{\partial g}{\partial t}(t, x) + b(t, x) \frac{\partial g}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 g}{\partial x^2}(t, x) = 0,$$

and terminal condition $g(T, x) = h(x)$.

Proof We apply the Itô's formula and obtain

$$\begin{aligned} dg(t, X_t) &= \frac{\partial g}{\partial t} dt + b \frac{\partial g}{\partial x} dt + \phi \frac{\partial g}{\partial x} dW_t + \frac{1}{2} \phi^2 \frac{\partial^2 g}{\partial x^2} dt \\ &= \left(\frac{\partial g}{\partial t} + b \frac{\partial g}{\partial x} + \frac{1}{2} \phi^2 \frac{\partial^2 g}{\partial x^2} \right) dt + \phi \frac{\partial g}{\partial x} dW_t. \end{aligned}$$

As $g(t, X_t)$ is a martingale (Proposition 13.1), the dt term should be zero. We obtain

$$\frac{\partial g}{\partial t}(t, X_t) + b(t, X_t) \frac{\partial g}{\partial x}(t, X_t) + \frac{1}{2} \phi^2(t, X_t) \frac{\partial^2 g}{\partial x^2}(t, X_t) = 0$$

along every path of X . Therefore,

$$\frac{\partial g}{\partial t}(t, x) + b(t, x) \frac{\partial g}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 g}{\partial x^2}(t, x) = 0$$

at every point (t, x) that can be reached by (t, X_t) . □

15.4 Kolmogorov Equations*

Consider the general diffusion process

$$dX_s = \theta(s, X_s)ds + \phi(s, X_s)dW_s.$$

Let $p(t, x; T, y)$ ($0 \leq t \leq T$) be the transition probability density of X_s . Kolmogorov forward equation assumes t and x “fixed” with $X_t = x$ and $p(t, x; T, y)$ is the probability density function with state variable y and a future time T . Conversely, Kolmogorov backward equation assumes T and y “fixed” with $X_T = y$ and shows the behaviour of $p(t, x; T, y)$ backwards in time t with state variable x .

Kolmogorov forward equation, also known as *Fokker–Planck equation*, governs the forward transition probability with t and x fixed.

Theorem 15.8 (Kolmogorov Forward Equation)

$$\frac{\partial}{\partial T} p(t, x; T, y) + \frac{\partial}{\partial y} (\theta(T, y) p(t, x; T, y)) - \frac{1}{2} \frac{\partial^2}{\partial y^2} (\phi^2(T, y) p(t, x; T, y)) = 0.$$

Proof Let $h(y)$ be any C^2 function with compact support (such a function vanishes outside a finite interval).¹ From Itô–Doeblin formula,

$$\begin{aligned} h(X_T) &= h(X_t) + \int_t^T \theta(s, X_s) h'(X_s) ds + \int_t^T \phi(s, X_s) h'(X_s) dW_s \\ &\quad + \frac{1}{2} \int_t^T \phi^2(s, X_s) h''(X_s) ds. \end{aligned}$$

Taking expectation with initial condition $X_t = x$, we have

$$\mathbb{E}^{t,x}[h(X_T)] = h(x) + \int_t^T \mathbb{E}^{t,x} \left[\theta(s, X_s) h'(X_s) + \frac{1}{2} \phi^2(s, X_s) h''(X_s) \right] ds.$$

On the other hand, the transition probability is the density that gives

$$\mathbb{E}^{t,x}[h(X_T)] = \int_{-\infty}^{\infty} h(y) p(t, x; T, y) dy.$$

Equating the above leads to

$$\begin{aligned} \int_{-\infty}^{\infty} h(y) p(t, x; T, y) dy \\ = h(x) + \int_t^T \left[\int_{-\infty}^{\infty} \left(\theta(s, y) h'(y) + \frac{1}{2} \phi^2(s, y) h''(y) \right) p(t, x; s, y) dy \right] ds. \end{aligned}$$

From integration by parts and the fact that h and h' vanish at ∞ :

$$\begin{aligned} \int_{-\infty}^{\infty} h(y) p(t, x; T, y) dy &= h(x) - \int_t^T \left[\int_{-\infty}^{\infty} h(y) \frac{\partial}{\partial y} (\theta(s, y) p(t, x; s, y)) dy \right] ds \\ &\quad + \frac{1}{2} \int_t^T \left[\int_{-\infty}^{\infty} h(y) \frac{\partial^2}{\partial y^2} (\phi^2(s, y) p(t, x; s, y)) dy \right] ds. \end{aligned}$$

¹ For example, the following function $h(y)$ vanishes for $y \notin (0, b)$: $h(y) =$

$$\begin{cases} e^{-\frac{1}{1-(\frac{2y-b}{b})^2}}, & 0 < y < b, \\ 0, & \text{otherwise.} \end{cases}$$

Differentiating w.r.t. T , we obtain that

$$\int_{-\infty}^{\infty} h(y) \left[\frac{\partial}{\partial T} p(t, x; T, y) + \frac{\partial}{\partial y} (\theta(T, y) p(t, x; T, y)) - \frac{1}{2} \frac{\partial^2}{\partial y^2} (\phi^2(T, y) p(t, x; T, y)) \right] dy = 0.$$

From the arbitrariness of $h(y)$, we get

$$\frac{\partial}{\partial T} p(t, x; T, y) + \frac{\partial}{\partial y} (\theta(T, y) p(t, x; T, y)) - \frac{1}{2} \frac{\partial^2}{\partial y^2} (\phi^2(T, y) p(t, x; T, y)) = 0.$$

□

The following is the backward case with future time T and state y fixed.

Theorem 15.9 (Kolmogorov Backward Equation)

$$\frac{\partial}{\partial t} p(t, x; T, y) + \theta(t, x) \frac{\partial}{\partial x} p(t, x; T, y) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2}{\partial x^2} p(t, x; T, y) = 0.$$

Proof Let $h(y)$ be any C^2 function with compact support. From Feynman–Kac formula, the transition function $g(t, x) = \mathbb{E}^{t,x}[h(X_T)] = \int_{-\infty}^{\infty} h(y) p(t, x; T, y) dy$ satisfies

$$\frac{\partial g}{\partial t}(t, x) + \theta(t, x) \frac{\partial g}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 g}{\partial x^2}(t, x) = 0,$$

and hence

$$\int_{-\infty}^{\infty} h(y) \left[\frac{\partial p}{\partial t}(t, x; T, y) + \theta(t, x) \frac{\partial p}{\partial x}(t, x; T, y) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 p}{\partial x^2}(t, x; T, y) \right] dy = 0.$$

From the arbitrariness of $h(y)$, we obtain

$$\frac{\partial p}{\partial t}(t, x; T, y) + \theta(t, x) \frac{\partial p}{\partial x}(t, x; T, y) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 p}{\partial x^2}(t, x; T, y) = 0.$$

□

15.5 Breeden–Litzenberger Formulas*

Theorem 15.10 (Breeden–Litzenberger Formulas) *Let $c(T, K)$ be the European call price with strike K and maturity T , and $f(T, K)$ be the forward transition probability density, i.e., $f(t, x) = p(0, S_0; t, x)$. Then,*

$$\begin{cases} f(T, K) = e^{rT} \frac{\partial^2 c}{\partial K^2}(T, K), \\ \tilde{\mathbb{E}}[e^{-rT} \mathbf{1}_{\{S_T > K\}}] = -\frac{\partial c}{\partial K}(T, K), \\ \tilde{\mathbb{E}}[e^{-rT} \delta(S_T - K)] = \frac{\partial^2 c}{\partial K^2}(T, K). \end{cases}$$

Proof From the risk-neutral representation of call option price,

$$c(T, K) = e^{-rT} \int_0^\infty (x - K)^+ f(T, x) dx$$

and hence

$$\begin{aligned} c(T, K) &= e^{-rT} \int_0^\infty (x - K)^+ f(T, x) dx = e^{-rT} \int_K^\infty (x - K) f(T, x) dx \\ &= e^{-rT} \int_K^\infty f(T, x) \int_K^x dy dx = e^{-rT} \int_K^\infty \int_y^\infty f(T, x) dx dy. \end{aligned}$$

Differentiating w.r.t. K gives

$$\frac{\partial c}{\partial K}(T, K) = -e^{-rT} \int_K^\infty f(T, x) dx = -\tilde{\mathbb{E}}[e^{-rT} \mathbf{1}_{\{S_T > K\}}],$$

and differentiating again leads to

$$\frac{\partial^2 c}{\partial K^2}(T, K) = e^{-rT} f(T, K).$$

and

$$\frac{\partial^2 c}{\partial K^2}(T, K) = \tilde{\mathbb{E}}[e^{-rT} \delta(S_T - K)]$$

□

15.6 Further Properties of Brownian Motion (BM)*

The properties to be introduced in this section are useful in many applications, including closed-form solutions for some exotic options, Monte Carlo simulation, tree method, etc.. However, first-time readers are advised to skip this section and to directly jump to the next chapter on risk-neutral pricing framework.

15.6.1 Covariance of Brownian Motions

Proposition 15.3 For any $s, t \geq 0$, $\text{Cov}(W_s, W_t) = \mathbb{E}[W_s W_t] = \min\{s, t\}$.

Proof We only prove the case $\mathbb{E}[W_s W_t] = s$ when $s < t$ because the other case is symmetric. With the independence between $W_t - W_s$ and W_s ,

$$\begin{aligned}\mathbb{E}[W_s W_t] &= \mathbb{E}[W_s(W_t - W_s + W_s)] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] \\ &= \mathbb{E}[W_s]\mathbb{E}[W_t - W_s] + s = s.\end{aligned}\quad \square$$

The result is used later in the analysis for a property of Asian option.

15.6.2 First Passage Time

Let x be any real number. The first passage time to level x , also called the first hitting time, describes the time when the Brownian motion hits the level x for the first time. It is defined as the following *random variable*:

$$\tau_x = \min\{t \geq 0 : W_t = x\}, \text{ which is a stopping time.}$$

If the Brownian motion never reaches the level x , we set $\tau_x = \infty$. Different sample paths of W_t may firstly hit x at different moments, making it a random variable $\tau_x(\omega)$ for each ω (sample path). See two realizations of τ_x shown in Fig. 15.1 (left).

15.6.3 Extremum to Date

The *maximum to date* for Brownian motion starting at 0 is defined as the following stochastic process:

$$M_t = \max_{0 \leq s \leq t} W_s.$$

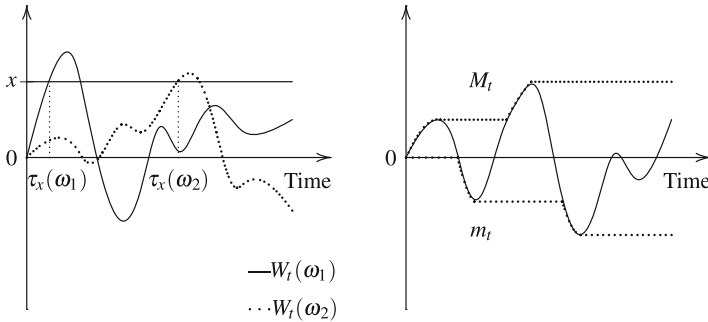


Fig. 15.1 First passage time and maximum/minimum to date

Note that M_t is a non-decreasing process, and visually consists of a considerable amount of flat lines. See Fig. 15.1 (right) for reference. Also $M_t \geq W_0 = 0$.

The *minimum to date* for Brownian motion starting from 0 is defined as the following stochastic process:

$$m_t = \min_{0 \leq s \leq t} W_s.$$

It is also easy to see that m_t is a non-increasing process with a lot of flat lines. See Fig. 15.1 (right) for reference. Moreover, $m_t \leq W_0 = 0$. Together, the processes M_t and m_t form some sort of envelope around W_t .

15.6.4 Reflection Principle

Brownian motion $W_t \sim \mathcal{N}(0, t)$ is symmetric, i.e., it has an equal chance of reaching a positive number or its negation at a future time (starting from time 0). This property can further be elevated and reveals how the Brownian motion has a mirror image *after* it hits a certain value; see Fig. 15.2. This can be mathematically described as follows. For $x > 0$, the set $\{\tau_x \leq t\}$ implies that the Brownian motion

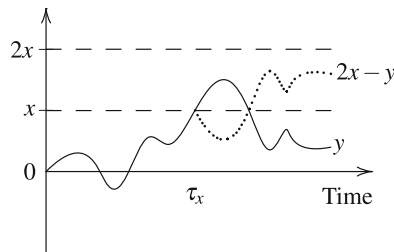


Fig. 15.2 The reflection principle: Brownian motion (solid line) and its reflection with respect to x (dotted line)

has already touched x at or before t . Then, the probability of W_t having retraced to a lower level $y \leq x$ from x , should be equal to the probability of W_t rising above y 's mirror image about x , i.e., $2x - y$. Thus, we first have the heuristic observation below without proof (see [11] for the proof).

Theorem 15.11 (Reflection Principle)

$$\mathbb{P}\{\tau_x \leq t, W_t \leq y\} = \mathbb{P}\{W_t \geq 2x - y\}, \quad y \leq x, \quad x > 0, \quad W_0 = 0.$$

The special case when $y = x$ is also useful:

$$\mathbb{P}\{\tau_x \leq t, W_t \leq x\} = \mathbb{P}\{W_t \geq x\}, \quad x > 0, \quad W_0 = 0.$$

Corollary 15.1 *We have $\mathbb{P}\{\tau_x \leq t\} = 2\mathbb{P}\{W_t \geq x\}$.*

Proof By the special case of reflection principle, we have

$$\mathbb{P}\{\tau_x \leq t, W_t \leq x\} = \mathbb{P}\{W_t \geq x\}.$$

Note that if $W_t \geq x$, then $\tau_x \leq t$ so that $\mathbb{P}\{\tau_x \leq t, W_t \geq x\} = \mathbb{P}\{W_t \geq x\}$. Therefore

$$\mathbb{P}\{\tau_x \leq t\} = \mathbb{P}\{\tau_x \leq t, W_t \leq x\} + \mathbb{P}\{\tau_x \leq t, W_t \geq x\} = 2\mathbb{P}\{W_t \geq x\}.$$

□

There is also another version of reflection principle that involves the maximum to date M_t . The important thing is to realize that the set $\{\tau_x \leq t\}$ is equivalent to $\{M_t \geq x\}$. Then,

$$\mathbb{P}\{M_t \geq x, W_t \leq y\} = \mathbb{P}\{\tau_x \leq t, W_t \leq y\} = \mathbb{P}\{W_t \geq 2x - y\}, \quad y \leq x, \quad x > 0,$$

where the second quality is from the reflection principle. From the corollary, we also have the following useful result:

$$\mathbb{P}\{M_t \geq x\} = \mathbb{P}\{\tau_x \leq t\} = 2\mathbb{P}\{W_t \geq x\}.$$

15.6.5 Distribution of First Passage Time

The first passage time w.r.t. x is a random variable when we consider a fixed x . It is interesting to study the distribution of this random variable.

Proposition 15.4 (First Passage Time Distribution) Suppose $t > 0$ and $\forall x \neq 0$. The random variable τ_x has the following cumulative distribution function and probability density function

$$\mathbb{P}\{\tau_x \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{\frac{|x|}{\sqrt{t}}}^{\infty} e^{-\frac{z^2}{2}} dz \quad \text{and} \quad f_{\tau_x}(t) = \frac{|x|}{t\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}.$$

Proof We first consider the case $x > 0$. By the Reflection Principle's corollary, and the normal density of Brownian motion, we have

$$\mathbb{P}\{\tau_x \leq t\} = 2\mathbb{P}\{W_t \geq x\} = \frac{2}{\sqrt{2\pi t}} \int_x^{\infty} e^{-\frac{u^2}{2t}} du.$$

Defining $z = \frac{u}{\sqrt{t}}$, we get $\mathbb{P}\{\tau_x \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{\frac{|x|}{\sqrt{t}}}^{\infty} e^{-\frac{z^2}{2}} dz$. For $x < 0$, we observe that τ_x and $\tau_{|x|}$ have the same distribution, and we have the same result. To obtain the probability density function, we recall that $f_{\tau_x}(t) = \frac{d}{dt}\mathbb{P}\{\tau_x \leq t\}$ and direct differentiation will give the result. $\square$

15.6.6 Joint Distribution for BM Extremum

The reflection principle also allows us to derive the joint density of the Brownian motion and its maximum or minimum.

Proposition 15.5 Let $t > 0$. The joint density of (M_t, W_t) is

$$f_{M_t, W_t}(x, y) = \frac{2(2x - y)}{t\sqrt{2\pi t}} e^{-\frac{(2x-y)^2}{2t}} \mathbf{1}_{x \geq \max\{y, 0\}}.$$

The joint density of (m_t, W_t) is

$$f_{m_t, W_t}(\bar{x}, \bar{y}) = \frac{2(\bar{y} - 2\bar{x})}{t\sqrt{2\pi t}} e^{-\frac{(\bar{y}-2\bar{x})^2}{2t}} \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}}.$$

Proof Note that by definition of distribution function

$$\mathbb{P}\{M_t \geq x, W_t \leq y\} = \int_x^{\infty} \int_{-\infty}^y f_{M_t, W_t}(u, v) dv du$$

and the LHS is equal to $\mathbb{P}\{W_t \geq 2x - y\}$ by the reflection principle with M_t . Moreover, by the normal density of W_t we also have

$$\mathbb{P}\{W_t \geq 2x - y\} = \frac{1}{\sqrt{2\pi t}} \int_{2x-y}^{\infty} e^{-\frac{z^2}{2t}} dz.$$

We get the proof by differentiating the right-hand side of both equations w.r.t. x and y . The joint density of (m_t, W_t) can be deduced from (M_t, W_t) . Defining

$$\overline{W}_t = -W_t, \quad \bar{x} = -x \quad \text{and} \quad \bar{y} = -y,$$

we have

$$\bar{m}_t = \min_{0 \leq s \leq t} \overline{W}_s = -\max_{0 \leq s \leq t} (-\overline{W}_s) = -\max_{0 \leq s \leq t} W_s = -M_t.$$

The following equivalence

$$\mathbb{P}\{\bar{m}_t \leq \bar{x}, \overline{W}_t \geq \bar{y}\} = \mathbb{P}\{M_t \geq x, W_t \leq y\}$$

would lead to $f_{m_t, W_t}(\bar{x}, \bar{y}) = f_{\bar{m}_t, \overline{W}_t}(\bar{x}, \bar{y}) = f_{M_t, W_t}(x, y)$, which is the result to prove after inserting $x = -\bar{x}$ and $y = -\bar{y}$ in $f_{M_t, W_t}(x, y)$. $\square$

Note that $M_t \geq 0$ and $m_t \leq 0$ and therefore their state variables inherit the same ranges. For the state variable of W_t , it cannot exceed that of M_t or stay under that of m_t . Therefore, the domain of the joint density is not the whole 2D plane.

15.6.7 Conditional Distribution for Drifted BM Extremum

Let a *Brownian motion with drift* be defined as $\widehat{W}_t = W_t + \mu t$, $0 \leq t \leq T$. Evidently, $\widehat{W}_t$ follows the normal distribution with mean μt and variance t . Its density is

$$f_{\widehat{W}_t}(y) = f_{W_t}(y - \mu t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-\mu t)^2}{2t}} = \frac{1}{\sqrt{2\pi t}} e^{\mu y - \frac{1}{2}\mu^2 t} e^{-\frac{y^2}{2t}}.$$

The maximum of Brownian motion with drift is defined as $\widehat{M}_T = \max_{0 \leq t \leq T} \widehat{W}_t$. The minimum of Brownian motion with drift is defined as $\widehat{m}_T = \min_{0 \leq t \leq T} \widehat{W}_t$.

Proposition 15.6 *The joint density of $(\widehat{M}_T, \widehat{W}_T)$ is*

$$f_{\widehat{M}_T, \widehat{W}_T}(x, y) = \frac{2(2x - y)}{T\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x-y)^2}{2T}} \mathbf{1}_{x \geq \max\{y, 0\}}.$$

The joint density of $(\widehat{m}_T, \widehat{W}_T)$ is

$$f_{\widehat{m}_T, \widehat{W}_T}(\bar{x}, \bar{y}) = \frac{2(\bar{y} - 2\bar{x})}{T\sqrt{2\pi T}} e^{\mu \bar{y} - \frac{1}{2}\mu^2 T - \frac{(\bar{y}-2\bar{x})^2}{2T}} \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}}.$$

The distribution functions for First Passage Time and Maximum of Brownian motion allow the derivation of closed-form solutions for some of the exotic options.

Proof We apply Girsanov Theorem to remove the drift. Defining $\widehat{Z}_t = e^{-\mu \widehat{W}_t - \frac{1}{2}\mu^2 t} = e^{-\mu \widehat{W}_t + \frac{1}{2}\mu^2 t}$ and $\widehat{\mathbb{P}} = \int_A \widehat{Z}_T d\mathbb{P}$, then, $\widehat{W}_t$ is a Brownian motion with zero drift under $\widehat{\mathbb{P}}$. Hence, for $x \geq \max\{y, 0\}$,

$$\begin{aligned} \mathbb{P}\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\} &= \mathbb{E}\left[\mathbf{1}_{\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\}}\right] = \widehat{\mathbb{E}}\left[\frac{1}{\widehat{Z}_T} \mathbf{1}_{\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\}}\right] \\ &= \widehat{\mathbb{E}}\left[e^{\mu \widehat{W}_T - \frac{1}{2}\mu^2 T} \mathbf{1}_{\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\}}\right] \\ &= \int_{-\infty}^y \int_{-\infty}^x e^{\mu v - \frac{1}{2}\mu^2 T} \left[\frac{2(2u - v)}{T\sqrt{2\pi T}} e^{-\frac{(2u-v)^2}{2T}} \right] dudv, \end{aligned}$$

where the last equality comes from Proposition 15.5. Differentiating w.r.t. y and x , we get

$$f_{\widehat{M}_T, \widehat{W}_T}(x, y) = \frac{2(2x - y)}{T\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x-y)^2}{2T}} \mathbf{1}_{x \geq \max\{y, 0\}}.$$

Similarly, we can prove the joint density of $(\widehat{m}_T, \widehat{W}_T)$. □

Proposition 15.7 (Conditional Distribution of $\widehat{M}_t$ and $\widehat{m}_t$) *The conditional distribution of $(\widehat{M}_t, \widehat{W}_t)$ is*

$$\begin{aligned} f_{\widehat{M}_t | \widehat{W}_t}(x|y) &= \frac{2(2x - y)}{t} \exp\left(-\frac{2x(x - y)}{t}\right) \mathbf{1}_{x \geq \max\{y, 0\}}, \\ \mathbb{P}\{\widehat{M}_t \leq x | \widehat{W}_t = y\} &= \left[1 - \exp\left(-\frac{2x(x - y)}{t}\right)\right] \mathbf{1}_{x \geq \max\{y, 0\}}. \end{aligned}$$

The conditional distribution of $(\widehat{m}_t, \widehat{W}_t)$ is

$$\begin{aligned} f_{\widehat{m}_t | \widehat{W}_t}(\bar{x}|\bar{y}) &= \frac{2(\bar{y} - 2\bar{x})}{t} \exp\left(-\frac{2\bar{x}(\bar{x} - \bar{y})}{t}\right) \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}}, \\ \mathbb{P}\{\widehat{m}_t \leq \bar{x} | \widehat{W}_t = \bar{y}\} &= \exp\left(-\frac{2\bar{x}(\bar{x} - \bar{y})}{t}\right) \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}}. \end{aligned}$$

Proof The conditional density $f_{\widehat{M}_t|\widehat{W}_t}(x|y)$ can be obtained with $f_{\widehat{M}_t|\widehat{W}_t}(x|y) = \frac{f_{\widehat{M}_t, \widehat{W}_t}(x, y)}{f_{\widehat{W}_t}(y)}$. Then,

$$\begin{aligned} \mathbb{P}\{\widehat{M}_t \leq x | \widehat{W}_t = y\} &= \int_0^x f_{\widehat{M}_t|\widehat{W}_t}(u, y) du \\ &= \left[1 - \exp\left(-\frac{2x(x-y)}{t}\right) \right] \mathbf{1}_{x \geq \max\{y, 0\}}. \end{aligned}$$

Similarly, we can obtain the conditional distribution for $(\widehat{m}_t, \widehat{W}_t)$. $\square$

This proposition is useful for Monte Carlo simulations (see Sect. 21) where both the terminal value and the extremum are needed. The simulation of the pair can be done with two steps: (1) generating the terminal value, and (2) generating the extremum using $\mathbb{P}^{-1}$ (e.g., by inverse sampling, we have $\widehat{m}_t = \frac{\bar{y}}{2} - \sqrt{\frac{\bar{y}^2}{4} - \frac{t \ln u}{2}}$, $u \in \mathcal{U}[0, 1]$ is a uniform random). We refer to [14] for generating the triplet $(\widehat{m}_t, \widehat{M}_t, \widehat{W}_t)$ with Monte Carlo.

For further illustration, we derive two practical and useful results on conditional minimum for option pricing under the BSM model that can be extended to the conditional maximum case and other models. The BSM model implies $\ln S_t = \ln S_0 + (r - \frac{1}{2}\sigma^2)t + \sigma W_t$. Hence, $\widehat{W}_t = (r - \frac{1}{2}\sigma^2)t/\sigma + W_t$ is a drifted Brownian and $S_T = S_0 e^{\sigma \widehat{W}_T}$. Consider the period $[0, T]$ and denote

$$x = \ln \frac{S_0}{H}/\sigma \quad \text{and} \quad y = \ln \frac{S_0}{S_T}/\sigma.$$

Corollary 15.2 (Conditional Barrier-Breach Probability)

$$\mathbb{P}[\min_{0 \leq t \leq T} S_t \leq H | S_T] = \exp\left(-\frac{2}{\sigma^2 T} \ln \frac{S_0}{H} \ln \frac{S_T}{H}\right).$$

Proof By Theorem 15.7, we have

$$\mathbb{P}[\min_{0 \leq t \leq T} S_t \leq H | S_T] = \mathbb{P}[\widehat{m}_T \leq x | \widehat{W}_T = y] = \exp\left(-\frac{2}{\sigma^2 T} \ln \frac{S_0}{H} \ln \frac{S_T}{H}\right).$$

$\square$

Corollary 15.3 (Conditional Minimum Simulation) Monte Carlo simulation for the minimum conditional on S_T can be performed with

$$S_0 \exp\left(\frac{\sigma y}{2} - \sigma \sqrt{\frac{y^2}{4} - \frac{T \ln u}{2}}\right), \quad u \in \mathcal{U}[0, 1]$$

Proof Let $g(x) = \mathbb{P}[\widehat{m}_T \leq x|y]$ as defined in Theorem 15.7. Inverting $g(x) = u$ shows

$$u = \exp\left(-\frac{2}{T}x(x-y)\right) \quad \text{and} \quad x = \frac{y}{2} - \sqrt{\frac{y^2}{4} - \frac{T \ln u}{2}}$$

Hence, the minimum conditional on S_T can be simulated with

$$S_0 e^{\sigma x} = S_0 \exp\left(\frac{\sigma y}{2} - \sigma \sqrt{\frac{y^2}{4} - \frac{T \ln u}{2}}\right).$$

□

Exercises

1. Consider a stochastic integral $I_t = \int_0^t f_u dW_u$ where f_u is *deterministic*, i.e., a non-random function of t .

(1) Show that $\mathbb{E}[I_t] = 0$ and $\text{Var}(I_t) = \int_0^t f_u^2 du$.

- (2) Show that the moment generating function of I_t for any $a \in \mathbb{R}$ is

$$\mathbb{E}[e^{aI_t}] = \exp\left\{\frac{1}{2}a^2 \int_0^t f_u^2 du\right\} = \exp\left\{a\mathbb{E}[I_t] + \frac{1}{2}a^2 \text{Var}(I_t)\right\},$$

i.e., I_t is normally distributed with mean 0 and variance $\int_0^t f_u^2 du$.

2. Let $X_t = W_t^2 - t$ that is an $\mathcal{F}_t$ -measurable martingale. Find the unique predictable process ζ_u as in the predictable martingale representation such that

$$X_t = X_0 + \int_0^t \zeta_u dW_u.$$

3. The following assumes the same notations as in the Feynman–Kac theorem. Let r be a constant. Define f as a function of t and x with

$$f(t, x) = e^{-r(T-t)} \mathbb{E}_t^x[h(X_T)],$$

assuming $\mathbb{E}_t^x[|h(X_T)|] < \infty, \forall x$ and $t < T$. Show that

- (1) $e^{-rt} f(t, X_t)$ is a martingale
- (2) $f(t, x)$ satisfies and terminal condition $f(T, x) = h(x)$ and the PDE

$$\frac{\partial f}{\partial t}(t, x) + \theta(t, x) \frac{\partial f}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 f}{\partial x^2}(t, x) = rf(t, x).$$

4. (1) Derive the Kolmogorov forward equation for Brownian motion W_t .
- (2) Verify that the following transition probability function

$$p(t, x; T, y) = \frac{1}{\sqrt{2\pi(T-t)}} e^{-\frac{(y-x)^2}{2(T-t)}} \text{ satisfies the equation in (1).}$$

5. Use the Breeden–Litzenberger formula and perform $\frac{\partial^2}{\partial K^2}$ on the Black–Scholes call price c (assuming zero dividend) to recover the log-normal probability density function of the underlier at maturity T , i.e.,

$$f(T, K) = \frac{1}{K\sigma\sqrt{2\pi T}} e^{-\frac{[\ln K - \ln S_0 - (r - \frac{1}{2}\sigma^2)T]^2}{2\sigma^2 T}}.$$

Chapter 16

Risk-Neutral Pricing Framework



The risk-neutral pricing framework is about the analysis and techniques for derivatives hedging and pricing. The pioneer work of Black, Scholes and Merton marked the beginning of the development of both theory and practice. The delta (i.e., the quantity of underlying assets for hedging) for some of the derivatives such as forward and futures contracts is constant and can be determined by arbitrage without stochastic modelling (see Part I), while the delta for other derivatives is dynamic and requires stochastic modelling (which also covers forward and futures). The derivative contracts with constant delta property are called *delta-one* products¹ and the other derivatives are often called options generally. As the hedging and pricing analysis is general and applicable to all derivatives, we interchangeably use the terms such as contingent claims, options, derivatives and structured products in this book.

For clarity and learning efficiency, the analysis in this chapter is carried out for a one-factor model with a single underlier, followed by extensions to the multiple underlier and/or multi-factor case.

16.1 Money Market Account and Discounting

Definition 16.1 (Money Market Account) Let r_t be instantaneous interest rate, which can be deterministic or stochastic. The money market account is defined by the following process

$$M_t = e^{\int_0^t r_s ds}.$$

¹ The word “one” comes from 100% that is applied to the contract’s notional to get the delta amount.

In particular, when r_t is a constant r , we get the familiar notations:

$$M_t = e^{rt} \quad \text{and} \quad \frac{1}{M_t} = e^{-rt}.$$

M_t can be considered as the value of a default-free cash deposit account and $\frac{1}{M_t}$ can be used to discount cash flows at time t . We observe the following properties

$$dM_t = r_t M_t dt \quad \text{and} \quad d\frac{1}{M_t} = -\frac{r_t}{M_t} dt. \quad (16.1)$$

The discounting of an asset price S_t is $\frac{S_t}{M_t}$, which can be understood as the number of units of money market account M_t . The role of money market account is the so-called *numéraire*, which is the asset used for financing and will be revisited in Chap. 17. By Eq. (16.1) and $dt dt = 0$, $dt dW_t = 0$, we have the following special results on cross variations with M_t (i.e., money market account numéraire)

$$d\langle \frac{1}{M_t}, S_t \rangle = 0, \quad d\langle M_t, S_t \rangle = 0, \quad d\langle \frac{1}{M_t}, M_t \rangle = 0 \quad \text{and} \quad d\langle M_t, M_t \rangle = 0. \quad (16.2)$$

where S_t is assumed to be an Itô process or a jump-diffusion process.

16.2 Self-Financing Portfolio

16.2.1 Properties of a Self-Financing Portfolio

Consider a portfolio of an asset with total return price S_t (see Sect. 5.2.5) and money market account M_t , i.e., $\Pi_t = \Delta_t S_t + \zeta_t M_t$ where Δ_t, ζ_t are *predictable processes* (which may take negative values) and represent respectively the units of the asset and the units in money market account.

Definition 16.2 (Self-Financing Portfolio) A *self-financing portfolio* is defined as the portfolio satisfying

$$\Pi_T = \Pi_0 + \int_0^T \Delta_t dS_t + \int_0^T \zeta_t dM_t, \quad \forall T.$$

For a self-financing portfolio, the re-balancing of the portions Δ_t and ζ_t in the asset and money market account respectively does not change the value of the portfolio, i.e., no money is brought in or taken out of the portfolio. The discounted portfolio process is $\frac{\Pi_t}{M_t} = \Delta_t \frac{S_t}{M_t} + \zeta_t$, which illustrates that the increment of a discounted self-financing portfolio process is only contributed by the increment of the discounted asset process $\frac{S_t}{M_t}$.

Proposition 16.1 *Portfolio $\Pi_t = \Delta_t S_t + \zeta_t M_t$ is self-financing if and only if*

$$\frac{\Pi_T}{M_T} = \frac{\Pi_0}{M_0} + \int_0^T \Delta_t d \frac{S_t}{M_t}, \quad \forall T, \quad \text{or equivalently } d \frac{\Pi_t}{M_t} = \Delta_t d \frac{S_t}{M_t}. \quad (16.3)$$

Proof It is a special case of Proposition 17.3 with M_t as numéraire. $\square$

Let Δ_t^i be the units of asset S^i at time t . The below proposition extends the above analysis to multi-asset portfolios.

Proposition 16.2 *A portfolio $\Pi_t = \sum_{i=1}^n \Delta_t^i S_t^i + \zeta_t M_t$ is self-financing if and only if*

$$\frac{\Pi_t}{M_t} = \frac{\Pi_0}{M_0} + \sum_{i=1}^n \int_0^t \Delta_u^i d \frac{S_u^i}{M_u}. \quad (16.4)$$

By replacing $\Delta_t[\cdot]$ by $\sum_{i=1}^n \Delta_t^i[\cdot]$ in the proof for the one-dimensional case, we can prove the above proposition.

16.2.2 Excess Return, Self-Financing and Portfolio Return

By $d \frac{S_t}{M_t} = \frac{1}{M_t} dS_t + S_t d \frac{1}{M_t} + d \langle \frac{1}{M_t}, S_t \rangle$ and Eqs. (16.2), the increment of discounted asset price for dt period is

$$d \frac{S_t}{M_t} = \frac{1}{M_t} (dS_t - \frac{S_t}{M_t} dM_t) = \frac{1}{M_t} (dS_t - r_t S_t dt) \quad (16.5)$$

We observe that

- $r_t S_t dt$ is the *financing cost* for one unit of asset;
- $(dS_t - r_t S_t dt)$ is the *self-financing return* and also the *excess return*, which corresponds to practical implementation in the financial market;
- $(dS_t - \frac{S_t}{M_t} dM_t)$ leads to a more general interpretation of the return from a self-financing implementation consisting of a long position of 1 unit of asset S_t financed by a short position of $\frac{S_t}{M_t}$ units of M_t (which is another asset).

By $d \frac{\Pi_t}{M_t} = \frac{1}{M_t} d\Pi_t + \Pi_t d \frac{1}{M_t} + d \langle \Pi_t, \frac{1}{M_t} \rangle$, $d \frac{\Pi_t}{M_t} = \Delta_t d \frac{S_t}{M_t}$ and Eqs. (16.2) and (16.5), we obtain

$$d\Pi_t = \Delta_t \left(dS_t - \frac{S_t}{M_t} dM_t \right) + \frac{\Pi_t}{M_t} dM_t \quad (16.6)$$

The increment of self-financing portfolio $d\Pi_t$ is hence composed of the self-financing return and portfolio carry (see Remark 17.1 for the generalization with other numéraires).

16.3 Risk-Neutral Probability Measure

To simplify the reading, we often use the term *asset* to indicate a tradable asset, security and fully funded contingent claim or financial instrument that does not incur liability for its owner or long party (i.e., its value is non-negative).

Definition 16.3 (Risk-Neutral Probability) A risk-neutral probability measure $\tilde{\mathbb{P}}$, is defined as the one under which the discounted total return price of any tradable asset S_t is a martingale, i.e.,

$$\tilde{\mathbb{E}} \left[\frac{S_t}{M_t} \middle| \mathcal{F}_0 \right] = \frac{S_0}{M_0}.$$

For a portfolio of tradable assets, as its price Π_t is a linear combination of asset total return prices, $\frac{\Pi_t}{M_t}$ must be a martingale under $\tilde{\mathbb{P}}$.

Now, we carry out the continuous-time modelling analysis on a total return asset price that follows the below general stochastic differential equation:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t, \text{ under } \mathbb{P},$$

where μ_t and $\sigma_t > 0$ are $\mathcal{F}_t$ -adapted processes.

Proposition 16.3 *There exists a risk-neutral probability measure $\tilde{\mathbb{P}}$ under which $\frac{S_t}{M_t}$ is a martingale and*

$$d \frac{S_t}{M_t} = \sigma_t \frac{S_t}{M_t} d\tilde{W}_t,$$

or equivalently, in a more popular notation (which does not directly show the martingale property),

$$dS_t = r_t S_t dt + \sigma_t S_t d\tilde{W}_t.$$

Proof By Itô's product rule

$$\begin{aligned} d \frac{S_t}{M_t} &= \frac{1}{M_t} dS_t + S_t d \frac{1}{M_t} + d \left\langle \frac{1}{M_t}, S_t \right\rangle = \frac{1}{M_t} (\mu_t S_t dt + \sigma_t S_t dW_t) - r_t \frac{S_t}{M_t} dt \\ &= (\mu_t - r_t) \frac{S_t}{M_t} dt + \sigma_t \frac{S_t}{M_t} dW_t = \sigma_t \frac{S_t}{M_t} \left(dW_t + \frac{\mu_t - r_t}{\sigma_t} dt \right). \end{aligned}$$

Applying Girsanov theorem with $Z_t = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp \left\{ - \int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right\}$, where $\theta_t = (\mu_t - r_t)/\sigma_t$, we obtain $d\tilde{W}_t = dW_t + \theta_t dt$ such that

$$d \frac{S_t}{M_t} = \sigma_t \frac{S_t}{M_t} d\tilde{W}_t,$$

which is a martingale under $\tilde{\mathbb{P}}$. From $d \frac{S_t}{M_t} = \frac{1}{M_t} (dS_t - r_t S_t dt)$, we obtain

$$dS_t = r_t S_t dt + \sigma_t S_t d\tilde{W}_t. \quad \square$$

Actually, probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$ are not abstract concepts. They are implied by the SDEs $dS_t = \mu_t S_t dt + \sigma_t S_t dW_t$ and $dS_t = r_t S_t dt + \sigma_t S_t d\tilde{W}_t$ with initial value S_0 . For example, the distribution of S_t under $\tilde{\mathbb{P}}$ can be simulated by Monte Carlo method (see Sect. 21) with its SDE, which provides the probability of S_t falling into any interval.

Remark 16.1 (Drift for Excess Return Asset Process) From $d\hat{S}_t = \sigma_t S_t d\tilde{W}_t$, where $d\hat{S}_t = dS_t - r_t S_t dt$, we notice that the price process drift of the excess return asset $\hat{S}_t$ is zero. If $r_t > 0, \forall t$, a bullish product linked to $\hat{S}_t$ is cheaper than the product linked to S_t because the forward of $\hat{S}_t$ is lower. $\square$

For the multi-factor case, i.e., $dS_t = \mu_t S_t dt + S_t \sum_i \sigma_t^i dW_t^i$, we can follow the same approach in the proof but with the multi-dimensional Girsanov theorem to obtain $d \frac{S_t}{M_t} = \frac{S_t}{M_t} \sum_i \sigma_t^i d\tilde{W}_t^i$.

Proposition 16.4 *Let Π_t be a self-financing portfolio, the process $\frac{\Pi_t}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale:*

$$\tilde{\mathbb{E}} \left[\frac{\Pi_t}{M_t} \middle| \mathcal{F}_s \right] = \frac{\Pi_s}{M_s}, \quad s \leq t.$$

Proof Take expectation under $\tilde{\mathbb{P}}$ on both sides of Eq. (16.4). The integral term is zero because $\frac{S_t^i}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale. $\square$

16.4 Pricing and Hedging of Derivatives

Let V_T be the terminal payoff at T of a derivative instrument on S_T . If there exists a self-financing process Π_t such that $\Pi_T = V_T$, then, the time- t ($0 \leq t \leq T$) price of the derivative instrument V_t can be defined as $V_t \equiv \Pi_t$ because Π_t can perfectly replicate the derivative's price process as described by Eq. (16.3). The pricing formula can be obtained by taking expectation on Eq. (16.3):

$$V_t = M_t \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right] \text{ (which is also implied by Proposition 16.4).}$$

Actually, the existence assumption of a replicating process can be relaxed. Let's define directly the continuous *price process* for a derivative by the integrable function V_t :

$$V_t = M_t \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right]. \quad (16.7)$$

In particular, if $M_t = e^{rt}$, then we have $M_0 = 1$ and

$$V_t = \tilde{\mathbb{E}} \left[e^{-r(T-t)} V_T | \mathcal{F}_t \right], \quad V_0 = \tilde{\mathbb{E}} \left[e^{-rT} V_T | \mathcal{F}_0 \right]$$

The following proposition confirms that V_t is equal to a self-financing hedging portfolio's value, which justifies the definition.

Proposition 16.5 $\forall 0 \leq t \leq T$,

- (1) $\frac{V_t}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale and there is a process Δ_t such that

$$\frac{V_T}{M_T} = \frac{V_t}{M_t} + \int_t^T \Delta_u d \frac{S_u}{M_u} \quad (16.8)$$

- (2) there exists a self-financing portfolio $\Pi_t = \Delta_t S_t + \zeta_t M_t$ such that $\Pi_t = V_t$.

Proof We prove the two results in the proposition separately.

- (1) By iterated conditioning, we can check that $\frac{V_t}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale:

$$\tilde{\mathbb{E}} \left[\frac{V_t}{M_t} | \mathcal{F}_u \right] = \tilde{\mathbb{E}} \left[\tilde{\mathbb{E}} \left[\frac{V_T}{M_T} | \mathcal{F}_t \right] | \mathcal{F}_u \right] = \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} | \mathcal{F}_u \right] = \frac{V_u}{M_u}, \quad \forall 0 \leq u < t \leq T.$$

Hence, by the predictable martingale representation property, there exists a unique predictable process ϕ_u , such that $\frac{V_T}{M_T} = \frac{V_t}{M_t} + \int_t^T \phi_u d\tilde{W}_u$. Setting $\Delta_u = \phi_u / (\sigma_u \frac{S_u}{M_u})$, we obtain

$$\frac{V_T}{M_T} = \frac{V_t}{M_t} + \int_t^T \Delta_u \sigma_u \frac{S_u}{M_u} d\tilde{W}_u = \frac{V_t}{M_t} + \int_t^T \Delta_u d \frac{S_u}{M_u} \quad (\text{by Proposition 16.3}).$$

- (2) Defining $\zeta_t = \frac{1}{M_t} (V_t - \Delta_t S_t)$ and $\Pi_t = \Delta_t S_t + \zeta_t M_t = V_t$. We have,

$$\frac{\Pi_T}{M_T} = \frac{\Pi_t}{M_t} + \int_t^T \Delta_u d \frac{S_u}{M_u}.$$

By Proposition 16.1, Π_t is a self-financing portfolio. □

This proposition shows that the pricing formula (16.7) must be satisfied. Otherwise, there will be an arbitrage by a self-financing portfolio. As we assume no risk-free arbitrage, there must be only *one price* for any tradable asset and its derivatives at time t , which implies that the risk-neutral probability is unique *almost-surely*.

The generalization of the proposition to a multi-asset derivative is

$$\frac{V_T}{M_T} = \frac{V_t}{M_t} + \sum_{i=1}^n \int_t^T \Delta_u^i d \frac{S_u^i}{M_u}, \quad 0 \leq t \leq T \quad (M_0 = 1). \quad (16.9)$$

The proof requires the multi-dimensional martingale representation and definition of $\Delta_u^i = \phi_u^i / (\sigma_u^i \frac{S_u^i}{M_u})$.

Now, we determine the hedging parameter Δ_t .

Proposition 16.6 *Let $dS_t = r_t S_t dt + \sigma_t S_t d\tilde{W}_t$. For $t < T$, if V_t is continuously differentiable w.r.t. t and twice continuously differentiable w.r.t. S_t , then*

- (1) *the delta amount is $\Delta_t = \frac{\partial V_t}{\partial S_t}$*
- (2) *while S_t/M_t and V_t/M_t are both $\tilde{\mathbb{P}}$ -martingales, V_t must satisfy the well-known **Black–Scholes equation***

$$\frac{\partial V_t}{\partial t} + r_t S_t \frac{\partial V_t}{\partial S_t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2} - r_t V_t = 0 \quad (16.10)$$

Proof Proposition 16.5 shows that both $\frac{S_t}{M_t}$ and $\frac{V_t}{M_t}$ are $\tilde{\mathbb{P}}$ -martingales. By Itô's product rule and Itô–Doebelin formula, the differential of $\frac{V_t}{M_t}$ is

$$\begin{aligned} d \frac{V_t}{M_t} &= \frac{1}{M_t} \left(-r_t V_t dt + \frac{\partial V_t}{\partial t} dt + \frac{\partial V_t}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 V_t}{\partial S_t^2} d\langle S_t, S_t \rangle_t \right) \\ &= \frac{1}{M_t} \left(-r_t V_t + \frac{\partial V_t}{\partial t} + r_t S_t \frac{\partial V_t}{\partial S_t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2} \right) dt + \frac{\partial V_t}{\partial S_t} \sigma_t \frac{S_t}{M_t} d\tilde{W}_t. \end{aligned}$$

Since $\frac{V_t}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale, the dt term should be zero, which leads to the Black–Scholes equation in (2). By Proposition 16.3, we have $d \frac{V_t}{M_t} = \frac{\partial V_t}{\partial S_t} \sigma_t \frac{S_t}{M_t} d\tilde{W}_t = \frac{\partial V}{\partial S} d \frac{S_t}{M_t}$, while Proposition 16.5 shows that $d \frac{V_t}{M_t} = \Delta_t d \frac{S_t}{M_t}$. Hence, we obtain the delta amount in (1) by comparing these two equations. $\square$

For multi-asset derivatives, the same analysis using multi-dimensional Itô–Doebelin formula and (16.9) leads to

$$\Delta_t^i = \frac{\partial V_t}{\partial S_t^i} (t, S_t^1, \dots, S_t^i, \dots).$$

16.5 Discussion on Hedging, Pricing and Risk-Neutral Framework

Self-financing must be satisfied in hedging transactions. Including a new self-financing position in a portfolio at time- t does not change the value of the portfolio at time- t , e.g., assets funded by borrowing in money market for delta-hedging,

funded options for vega hedging. As shown in Sect. 16.2, a self-financing investment return can be derived from the increments of the discounted asset prices.

The purpose of a risk-neutral probability $\tilde{\mathbb{P}}$ is to find a self-financing hedging process for derivatives, which is achieved by constructing $\tilde{\mathbb{P}}$ in such a way that both the discounted underlying asset price and the discounted option price are $\tilde{\mathbb{P}}$ -martingales. $\tilde{\mathbb{P}}$ is a mere mathematical tool and there is no use to assign an economic meaning to it. To illustrate the hedging process, we rewrite the formula (16.8)

$$V_T = \frac{M_T}{M_t} V_t + \int_t^T M_T \Delta_u d \frac{S_u}{M_u}. \quad (16.11)$$

Expanding $d \frac{S_u}{M_u}$ with $M_u = e^{ru}$, we obtain the below equivalent self-financing hedging process

$$V_T = e^{r(T-t)} V_t + \int_t^T e^{r(T-u)} \Delta_u (dS_u - S_u r du). \quad (16.12)$$

where

- the term $e^{r(T-t)} V_t$ is the accrued value of the initial option price V_t ;
- the integral term is the accrued hedging P/L, in which, $r S_u du$ represents the financing cost for holding one underlying asset unit over du period and $(dS_u - S_u r du)$, or equivalently $(dS_u - \frac{S_u}{M_u} dM_u)$, is the self-financing return for the period; The hedging process consists of holding Δ_u asset units at u by self-financing:
 - if $\Delta_u \geq 0$, the cash amount of $\Delta_u S_u$ is borrowed to purchase Δ_u asset units;
 - otherwise, $|\Delta_u|$ asset units are borrowed and sold at the prevailing price S_u with the selling proceeds $|\Delta_u| S_u$ lent to money market.

Given a hedging process as described by (16.12) or (16.11), it seems to be possible to derive the value of V_t if the expectation of the integral term is zero under a certain probability. Actually, a risk-neutral probability $\tilde{\mathbb{P}}$ is the one to make it happen: under $\tilde{\mathbb{P}}$, as $\frac{S_u}{M_u}$ is a martingale, the expectation of the increment of $\frac{S_u}{M_u}$ must be zero and the expectation of the integral of increments of $\frac{S_u}{M_u}$ must also be zero. This leads to the general pricing formula:

$$V_t = \tilde{\mathbb{E}} \left[e^{-r(T-t)} V_T \middle| \mathcal{F}_t \right], \text{ or more generally, } V_t = M_t \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right].$$

To show the hedging over the period dt , we can derive formula (16.8) to obtain the below formula (see also formula (16.6) and Remark 17.1):

$$dV_t = \Delta_t \left(dS_t - \frac{S_t}{M_t} dM_t \right) + \frac{V_t}{M_t} dM_t. \quad (16.13)$$

It shows that dV_t is the sum of the self-financing delta-hedging P/L and the carry of the cash value of the hedging portfolio in money market. We note that $dM_t (= M_t r dt)$ can be understood as the interest produced by the cash amount of M_t in money market.

Δ_t is equal to $\frac{\partial V_t}{\partial S_t}$, which is the ratio of option price change w.r.t. asset price change over dt period. Consider the short position of one call option on a stock as illustration example. If the ratio is 0.5 at time- t , it is intuitive that the holding of 0.5 unit of the stock may roughly replicate the option price change by the stock price change: if the stock price increases by \$1, the option price will increase by roughly \$0.5 while the holding of 0.5 stock will also have a gain of \$0.5. However, the option price convexity and self-financing cost may make people doubt about the delta-hedging accuracy. Magically, the suspected “error” is actually perfectly offset by the time decay of the option price (see discussion on Gamma/Theta in Sect. 14.4).

16.6 The Black–Scholes–Merton Model Revisited

Recall that under the BSM (Black–Scholes–Merton) model, an asset’s total return price follows $dS_t = \mu S_t dt + \sigma S_t dW_t$ under $\mathbb{P}$ where σ is the constant volatility and μ is the drift. Let r be the constant money market rate. By Proposition 16.3, we have $d\frac{S_t}{M_t} = \sigma \frac{S_t}{M_t} d\tilde{W}_t$, where $Z_t = \left. \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp \left\{ -\theta W_t - \frac{1}{2} \theta^2 t \right\}$, with $\theta = (\mu - r)/\sigma$. Under $\tilde{\mathbb{P}}$, $\frac{S_t}{M_t}$ is a martingale and the SDE for asset price process is

$$dS_t = r S_t dt + \sigma S_t d\tilde{W}_t.$$

According to Proposition 14.2, the solution to the SDE starting at S_t , $0 \leq t < T$, is

$$S_T = S_t e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma \tilde{W}_{T-t}}. \quad (16.14)$$

16.6.1 Closed-Form Formulas for European Vanilla Options

Closed-form formulas for European call and put options can be derived by solving the pricing formula. We take a call option paying out $c_T = (S_T - K)^+$ at maturity T for illustration. Denote $\tau = T - t$, the time- t option price is

$$\begin{aligned} c_t &= M_t \tilde{\mathbb{E}} \left[\left. \frac{c_T}{M_T} \right| \mathcal{F}_t \right] = e^{-r\tau} \tilde{\mathbb{E}} [(S_T - K)^+ | \mathcal{F}_t] \\ &= e^{-r\tau} \tilde{\mathbb{E}} [S_T \mathbf{1}_{\{S_T > K\}} | \mathcal{F}_t] - K e^{-r\tau} \tilde{\mathbb{E}} [\mathbf{1}_{\{S_T > K\}} | \mathcal{F}_t] = A - B. \end{aligned}$$

Firstly, by Eq. (16.14), the condition $S_T > K$ is equivalent to $\tilde{W}_\tau > -d_2\sqrt{\tau}$, where $d_2 = (\ln \frac{S_t}{K} + (r - \frac{1}{2}\sigma^2)\tau) / \sigma\sqrt{\tau}$. Denoting $\epsilon \sim \mathcal{N}(0, 1)$, it follows that

$$\begin{aligned} B &= e^{-r\tau} K \tilde{\mathbb{E}}[\mathbf{1}_{\{\tilde{W}_\tau > -d_2\sqrt{\tau}\}} | \mathcal{F}_t] = e^{-r\tau} K \tilde{\mathbb{E}}[\mathbf{1}_{\{\tilde{W}_\tau > -d_2\sqrt{\tau}\}}] \\ &\quad (\tilde{W}_\tau \text{ is independent of } \mathcal{F}_t) \\ &= e^{-r\tau} K \tilde{\mathbb{P}}\{\tilde{W}_\tau > -d_2\sqrt{\tau}\} = e^{-r\tau} K \tilde{\mathbb{P}}\{\epsilon > -d_2\} \\ &= e^{-r\tau} K \tilde{\mathbb{P}}\{\epsilon < d_2\} \quad (\text{by symmetry of normal distribution}) \\ &= e^{-r\tau} K N(d_2). \end{aligned}$$

We can see that $N(d_2)$ is the $\tilde{\mathbb{P}}$ -probability for a call option to end up in-the-money.

$$A = e^{-r\tau} \tilde{\mathbb{E}}[S_T \mathbf{1}_{\{S_T > K\}} | \mathcal{F}_t] = \frac{S_t}{Z_t} \tilde{\mathbb{E}}\left[Z_T \mathbf{1}_{\{\tilde{W}_\tau > -d_2\sqrt{\tau}\}} \middle| \mathcal{F}_t\right],$$

$$\text{where } Z_u = e^{-\frac{1}{2}\sigma^2 u + \sigma \tilde{W}_u}.$$

Z_T in the expectation can be eliminated by the probability change with $Z_t = \frac{d\mathbb{Q}}{d\tilde{\mathbb{P}}} \Big|_{\mathcal{F}_t}$.

$$\begin{aligned} A &= \frac{S_t}{Z_t} \tilde{\mathbb{E}}[Z_T \mathbf{1}_{\{\tilde{W}_\tau > -d_2\sqrt{\tau}\}} | \mathcal{F}_t] = S_t \tilde{\mathbb{E}}^{\mathbb{Q}}[\mathbf{1}_{\{\tilde{W}_\tau > -d_2\sqrt{\tau}\}} | \mathcal{F}_t] \quad (\text{by Bayes formula}) \\ &= S_t \tilde{\mathbb{E}}^{\mathbb{Q}}[\mathbf{1}_{\{\tilde{W}_\tau^{\mathbb{Q}} + \sigma\tau > -d_2\sqrt{\tau}\}} | \mathcal{F}_t] \quad (\text{by Girsanov theorem}) \\ &= S_t \mathbb{Q}\{\tilde{W}_\tau^{\mathbb{Q}} > -d_2\sqrt{\tau} - \sigma\tau\} = S_t \mathbb{Q}\{\epsilon > -d_2 - \sigma\sqrt{\tau}\} \\ &= S_t \mathbb{Q}\{\epsilon < d_2 + \sigma\sqrt{\tau}\} = S_t N(d_2 + \sigma\sqrt{\tau}). \end{aligned}$$

We note that the change of probability simplifies the derivation of A , but we can also directly perform integration to get the same solution. From $c_t = A - B$, we obtain the well-known *Black-Scholes formula* for European call options

$$c_t = S_t N(d_1) - e^{-r(T-t)} K N(d_2),$$

where $d_1 = d_2 + \sigma\sqrt{T-t}$ and $d_2 = \left[\ln \frac{S_t}{K} + (r - \frac{1}{2}\sigma^2)(T-t) \right] / (\sigma\sqrt{T-t})$.

16.6.2 Vega-Gamma Relationship*

With risk-neutral pricing framework, we can prove the following relationship.

Proposition 16.7 (Vega-Gamma Relationship) *Consider a European option under the BSM model. Let σ be the volatility of the underlier, $\tau = T - t$ the*

time to maturity of the option and $\zeta_t = \sigma^2 \tau$. The option admits the following Vega-Gamma relationship:

$$\frac{\partial V_t}{\partial \sigma} = \sigma \tau S_t^2 \frac{\partial^2 V_t}{\partial S_t^2}, \quad \text{or equivalently} \quad \frac{\partial V_t}{\partial \zeta_t} = \frac{1}{2} S_t^2 \frac{\partial^2 V_t}{\partial S_t^2}.$$

Proof Using the usual notations for the BSM model, we have $\ln S_T = \ln S_t + r\tau - \frac{1}{2}\sigma^2\tau + \sigma\tilde{W}_\tau$. Denoting $Y = \ln S_T$ and $m = \ln S_t + r\tau - \frac{1}{2}\sigma^2\tau$, we have $Y \sim \mathcal{N}(m, \zeta_t)$ with density $f(y)$. Since $\tilde{W}_\tau$ is independent of $\mathcal{F}_t$, the general risk-neutral pricing formula for European options can be written as

$$\begin{aligned} V_t &= \tilde{\mathbb{E}}[e^{-r\tau} V_T | \mathcal{F}_t] = \tilde{\mathbb{E}}[e^{-r\tau} h(Y)] \\ &= \int_{-\infty}^{+\infty} e^{-r\tau} h(y) f(y) dy \equiv \int_{-\infty}^{+\infty} e^{-r\tau} h(y) \phi(x) dy, \end{aligned}$$

where $\phi(x) = \frac{1}{\sqrt{2\pi\zeta_t}} e^{-\frac{x^2}{2\zeta_t}}$ is the p.d.f. of $\mathcal{N}(0, \zeta_t)$ with $x = y - m = y - \ln S_t - r\tau + \frac{1}{2}\sigma^2\tau$. The vega and gamma can be determined respectively as follows:

$$\frac{\partial V_t}{\partial \sigma} = \int_{-\infty}^{+\infty} e^{-r\tau} h(y) \frac{\partial \phi}{\partial \sigma} dy, \quad \frac{\partial^2 V_t}{\partial S_t^2} = \int_{-\infty}^{+\infty} e^{-r\tau} h(y) \frac{\partial^2 \phi}{\partial S_t^2} dy.$$

To simplify the proof, we use Plackett's identity $p = \frac{\partial \phi}{\partial \zeta_t} = \frac{1}{2} \frac{\partial^2 \phi}{\partial x^2}$ (see Appendix A.5). Since ϕ is a function of S_t and σ implicitly, we can calculate the total derivatives,

$$\begin{aligned} \frac{\partial \phi}{\partial \sigma} &= \frac{\partial \phi}{\partial \zeta_t} \frac{\partial \zeta_t}{\partial \sigma} + \frac{\partial \phi}{\partial x} \frac{\partial x}{\partial \sigma} = \sigma \tau \left(2p + \frac{\partial \phi}{\partial x} \right), \\ \frac{\partial^2 \phi}{\partial S_t^2} &= \frac{\partial}{\partial S_t} \left(\frac{\partial \phi}{\partial x} \frac{\partial x}{\partial S_t} \right) = \frac{\partial}{\partial S_t} \left(\frac{\partial \phi}{\partial x} \frac{-1}{S_t} \right) = \frac{1}{S_t^2} \left(2p + \frac{\partial \phi}{\partial x} \right). \end{aligned}$$

Hence, we obtain $\frac{\partial \phi}{\partial \sigma} = \sigma \tau S_t^2 \frac{\partial^2 \phi}{\partial S_t^2}$, which, combined with $\frac{\partial V_t}{\partial \sigma} = \frac{\partial V_t}{\partial \zeta_t} \frac{\partial \zeta_t}{\partial \sigma} = 2\sigma \tau \frac{\partial V_t}{\partial \zeta_t}$, proves the proposition.

Instead of using Plackett's identity, direct (tedious) calculations of the partial derivatives of ϕ w.r.t. σ and S_t^2 can also lead to the result to be proved. $\square$

16.7 Dividend Modelling

16.7.1 Dividend Types

Many assets, such as equities, pay out dividends. The price of an option on an asset that pays dividends is affected by the payments. There are mainly three types of dividend modelling:

Cash dividend: dividends are paid as fixed cash amounts at discrete times. This is the most common dividend form;

Proportional dividend: dividend amount of $b_{t_i} S_{t_i}^-$ are paid at discrete times;

Continuous dividend: continuous dividend does not exist in reality, but it is the easiest to model. It is applied when this approximation does not have a significant impact.

Applying fixed cash dividend for long term is not a viable dividend modelling approach (a stock of \$100 paying \$3 dividend will not still pay \$3 when its price drops to \$10). In practice, the first cash dividend (if applicable) is more or less predictable as it is linked to the forthcoming profits of the company to which analysts provide valuable forecasts. Hence, the near term cash dividend may be used in modelling, as shown in the below example:

- cash dividends for short term (e.g., ≤ 2 years)
- proportional dividends for long term (e.g., ≥ 5 years)
- mixture of cash dividends and proportional dividends in between.

The general principle to derive the risk-neutral process of a dividend-paying asset is that the discounted total return price incorporating all the dividends and repo payments is a $\tilde{\mathbb{P}}$ -martingale.

16.7.2 Continuous Dividend Modelling

The asset price model with repo and continuous dividend is

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t - q_t S_t dt,$$

where q_t is the sum of dividend yield and repo rate. The term $q_t S_t dt$ can be understood as the value reduction in asset price after dividend. Let $S_t^* = e^{\int_0^t q_s ds} S_t$ denote the total return price of the asset with dividends reinvested (where $e^{\int_0^t q_s ds}$ represents the asset units increased by the dividend and repo reinvestment, see

Sect. 5.2.5). S_t^* can be considered as a non-dividend paying asset, its discounted value S_t^*/M_t should be a martingale under risk-neutral probability $\tilde{\mathbb{P}}$:

$$d \frac{S_t^*}{M_t} = \sigma_t \frac{S_t^*}{M_t} d\tilde{W}_t \quad (16.15)$$

As $d \frac{S_t^*}{M_t} = e^{\int_0^t (q_s - r_s) ds} [(q_t - r_t) S_t dt + dS_t]$ by Itô's product rule and $\frac{S_t^*}{M_t} = e^{\int_0^t (q_s - r_s) ds} S_t$, Eq. (16.15) is equivalent to the below SDE

$$dS_t = (r_t - q_t) S_t dt + \sigma_t S_t d\tilde{W}_t \quad \text{under } \tilde{\mathbb{P}}. \quad (16.16)$$

The Black–Scholes equations for derivatives price V_t becomes

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + (r_t - q_t) S_t \frac{\partial V}{\partial S} - r_t V = 0,$$

which is obtained by following the proof in Proposition 16.6.

European call price c_t and put option price p_t can be derived from Eq. (16.16). In particular, with constant continuous dividend q , we have

$$\begin{aligned} c_t &= e^{-q(T-t)} S_t N(d_1) - K e^{-r(T-t)} N(d_2), \\ p_t &= K e^{-r(T-t)} N(-d_2) - e^{-q(T-t)} S_t N(-d_1), \end{aligned} \quad (16.17)$$

where

$$d_1 = d_2 + \sigma \sqrt{T-t}, \quad d_2 = \frac{\ln(S_t/K) + (r - q - \frac{1}{2}\sigma^2)(T-t)}{\sigma \sqrt{T-t}}.$$

The above formulas lead to the modified version of put-call parity:

$$S_t e^{-q(T-t)} + p_t - c_t = K e^{-r(T-t)},$$

which can also be verified by a no-arbitrage argument.

16.7.3 Discrete Dividend Modelling*

A discrete dividend paid at t_i can be generally represented by $d_i = b_i S_{t_i^-} + a_i$ where b_i is the proportional dividend yield and a_i the cash dividend part. We assume that b_i and a_i are not random.

Considering the value decrease due to discrete dividend payments, the asset price S_t and the dividend assumption d_i follow the below SDE under $\tilde{\mathbb{P}}$:

$$dS_t = (r_t - q_t)S_t dt + \sigma_t S_t d\tilde{W}_t - \sum_i d_i \delta(t - t_i), \quad \text{s.t. } d_i < S_{t_i}^-,$$

where $\delta(\cdot)$ is the Dirac delta function (see Appendix A.6) and the condition $d_i < S_{t_i}^-$ is to let S_t stay non-negative. If d_i only contains proportional dividend, S_t will always be non-negative. But with the presence of cash dividend, adjustment should be made in numerical implementations to prevent negative S_t values.

In modelling, instead of making up a complex stochastic process to tolerate unreasonable dividend assumptions, it is better to focus on the dividend assumption improvement and to use a simple stochastic process with adjusted cash dividend (e.g., $\max\{0, \min\{a_i, S_{t_i}^-(1 - b_i) - \epsilon\}\}$, ϵ being a small positive number to prevent negative asset price) in numerical implementation. As explained earlier, fixed cash dividend assumption for the long-term part does not make economic sense as both very low and very high S_t levels have non-negligible probabilities and proportional dividend should be used instead. For the short-term part, the technical adjustment on cash dividend to keep S_t positive will not have a significant impact on pricing anyway, as the probability is very low.

It is often useful to use a “synthetic total return asset” to work on. If we choose to re-invest repo return into the stock and to re-invest cash dividends into money market, then the discounted total return process is

$$X_t = \frac{1}{M_t} \beta(t) [S_t + \alpha(t)] \quad (16.18)$$

where $\beta(t)$ and $\alpha(t)$ are defined by formula (5.5). X_t should be a positive $\tilde{\mathbb{P}}$ -martingale that can be written as:

$$dX_t = X_t \sigma_X d\tilde{W}_t. \quad (16.19)$$

We note that the pricing can be conducted with X_t instead of S_t . For instance,

$$(S_T - K)^+ = \left(\frac{M_T}{\beta(T)} X_T - \alpha(T) - K \right)^+ = \frac{M_T}{\beta(T)} (X_T - K')^+ \quad (16.20)$$

where $K' = \frac{\beta(T)}{M_T} [K + \alpha(T)]$. Therefore, the call option prices for X_t can be determined from the call options prices for S_t with adjusted strike and notional. In other terms, the “implied volatilities” for X_t can be determined from the implied volatilities for S_T . A direct application of the above analysis will be presented in Sect. 25.1.3 for the local volatility model.

16.7.4 Adjustment to Derivatives for Corporate Actions*

In option and forward contract pricing, assumptions are made for dividends. The parties of an option trade take the dividend risk that the effective paid dividends may differ from the assumed levels.

As explained in Sect. 5.2.4, for the below situations, the parties of the trade usually choose to keep valuation continuity by adjusting some of the contract parameters/terms upon a corporate action:

- exceptional corporate actions such as dividend, rights issue or split that are not scheduled at the inception of the contract;
- large size or illiquid trades: it is a common practice to price such derivative contracts with no corporate action assumption and to adjust the contract terms upon event, scheduled or not. For instance, zero dividend assumption can be used in the contract, which leads to adjustment on the strike and share notional at each dividend payment (i.e., treated as exceptional dividend). This method is often used for convertible bonds (see Sect. 31.8.1) and strategic equity derivatives trades due to their large sizes.

Valuation continuity can be achieved by adjusting the notional amount as well as the initial spot related parameters such as strike and barrier of the option so that the option valuation remains unchanged by the corporate action. This type of adjustment is covered in the term sheet and agreed by both parties of the option transaction, which follows the underlier's price adjustment explained in Sect. 5.2.6 for corporate actions.

For example, the adjustment methodology for a European single stock call option following an exceptional corporate action (e.g., exceptional dividend) can be illustrated by

$$N\tilde{\mathbb{E}}_t[(S_T^* - K)^+] = N\tilde{\mathbb{E}}_t[(S_T \frac{1}{1-d} - K)^+] = \hat{N}\tilde{\mathbb{E}}_t[(S_T - \hat{K})^+] \quad (16.21)$$

where S_T^* is the adjusted underlier price at T , d the value change ratio as defined in Sect. 5.2.6, $\alpha = 1 - d$, $\hat{N} = \frac{N}{\alpha}$ and $\hat{K} = \alpha K$. Hence, the adjustment² consists of changing the *share notional* to N/α and *strike* to αK .

Obviously, it can be applied to put options and forward contracts. Similar methods can be derived for the adjustment factors for other options, such as an Asian option that has already started with some known fixings.

² Here, forward adjustment method has been applied to the final price S_T . Alternatively, we can use backward adjustment by rewriting the option in terms of cash notional and adjusting S_0 : $N S_0 \tilde{\mathbb{E}}_t[(\frac{S_T}{S_0} - \frac{K}{S_0})^+]$ with $S_0^* = S_0(1 - d)$, which leads to the same result.

16.8 Collateralized Derivative Pricing and FVA*

As introduced in Chap. 1, an OTC derivative may be traded under a master agreement such as ISDA/CSA where the CSA (Credit Support Annex) agreement sets out the collateral posting terms and conditions. The derivative pricing framework presented so far has not considered the impact of collateral. In the following, we will adjust the standard derivative pricing framework to account for collateral posting.

We consider the situation for an OTC derivative where the CSA agreement between the two counterparties has the following terms:

- zero threshold for posting collateral
- zero MTA (Minimum Transfer Amount)
- collateral is cash in the same currency as the denomination for the derivative transaction
- the interest rate applied to the collateral is r^c (e.g., risk-free rate, see Chap. 3)

Under such condition, the party having positive mark-to-market V_t (say, party A) will receive cash collateral from the other party (say, party B) and will pay interest on the cash collateral at the rate of r^c . If the normal financing rate for party A is r (which could be internal financing rate or unsecured funding rate), then the cash collateral with interest rate r^c implies a PnL of $(r_t - r_t^c) \times V_t \times \delta_t$ for a small period δ_t to party A. Recall that V_t represents the value of the self-financing portfolio replicating the payoff of the derivative. To take into account this PnL impact from r^c , we set $V_t^* = V_t e^{\int_0^t (r_u - r_u^c) du}$, which is the “total price” of the derivative. It implies that any interest received/paid will be invested in/divested from V_t . Under $\mathbb{P}$, its discounted value V_t^*/M_t should be a martingale. Hence, by martingale representation property, we have

$$\frac{V_T^*}{M_T} = \frac{V_0^*}{M_0} + \int_0^T \phi_u d\tilde{W}_u,$$

where ϕ_u is a predictable process, $\frac{V_T^*}{M_T} = \frac{V_T e^{\int_0^T (r_u - r_u^c) du}}{M_T} = V_T e^{-\int_0^T r_u^c du}$ and $\frac{V_0^*}{M_0} = V_0$. Taking conditional expectation, we obtain

$$V_0 = \tilde{\mathbb{E}} \left[e^{-\int_0^T r_u^c du} V_T \mid \mathcal{F}_0 \right]. \quad (16.22)$$

We highlight that r^c appears only in the discounting term. The interest rate in the risk-neutral SDE for the underlier is still the financing rate r , i.e., for the BSM model, $dS_t = rS_t dt + \sigma S_t d\tilde{W}_t$.

This pricing framework is known as the method of *CSA discounting* or *OIS discounting*. Its application needs to take into account the CSA terms and conditions

on the collateral type, threshold/MTA and netting, etc. There is an abundance of literature on this topic after the financial crisis of 2008, e.g., [91].

For counterparties with CSA, the **FVA (Funding Value Adjustment)** is caused by the difference between the interest rate for hedging and the interest rate applied on the collateral for the trade's MtM exposure. The value of FVA is actually the expected funding P/L over the life of the trade. It can be calculated as the difference between the pricing with CSA discounting and the pricing without CSA discounting on the future cash flows of the transaction.

16.9 Futures and Forward Modelling*

In this section, we analyze the modeling of futures and forward prices for contracts maturing at T assuming that the margining model is STM (settled-to-market) for futures, but CTM (collateralized-to-market) for forward contracts (see Sect. 2.2). In the below, U_t represents the underlier's time- t price or rate ($t < T$).

The underlier's price in a contract linked to equity, commodity, or FX is a tradable asset price S_t (FX rate is considered as the price of ccy1 in terms of ccy2). But for FRA and interest rate futures, the underlier's price is the interest rate R_t that is not an asset.

We use F_t to denote the time- t futures price for all types of futures except interest rate futures for which, F_t denotes $(1 - R_t)$ where R_t is the time- t underlying interest rate.

Denote the discount factor by $D(t, T) = \exp(-\int_t^T r_s ds)$. We know that $P_t(T) = \tilde{\mathbb{E}}_t[D(t, T)]$ where $P_t(T)$ is the zero coupon bond price at t (see formula (27.1)). As a forward or futures is equivalent to a portfolio of call and put options (both are tradable assets) by put/call parity, the discounted payoffs should be martingale under risk-neutral measure $\tilde{\mathbb{P}}$.

16.9.1 Futures

By its design, a futures contract has the following features:

- futures' final price converges to underlier's price: $F_T = 1 - R_T$ for interest rate futures and $F_T = S_T$ for all the other futures;
- taking a new position with futures contracts involves no financing as the valuation is zero at the time of transaction;³

³ The revenue from the initial margin, either cash or securities, belongs to the margin poster, which approximately offsets the financing cost. The optimization of collateral management may have a significant economic impact in practice.

- the P/L of a futures position is settled daily with the margin account of the participant;

Now, let t^+ represent the closest instant after t that the futures contract can be traded. Then, the holding of a long futures position can be considered as the following repeating transaction process until T : opening of a long position at $t \rightarrow$ closing of the position at t^+ , which leads to the settlement of the P/L linked to $(F_{t^+} - F_t) \rightarrow$ opening a new position at t^+ at $F_{t^+} \rightarrow, \dots$, and so on, with $F_T = S_T$.

Under risk-neutral probability $\tilde{\mathbb{P}}$, the discounted value of the payoff proportional to $(F_{t^+} - F_t)$ settled at t^+ should be a martingale with an expected value of zero at t , i.e., $0 = \tilde{\mathbb{E}}_t[D(t, t^+)(F_{t^+} - F_t)] \approx D(t, t^+)\tilde{\mathbb{E}}_t[F_{t^+} - F_t]$ as $(t^+ - t)$ is small (i.e. $D(t, t^+)$ is considered to be $\mathcal{F}_t$ -measurable). This observation leads to the modelling of F_t with the following features:

- $F_t = \tilde{\mathbb{E}}_t[F_{t^+}]$ and it is natural to consider F_t as a martingale, i.e., $F_t = \tilde{\mathbb{E}}_t[F_T]$;
- $F_t = \tilde{\mathbb{E}}_t[S_T]$; Actually, $F_t = \tilde{\mathbb{E}}_t[F_T] = \tilde{\mathbb{E}}_t[\tilde{\mathbb{E}}_T[F_T]] = \tilde{\mathbb{E}}_t[S_T]$;
- F_t has zero drift; A popular modeling is

$$dF_t = \sigma F_t d\tilde{W}_t. \quad (16.23)$$

This reflects the fact that there is no financing cost in trading futures. The corresponding pricing formulas for vanilla options on futures are provided in Appendix B.

16.9.2 Forward

The pricing of forward contracts has been fully analyzed in Part-I by no-arbitrage arguments. In this section, we provide the risk-neutral modelling of forward, which is helpful for further analysis. As one can expect, the results of the two approaches are consistent.

At its inception, the forward price⁴ K satisfies $\tilde{\mathbb{E}}[D(0, T)(U_T - K)] = 0$. Hence, we can obtain the general forward price/rate formula:

$$K = \frac{\tilde{\mathbb{E}}[D(0, T)U_T]}{P(T)},$$

where we recall that $P_0(T)$ can be denoted by $P(T)$.

⁴ The forward contract considered in this section is a non-centrally cleared OTC derivative. We note that the initial margin and variation margin do not provide economic benefits to the receiver, as the generated interest/revenue belongs to the margin poster.

For an equity underlier (assuming no discrete dividend), U_T is the asset price S_T and we obtain

$$K = \frac{\tilde{\mathbb{E}}[D(0, T)S_T]}{P(T)} = \frac{S_0 e^{-qT}}{P(T)},$$

where q is the sum of repo rate and continuous dividend rate. We have similar results on commodities and FX forward prices.

For FRA, U_T is the interest rate R_T applied to the future period $[T, T_m]$ with τ as the day count fraction, and R_T is fixed at T for IBOR. For a rear-fixed rate such as SOFR, even though it is fixed at T_m for the period $[T, T_m]$, we can still consider it to be known at T thanks to the OIS market that allows to lock in the rate at T . Hence, we can unify the modelling analysis for FRA as follows:

$$\begin{aligned} K &= \frac{1}{P(T_m)} \tilde{\mathbb{E}}[D(0, T_m)R_T] = \frac{1}{P(T_m)} \tilde{\mathbb{E}}[D(0, T) \tilde{\mathbb{E}}_T[D(T, T_m)]R_T] \\ &= \frac{1}{P(T_m)} \tilde{\mathbb{E}}[D(0, T)P_T(T_m)R_T] = \frac{1}{P(T_m)} \tilde{\mathbb{E}}[D(0, T)(P_T(T) - P_T(T_m))/\tau] \\ &= \frac{1}{P(T_m)} (P(T) - P(T_m))/\tau = \left(\frac{P(T)}{P(T_m)} - 1 \right) / \tau, \end{aligned}$$

where we have used $P_T(T) = 1$ and $\frac{1}{1+R_T\tau} = P_T(T_m)$ with τ being the day count fraction between T and T_m .

16.9.3 Futures/Forward Convexity Adjustment

A futures contract and a forward contract provide similar exposure to the underlier. However, the difference in their payout mechanism may lead to a difference in pricing that is known as *futures/forward convexity adjustment* that we denote by $\lambda = K - F_0$ where $F_0 = \tilde{\mathbb{E}}[U_T]$. Using the model for pricing/hedging, λ can be calculated directly (e.g., by Monte Carlo method). For certain models, a closed-form formula can be found. For instance, under the Ho–Lee interest rate model, $\lambda = \frac{1}{2}\sigma^2 T T_m$ for FRA (see [50]).

The below analysis [53] provides a closer view on the convexity adjustment:

$$\begin{aligned} \lambda &= K - F_0 = \frac{\tilde{\mathbb{E}}[D(0, T)U_T]}{P(T)} - \tilde{\mathbb{E}}[U_T] \\ &= \frac{1}{P(T)} (\tilde{\mathbb{E}}[D(0, T)U_T] - \tilde{\mathbb{E}}[D(0, T)]\tilde{\mathbb{E}}[U_T]) = \frac{1}{P(T)} \text{Cov}(D(0, T), U_T). \end{aligned}$$

If the interest rate is assumed to be non-stochastic, which is often the case for equity models for products with a short tenor, the correlation between $D(0, T)$ and S_T is zero, hence $\lambda = 0$. For interest rate forward and futures, we have $\lambda < 0$ whereby the short party of the futures (i.e., long interest rate) will reinvest the daily profits at higher rate levels and finance the daily losses at lower rate levels.

16.10 Overview on Numerical Methods for Option Pricing

Closed-form solutions (e.g., Black–Scholes formula) are accurate and fast to calculate. However, only very few options have had closed-form solutions found under simple models such as the Black–Scholes–Merton. Therefore, numerical methods in the following are necessary for option pricing.

- *Numerical integration*, which is an efficient method. However, it is only applicable when the underlier's probability of transition is known.
- *Tree methods*, which assume two or more possible moves for underlier price for each small time step with certain probabilities. A tree can be constructed for all possible underlier prices until maturity date of the considered option. The option price can be calculated from the tree. Tree methods are easy to implement for one dimensional case, but are not suitable for some path-dependent options.
- *PDE methods*, which make use of the link between SDE and PDE. Numerical methods for PDEs are well studied in mathematics and physics. PDE methods are superior to tree methods for both the scope of suitable options to price and the convergence. However, similar to the tree methods, the numerical complexity grows exponentially with the dimensionality.
- *Monte Carlo method*, which is the simplest but the most universal method for pricing all types of options (excluding American options that require special treatment). It consists of simulating the random paths and then calculating the averaged discounted payoff value as option price. Its drawback is the accuracy issue.

In this book, we will focus on the methods of Monte Carlo, finite difference for PDE and binomial/trinomial trees in later dedicated chapters.

Exercises

1. Given the asset process $dS_t = \mu_t S_t dt + \sigma_t S_t dW_t$, where μ_t and σ_t are non-random functions of t . The risk-free interest rate r_t is also a deterministic function of t .

Derive the SDE and PDE for European options and obtain the closed-form solution for the time- t price of a vanilla put option paying out $(K - S_T)^+$ at T .

2. Consider the futures price with maturity T at time t , $F_t(T) = \widetilde{\mathbb{E}}[S_T | \mathcal{F}_t]$. Assume by the martingale representation theorem that $dF_t(T) = \sigma_t F_t(T) d\widetilde{W}_t$, where σ is a constant. For $0 \leq s \leq t$, show the *Black's formula*:

$$\widetilde{\mathbb{E}}[(F_t(T) - K)^+ | \mathcal{F}_s] = F_s(T)N(d_1) - KN(d_2)$$

where $d_1 = \left(\ln\left(\frac{F_s(T)}{K}\right) + \frac{1}{2}\sigma^2(t-s) \right) / (\sigma\sqrt{t-s})$ and $d_2 = d_1 - \sigma\sqrt{t-s}$.

Chapter 17

Numéraires and Vanilla Interest Rate Options Pricing*



17.1 Introduction of Numéraire

In this chapter, we introduce the concept of *numéraire* and its application in the modelling and pricing of derivatives. In financial mathematics, a numéraire is the asset (e.g., money market account unit) used for the carry of a self-financing portfolio and the funding for the exposure to other assets (i.e., underlying assets for derivatives): a long (short) position of an asset S_t is funded by a short (long) position of the numéraire (see Remark 17.1). Without loss of generality, the asset prices considered in this chapter are all *total return assets* (see Sect. 5.2.5). In this chapter, all the price processes are assumed to be $\mathcal{F}_t$ -measurable.

Any asset with strictly positive price process can be used as a numéraire. The *risk-neutral measure for a given numéraire N* is the probability measure $\tilde{\mathbb{P}}^N$ under which any tradable asset's total return price S_t and all tradable derivative price denominated in units of the numéraire (i.e., S_t/N_t and V_t/N_t , known as N_t -**deflated** prices) are $\tilde{\mathbb{P}}^N$ -martingales. The techniques related to numéraire may provide simplification and clarity in financial derivatives modelling.

Proposition 17.1 provides the construction for $\tilde{\mathbb{P}}^N$. Under the assumption of no risk-free arbitrage for tradable instruments, there is only one price for any tradable instrument (known as the Law of One Price) and, hence, $\tilde{\mathbb{P}}^N$ is unique almost-surely.

Actually, we have already encountered a numéraire in previous content, which is the money market account: $M_t = e^{\int_0^t r_s ds}$. Its value at t is the total return of investing \$1 with continuous interest rate r_s . Under money market risk-neutral probability $\tilde{\mathbb{P}}$, all tradable asset's total return price denominated in units of M_t , i.e., $S_t/M_t = e^{-\int_0^t r_s ds} S_t$, is a $\tilde{\mathbb{P}}$ -martingale.

17.2 Change of Numéraire

Now, we consider the change of numéraire for derivatives pricing. Let N be the original numéraire and $\tilde{\mathbb{P}}^N$ be its risk-neutral probability measure. U is an asset with strictly positive price process U_t , which can be used as a numéraire.

Proposition 17.1 *Let N_t be a numéraire and $\tilde{\mathbb{P}}^N$ be its risk-neutral probability measure. Let S_t be the price process of any tradable asset and V_t be the price process of a derivative instrument on S_t . U_t is a strictly positive process. Then*

- $Z_t = \frac{U_t/U_0}{N_t/N_0}$ can be used as Radon–Nikodym derivative to define a probability $\tilde{\mathbb{P}}^U$
- S_t/U_t and V_t/U_t are $\tilde{\mathbb{P}}^U$ -martingales, i.e., $\tilde{\mathbb{P}}^U$ is a risk-neutral probability.

Proof As $\tilde{\mathbb{P}}^N$ is a risk-neutral probability, Z_t is a $\tilde{\mathbb{P}}^N$ -martingale and that $\tilde{\mathbb{E}}^N[Z_t] = \frac{N_0}{U_0} \tilde{\mathbb{E}}^N\left[\frac{U_t}{N_t}\right] = 1$. Hence, we can define $\tilde{\mathbb{P}}^U$ with $\frac{d\tilde{\mathbb{P}}^U}{d\tilde{\mathbb{P}}^N}|_{\mathcal{F}_t} = Z_t$. Now, denote X_t as either S_t or V_t , we have

$$\tilde{\mathbb{E}}_s^U\left[\frac{X_t}{U_t}\right] = \frac{1}{Z_s} \tilde{\mathbb{E}}_s^N\left[\frac{X_t}{U_t} Z_t\right] = \frac{N_s}{U_s} \tilde{\mathbb{E}}_s^N\left[\frac{X_t}{N_t}\right] = \frac{N_s}{U_s} \frac{X_s}{N_s} = \frac{X_s}{U_s}, \quad \forall s < t.$$

Actually, from money market numéraire, we can construct the risk-neutral probability for any numéraire with the approach described in the above. $\square$

We call the below formula set **the toolkit for numéraire change**:

$$\left\{ \begin{array}{ll} Z_t = \frac{U_t/U_0}{N_t/N_0} \quad (= \frac{d\tilde{\mathbb{P}}^U}{d\tilde{\mathbb{P}}^N}|_{\mathcal{F}_t}) & \text{(Radon–Nikodym derivative process)} \\ \tilde{\mathbb{E}}^U[Y|\mathcal{F}_s] = \frac{1}{Z_s} \tilde{\mathbb{E}}^N[Y Z_t|\mathcal{F}_s] & \text{(Bayes' Formula)} \\ d\tilde{W}_t^U = d\tilde{W}_t^N - \frac{d\langle Z, \tilde{W}^N \rangle_t}{Z_t} & \text{(Girsanov Theorem)} \end{array} \right.$$

for all $0 \leq s \leq t$ and any $\mathcal{F}_t$ -measurable variable Y .

Numéraire change technique may be used for calculation simplification. For a pricing problem that requires evaluation under $\tilde{\mathbb{P}}^N$, as in RHS in Bayes' formula, there will be one term less (i.e., Z_t) inside $\tilde{\mathbb{E}}^U$ after numéraire change.

Example 17.1 (Underlier Numéraire) In Sect. 16.6.1, in order to take S_T out of $\tilde{\mathbb{E}}_t[S_T \mathbf{1}_{\{S_T > K\}}]$ to derive Black–Scholes formulas, a numéraire change is applied with $Z_t = \frac{S_t/S_0}{M_t/M_0} (= e^{-\frac{1}{2}\sigma^2 t + \sigma \tilde{W}_t})$. Then, under $\tilde{\mathbb{P}}^S$ and with $\tau = T - t$, we have

$$V_t = M_t \tilde{\mathbb{E}}_t\left[\frac{S_T}{M_T} \mathbf{1}_{\{S_T > K\}}\right] = \frac{S_t}{Z_t} \tilde{\mathbb{E}}_t[Z_T \mathbf{1}_{\{\tilde{W}_\tau > -d_2 \sqrt{\tau}\}}] = S_t \tilde{\mathbb{E}}_t^S[\mathbf{1}_{\{\tilde{W}_\tau^S + \sigma \tau > -d_2 \sqrt{\tau}\}}].$$

17.3 Generalized Risk-Neutral Framework

Proposition 17.2 (Generalized Pricing Formula) *The pricing of a European derivative paying out V_T at T can be performed under any risk-neutral probability $\tilde{\mathbb{P}}^N$, i.e.,*

$$V_t = N_t \tilde{\mathbb{E}}_t^N \left[\frac{V_T}{N_T} \right], \text{ where } N_t \text{ is an arbitrary numéraire.}$$

Proof By Proposition 17.1, $\frac{V_T}{N_T}$ is a $\tilde{\mathbb{P}}^N$ -martingale, i.e., $\frac{V_t}{N_t} = \tilde{\mathbb{E}}_t^N \left[\frac{V_T}{N_T} \right]$, which leads to the pricing formula to prove. Alternative, we can show it by the change of numéraire from money market numéraire M_t to N_t and with the Radon-Nikodym derivative $Z_t = \frac{M_t/M_0}{N_t/N_0}$:

$$V_t = M_t \tilde{\mathbb{E}}_t \left[\frac{V_T}{M_T} \right] = M_t \frac{1}{Z_t} \tilde{\mathbb{E}}_t^N \left[\frac{V_T}{M_T} Z_T \right] = N_t \tilde{\mathbb{E}}_t^N \left[\frac{V_T}{N_T} \right]$$

□

Next, we generalize the definition of self-financing in Sect. 16.2.1 for a portfolio of assets and a given numéraire N_t , i.e., $\Pi_t = \Delta_t S_t + \zeta_t N_t$ where Δ_t, ζ_t are *predictable processes* (that may take negative values) and represent respectively the units of the asset and the units of the numéraire.

Definition 17.1 (Self-Financing Portfolio) A *self-financing portfolio* with numéraire N_t is defined as a portfolio satisfying

$$\Pi_T = \Pi_0 + \int_0^T \Delta_t dS_t + \int_0^T \zeta_t dN_t, \forall T, \text{ or equivalently } d\Pi_t = \Delta_t dS_t + \zeta_t dN_t,$$

The N_t -deflated portfolio is $\frac{\Pi_t}{N_t} = \Delta_t \frac{S_t}{N_t} + \zeta_t$.

Proposition 17.3 (Self-Financing Condition with a Given Numéraire) *Let N be a numéraire. Portfolio $\Pi_t = \Delta_t S_t + \zeta_t N_t$ is self-financing if and only if*

$$d \frac{\Pi_t}{N_t} = \Delta_t d \frac{S_t}{N_t}.$$

Hence, if $\frac{S_t}{N_t}$ is a $\tilde{\mathbb{P}}^N$ -martingale, $\frac{\Pi_t}{N_t}$ is also a $\tilde{\mathbb{P}}^N$ -martingale.

Proof As Δ_t, ζ_t are predictable processes, we have the below equation

$$d \langle \Pi_t, \frac{1}{N_t} \rangle = \Delta_t d \langle S_t, \frac{1}{N_t} \rangle + \zeta_t d \langle N_t, \frac{1}{N_t} \rangle \quad (17.1)$$

If Π_t is self-financing, then $d\Pi_t = \Delta_t dS_t + \zeta_t dN_t$ and

$$\begin{aligned}
 d\frac{\Pi_t}{N_t} &= \Pi_t d\frac{1}{N_t} + \frac{1}{N_t} d\Pi_t + d\langle \Pi_t, \frac{1}{N_t} \rangle \\
 &= (\Delta_t S_t + \zeta_t N_t) d\frac{1}{N_t} + \frac{1}{N_t} (\Delta_t dS_t + \zeta_t dN_t) + d\langle \Pi_t, \frac{1}{N_t} \rangle \\
 &= \Delta_t \left(S_t d\frac{1}{N_t} + \frac{1}{N_t} dS_t + d\langle S_t, \frac{1}{N_t} \rangle \right) \\
 &\quad + \zeta_t \left(N_t d\frac{1}{N_t} + \frac{1}{N_t} dN_t + d\langle N_t, \frac{1}{N_t} \rangle \right) \\
 &= \Delta_t d\left(S_t \frac{1}{N_t}\right) + \zeta_t d\left(N_t \frac{1}{N_t}\right) = \Delta_t d\frac{S_t}{N_t} + \zeta_t d(1) = \Delta_t d\frac{S_t}{N_t}.
 \end{aligned}$$

This proves the necessary condition.

To prove the sufficient condition, we assume $d\frac{\Pi_t}{N_t} = \Delta_t d\frac{S_t}{N_t}$ and first derive a few useful intermediary results. By $dS_t = d(N_t \frac{S_t}{N_t}) = N_t d\frac{S_t}{N_t} + \frac{S_t}{N_t} dN_t + d\langle N_t, \frac{S_t}{N_t} \rangle$, we obtain

$$d\frac{\Pi_t}{N_t} = \Delta_t d\frac{S_t}{N_t} = \frac{\Delta_t}{N_t} \left(dS_t - \frac{S_t}{N_t} dN_t - d\langle N_t, \frac{S_t}{N_t} \rangle \right). \quad (17.2)$$

We also have

$$d\langle N_t, \frac{\Pi_t}{N_t} \rangle = dN_t d\frac{\Pi_t}{N_t} = \Delta_t dN_t d\frac{S_t}{N_t} = \Delta_t d\langle N_t, \frac{S_t}{N_t} \rangle. \quad (17.3)$$

Now, using Eqs. (17.2) and (17.3), we have

$$\begin{aligned}
 d\Pi_t &= d\left(\frac{\Pi_t}{N_t} N_t\right) = N_t d\frac{\Pi_t}{N_t} + \frac{\Pi_t}{N_t} dN_t + d\langle N_t, \frac{\Pi_t}{N_t} \rangle \\
 &= N_t \Delta_t d\frac{S_t}{N_t} + \frac{\Pi_t}{N_t} dN_t + \Delta_t d\langle N_t, \frac{S_t}{N_t} \rangle \\
 &= \Delta_t \left(dS_t - \frac{S_t}{N_t} dN_t - d\langle N_t, \frac{S_t}{N_t} \rangle \right) + \frac{\Pi_t}{N_t} dN_t + \Delta_t d\langle N_t, \frac{S_t}{N_t} \rangle \\
 &= \Delta_t \left(dS_t - \frac{S_t}{N_t} dN_t \right) + \frac{\Pi_t}{N_t} dN_t \\
 &= \Delta_t \left(dS_t - \frac{S_t}{N_t} dN_t \right) + \frac{1}{N_t} (\Delta_t S_t + \zeta_t N_t) dN_t \\
 &= \Delta_t dS_t + \zeta_t dN_t.
 \end{aligned}$$

□

The proposition can be extended to a portfolio with n assets ($n \geq 2$), i.e., $\Pi_t = \sum_{i=1}^n \Delta_t^i S_t^i + \zeta_t N_t$. The proof is similar by replacing S_t^i -related parts with a linear combination.

Remark 17.1 (Self-financing by a Long/Short Position) The third line from the bottom in the above proof reads

$$d\Pi_t = \Delta_t \left(dS_t - \frac{S_t}{N_t} dN_t \right) + \frac{\Pi_t}{N_t} dN_t, \quad (17.4)$$

which indicates that the increment of a self-financing portfolio is composed of

- the self-financing return $\Delta_t(dS_t - \frac{S_t}{N_t}dN_t)$ by the long position of 1 unit of S_t financed by the proceeds of the short position of $\frac{S_t}{N_t}$ units of N_t
- the return from the carry of the portfolio $\frac{\Pi_t}{N_t}dN_t$ □

Propositions 16.4, 16.5 and 16.6 can all be generalized to any numéraire except the underlier numéraire for a contingent claim (see Example 17.1) because the martingale representation theorem cannot be applied to find a hedging process. Actually, for any self-financing portfolio $\Pi_t = \Delta_t S_t + \zeta_t N_t$, we have $d(\Pi_t/N_t) \equiv 0$ if $N_t = S_t$. Hence, $d(\Pi_t/N_t)$ cannot be expressed by a martingale of the form $\phi_t d\tilde{W}_t$.

17.4 Usual Numéraires and Vanilla Interest Rate Options Pricing

In the following, we introduce some widely used numéraires and the corresponding measures. The application of numéraires helps to derive explicit formulas for vanilla interest rate options under Black models (lognormal models).

17.4.1 Money Market Account Numéraire

Money market numéraire is $M_t = e^{\int_0^t r_s ds}$, where r_s is the instantaneous interest rate that may be deterministic or stochastic. The corresponding measure is the risk-neutral measure $\tilde{\mathbb{P}}$ used so far in this book. The pricing of a European derivative paying out V_T at T is

$$V_t = M_t \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right] \quad (\tilde{\mathbb{E}} \text{ is the expectation under } \tilde{\mathbb{P}}). \quad (17.5)$$

17.4.2 Zero-Coupon Bond Numéraire and Cap/Floor Pricing

A zero-coupon bond paying 1 at maturity T is denoted as $P_t(T)$ for its value at time t . The probability measure with zero-coupon bond as numéraire is called T -forward measure. The pricing framework under T -forward measure admits a simple form that does not contain a discounting term inside the expectation operator. Let V_t be the time- t price of a European-style derivative with maturity T . According to Proposition 17.1, $V_t/P_t(T)$ is $\tilde{\mathbb{P}}^T$ -martingale, i.e. $V_t/P_t(T) = \tilde{\mathbb{E}}^T[V_T/P_T(T)|\mathcal{F}_t]$. As $P_T(T) = 1$, we have

$$V_t = P_t(T) \tilde{\mathbb{E}}^T[V_T|\mathcal{F}_t]. \quad (17.6)$$

Consider a tenor structure $0 \leq T_0 < T_1 < \dots < T_n$. Denote R_t^i as the rate for the period $\tau_i = T_{i+1} - T_i$ that is fixed at T_i . From the arbitrage relationship seen in FRA rate calculation, we have $R_t^i = \frac{P_t(T_i) - P_t(T_{i+1})}{\tau_i P_t(T_{i+1})}$.

Proposition 17.4 *The rate R_t^i is a martingale under $\tilde{\mathbb{P}}^{T_{i+1}}$.*

Proof Let $0 < s < t$. Note that $G_t = \frac{P_s(T_i) - P_s(T_{i+1})}{\tau_i}$ is the price process of a portfolio of assets and $\frac{G_t}{P_t(T_{i+1})} = R_t^i$. Hence, the martingale property applies to deflated G_t and we have

$$\tilde{\mathbb{E}}^{T_{i+1}}[R_t^i|\mathcal{F}_s] = \tilde{\mathbb{E}}^{T_{i+1}}\left[\frac{G_t}{P_t(T_{i+1})}|\mathcal{F}_s\right] = \frac{G_s}{P_s(T_{i+1})} = R_s^i.$$

□

Pricing of Cap or Floor

An interest rate cap contract is a strip of European call options (each option is called a caplet) on interest rate paying out $N \times (R_{T_i}^i - K)^+ \tau_i$ at time T_{i+1} where $R_{T_i}^i$ is the effective reference rate for the period $[T_i, T_{i+1}]$ which is fixed at T_i .

In the Black model, R_t^i is non-negative. Then, under $\tilde{\mathbb{P}}^{T_{i+1}}$, R_t^i follows

$$dR_t^i = R_t^i \sigma_i d\tilde{W}_t^{T_{i+1}},$$

in which the volatility σ_i is deterministic. From Proposition 17.4, the pricing of a caplet can be performed under T_{i+1} -forward measure, which can be solved by Black's formula (similar to Black-Scholes formula derivation):

$$V_i(t) = P_t(T_{i+1}) \tilde{\mathbb{E}}^{T_{i+1}}[N(R_{T_i}^i - K)^+ \tau_i|\mathcal{F}_t] = N \tau_i P_t(T_{i+1}) [R_t^i N(d_1) - K N(d_2)],$$

with $d_1 = d_2 + \sigma_i \sqrt{T_i - t}$ and $d_2 = \left(\ln \frac{R_t^i}{K} - \frac{1}{2} \sigma_i^2 (T_i - t)\right) / (\sigma_i \sqrt{T_i - t})$. Similar results can be obtained for floor contract that is a strip of European put options on interest rate (i.e., floorlet).

17.4.3 Annuity Factor Numéraire and Swaption Pricing

We recall from Sect. 4.3 that Annuity factor is defined as

$$A_t = \sum_{i=0}^{n-1} \tau_i P_t(T_{i+1}),$$

where T_i 's represent the interest rate fixing/payment dates. It is a positive linear combination of zero coupon bonds and hence can be used as a numéraire. By Proposition 17.1 with the risk neutral probability $\tilde{\mathbb{P}}^A$, the time- t price of a European-style derivative with maturity T is

$$V_t = A_t \tilde{\mathbb{E}}^A \left[\frac{V_T}{A_T} \middle| \mathcal{F}_t \right] \quad (\tilde{\mathbb{E}}^A \text{ is the expectation under } \tilde{\mathbb{P}}^A) \quad (17.7)$$

Proposition 17.5 *The swap rate of a standard contract starting at t , $s_t = \frac{P_t(T) - P_t(T_n)}{A_t}$, is a martingale under $\tilde{\mathbb{P}}^A$.*

Proof Let $u < t$. Note that $P_u(T) - P_u(T_n)$ is a portfolio of assets. Then, the deflated price process is a $\tilde{\mathbb{P}}^A$ -martingale. Hence,

$$\tilde{\mathbb{E}}^A[s_t | \mathcal{F}_u] = \tilde{\mathbb{E}}^A \left[\frac{P_t(T) - P_t(T_n)}{A_t} \middle| \mathcal{F}_u \right] = \frac{P_u(T) - P_u(T_n)}{A_u} = s_u.$$

□

Pricing of Swaption

Here we consider the pricing of a basic (vanilla) swaption that is a European call option on a swap. Assume T is the time when the swaption expires and the underlying swap starts. The swaption's payoff function is

$$N \times A_T (s_T - K)^+.$$

Under $\tilde{\mathbb{P}}^A$ with A_t as numéraire, for $t < T$, the pricing formula is

$$V_t = A_t \tilde{\mathbb{E}}^A \left[\frac{N A_T (s_T - K)^+}{A_T} \middle| \mathcal{F}_t \right] = N A_t \tilde{\mathbb{E}}^A [(s_T - K)^+ | \mathcal{F}_t].$$

By Proposition 17.5, the swap rate is a martingale under $\tilde{\mathbb{P}}^A$. In the Black model, the swap rate is assumed to be non-negative with deterministic volatility σ_t . Hence,

$$ds_t = s_t \sigma_t d\tilde{W}_t^A.$$

The pricing formula for call swaption is the below Black's formula:

$$V_t = N A_t \tilde{\mathbb{E}}^A[(s_T - K)^+ | \mathcal{F}_t] = N A_t [s_t N(d_1) - K N(d_2)].$$

with $d_1 = d_2 + \sigma \sqrt{T-t}$ and $d_2 = \left(\ln \frac{s_t}{K} - \frac{1}{2} \sigma^2 (T-t) \right) / (\sigma \sqrt{T-t})$.

Remark 17.2 (Negative Interest Rate) Negative interest rates appeared around 2010s with the quantitative easing policies for several major currencies¹. For vanilla interest rate options (cap/floor, swaption), some practitioners use the Bachelier model ($dR_t = \mu dt + \sigma dW_t$) which assumes normal distribution allowing negative interest rate values. $\square$

17.4.4 SABR Model for Vanilla Interest Rate Options

Similar to equity options, the market implied volatilities (using the Black model or the Bachelier model) for swaptions and caps/floors vary with strike levels. Among the models allowing to take into account the volatility skew, we mention the SABR (Stochastic Alpha Beta Rho) model, which is a *CEV (Constant Elasticity of Variance)* type model with stochastic volatility:

$$\begin{cases} dR_t = \sigma_t R_t^\beta d\tilde{W}_t \\ d\sigma_t = \alpha \sigma_t d\tilde{W}_t^\sigma \end{cases}$$

where $d\tilde{W}_t d\tilde{W}_t^\sigma = \rho dt$, $\alpha \geq 0$, and $0 < \beta \leq 1$. Volatility skew can be produced by both β and ρ .

Exercises

1. Let R_t^i be modelled as a positive martingale under $\tilde{\mathbb{P}}^{T_{i+1}}$,

$$dR_t^i = R_t^i \sigma_i d\tilde{W}_t^{T_{i+1}}, \text{ where } \sigma_i \text{ is deterministic.}$$

Derive the price of a floorlet paying out $N \tau_i (K - R_{T_i}^i)^+$ at time T_{i+1} with the fixing of $R_{T_i}^i$ at T_i .

¹ Actually, the negative interest rate on JPY appeared much earlier.

Chapter 18

Foreign Exchange Modelling



In this chapter, we study the modelling of foreign exchange rate for FX options as well as other derivatives linked to foreign underliers implying FX risk exposure. Quanto and composite options are analyzed in detail. Lastly, we discuss the hedging of FX risk for derivatives products.

18.1 Stochastic Model for Foreign Exchange Rate

To simplify the analysis, we assume constant foreign and domestic interest rates r^f and r^d . From the domestic investor's perspective, foreign currency rate X_t can be considered as an asset paying continuous dividend and $e^{r^f t} X_t$ is a total return process (see formula 5.3). Hence, under the **domestic risk-neutral measure** $\tilde{\mathbb{P}}^d$ (denoted as $\tilde{\mathbb{P}}$ in previous context), the process

$$\frac{e^{r^f t} X_t}{M_t} = e^{-(r^d - r^f)t} X_t, \quad (M_t = e^{r^d t})$$

is a martingale. Let's further assume that X_t follows geometric Brownian motion. Then, by (16.16), we obtain the SDE for X_t under $\tilde{\mathbb{P}}^d$:

$$dX_t = (r^d - r^f)X_t dt + \sigma_X X_t d\tilde{W}_t^{d,X}. \quad (18.1)$$

This model is known as the *Garman–Kohlhagen model*. The notation $\tilde{W}_t^{d,X}$ is nothing but a Brownian motion under $\tilde{\mathbb{P}}^d$. The superscript X is used to specify that

the Brownian motion is about X_t as there will be other $\tilde{\mathbb{P}}^d$ -Brownians in this chapter. The Radon–Nikodym derivative process for the numéraire change from $\tilde{\mathbb{P}}^d$ to $\tilde{\mathbb{P}}^f$ is

$$Z_t = \frac{U_t/U_0}{M_t/M_0} = e^{-(r^d - r^f)t} X_t/X_0, \quad \text{where } U_t = e^{r^f t} X_t. \quad (18.2)$$

Actually, “domestic” and “foreign” are only relative concepts. Denoting $\bar{X}_t = \frac{1}{X_t}$, the same analysis as the above leads to

$$d\bar{X}_t = (r^f - r^d)\bar{X}_t dt - \sigma_X \bar{X}_t d\tilde{W}_t^{f,\bar{X}},$$

where $\tilde{\mathbb{P}}^f$ is the **foreign risk-neutral measure**.

18.1.1 Cross Rate Volatility

Suppose two foreign exchange rates are given as X_t^1 and X_t^2 , say USD/JPY and USD/CHF, under the Garman–Kohlhagen model. The cross rate CHF/JPY, which is equal to $X_t = X_t^1/X_t^2$, satisfies

$$\ln X_t = \ln X_t^1 - \ln X_t^2.$$

From the normality of $\ln X_t^1$ and $\ln X_t^2$, it is easy to find out that the volatility σ of the cross rate is

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2,$$

where σ_1 and σ_2 are the volatilities for X_t^1 and X_t^2 while ρ is their correlation.

18.2 Pricing Formulas for Vanilla Options and FX Option Duality

Consider simple call or put options on the FX rate X_t under the Garman–Kohlhagen model. Their payoffs at maturity T are

$$N^f(X_T - K)^+ \quad \text{or} \quad N^f(K - X_T)^+$$

respectively. Here N^f is the notional amount in *foreign currency* written on the contract, and the payoff is settled in domestic currency. We will drop N^f in the following because it is just a constant. Closed-form solutions of call or put prices

can be derived in the same way as the BSM formula for stocks paying continuous dividends. They are also known as the Garman–Kohlhagen formulas:

$$\begin{aligned} c_{\text{FX}} &= e^{-r^d T} \tilde{\mathbb{E}}^d[(X_T - K)^+ | \mathcal{F}_0] = e^{-r^f T} X_0 N(d_1) - e^{-r^d T} K N(d_2), \\ p_{\text{FX}} &= e^{-r^d T} \tilde{\mathbb{E}}^d[(K - X_T)^+ | \mathcal{F}_0] = e^{-r^d T} K N(-d_2) - e^{-r^f T} X_0 N(-d_1), \end{aligned}$$

where

$$d_1 = d_2 + \sigma_X \sqrt{T} \quad \text{and} \quad d_2 = \frac{\ln \frac{X_0}{K} + \left(r^d - r^f - \frac{1}{2}\sigma_X^2\right) T}{\sigma_X \sqrt{T}}.$$

18.2.1 FX Option Duality

As explained in Sect. 10.3.2, N call option of strike K on ccy1/ccy2 (option premium in ccy2) is actually $K \times N$ put option of strike $\bar{K} = 1/K$ on ccy2/ccy1 (option premium in ccy1) (known as FX Option Duality). It can also be verified formally with the numéraire change (assuming $K \neq 0$):

$$\begin{aligned} \tilde{\mathbb{E}}^d[e^{-r^d T} (X_T - K)^+] &= \tilde{\mathbb{E}}^d \left[e^{-r^d T} X_T \left(1 - \frac{K}{X_T}\right)^+ \right] \\ &= X_0 K \tilde{\mathbb{E}}^d \left[e^{-r^f T} Z_T (\bar{K} - \bar{X}_T)^+ \right] \quad (Z_T = e^{-(r^d - r^f)T} X_T / X_0) \\ &= X_0 K \tilde{\mathbb{E}}^f \left[e^{-r^f T} (\bar{K} - \bar{X}_T)^+ \right]. \end{aligned} \tag{18.3}$$

X_0 term in the last line above ensures the option premium equivalence for call and put brought in ccy2.

18.3 SDE for Foreign Asset Under Domestic Risk-Neutral Probability

For pricing certain options linked to foreign assets, it is often necessary to use the SDE of the foreign assets under the domestic risk-neutral probability measure $\tilde{\mathbb{P}}^d$. For instance, for options linked to a basket of assets containing foreign assets, the option pricing should be using a *common* risk-neutral measure that is often chosen as the domestic one. In fact, a risk-neutral probability measure is characterized by the currency used in self-financing hedging process: e.g., for funding a foreign stock S_t in delta hedging, $X_t S_t$ units of the domestic currency is borrowed. In this section, we will make the analysis and provide the SDE followed by a foreign asset under $\tilde{\mathbb{P}}^d$.

Let S_t be a foreign asset price process following the below SDE under the foreign risk-neutral probability measure $\mathbb{P}^f$:

$$dS_t = r^f S_t dt + \sigma_S S_t d\tilde{W}_t^{f,S}, \quad (18.4)$$

where $\tilde{W}_t^{f,S}$ is a $\mathbb{P}^f$ -Brownian motion. From Girsanov Theorem, there exists θ such that

$$\tilde{W}_t^{d,S} = \tilde{W}_t^{f,S} + \theta t \quad (18.5)$$

is a $\mathbb{P}^d$ -Brownian. So, it suffices to determine θ to obtain the SDE of S_t under $\mathbb{P}^d$. For the foreign exchange process X_t , we assume that it follows the SDE (18.1) and $d\tilde{W}_t^{d,X} d\tilde{W}_t^{d,S} = \rho dt$.

Proposition 18.1 *Under $\mathbb{P}^d$, the process $Y_t = X_t S_t$, which can be regarded as a domestic asset, is given by*

$$dY_t = Y_t(r^d dt + \sigma_Y d\tilde{W}_t^{d,Y}),$$

where $\sigma_Y = \sqrt{\sigma_X^2 + \sigma_S^2 + 2\rho\sigma_X\sigma_S}$ and $\tilde{W}_t^{d,Y}$ is a $\mathbb{P}^d$ -Brownian.

Proof By Itô's product rule,

$$\begin{aligned} dY_t &= S_t dX_t + X_t dS_t + d\langle X, S \rangle_t \\ &= X_t S_t [(r^d - r^f)dt + \sigma_X d\tilde{W}_t^{d,X} + r^f dt + \sigma_S d\tilde{W}_t^{f,S} + \rho\sigma_X\sigma_S dt] \\ &= Y_t [r^d dt + \sigma_X d\tilde{W}_t^{d,X} + \sigma_S (d\tilde{W}_t^{d,S} - \theta dt) + \rho\sigma_X\sigma_S dt] \\ &= Y_t [(r^d + \rho\sigma_X\sigma_S - \theta\sigma_S)dt + \sigma_X d\tilde{W}_t^{d,X} + \sigma_S d\tilde{W}_t^{d,S}], \end{aligned}$$

where $d\langle X, S \rangle_t = \sigma_X\sigma_S X_t S_t d\langle \tilde{W}^{d,X}, \tilde{W}^{d,S} \rangle_t = \rho\sigma_X\sigma_S dt$. Under $\mathbb{P}^d$, Y_t should earn the domestic interest rate r^d . From the drift term, we obtain

$$r^d + \rho\sigma_X\sigma_S - \theta\sigma_S = r^d. \quad (18.6)$$

Furthermore, as $\tilde{W}_t^{d,X}$ and $\tilde{W}_t^{d,S}$ follow normal distribution, we have the below result by checking the expectation and variance of both sides of the equation:

$$\sigma_X \tilde{W}_t^{d,X} + \sigma_S \tilde{W}_t^{d,S} = \sqrt{\sigma_X^2 + \sigma_S^2 + 2\rho\sigma_X\sigma_S} \tilde{W}_t^{d,Y} = \sigma_Y \tilde{W}_t^{d,Y}.$$

Then, we obtain the result. $\square$

Corollary 18.1 *The drift coefficient θ in Eq. (18.5) is determined by*

$$\theta = \rho\sigma_X. \quad (18.7)$$

This is obtained directly from Eq. (18.6).

Now we are ready to give the SDE of foreign asset under $\tilde{\mathbb{P}}^d$.

Proposition 18.2 *Under the domestic risk-neutral measure $\tilde{\mathbb{P}}^d$, the SDE of the foreign asset S_t is*

$$dS_t = (r^f - \rho\sigma_X\sigma_S)S_t dt + \sigma_S S_t d\tilde{W}_t^{d,S}. \quad (18.8)$$

Proof From Eqs. (18.4), (18.5) and (18.7), we have

$$\begin{aligned} dS_t &= r^f S_t dt + \sigma_S S_t d\tilde{W}_t^{f,S} = r^f S_t dt + \sigma_S S_t d(\tilde{W}_t^{d,S} - \rho\sigma_X dt) \\ &= (r^f - \rho\sigma_X\sigma_S)S_t dt + \sigma_S S_t d\tilde{W}_t^{d,S}. \end{aligned}$$

□

Corollary 18.2 (SDE for a Dividend-Paying Foreign Asset) *Consider a foreign asset S_t with repo and/or continuous dividend. Let q represent the sum of the repo rate and the continuous dividend rate. The SDE of S_t under $\tilde{\mathbb{P}}^d$ is*

$$dS_t = (r^f - q - \rho\sigma_X\sigma_S)S_t dt + \sigma_S S_t d\tilde{W}_t^{d,S}.$$

Proof We observe that $e^{qt} S_t$ can be considered as a non-repo/non-dividend paying asset (e^{qt} is the increased asset units by repo/dividend reinvestment). Applying Proposition 18.2 on $e^{qt} S_t$ leads to the result to be proved. □

With Eq. (18.7), we can revisit the consistency between domestic and foreign risk-neutral measures.

Proposition 18.3 *Let X_t be the exchange rate that follows SDE (18.1) under $\tilde{\mathbb{P}}^d$. Denoting $\bar{X}_t = \frac{1}{X_t}$ as the inverse exchange rate, we have*

- (1) under $\tilde{\mathbb{P}}^f$, $d\bar{X}_t/\bar{X}_t = (r^f - r^d)dt + \sigma_X d\tilde{W}_t^{f,\bar{X}}$
- (2) under $\tilde{\mathbb{P}}^d$, $d\bar{X}_t/\bar{X}_t = (r^f - r^d + \sigma_X^2)dt + \sigma_X d\tilde{W}_t^{d,\bar{X}}$ (known as Siegel's Paradox).

Proof We first prove Siegel's Paradox. Applying Itô–Doeblin formula to $f(X_t) = \frac{1}{X_t}$, we have

$$d\frac{1}{X_t} = \frac{1}{X_t}[(r^f - r^d + \sigma_X^2)dt - \sigma_X d\tilde{W}_t^{d,X}].$$

Setting $\tilde{W}_t^{d,\bar{X}} = -\tilde{W}_t^{d,X}$, we obtain the SDE of $\bar{X}_t$ under $\tilde{\mathbb{P}}^d$ (Siegel's Paradox):

$$\frac{d\bar{X}_t}{\bar{X}_t} = (r^f - r^d + \sigma_X^2)dt + \sigma_X d\tilde{W}_t^{d,\bar{X}}.$$

Consider $\bar{X}_t$ as a foreign asset paying “continuous dividend” at the interest rate r^d . To prove (1), we apply Girsanov Theorem $\tilde{W}_t^{d,\bar{X}} = \tilde{W}_t^{f,\bar{X}} + \theta t$ in (18.5), i.e., replacing $\tilde{W}_t^{d,\bar{X}}$ by $\tilde{W}_t^{f,\bar{X}} + \theta t$. Obviously, the correlation between $\tilde{W}_t^{d,\bar{X}}$ and $\tilde{W}_t^{d,X}$ is $\rho = -1$:

$$\rho dt = d\tilde{W}_t^{d,\bar{X}} d\tilde{W}_t^{d,X} = d(-\tilde{W}_t^{d,X}) d\tilde{W}_t^{d,X} = -dt,$$

which can also be observed easily from $\ln \bar{X}_t = -\ln X_t$. By Corollary 18.1, we have $\theta = \rho\sigma_X = -\sigma_X$, which leads to SDE (2) for $\bar{X}_t$ under $\tilde{\mathbb{P}}^f$. $\square$

18.4 Quanto Option

In a *quanto option* (quantity adjusted option), the payoff is linked to the performance of a foreign currency denominated underlier S_t (such as $\frac{S_t}{S_0}$) but *paid in domestic currency* (instead of foreign currency for a vanilla option). For instance, the instrument paying the amount $N \left(\frac{S_T}{S_0} - \hat{K} \right)^+$ in domestic currency at maturity T is a quanto call option of notional N in domestic currency and strike $\hat{K}$ in percentage.¹ Equivalently, the option can be expressed as $N^S \times (S_T - K)^+$ paying out in domestic currency, where $N^S = N/S_0$ represents the notional amount in number of asset units and $K = \hat{K} S_0$ is the strike price in foreign currency.

Quanto mechanism is a natural feature in financial products with returns linked to the performance of foreign underliers (denominated in one or more foreign currencies). This is because investors' view on the performance of a foreign underlier is always relative to its initial price expressed in its currency. It is much harder to have a view on the performance of a foreign underlier's price converted in domestic currency due to the impact of FX rate changes.

¹ The quanto feature can be equivalently formulated under “pure domestic” probability. For instance, a quanto call option can be equivalently written as the exotic option $N \left(\frac{Y_T/X_T}{Y_0/X_0} - \hat{K} \right)^+$ where $Y_t (= X_t S_t)$ and X_t are asset prices in domestic currency.

To illustrate the pricing of a quanto call option (a similar approach can also be applied to quanto put option), we make use of the stock price process under $\mathbb{P}^d$ as described in Proposition 18.2.

$$\begin{aligned}
c_{\text{quanto}} &= e^{-r^d T} \widetilde{\mathbb{E}}^d[(S_T - K)^+ | \mathcal{F}_0] = e^{-r^d T} \widetilde{\mathbb{E}}^d[(S_T - K)^+] \\
&= e^{-r^d T} \widetilde{\mathbb{E}}^d[(S_0 e^{(r^f - \rho \sigma_X \sigma_S)T - \frac{1}{2} \sigma_S^2 T + \sigma_S \widetilde{W}_T^{d,S}} - K)^+] \\
&= e^{(-r^d + r^f - \rho \sigma_X \sigma_S)T} \widetilde{\mathbb{E}}^d[e^{-(r^f - \rho \sigma_X \sigma_S)T} (S_0 e^{(r^f - \rho \sigma_X \sigma_S)T - \frac{1}{2} \sigma_S^2 T + \sigma_S \widetilde{W}_T^{d,S}} - K)^+] \\
&= e^{(-r^d + r^f - \rho \sigma_X \sigma_S)T} [S_0 N(d_1) - e^{-(r^f - \rho \sigma_X \sigma_S)T} K N(d_2)] \\
&= e^{-(r^d - r^f)T} (e^{-\rho \sigma_X \sigma_S T} S_0 N(d_1) - e^{-r^f T} K N(d_2)),
\end{aligned}$$

where

$$d_1 = d_2 + \sigma_S \sqrt{T} \quad \text{and} \quad d_2 = \frac{\ln \frac{S_0}{K} + (r^f - \rho \sigma_X \sigma_S - \frac{1}{2} \sigma_S^2)T}{\sigma_S \sqrt{T}}.$$

18.5 Composite Option

In a *composite option*, the product of a foreign asset price S_t and the exchange rate X_t appears in the payoff formula to represent the price of the foreign asset in domestic currency. For instance, the payoff at maturity T for a composite call option is $N(\frac{X_T S_T}{X_0 S_0} - \hat{K})^+$, where the notional N is in domestic currency and strike $\hat{K}$ is a percentage. Equivalently, the option can be expressed as $N^S(X_T S_T - K)^+$ paying out in domestic currency, where $N^S = N/(X_0 S_0)$ represents the notional amount in number of asset units and $K = \hat{K} X_0 S_0$ is the strike price in domestic currency.

Obviously, the SDE for $Y_t = S_t X_t$ as described in Proposition 18.1 can be used for option pricing. For example,

$$c_{\text{compo}} = e^{-r^d T} \widetilde{\mathbb{E}}^d[(X_T S_T - K)^+ | \mathcal{F}_0] = X_0 S_0 N(d_1) - e^{-r^d T} K N(d_2),$$

where

$$d_1 = d_2 + \sigma_Y \sqrt{T} \quad \text{and} \quad d_2 = \frac{\ln \frac{X_0 S_0}{K} + (r^d - \frac{1}{2} \sigma_Y^2)T}{\sigma_Y \sqrt{T}}.$$

One can notice from the composite call option formula that there is no r^f term. Hence, there is no foreign interest rate risk in composite options. This is an important feature for making options on the foreign underlier when there is no liquid financial instruments in the market to hedge the foreign interest rate risk. In practice, the composite feature is not often used as it requires the investor to have views on both the underlier and the FX rate.

18.6 Discussions on Hedging*

We use $V(t, S_t, X_t)$ to denote the price of an option subject to FX risk in hedging in general, even though the exchange rate X_t does not appear in the payoff. For instance, the FX rate X_t does not appear explicitly in the payoff function for quanto options, but it does appear in the hedging process involving the financing in domestic currency for the delta positions with units of $(X_t S_t)$ and X_t .

Following the multi-asset option hedging approach, the hedging of a short position of a derivative V is realized through the following self-financing portfolio $\Pi(t, S_t, X_t)$:

- holding Δ_t^S units of the asset (stock) S_t ;
- holding Δ_t^X units of the foreign currency;
- financing the position with domestic interest rate r^d ;
- investing the portfolio value $\Pi(t, S_t, X_t)$ in money market with rate r^d .

Denote $M_t = e^{r^d t}$, $\widehat{\Pi}_t = \frac{\Pi_t}{M_t} = e^{-r^d t} \Pi_t$ and define $X_t^* = e^{r^f t} X_t$ as the “total return asset” for the foreign currency. Let Δ_t^* be the quantity of the “total return asset” used for hedging and Δ_t^X be the quantity of the foreign currency of the same value, i.e., $\Delta_t^X X_t = \Delta_t^* X_t^*$. Obviously, we have $\Delta_t^* = e^{-r^f t} \Delta_t^X$. Under $\widetilde{\mathbb{P}}^d$, the discounted self-financing hedging portfolio $\widehat{\Pi}_t$ follows

$$\begin{aligned} d\widehat{\Pi}_t &= \Delta_t^S d\left(\frac{X_t S_t}{M_t}\right) + \Delta_t^* d\left(\frac{X_t^*}{M_t}\right) \\ &= \Delta_t^S d(e^{-r^d t} X_t S_t) + e^{-r^f t} \Delta_t^X d(e^{-(r^d - r^f)t} X_t) \\ &= e^{-r^d t} [\Delta_t^S X_t dS_t + (\Delta_t^X + \Delta_t^S S_t) dX_t] + [\dots]dt. \end{aligned} \quad (18.9)$$

The value of a derivative paying out $V(T, S_T, X_T)$ can be delta-hedged by setting $V_t = \Pi(t, S_t, X_t)$. Applying Itô–Doebelin formula to $\widehat{V}_t = e^{-r^d t} V_t$, we get

$$d\widehat{V}_t = \frac{\partial \widehat{V}_t}{\partial S_t} dS_t + \frac{\partial \widehat{V}_t}{\partial X_t} dX_t + [\dots]dt = e^{-r^d t} \left(\frac{\partial V_t}{\partial S_t} dS_t + \frac{\partial V_t}{\partial X_t} dX_t \right) + [\dots]dt. \quad (18.10)$$

Equating the dS_t and dX_t terms in (18.10) and (18.9), we obtain:

$$\Delta_t^S = \frac{1}{X_t} \frac{\partial V_t}{\partial S_t} \quad \text{and} \quad \Delta_t^X = \frac{\partial V_t}{\partial X_t} - \Delta_t^S S_t.$$

Intuitively, $X_t \partial S_t$ in Δ_t^S implies the infinitesimal increment of S_t in domestic currency, and $\Delta_t^S S_t$ in the expression for Δ_t^X represents the FX exposure from the foreign asset position.

Exercises

1. Consider a EUR-denominated stock S^1 , and a USD-denominated stock S^2 . EUR/USD represents the exchange rate (units of USD per EUR). The stocks S^1 and S^2 follow the Black–Scholes–Merton model and EUR/USD rate follows the Garman–Kohlhagen model.

Assume 3 independent trials for standard normal variates (ϵ_1, ϵ_2) , where $\epsilon_1, \epsilon_2 \sim \mathcal{N}(0, 1)$: $(1.02, 0.12) \mid (-0.50, -0.10) \mid (0.6, 0.70)$.

Use these 3 trials to construct 3 paths by Monte Carlo simulation to price the following 1Y tenor, USD10m notional amount (N) options:

Market data					
Name	S^1	S^2	EUR/USD	EUR	USD
Volatility	30%	25%	10%		
Correlation S^1	1				
Correlation S^2	0.60	1			
Correlation EUR/USD		-0.20	1		
Rate (continuous)				0.2%	0.4%

(1) Call option: $N \times \max \left\{ \frac{S_T^2}{S_0^2} - 1, 0 \right\}$

(2) Quanto call option: $N \times \max \left\{ \frac{S_T^1}{S_0^1} - 1, 0 \right\}$

(3) Quanto worst-of call option: $N \times \max \left\{ \min \left\{ \frac{S_T^1}{S_0^1}, \frac{S_T^2}{S_0^2} \right\} - 1, 0 \right\}$.

2. Assume that the FX rate X_t follows the Garman–Kohlhagen model (18.1) and the foreign asset price process S_t follows (18.8). Show that for $t < T$,

$$\tilde{\mathbb{E}}^d[X_T S_T | \mathcal{F}_t] = \tilde{\mathbb{E}}^d[X_T | \mathcal{F}_t] \tilde{\mathbb{E}}^d[S_T | \mathcal{F}_t] e^{\rho \sigma_X \sigma_S (T-t)}.$$

Chapter 19

American and Exotic Options*



In this chapter, we first introduce American option modelling in a simple way through Bermudan options that have discrete exercise times. A general Partial Differential Inequality is derived in a heuristic way without using advanced mathematical concepts. Then, some well-known closed-form solutions for exotic options are presented under the BSM model.

19.1 American Options

19.1.1 General Analysis on American Options

An *American option* gives the option holder the right to early terminate the option before its maturity. Upon the option exercise, the option holder will be paid the intrinsic value or other predefined payout. A derivative product with discretionary feature is said to be *callable* if the exercise belongs to the product issuer, and *puttable* otherwise. The American option style with discrete exercise dates is also known as *Bermudan option style*. Technically, the early termination time is a stopping time.

To illustrate some general results on American options, we consider only Bermudan options without loss of generality for practical applications.

Let $0 = t_0 < t_1 < \dots < t_n$ be the exercise dates with t_n as the expiration date of the option. Denote

S_i : underlier price at t_i ;

H_i : exercise payoff at t_i (H_n for the final payoff if no early exercise happened, for example: $H_n = (K - S_n)^+$ for put);

V_i : option value at time t_i , $t_0 \leq t_i \leq t_n$, if the option is not exercised before t_i ;

M_i : money market account numéraire from t_0 to t_i .

At maturity t_n , we have $V_n = H_n$. At the preceding exercise date t_{n-1} , there can be two cases:

- (1) if the option is exercised at t_{n-1} , the option value is H_{n-1} (the payout);
- (2) if the option is not exercised at t_{n-1} , the option value V_{n-1} will become V_n at time t_n . Because there is no exercise date *between* t_{n-1} and t_n , $\frac{V_t}{M_t}$ ($t_{n-1} \leq t \leq t_n$) *should be a martingale* under risk-neutral probability $\tilde{\mathbb{P}}$ and there is a self-financing strategy, following which, the amount of $M_{n-1} \tilde{\mathbb{E}}[\frac{V_n}{M_n} | \mathcal{F}_{n-1}]$ at time t_{n-1} will become V_n at time t_n .

A rational option holder will make the exercise decision to get the maximum value:

$$V_{n-1} = \max \left\{ H_{n-1}, M_{n-1} \tilde{\mathbb{E}} \left[\frac{V_n}{M_n} | \mathcal{F}_{n-1} \right] \right\}.$$

Denoting the discounted processes by $\hat{V}_t = \frac{V_t}{M_t}$, $\hat{H}_t = \frac{H_t}{M_t}$, we get

$$\hat{V}_{n-1} = \max \{ \hat{H}_{n-1}, \tilde{\mathbb{E}}[\hat{V}_n | \mathcal{F}_{n-1}] \}.$$

Doing similar analysis recursively backward, we obtain the general result for all $i \in \{0, \dots, n-1\}$:

$$\hat{V}_i = \max \{ \hat{H}_i, \tilde{\mathbb{E}}[\hat{V}_{i+1} | \mathcal{F}_i] \}, \quad (19.1)$$

and V_i is then the option price at time t_i . For a given exercise date t_i , we call

- $M_i \tilde{\mathbb{E}} \left[\frac{V_{i+1}}{M_{i+1}} | \mathcal{F}_i \right]$ the *continuation value*, and
- H_i the *exercise value*.

In summary, the **optimal exercise decision** can be implemented by computing, at each exercise time point,

- *continuation value*: option value if we do not exercise the option;
- *exercise value*: payout of the option with early exercise
- *option price*:

$$\text{option price} = \max\{\text{continuation value}, \text{exercise value}\}.$$

So far, no closed-form solution has been found for American options and numerical methods are used. *Backward induction* procedure (also known as *dynamic programming* in PDE and tree methods (see Sects. 22.1.4 and 23.4) provides naturally the *continuation value* at each discretization node. However, PDE and tree methods have the limitation on the number of dimensions. Monte Carlo method deals well with high dimensional problems. But as a forward method, it does not provide the continuation value for exercise decision. Special Monte Carlo methods have

been studied by researchers, and Longstaff-Schwartz's least-squares method (see Sect. 21.7.2) is the most popular one used by practitioners so far for issuer callable (not puttable) products.

19.1.2 American Option Price Process*

Consider a *finite time horizon* $[t_0, t_n]$ and *discrete stopping times* taking values in $\{t_0, \dots, t_n\}$. Let h_t be an $\mathcal{F}_t$ -adapted process. Define a process v_t , starting from v_0 , as follows:

$$\begin{cases} v_n = h_n, \\ v_i = \max\{h_i, \mathbb{E}[v_{i+1}|\mathcal{F}_i]\}, & \forall i \in \{0, \dots, n\}, \\ v_t = \mathbb{E}[v_j|\mathcal{F}_t], & \forall t_{j-1} < t < t_j, \quad j \in \{1, \dots, n\}. \end{cases}$$

Proposition 19.1 *The process v_t is a supermartingale, i.e., $\forall s, t \in \mathbb{R}$ such that $t_0 \leq s < t \leq t_n$, $\mathbb{E}[v_t|\mathcal{F}_s] \leq v_s$.*

Proof First, we show that $\mathbb{E}[v_i|\mathcal{F}_t] \geq \mathbb{E}[v_m|\mathcal{F}_t]$, $\forall t_0 \leq t \leq t_i \leq t_m$, $t \in \mathbb{R}$, $i, m \in \{0, \dots, n\}$. From definition, we have

$$v_i = \max\{h_i, \mathbb{E}[v_{i+1}|\mathcal{F}_i]\} \geq \mathbb{E}[v_{i+1}|\mathcal{F}_i].$$

Taking conditional expectation on both sides with $\mathcal{F}_t$, we get

$$\mathbb{E}[v_i|\mathcal{F}_t] \geq \mathbb{E}[\mathbb{E}[v_{i+1}|\mathcal{F}_i]|\mathcal{F}_t] = \mathbb{E}[v_{i+1}|\mathcal{F}_t].$$

By iteration, we prove that $\mathbb{E}[v_i|\mathcal{F}_t] \geq \mathbb{E}[v_m|\mathcal{F}_t]$.

Now, $\forall t_0 \leq s < t \leq t_n$, let $j, k \in \{1, \dots, n\}$ such that $t_{j-1} \leq s \leq t_j$ and $t_{k-1} \leq t \leq t_k$. We have

$$\mathbb{E}[v_t|\mathcal{F}_s] = \mathbb{E}[\mathbb{E}[v_k|\mathcal{F}_t]|\mathcal{F}_s] = \mathbb{E}[v_k|\mathcal{F}_s] \leq \mathbb{E}[v_j|\mathcal{F}_s] = v_s. \quad \square$$

Proposition 19.2 *Let $t_\eta = \min\{t_i \in \{t_0, \dots, t_n\} | v_i = h_i\}$, $i, \eta \in \{0, \dots, n\}$. Then, (1) t_η is a stopping time; (2) $\forall t \in \mathbb{R}$ such that $t_0 \leq t \leq t_n$, the stopped process $v_{t \wedge t_\eta}$ is a martingale.*

Proof For convenience, we denote v_t^η for $v_{t \wedge t_\eta}$, v_i for v_{t_i} and $\mathcal{F}_i$ for $\mathcal{F}_{t_i}$.

- (1) η is a stopping time because $\{\eta = 0\} = \{v_0 = h_0\} \in \mathcal{F}_0$, and for $i \in \{1, \dots, n\}$, $\{\eta = i\} = \{v_0 > h_0\} \cap \dots \cap \{v_{i-1} > h_{i-1}\} \cap \{v_i = h_i\} \in \mathcal{F}_i$.
- (2) We first prove that for any discrete points $j, k \in \{0, \dots, n\}$ such that $\forall j < k$, $v_j^\eta = \mathbb{E}[v_k^\eta|\mathcal{F}_j]$ and it suffices to prove that $v_j^\eta = \mathbb{E}[v_{j+1}^\eta|\mathcal{F}_j]$ (as for $k > j+1$, it leads to $\mathbb{E}[v_{j+1}^\eta|\mathcal{F}_j] = \mathbb{E}[\mathbb{E}[v_{j+2}^\eta|\mathcal{F}_{j+1}]|\mathcal{F}_j] = \mathbb{E}[v_{j+2}^\eta|\mathcal{F}_j] = \dots = \mathbb{E}[v_k^\eta|\mathcal{F}_j]$).

Actually, it can be checked that

- for $\eta \leq j$, we have $v_j^\eta = h_\eta = \mathbb{E}[h_\eta | \mathcal{F}_j] = \mathbb{E}[v_{j+1}^\eta | \mathcal{F}_j]$;
- for $\eta > j$, we have $v_j^\eta = v_j = \max\{h_j, \mathbb{E}[v_{j+1} | \mathcal{F}_j]\}$, and $v_j > h_j$ because $v_j \leq h_j$ implies $\eta \leq j$. Hence, $v_j^\eta = \mathbb{E}[v_{j+1} | \mathcal{F}_j] = \mathbb{E}[v_{j+1}^\eta | \mathcal{F}_j]$.

It follows from the above two cases that

$$v_j^\eta = v_j^\eta \mathbf{1}_{\eta \leq j} + v_j^\eta \mathbf{1}_{\eta > j} = \mathbb{E}[v_{j+1}^\eta | \mathcal{F}_j] \mathbf{1}_{\eta \leq j} + \mathbb{E}[v_{j+1}^\eta | \mathcal{F}_j] \mathbf{1}_{\eta > j} = \mathbb{E}[v_{j+1}^\eta | \mathcal{F}_j].$$

Now, we consider the general case with $\forall s, t \in \mathbb{R}$ such that $t_0 \leq s < t \leq t_n$. Let $j, k \in \{1, \dots, n\}$ such that $t_{j-1} \leq s \leq t_j$ and $t_{k-1} \leq t \leq t_k$, then

- for $t_\eta \geq t$, we have

$$\mathbb{E}[v_t^\eta | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[v_k^\eta | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}[v_k^\eta | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[v_k^\eta | \mathcal{F}_j] | \mathcal{F}_s] = \mathbb{E}[v_j^\eta | \mathcal{F}_s] = v_s^\eta;$$

- for $t_\eta < t$, we have

$$\mathbb{E}[v_t^\eta | \mathcal{F}_s] = \mathbb{E}[v_k^\eta | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[v_k^\eta | \mathcal{F}_j] | \mathcal{F}_s] = \mathbb{E}[v_j^\eta | \mathcal{F}_s] = v_s^\eta.$$

Hence,

$$v_s^\eta = v_s^\eta \mathbf{1}_{t_\eta \geq t} + v_s^\eta \mathbf{1}_{t_\eta < t} = \mathbb{E}[v_t^\eta | \mathcal{F}_s] \mathbf{1}_{t_\eta \geq t} + \mathbb{E}[v_t^\eta | \mathcal{F}_s] \mathbf{1}_{t_\eta < t} = \mathbb{E}[v_t^\eta | \mathcal{F}_s]. \quad \square$$

Proposition 19.3 *Let $\mathcal{T}_{t,n}$ be the set of all stopping times taking values in $\{t_k, \dots, t_n\}$ and $t \in \mathbb{R}$ such that $t_0 \leq t \leq t_k$. Then,*

$$v_t = \sup_{\tau \in \mathcal{T}_{t,n}} \mathbb{E}[h_\tau | \mathcal{F}_t].$$

Proof According to Proposition 19.1, v_t is a supermartingale. The stopped process $v_{t \wedge \tau}$ is also a supermartingale by *Optional Sampling Theorem*. Hence,

$$v_t = v_{t \wedge \tau} \geq \mathbb{E}[v_{n \wedge \tau} | \mathcal{F}_t] = \mathbb{E}[v_\tau | \mathcal{F}_t] = \mathbb{E}[h_\tau | \mathcal{F}_t], \quad \forall \tau \in \mathcal{T}_{t,n}.$$

In particular, let t_η be the first time that $v_{t_\eta} = h_{t_\eta}$ as defined in Proposition 19.2. The stopped process $v_{t \wedge t_\eta}$ is a martingale,

$$v_t = v_{t \wedge t_\eta} = \mathbb{E}[v_{n \wedge t_\eta} | \mathcal{F}_t] = \mathbb{E}[v_{t_\eta} | \mathcal{F}_t] = \mathbb{E}[h_{t_\eta} | \mathcal{F}_t].$$

Hence, necessarily,

$$v_t = \sup_{\tau \in \mathcal{T}_{t,n}} \mathbb{E}[h_\tau | \mathcal{F}_t]. \quad \square$$

Comparing the definitions of v_t and $\widehat{V}_t$ in Sect. 19.1.1, we can see that v_t is applicable to price Bermudan options by setting $v_t = \widehat{V}_t$, $h_t = \widehat{H}_t$ and $\mathbb{E}$ as $\widetilde{\mathbb{E}}$.

It can be shown that, for an American option with continuous time exercise, the discounted price of the option is

$$\widehat{V}_t = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}[\widehat{H}_\tau | \mathcal{F}_t], \quad (19.2)$$

where $\mathcal{T}_{t,T}$ is the set of stopping times taking values in $[t, T]$. The process $\widehat{V}_t$ is a supermartingale and the stopped process $\widehat{V}_{t \wedge \tau}$ is a martingale (see [59]). For an in-depth analysis on American options, we refer to [66].

19.1.3 Partial Differential Inequality*

Because of the early exercise feature, the Black–Scholes PDE does not apply in general for American options under the BSM model. However, the price of an American option satisfies a system of inequalities as described below with a heuristic explanation.

The price of an option with exercise payoff of $H_t = H(t, S_t)$ ¹ is $\widehat{V}_t = \widehat{V}(t, S_t) = \max_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}[\widehat{H}_\tau | \mathcal{F}_t]$. By Itô–Doebelin formula, it satisfies

$$\begin{aligned} \frac{V_T}{M_T} &= \frac{V_0}{M_0} + \int_0^T \frac{1}{M_t} \left[\frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - r V \right] dt \\ &\quad + \int_0^T \frac{1}{M_t} \frac{\partial V}{\partial S} \sigma S_t d\widetilde{W}_t. \end{aligned}$$

As $\frac{V_t}{M_t}$ is a supermartingale, the term before dt should be ≤ 0 . We obtain a partial differential system for American options:

$$\begin{cases} \frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - r V \leq 0, \\ V \geq H, \\ \left(\frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - r V \right) (V - H) = 0, \\ V_T = H_T. \end{cases} \quad (19.3)$$

¹ For non-path-dependent options such as American call or put options, $H_t = H(t, S_t)$; for path-dependent options, $H_t = H(t, S_t, Y_t)$ where Y_t is an additional state variable.

The third equation, called linear complementarity problem, implies that if $V_t > H_t$, the option price follows the same PDE as the corresponding European option:

$$\frac{\partial V}{\partial t} + rS_t \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

19.2 Pricing of Some Exotic Options under the BSM Model*

In this section, we present some well-known results on binary, barrier, and Asian options under the BSM model. For other path-dependent options, we refer to [65, 105].

We note that the discontinuity in a binary or barrier payoff causes hedging issues such as very large delta and gamma when the underlier's price is close to the barrier level. Hence, the closed-form solutions presented in this section may not be directly used in option hedging/pricing. Practitioners usually apply the so-called payoff smoothing, which will be explained in Sect. 20.1.

19.2.1 European Binary Options

Binary option is the simplest option with a discontinuous payoff. Its holder receives assets or cash if the underlier ends up in-the-money, or nothing if out-of-the money. Here we consider the normalized *cash-or-nothing* call option, and its payoff at maturity T is given by

$$\mathbf{1}_{\{S_T > K\}}.$$

There is a closed-form formula for theoretically pricing cash-or-nothing call option under the BSM model. One can quickly realize that the desired price appears as the second term in the derivation of the Black–Scholes formula in Sect. 16.6.1,

$$\tilde{\mathbb{E}} \left[\frac{1}{M_T} \mathbf{1}_{\{S_T > K\}} \right] = e^{-rT} \tilde{\mathbb{P}}\{S_T > K\} = e^{-rT} N(d_2),$$

where we recall that $d_2 = \frac{\ln \frac{S_0}{K} + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_1 = d_2 + \sigma\sqrt{T}$. As a side note, the price of an *asset-or-nothing* call option with payoff $S_T \mathbf{1}_{\{S_T > K\}}$ is given by

$$\tilde{\mathbb{E}} \left[\frac{1}{M_T} S_T \mathbf{1}_{\{S_T > K\}} \right] = S_0 N(d_1).$$

Together, they form a typical European call option.

Another interesting property regarding binary options is the relation between a cash-or-nothing ATMS call option with payoff $1_{\{S_T > S_0\}}$ and a cash-or-nothing ATMS put option with payoff $1_{\{S_T \leq S_0\}}$. We know that their prices are

$$e^{-rT} \tilde{\mathbb{P}}\{S_T > S_0\} \text{ and } e^{-rT} \tilde{\mathbb{P}}\{S_T \leq S_0\}$$

respectively. Assuming interest rate is zero, the forward price of a non-dividend paying asset will be equal to its initial price (i.e., $\mathbb{E}[S_T] = S_0$). One may expect the aforementioned probabilities will be equal. Actually, since S_T is log-normally distributed in the BSM model, its probability density function is skewed to the left. So, $\tilde{\mathbb{P}}\{S_T \leq S_0\}$ is higher than $\tilde{\mathbb{P}}\{S_T > S_0\}$, and hence the binary put is worth more than the binary call.

19.2.2 American Binary and Barrier Options

American barrier options are widely used path-dependent options with payoff related to the underlier price history during the option life. There are two main categories of American barrier options. A *knock-in* (also known as *kick-in*) option becomes existent when the underlier price breaches a certain barrier level, while a *knock-out* option becomes void when the underlier price breaches the barrier at any time before the option maturity. The “in” options and “out” options respect the *in-out parity* such that one can induce the other (see Sect. 10.4.3). Therefore, we only take an up-and-out call option to illustrate the pricing and hedging of barrier options. The payoff at maturity T can be written as

$$(S_T - K)^+ 1_{\{\max_{0 \leq t \leq T} S_t < H\}},$$

where $H (> K)$ is the fixed barrier level. Notice that

$$(S_T - K)^+ 1_{\{\max_{0 \leq t \leq T} S_t < H\}} \leq (S_T - K)^+.$$

Therefore, the up-and-out call option is cheaper than the vanilla European call option, which is attractive to investors who believe in limited upside price moves.

The risk-neutral pricing formula for an up-and-out call is

$$V_0 = \tilde{\mathbb{E}}[e^{-rT} \max\{S_T - K, 0\} 1_{\max_{t \in [0, T]} S_t \leq H}].$$

Let $\hat{W}_T = \tilde{W}_T + \mu T$ where $\mu = \frac{1}{\sigma}(r - \frac{1}{2}\sigma^2)$. Then, $S_t = S_0 e^{\sigma \hat{W}_t}$. The joint distribution of S_T and $\max_{t \in [0, T]} S_t$ can be represented by the joint distribution of $\hat{W}_T, \hat{M}_T$.

The region for the integration implied by the expectation is $S_T \leq \max_{t \in [0, T]} S_t \leq H$, where $K \leq S_T \leq H$, which is equivalent to

$$\max\{\widehat{W}_T, 0\} \leq \widehat{M}_T \leq h \quad \text{and} \quad l \leq \widehat{W}_T \leq h$$

where $l = \ln \frac{K}{S_0} / \sigma$ and $h = \ln \frac{H}{S_0} / \sigma$. Hence,

$$\begin{aligned} V_0 &= \widetilde{\mathbb{E}}[e^{-rT} \max\{S_T - K, 0\} \mathbf{1}_{\max_{t \in [0, T]} S_t \leq H}] \\ &= \int_l^h \int_{\max\{y, 0\}}^h e^{-rT} (S_0 e^{\sigma y} - K) \frac{2(2x - y)}{T \sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x - y)^2}{2T}} dx dy \\ &= \int_l^h e^{-rT} (S_0 e^{\sigma y} - K) \frac{1}{\sqrt{2\pi T}} [-e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x - y)^2}{2T}}]_{\max\{y, 0\}}^h dy \\ &= A + B + C + D, \end{aligned}$$

where

$$\begin{aligned} A &= \int_l^h e^{-rT} S_0 e^{\sigma y} \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{y^2}{2T}} dy \\ B &= - \int_l^h e^{-rT} K \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{y^2}{2T}} dy \\ C &= - \int_l^h e^{-rT} S_0 e^{\sigma y} \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2h - y)^2}{2T}} dy \\ D &= \int_l^h e^{-rT} K \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2h - y)^2}{2T}} dy. \end{aligned}$$

The first term is

$$\begin{aligned} A &= S_0 \int_l^h e^{-rT + \sigma y} \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{y^2}{2T}} dy = S_0 \int_l^h \frac{1}{\sqrt{2\pi T}} e^{-\frac{[y - (\sigma + \mu)T]^2}{2T}} dy \\ &= S_0 \int_{\frac{l - (\sigma + \mu)T}{\sqrt{T}}}^{\frac{h - (\sigma + \mu)T}{\sqrt{T}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = S_0 [N(a_1) - N(a_2)], \end{aligned}$$

where $z = \frac{y - (\sigma + \mu)T}{\sqrt{T}}$, $a_1 = \frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $a_2 = \frac{\ln \frac{S_0}{H} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ by some algebraic operations and the property $N(-x) = 1 - N(x)$.

Similarly, we can obtain B, C, D , which leads to the formula for an up-and-out call option:

$$V_0 = S_0[N(a_1) - N(a_2)] - Ke^{-rT}[N(a_3) - N(a_4)] \\ - H \left(\frac{S_0}{H} \right)^{-\frac{2r}{\sigma^2}} [N(a_5) - N(a_6)] + Ke^{-rT} \left(\frac{S_0}{H} \right)^{-\frac{2r}{\sigma^2}+1} [N(a_7) - N(a_8)],$$

where $a_3 = a_1 - \sigma\sqrt{T}$, $a_4 = a_2 - \sigma\sqrt{T}$, $a_5 = \frac{\ln \frac{H^2}{KS_0} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $a_6 = \frac{\ln \frac{H}{S_0} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $a_7 = a_5 - \sigma\sqrt{T}$ and $a_8 = a_6 - \sigma\sqrt{T}$.

19.2.3 Asian Options with Continuous Sampling*

The most common averaging methods are arithmetic average and geometric average. Their mathematical representations take the following forms for $0 < t_1 < \dots < t_m = T$:

- Arithmetic average: $Y_T = \frac{1}{m} \sum_{j=1}^m S_j$
- Geometric average: $Y_T = \sqrt[m]{S_1 S_2 \dots S_m}$.

Their continuous-time approximations can be

- Arithmetic average: $Y_T = \frac{1}{T} \int_0^T S_t dt$
- Geometric average: $Y_T = \exp(\frac{1}{T} \int_0^T \ln S_t dt)$.

In Part I, we have discussed the purpose of Asian options in practice. One thing worth noticing is that averaging mechanism helps to control the delta at the sampling dates because of less volatility. However, averaging does not provide control on gamma during the life of the option.

Closed-form formulas of Asian options are not known for most cases, except when the payoff uses geometric average and the underlier follows log-normal distribution. The pricing utilizes the fact that the product of two log-normal variates still follows a log-normal distribution. On the other hand, arithmetic average makes use of addition operator and the sum of log-normal variates is no longer a log-normal variate. Therefore, numerical approaches are used for these cases.

Due to the path-dependency feature of the payoff, the option price at time t depends not only on (t, S_t) but also on $\{S_u, u < t\}$, leading to a non-Markovian problem. We need to add a new spatial dimension with the variable I_t :

$$I_t = \int_0^t S_u du, \text{ hence } dI_t = S_t dt \quad (19.4)$$

with $\frac{1}{T}I_t$ known as *running average* so that the option price will be dependent only on (t, S_t, I_t) and the problem becomes Markovian. We show now the derivation of the PDE for Asian Option with Continuous Sampling with payoff $V_T(S_T, Y_T)$.

Under $\widetilde{\mathbb{P}}$, Itô-Doeblin formula implies that the differential $d(e^{-rt}V(S_t, Y_t))$ is

$$e^{-rt} \left(\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + S \frac{\partial V}{\partial I} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV \right) dt + \sigma S \frac{\partial V}{\partial S} d\widetilde{W}_t$$

where Eq. (19.4) is used. The martingale property of $e^{-rt}V(S_t, Y_t)$ requires that the dt term should be zero, which leads to the PDE for average option with continuous sampling:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + S \frac{\partial V}{\partial I} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \quad (19.5)$$

The terminal and boundary conditions depend on the option payoff. For example, the conditions for the payoff $\max\{\frac{1}{T}I_T - K, 0\}$ are

$$\begin{aligned} V_T(S, I) &= \max\left\{\frac{1}{T}I_T - K, 0\right\}, \\ \lim_{I \rightarrow 0} V_t(S, I) &= 0, \\ \lim_{S \rightarrow 0} V_t(S, I) &= e^{-r(T-t)} \max\left\{\frac{1}{T}I_T - K, 0\right\}. \end{aligned} \quad (19.6)$$

The resolution of this PDE is tricky for the convergence (see [119]). Actually, the Asian options used in practice involve rather discrete sampling than continuous sampling, which will be discussed in Sect. 23.9. However, the methodology of adding state variables remains useful for implementing applicable numerical solutions for path dependent options in practice (see Sect. 23.15). Interested readers can refer to [105] for the PDE derivation for other lookback options.

A Useful Result

Practitioners often make a quick estimation of Asian option price with the pricer of vanilla options by applying the volatility of $\frac{\sigma}{\sqrt{3}}$. This is based on the below property of Brownian motion.

Proposition 19.4 *Consider a process X_t satisfying $dX_u = \sigma dW_u$. Then,*

$$\text{Var} \left(\frac{1}{t-s} \int_s^t (X_u - X_s) du \right) = \frac{\sigma^2(t-s)}{3}.$$

Proof It is equivalent to proving

$$\text{Var} \left(\int_s^t (W_u - W_s) du \right) = \frac{(t-s)^3}{3}.$$

Covariance of Brownian motion and direct calculations give

$$\begin{aligned} \text{Var} \left(\int_s^t (W_u - W_s) du \right) &= \mathbb{E} \left[\int_s^t \int_s^t (W_u - W_s)(W_v - W_s) dudv \right] \\ &= \int_s^t \int_s^t \mathbb{E}[(W_u - W_s)(W_v - W_s)] dudv = \int_s^t \int_s^t (\min\{u, v\} - s) dudv \\ &= \int_s^t \int_s^t (u - s) \mathbf{1}_{\{u \leq v\}} dudv + \int_s^t \int_s^t (v - s) \mathbf{1}_{\{u > v\}} dudv \\ &= 2 \int_s^t \int_s^y (x - s) dx dy = 2 \int_s^t \int_0^{y-s} z dz dy = \int_s^t (y - s)^2 dy = \frac{(t-s)^3}{3}. \end{aligned}$$

□

Let Y_t denote the arithmetic average of X_t with continuous sampling, i.e., $Y_t = \frac{1}{t} \int_0^t X_u du$ with $Y_0 = 0$. Since $\text{Var}(X_t) = \sigma^2 t$, we know from above that variance of Y_t becomes one third of the original, i.e.,

$$\text{Var}(Y_t) = \text{Var} \left(\frac{1}{t} \int_0^t X_u du \right) = \frac{\sigma^2 t}{3},$$

by taking $s = 0$. Equivalently, the volatility of Y_t for the time period is $\frac{\sigma}{\sqrt{3}}$.

Chapter 20

Hedging/Pricing Options and Structured Products in Practice*



In this chapter, we present some hedging issues for options and structured products in practice. Pricing solution and risk mitigation methods are introduced for commonly encountered product features and risks: barrier risk, correlation risk and large delta issue. We note that one or more risks may exist in a single derivative product.

20.1 Barrier Risk and the Mitigation Methods

Some exotic options and structured products offer sizable payoff contingent on the underlier price being above or below a predefined level known as barrier, such as binary options. When the underlier price is close to the barrier level, the delta tends to infinity for American barrier options, as well as European options with a short time to maturity. This makes the delta hedging quite impossible to implement. Furthermore, the execution of a large delta amount may impact the market price of the underlier. In addition to these issues, the barrier breach may also lead to a negative gamma situation for the trading book. The issue of hedging a product with binary payoff near the barrier is known as *gap risk*.

The techniques to mitigate the gap risk usually consist of transforming the *contract payoff* with barrier into a smoother *hedging payoff* that covers the contract payoff. The application of smoothing techniques is known as *over-hedging* because the impact of the product's price is always favorable to the hedging party and the over-hedged payoff also goes to the hedging party.

Maximum delta amount is a key measure for market risk management that is typically set at a target below 20% of the underlier's ADTV (average daily trading volume) to minimize the impact of delta hedging trades (buy or sell) on

the market price of the underlier. The parameter Δ_{max} as defined in the below can be determined to control maximum delta amount:

$$N \times \Delta_{max} \leq 20\% \text{ADTV}.$$

The focus of this section is about the smoothing techniques used in practice. As exotic options and structured products are usually offered by a hedging party to investors who do not hedge their exposure in general, we will use the word *trader* to designate the hedging party in the sequel. Generally, we denote H (in percentage of S_0) for the barrier level in a derivative contract and P_t for the performance of the underlier price ($P_t = S_t/S_0$). The basic techniques used in practice for hedging derivatives and structured products with gap risk are

- *call/put spread smoothing*: when the barrier is observed at a fixed time point (European barrier);
- *barrier shift*: when the barrier is observed continuously or nearly continuously (American barrier).

20.1.1 Smoothing for European Barrier

For a European barrier, the delta can be smoothed by applying a *call/put spread* to the contract payoff. Let us take a contract paying $N \times \mathbf{1}_{\{P_T > H\}}$ as an example to illustrate the method. If the investor is the long party, call spread smoothing means that instead of hedging the contract payoff, the trader hedges the below transformed option that does not have gap risk but still covers the investor's payoff as illustrated by the left graphic on Fig. 20.1:

$$N \times \min\{1, \max\{0, \Delta_{max} \times (P_T - H')\}\},$$

where

- $N \times \Delta_{max}$ is the maximum delta;
- $\text{spread} = 1/\Delta_{max}$ and $H' = H - \text{spread}$.

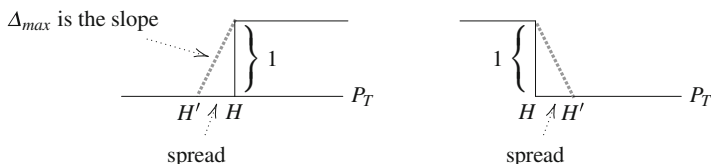


Fig. 20.1 Smoothing gap risk by call or put spread

We note that the product terms for the investor are not changed by the over-hedging. If the final performance ends up at a level between H and H' , the “over-hedged payout” will belong to the hedging party, but not to the investor.

If the investor is the short party, call spread smoothing implies the hedging with the below transformed payoff

$$N \times \min\{1, \max\{0, \Delta_{max} \times (P_T - H)\}\},$$

and we notice that the client’s final payment will cover the hedging party’s contractual payoff. The put spread smoothing case can be deduced following the same rationale. For worst-of or best-of observations, the call/put spread can be applied to the worst/best performance.

Example 20.1 Consider a USD1m 110% strike European binary call option linked to a stock with ADTV of USD50m. The trader wants to use $\Delta_{max} = \text{USD10m}$ that corresponds to 20% of ADTV. The hedging call-spread $H' - 110\%$ should satisfy $1/(110\% - H') = 10$, which leads to $H' = 100\%$ (see Fig. 20.2). $\square$

For large size trades with basket underliers of low liquidity, the following trick may be considered to obtain more aggressive pricing without losing control on Δ_{max} . The idea is to transform a payoff conditioned by one performance, e.g., the worst performance, into a sum of payoffs linked to the performance of all the underliers. For instance, the transformation to the contract payoff $\mathbf{1}_{\{\min_i P_T^i > H\}}$ is as follows:

$$\mathbf{1}_{\{\min_i P_T^i > H\}} = 1 - \mathbf{1}_{\{\min_i P_T^i < H\}} = 1 - \min \left\{ 1, \sum_{i=1}^n \mathbf{1}_{\{P_T^i < H\}} \right\}.$$

For each $\mathbf{1}_{\{P_T^i < H\}}$, we apply put spread for smoothing:

$$\text{Smoothed Payoff} = 1 - \min\{1, \sum_{i=1}^n \text{Put Spread}(i)\},$$

where $\text{Put Spread}(i) = \max\{0, \min\{1, \Delta_{max}^i \times (H - P_T^i)\}\}$.

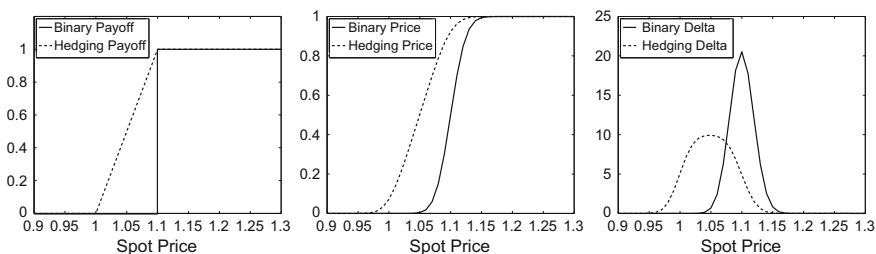


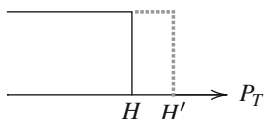
Fig. 20.2 Delta-hedging a binary call with call spread

Imagine that the current worst performing underlier has very poor liquidity, but the second worst underlier has very high liquidity. The poor liquidity of the worst underlier results in a large spread that makes the price quite expensive. However, the second worst has a very tight spread thanks to its liquidity. In the normal smoothing method, only the worst performance is used for pricing. But with our method, the second worst performance may also play a role in pricing, which cheapens the price of the product.

For options with early termination based on European barrier, the gap risk value is the difference between the knock-out or exercise payoff and the continuation value if it does not early terminate. We note that for some of structures, the sign of the gap risk value may be dependent on the market parameters. For the normal forward Monte Carlo method, the continuation value needs to be estimated at each barrier observation time and American Monte Carlo or any other method can be used for this purpose. Another simple approach, known as backward smoothing method, consists of applying call/put spread smoothing path-by-path in a backward way. It is easy to implement, but usually over estimates the smoothing cost.

20.1.2 Smoothing for American Barrier

For smoothing the gamma/delta near an American barrier, *barrier shift* technique can be applied, which consists of hedging/pricing the product with a shifted barrier that is not the contract barrier of the product. The natural barrier shift should move the barrier further away from the spot price, so that the option will be knocked out by the contract barrier before reaching the shifted barrier level. See below graphic for reference. For example, the hedging party selling an up-and-out binary option with contract barrier at 135% of initial price may shift the barrier to 140% for hedging/pricing. The pricing will be on the conservative side to the hedging party.



Barrier shift for smoothing gamma or delta

However, if the hedging party is selling the up-and-in binary option, the hedging barrier cannot be simply shifted closer to the current price because this will not smooth out gamma/delta at the barrier. In this case, the hedging party may break the trade into n smaller size trades over a series of shifted barrier levels between the current price and the contract barrier, as illustrated by the right graphic on Fig. 20.3.

Instead of making barrier shift, another method consists of applying *call/put spread on the extremum* (i.e., P_t^i replaced by the extremum for the whole barrier observation period, either minimum or maximum, according to the product) as

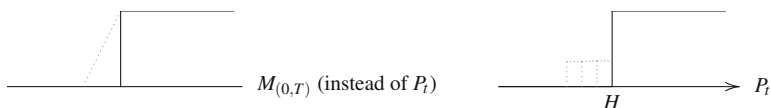


Fig. 20.3 Hedging up-and-in binary option ($M_{(0,T)}$ is the maximum performance in $[0, T]$)

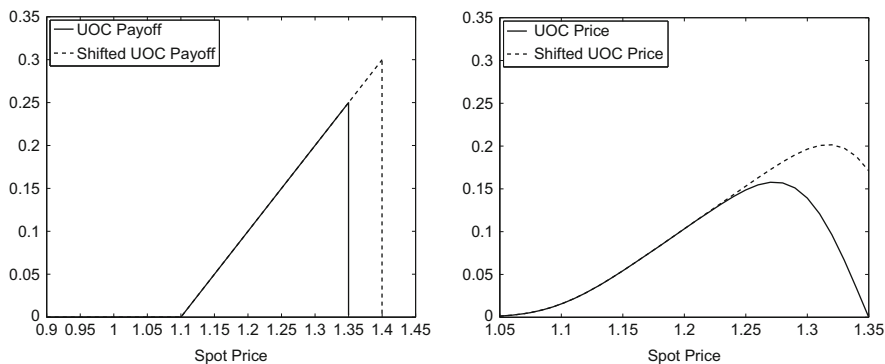


Fig. 20.4 Delta-hedging up-and-out calls with two different barriers

shown by the left graphic on Fig. 20.3. The method works for both buying and selling positions.

Example 20.2 Figure 20.4 illustrates *barrier shift* method applied to an up-and-out call with contract barrier H . The option writer used H' ($H' > H$) for pricing and hedging. We can notice in the graphic that over-hedging with a shifted barrier does not bound the delta as the case in European binary option. The delta and gamma may be still large near H' , but it no longer matters because the contract is already void after breaking H , and that the hedging stops. $\square$

20.2 Correlation Risk

20.2.1 Sources of Correlation Risk

The correlation sensitivity in derivative products may come from various sources, including

- multi-asset feature: worst-of type of structures are widely used in practice, which leads to one-way correlation exposures to the trading book of the product issuer. In the market, implied volatilities are quoted, and the volatility risk can be hedged for a large range of underliers. However, it is not the case for correlations;

- quanto feature: as highlighted in Sect. 11.4, quanto is a natural feature for derivative products. It involves the correlation risk between the foreign exchange rate and underlier prices, although the foreign exchange rate may not appear in the payoff formula. The modelling for FX risk is presented in Sect. 18.3 that covers both the FX rate volatility risk and the correlation risk. It also reveals the foreign exchange position in hedging.
- multi-factor modelling: multi-factor models such as the Bergomi stochastic volatility model and the BGM model have the correlations as part of model parameters.

20.2.2 Correlation Risk Management

In general, correlation risk is difficult to hedge in the market. Correlations between equity underliers are normally positive and tend to be high in a bearish market and low in a bullish market. This phenomenon is called *correlation skew*. Sometimes, correlation exposure may be obtained through outperformance options, correlation swaps, etc. on liquid names in the broker market. But the correlation risks from structured business are mainly warehoused in the trading book.

To estimate the correlations to be used in pricing, practitioners usually look at the chart of historical correlations over one or more time windows, e.g., 3 months and 6 months, rolling over an observation period of ~ 3 years. To produce a chart with a rolling time window, we start with the first date of the observation period as the time window beginning date and calculate the correlation value with the formula in Sect. 10.5.3, which leads to the first point in the chart. Then, we choose the second date in the observation period as the new time window beginning date and calculate the correlation of the new time window. This process is repeated as far as possible, and we obtain a historical correlation chart for the observation period.

For the majority of structured products, the trading books are selling correlation, i.e., an increase in correlation between the underliers in a product will increase its valuation if the other market parameters remain stable.

Trading books take reserves on correlation by conservatively shifting the estimated correlation for pricing and hedging. If shifted correlations are also used to produce the Greeks for hedging, this reserve will be gradually released through the time decay.

For shifting correlations, a simple method used by practitioners consists of interpolating between the estimated historical correlation matrix $\Sigma_{\text{Historical}}$ and an interpolation matrix E^+ , E^0 or E^- that are matrices with 1 on the main diagonal and

respectively, 1, 0 and -1 everywhere else. For example, to bump up the correlation matrix, we can apply the below method:

$$\text{shifted correlation} = \alpha \Sigma_{\text{Historical}} + (1 - \alpha) \mathbf{E}^+, \alpha \in [0, 1].$$

As the trading books for equity structured products are selling correlations in general, traders are happy to buy correlations at historical levels.

In case there is a need to shift down some correlations, e.g., FX-equity correlation, a more general interpolation matrix can be considered, such as

$$\mathbf{E}_{n \times n} = \begin{pmatrix} \mathbf{A} & \mathbf{B}^\top \\ \mathbf{B} & \mathbf{C} \end{pmatrix}, \quad 0 < m < n$$

where $\mathbf{A} = \mathbf{E}_{m \times m}^+$, $\mathbf{B}$ is $(n - m) \times m$ matrix with -1 for all entries, and $\mathbf{C} = \mathbf{E}_{(n-m) \times (n-m)}^+$. That is,

$$\text{shifted correlation} = \alpha \Sigma_{\text{Historical}} + (1 - \alpha) \mathbf{E}, \alpha \in [0, 1].$$

Practitioners may also “manually” set some correlation values directly in the matrix.

The historical correlation matrix shifted by the interpolation method with $\mathbf{E}^+$ or $\mathbf{E}^0$ is always positive semi-definite as $\mathbf{E}^+$ and $\mathbf{E}^0$ are both positive definite and so is the interpolated matrix. However, the shift with $\mathbf{E}^-$ may not keep the positive semi-definite property for $n \geq 3$ as $\mathbf{E}^-$ is not positive semi-definite in this case.¹ If the correlation matrix after shift or “manual” change is no more positive semi-definite, we can apply a repair method to make it positive semi-definite (e.g., spectral decomposition method in [58]²).

20.3 Large Delta Issue

For large delta amount (compared to the liquidity of the underlier) at product expiration or at initial hedging time, the execution of the delta amount in the market is exposed to slippage risk. Averaging is often used to mitigate such risk. It consists of taking the averaged price over a few days (e.g., 3 days) as reference (either final reference price or initial reference price) to determine the payoff amount, which is specified in contract terms. We note that while averaging reduces the delta at expiration, it does not help much for reducing gamma before expiration unless the averaging is applied over a significantly long period.

¹ Actually, $\mathbf{v} \mathbf{E}^- \mathbf{v}^\top < 0$ where $\mathbf{v} = \text{diag}(1, 1, 1, 0, \dots, 0)$.

² The idea of this method consists of (1) diagonalizing the correlation matrix into eigenvectors and eigenvalues, (2) replacing the negative eigenvalues by zero, (3) using the updated eigenvalues with the eigenvectors and a normalization factor to regenerate the correlation matrix.

Exercises

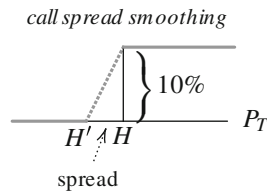
1. What is a correlation skew?
2. Assume the underlier follows the BSM with no dividend. Let c denote the time- t Black–Scholes European call price with strike price K and maturity T . Show that

$$-\frac{\partial c}{\partial K} = e^{-r(T-t)} N(d_2).$$

Explain how a bullish call spread and a binary cash-or-nothing call option are related through the above equation.

3. Assume the underlier follows the BSM with no dividend. Derive the time- t delta of a binary cash-or-nothing call option with payoff $1_{\{S_T > K\}}$ where S_T is the underlier price at maturity T and K is the strike. Show that the delta approaches $+\infty$ when $t \rightarrow T$ for $S_t = K$.
4. Consider a stock with ADTV (Average Daily Trading Volume) being USD80m. A client of an investment bank wants to buy a 3 month, 105% strike binary call option (European style) that pays a coupon of 10% of the notional amount of USD10m.

The trader needs to control the maximum delta to be at or below 20% of ADTV by applying a call-spread type of smoothing as illustrated below:



Determine the width of *spread* in percentage (%) of the initial stock price.

Chapter 21

Numerical Method (1): Monte Carlo Simulation



21.1 Monte Carlo Method for Pricing European Options

Risk-neutral pricing analysis suggests that the European option price is the expectation of discounted payoff under $\tilde{\mathbb{P}}$,

$$V_0 = \tilde{\mathbb{E}}[e^{-rT} V_T].$$

Monte Carlo simulation method consists of estimating the expectation by repeatedly sampling the involved random variables.¹ It relies on the Strong Law of Large Numbers implying that for *i.i.d.* (independent and identically distributed) random variables $X_1, X_2, \dots$ with expected value $\mathbb{E}[X]$ ($\mathbb{E}[|X|] < +\infty$), we have²

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N X_k = \mathbb{E}[X] \quad \text{a.s.}$$

Hence, the price of a European option can be approximated by taking the arithmetic average of the simulated values of discounted payoff, $e^{-rT} V_T^k$, where $k = 1, \dots, N$, under risk-neutral probability measure $\tilde{\mathbb{P}}$. This can be expressed as

$$V_0 = \tilde{\mathbb{E}}[e^{-rT} V_T] \approx \frac{1}{N} \sum_{k=1}^N e^{-rT} V_T^k.$$

¹ Monte Carlo simulation is also a general numerical integration method, which can be understood by $\int_a^b f(x)dx = (b-a) \int_a^b f(x) \frac{1}{b-a} dx = (b-a) \mathbb{E}[f(u)]$ where u is a $[a, b]$ -uniform random variable with probability density $\frac{1}{b-a}$.

² For any integrable function $h(X)$, $h(X_k)$ are also *i.i.d.* random variables. Hence, $h(X)$ can be approximated by the average of the simulated $h(X_k)$ values.

In this chapter, we focus on the methods used to simulate discounted payoff value in Monte Carlo simulation. For a comprehensive presentation of Monte Carlo implementation in financial engineering, please refer to [43].

Consider the general Itô process³ with known initial value for the underlier's price process under risk-neutral probability measure:

$$dS_t = \theta(t, S_t)dt + \sum_{j=1}^d \phi_j(t, S_t)dW_t^j.$$

The Monte Carlo simulation for European options can be described as follows :

- (1) discretize the time period $[t, T]$ into m intervals: $t = t_0 < t_1 < \dots < t_m = T$. The intervals $(\Delta t_i = t_i - t_{i-1})$ depend on the option payoff and the model parameters. It is not necessary to have equally spaced intervals. All the sampling points used in the option payoff description should be considered to define the time intervals;
- (2) generate the independent Brownian increments $\Delta W_i^j = \epsilon_i^j \sqrt{\Delta t_i}$, $\forall j = 1, \dots, d$, where the ϵ_i^j follow a d -dimensional normal distribution; The method to generate ϵ_i^j will be presented in the next sections;
- (3) simulate the sample path with *Euler–Maruyama scheme*,⁴ i.e.,

$$S_i = S_{i-1} + \theta(t_{i-1}, S_{i-1})\Delta t_i + \sum_{j=1}^d \phi_j(t_{i-1}, S_{i-1})\Delta W_i^j$$

where S_i stands for S_{t_i} for notation simplicity;

- (4) compute the option payoff V_T for the path with the simulated S_T , e.g., for a European call option, it is $\max\{S_T - K, 0\}$;
- (5) repeat steps (2)–(4) N times. Let V_T^k ($1 \leq k \leq N$) be the k -th simulated option payoff. The option price can be approximated by $V_t = \frac{1}{N} \sum_{i=1}^N e^{-r(T-t)} V_T^k$.

Monte Carlo simulation has the following advantages:

- it is numerically efficient for high-dimensional cases because the computation complexity increases linearly with the number of underliers. In contrast, the computation time for other numerical methods tends to increase exponentially with the number of dimensions;
- it can accommodate complex payoffs and complex stochastic processes;
- the convergence can be improved by increasing the simulation numbers (see Sect. 21.4)

³ The Monte Carlo methods for processes other than Itô process follow the same methodology presented here.

⁴ There exist more accurate schemes, e.g., Milstein scheme that contains a second order term.

Monte Carlo approach is suitable for pricing path-dependent options such as Asian options or lookback options.

21.1.1 Simulation Under the Black–Scholes–Merton Model

For the Black–Scholes–Merton model, there is no need to simulate with small time steps. This is because the model parameters are constant (i.e., volatility and interest rate) and that the future price S_T has a solution under risk-neutral measure

$$S_T = S_t e^{(r-\sigma^2/2)(T-t) + \sigma \Delta W_{T-t}}$$

where S_t is the current underlier price. So, the simulation can be done with one time step only from t to T for European options.

If we need to simulate some underlier prices between t and T (e.g., for pricing path-dependent options), the above equation can be used between two consecutive required dates. The k -th path from t to T is generated by:

$$S_i^k = S_{i-1}^k e^{(r-\sigma^2/2)\Delta t_i + \sigma \Delta W_i^k}, \quad i = 1, \dots, m, \quad k = 1, \dots, N, \quad (21.1)$$

where $\Delta W_i^k = \epsilon_i^k \sqrt{T-t}$ and ϵ_i^k are independent random numbers distributed as $\mathcal{N}(0, 1)$. The number of paths N depends on the requirement on accuracy (see Sect. 21.4).⁵

Example 21.1 Consider a European put with $K = 10$, $S_0 = 5$, $r = 0.01$ and $\sigma = 0.25$. Table 21.1 shows the comparison of Monte Carlo method and the accurate solution of the Black–Scholes equation for different expiry time and number of paths. The results are more accurate when we increase the number of paths N . $\square$

Table 21.1 Monte Carlo simulation for a European put

$T - t$	Accurate value	$N = 10$	$N = 10^2$	$N = 10^3$	$N = 10^4$	$N = 10^5$
0.25	4.9750	4.4913	4.9211	5.0018	4.9727	4.9751
0.50	4.9501	4.8818	4.9796	4.9510	4.9597	4.9495
0.75	4.9256	5.1352	4.7105	4.9073	4.9094	4.9284
1.00	4.9022	4.4807	4.8380	4.9072	4.8957	4.9015

⁵ For the exotic option books of investment banks, the number of simulation paths N is more than 2^{14} when Quasi-Monte Carlo is used.

21.2 Generating One-Dimensional Normal Variate

We introduce two popular methods to generate one-dimensional normal random variates.

21.2.1 Inverse Sampling

Inverse sampling is a general method to generate random variables with an invertible and continuous distribution function F .

Denote $\mathcal{U}[0, 1]$ as the uniform distribution over $[0, 1]$. By definition, for $u \sim \mathcal{U}[0, 1]$, we have

$$\mathbb{P}\{u \leq z\} = \begin{cases} 0, & z < 0, \\ z, & 0 \leq z \leq 1, \\ 1, & z > 1. \end{cases}$$

Inverse sampling method consists of applying the below steps to generate a random variable X with distribution function F :

- (1) generate⁶ u from $\mathcal{U}[0, 1]$
- (2) calculate $X = F^{-1}(u)$.

To generate a standard normal random variable⁷ ϵ , we use $F = N$ to get $\epsilon = N^{-1}(u)$. Monte Carlo simulation for Poisson process based jump models will be presented in Chap. 26.

Now, we check the rationale of the inverse sampling method. As $F(x)$ is an increasing continuous function of x , we have $g = F^{-1}(u) < x$ if and only if $u < F(x)$. It follows that

- $F(x)$ follows $[0, 1]$ uniform distribution: $\forall u \in [0, 1]$,

$$\begin{aligned} \mathbb{P}\{F(x) \leq u\} &= 1 - \mathbb{P}\{u < F(x)\} = 1 - \mathbb{P}\{g < x\} \\ &= \mathbb{P}\{x \leq g\} = F(g) = F(F^{-1}(u)) = u; \end{aligned}$$

- $X (= F^{-1}(u))$ follows the distribution of F :

$$\mathbb{P}\{X \leq x\} = \mathbb{P}\{F^{-1}(u) \leq x\} = \mathbb{P}\{u \leq F(x)\} = F(x).$$

⁶ The method to generate uniform variates can be found in [93]. In MS-Excel, it is RAND().

⁷ The inverse function for normal variates in MS-Excel is NORMSINV or NORMINV.

21.2.2 Box–Muller Method*

Another popular numerical method to generate normal randoms is the Box–Muller transform, which is based on the result that if u_1 and u_2 are two independent $[0, 1]$ uniform random variables, then ϵ_1 and ϵ_2 defined below are independent normal random variables;

$$\epsilon_1 = \sqrt{-2 \ln u_1} \cos(2\pi u_2), \quad \epsilon_2 = \sqrt{-2 \ln u_1} \sin(2\pi u_2).$$

See Appendix A.7 for the proof.

21.3 Generating Multi-Dimensional Normal Variates*

For multi-asset products or multi-factor models that require the simulation of multiple assets, we also need to take the correlation among assets into consideration. For example, it may involve generating correlated Brownian motions, or essentially a normally distributed vector $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with mean vector $\boldsymbol{\mu}$, covariance matrix $\boldsymbol{\Sigma}$ and correlation matrix $\mathbf{C}$ ($\boldsymbol{\Sigma} = \mathbf{D}\mathbf{C}\mathbf{D}^\top$, $\mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$, $\mathbf{D}^\top = \mathbf{D}$; see Sect. 12.3.3). Let $\boldsymbol{\epsilon}$ be an n -vector of independent standard normal variables, i.e., $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ where $\mathbf{0}$ is an n -vector of zeros and $\mathbf{I}$ identity matrix. We have the following result for generating a normal vector from a standard normal vector.

Proposition 21.1 *Normally distributed $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ can be generated through either one of the below methods*

- $\mathbf{X} = \boldsymbol{\mu} + \mathbf{A}\boldsymbol{\epsilon}$
- $\mathbf{X} = \boldsymbol{\mu} + \mathbf{D}\mathbf{L}\boldsymbol{\epsilon}$

where $\mathbf{A}$ and $\mathbf{L}$ are $n \times n$ matrices that decompose, respectively, the covariance matrix $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top$ and correlation matrix $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$.

Proof By the linear transformation property for multivariate normal vectors (see Proposition A.2 in Appendix), $\mathbf{A}\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{A}\mathbf{0}, \mathbf{A}\mathbf{I}\mathbf{A}^\top) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$. Therefore, $\mathbf{X} = \boldsymbol{\mu} + \mathbf{A}\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Similarly, we have $\mathbf{D}\mathbf{L}\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{D}\mathbf{L}\mathbf{0}, \mathbf{D}\mathbf{L}\mathbf{L}^\top\mathbf{D}^\top) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{D}\mathbf{L}\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. $\square$

The following two methods can be used to find the decomposition of a covariance matrix $\boldsymbol{\Sigma}$ or a correlation matrix $\mathbf{C}$.

21.3.1 Cholesky Factorization

Cholesky factorization decomposes Σ or $\mathbf{C}$ into the product of a lower triangular matrix and its transpose. Consider a positive definite covariance matrix Σ ,

$$\Sigma = \mathbf{A}\mathbf{A}^\top = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ 0 & a_{22} & \cdots & a_{n2} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{bmatrix}.$$

Given the entries σ_{ij} of Σ , we can solve $\mathbf{A}$ recursively through the following system of equations:

$$\begin{cases} a_{11}^2 = 1, \\ a_{11}a_{21} = \sigma_{12}, \\ \vdots \\ a_{n1}a_{11} = \sigma_{1n}, \\ a_{21}^2 + a_{22}^2 = 1, \\ \vdots \\ a_{n1}^2 + a_{n2}^2 + \cdots + a_{nn}^2 = 1. \end{cases}$$

Example 21.2 Cholesky decomposition of a two-dimensional correlation matrix $\mathbf{C}$ is

$$\mathbf{C} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \begin{bmatrix} 1 & \rho \\ 0 & \sqrt{1-\rho^2} \end{bmatrix} = \mathbf{L}\mathbf{L}^\top.$$

To generate two correlated normal variates X_1 and X_2 such that

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right),$$

we can carry out

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix}$$

where ϵ_1 and ϵ_2 are independent standard normal variates. Readers can compare this example with the definition of correlated Brownian motions in Chap. 13 by setting $\mu_1 = \mu_2 = 0$ and $\sigma_1 = \sigma_2 = 1$. $\square$

21.3.2 Eigenvector Factorization*

The Cholesky decomposition is not the only way to have $\Sigma = \mathbf{A}\mathbf{A}^\top$. According to Spectral Theorem in linear algebra, a symmetric and real valued matrix such as Σ can be diagonalized as

$$\Sigma = \mathbf{U}^\top \mathbf{A} \mathbf{U},$$

where $\mathbf{A}$ is a diagonal matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, and $\mathbf{U}$ is an orthogonal matrix ($\mathbf{U}^\top \mathbf{U} = \mathbf{I}$) (see Appendix Proposition A.3). Setting

$$\mathbf{A} = \mathbf{U}^\top \mathbf{A}^{1/2} = \mathbf{U} \begin{bmatrix} \sqrt{\lambda_1} & & & \\ & \sqrt{\lambda_2} & & \\ & & \ddots & \\ & & & \sqrt{\lambda_n} \end{bmatrix},$$

we have $\mathbf{A}\mathbf{A}^\top = \mathbf{U}^\top \mathbf{A}^{1/2} \mathbf{A}^{1/2} \mathbf{U} = \mathbf{U}^\top \mathbf{A} \mathbf{U} = \Sigma$.

The ranking of the eigenvalues from such decomposition method indicates the importance order for the dimensions which are used in the principal component method for Quasi-Monte Carlo.

21.4 Simulation Accuracy Measure*

Confidence interval (see Sect. 12.4.3) can be used to measure the accuracy of a simulated result. Practitioners can increase the number of simulations until a satisfactory confidence interval is achieved. For example, the 95% confidence interval for V is given by

$$\bar{\mu} - \frac{1.96\bar{\sigma}}{\sqrt{N}} < V < \bar{\mu} + \frac{1.96\bar{\sigma}}{\sqrt{N}},$$

where $\bar{\mu}$ and $\bar{\sigma}$ denote the sample mean and standard deviation of N simulations for the option price V . The Central Limit Theorem states that V should be distributed as $\mathcal{N}(\bar{\mu}, \frac{\bar{\sigma}^2}{N})$ implying that the convergence depends on $O(\frac{1}{\sqrt{N}})$. Hence, to halve the confidence interval, we must quadruple the number of simulations.

To improve convergence efficiency, researchers have studied various approaches, such as Quasi-Monte Carlo methods and variance reduction techniques that we will present in the following sections.

21.5 Variance Reduction Techniques*

In this section, we will introduce several well-known variance reduction techniques, including problem-specific methods and universal methods. It can be easily checked that they all provide unbiased estimate for the derivative value V . The combination of the methods may further improve the result.

Variance reduction analysis is provided to compare with the variance of the normal Monte Carlo, which is $\text{Var}(V) = \text{Var}(\frac{1}{N} \sum_{i=1}^N V_i) = \frac{\sigma^2}{N}$, i.e., convergence of $O(\frac{1}{\sqrt{N}})$ as indicated by the Central Limit Theorem.

21.5.1 Antithetic Sampling

A pair of identically distributed random variables X and X^* are called antithetic if they are negatively correlated. To simulate a Brownian motion, the antithetic sampling method consists of using both ϵ_i and $-\epsilon_i$ where $\epsilon_i \sim \mathcal{N}(0, 1)$.

The simulated value for a derivative by antithetic sampling with N iterations is

$$\tilde{V}^{AS} = \frac{1}{2}(\tilde{V} + \tilde{V}^*) = \frac{1}{N} \sum_{i=1}^N \frac{V_i + V_i^*}{2}$$

where

- V_i and V_i^* are respectively the derivative values using ϵ_i and $-\epsilon_i$ for the i -th iteration;
- $\tilde{V} = \frac{1}{N} \sum_{i=1}^N V_i$ and $\tilde{V}^* = \frac{1}{N} \sum_{i=1}^N V_i^*$ are respectively the simulated derivative values with ϵ_i 's and $-\epsilon_i$'s respectively.

Intuitively, for most of the derivatives, if $\tilde{V}$ overestimates V , $\tilde{V}^*$ may tend to underestimate it, and vice versa.

We note that each V_i and V_i^* is distributed as V . Let σ^2 be the variance of V and ρ the correlation between V_i and V_i^* . The variance of $\tilde{V}^{AS}$ is

$$\tilde{V}^{AS} = \frac{N}{4N^2} [\text{Var}(V) + \text{Var}(V^*) + 2\text{Cov}(V, V^*)] = \frac{\sigma^2}{2N}(1 + \rho).$$

For comparison, the variance of the derivative value by normal Monte Carlo is $\frac{\sigma^2}{2N}$ with $2N$ iterations and $\frac{\sigma^2}{N}$ with N iterations. The cost of computation of antithetic sampling is closer to that for $2N$ iterations. For a directional payoff, ρ is often negative, which provides variance reduction. However, there exist positive correlation cases (e.g., straddle option, some payoffs depending on higher order moments of the underlier price), whereby antithetic sampling method fails.

21.5.2 Control Variables

The variance reduction by control variable technique for estimating the price of derivative H exploits the information about the errors in estimating another known quantity H_c that is similar to H . It consists of simulating both H and H_c path-by-path with the same generated random values. The final estimation of H by control variate technique is

$$\tilde{V}^{CV} = \tilde{V} + \beta(V_c - \tilde{V}_c)$$

where $\tilde{V}$ is the Monte Carlo estimation of H , V_c is the accurate value of H_c (by an analytical or any other numerical method), $\tilde{V}_c$ is the Monte Carlo estimation of H_c . β is a value that can be set at 1 or an optimal value that is specific to the problem. With similar analysis for antithetic sampling, the variance of $\tilde{V}^{CV}$ is found to be

$$\text{Var}(\tilde{V}^{CV}) = \frac{1}{N} \left(\sigma^2 + \beta^2 \sigma_c^2 - 2\beta \text{Cov}(V, V_c) \right)$$

where σ^2 and σ_c^2 are respectively the variance of V and V_c . The optimal value of β minimizing $\text{Var}(\tilde{V}^{CV})$ can be obtained by the first order condition (i.e., $\frac{d}{d\beta} \text{Var}(\tilde{V}^{CV})$), which leads to

$$\beta = \frac{\text{Cov}(V, V_c)}{\sigma_c^2}$$

where $\text{Cov}(V, V_c)$ can be estimated from the samples. Hence, we obtain

$$\text{Var}(\tilde{V}^{CV}) = \frac{1}{N} \sigma^2 \left(1 - \frac{\text{Cov}(V, V_c)^2}{\sigma^2 \sigma_c^2} \right) = \frac{1}{N} \sigma^2 (1 - \rho^2)$$

We can see that choosing an H_c highly correlated with H may result in significant variance reduction. For example, for an arithmetic Asian option H , H_c can be the call option with the same maturity and strike.

21.5.3 Importance Sampling

Certain values of the random variables in a simulation have more impact than others in Monte Carlo simulation. The importance sampling methodology consists

of choosing a distribution that emphasizes the important values. It is implemented with the change of probability measure:

$$\mathbb{E}^f[H(x)] = \int H(x)f(x)dx = \int H(x)\frac{f(x)}{g(x)}g(x)dx = \mathbb{E}^g[H(x)Z(x)]$$

where the Radon-Nikodym derivative $Z(x) = \frac{f(x)}{g(x)}$ is known as *likelihood*. The idea of importance sample for reducing the variance of $H(x)Z(x)$ is to make $H(x)Z(x)$ less variable, i.e., to find small $Z(x)$ (implying large $g(x)$) for large $H(x)$ and large $Z(x)$ for small $H(x)$.

For illustration purpose, we take the example of Monte Carlo pricing of a put option with deeply out-of-the-money strike K under the BSM model. Denoting S_T for the underlier's terminal price, we know that $\ln(S_T) = x \sim \mathcal{N}(\mu_f, \sigma^2 T)$ where $\mu_f = \ln(S_0) + (r - \frac{1}{2}\sigma^2)T$. In order to generate more paths ending close to the strike, we take $\mu_g = \ln(K)$. Then,

$$Z(x) = \left[\frac{1}{\sqrt{2\pi}\sigma\sqrt{T}} e^{-\frac{1}{2}\frac{(x-\mu_f)^2}{\sigma^2 T}} \right] / \left[\frac{1}{\sqrt{2\pi}\sigma\sqrt{T}} e^{-\frac{1}{2}\frac{(x-\mu_g)^2}{\sigma^2 T}} \right].$$

By importance sampling method, the option pricing is changed from $\widetilde{\mathbb{E}}^f[e^{-rT}(K - e^x)^+]$ to $\widetilde{\mathbb{E}}^g[e^{-rT}(K - e^x)^+ Z(x)]$. Each simulation sample of the latter consists of (i) taking a $[0, 1]$ uniform variate u , (ii) finding $x = N^{-1}(u, \mu_g, \sigma^2 T)$, and (iii) computing $(K - e^x)^+ Z(x)$. We refer to [43] for further examples and details.

21.5.4 Stratified Sampling*

For a one-dimensional case, instead of using $u \in \mathcal{U}[0, 1]$, stratified sampling divides $[0, 1]$ into n intervals (strata) of equal width I_k ($\sum_{k=1}^n I_k = [0, 1]$) and take m samples uniformly from every I_k . The total number of samples is $N = nm$. Denote

V_k : expected value of V conditional on the uniform random variates in I_k

$\tilde{V}_k$: simulated value for V_k

μ, σ : respectively the mean and variance of V

$\tilde{V}^{SS}$ is the derivative value calculated with all the samples

We have

$$\tilde{V}^{SS} = \frac{1}{n} \sum_{k=1}^n \tilde{V}_k.$$

The variance of $\tilde{V}^{SS}$ is

$$\text{Var}(\tilde{V}^{SS}) = \frac{1}{n^2} \sum_{k=1}^n \text{Var}(\tilde{V}_k) = \frac{1}{n^2 m} \sum_{k=1}^n \sigma_k^2 = \frac{1}{nN} \sum_{k=1}^n \sigma_k^2.$$

Denote $\text{Var}(\tilde{V}^{MC})$ for the variance of normal Monte Carlo with N samples. Now, we show that stratified sampling provides variance reduction:

$$\begin{aligned}\text{Var}(\tilde{V}^{MC}) &= \frac{1}{N}\sigma^2 = \frac{1}{N}(\mathbb{E}[V^2] - \mu^2) = \frac{1}{N}\left(\frac{1}{n}\sum_{k=1}^n \mathbb{E}[V_k^2] - \mu^2\right) \\ &= \frac{1}{N}\left(\frac{1}{n}\sum_{k=1}^n (\mu_k^2 + \sigma_k^2) - \mu^2\right) = \frac{1}{nN}\sum_{k=1}^n [(\mu_k - \mu)^2 + \sigma_k^2] \\ &\geq \frac{1}{nN}\sum_{k=1}^n \sigma_k^2 = \text{Var}(\tilde{V}^{SS}).\end{aligned}$$

Actually, the method is similar to the Quasi-Monte Carlo method with van der Corput sequence (see Sect. 21.6.1). However, unlike the Quasi-Monte Carlo, its straightforward application to higher dimensions suffers from the curse of dimensionality.

21.5.5 Moment Matching

The moment matching method consists of adjusting the sample for a random variable so that the moments of the sample match the moments of the random variable. For instance, the sample for a standard normal variable can be adjusted as

$$\frac{\epsilon_i - \bar{\mu}}{\bar{\sigma}}$$

where $\bar{\mu}$ and $\bar{\sigma}$ are respectively the sample mean and the sample standard deviation.

21.6 Quasi-Monte Carlo*

Quasi-Monte Carlo (QMC) method is a technique used to achieve a fast rate of convergence. Instead of using a random sequence, as in a regular Monte Carlo method, it uses a low-discrepancy sequence, such as the Sobol or Faure sequence, to simulate random variables. Multi-dimensional problems may arise either from the number of underliers and model factors or from time discretization.

An important advantage of Quasi-Monte Carlo method is its generality to all applications as it is implemented at the $[0, 1]$ uniform numbers generation level, instead of problem-specific implementations that are costly and time-consuming. Nowadays, all major investment banks have implemented Quasi-Monte Carlo methods in the pricing engine for option derivatives.

However, it is known that the coordinates for very high dimensions tend to be “correlated” and the lower dimension coordinates provide better uniformity among the coordinates.⁸ To address this issue, two popular methods are being used in practice for applying the “best” sequences (i.e., from lower coordinates) to the most important factors (principal component method) and time points (Brownian bridge method). These two methods can also be used together.

21.6.1 Low-Discrepancy Sequences

One Dimensional Low-Discrepancy Sequence

The base-2 van der Corput (vdC) sequence⁹ provides uniform points in $[0, 1]$ progressively. The method is intuitive and easy to implement. Any non-negative integer $\mathbf{i}$ has a unique binary representation “ $a_n \dots a_j \dots a_1 a_0$ ” that is determined as follows:

$$\mathbf{i} = \sum_{j=0}^{\infty} a_j(\mathbf{i}) 2^j, \text{ where } a_j(\mathbf{i}) \in \{0, 1\}; \text{ e.g., } 1, 10, 11, 100 \text{ for decimal } 1, 2, 3, 4.$$

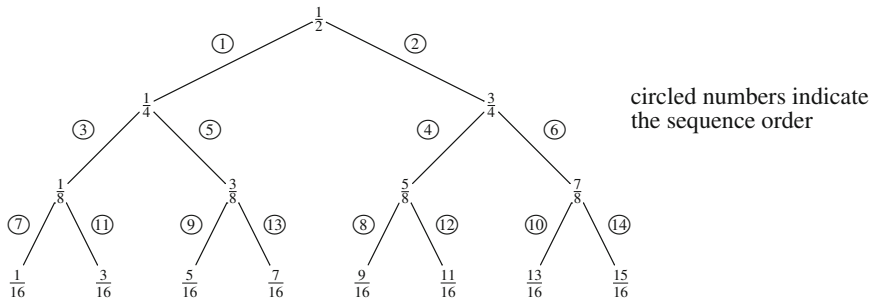
The base-2 van der Corput sequence $\phi(1), \phi(2), \dots$ is produced by reversing the digits of each binary representation of the consecutive whole numbers $\mathbf{i} = 0, 1, 2, \dots$ from “ $a_n \dots a_j \dots a_1 a_0$ ” into the form “ $.a_0 a_1 \dots a_j \dots a_n$ ”, which is obtained by

$$\phi(\mathbf{i}) = \sum_{j=0}^n \frac{a_j(\mathbf{i})}{2^{j+1}}; \text{ e.g., } \begin{array}{c|c} \text{binary number} & \text{van der Corput number} \\ \hline 1110010 & .0100111 \end{array}$$

⁸ For a multi-dimensional Sobol sequence of $2^N - 1$ points, its projection to each dimension still provides perfect uniformity (see Appendix C).

⁹ Van der Corput sequence can be applied to any base, but the lower the base, the better the progressive uniformity for the generated points.

Readers can observe the progressive positioning of the points by the below tree:



Due to its binary feature and the order for new points, the base-2 van der Corput sequence with $2^n - 1$ non-zero points from the start provides perfect progressive uniform points.

Multi-Dimensional Low-Discrepancy Sequence

For d ($d > 1$) dimensional sequences, one cannot simply apply van der Corput (vdC) sequences to each dimension as it would produce perfectly correlated points, e.g., aligned points on the $[0, 1) \times [0, 1)$ plane when $d = 2$. In order to obtain dispersed uniform points, one approach consists of permuting differently the van der Corput sequence for each dimension.¹⁰ k . Faure, Sobol and Niederreiter sequences (see [83]) are such methods by making a linear transformation to each van der Corput number with a *generator matrix* $V^k(\mathbf{i})$ to generate a sequence of up to $2^n - 1$ non-zero points:

$$\mathbf{s}(\mathbf{i}) = V^k(\mathbf{i})\mathbf{a}(\mathbf{i}) \pmod{b}, \quad (21.2)$$

where

- $\mathbf{a}(\mathbf{i}) = [a_0, a_1, \dots, a_n]^T$ to represent the binary vdC number $.a_0a_1\dots a_n$;
- $\mathbf{s}(\mathbf{i}) = [s_0, s_1, \dots, s_n]^T$ to represent the binary Sobol sequence number $.s_0s_1\dots s_n$;
- $V^k(\mathbf{i})$ is a $n \times n$ matrix. n is the length of $\mathbf{a}(\mathbf{i})$;
- b is the base used in the method ($b = 2$ for Sobol method) and $(\text{mod } b)$ means computation with modulus b .

In Appendix C, we introduce the generation method for Sobol sequence, which is currently the most implemented one in financial industry. Interested readers may

¹⁰ Having one point in each d -dimensional volume of $(\frac{1}{n})^d$ to produce a uniform sequence is not doable because the number of points n^d increases exponentially with d .

refer to [93], [43] for more detailed presentations on low-discrepancy sequence generation methods.

21.6.2 Principal Component Method*

This method consists of sorting the dimensions in decreasing order according to their importance and then applying the coordinates from low to high following this order.

The correlation diagonalization method by principal component produces non-negative eigenvalues representing the importance of the dimensions. Hence, the sorting is made according to the decreasing order of eigenvalues, i.e., the dimension with the highest eigenvalue is the most important one.

21.6.3 Brownian Bridge Method*

The value for a *Brownian Bridge* process $X_t^{a,b}(t_1, t_2)$, is defined as the Brownian motion value at time- t ($t_1 < t < t_2$) conditional on its values on t_1 and t_2 that are a and b , i.e., $X_t^{a,b}(t_1, t_2) = W_t | W_{t_1}=a, W_{t_2}=b$. When there is no confusion, we denote $X_t^{a,b}$ for $X_t^{a,b}(0, T)$, $X_t(t_1, t_2)$ for $X_t^{\bar{X}_{t_1}, \bar{X}_{t_2}}(t_1, t_2)$, and X_t for $X_t^{a,b}(t_1, t_2)$. The Brownian Bridge method for Quasi-Monte Carlo consists of simulating the points on a path by starting from the most important time point, followed by the second important one, and so on. At each point except the terminal one, instead of using a Brownian motion value, we use a simulated Brownian Bridge value conditional on the values of the neighbouring points. The best quality QMC sequences (i.e. the low coordinate ones) are applied to the most important time points. For the time interval $[0, 1]$, the terminal point, e.g. time-1, is usually the most important one as X_1 frames the whole path, and the typical order is $0, 1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \dots$ with the values $X_0, X_1, X_{\frac{1}{2}}(0, 1), X_{\frac{1}{4}}(0, \frac{1}{2}), X_{\frac{3}{4}}(\frac{1}{2}, 1), \dots$, where $X_0 = 0$ and $X_1 = W_1$.

We now analyze the distribution of Brownian Bridge (see [105] for more details). Without loss of generality, we consider $X_t^{a,b}(0, T)$. It follows the same distribution as $a + X_t^{0,b-a}$ and the probability density of $X_t^{0,b-a}$ is the following conditional transition probability density

$$p(0, 0; t, x) |_{X_T^{0,b-a}} = p(0, 0; t, x) p(t, x; T, b - a) / p(0, 0; T, b - a)$$

where $p(s, x_s; u, x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x-x_s}{\sigma})^2}$, $u > s$ and $\sigma = \sqrt{u-s}$. Direct calculation leads to

$$p(0, 0; t, x)|_{X_t^{0,b-a}} = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x-m}{\sigma})^2}, \quad m = \frac{t}{T}(b-a), \quad \sigma = \sqrt{\frac{t(T-t)}{T}},$$

which implies

$$X_t^{0,b-a} \sim \mathcal{N}\left(\frac{t}{T}(b-a), \frac{t(T-t)}{T}\right).$$

Hence, we have

$$X_t^{a,b} = a + X_t^{0,b-a} = a + \frac{t}{T}(b-a) + X_t^{0,0} = a \frac{T-t}{T} + b \frac{t}{T} + X_t^{0,0}.$$

where $X_t^{0,0} \sim \mathcal{N}\left(0, \frac{t(T-t)}{T}\right)$.

For instance, if the simulated number of points n is a power of 2, the simulation can be done as follows with independent $\epsilon_i \sim \mathcal{N}(0, 1)$, $\forall i$:

$$\begin{aligned} X_n &= \sqrt{T} \epsilon_1 \\ X_{n/2} &= \frac{1}{2}(X_0 + X_n) + \sqrt{T/4} \epsilon_2 \\ X_{n/4} &= \frac{1}{2}(X_0 + X_{n/2}) + \sqrt{T/8} \epsilon_3 \\ X_{3n/4} &= \frac{1}{2}(X_{n/2} + X_{3n/4}) + \sqrt{T/8} \epsilon_4 \\ &\dots \end{aligned}$$

The first coordinate of Quasi-Monte Carlo sequences can be applied to generate ϵ_1 .

21.7 Monte Carlo Approach for American Options*

21.7.1 Introduction

To price American options with one dimension¹¹ the preferred approach is PDE or Tree method in practice. To overcome the dimensionality limitation, Monte Carlo pricing solutions are highly expected. As the usual Monte Carlo method for

¹¹ The dimension number may come from the multiple underliers and factors in modelling.

European options cannot provide the continuation value for the optimal exercise decision, special Monte Carlo methods need to be elaborated.

In this section, we introduce two Monte Carlo methods for pricing American options with discrete exercise times, which are summarized in [43]:

- the first one is known as Least-Squares Monte Carlo, proposed by Longstaff and Schwartz [72]. It is widely used in the industry due to its generality, the linear computation complexity as well as its ability to meet the accuracy requirement. However, it only provides a lower bound to the option price. Consequently, it can be used to price callable structures in which the option issuer pays premium for the option right;
- the second approach, known as *martingale methods* (see [97],[47] and [2]), provides an upper bound of the option price. Hence, it is suitable for pricing puttable structures, which is conservative for the option issuer as it is short of the exercise right in such structures (hence, it receives the option price).

Evidently, pricing accuracy is better assessed if both lower and upper bounds are provided.

In the sequel, we consider multi-dimensional American options with maturity t_m and discrete exercise times before maturity $[t_1, \dots, t_{m-1}]$, i.e., Bermuda options. To lighten the notation, we sometimes use i instead of t_i as the index for time. Specifically, we use

- $\mathbf{X}_i = [X_i^1, \dots, X_i^d]^\top$ to denote the time- t_i underliers prices and the values of stochastic state variables that are chosen by the user according to the model/option;
- H_i for the discounted exercise value at t_i (instead of our usual notation $\widehat{H}_i$);
- V_i for the discounted option value at t_i (instead of our usual notation $\widehat{H}_i$);
- $\mathcal{F}_i$ for $\mathcal{F}_{t_i}$ that is the σ -algebra generated by $\mathbf{X}_i$;
- $\mathbb{E}_i[\cdot]$ for $\mathbb{E}[\cdot | \mathcal{F}_i]$ or equivalently $\mathbb{E}[\cdot | \mathbf{X}_i]$.

21.7.2 Least-Squares Method for Lower Bound

Least-Squares Method (LSM) [72] estimates, by Monte Carlo simulation, the stopping rule (i.e., exercise decisions) optimizing the option value at each exercise point along the path, and then uses the obtained stopping rule to compute the option price. As the estimated stopping rule may not be optimal, the computed option price is hence a lower bound to the long party of the option.

According to Eq. (19.1), the discounted *continuation value* at time t_i is

$$C_i(\mathbf{x}) = \mathbb{E}[V_{i+1} | \mathbf{X}_i = \mathbf{x}] = \mathbb{E}_i[V_{i+1}(\mathbf{X}_{i+1})].$$

The conditional expectation $C_i(\mathbf{x})$ can be estimated by multiple least-squares regression (see Remark 13.1) with basis functions¹² $\boldsymbol{\phi} = [\phi_1, \dots, \phi_h]^\top$ and a real vector $\boldsymbol{\beta}_i = [\beta_{i1}, \dots, \beta_{ih}]^\top$:

$$C_i(\mathbf{x}) = \boldsymbol{\beta}_i^\top \boldsymbol{\phi}(\mathbf{x}) \quad \text{with} \quad \boldsymbol{\beta}_i = \mathbb{E}_i[\boldsymbol{\phi}(\mathbf{x})\boldsymbol{\phi}(\mathbf{x})^\top]^{-1} \mathbb{E}_i[V_{i+1}(\mathbf{X}_{i+1})\boldsymbol{\phi}(\mathbf{x})]. \quad (21.3)$$

Let $\mathbf{X}_i^k$ be the k -th simulated value for $\mathbf{X}_i$ and V_{i+1}^k be the estimated option price at t_{i+1} conditional on $\mathbf{X}_{i+1}^k$. The entries of the expectation terms in (21.3) can be approximated by the pairs $(V_{i+1}^k(\mathbf{X}_{i+1}^k), \mathbf{X}_i^k)$, $k = 1, \dots, N$:

$$\frac{1}{N} \sum_{k=1}^N \boldsymbol{\phi}(\mathbf{X}_i^k) \boldsymbol{\phi}(\mathbf{X}_i^k)^\top \quad \text{and} \quad \frac{1}{N} \sum_{k=1}^N V_{i+1}^k(\mathbf{X}_{i+1}^k) \boldsymbol{\phi}(\mathbf{X}_i^k)^\top.$$

The American option pricing steps used in [72] are as follows

- (1) generate N paths for each $\mathbf{X}_i$: $\mathbf{X}_i^k$, $i = 1, \dots, m$, $k = 1, \dots, N$;
- (2) set $V_m^k = H_m(\mathbf{X}_m^k)$;
- (3) perform backward induction over i from $m - 1$ to 1:
 - make regression for β_i with selected pairs $(V_{i+1}^k, \mathbf{X}_i^k)$, $k = 1, \dots, N$, where $\mathbf{X}_i^k$ must be in-the-money, i.e., with positive $H_i(\mathbf{X}_i^k)$. Then, the continuation value $C_i(\mathbf{X}_i^k)$ is obtained by (21.3);
 - set $V_i^k = H_i \mathbf{1}_A + V_{i+1}^k \mathbf{1}_{A^c}$ where A is the exercise event $\{H_i \geq C_i(\mathbf{X}_i^k)\}$;
- (4) compute the option price by: $V_0 = \frac{1}{N} \sum_{k=1}^N V_1^k$.

According to [72], using the selected pairs only (as described in step (3)) instead of all the pairs for regression improves the efficiency of the algorithm. The continuation value $C_i(\mathbf{X}_i^k)$ is only used for the stopping rule and it is not propagated back as option value to avoid upward bias.

Example 21.3¹³ Consider a 3-year put option on a stock with two exercise dates (year 1 and year 2) with *initial spot* = 1, *strike* = 1.1 and *one year discount factor* = 0.94176. Table 21.2 shows 8 simulated paths at inception and the end of each year. We can see the option payoff at year 3 conditional on no early exercise in the last column of the table.

Let X denote the stock price and Y be the discounted cash flow received at the considered year, conditional on no exercise before that year. In this example, it is assumed that $\mathbb{E}[Y|X] = a + bX + cX^2$, where X is the state variable for regression.

¹² Basis functions are mathematical functions that can be used to construct other functions by taking combinations of them. The simplest basis functions are $1, x, x^2, x^3, \dots$, but more efficient ones can be chosen from orthogonal polynomials such as Laguerre, Hermite, etc.

¹³ This example is extracted from the original paper of Longstaff–Schwartz [72], which adopts non-discounted exercise and continuation values.

Table 21.2 Simulated stock price paths and Cash-flow matrix at Year 3 (conditional on no exercise before year 3)

Path	Simulated price paths				Cash-flow matrix at year 3		
	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 1$	$t = 2$	$t = 3$
1	1	1.09	1.08	1.34			
2	1	1.16	1.26	1.54			
3	1	1.22	1.07	1.03			0.07
4	1	0.93	0.97	0.92			0.18
5	1	1.11	1.56	1.52			
6	1	0.76	0.77	0.90			0.20
7	1	0.92	0.84	1.01			0.09
8	1	0.88	1.22	1.34			

Table 21.3 Year 2

Path	Regression		Exercise decision		Cash-flow matrix		
	Y	X	Exercise	Continuation	$t = 1$	$t = 2$	$t = 3$
1	0×0.94176	1.08	0.02	0.0369			
2							
3	0.07×0.94176	1.07	0.03	0.0461			0.07
4	0.18×0.94176	0.97	0.13	0.1176		0.13	
5							
6	0.20×0.94176	0.77	0.33	0.1520		0.33	
7	0.09×0.94176	0.84	0.26	0.1565		0.26	
8							

The pairs (Y, X) for regression at year 2 are listed in Table 21.3. The dependent variable Y contains the discounted values from Table 21.2. Only in-the-money paths where $X < 1.1$ are used for regression. The regression result is $\tilde{\mathbb{E}}[Y|X] = -1.070 + 2.983X - 1.813X^2$, which is used to compute the continuation values for the stopping rule (i.e., exercise decisions). For instance, Path 7 implies stopping as the exercise value 0.26 is higher than the continuation value 0.1565.

We repeat the process at year 1 similarly to year 2. Table 21.4 shows the independent variables for X that are in-the-money and the dependent variables for Y . The corresponding regression result is $\tilde{\mathbb{E}}[Y|X] = 2.038 - 3.335X + 1.356X^2$. It is used to produce the cash-flow matrix at year 1.

The option price ($= 0.1144$) is the average of the discounted payoffs shown on the cash-flow matrix in Table 21.4. The stopping rules are also indicated by the entries with positive values on this cash-flow matrix. $\square$

Table 21.4 Year 1

Path	Regression		Exercise decision		Cash-flow matrix		
	Y	X	Exercise	Continuation	$t = 1$	$t = 2$	$t = 3$
1	0×0.94176	1.09	0.01	0.0139			
2							
3							0.07
4	0.13×0.94176	0.93	0.17	0.1092	0.17		
5							
6	0.33×0.94176	0.76	0.34	0.2866	0.34		
7	0.26×0.94176	0.92	0.18	0.1175	0.18		
8	0×0.94176	0.88	0.22	0.1533	0.22		

21.7.3 Martingale Method for Upper Bound*

Several methods based on martingale approach (see [97], [47] and [2]) have been proposed for upper bounds to American option prices. Here, we introduce the one by Andersen and Broadie [2].

Based on the analysis in Chap. 19, the American option pricing is given by the formula (19.2) that we recall here:

$$V_0 = \sup_{\tau \in \mathcal{T}} \tilde{\mathbb{E}}[H_\tau | \mathcal{F}_0], \quad (21.4)$$

where $\mathcal{T}$ is the set of stopping times taking values in $[t_1, t_m]$. Let M_t be a martingale with $M_0 = 0$. Then, we have

$$V_0 = \sup_{\tau \in \mathcal{T}} \tilde{\mathbb{E}}_0[H_\tau] = \sup_{\tau \in \mathcal{T}} \tilde{\mathbb{E}}_0[H_\tau - M_\tau] \leq \tilde{\mathbb{E}}_0[\max_{i=1, \dots, m} \{H_i - M_i\}].$$

Hence, an upper bound for an American option price can be expressed as the below minimization problem over all martingales M , which is called the *dual problem*:

$$V_0 \leq \inf_M \tilde{\mathbb{E}}_0[\max_{i=1, \dots, m} \{H_i - M_i\}]$$

while the maximization problem in (21.4) is known as the *primal problem*. Given the functions to estimate the *continuation values* $C_i(\mathbf{X}_i)$ with $C_m(\mathbf{X}_m) \equiv 0$, e.g., by LSM introduced previously, we define τ_i as

$$\tau_i = \min\{t_j : H_j(\mathbf{X}_j) \geq C_j(\mathbf{X}_j), j = i, \dots, m\}.$$

Given a simulation path, τ_i can be determined. Next, we can define a martingale as follows with $M_0 = 0$, which is used in [2]:

$$M_i = \sum_{j=1}^i \Delta_j, \text{ where } \Delta_j = \tilde{\mathbb{E}}_j[H_{\tau_j}(\mathbf{X}_{\tau_j})] - \tilde{\mathbb{E}}_{j-1}[H_{\tau_j}(\mathbf{X}_{\tau_j})]. \quad (21.5)$$

Actually, the first expectation term in Δ_j can be broken down into two terms:

$$\tilde{\mathbb{E}}_j[H_{\tau_j}(\mathbf{X}_{\tau_j})] = H_j(\mathbf{X}_j)\mathbf{1}_{\tau_j=t_j} + \tilde{\mathbb{E}}_j[H_{\tau_{j+1}}(\mathbf{X}_{\tau_{j+1}})]\mathbf{1}_{\tau_j>t_j}. \quad (21.6)$$

The simulation of M_i on each Monte Carlo path requires the second expectation terms in equations (21.6) and (21.5), which can be simulated by the so-called *nested* Monte Carlo simulations¹⁴, consisting of replacing $\tilde{\mathbb{E}}_i[H_{\tau_{i+1}}(\mathbf{X}_{\tau_{i+1}})]$ by

$$\frac{1}{n} \sum_{\ell=1}^n H_{\tau_{i+1}}(\mathbf{X}_{\tau_{i+1}}^\ell) \mid \mathbf{X}_i. \quad (21.7)$$

Based on the *continuation value* functions provided by a lower bound method, this dual method estimates the upper bound of an American option price with the following steps

- (1) simulate the k -th path to obtain $\mathbf{X}_i^k, i = 1, \dots, m, k = 1, \dots, N$;
- (2) construct M_i^k for $i = 1, \dots, m$ as defined in (21.5) using (21.7) for computing $\tilde{\mathbb{E}}_i[H_{\tau_{i+1}}(\mathbf{X}_{\tau_{i+1}}^k)]$;
- (3) evaluate $V_0^k = \max_{i=1, \dots, m} \{H_i(\mathbf{X}_i^k) - M_i^k\}$;
- (4) repeat the steps (1) to (3) N times and compute the American option price by $\frac{1}{N} \sum_{k=1}^N V_0^k$.

Exercises

1. Under the BSM model for a non-dividend-paying stock, let $S_0 = 1, r = 0.01, \sigma = 0.2$. Calculate the price of a 1-year ATM call option with the below methods and compare the results:

- (1) Black–Scholes formula;
- (2) Monte Carlo based on the direct solution:

$$S_T = S_0 e^{(r - \frac{\sigma^2}{2})T + \sigma \tilde{W}_T};$$

¹⁴ In [2], over a few thousand paths are used for each nested Monte Carlo simulation while the number of Monte Carlo iterations is over a million.

(3) Euler–Maruyama scheme (step size $\Delta t = 0.005$):

$$S_i = S_{i-1} + r S_{i-1} \Delta t + \sigma S_{i-1} \tilde{W}_{\Delta t}.$$

2. What type of American option to which Longstaff–Schwartz (Least-Squares) method is not suitable to apply?

Chapter 22

Numerical Method (2): Binomial and Trinomial Trees



As a lattice approach, tree methods, pioneered by Cox, Ross, and Rubinstein in 1979 [32], employ a discrete multi-period representation for future possible asset prices for option pricing. Tree methods are easy to understand and to implement for simple options, in particular for American options, although its application scope in practice is limited. The tree approach provides a good introduction to the more powerful and efficient lattice approach—PDE finite difference method.

In this chapter, tree methods are introduced for the Black–Scholes–Merton (BSM) model in which the underlier price S_t follows $dS_t = rS_t dt + \sigma S_t d\tilde{W}_t$ under risk-neutral probability $\tilde{\mathbb{P}}$. The analysis methodology can be extended to other models.

22.1 Binomial Tree Method

22.1.1 Tree Construction

We first construct an m -step binomial tree by dividing the considered period $[0, T]$ into m subperiods: $0 = t_0 < t_1 < \dots < t_m = T$, $t_k = k\Delta t$, $k = 0, 1, \dots, m$ and $\Delta t = T/m$. Given an asset price S at time t , binomial method assumes that there are only two possible prices for the next time point $t + \Delta t$:

- “up” by a proportion of u with risk-neutral probability p to be Su , or
- “down” by a proportion of d with probability $1 - p$ to become Sd .

Starting from the initial underlier price S_0 , we repeat the above branching process m times and a “recombining tree” representation is formed with the transition probabilities between the nodes, which provides enough information to conduct option pricing on the tree. Denoting S_k^i ($S_0^0 = S_0$) for the i -th underlier price at time t_k , Fig. 22.1 illustrates the recombining tree with $m + 1$ possible underlier prices. If

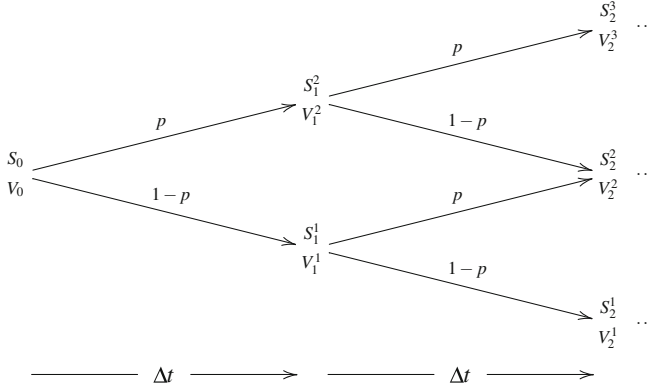


Fig. 22.1 *m*-step binomial method

p is provided, then the time- k option price at i -node that we denote by V_k^i (i.e., the option price conditional on S_k^i) can be determined on the binomial tree.

22.1.2 Determination of the Parameters

The martingale property can be written as $S = \tilde{\mathbb{E}}[e^{-r\Delta t} S_\tau]$ where S is a time- t underlier price at a tree node, $\tau = t + \Delta t$ and S_τ represents its possible values after Δt time. Hence, $S = pe^{-r\Delta t} Su + (1-p)e^{-r\Delta t} Sd$, or equivalently,

$$e^{r\Delta t} = pu + (1-p)d. \quad (22.1)$$

From Corollary 14.1, we know the variance of $S_{\Delta t}$: $\tilde{\text{Var}}(S_{\Delta t}) = S^2 e^{2r\Delta t} (e^{\sigma^2 \Delta t} - 1)$. Since $\tilde{\text{Var}}(S_\tau) = \tilde{\mathbb{E}}[S_\tau^2] - (\tilde{\mathbb{E}}[S_\tau])^2$ and $\tilde{\mathbb{E}}[S_\tau^2] = p(Su)^2 + (1-p)(Sd)^2$, it follows that

$$S^2 e^{2r\Delta t} (e^{\sigma^2 \Delta t} - 1) = pS^2 u^2 + (1-p)S^2 d^2 - S^2 (pu + (1-p)d)^2,$$

which simplifies to

$$e^{2r\Delta t + \sigma^2 \Delta t} = pu^2 + (1-p)d^2. \quad (22.2)$$

Equations (22.1) and (22.2) impose two conditions on p , u and d . The third condition can be

$$u = \frac{1}{d}. \quad (22.3)$$

Solving (22.1), (22.2) and (22.3), we obtain

$$p = \frac{e^{r\Delta t} - d}{u - d}, \quad u = A + \sqrt{A^2 - 1}, \quad d = A - \sqrt{A^2 - 1},$$

where $A = \frac{1}{2} \left(e^{-r\Delta t} + e^{(r+\sigma^2)\Delta t} \right)$. The choice of the binomial tree of Cox, Ross, and Rubinstein [32] is

$$p = \frac{e^{r\Delta t} - d}{u - d}, \quad u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u}, \quad (22.4)$$

which is based on the fact that the first three terms of the Taylor expansion of $u = A + \sqrt{A^2 - 1}$ coincides with those of $u = e^{\sigma\sqrt{\Delta t}}$, i.e., $u = 1 + \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t + O(\Delta t^{3/2})$. Actually, the choice is not unique. For instance, instead of applying $u = \frac{1}{d}$, we may choose $p = \frac{1}{2}$, which leads to $u = e^{r\Delta t}(1 + \sqrt{e^{\sigma^2\Delta t} - 1})$ and $d = e^{r\Delta t}(1 - \sqrt{e^{\sigma^2\Delta t} - 1})$.

Remark 22.1 Given the time period on a binomial tree, the underlier value increments are not homogeneous. However, the increments on the log values are the same $\Delta \ln S = \ln Su - \ln S = \ln u = \sigma\sqrt{\Delta t} = -\ln d = \ln Sd - \ln S$, which is solely determined by Δt (while it is not the case for trinomial tree). $\square$

Binomial tree method is memory efficient as it is recombining. The largest memory requirement is at the last time-step m , where $2(m+1)$ memory locations are required for the underlier and option prices. As there are $O(m^2)$ nodes, the execution time is $O(m^2)$.

22.1.3 European Option Pricing

Let V_k^i be the price on the i -th node at the k -period for a European option. Under $\tilde{\mathbb{P}}$, the discounted option price V_k^i is a martingale:

$$V_k^i = e^{-r\Delta t} [pV_{k+1}^{i+1} + (1-p)V_{k+1}^i], \quad i = 0, \dots, k. \quad (22.5)$$

Hence, the pricing of a European option at time- t consists of

- (1) applying the payoff function for the option on S_m^i , the last layer of the tree. For instance, in case of a call or put option, we have,

$$V_m^i = \max\{S_m^i - K, 0\} \quad \text{or} \quad V_m^i = \max\{K - S_m^i, 0\}, \quad i = 0, \dots, m,$$

where K is the exercise price;

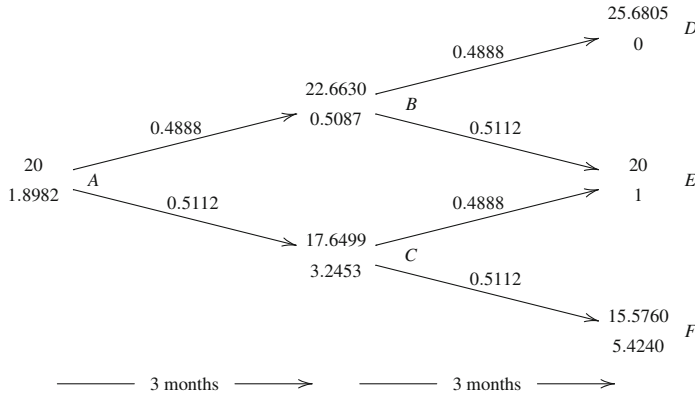


Fig. 22.2 Two-step binomial method for a European Option

- (2) evaluating the option prices in between, V_k^i , $k = 0, \dots, m - 1$, backward one step at a time using the transition probabilities p and $1 - p$;
- (3) taking the time-0 price V_0^0 as the option price.

Example 22.1 Let $S_0 = 20$, $r = 2\%$ and $\sigma = 25\%$. Consider a European put option with $K = 21$ and $T = 0.5$ (6 months). For illustration purpose, we only break $[0, T]$ into two time-steps with $\Delta t = 0.25$ and construct a simplified two-step binomial tree, see Fig. 22.2.

By formulas in (22.4), we get $u = 1.1331$, $d = 0.8825$, and $p = 0.4888$. We compute the value of the option at node B using the option values at nodes D and E

$$e^{-0.02 \times 0.25} (0.4888 \times 0 + 0.5112 \times 1) = 0.5087.$$

Similarly, option values at nodes E and F give the option value at node C. We repeat the computation once again, using the value of the options at nodes B and C, we get the option price at node A:

$$e^{-0.02 \times 0.25} (0.4888 \times 0.5087 + 0.5112 \times 3.2453) = 1.8982.$$

□

22.1.4 American Option Pricing

The tree for an American option pricing is constructed in the same way as the one for a European option, but the option calculation is different. With backward induction,

we compute, at each node:

- *continuation value* : option value propagated from previous time step;
- *exercise value* : payout of the option upon exercise
- *option price* : determined by the optimal exercise strategy, which is

$$\text{option price} = \max\{\text{exercise value}, \text{continuation value}\}.$$

For instance, the *exercise value* at the node k, i for an American put option is

$$H_k^i = \max\{K - S_k^i, 0\}, \quad i = 0, \dots, k.$$

The *continuation value* is

$$e^{-r\Delta t} (pV_{k+1}^{i+1} + (1-p)V_{k+1}^i), \quad i = 0, \dots, k.$$

Hence, the value of the American option is

$$V_k^i = \max \left\{ H_k^i, e^{-r\Delta t} (pV_{k+1}^{i+1} + (1-p)V_{k+1}^i) \right\}, \quad i = 0, \dots, k.$$

Example 22.2 Let us compute an American put with the 2-step binomial tree used previously, i.e., $S_0 = 20$, $K = 21$, $T = 0.5$, $r = 2\%$, $\sigma = 25\%$, $\Delta t = 0.25$, $u = 1.1331$, $d = 0.8825$, and $p = 0.4888$. The binomial tree is given in Fig. 22.3. At node B, if we exercise, we get 0. If we do not exercise, the option value is

$$e^{-0.02 \times 0.25} (0.4888 \times 0 + 0.5112 \times 1) = 0.5087.$$

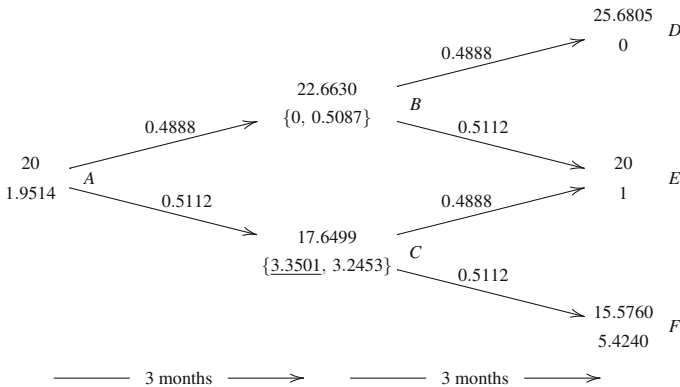


Fig. 22.3 Two-step binomial method for an American option

Clearly, we do not exercise, and the option price is 0.5087. At node C, if we exercise, we get 3.3501. If we do not exercise, the option value is

$$e^{-0.02 \times 0.25} (0.4888 \times 1 + 0.5112 \times 5.4240) = 3.2453.$$

In this case, we exercise, and the option price is 3.3501. Finally, at $t = 0$, the option price is given by

$$e^{-0.02 \times 0.25} (0.4888 \times 0.5087 + 0.5112 \times 3.3501) = 1.9514,$$

which is higher than the exercise value of 1. □

22.1.5 Greeks*

By finite difference method (see Sect. 23.1.1), the Greeks can be computed by repricing the option with shifted parameters. But with the information provided on binomial tree, some Greeks (e.g., theta, delta, and gamma) can be estimated without repricing the option. For instance, the current delta can be estimated with the delta at time Δt for the point $S_1 = \frac{1}{2}(S_1^2 + S_1^1)$:

$$\frac{V_1^2 - V_1^1}{S_1^2 - S_1^1}$$

and the gamma can be approximated with the rate of change in delta at time $2\Delta t$ for two points, $\frac{1}{2}(S_2^3 + S_2^2)$ and $\frac{1}{2}(S_2^2 + S_2^1)$:

$$\left(\frac{V_2^3 - V_2^2}{S_2^3 - S_2^2} - \frac{V_2^2 - V_2^1}{S_2^2 - S_2^1} \right) / [(S_2^3 - S_2^1)/2].$$

22.1.6 Accuracy and Stability Analysis*

Accuracy Analysis

We follow [65] to show that with $\Delta t \rightarrow 0$, the result obtained from binomial tree converges to that of Black–Scholes equation with accuracy $O(\Delta t)$ and $O((\Delta \ln S)^2)$. Firstly, we can check by Taylor expansion that

$$\begin{aligned} u &= 1 + \sigma\sqrt{\Delta t} + O(\Delta t); & p &= 1/2 + O(\sqrt{\Delta t}); \\ d &= 1 - \sigma\sqrt{\Delta t} + O(\Delta t); & e^{-r\Delta t} &= 1 - r\Delta t + O(\Delta t^2). \end{aligned} \tag{22.6}$$

Let $f(S, t)$ be a continuous function satisfying the binomial tree option value propagation formula (22.5):

$$f(S, t - \Delta t) = e^{-r\Delta t} [pf(Su, t) + (1 - p)f(Sd, t)]. \quad (22.7)$$

Let $\Delta S_u = Su - S = (u - 1)S$, $\Delta S_d = Sd - S = (d - 1)S$ and

$$P_i = e^{-r\Delta t} (p\Delta S_u^i + (1 - p)\Delta S_d^i), \quad \forall i \geq 1.$$

Then,

$$\begin{aligned} (1) \quad P_1 &= e^{-r\Delta t} [p(u - 1) + (1 - p)(d - 1)]S = rS\Delta t + O(\Delta t^2), \\ (2) \quad P_2 &= e^{-r\Delta t} [p(u - 1)^2 + (1 - p)(d - 1)^2]S^2 = \sigma^2 S^2 \Delta t + O(\Delta t^2), \\ (3) \quad P_i &= e^{-r\Delta t} [p(u - 1)^i + (1 - p)(d - 1)^i]S^i \\ &= 0 + O(\Delta t^{k(i)}), \quad k(i) \geq 2, \quad \forall i \geq 3, \end{aligned}$$

where (1) can be checked with Eq. (22.1), (2) with Eq. (22.1) and (3) by the relations (22.6).¹ Performing Taylor expansion to $f(S, t - \Delta t)$, $f(Su, t)$ and $f(Sd, t)$ in Eq. (22.7) at $f(S, t)$ (denoted as f to lighten the notation) and using (1), (2) and (3), we obtain

$$\begin{aligned} 0 &= -f(S, t - \Delta t) + e^{-r\Delta t} [pf(Su, t) + (1 - p)f(Sd, t)] \\ &= -f + \frac{\partial f}{\partial t} \Delta t + O(\Delta t^2) + e^{-r\Delta t} f + P_1 \frac{\partial f}{\partial S} + \frac{1}{2!} P_2 \frac{\partial^2 f}{\partial S^2} + \frac{1}{3!} P_3 \frac{\partial^3 f}{\partial S^3} + \dots \\ &= -f + \frac{\partial f}{\partial t} \Delta t + e^{-r\Delta t} f + rS \frac{\partial f}{\partial S} \Delta t + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} \Delta t + O(\Delta t^2), \end{aligned}$$

which leads to the below equation as $(e^{-r\Delta t} - 1)f = -rf\Delta t + O(\Delta t^2)$:

$$\frac{\partial f}{\partial t} + rS \frac{\partial f}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} - rf = 0 + O(\Delta t)$$

Hence, f converges to the solution of Black-Scholes equation with accuracy $O(\Delta t)$. By Remark 22.1, the accuracy of $O(\Delta t)$ implies $O((\Delta \ln S)^2)$ as $\sigma \sqrt{\Delta t} = \Delta \ln S$.

¹ For $i = 3$, it is a bit tedious to check that $P_i = O(\Delta t^2)$. (2) can be used in the calculation.

Stability Analysis

For stability analysis, we use the von Neumann's method (see Sect. 23.11 for the method as well as the notations) to apply discrete Fourier transform on both side of (22.5), which leads to the below amplification factor (see Eq. (23.19)). Denoting $i = \sqrt{-1}$, $0 \leq \omega \leq \pi$ and using Euler's formula, we have

$$\begin{aligned} G^2(\omega) &= e^{-2r\Delta t} \|pe^{i\omega} + (1-p)e^{-i\omega}\|^2 \\ &= e^{-2r\Delta t} \|p(e^{i\omega} - e^{-i\omega}) + e^{-i\omega}\|^2 \\ &= e^{-2r\Delta t} \|i2p \sin(\omega) + \cos(-\omega) + i \sin(-\omega)\|^2 \\ &= e^{-2r\Delta t} (1 - (1 - (2p - 1)^2) \sin(\omega)). \end{aligned}$$

We can see that if r is non-negative or slightly negative, $G(\omega)$ is not greater than 1, i.e., the numerical method is stable.

22.2 Trinomial Tree Method*

A trinomial tree is similar to a binomial tree but assumes that from a node (t, S) , there are three possible underlier values for the next time step $t + \Delta t$, S_u , S and S_d with respectively probabilities p_u , p_m and p_d such that

$$p_u + p_m + p_d = 1. \quad (22.8)$$

Trinomial tree is also made to be recombining. Now, we show the parameters determination for the BSM model in which $\ln S_{t+\Delta t} - \ln S_t = (r - \frac{1}{2}\sigma^2)\Delta t + \sigma d\tilde{W}_t$. It follows that $\tilde{\mathbb{E}}[\Delta \ln S] = (r - \frac{1}{2}\sigma^2)\Delta t$ and $\tilde{\text{Var}}(\Delta \ln S) = \sigma^2\Delta t$. In [63], it is proposed to set $g = \lambda\sigma\sqrt{\Delta t}$ with $\lambda \geq 1$, $\ln u = g$ and $\ln d = -g$. Then,

$$\tilde{\mathbb{E}}[\Delta \ln S] = p_u\lambda\sigma\sqrt{\Delta t} + p_m \times 0 - p_d\lambda\sigma\sqrt{\Delta t} = (r - \frac{1}{2}\sigma^2)\Delta t, \quad (22.9)$$

which implies that $\tilde{\mathbb{E}}[\Delta \ln S]^2 = 0 + O(\Delta t^2)$. Hence

$$\tilde{\text{Var}}(\Delta \ln S) = \tilde{\mathbb{E}}[(\Delta \ln S)^2] - \tilde{\mathbb{E}}[\Delta \ln S]^2 \approx (p_u + p_d)(\lambda\sigma\sqrt{\Delta t})^2 \approx \sigma^2\Delta t. \quad (22.10)$$

The resolution of (22.8), (22.9) and (22.10) shows, with accuracy $O(\Delta t^2)$,

$$\begin{aligned} p_u &= \frac{1}{2\lambda^2} + \frac{1}{2\lambda\sigma}\left(r - \frac{1}{2}\sigma^2\right)\sqrt{\Delta t}; \\ p_m &= 1 - \frac{1}{\lambda^2}; \\ p_d &= \frac{1}{2\lambda^2} - \frac{1}{2\lambda\sigma}\left(r - \frac{1}{2}\sigma^2\right)\sqrt{\Delta t}. \end{aligned} \tag{22.11}$$

One advantage of trinomial tree over binomial tree is the flexibility for u (and d) by the choice of λ that allows to set S_t value to a desired level without having to adjust Δt (see Remark 22.1).

The option pricing method on a trinomial tree is similar to the binomial case. Reportedly, the trinomial tree method offers better performance than the binomial tree method with a similar computation effort. Actually, by choosing $\lambda = \frac{2\Delta x}{\sigma\sqrt{\Delta t}}$ and comparing with (23.8), we can see that trinomial tree method is equivalent to the explicit finite difference scheme in PDE approach that is well analyzed for convergence.

22.3 Discrete Dividends*

For a stock underlier paying proportional dividends, the tree construction is straightforward by shifting down the stock price on each node at or after an ex-dividend date by its dividend amount. More formally, let τ be an ex-dividend date with proportional dividend yield b (see Sect. 5.2.1) and k represents the time point on the tree just prior to τ , i.e., $k\Delta t < \tau$. The stock prices on the next two nodes from the node at k with stock value S will be $S(1-b)u$ and $S(1-b)d$. It can be checked that the tree is still recombining. The option pricing process on the tree remains the same.

For a stock paying fixed cash dividend, the tree construction is less straightforward and requires some extra modelling assumptions. We refer to [65], [49] for interested readers. However, as the purpose of a tree method is more for a quick and simple solution rather than a robust and efficient implementation (i.e., PDE or Monte Carlo), proportional dividend assumption (which is more reasonable than the fixed cash assumption for long-term dividends) should satisfy most of the needs in practice.

22.4 Barrier and Path-Dependent Options Pricing Issues*

22.4.1 Barrier Options

Tree methods can be used to price European barrier options in practice, provided that the payoff discontinuity is smoothed by the approach presented in Sect. 20.1.1. However, for pricing American barrier options, the tree method suffers stability and accuracy issues. The lack of grid points is one of the causes. To improve the convergence, the following methods can be considered:

- adjust the tree (i.e., by adjusting Δt for binomial tree, λ alone and/or Δt for trinomial tree) so that the barrier value(s) is(are) on the nodes of the tree;
- use the conditional probabilities for drifted Brownian extremum introduced in Sect. 15.6.6 to capture the barrier-breach probability between two successive nodes for continuous or near continuous barrier options. For example, if H is a down-and-out barrier and $S_{k+1}^{down} \leq H < S_k$, then the barrier-breach probability q is given in Corollary 15.2:

$$q = \mathbb{P}[\min_{t_k \leq t \leq t_{k+1}} S_t \leq H | S_{k+1}^{up}] = \exp\left(-\frac{2}{\sigma^2 \Delta t} \ln \frac{S_k}{H} \ln \frac{S_{k+1}^{up}}{H}\right).$$

Hence, the option price propagation to this node at k can be adjusted as

$$V_k^i = e^{-r \Delta t} [p(1 - q)V_{k+1}^{up} + (1 - p) \times 0]$$

(see [12]).

Nevertheless, it is not recommended to use tree methods for barrier options for professional use.

22.4.2 Other Path-Dependent Options

For other path-dependent options such as Asian options, lookback options alike, the implementation is sometimes complicated and tricky due to the required auxiliary state variables. Interested readers can refer to [65] for more details. We note that Monte Carlo implementation is simple and straightforward for path-dependent options. PDE methods can also be used for low dimensional problems.

Exercises

1. Compute an American call by a 2-step binomial method with $S_0 = 40$, $K = 42$, $T = 0.5$, $r = 0.02$, $\sigma = 0.25$, $\Delta t = 0.25$. Show that at each node of the tree it is not optimal to early exercise the American call since the stock does not pay dividend;
2. Assume $p = 0.5$ is used in the binomial method.
 - (1) Prove that under risk-neutral probability,

$$u = e^{r\Delta t} \left(1 + \sqrt{e^{\sigma^2 \Delta t} - 1} \right) \quad \text{and} \quad d = e^{r\Delta t} \left(1 - \sqrt{e^{\sigma^2 \Delta t} - 1} \right).$$

- - (2) Find the upper bound for Δt such that the binomial method with $p = 0.5$ is feasible.
 - (3) Show that $ud \rightarrow 1$ as $\Delta t \rightarrow 0$.

Chapter 23

Numerical Method (3): PDE Approach*



Feynman–Kac theorem shows the equivalence of an expectation calculation and the solution to the related PDE for option pricing. In this chapter, we discuss how to solve PDEs with finite difference approach by starting with a one-dimensional PDE followed by the multi-dimensional case. For more advanced analysis, we refer to [111] and [37].

Without loss of generality, the Black–Scholes–Merton model is used to illustrate the finite difference method. We recall that the one-dimensional Black–Scholes equation¹ (see Sect. 14.2) is

$$\frac{\partial V_t}{\partial t} + \frac{1}{2}\sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2} + r S_t \frac{\partial V_t}{\partial S_t} = r V_t. \quad (23.1)$$

According to PDE categorization (that can be found in any textbook on the subject), this equation belongs to the category of *linear parabolic* PDE.² Unlike the other two categories (namely, *elliptic* and *hyperbolic* PDEs), a parabolic PDE has a preferred direction of propagation of the solution that is “time” in the applications, leading to time-marching numerical algorithms.

For partial derivatives, the usual notations in mathematics are used in this book, e.g., $f'(x)$, f'_x or ∂_x for $\frac{\partial f}{\partial x}$, f'' for $\frac{\partial^2 f}{\partial x^2}$ and f''_{xy} or ∂_{xy} for $\frac{\partial^2 f}{\partial x \partial y}$.

¹ This PDE is known as a convection–diffusion equation in physics. Its first-order term $r S_t \frac{\partial V_t}{\partial S_t}$ is called *convection* or *advection* term while the second-order term $\frac{1}{2}\sigma_t^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2}$ is called *diffusion* term and $r V_t$ is called *source* term.

² A PDE is said to be linear if it is linear for the unknown function and its partial derivatives.

23.1 Introduction to Finite Difference Method

23.1.1 Derivatives Approximation

Finite difference approaches consist of approximating the differential operators in differential equations by difference operators. Given a function $u(x)$, using Taylor's expansion, we have

$$\begin{aligned} u(x + \Delta x) &= u(x) + u'(x)\Delta x + \frac{1}{2!}u''(x)\Delta x^2 + \frac{1}{3!}u'''(x)\Delta x^3 + O(\Delta x^4) \\ u(x - \Delta x) &= u(x) - u'(x)\Delta x + \frac{1}{2!}u''(x)\Delta x^2 - \frac{1}{3!}u'''(x)\Delta x^3 + O(\Delta x^4). \end{aligned} \quad (23.2)$$

If Δx is small, we can approximate $u'(x)$ and $u''(x)$ at x by the non-derivative terms in (23.2):

- *forward difference* for $u'(x)$:

$$u'(x) = \frac{u(x + \Delta x) - u(x)}{\Delta x} + O(\Delta x). \quad (23.3)$$

It provides first-order accuracy $O(\Delta x)$ and is usually used for the differential w.r.t. time in problems of physics or finance.

- *first-order central difference* for $u'(x)$:

$$u'(x) = \frac{u(x + \Delta x) - u(x - \Delta x)}{2\Delta x} + O(\Delta x^2). \quad (23.4)$$

It provides second-order accuracy $O(\Delta x^2)$.

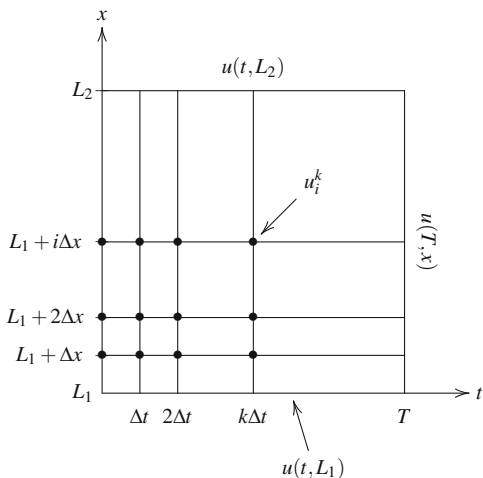
- *second-order central difference* for $u''(x)$:

$$u''(x) = \frac{u(x + \Delta x) - 2u(x) + u(x - \Delta x)}{\Delta x^2} + O(\Delta x^2). \quad (23.5)$$

23.1.2 Discretization Grid

As the Black–Scholes–Merton model is a log-normal model, it is more efficient to work with the case after the variable change $x = \ln S$, which results in less spatial grid points. To align with the usual notation in PDE literature, we also change the option price notation from $V(t, x)$ to $u(t, x)$ or u if it does not create confusion. The

Fig. 23.1 Solution domain of European options and the grid



Black–Scholes Eq. (23.1) becomes the below PDE with constant coefficients:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2} + \left(r - \frac{\sigma^2}{2}\right) \frac{\partial u}{\partial x} - ru|_{(x,t)} = 0, \quad (23.6)$$

with European-type payoff $u(T, x)$ for $-\infty < x < \infty$. Since it is impossible to work with infinity numerically, we first truncate the spatial domain to $x \in [L_1, L_2]$. Then we replace the boundary conditions $u(t, -\infty)$ by $u(t, L_1)$ and $u(t, \infty)$ by $u(t, L_2)$.

Next we partition the solution domain $[L_1, L_2] \times [0, T]$ by grid lines: $[L_1, L_2]$ into n equal sub-intervals, each of length Δx and $[0, T]$ into m equal sub-intervals, each of length Δt , see Fig. 23.1. Denote $x_i = L_1 + i\Delta x$ and $t_k = L_1 + k\Delta t$ for $0 \leq i \leq n$ and $0 \leq k \leq m$. As the standard deviation of $\Delta \ln S$ is $\sigma\sqrt{\Delta t}$, it is reasonable to make Δt and Δx^2 of the same order for the values.

Remark 23.1 (Non-Uniform Grid) Irregular time steps can be added to match the specific features of the problem, e.g., discrete dividend dates, observation dates for the payoff. Non-uniform spatial steps may be introduced to improve the convergence in some special situations, e.g., smaller spatial value steps near the barrier levels for barrier options. This can be implemented by either a direct approach with Taylor expansion for the finite difference schemes, which may result in a non-tridiagonal matrix, or by *coordinate transformation* (see [111]). $\square$

23.1.3 Resolution Equations

Let u_i^k denote $u(t_k, x_i)$ for $i = 1, \dots, n-1, k = 1, \dots, m$, and $\mathbf{u}^k = [u_1^k, \dots, u_{n-1}^k]^\top$. The boundary values $u(t, x_0)$ and $u(t, x_n)$ will be discussed in the next section. Applying the second-order accurate finite difference schemes (23.4) and (23.5) for $\frac{\partial u}{\partial x}$ and $\frac{\partial^2 u}{\partial x^2}$ in (23.6) gives

$$\frac{\partial u_i^k}{\partial t} + \frac{1}{2}\sigma^2 \frac{u_{i+1}^k - 2u_i^k + u_{i-1}^k}{\Delta x^2} + \left(r - \frac{\sigma^2}{2}\right) \frac{u_{i+1}^k - u_{i-1}^k}{2\Delta x} - ru_i^k = 0$$

that can be written as a system of equations

$$\begin{bmatrix} \partial_t u_1^k \\ \partial_t u_2^k \\ \vdots \\ \partial_t u_{n-2}^k \\ \partial_t u_{n-1}^k \end{bmatrix} = \begin{bmatrix} \mathbf{b} & \mathbf{c} & & & \\ \mathbf{a} & \mathbf{b} & \mathbf{c} & & \\ & \ddots & \ddots & \ddots & \\ & & \mathbf{a} & \mathbf{b} & \mathbf{c} \\ & & & \mathbf{a} & \mathbf{b} \end{bmatrix} \begin{bmatrix} u_1^k \\ u_2^k \\ \vdots \\ u_{n-2}^k \\ u_{n-1}^k \end{bmatrix} + \begin{bmatrix} \mathbf{a}u(x_0, t) \\ 0 \\ \vdots \\ 0 \\ \mathbf{c}u(x_n, t) \end{bmatrix},$$

or

$$\frac{\partial \mathbf{u}^k}{\partial t} = \mathbf{A}\mathbf{u}^k + \mathbf{f}(t), \quad (23.7)$$

where

$$\mathbf{a} = \frac{r - \sigma^2/2}{2\Delta x} - \frac{\sigma^2}{2(\Delta x)^2}, \quad \mathbf{b} = r + \frac{\sigma^2}{(\Delta x)^2}, \quad \mathbf{c} = -\frac{r - \sigma^2/2}{2\Delta x} - \frac{\sigma^2}{2(\Delta x)^2} \quad (23.8)$$

and $\mathbf{f}(t)$ is a known vector at time t by the boundary conditions. Matrix $\mathbf{A}$ has constant diagonals, and it is known as a *Toeplitz matrix*. Its special structure enjoys a number of fast numerical algorithms [25, 26].

23.1.4 Boundary Conditions

Specific Boundary Conditions

For a specific problem, the boundary conditions can be approximated. For instance, for a European call option, we can use $u(t, x_0) = u(t, L_1) \approx 0$ and $u(t, x_n) = u(t, L_2) \approx e^{L_2} - Ke^{-r(T-t)}$.

General Boundary Conditions with Zero Gamma

Zero gamma on boundaries is a general condition for all options with natural smoothness ($\partial_{xx}u = 0$). It results in $f(t) = 0$ with changed values on the first and last rows of $\mathbf{A}$. The finite difference equations for zero gamma are

$$\frac{u(t, x_0) - 2u_1^k + u_2^k}{\Delta x} = 0 \quad \text{and} \quad \frac{u(t, x_n) - 2u_{n-1}^k + u_{n-2}^k}{\Delta x} = 0,$$

which leads to the boundary conditions for the upper and lower bounds:

$$u(t, x_0) = 2u_1^k - u_2^k \quad \text{and} \quad u(t, x_n) = 2u_{n-1}^k - u_{n-2}^k.$$

23.1.5 Terminal Condition

For a European option, the payoff is known at maturity T or at non-discretionary early termination.³ Therefore, the problem solving uses backward *time-marching* for option pricing. However, it suffices to use $\bar{t}_i = T - t_i$ to switch between forward and backward time-marching. In mathematics, a PDE with initial or termination condition is known as an *initial-value problem*.

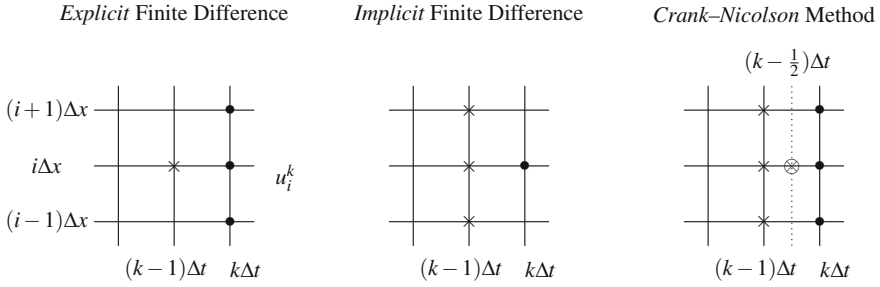
23.2 Finite Difference Schemes

Let $\mathbf{u}^k$ denote the vector $\mathbf{u}(t_k)$. Applying forward difference (23.3) for $\frac{\partial \mathbf{u}^k}{\partial t}$ in (23.7) with different treatment on the RHS results in different numerical schemes. The most popular ones are

$$\begin{aligned} \text{Explicit scheme :} \quad & \frac{\mathbf{u}^k - \mathbf{u}^{k-1}}{\Delta t} = \mathbf{A}\mathbf{u}^k \\ \text{Implicit scheme :} \quad & \frac{\mathbf{u}^k - \mathbf{u}^{k-1}}{\Delta t} = \mathbf{A}\mathbf{u}^{k-1} \\ \text{Crank-Nicolson scheme :} \quad & \frac{\mathbf{u}^k - \mathbf{u}^{k-1}}{\Delta t} = \frac{1}{2}(\mathbf{A}\mathbf{u}^k + \mathbf{A}\mathbf{u}^{k-1}). \end{aligned}$$

Figure 23.2 contains the graphical representations of the three discretization schemes for the time domain.

³ By non-discretionary early termination, we mean the termination by the underliers' price performance as opposed to discretionary termination in the case of American options.



Note: the value of $u(t, x)$ at the point $\bullet$ on the grid is known for the current time step.

Fig. 23.2 Explicit, implicit and Crank–Nicolson schemes

23.2.1 Explicit Scheme

Explicit scheme is equivalent to, for $k = 0, 1, \dots, m$,

$$\mathbf{u}^{k-1} = (\mathbf{I} - \Delta t \mathbf{A}) \mathbf{u}^k,$$

where $\mathbf{I}$ is an identity matrix of size $n - 1$. Since $\mathbf{u}^m = [u_1^m, \dots, u_{n-1}^m]^\top$ are known values at maturity, a time-marching operation is performed to obtain $\mathbf{u}^{m-1}$, $\mathbf{u}^{m-2}$, and after m times, $\mathbf{u}^0$ at time 0.

The accuracy of explicit scheme is $O(\Delta t)$ and $O(\Delta x^2)$, which is shown in Sect. 23.10. Explicit scheme is conditionally stable (see Sect. 23.11).

23.2.2 Implicit Scheme

Implicit scheme is equivalent to, for $k = 0, 1, \dots, m$,

$$(\mathbf{I} + \Delta t \mathbf{A}) \mathbf{u}^{k-1} = \mathbf{u}^k.$$

The time-marching process starts from the known $\mathbf{u}^m$ and a tridiagonal linear system needs to be solved at each time step, which requires a computational cost of $O(n)$ by Thomas algorithm in numerical linear algebra.

The accuracy of implicit scheme is $O(\Delta t)$ and $O(\Delta x^2)$ (see Sect. 23.10). Compared to explicit method, implicit method is more robust. It does not suffer from the choice of Δt and Δx , which is called unconditional stability in numerical PDE approaches (see Sect. 23.11).

23.2.3 Crank–Nicolson Scheme and θ -Scheme*

Crank–Nicolson scheme [35] is the combination of implicit and explicit schemes. It is equivalent to

$$\left(\mathbf{I} + \frac{\Delta t}{2}\mathbf{A}\right)\mathbf{u}^{k-1} = \left(\mathbf{I} - \frac{\Delta t}{2}\mathbf{A}\right)\mathbf{u}^k,$$

and a tridiagonal system is solved at each time-step during time-marching. As shown in Sect. 23.10, the scheme is second-order accurate in both space and time (i.e., $O(\Delta t^2)$ and $O(\Delta x^2)$).

Crank–Nicolson scheme is known to suffer from the non-smoothness of payoff function, and a technique called Rannacher time stepping [95] is usually used for the initial time-steps. However, the payoff non-smoothness causes hedging issues as explained in Sect. 20.1, and practitioners smooth out discrete payoffs in option hedging and pricing.

Actually, explicit, implicit and Crank–Nicolson schemes can be seen as special cases of the so-called θ -scheme, with $\theta = 1$, $\theta = 0$, and $\theta = 1/2$ respectively. It is expressed as

$$\frac{\mathbf{u}^k - \mathbf{u}^{k-1}}{\Delta t} = \theta\mathbf{A}\mathbf{u}^k + (1 - \theta)\mathbf{A}\mathbf{u}^{k-1}.$$

By stability analysis (see Sect. 23.11), the θ -scheme is unconditionally stable for $1/2 \leq \theta \leq 1$.

23.3 Numerical Resolution for Finite Difference Methods*

Among finite difference methods previously presented, the resolution of explicit method can be performed by straightforward calculation for *each step of backward time-marching*. For the numerical methods with an implicit scheme, a linear equation system needs to be resolved for each time-marching. We introduce two main resolution approaches: *direct* method and *iterative* method involving iterations for each backward time-marching step. Let us use general notations for the linear system that covers an implicit finite difference equation for one time-marching step:

$$\mathbf{A} = \begin{pmatrix} b_1 & c_1 & 0 & 0 & \dots & 0 \\ a_2 & b_2 & c_2 & 0 & \dots & 0 \\ 0 & a_3 & b_3 & c_3 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & a_{n-1} & b_{n-1} & c_{n-1} \\ 0 & 0 & \dots & 0 & a_n & b_n \end{pmatrix}, \quad \mathbf{A}\mathbf{u} = \mathbf{g}.$$

23.3.1 Direct Method

Thomas algorithm is an efficient direct resolution method for linear equation system with tridiagonal matrix, which is the case for our finite difference choices. It is conducted in two steps:

- (1) transform the tridiagonal matrix into a lower diagonal matrix by Gauss elimination method:

- (i) $b'_n = b_n, g'_n = g_n;$

- (ii) from $i = n - 1$ to 1: $b'_i = b_i - c_i a_{i+1}/b'_{i+1}, g'_i = g_i - c_i g'_{i+1}/b'_{i+1};$

- (2) resolve the lower diagonal equation system:

- (i) $u_1 = g'_1/b'_1;$

- (ii) from $i = 2$ to n : $u_i = (g'_i - a_i u_{i-1})/b'_i.$

$$\mathbf{A}' = \begin{pmatrix} b'_1 & 0 & 0 & 0 & \dots & 0 \\ a_2 & b'_2 & 0 & 0 & \dots & 0 \\ 0 & a_3 & b'_3 & 0 & \dots & 0 \\ & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & a_{n-1} & b'_{n-1} & 0 \\ 0 & 0 & \dots & 0 & a_n & b'_n \end{pmatrix}, \quad \mathbf{A}'\mathbf{u} = \mathbf{g}'.$$

23.3.2 Iterative Method

As a general linear equation resolution method, iterative methods can be more efficient than Thomas algorithm for large and diagonal dominant matrix $\mathbf{A}$. We present the approach for the general case (i.e., not limited to tridiagonal matrices).

For the i -th equation $a_{i1}u_1 + a_{i2}u_2 + \dots + a_{in}u_n = g_i$, we can write

$$u_i = \frac{1}{a_{ii}} \left(g_i - \sum_{j=1}^{i-1} a_{ij}u_j - \sum_{j=i+1}^n a_{ij}u_j \right). \quad (23.9)$$

Several iterative methods can be constructed from (23.9) for $i = 1$ to n with increasing order of convergence speed in general:

$$\left\{ \begin{array}{l} \text{Jacobi: } u_i^{(h+1)} = \frac{1}{a_{ii}} [g_i^{(h)} - \sum_{j=1}^{i-1} a_{ij} u_j^{(h)} - \sum_{j=i+1}^n a_{ij} u_j^{(h)}]; \\ \text{Gauss-Seidel: } u_i^{(h+1)} = \frac{1}{a_{ii}} [g_i^{(h)} - \sum_{j=1}^{i-1} a_{ij} u_j^{(h+1)} - \sum_{j=i+1}^n a_{ij} u_j^{(h)}]; \\ \text{SOR: } u_i^{(h+1)} = u_i^{(h)} + \frac{\omega}{a_{ii}} [g_i^{(h)} - \sum_{j=1}^{i-1} a_{ij} u_j^{(h+1)} - \sum_{j=i}^n a_{ij} u_j^{(h)}], \quad 0 < \omega < 2. \end{array} \right.$$

Starting from an initial value for u_i^0 , the iteration process stops when $\sum_{i=1}^n (u_i^{(h+1)} - u_i^{(h)})^2$ is smaller than a certain accuracy level ϵ^2 . The optimal value for the relaxation parameter ω in SOR method is problem-specific, but it must be between 0 and 2 ($\omega = 1$ is equivalent to Gauss-Seidel method). For further analysis on convergence conditions, we refer to advanced books or references dedicated to numerical analysis, e.g., [23], page 348.

23.4 Numerical Resolution for American Options*

For American options, the partial differential inequality (19.3) needs to be satisfied. Although, it is a free-boundary problem, simple resolution algorithms have been found. We now introduce the most popular ones by considering any step k ($k < m$).

For explicit method, American option pricing is similar to tree-method: after the calculation of each backward solution u_i from the known values at the previous time-marching step, we take $\max\{u_i, F_i\}$ as the American option price for the grid point (k, i) where F_i is the exercise payoff at S_i .

For implicit method, we need a component-wise method to solve the equation, i.e., for each grid point (k, i) , a continuation value $\tilde{u}_i$ can be obtained individually and progressively. Then the American option price at the grid point will be $\max\{\tilde{u}_i, F_i\}$. Both the direct method and iterative methods presented in Sect. 23.3 allow such implementation. For notation convenience, we drop off the superscript of time-marching step k for the PDE solution u .

For the direct method, we can modify the Thomas algorithm as

- (1) transform the tridiagonal matrix into a lower diagonal matrix by Gauss elimination method:

- (i) $b'_n = b_n, g'_n = g_n$;
- (ii) from $i = n - 1$ to 1: $b'_i = b_i - c_i a_{i+1} / b'_{i+1}, g'_i = g_i - c_i g'_{i+1} / b'_{i+1}$;

(2) resolve the lower diagonal equation system:

- (i) $u_1 = g'_1/b'_1$;
- (ii) from $i = 2$ to n : $u_i = \max\{(g'_i - a_i u_{i-1})/b'_i, F_i\}$.

Readers can refer to [118] [67] for further details and convergence analysis.

For the iterative methods, we take the SOR method as an example

$$u_i^{(h+1)} = \max\{u_i^{(h)} + \frac{\omega}{a_{ii}}[g_i^{(h)} - \sum_{j=1}^{i-1} a_{ij}u_j^{(h+1)} - \sum_{j=i}^n a_{ij}u_j^{(h)}], F_i\},$$

which is known as *projected* SOR.

We refer to [118] for numerical methods on American option pricing in the context of a jump-diffusion model.

23.5 Multi-Dimensional PDE Derivation*

Multi-dimensional PDEs originate from various context, including

- (a) multi-asset options commonly found in structured products;
- (b) models involving risks that are not related to underlier prices, such as stochastic volatility, jumps; For such non-complete market models,⁴ risk premia are introduced that may be calibrated by the market price of tradable options (see Sect. 25.2.1 for the Heston model example);
- (c) path-dependent options where state variables (e.g., running average) are used to facilitate numerical solutions. See equation (19.5).

To illustrate the PDE derivation, we introduce two typical approaches.

23.5.1 Dynamic Hedging Portfolio Approach

This PDE derivation approach consists of applying

- (1) Itô–Doebelin formula to the portfolio of the option and the hedging instruments;
- (2) hedging strategy to eliminate the risk term (e.g., Brownian terms);
- (3) no-arbitrage and terminal/boundary conditions (see Sect. 14.2 as an example).

The 1st derivation method for Heston PDE presented in Sect. 25.2.1 provides a typical illustration of the method.

⁴ Roughly speaking, a *complete market* model is the one under which every derivative instrument can be replicated by a self-financing (dynamic) portfolio of underlying assets and cash.

23.5.2 Risk-Neutral Framework Approach

Under risk-neutral probability $\tilde{\mathbb{P}}$, the discounted values for the underlying assets and the related options are all martingales. This is possible only if a PDE relating the underlying asset prices and the option price is satisfied. Here, we only introduce the PDE derivation for Brownian motion driven models, which is actually equivalent to the direct application of Feynman–Kac theorem. Readers can refer to Sect. 26.6 for the PIDE (partial integro-differential equation) derivation for a jump-diffusion model involving the Poisson process.

Generally, a multi-factor model driven by Brownian motions can be presented as

$$\begin{cases} dx_1 = a_1 dt + b_1 d\tilde{W}_t^1, \\ \vdots \\ dx_d = a_d dt + b_d d\tilde{W}_t^d. \end{cases} \quad \forall i, j : d\tilde{W}_t^i d\tilde{W}_t^j = \rho_{ij} dt, \quad \rho_{ii} = 1. \quad (23.10)$$

The stochastic quantities $x_1, \dots, x_d$ impact directly or indirectly the option price $u(t, x_1, \dots, x_d)$. Applying multi-dimensional Itô–Doebelin formula to the discounted option price:

$$d(e^{-rt} u(t, x_1, \dots, x_d)) = e^{-rt} \left(\left[\frac{\partial u}{\partial t} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d b_i b_j \rho_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^d a_i \frac{\partial u}{\partial x_i} - ru \right] dt + \sum_{i=1}^d b_i \frac{\partial u}{\partial x_i} d\tilde{W}_t^i \right).$$

As the discounted option price is a $\tilde{\mathbb{P}}$ -martingale, the dt term inside $[\cdot]$ must be zero, which leads to be below PDE (*parabolic* and *linear*):

$$\frac{\partial u}{\partial t} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \rho_{ij} b_i b_j \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^d a_i \frac{\partial u}{\partial x_i} - ru = 0. \quad (23.11)$$

For its resolution by finite difference method, explicit scheme can be applied directly. However, implicit scheme may suffer from the *curse of dimensionality*, i.e., the computation complexity grows as $O(mn^d)$ (m is the number of time steps and n is the number of steps for each space dimension). There exist methods decomposing the multi-dimensional problem into a series of 1-D computations, and we will introduce a popular one in the sequel for a 2-D PDE without loss of generality. We note that in financial industry, problems of more than two spatial dimensions are mostly resolved by Monte Carlo method.

23.6 Alternating Direction Implicit (ADI) Method*

For options with two factors, the PDE is 2-dimensional and the ADI method is a quite efficient method that consists of using two half time-steps and applying implicit scheme on one spatial dimension while using explicit scheme on the other spatial dimension in the first half time-step, then alternating the implicit/explicit schemes for the second half time-step. In this method, the dimension of the matrix to be inverted at each time is of $O(n)$, similar to the 1-D case.

Here, we illustrate the situation without the mixed derivative term. Let us keep and extend the notations used in the 1-D case with n grid points for both spatial dimensions. Now, $\mathbf{u}^k$ denotes the $n \times n$ option value vector for each time point t_k . We decompose the coefficient matrix $\mathbf{A}$ by $\mathbf{A}_1$ for the finite difference w.r.t. x_1 and $\mathbf{A}_2$ for x_2 . We have

$$\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2.$$

For each time-step, ADI method applies the below steps

$$\begin{cases} \frac{\mathbf{u}^k - \mathbf{u}^{k-\frac{1}{2}}}{\Delta t/2} = (\mathbf{A}_2 \mathbf{u}^k + \mathbf{A}_1 \mathbf{u}^{k-\frac{1}{2}}) \\ \frac{\mathbf{u}^{k-\frac{1}{2}} - \mathbf{u}^{k-1}}{\Delta t/2} = (\mathbf{A}_1 \mathbf{u}^{k-\frac{1}{2}} + \mathbf{A}_2 \mathbf{u}^{k-1}). \end{cases} \quad (23.12)$$

ADI can also be implemented with the following “predictor-corrector” steps with $0 < \theta \leq 1$:

$$\begin{aligned} \mathbf{y}_0 &= \mathbf{u}^k - \Delta t \mathbf{A} \mathbf{u}^k \\ \mathbf{y}_1 &= \mathbf{y}_0 - \theta \Delta t (\mathbf{A}_1 \mathbf{y}_1 - \mathbf{A}_1 \mathbf{u}^k); \quad \mathbf{y}_2 = \mathbf{y}_1 - \theta \Delta t (\mathbf{A}_2 \mathbf{y}_2 - \mathbf{A}_2 \mathbf{u}^k) \\ \mathbf{u}^{k-1} &= \mathbf{y}_2. \end{aligned} \quad (23.13)$$

The first line in (23.13) is a rough estimation with explicit method and the second line corresponds to “corrections” with implicit method for each space directions. Actually, this approach is equivalent to (23.12) with $\theta = 1/2$.⁵

⁵ From Eq. (23.12) with $\theta = 1/2$, we obtain $(\mathbf{I} + \theta \Delta t \mathbf{A}_1) \mathbf{u}^{k-\frac{1}{2}} = (\mathbf{I} - \theta \Delta t \mathbf{A}_2) \mathbf{u}^k$ and $(\mathbf{I} - \theta \Delta t \mathbf{A}_1) \mathbf{u}^{k-\frac{1}{2}} = (\mathbf{I} + \theta \Delta t \mathbf{A}_2) \mathbf{u}^{k-1}$, leading to $(\mathbf{I} - \theta \Delta t \mathbf{A}_1)(\mathbf{I} - \theta \Delta t \mathbf{A}_2) \mathbf{u}^k = (\mathbf{I} + \theta \Delta t \mathbf{A}_1)(\mathbf{I} + \theta \Delta t \mathbf{A}_2) \mathbf{u}^{k-1}$. By rewriting (23.13) into one formula with $\mathbf{u}^k$ and $\mathbf{u}^{k-1}$ only, we obtain the same result.

23.7 ADI Method with Mixed Derivative Terms*

Mixed derivative (or cross derivative) terms can be directly dealt with numerically. Explicit finite difference is a straight-forward method for PDEs with mixed derivative terms. For improving the convergence, there are various partially implicit finite difference methods well studied and applied. Here, we introduce a popular one known as Craig–Sneyd scheme in which a “predictor-corrector” approach is applied to a 2-D PDE. For higher dimensions and complete analysis, we refer to [34].

For mixed derivatives, we can apply 2-D Taylor series on $u(x_1 \pm \Delta x_1, x_2 \pm \Delta x_2)$ to get the finite difference with accuracy $O(\Delta x_1^2, \Delta x_2^2)$:

$$\frac{\partial^2 u}{\partial x_1 \partial x_2}(i, j) \approx \frac{1}{4\Delta x_1 \Delta x_2} (u_{i+1, j+1} - u_{i+1, j-1} - u_{i-1, j+1} + u_{i-1, j-1}).$$

Let us split the finite difference coefficient matrix $\mathbf{A}$ into matrices:

$$\mathbf{A} = \mathbf{A}_0 + \mathbf{A}_1 + \mathbf{A}_2,$$

where $\mathbf{A}_0$ is for mixed derivatives $\frac{\partial^2 u}{\partial x_1 \partial x_2}$, $\mathbf{A}_1$ for x_1 -derivatives and $\mathbf{A}_2$ for x_2 -derivatives. Craig–Sneyd scheme involves several intermediary calculations with 1-D complexity for each time step by using explicit scheme to get a rough estimation “prediction” followed by a “correction” with implicit scheme. Explicit scheme is applied to mixed derivative terms:

$$\begin{aligned} \mathbf{y}_0 &= \mathbf{u}^k - \Delta t \mathbf{A} \mathbf{u}^k \\ \mathbf{y}_1 &= \mathbf{y}_0 - \theta \Delta t (\mathbf{A}_1 \mathbf{y}_1 - \mathbf{A}_1 \mathbf{u}^k); \quad \mathbf{y}_2 = \mathbf{y}_1 - \theta \Delta t (\mathbf{A}_2 \mathbf{y}_2 - \mathbf{A}_2 \mathbf{u}^k) \\ \mathbf{y}_3 &= \mathbf{y}_0 - \frac{1}{2} \Delta t (\mathbf{A}_0 \mathbf{y}_2 - \mathbf{A}_0 \mathbf{u}^k) \\ \mathbf{y}_4 &= \mathbf{y}_3 - \theta \Delta t (\mathbf{A}_1 \mathbf{y}_4 - \mathbf{A}_1 \mathbf{u}^k); \quad \mathbf{y}_5 = \mathbf{y}_4 - \theta \Delta t (\mathbf{A}_2 \mathbf{y}_5 - \mathbf{A}_2 \mathbf{u}^k) \\ \mathbf{u}^{k+1} &= \mathbf{y}_5 \end{aligned}$$

As $\mathbf{A}_0$ is no longer a tridiagonal matrix with 3 points involved, the numerical schemes for mixed terms are normally explicit in the various methods.

23.8 Discrete Dividend*

At the ex-dividend date τ for a scheduled discrete dividend, the option value does not change just before or after τ . Denoting $V(t, S)$ for $u(t, x)$ for showing the treatment of dividend impact on S , the solution propagation is

$$V(\tau^-, S) = V(\tau, S - \text{div}), \quad (23.14)$$

which needs to be taken as an extra condition to be added to the PDE equation system. As mentioned in Remark 23.1, additional time steps can be added to ensure that all the ex-div dates are on the grid. However, $S - \text{div}$ may not be on the grid and interpolation needs to be implemented for the value of $V(\tau, S - \text{div})$.

The above approach applies to other predictable corporate actions to avoid the impact on the value of the derivative.

23.9 Discrete Observation/Sampling*

For path-dependent or barrier options, the natural observation/sampling on underlier prices is discrete.

23.9.1 No-Arbitrage Condition for Discrete Sampling*

Consider a discrete sampling time and denote

- τ as the sampling time or, if such time does not appear on the finite difference grid, the nearest time immediately after the sampling date;
- τ^- as the time on the grid immediately prior to τ .

The option prices V should satisfy the *no-arbitrage condition*, which keeps the price continuity that we illustrate with the below equation for a one-underlier option:

$$V(\tau^-, S, F_{\tau^-}) = V(\tau, S, F(S, F_{\tau^-})), \quad (23.15)$$

where $F(S, F_{\tau^-})$ is a state variable, which can be an added spatial dimension for pricing, e.g.,

- Asian option: $V(\tau^-, S, I_{\tau^-}) = V(\tau, S, I_{\tau})$, where I_t is the running sum used in the option payoff and $I_{\tau} = I_{\tau^-} + S$;
- look-back spot option: $V(\tau^-, S, S_{\tau^-}^*) = V(\tau, S, S_{\tau}^*)$ where S_t^* is the running maximum used in the option payoff and $S_{\tau}^* = \max\{S_{\tau^-}^*, S\}$.

23.9.2 Discrete Sampling for Backward Pricing Approach

For one-factor models such as BSM or local volatility models, PDE pricer can be used for path-dependent options with discrete sampling. Similar to the continuous case, the state variables such as running average or running minimum/maximum are necessary for some path-dependent structures for PDE methods.

For the path-dependent options requiring state variables, the PDE between two successive sampling times will not contain the state variables (e.g., normal Black–Scholes equation $\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$). Hence, the option price should behave like a European option between two sampling dates. However, the pricing algorithm must apply the no-arbitrage condition at each sampling time to all the discretized values of the state variables.

23.10 Accuracy and Consistency Analysis*

Truncation error reflects the approximation error by a finite difference scheme. Denote all the terms of the considered PDE by the operator $L[u(t, x)]$ with $L[u(t, x)] = 0$, its approximation by a finite difference method by $\tilde{L}[u(t, x)]$ where $u(t, x)$ is an exact solution. The truncation error of a finite difference scheme at any discretization point (t, x) is measured by $\tilde{L}[u(t, x)] - L[u(t, x)] = \tilde{L}[u(t, x)]$. Taylor analysis is the natural tool for truncation error analysis.

A finite difference approximation is said to be *consistent* if the truncation error at any discretization point approaches zero with vanishing discretization steps. It is a necessary condition for the convergence of the finite difference method.

Consider a general 1-D parabolic PDE with central finite difference for spatial dimension and forward finite difference for time steps. We have then

$$L_{k,i} = \frac{\partial u_i^k}{\partial t} + H_i^k = 0, \forall k, i \text{ where } H_i^k = \alpha \frac{\partial^2 u_i^k}{\partial x^2} + \beta \frac{\partial u_i^k}{\partial x} + \gamma u_i^k.$$

For the approximation of H_i^k , we apply the central finite difference schemes for the partial derivatives. From (23.5) and (23.4), its error in the spatial dimension is

$$\tilde{H}_i^k - H_i^k = O(\Delta x^2), \forall k, i. \quad (23.16)$$

For explicit finite difference scheme, the truncation error at the point (t_k, x_i) is

$$\tilde{L}_{k,i}^{ex} = \frac{u_i^k - u_i^{k-1}}{\Delta t} + \tilde{H}_i^k = \frac{\partial u_i^k}{\partial t} - \frac{1}{2} \frac{\partial^2 u_i^k}{\partial t^2} \Delta t + O(\Delta t^2) + \tilde{H}_i^k.$$

Subtracting it by $L_{k,i} (= 0)$ gives

$$\tilde{L}_{k,i}^{ex} - L_{k,i} = -\frac{1}{2} \frac{\partial^2 u_i^k}{\partial t^2} \Delta t + O(\Delta t^2) + O(\Delta x^2) = O(\Delta t) + O(\Delta x^2).$$

Hence, the order of accuracy for explicit scheme is $O(\Delta t, \Delta x^2)$. Similarly, we can check that the accuracy for implicit scheme is $O(\Delta t, \Delta x^2)$.

For Crank–Nicolson scheme, the truncation error is

$$\begin{aligned}
 \tilde{L}_{k,i}^{cn} &= \frac{1}{2} \left(\frac{u_i^k - u_i^{k-\frac{1}{2}}}{\epsilon} + \tilde{H}_i^k \right) + \frac{1}{2} \left(\frac{u_i^{k-\frac{1}{2}} - u_i^{k-1}}{\epsilon} + \tilde{H}_i^{k-1} \right) \\
 &= \frac{1}{2} \left(\frac{\partial u_i^{k-\frac{1}{2}}}{\partial t} + \frac{1}{2} \frac{\partial^2 u_i^{k-\frac{1}{2}}}{\partial t^2} \epsilon + \tilde{H}_i^k \right) \\
 &\quad + \frac{1}{2} \left(\frac{\partial u_i^{k-\frac{1}{2}}}{\partial t} - \frac{1}{2} \frac{\partial^2 u_i^{k-\frac{1}{2}}}{\partial t^2} \epsilon + \tilde{H}_i^{k-1} \right) + O(\epsilon^2) \\
 &= \frac{\partial u_i^{k-\frac{1}{2}}}{\partial t} + \frac{1}{2} (\tilde{H}_i^k + \tilde{H}_i^{k-1}) + O(\epsilon^2)
 \end{aligned} \tag{23.17}$$

where $t_{k-\frac{1}{2}} = \frac{1}{2}(t_{k-1} + t_k)$, $\epsilon = \Delta t/2$. Taylor expansion of $\frac{\partial u}{\partial t}$ at $(t_{k-\frac{1}{2}}, x_i)$ implies

$$\begin{cases} \frac{\partial u_i^k}{\partial t} = \frac{\partial u}{\partial t}(t_{k-\frac{1}{2}} + \epsilon, x_i) = \frac{\partial u_i^{k-\frac{1}{2}}}{\partial t} + \frac{\partial^2 u_i^{k-\frac{1}{2}}}{\partial t^2} \epsilon + O(\epsilon^2) \\ \frac{\partial u_i^{k-1}}{\partial t} = \frac{\partial u}{\partial t}(t_{k-\frac{1}{2}} - \epsilon, x_i) = \frac{\partial u_i^{k-\frac{1}{2}}}{\partial t} - \frac{\partial^2 u_i^{k-\frac{1}{2}}}{\partial t^2} \epsilon + O(\epsilon^2). \end{cases}$$

Adding both sides of the above equations leads to

$$\frac{\partial u_i^{k-\frac{1}{2}}}{\partial t} = \frac{1}{2} \left(\frac{\partial u_i^k}{\partial t} + \frac{\partial u_i^{k-1}}{\partial t} \right) + O(\epsilon^2). \tag{23.18}$$

Subtracting $\tilde{L}_{k,i}^{cn}$ by $\frac{1}{2}L_{k,i}(=0)$ and $\frac{1}{2}L_{k-1,i}(=0)$, and applying Eqs. (23.16), (23.18), we obtain $\tilde{L}_{k,i}^{cn} = O(\epsilon^2) + O(\Delta x^2) = O(\Delta t^2) + O(\Delta x^2)$. Therefore, the accuracy for Crank–Nicolson scheme is $O(\Delta t^2, \Delta x^2)$.

23.11 Stability and Convergence Analysis*

Many research works have been done on the stability and convergence analysis for finite difference methods and von Neumann's stability analysis is the most used method for financial problems.⁶ Here, we only outline it informally by using

⁶ The matrix/spectral stability analysis is more general, but it is less easy to apply. See [23].

forward time-marching and the notations commonly used in the literature. See [23, 37, 112] for further and more rigorous presentations.

A finite difference approximation is *stable* to the norm $\|\cdot\|$ (we assume Euclidean and L^2 norms in the below analysis) if there exists a constant $K(T) > 0$ such that

$$\|u^{k+1}\| \leq K(T)\|u^k\|.$$

For linear initial-value problems (that cover most of the PDEs for option pricing), Lax Equivalence Theorem (see [109]) confirms that a finite difference approximation converges to the exact solution if and only if it is consistent and stable. Parseval's Identity theorem relates $u(x)$ with its Fourier transform $\mathcal{F}[u(x)](\phi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\phi x \sqrt{-1}} u(x) dx$, $\forall \phi \in \mathbb{R}$ by the below equality:

$$\int_{-\infty}^{\infty} |u(x)|^2 dx = \int_{-\infty}^{\infty} |\mathcal{F}[u(x)](\phi)|^2 d\phi.$$

It can be proved that the discrete Fourier transform defined in the below also satisfies the Parseval's identity (see [112] for the proof):

$$\hat{u}(\omega) = \frac{1}{\sqrt{2\pi}} \sum_{i=-\infty}^{\infty} e^{-i\omega\sqrt{-1}} u_i, \quad \forall \omega \in [-\pi, \pi].$$

Parseval's identity shows that the Fourier transform is norm-preserving. **Von Neumann's method** uses $\|\hat{u}^{k+1}\| \leq K(T)\|\hat{u}^k\|$ as stability criteria. Let's denote

$$\hat{u}_i = \frac{1}{\sqrt{2\pi}} \sum_{i=-\infty}^{\infty} e^{-i\omega\sqrt{-1}} u_i \quad \text{and} \quad \hat{u}_{i\pm 1} = \frac{1}{\sqrt{2\pi}} \sum_{i=-\infty}^{\infty} e^{-i\omega\sqrt{-1}} u_{i\pm 1}.$$

We can check by variable change on the index that:⁷

$$\hat{u}_{i+1} = e^{\omega\sqrt{-1}} \hat{u}_i, \quad \hat{u}_{i-1} = e^{-\omega\sqrt{-1}} \hat{u}_i. \quad (23.19)$$

The iteration equations in finite difference methods take the below form for some $G(\omega)$ known as the *amplification factor*

$$\hat{u}_i^{k+1} = G(\omega) \hat{u}_i^k \quad \text{and} \quad \hat{u}_i^k = G(\omega)^k \hat{u}_i^0 \quad \text{by iterations.}$$

⁷ e.g., $\hat{u}_{i+1} = \frac{1}{\sqrt{2\pi}} \sum_{i=-\infty}^{\infty} e^{-i\omega\sqrt{-1}} u_{i+1} = e^{\omega\sqrt{-1}} \frac{1}{\sqrt{2\pi}} \sum_{i=-\infty}^{\infty} e^{-(i+1)\omega\sqrt{-1}} u_{i+1} = e^{\omega\sqrt{-1}} \hat{u}_i$.

As u^k can be considered as the sum of the exact solution and the error term, it is intuitive that $|G(\omega)| \leq 1$ prevents the growth of the error term (by linearity of matrix calculation) at each step, which turns out to be a necessary and sufficient condition for the stability (see [112] for the proof).

Now, let's apply von Neumann's stability analysis to the explicit scheme for the 1-D PDE (23.6). The backward iteration is

$$u_i^{k+1} = -au_{i-1}^k + (1-b)u_i^k - cu_{i+1}^k.$$

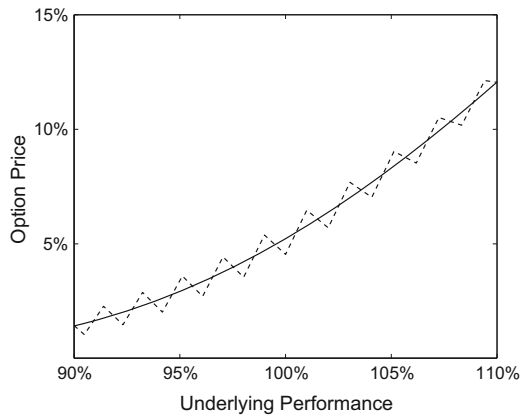
where a , b and c are defined in (23.8). Apply the discrete Fourier transform on both sides of the equation. By (23.19) and Euler's formula, we obtain

$$\begin{aligned} G &= -ae^{-\omega\sqrt{-1}} + (1-b) - ce^{\omega\sqrt{-1}} \\ &= 1 - r\Delta t + \frac{\sigma^2\Delta t}{\Delta x^2}(\cos(\omega) - 1) + \sqrt{-1}\left(r - \frac{\sigma^2}{2}\right)\frac{\Delta t}{\Delta x}\sin(\omega). \end{aligned}$$

We can see that $|G| \leq 1$ is not satisfied unconditionally. Therefore, explicit scheme is only conditionally stable. It has been found that $\alpha = \Delta t/(\Delta x)^2$ has to be less than a certain constant α_0 to satisfy the stability criteria (see [116]) that may lead to a very small Δt (hence a large m) for certain problems. For implicit and Crank–Nicolson schemes, it turns out that $|G| \leq 1$ is always true. Hence, the stability is unconditional for both schemes.

Example 23.1 Consider pricing a European call option by explicit scheme with $T = 0.25$, $K = 1$, $r = 0.02$, $\sigma = 0.25$, $L_1 = -0.4$, $L_2 = 0.3$. We fix $n = 70$ and test $m = 154$ or $m = 200$, corresponding to $\alpha = 16.23$ or $\alpha = 12.5$ respectively. The solutions with the two settings are plotted in Fig. 23.3. We see that the solution is oscillatory with $\alpha = 16.23$.

Fig. 23.3 Option price (close-up around strike) obtained from explicit scheme with $\alpha = 12.5$ (solid line); oscillatory when $\alpha = 16.23$ (dashed line)



Exercises

1. Consider pricing a European call option using finite difference method with explicit scheme. The parameters are $K = 1$, $T = 0.25$, $r = 0.02$, $\sigma = 0.25$, $L_1 = -0.5$, $L_2 = 0.5$, and the number of time-steps $m = 5000$. Zero gamma boundary conditions are applied. Let c_1 and c_2 denote the finite difference solutions at $x = 0$ ($S = 1$) using $n = 50$ and $n = 100$ respectively. Compare the error of c_1 and c_2 respectively against the Black–Scholes formula at $x = 0$. How fast is the error reducing? Explain this phenomenon.

Part III

Extensions to Financial Modelling

Chapter 24

Static Hedging, Variance Swap and Volatility Index*



At the beginning of this chapter, we first discuss how to replicate a European-style option with call and put options of different strikes. From there, a special contract called variance swap will be covered, together with its valuation and hedging. The static hedging formula with calls/puts can link implied volatilities with variance swap prices.

24.1 Static Hedging for European Style Options

Static hedging, which is opposite to dynamic hedging, does not adjust the hedging portfolio once it has been set up. In this section, an important result is first presented, i.e., any European style option can be hedged statically by calls and puts provided that there are calls and puts with any strike price in the market.

Proposition 24.1 *Any European style option can be (theoretically) hedged with a static portfolio of calls and puts:*

$$\begin{aligned} & \widetilde{\mathbb{E}}_t[e^{-r(T-t)}h(S_T)] \\ &= e^{-r(T-t)}(h(\lambda) + h'(\lambda)(F - \lambda)) + \int_{\lambda}^{\infty} h''(K)c(K)dK + \int_0^{\lambda} h''(K)p(K)dK, \end{aligned} \quad (24.1)$$

where h is a twice differentiable payoff function, S_t is the underlier price at time t , $F = \widetilde{\mathbb{E}}_t[S_T]$ is the forward price of the underlier, r is the constant interest rate, $c(K)$ and $p(K)$ are European call and put prices of strike K , and λ is any fixed level of the underlier price. In particular, when $\lambda = F$, the formula is reduced as

$$e^{-r(T-t)}h(F) + \int_F^{\infty} h''(K)c(K)dK + \int_0^F h''(K)p(K)dK.$$

Proof We first prove a general decomposition of a twice differentiable function h :

$$\begin{aligned}
 h(S) &= h(\lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S h'_u du - \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} h'_u du \\
 &= h(\lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S \left(h'(\lambda) + \int_{\lambda}^u h''(v) dv \right) du \\
 &\quad - \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} \left(h'(\lambda) - \int_u^{\lambda} h''(v) dv \right) du \\
 &= h(\lambda) + h'(\lambda)(S - \lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S \int_{\lambda}^u h''(v) dv du + \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} \int_u^{\lambda} h''(v) dv du \\
 &= h(\lambda) + h'(\lambda)(S - \lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S \int_v^S h''(v) du dv + \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} \int_S^v h''(v) du dv \\
 &= h(\lambda) + h'(\lambda)(S - \lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S h''(v)(S - v) dv + \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} h''(v)(v - S) dv \\
 &= h(\lambda) + h'(\lambda)(S - \lambda) + \int_{\lambda}^{\infty} h''(v)(S - v)^+ dv + \int_0^{\lambda} h''(v)(v - S)^+ dv.
 \end{aligned}$$

Now, setting $S = S_T$, denoting $v = K$, multiplying both sides by the discount factor $e^{-r(T-t)}$, and taking conditional expectation, we obtain the static hedging for a generalized European option:

$$\begin{aligned}
 &\widetilde{\mathbb{E}}_t[e^{-r(T-t)}h(S_T)] \\
 &= e^{-r(T-t)}[h(\lambda) + h'(\lambda)(F - \lambda)] + \int_{\lambda}^{\infty} h''(K)c(K)dK + \int_0^{\lambda} h''(K)p(K)dK. \quad \square
 \end{aligned}$$

From the above formula, in order to replicate the contingent claim on the left-hand side, we need a continuously changing value K to complete the two integrals on the right-hand side. In practical implementation, approximations will be applied because the calls and puts in the market only have a discrete set of strike prices.

A useful special case is the payoff of a log contract, i.e., $h(S_T) = \ln(S_T)$. The static hedging representation is given by

$$\widetilde{\mathbb{E}}_t[\ln(S_T)] = \ln(\lambda) + \frac{F - \lambda}{\lambda} - e^{r(T-t)} \left[\int_{\lambda}^{\infty} \frac{c(K)}{K^2} dK + \int_0^{\lambda} \frac{p(K)}{K^2} dK \right]. \quad (24.2)$$

The Eq. (24.2) will be used for pricing and hedging variance swaps.

24.2 Variance Swap

A variance swap is an agreement to exchange the realized variance V during a fixed maturity T with n business days for a pre-specified variance strike K_{var} in the contract. It provides a pure exposure to variance (hence volatility) useful for risk management. However, as the vega profile of options usually depends on the underlier's reference levels such as strike and barrier, a pure volatility instruments such as variance swap or volatility index futures can only provide macro hedging for a derivatives book. The payoff at maturity is given by

$$N_{\text{var}}(V - K_{\text{var}}),$$

where N_{var} is the notional amount that converts the units from variance to denominated currency on the contract. In the contract term sheets, V is usually the estimator for sample variance

$$V = \frac{252}{n} \sum_{i=1}^n \left(\ln \frac{S_i}{S_{i-1}} \right)^2,$$

in which the sample mean is negligible and set at 0. The number 252 is the annualization factor representing the number of business days in a year.

The contract often contains volatility linked concepts that people are more familiar with (but do not confuse it with volatility swap). Suppose a volatility swap has volatility strike K_{σ} , notional amount N_{vega} , and payoff $N_{\text{vega}}(\sqrt{V} - K_{\sigma})$. The term N_{vega} is also known as the vega notional because the volatility swap payoff would be N_{vega} if the $\sqrt{V} - K_{\sigma}$ is 1 volatility point. Then we have the following two relations between variance and volatility swaps:¹

$$K_{\sigma} = \sqrt{K_{\text{var}}} \quad \text{and} \quad N_{\text{vega}} = N_{\text{var}} \times 2K_{\sigma}.$$

Given N_{vega} , one can determine the notional N_{var} according to the above relation. Note that variance swaps usually have a cap in the payoff formula (especially for single stocks) for controlling the risk to an acceptable level (stock price could become zero).

¹ It is obtained by setting an equal sensitivity to volatility: $\frac{d}{d\sqrt{V}}(N_{\text{vega}}(\sqrt{V} - \sqrt{K_{\text{var}}}))|K_{\text{var}} = \frac{d}{d\sqrt{V}}(N_{\text{var}}(V - K_{\text{var}}))|K_{\text{var}}$ with $K_{\sigma} = \sqrt{K_{\text{var}}}$.

24.2.1 Valuation of Variance Swap

Proposition 24.2 Assume that the underlier of a variance swap follows the model

$$dS_t = rS_t dt + \sigma_t S_t d\tilde{W}_t,$$

where there is no dividend, but σ_t can be stochastic itself. We have the valuation of variance swap that is independent of the modelling for the volatility σ_t :

$$\text{Price of Variance Swap} = N_{\text{var}}(\tilde{\mathbb{E}}[V] - K_{\text{var}})e^{-rT},$$

where $V = \frac{1}{T} \int_0^T \sigma_t^2 dt$,

$$\tilde{\mathbb{E}}[V] = \frac{2}{T} \left[\ln\left(\frac{F}{\lambda}\right) + \frac{\lambda - F}{\lambda} + e^{rT} \int_{\lambda}^{\infty} \frac{c(K)}{K^2} dK + e^{rT} \int_0^{\lambda} \frac{p(K)}{K^2} dK \right]. \quad (24.3)$$

In other words, the variance swap can be theoretically priced and hedged by a series of calls and puts.

Proof Applying Itô–Doebelin formula to $\ln(S_t)$, we get

$$d \ln(S_t) = \frac{dS_t}{S_t} - \frac{1}{2} \frac{(dS_t)^2}{S_t^2} = \frac{dS_t}{S_t} - \frac{1}{2} \sigma_t^2 dt.$$

After integration, it becomes

$$\ln(S_T) = \ln(S_0) + \int_0^T \frac{dS_t}{S_t} - \frac{1}{2} \int_0^T \sigma_t^2 dt.$$

Then,

$$\begin{aligned} V &= \frac{1}{T} \int_0^T \sigma_t^2 dt = \frac{2}{T} \left[\ln(S_0) + \int_0^T \frac{dS_t}{S_t} - \ln(S_T) \right] \\ &= \frac{2}{T} \left[\ln(S_0) + rT + \int_0^T \sigma_t d\tilde{W}_t - \ln(S_T) \right] = \frac{2}{T} \left[\ln(F) + \int_0^T \sigma_t d\tilde{W}_t - \ln(S_T) \right] \end{aligned}$$

since the forward value is $F = S_0 e^{rT}$ with constant r . Taking the risk-neutral expectation on both sides and making the substitution (24.2), we obtain

$$\begin{aligned} \tilde{\mathbb{E}}[V] &= \frac{2}{T} [\ln(F) - \tilde{\mathbb{E}}[\ln(S_T)]] \\ &= \frac{2}{T} \left[\ln\left(\frac{F}{\lambda}\right) + \frac{\lambda - F}{\lambda} + e^{rT} \int_{\lambda}^{\infty} \frac{c(K)}{K^2} dK + e^{rT} \int_0^{\lambda} \frac{p(K)}{K^2} dK \right]. \quad \square \end{aligned}$$

Usually, the value λ is chosen to be $F = S_0 e^{rT}$ such that the formula is simplified to be

$$\tilde{\mathbb{E}}[V] = \frac{2e^{rT}}{T} \left[\int_F^\infty \frac{c(K)}{K^2} dK + \int_0^F \frac{p(K)}{K^2} dK \right].$$

Dividends impact the pricing of variance swap, especially for single stock underlier. One way to deal with dividends is to embed the clause to make adjustment to the return calculation at ex-dividend dates to produce the total returns to be used by variance swaps. For index linked variance swaps, the dividend impact is not significant due to the small weight of each single stock and the dispersed ex-dividend dates.

24.3 Volatility Index*

A volatility index (e.g., VIX, VHSI, S&P/JPX JGB VIX, CBOE/CME FX Euro Volatility, etc.) targets to be an indicator of the implied volatility for an index.

VIX index is the most popular volatility index, which is related to 30-day options on S&P500 index. CBOE listed VIX index futures in 2004 and options in 2006. VIX index is seen as a barometer of investor sentiment and nicknamed *Wall Street's fear index*. The index tends to increase sharply upon bad news and tends to be low in a bullish market. Due to its likely negative correlation to the stock market in extreme market conditions, VIX futures can be used in an investment portfolio as a diversification factor, to macro hedge stock market downside risk, volatility risk and to serve as a speculation tool. However, the volatility of VIX index itself can sometimes be very high, making it difficult to use VIX as a signal for algorithmic trading strategies.

Technically, VIX index is a 30-day log contract price volatility calculated from a wide range of the listed vanilla options on S&P500 index with maturities of around 30 days. The calculation formula for the index level σ is²

$$\frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2,$$

where

r : interest rate

F : forward index level derived from index option prices

K_0 : first strike below the forward index level F

² Its methodology is explained in <https://cdn.cboe.com/resources/vix/vixwhite.pdf>.

K_i : strike price of the i -th out-of-the-money option; It is for a call if $K_i > K_0$, for a put if $K_i < K_0$, for both put and call if $K_i = K_0$

ΔK_i : $\Delta K_i = \frac{1}{2}(K_{i+1} - K_{i-1})$

$Q(K_i)$: midpoint of the bid-ask spread for each option with strike K_i .

We notice that the calculation formula of σ is actually the discrete approximation of Eq. (24.3):

- the summation term is the discrete approximation of the integral part of (24.3);
- $-\frac{1}{T}(\frac{F}{K_0} - 1)^2$ is the approximation of the non-integral part of (24.3), i.e., $\frac{2}{T} \left[\ln(\frac{F}{K_0}) + \frac{K_0 - F}{K_0} \right]$ with λ replaced by K_0 and using the second order Taylor expansion of $\ln x$ about $x = 1$ that is $(x - 1) - \frac{1}{2}(x - 1)^2$.

We refer to [15] for VIX modelling.

Exercises

1. Assume that the underlier of a variance swap follows the classic Black–Scholes–Merton model

$$dS_t = rS_t dt + \sigma S_t d\tilde{W}_t.$$

Let

$$f(S, \sigma) = \frac{2e^{rT}}{T} \left[\int_{Se^{rT}}^{\infty} \frac{c(S, K, \sigma)}{K^2} dK + \int_0^{Se^{rT}} \frac{p(S, K, \sigma)}{K^2} dK \right]$$

where S is the spot price and σ is the implied volatility of $c(S, K, \sigma)$ and $p(S, K, \sigma)$, the Black–Scholes call and put prices respectively with strike price K and time to maturity T .

- (1) Show that $\frac{\partial c}{\partial \sigma} = \frac{\partial p}{\partial \sigma}$ and then derive $\frac{\partial f}{\partial \sigma}$.
- (2) Show that $\frac{\partial^2 f}{\partial S \partial \sigma} = 0$.

Chapter 25

Local and Stochastic Volatility Models



With the fast development of derivatives and structured products, advanced models are needed to improve the pricing and valuation accuracy as well as the hedging efficiency. As indicated in Sect. 14.5.4, the P/L of a delta-hedged option is not flat but is related to the realized gamma and variance. Gamma-hedging improves the hedging P/L but makes the option's price dependent on the implied volatility that changes during the hedging period. Stochastic volatility models (modelling implied volatility) offer a more consistent pricing and risk management framework compared to simple models such as the BSM that cannot capture the second order and cross derivatives impacts involving volatility.

We refer to [15] for an in-depth analysis on hedging under the local volatility model and stochastic volatility models.

25.1 Local Volatility Model

Compared to the BSM model, the *local volatility model* takes into account the skew and kurtosis of the distribution of the underlier prices by applying $\sigma(t, S_t)$ instead of a constant σ . It prices European options exactly to the market prices. Local volatility is widely used to price equity derivatives. The SDE for the local volatility model is

$$dS_t = (r - q)S_t dt + \sigma(t, S_t)S_t d\tilde{W}_t,$$

under risk-neutral probability measure $\tilde{\mathbb{P}}$, and q is the sum of repo rate and continuous dividend yield.

Let X_t be a process that follows $dX_t = \mu_t dt + \sigma_t dW_t$ where σ_t is a stochastic Itô process. According to Gyöngy [46], there exists a process Y_t , $dY_t = \mu_t dt + \sigma(t, X_t)dW_t$ where $\sigma^2(t, x) = \mathbb{E}[\sigma_t^2 | X_t = x]$, known as the Markovian projection of X_t , such that Y_t and X_t have the same marginal distribution $\forall t > 0$. From this

angle, the local volatility model can be considered as the Markovian projection of all the stochastic volatility models based on Itô processes.

25.1.1 Local Volatility Calculation by Call Option Prices

Proposition 25.1 (Dupire's Formula) *For the local volatility model,*

$$\sigma^2(T, K) = 2 \frac{\partial_T c(T, K) + qc(T, K) + (r - q)K \partial_K c(T, K)}{K^2 \partial_{KK} c(T, K)}.$$

Proof¹ We apply Itô–Doebelin formula² to $e^{-rT}h$ with $h = (S_T - K)^+$, $\frac{\partial h}{\partial S_T} = \mathbf{1}_{\{S_T \geq K\}}$ and $\frac{\partial^2 h}{\partial S_T^2} = \delta(S_T - K)$, where $\delta(\cdot)$ is the Dirac delta function. Then

$$d(e^{-rT}h) = -re^{-rT}hdT + e^{-rT}[\mathbf{1}_{\{S_T \geq K\}}dS_T + \frac{1}{2}\delta(S_T - K)\sigma_T^2 S_T^2 dT].$$

Without loss of generality, we set $t = 0$ to lighten the notation. Taking expectations on both sides at time t , and noting that $c = \mathbb{E}_t[e^{-rT}h]$, $\mathbb{E}_t[dS_T] = (r - q)S_T dT$ (as $\mathbb{E}_t[\sigma_T S_T d\tilde{W}_T] = 0$) and $\mathbb{E}_t[e^{-rT}S_T \mathbf{1}_{\{S_T \geq K\}}] = c + \mathbb{E}_t[e^{-rT}K \mathbf{1}_{\{S_T \geq K\}}]$, we have

$$\begin{aligned} dc &= \left\{ -rc + (r - q)\mathbb{E}_t[e^{-rT}S_T \mathbf{1}_{\{S_T \geq K\}}] + \frac{1}{2}\mathbb{E}_t[e^{-rT}\delta(S_T - K)\sigma_T^2 S_T^2] \right\} dT \\ &= \left\{ -qc + (r - q)\mathbb{E}_t[e^{-rT}K \mathbf{1}_{\{S_T \geq K\}}] + \frac{1}{2}K^2\mathbb{E}_t[e^{-rT}\delta(S_T - K)\sigma_T^2] \right\} dT. \end{aligned}$$

We obtain hence,

$$\mathbb{E}_t[\delta(S_T - K)\sigma_T^2] = 2 \left(\frac{\partial c}{\partial T} + qc - (r - q)K\mathbb{E}_t[e^{-rT}\mathbf{1}_{\{S_T \geq K\}}] \right) / (e^{-rT}K^2).$$

From the representation of conditional expectation with Dirac delta function (12.3), we have

$$\mathbb{E}_t[\sigma_T^2 | S_T = K] = \frac{\mathbb{E}_t[\sigma_T^2 \delta(S_T - K)]}{\mathbb{E}_t[\delta(S_T - K)]} = 2 \frac{\frac{\partial c}{\partial T} + qc + (r - q)K \frac{\partial c}{\partial K}}{K^2 \frac{\partial^2 c}{\partial K^2}},$$

¹ In Edition 1 of this book, Kolmogorov Forward equation was used for the proof following the original work of Dupire. The proof provided here is helpful for a more in-depth understanding of the local volatility and local-stochastic volatility (LSV) models.

² More precisely, it is Tanaka formula (see [60]). Leblanc [68] may have been the first to use it to provide an alternative proof to Dupire equation.

where we have applied Breeden–Litzenberger formulas $\widetilde{\mathbb{E}}_t[e^{-rT} \mathbf{1}_{\{S_T \geq K\}}] = -\frac{\partial c}{\partial K}$ and $\widetilde{\mathbb{E}}_t[\delta(S_T - K)] = \frac{\partial^2 c}{\partial K^2} e^{rT}$.

□

25.1.2 Local Volatility Calculation by Implied Volatilities*

The Dupire’s formula requires numerical calculation of the derivatives of call prices. Actually, Dupire’s formula can be transformed into a function involving the calculation with implied volatilities instead of the calculation with call option prices (see [41, 115, 116]). The calculation with implied volatilities has the advantage of being more stable.

To calibrate the local volatilities surface with implied volatilities, it is convenient to work on an alternative process defined as $\widehat{S}_t = \frac{S_t}{F_t}$ where $F_t = S_0 e^{(r-q)t}$. It is easy to see that the SDE for $\widehat{S}_t$ is

$$d\widehat{S}_t = \hat{\sigma}(t, \widehat{S}_t) \widehat{S}_t d\widetilde{W}_t,$$

where $\hat{\sigma}(t, \widehat{S}_t) = \sigma(t, F_t \widehat{S}_t)$ and σ is the local volatility for the process S_t . Let c denote the call option price with payoff $\max\{0, S_T - K\}$ at maturity T , and $\hat{c}$ denote $\widetilde{\mathbb{E}}_t\left[\frac{1}{F_T} \max\{0, S_T - K\}\right] = \widetilde{\mathbb{E}}_t[\max\{0, \widehat{S}_T - \widehat{K}\}]$ where $\widehat{K} = \frac{K}{F_T}$ is the “moneyness” of the option. The time-0 prices of c and $\hat{c}$ are then related by $c(T, K) = F_T e^{-rT} \hat{c}(T, \widehat{K})$. The Dupire’s formula with $\hat{c}$ is

$$\hat{\sigma}^2(T, \widehat{K}) = \frac{2\partial_T \hat{c}(T, \widehat{K})}{\widehat{K}^2 \partial_{\widehat{K}} \hat{c}(T, \widehat{K})}.$$

In the following, we use t and x to represent T and $\widehat{K}$ respectively for brevity. There exists an implied volatility function $\Sigma(t, x)$ in terms of maturity and moneyness such that $\hat{c}(t, x)$ satisfies the formula

$$\hat{c}(t, x) = N(d_1(t, x)) - xN(d_2(t, x)),$$

where $d_1(t, x) = \frac{-\ln x + \frac{1}{2}\Sigma(t, x)^2 t}{\Sigma(t, x)\sqrt{t}}$ and $d_2(t, x) = d_1(t, x) - \Sigma(t, x)\sqrt{t}$. Note that $\Sigma(t, x)$ equals to the implied volatility parameterized by strike K .

The call price $\hat{c}$ can be regarded as a function of t , x , and $\Sigma(t, x)$. Then, the Dupire’s formula can be revised as $\hat{\sigma}^2(t, x) = 2 \frac{d\hat{c}}{dt} \bigg/ x^2 \frac{d^2 \hat{c}}{dx^2}$, where the operator d stands for total differentiation. The total derivative w.r.t. t is given by

$$\frac{d\hat{c}}{dt} = \frac{\partial \hat{c}}{\partial t} + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial \Sigma}{\partial t}$$

and the second-order total derivative w.r.t. x is

$$\begin{aligned}
 \frac{d^2 \hat{c}}{dx^2} &= \frac{d}{dx} \left(\frac{\partial \hat{c}}{\partial x} + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial \Sigma}{\partial x} \right) = \frac{d}{dx} \left(\frac{\partial \hat{c}}{\partial x} \right) + \frac{d}{dx} \left(\frac{\partial \hat{c}}{\partial \Sigma} \right) \frac{\partial \Sigma}{\partial x} + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial^2 \Sigma}{\partial x^2} \\
 &= \frac{\partial^2 \hat{c}}{\partial x^2} + \frac{\partial^2 \hat{c}}{\partial x \partial \Sigma} \frac{\partial \Sigma}{\partial x} + \frac{\partial^2 \hat{c}}{\partial x \partial \Sigma} \frac{\partial \Sigma}{\partial x} + \frac{\partial^2 \hat{c}}{\partial \Sigma^2} \left(\frac{\partial \Sigma}{\partial x} \right)^2 + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial^2 \Sigma}{\partial x^2} \\
 &= \frac{\partial^2 \hat{c}}{\partial x^2} + 2 \frac{\partial^2 \hat{c}}{\partial x \partial \Sigma} \frac{\partial \Sigma}{\partial x} + \frac{\partial^2 \hat{c}}{\partial \Sigma^2} \left(\frac{\partial \Sigma}{\partial x} \right)^2 + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial^2 \Sigma}{\partial x^2}.
 \end{aligned}$$

The partial derivatives of $\hat{c}$ can be obtained from the formula $\hat{c}(t, x) = N(d_1) - xN(d_2)$, e.g.,

$$\frac{\partial \hat{c}}{\partial t} = N'(d_1) \left(\frac{\Sigma}{2\sqrt{t}} + \frac{\partial d_2}{\partial t} \right) - xN'(d_2) \frac{\partial d_2}{\partial t} = N'(d_1) \frac{\Sigma}{2\sqrt{t}},$$

where $N'(d_1) = xN'(d_2)$ (see Exercise 2 in Chap. 14) is used. Similarly

$$\frac{\partial \hat{c}}{\partial \Sigma} = N'(d_1) \left(\sqrt{t} + \frac{\partial d_2}{\partial \Sigma} \right) - xN'(d_2) \frac{\partial d_2}{\partial \Sigma} = N'(d_1) \sqrt{t},$$

$$\frac{\partial \hat{c}}{\partial x} = N'(d_1) \frac{\partial d_1}{\partial x} - N(d_2) - xN'(d_2) \frac{\partial d_2}{\partial x} = -N(d_2).$$

And the second-order derivatives are

$$\begin{aligned}
 \frac{\partial^2 \hat{c}}{\partial \Sigma^2} &= N''(d_1) \frac{\partial d_1}{\partial \Sigma} \sqrt{t} = -d_1 N'(d_1) \frac{\partial d_1}{\partial \Sigma} \sqrt{t} = \frac{d_1 d_2 N'(d_1) \sqrt{t}}{\Sigma}, \\
 \frac{\partial^2 \hat{c}}{\partial x \partial \Sigma} &= -N'(d_2) \frac{\partial d_2}{\partial \Sigma} = -\frac{N'(d_1)}{x} \left(-\frac{d_1}{\Sigma} \right) = \frac{d_1 N'(d_1)}{x \Sigma}, \\
 \frac{\partial^2 \hat{c}}{\partial x^2} &= -N'(d_2) \frac{\partial d_2}{\partial x} = -\frac{N'(d_1)}{x} \left(-\frac{1}{x \Sigma \sqrt{t}} \right) = \frac{N'(d_1)}{x^2 \Sigma \sqrt{t}}.
 \end{aligned}$$

Replacing the partial derivatives of $\hat{c}$ in $\frac{d\hat{c}}{dt}$ and $\frac{d^2 \hat{c}}{dx^2}$, we have

$$\begin{aligned}
 \frac{d\hat{c}}{dt} &= N'(d_1) \frac{\Sigma}{2\sqrt{t}} + N'(d_1) \sqrt{t} \frac{\partial \Sigma}{\partial t} = \frac{N'(d_1)}{2\Sigma\sqrt{t}} \left(\Sigma^2 + 2\Sigma t \frac{\partial \Sigma}{\partial t} \right), \\
 \frac{d^2 \hat{c}}{dx^2} &= \frac{N'(d_1)}{x^2 \Sigma \sqrt{t}} \left[1 + 2xd_1 \sqrt{t} \frac{\partial \Sigma}{\partial x} + x^2 d_1 d_2 t \left(\frac{\partial \Sigma}{\partial x} \right)^2 + x^2 \Sigma t \frac{\partial^2 \Sigma}{\partial x^2} \right].
 \end{aligned}$$

Hence, the calculation formula using implied volatilities is

$$\hat{\sigma}^2(t, x) = \frac{2 \frac{d\hat{c}}{dt}}{x^2 \frac{d^2 \hat{c}}{dx^2}} = \frac{\Sigma^2 + 2\Sigma t \frac{\partial \Sigma}{\partial t}}{1 + 2x d_1 \sqrt{t} \frac{\partial \Sigma}{\partial x} + x^2 d_1 d_2 t \left(\frac{\partial \Sigma}{\partial x} \right)^2 + x^2 \Sigma t \frac{\partial^2 \Sigma}{\partial x^2}}.$$

25.1.3 Local Volatility Calculation with Discrete Dividends*

The local volatility calculation formula presented above has assumed no discrete dividends (i.e., proportional dividends and/or cash dividends).

Following Sect. 16.7.3, a discrete dividend paid at t_i can be generally represented by $d_i = b_i S_{t_i^-} + a_i$ where b_i is the proportional dividend yield and a_i the cash dividend part. The manufactured non-dividend paying asset process is

$$X_t = [S_t + \alpha(t)]/\beta(t),$$

where $\alpha(t)$ and $\beta(t)$ are defined in Sect. 16.7.3. We recall that X_t represents the discounted total return of an investment into one stock at time-0 and in the local volatility model, it should be a positive $\mathbb{P}$ -martingale of the form:

$$dX_t = X_t \sigma_X(t, X_t) d\tilde{W}_t.$$

The “implied volatilities” for X_t can be determined from the implied volatilities for S_T as $(S_T - K)^+ = (\beta(T)X_T - \alpha(T) - K)^+ = \beta(T)(X_T - K')^+$ where $K' = \frac{K + \alpha(T)}{\beta(T)}$. The local volatilities of the non-dividend paying asset X_t , i.e., $\sigma_X(t, X_t)$, can be calculated with the formula presented previously with the “implied volatilities” for X_t .

Given $\sigma_X(t, X_t)$, the local volatility $\sigma(t, S_t)$ can be derived from $S_t = \beta(t)X_t - \alpha(t)$. Actually, the multipliers for the Brownian motion terms in dS_t and $d(\beta(t)X_t - \alpha(t))$ should be equal. They are respectively $S_t \sigma(t, S_t)$, and $\beta(t)X_t \sigma_X(t, X_t) = [S_t + \alpha(t)] \sigma_X \left(t, \frac{S_t + \alpha(t)}{\beta(t)} \right)$. Hence, the local volatilities for S_t can be determined by the formula

$$\sigma(t, S_t) = \frac{S_t + \alpha(t)}{S_t} \sigma_X \left(t, \frac{S_t + \alpha(t)}{\beta(t)} \right).$$

For further details on dividend modelling in practice, we refer to [15].

25.1.4 Practical Implementation

The local volatility model is an improvement to the BSM model and provides a consistent pricing and risk management framework. However, it is worth noting that the local volatilities are state variables relating the implied BSM volatilities observed in the market and should be recalibrated frequently (often on-the-fly, i.e., automatic recalibrated before their usage). For risk management, the focus is still on the implied volatilities. The vega is not numerically calculated by shifting the implied volatility, but not the local volatilities. Regarding the delta, as the spot shift may impact the implied volatilities (or, equivalently, call options), the delta to be used should extract out this impact from the calculated sticky-strike delta (see Sect. 10.3.6 and [15]). The impact of the implied volatilities change is captured by vega hedging.

The continuum of implied volatilities is needed to calculate the local volatilities. However, only a set of discrete values is quoted in the market. The following procedures may be considered for implementation:

- (1) for each considered maturity, a smooth, twice differentiable function with few parameters (say, a quadratic function) is calibrated to represent the implied volatilities of all strikes. The volatility on both tails can be capped at reasonable levels. The volatility dynamics with spot may be included, e.g., with a term linked to S_{ref}/S_t where S_{ref} is a constant for sticky-strike regime.
- (2) the local volatilities are then calculated according to Dupire's formula with the parametric function obtained from above.

Because of the asymmetric form of volatility smile, the parametric function for implied volatilities may be composed of two segments, a quadratic function calibrated for the left side and another quadratic function calibrated for the right side, with eventually the forward as the pivoting point.

Remark 25.1 Although the local volatility model is the market standard model for equity options, it is not suitable for pricing products with forward skew sensitivities (e.g., forward starting call-spread or cliquet options). Suppose the maturity is T , the option's starting time is t , and the pricing time is 0. The local volatility surface is known to become flatter as time goes by. Therefore, the skew will be progressively lost for the time period $[t, T]$ with increasing t . $\square$

25.2 Stochastic Volatility Models*

Stochastic volatility models take into account the skew/kurtosis of the underlier's distribution. They show better dynamics compared to the local volatility model, which makes them suitable for pricing products with forward skew sensitivities such as cliquet options. Stochastic volatility models provide more accurate delta with the theta mitigating the second order derivatives linked to the volatility, which improves the carry of derivatives position.

25.2.1 Heston Model

We outline here the *Heston model*, which is one of the most popular stochastic volatility models [48]. The SDEs of the Heston model are

$$\begin{cases} dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t, \\ dv_t = \kappa(\theta - v_t)dt + \sigma \sqrt{v_t} dX_t, \end{cases}$$

where W_t and X_t are correlated Brownians with $d\langle W, X \rangle_t = \rho dt$, κ is the mean reversion speed for the variance v_t , θ is the mean reversion level, and σ is the volatility of volatility (often read as vol of vol).

Denote $U(S, v, t)$ as an option value and $\lambda(S, v, t)$, a parameter known as *market price of volatility risk*. $U(S, v, t)$ satisfies the below *Heston PDE*:

$$\begin{aligned} & \frac{\partial U}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 U}{\partial S^2} + \rho \sigma v S \frac{\partial^2 U}{\partial S \partial v} + \frac{1}{2} \sigma^2 v \frac{\partial^2 U}{\partial v^2} + r S \frac{\partial U}{\partial S} - r U \\ &= [-\kappa(\theta - v) + \lambda(S, v, t)] \frac{\partial U}{\partial v}. \end{aligned}$$

All the parameters in the model are calibrated by the market prices of (vanilla) options.

We will show by two usual methods the derivation of the PDE. For lighten the notations, we define an operator

$$A(u) = \frac{\partial u}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 u}{\partial S^2} + \rho \sigma v S \frac{\partial^2 u}{\partial S \partial v} + \frac{1}{2} \sigma^2 v \frac{\partial^2 u}{\partial v^2}.$$

1st PDE Derivation Method

This approach follows the Black–Scholes PDE derivation presented in Sect. 14.2. As the volatility of S_t is now stochastic, we hedge the volatility risk by holding a hedging option linked to the same underlier to offset the vega of option U_t . We arbitrarily chose a hedging option with price U_t^h . Let $\Pi_t = -U_t + \Delta_t S_t + q_t^h U_t^h$ be the self-financing portfolio with Δ_t amount of the underlier and q_t^h amount of the hedging option. Itô–Doebelin formula shows

$$\begin{aligned} d\Pi_t &= -\left(A(U) - q^h A(U^h)\right) dt + \left(-\frac{\partial U}{\partial S} + \Delta + q^h \frac{\partial U^h}{\partial S}\right) dS \\ &\quad + \left(-\frac{\partial U}{\partial v} + q^h \frac{\partial U^h}{\partial v}\right) dv. \end{aligned}$$

The last two terms in the above equation will become zero by setting

$$q^h = \frac{\partial U}{\partial v} / \frac{\partial U^h}{\partial v} \quad \text{and} \quad \Delta = \frac{\partial U}{\partial S} - q^h \frac{\partial U^h}{\partial S}.$$

Under no-arbitrage assumption, we should have

$$d\Pi_t = \left[-A(U) + q^h A(U^h) \right] dt = r\Pi_t dt = (-rU + r\Delta_t + rq^h U^h) dt.$$

implying

$$\left(A(U) + rS \frac{\partial U}{\partial S} - rU \right) / \frac{\partial U}{\partial v} = \left(A(U^h) + rS \frac{\partial U^h}{\partial S} - rU^h \right) / \frac{\partial U^h}{\partial v}.$$

By the arbitrariness of the hedging option, the above equation should only depend on S , v , and t but not the option payoffs. Therefore, we can set the

$$\left(A(U) + rS \frac{\partial U}{\partial S} - rU \right) / \frac{\partial U}{\partial v} = -\kappa(\theta - v) + \lambda(S, v, t),$$

which is the Heston PDE.

2nd PDE Derivation Method

The PDE can be obtained directly by applying the following risk premia to produce a “risk-neutral” measure $\tilde{\mathbb{P}}$ defined by

$$d\tilde{W}_t = dW_t + \frac{\mu - r}{\sqrt{v_t}} dt, \quad d\tilde{X}_t = dX_t + \frac{\lambda}{\sigma \sqrt{v_t}} dt,$$

which implies

$$dS_t = -rS_t dt + \sqrt{v_t} S_t d\tilde{W}_t, \quad dv_t = (\kappa(\theta - v_t) - \lambda) dt + \sigma \sqrt{v_t} d\tilde{X}_t. \quad (25.1)$$

Applying Itô–Doëblin formula and equations (25.1),

$$\begin{aligned} d(e^{-rt} U_t) &= e^{-rt} \left[(A(U) - ru) dt + \frac{\partial U}{\partial S} dS_t + \frac{\partial U}{\partial v} dv_t \right] \\ &= e^{-rt} \left[\left(A(U) - ru + rS_t \frac{\partial U}{\partial S} + (\kappa(\theta - v_t) - \lambda) \frac{\partial U}{\partial v} \right) dt + M_t \right]. \end{aligned}$$

where $M_t = [\cdot] d\tilde{W}_t + [\cdot] d\tilde{X}_t$. As $e^{-rt} U_t$ is a $\tilde{\mathbb{P}}$ -martingale, the term in front of dt must be zero, which leads to the Heston PDE.

The meaning of $\lambda(S, v, t)$ can be shown in the below. Suppose that we only delta-hedge the option U with the underlier: $\Pi_t = U_t - \Delta_t S_t$, then

$$d\Pi_t = A(U)dt + \left(\frac{\partial U}{\partial S} - \Delta \right) dS_t + \frac{\partial U}{\partial v} dv_t.$$

Hence, by setting $\Delta_t = \frac{\partial U}{\partial S}$ and combining with Heston PDE, we get

$$d\Pi_t - r\Pi_t dt = \frac{\partial U}{\partial v} dv_t + [-\kappa(\theta - v) + \lambda] \frac{\partial U}{\partial v} dt = \frac{\partial U}{\partial v} [\lambda(S, v, t)dt + \sigma \sqrt{v_t} dX_t],$$

where we have replaced dv_t under the actual measure in the Heston model. We observe that λ represents the extra return on volatility risk through the term $\frac{\partial U}{\partial v}$.

Closed-Form Solution for European Options Under Heston Model

We outline Heston's method for a closed-form solution of European call option price without going into the details. It is based on the Fourier transform of the transition probability. Scott, L. generalized the method to a jump-diffusion model with stochastic volatility [103].

Denote $\tau = T - t$.

- (1) Transform the pricing problem into two probability calculations

$$\begin{aligned} U_t &= e^{-r\tau} \widetilde{\mathbb{E}}_t^{S,v}[(S_T - K)^+] \\ &= e^{-r\tau} \widetilde{\mathbb{E}}_t^{S,v}[S_T \mathbf{1}_{\{S_T > K\}}] - K e^{-r\tau} \widetilde{\mathbb{E}}_t^{S,v}[\mathbf{1}_{\{S_T > K\}}] \\ &= S\mathbb{Q}\{S_T > K\} - K e^{-r\tau} \widetilde{\mathbb{P}}\{S_T > K\} \\ &\quad \text{(already presented in the BSM model)} \\ &= e^x P_1(x, v, \tau) - K e^{-r\tau} P_2(x, v, \tau) \end{aligned}$$

where $x = \ln S$.

- (2) P_1 and P_2 are the probabilities (conditional expectations) for the option to finish in the money where x_t and v_t follow stochastic processes in $\mathbb{Q}$ and $\mathbb{P}$. As Feynman–Kac Proposition is applicable to conditional expectations that are Markovian, we can obtain the PDEs followed by P_1 and P_2 respectively.
- (3) Determine the characteristic functions of x at T for P_j ($j = 1, 2$):

$$f_j(x, v; \psi) = \widetilde{\mathbb{E}}_j[e^{i\psi x}].$$

As $f_j(x, v; \psi)$ ($j = 1, 2$) is a conditional expectation and Markovian, Feynman–Kac Proposition can be applied to obtain the PDE followed by $f_j(x, v; \psi)$. It can be proved that $f_j(x, v; \psi)$ takes the form of

$e^{C_j(\tau; \psi) + D_j(\tau; \psi)v + i\psi x}$. To obtain $C_j(\tau; \psi)$ and $D_j(\tau; \psi)$, we apply $f_j(x, v; \psi)$ to the PDE, which results in two ODEs (Riccati equations) that can be solved analytically.

- (4) Calculate P_j with the characteristic functions $f_j(x, v; \psi)$ (see Appendix A.9 for the inversion formula),

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left(\frac{e^{-i\psi \ln(K)} f_j(x, v; \psi)}{i\psi} \right) d\psi.$$

Numerical Approaches for Closed-Form Solution

The main task is to evaluate P_1 and P_2 . The integrand is calculated using f_j at each discretized ψ and the integral can be approximated by numerical integration approach. The two integrals in P_1 and P_2 can also be combined to reduce the computational cost.

An alternative approach is to apply the Carr–Madan formula [22], in which the fast Fourier transform is carried out for the numerical integration and a damping factor for the option price is used to handle the singularity near $\psi = 0$.

25.2.2 Forward Variance Model

Forward variance models can offer more flexibility than the early instantaneous variance based models in the literature, such as the Heston model. Several advanced investment banks in the derivatives business began implementing models with stochastic forward variance a few years before 2008 for pricing and hedging structured products and volatility/variance-linked derivatives. In this section, we provide a brief introduction, without providing details, to Bergomi's forward variance models that are presented and analyzed comprehensively in his book *Stochastic Volatility Modeling* [15]. The contents of this section are mostly taken from the book.

Bergomi's Multi-Factor Forward Variance Model

Let ξ_t^T be the time- T instantaneous forward variance observed at t ($t < T$). It is defined as $\xi_t^T = \frac{d}{dT}(\sigma_{t,T}^2(T-t))$, which is the limiting case of the *discrete forward variance*

$$\xi_t^{T',T} = \frac{(T-t)\sigma_{t,T}^2 - (T'-t)\sigma_{t,T'}^2}{T - T'} \quad \text{as } T' \rightarrow T,$$

where $\sigma_{t,u}$ represents the forward variance swap (VS) volatility for maturity u observed at t . The Bergomi's multi-factor forward variance model for maturity T can be described as

$$dS_t = (r - q)S_t dt + \sqrt{\xi_t^t} S_t d\tilde{W}_t^S, \quad 0 < t \leq T, \quad (25.2)$$

$$d\xi_t^u = \omega \alpha \xi_t^u \sum_n w_n e^{-k_n(u-t)} d\tilde{W}_t^n, \quad t \leq u \leq T, \quad (25.3)$$

where $\alpha = \left(\sqrt{\sum_{n,m} w_n w_m \rho_{nm}} \right)^{-1}$ is the normalization factor such that the instantaneous volatility of ξ_t^t is ω , and ρ_{nm} is the correlation of Brownian motions $\tilde{W}_t^n$ and $\tilde{W}_t^m$.

The solution to (25.3) is

$$\xi_t^u = \xi_0^u \exp \left\{ \omega \alpha \sum_n w_n e^{-k_n(u-t)} X_t^n - \frac{1}{2} \omega^2 \alpha^2 \sum_{n,m} w_n w_m e^{-(k_n+k_m)(u-t)} \mathbb{E}[X_t^n X_t^m] \right\},$$

where $dX_t^n = -k_n X_t^n dt + d\tilde{W}_t^n$, $X_0^n = 0$ (OU processes).

The values for ξ_0^T , correlations between $\tilde{W}_t^S$ and $\tilde{W}_t^n$ and all the other parameters of the model are calibrated to the curve of VS fair prices from the market if there are liquid VS quotations, or determined from the term structure and smile of option implied volatilities. The multi-factor model can be used without (25.2) for instruments with payoff linked to volatility or variance only, such as VIX and VS based derivatives. Unlike the local volatility model, the dynamics of ξ_t^u is time-homogeneous, i.e., $d\xi_t^u$ does not depend on t , but depends on $u - t$.

In practice, the Bergomi model with two to three factors is good enough for most of the structured products/exotic options driven by the underlier prices.

In the sections on static hedging and variance swap, we have seen that the volatility in a variance swap is the fair volatility of a log contract (under Brownian motion based modelling), which can be determined by the implied volatilities of European call and put options.

Some banks have implemented similar forward variance models, occasionally combined with leverage functions (in the sense of the LSV model to be introduced next) and jumps, for pricing and valuation. Due to the computation complexity, simpler models, e.g., the local volatility model, are used for daily risk management, i.e., hedging for large structured product books. In such a case, it is necessary to adjust periodically the valuation and the hedging greeks with forward variance models.

Bergomi's Discrete Forward Variance Model

The Bergomi's discrete forward variance model consists of modelling directly $\xi_t^{T_i, T_{i+1}}$ for the time period $[T_i, T_{i+1}]$ of interest for the product. It offers flexibility on the model parameters specific to each time period such as forward skew, which is important for cliquet or forward starting options.

The SDE for each $\xi_t^{T_i, T_{i+1}}$ is of the same form as the ones for the Forward Variance Model. For instance, the two-factor model for each time period is

$$d\xi_t^{T_i, T_{i+1}} = \omega_i \alpha_i \xi_t^{T_i, T_{i+1}} \left[(1 - \theta_i) e^{-k_1(T_i - t)} d\tilde{W}_t^1 + \theta_i e^{-k_2(T_i - t)} d\tilde{W}_t^2 \right].$$

For a multi-factor model, the same parameters k_n, ρ_{nm} should be applied to all the periods while the other parameters can be period-specific. For the consistency with the continuous model, we should have $\xi_t^{T_i, T_{i+1}} = \frac{1}{T_{i+1} - T_i} \int_{T_i}^{T_{i+1}} \xi_t^{T_i, u} du$, which can be achieved by relating ω_i and θ_i to, respectively, ω and θ , their peer parameters in the 2-factor continuous model, with the formulas provided in [15].

For products with payoff linked to the underlier's price performance for the considered time periods, the modelling is directly on $\tilde{S}_t = \frac{S_t}{S_{T_i}}$:

$$d\tilde{S}_t = (r - q)\tilde{S}_t dt + \sigma_i(\tilde{S}_t)\tilde{S}_t d\tilde{W}_t^{\tilde{S}}, \quad \tilde{S}_{T_i} = 1,$$

where $\sigma_i(\tilde{S}_t)$ described in [15] is a function with 3 parameters $(\sigma_{0,i}, \beta_i, N_i)$ to produce the required dynamics, in particular the forward skew, specific to this time period under the condition that the variance for the period is $\xi_{T_i}^{T_i, T_{i+1}}(T_{i+1} - T_i)$. In this model, the correlations between S_t and $\xi_t^{T_i, T_{i+1}}$ only impact the spot-starting volatility skew but not the skew for forward-starting periods.

25.2.3 Local-Stochastic Volatility (LSV) Model

The local volatility can be calibrated to the market implied volatilities exactly for any given time by the Dupire equation. But it does not model well the dynamics and forward skew of the volatilities. While the stochastic volatility models offer a greater flexibility to model the dynamics and forward skew of the volatilities, there is no guarantee that they can be calibrated to the market spot implied volatilities exactly.

The local-stochastic volatility model (LSV) combines the local volatility model with a stochastic volatility model. It can be calibrated to the market implied volatilities exactly, just like the local volatility model. It is used in practice, in

particular, for FX and equity exotic options and structured products. The risk-neutral underlier price process under the LSV model follows

$$dS_t = (r - q)S_t dt + l(t, S_t)\sqrt{\zeta_t}S_t d\tilde{W}_t^S,$$

where ζ_t follows a stochastic process (e.g., the two-factor Bergomi model, the Heston model, etc.), $l(t, S_t)$ is termed as a *leverage function*. The LSV model's volatility of S_t is actually $\sigma_t = l(t, S_t)\sqrt{\zeta_t}$.

In the LSV model, the volatility σ_t is exactly calibrated to the market implied local volatilities $\sigma(t, S_t)_{\text{mkt}}$ through the Dupire equation:

$$\begin{aligned}\tilde{\mathbb{E}}[\sigma_t^2 | S_t = K] &= \tilde{\mathbb{E}}[l^2(t, S_t)\zeta_t | S_t = K] \\ &= 2 \frac{\frac{\partial c}{\partial T} + qc + (r - q)K \frac{\partial c}{\partial K}}{K^2 \frac{\partial^2 c}{\partial K^2}} \bigg|_{T=t} = \sigma^2(t, K)_{\text{mkt}}.\end{aligned}$$

Hence,

$$l^2(t, K) = \frac{\sigma^2(t, K)_{\text{mkt}}}{\tilde{\mathbb{E}}[\zeta_t | S_t = K]} = \frac{\sigma^2(t, K)_{\text{mkt}}}{\int \zeta f(K, \zeta, t) d\zeta / \int f(K, \zeta, t) d\zeta},$$

where $f(S, \zeta, t)$ is the density function of the joint distribution of S_t and ζ_t (or its state variables).

For model calibration, PDE method can be used for one-factor models on ζ_t in which the forward Kolmogorov equation is applied to determine the density $f(S, \zeta, t)$ used in calculating $\tilde{\mathbb{E}}[\zeta_t | S_t = K]$ (see [15]). When a multi-factor model is used for ζ_t , a special Monte Carlo technique known as *particle method* in physics, which uses the empirical density obtained in the Monte Carlo simulation (see [45], [30]), would be efficient.

The LSV model offers rich degrees of freedom through the model parameters. It is natural to determine the stochastic model parameters first and then, calibrate the other parameters to market implied volatilities.

Exercises

1. Is the local volatility model suitable for pricing a forward starting call spread?
2. Show that, in the proof of the Dupire formula

$$\tilde{\mathbb{E}}_t[e^{-rT} \delta(S_T - K)] = \frac{\partial^2 c}{\partial K^2}.$$

3. Consider the CIR process in the Heston model,

$$dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dX_t.$$

Show that the forward variance, for $t < T$, in the Heston model is modelled by

$$d\xi_t^T = \sigma e^{-\kappa(T-t)}\sqrt{\xi_t^t}dX_t,$$

with no drift, i.e., the Heston model is a one-factor model for forward variances.

Chapter 26

Jump-Diffusion Models*



Market prices of financial assets often show jumps caused by unpredictable events or news. The market closing-opening is also a source of price jumps. The pure Brownian motion based diffusion models do not admit large asset price moves in a short period of time. Adding jumps to diffusion can show skewed distributions with fat tail that are difficult to produce by the BSM model.

In this chapter, we give an introduction to jump-diffusion processes and jump-diffusion models for European option pricing. For American option pricing in a jump-diffusion model, we refer to [118]. We recommend [105] for more detailed mathematical analysis on jump-diffusion processes.

26.1 Compound Poisson Process

The jump size for a Poisson process N_t is always 1. A *Compound Poisson process* G_t extends the Poisson process by taking *i.i.d.* random variables for the jump sizes:

$$G_t = \sum_{i=1}^{N_t} v_i, \quad t \geq 0,$$

where v_i represents the random variable for the i -th jump with $\mathbb{E}[v_i] = \eta, \forall i$. Let λ denote the intensity for the Poisson process. Here are some properties of compound Poisson process:

- (1) Mean: $\mathbb{E}[G_t] = \eta\lambda t$
- (2) Martingale property: $G_t - \eta\lambda t = G_t - \mathbb{E}[G_t]$ is a martingale
- (3) Independent and stationary increments: $G_t - G_s \stackrel{(d)}{\Longleftrightarrow} G_{t-s}$.

26.2 Simulating a Poisson Process and Compound Poisson Process

The key to simulating a Poisson process N_t lies in the arrival times T_i , where their differences are i.i.d. exponentially distributed random variables. Consider generating N_T , $T > 0$ starting from $N_0 = 0$. The procedure is as follows (assuming $T_0 = 0$):

- (1) generate a uniform variable $u_i \in \mathcal{U}(0, 1)$
- (2) use u_i to generate an exponential variable $\tau_i = -\frac{1}{\lambda} \ln u_i$ and calculate an arrival time by $T_i = T_{i-1} + \tau_i$ (if $T_i > T$ then stop the procedure)
- (3) let $N_t = i$ for $[T_i, T_{i+1})$.

Here the inverse sampling is performed and two facts are used: firstly u_i and $1 - u_i$ follow the same distribution, and the inverse c.d.f. of exponential distribution is explicit.

Simulating a compound Poisson process G_t is similarly done once the distribution of jump sizes v_i is specified.

26.3 Stochastic Calculus for Jump-Diffusion Processes

Let $\mathcal{F}_t$ be the natural filtration generated by W_t and G_t . Next, we introduce the stochastic calculus for jump processes. A *pure jump process*, denoted by J_t , has finite jumps in any time interval and is constant between jumps. Also, it is assumed that no jump takes place at time $t = 0$ and $J_0 = 0$ is the initial value. Poisson process N_t and compound Poisson process G_t are pure jump processes. For pure jump processes, we have

$$dJ_t = \Delta J_t = J_t - J_{t-} = (J_t - J_{t-})dN_t.$$

We define a *jump-diffusion process* as a right continuous process as follows:

$$X_t = X_0 + \int_0^t \theta_u du + \int_0^t \phi_u dW_u + J_t,$$

where θ_u and ϕ_u are adapted processes. Note that $X_t^c \equiv X_0 + \int_0^t \theta_u du + \int_0^t \phi_u dW_u$ is the continuous Itô process in the earlier context. The addition of J_t introduces a finite number of jumps to X_t^c in finite time. Since $\Delta X_t = \Delta J_t$, the jump-diffusion process can also be written as

$$X_t = X_t^c + J_t = X_t^c + \sum_{0 < u \leq t} \Delta J_u = X_t^c + \sum_{0 < u \leq t} \Delta X_u.$$

In differential form, we have $dX_t = dX_t^c + dJ_t$.

Previously, it is shown that a stochastic integration with continuous martingale integrator and predictable integrand gives a martingale. This property can be extended when the integrator is a martingale with jumps. See [94] for more details. The calculus of jump-diffusion processes is given below. The proofs can be found in [105].

Proposition 26.1 *The cross variation for two jump-diffusion processes $X_t = X_t^c + J_t^X$ and $Y_t = Y_t^c + J_t^Y$ is given by*

$$\langle X, Y \rangle_t = \langle X^c, Y^c \rangle_t + \langle J^X, J^Y \rangle_t = \int_0^t \phi_u^X \phi_u^Y du + \sum_{0 < u \leq t} \Delta J_u^X \Delta J_u^Y,$$

or in differential form, $d\langle X, Y \rangle_t = d\langle X^c, Y^c \rangle_t + d\langle J^X, J^Y \rangle_t$. In particular, the cross variation of any continuous process X_t^c and pure jump process J_t is zero, i.e.,

$$\langle X^c, J \rangle_t = 0. \quad (26.1)$$

Theorem 26.1 (One-Dimensional Itô–Doebelin Formula for Jump-Diffusion Process) *Let $f(u, x)$ be a sufficiently smooth function of real variables u and x . Suppose X_u is a jump-diffusion process. Then for $u < t$,*

$$\begin{aligned} f(t, X_t) &= f(0, X_0) + \int_0^t f_u(u, X_u) du + \int_0^t f_x(u, X_u) dX_u^c \\ &\quad + \frac{1}{2} \int_0^t f_{xx}(u, X_u) d\langle X^c, X^c \rangle_u + \sum_{0 < u \leq t} [f(u, X_u) - f(u, X_{u-})], \end{aligned}$$

where $f_u(u, X_u) = \frac{\partial f}{\partial u}(u, X_u)$, $f_x(u, X_u) = \frac{\partial f}{\partial x}(u, X_u)$ and $f_{xx}(u, X_u) = \frac{\partial^2 f}{\partial x^2}(u, X_u)$.

The difference from the ordinary Itô's formula lies in the last term.

Theorem 26.2 (Two-Dimensional Itô–Doebelin Formula for Jump-Diffusion Process) *Let X_u and Y_u be jump-diffusion processes and $f(u, x, y)$ whose derivatives appearing in the below are defined and continuous. Then*

$$\begin{aligned} f(t, X_t, Y_t) &= f(0, X_0, Y_0) + \int_0^t f_u(u, X_u, Y_u) du + \int_0^t f_x(u, X_u, Y_u) dX_u^c \\ &\quad + \int_0^t f_y(u, X_u, Y_u) dY_u^c + \frac{1}{2} \int_0^t f_{xx}(u, X_u, Y_u) d\langle X^c, X^c \rangle_u \\ &\quad + \int_0^t f_{xy}(u, X_u, Y_u) d\langle X^c, Y^c \rangle_u + \frac{1}{2} \int_0^t f_{yy}(u, X_u, Y_u) d\langle Y^c, Y^c \rangle_u \\ &\quad + \sum_{0 < u \leq t} (f(u, X_u, Y_u) - f(u, X_{u-}, Y_{u-})). \end{aligned}$$

The two dimensional Itô's formula leads to the following Itô's product rule.

Corollary 26.1 (Itô's Product Rule for Jump Processes) *Let X_t and Y_t be two jump processes. Then,*

$$X_t Y_t = X_0 Y_0 + \int_0^t Y_u dX_u^c + \int_0^t X_u dY_u^c + \langle X^c, Y^c \rangle_t + \sum_{0 < u \leq t} (X_u Y_u - X_{u-} Y_{u-}),$$

which can be equivalently written as integration by parts:

$$X_t Y_t = X_0 Y_0 + \int_0^t Y_{u-} dX_u + \int_0^t X_{u-} dY_u + \langle X, Y \rangle_t.$$

26.4 Jump-Diffusion Model for Option Pricing

The jump-diffusion model assumes that, under the original probability measure $\mathbb{P}$, the SDE for the asset is

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_{t-} d(G_t - \eta \lambda t).$$

The diffusion part is the same as the Black–Scholes model. The additional jump part is given by the *compensated* compound Poisson process $G_t - \eta \lambda t$, which is a $\mathbb{P}$ -martingale. The jumps v_i 's are *i.i.d.* random variables and we assume that they have continuous distribution with density $f(y)$ satisfying $v_i \geq -1$ (as S_t becomes 0 for $v_i = -1$). For example, in the Merton's model [80], $v_i = X_i - 1$ where X_i follows log-normal distribution. The term η is the expected jump size with $\eta = \mathbb{E}[v_i] = \int_{-1}^{\infty} y f(y) dy$.

Define a measure $\tilde{\mathbb{P}}$ with a Radon–Nikodym derivative Z_t :

$$\left. \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = Z_t = e^{-\theta W_t - \frac{1}{2} \theta^2 t}, \text{ with } \theta = \frac{\mu - r}{\sigma}.$$

Then, under $\tilde{\mathbb{P}}$, $\tilde{W}_t = W_t + \theta t$ is a Brownian and the discounted asset price is a martingale. The measure $\tilde{\mathbb{P}}$ implies

$$dS_t = r S_t dt + \sigma S_t d\tilde{W}_t + S_{t-} d(G_t - \eta \lambda t). \quad (26.2)$$

Note that $\tilde{\mathbb{P}}$ is not a unique risk-neutral measure. The case presented here states that the real-world and risk-neutral distribution for the jumps are identical, thus keeping the intensity and expected jump size, λ and η unchanged.

Proposition 26.2 *The solution to (26.2) under $\tilde{\mathbb{P}}$ is*

$$S_t = S_0 e^{\sigma \tilde{W}_t + (r - \eta\lambda - \frac{1}{2}\sigma^2)t} \prod_{i=1}^{N_t} (v_i + 1).$$

Proof Consider the pure jump process $Z_t = \prod_{i=1}^{N_t} (v_i + 1) = \prod_{0 \leq s \leq t} (1 + \Delta G_s)$ that satisfies $dZ_t = Z_{t-} dG_t$. Define a continuous process $S_t^c = S_0 e^{\sigma \tilde{W}_t + (r - \eta\lambda - \frac{1}{2}\sigma^2)t}$, which is a geometric Brownian motion solving

$$dS_t^c = (r - \eta\lambda) S_t^c dt + \sigma S_t^c d\tilde{W}_t.$$

From (26.1), $\langle S^c, Z \rangle_t = 0$. Applying Itô's product rule to $S_t = S_t^c Z_t$,

$$\begin{aligned} S_t &= S_t^c Z_t = S_0 + \int_0^t S_u^c dZ_u + \int_0^t Z_{u-} dS_u^c \\ &= S_0 + \int_0^t S_u^c Z_{u-} dG_u + (r - \eta\lambda) \int_0^t Z_{u-} S_u^c du + \sigma \int_0^t Z_{u-} S_u^c d\tilde{W}_u \\ &= S_0 + r \int_0^t S_u du + \sigma \int_0^t S_u d\tilde{W}_u + \int_0^t S_{u-} d(G_u - \eta\lambda u). \end{aligned}$$

The differential form of the above is the SDE (26.2). □

The solution S_t has the jump size of $\Delta S_t = S_t - S_{t-} = S_{t-} \Delta G_t$, which leads to $(S_t - S_{t-}) dN_t = S_{t-} dG_t$. This will be used later for discussion of hedging.

26.5 European Call or Put Option Pricing

Proposition 26.3 *Consider a European call or put option with strike price K and maturity T . If the underlier price follows the jump-diffusion model (26.2), the option price will be*

$$V_t = \tilde{\mathbb{E}}[e^{-r(T-t)} h(S_T) | \mathcal{F}_t], \quad h(S_T) = (S_T - K)^+ \text{ or } (K - S_T)^+,$$

is given by

$$V_t = \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[BS(\tau, \bar{S}(v_i, n))],$$

where the time to maturity $T - t$ is denoted as τ ,

$$\bar{S}(v_i, n) = S_t e^{-\eta\lambda\tau} \prod_{i=1}^n (v_i + 1),$$

and $BS(\tau, \bar{S})$ is the BSM call or put option formula with τ as tenor and $\bar{S}$ as spot.

Proof

$$\begin{aligned} V_t &= \tilde{\mathbb{E}}[e^{-r\tau} h(S_T) | \mathcal{F}_t] \\ &= \tilde{\mathbb{E}} \left[e^{-r\tau} h \left(S_t e^{\sigma \tilde{W}_\tau + (r - \eta\lambda - \frac{1}{2}\sigma^2)\tau} \prod_{i=1}^{N_\tau} (v_i + 1) \right) \right], \end{aligned}$$

where the properties of independent and stationary increments are used. Using the law of total expectation with the probability of n jumps,

$$\begin{aligned} V_t &= \tilde{\mathbb{E}} \left[\tilde{\mathbb{E}} \left[e^{-r\tau} h \left(S_t e^{\sigma \tilde{W}_\tau + (r - \eta\lambda - \frac{1}{2}\sigma^2)\tau} \prod_{i=1}^{N_\tau} (v_i + 1) \right) \middle| N_\tau = n \right] \right] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}} \left[e^{-r\tau} h \left(S_t e^{\sigma \tilde{W}_\tau + (r - \eta\lambda - \frac{1}{2}\sigma^2)\tau} \prod_{i=1}^n (v_i + 1) \right) \right] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[e^{-r\tau} h(\bar{S}(v_i, n)) e^{\sigma \tilde{W}_\tau + (r - \frac{1}{2}\sigma^2)\tau}] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[\tilde{\mathbb{E}}[e^{-r\tau} h(\bar{S}(v_i, n)) e^{\sigma \tilde{W}_\tau + (r - \frac{1}{2}\sigma^2)\tau} | \mathcal{F}_t]] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[BS(\tau, \bar{S}(v_i, n))]. \end{aligned}$$

□

One can see that the pricing formula is a summation of Black–Scholes prices multiplied by the corresponding probability of jumps, or in other words, an expectation over all jump occurrences. The pricing formula can be further simplified as a closed-form solution under the Merton's lognormal jump-diffusion model; see [80] for reference.

26.6 PIDE for European Style Options*

26.6.1 Derivation of PIDE

Consider a T -maturity European option price $V(t, S_t)$ with underlier S_t at time t . We have the following proposition for the governing equation of V .

Proposition 26.4 *Assume the underlier follows the jump-diffusion model (26.2). The price $V(t, S_t)$ for a European style option paying out $V(T, S_T)$ at maturity T satisfies the PIDE (partial integro-differential equation):*

$$\frac{\partial V}{\partial t} + (r - \eta\lambda)S_t \frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} = rV - \lambda\{\tilde{\mathbb{E}}[V(t, (\nu + 1)S_{t-})] - V(t, S_{t-})\},$$

where¹ $\tilde{\mathbb{E}}[V(t, (\nu + 1)S_{t-})] = \int_{-1}^{\infty} V(t, (z + 1)S_{t-})\tilde{f}(z)dz$ if jump density $\tilde{f}$ is known.

Proof Denote the process $A_t = -rV + \frac{\partial V}{\partial t} + (r - \eta\lambda)S_t \frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2}$.

Applying Itô's formula and product rule, we have

$$e^{-rt}V_t = V_0 + \int_0^t e^{-ru}A_u du + \int_0^t e^{-ru}\sigma S_u \frac{\partial V}{\partial S_u} d\tilde{W}_u + \sum_{0 < u \leq t} e^{-ru}[V(u, S_u) - V(u, S_{u-})].$$

If the jump happens at time u , then $S_u = (\nu + 1)S_{u-}$. Therefore, the last term is

$$\sum_{0 < u \leq t} e^{-ru}[V(u, S_u) - V(u, S_{u-})] = \int_0^t e^{-ru}[V(u, (\nu + 1)S_{u-}) - V(u, S_{u-})]dN_u.$$

Splitting the integrator into $dN_u = d(N_u - \lambda u) + \lambda du$,

$$\begin{aligned} e^{-rt}V_t &= V_0 + \int_0^t e^{-ru}A_u du + \int_0^t e^{-ru}\sigma S_u \frac{\partial V}{\partial S_u} d\tilde{W}_u \\ &\quad + \int_0^t e^{-ru}[V(u, (\nu + 1)S_{u-}) - V(u, S_{u-})]d(N_u - \lambda u) \\ &\quad + \int_0^t e^{-ru}[V(u, (\nu + 1)S_{u-}) - V(u, S_{u-})]\lambda du. \end{aligned}$$

¹ The term $\tilde{\mathbb{E}}[V(t, (\nu + 1)S_{t-})]$ is called *convolution* term and $rV - \lambda\{\tilde{\mathbb{E}}[V(t, (\nu + 1)S_{t-})] - V(t, S_{t-})\}$ is called *source* term.

Taking expectation, terms with $d\tilde{W}_u$ and $d(N_u - \lambda u)$ become 0. Then,

$$\tilde{\mathbb{E}}[e^{-rt} V_t | \mathcal{F}_0] = V_0 + \int_0^t e^{-ru} A_u du + \int_0^t e^{-ru} \{\tilde{\mathbb{E}}[V(u, (v+1)S_{u-})] - V(u, S_{u-})\} \lambda du.$$

Because $e^{-rt} V_t$ should be a martingale under $\tilde{\mathbb{P}}$, the du term should be 0, which proves the result. $\square$

Applying the PIDE, we obtain

$$\begin{aligned} e^{-rt} V_t = V_0 - \int_0^t e^{-ru} \lambda \{\tilde{\mathbb{E}}[V(u, (v+1)S_{u-})] - V(u, S_{u-})\} du \\ + \int_0^t e^{-ru} \sigma S_u \frac{\partial V}{\partial S_u} d\tilde{W}_u + \int_0^t e^{-ru} [V(u, S_u) - V(u, S_{u-})] dN_u, \end{aligned} \quad (26.3)$$

which will be used later in the context of hedging.

26.6.2 Finite Difference Method for PIDE

Finite difference method can be used for the resolution of the PIDE (partial integro-differential equation) in which there is the present of integral terms. The computation of the integral needs to be performed with an efficient numerical integration algorithm based on the options values that are found on the finite difference grid. The easiest method (lagging method) consists of using the known option values from the previous time step to compute the integral. This simplicity is at the expense of lowered accuracy, from $O(\Delta t^2)$ to $O(\Delta t)$ if Crank-Nicolson is used as finite difference method. To improve the accuracy, an iterative method similar to Jacobi algorithm 23.3.2 may be applied to each time step, which starts with the lagged option values at the previous time step and uses the newly computed option values at each iteration until the convergence level is achieved. Interested readers can refer to [111] for detailed analysis.

26.7 Discussion on Hedging Under Jump-Diffusion Model*

We conduct the analysis on a short call position. The delta hedging portfolio for the call $c(t, S_t)$ is Π_t that satisfies

$$d\Pi_t = r\Pi_t dt + \Delta_t(dS_t - rS_t dt),$$

with $\Pi_0 = c(0, S_0)$. Then,

$$\begin{aligned} d(e^{-rt} \Pi_t) &= e^{-rt} (-r \Pi_t dt + d\Pi_t) = e^{-rt} (\Delta_t dS_t - r \Delta_t S_t dt) \\ &= e^{-rt} [\Delta_t \sigma S_t d\tilde{W}_t + \Delta_t S_{t-} d(G_t - \eta \lambda t)] \\ &= e^{-rt} [\Delta_t \sigma S_t d\tilde{W}_t + \Delta_t (S_t - S_{t-}) dN_t - \Delta_t S_{t-} \eta \lambda dt], \end{aligned}$$

where we use $dG_t = \frac{S_t - S_{t-}}{S_{t-}} dN_t$. From (26.3), we have

$$\begin{aligned} d(e^{-rt} c(t, S_t)) &= e^{-rt} \left\{ -\lambda \left[\int_{-1}^{\infty} [c(t, (z+1)S_{t-}) - c(t, S_{t-})] \tilde{f}(z) dz \right] dt \right. \\ &\quad \left. + \sigma S_t \frac{\partial c}{\partial S_t}(t, S_{t-}) d\tilde{W}_t + [c(t, S_t) - c(t, S_{t-})] dN_t \right\}. \end{aligned}$$

Applying $\Delta_t = \frac{\partial c}{\partial S_t}(t, S_{t-})$ and $\eta = \int_{-1}^{\infty} z \tilde{f}(z) dz$, the hedging result for time interval dt is

$$\begin{aligned} &d(e^{-rt} \Pi_t - e^{-rt} c(t, S_t)) \\ &= e^{-rt} \left\{ \lambda \left[\int_{-1}^{\infty} [c(t, (z+1)S_{t-}) - c(t, S_{t-}) - z S_{t-} \frac{\partial c}{\partial S_t}(t, S_{t-})] \tilde{f}(z) dz \right] dt \right. \\ &\quad \left. - [c(t, S_t) - c(t, S_{t-}) - \frac{\partial c}{\partial S_t}(t, S_{t-})(S_t - S_{t-})] dN_t \right\}. \end{aligned}$$

As the function $c(t, x)$ is strictly convex in x , we have

$$c(t, x_2) - c(t, x_1) > (x_2 - x_1) \frac{\partial c}{\partial x}(t, x_1), \quad \text{for all } x_2 > x_1 \geq 0.$$

Therefore,

- between jumps, the hedging portfolio outperforms the option
(in this case, dN_t is 0 and $c(t, (z+1)S_{t-}) - c(t, S_{t-}) > z S_{t-} \frac{\partial c}{\partial S_t}(t, S_{t-})$ so that $d(e^{-rt} \Pi_t - e^{-rt} c(t, S_t)) > 0$)
- at jump times, the option outperforms the hedging portfolio
(because $c(t, S_t) - c(t, S_{t-}) > \frac{\partial c}{\partial S_t}(t, S_{t-})(S_t - S_{t-})$)
- the expected difference is zero.

Exercises

1. Let X_t and Y_t be two jump-diffusion processes. Show that

$$\begin{aligned} \int_0^t Y_u dX_u + \int_0^t X_u dY_u + \langle X, Y \rangle_t &= \int_0^t Y_u dX_u^c + \int_0^t X_u dY_u^c + \langle X^c, Y^c \rangle_t \\ &\quad + \sum_{0 < u \leq t} (X_u Y_u - X_{u-} Y_{u-}). \end{aligned}$$

2. Consider a stock modelled by the Merton's default model [81], which is a special jump diffusion model with one jump to zero value, i.e., $\eta = -1$ in (26.2) and also $\nu = -1$.

- (1) Derive the governing PIDE for a European call option with strike K and maturity T on this stock.
- (2) Obtain its time- t price.

3. Consider a stock modelled by the Merton's jump-diffusion model [80] where $v_i = X_i - 1$ and X_i follows i.i.d. log-normal distribution, i.e., $\log(X_i) \sim \mathcal{N}(\mu_J, \sigma_J^2)$.

- (1) Show that $1 + \eta = e^{\mu_J + \frac{1}{2}\sigma_J^2}$.
- (2) Show that the European call/put option price is given as

$$\sum_{n=0}^{\infty} e^{-\bar{\lambda}\tau} \frac{(\bar{\lambda}\tau)^n}{n!} BS(\tau, S_t, \sigma_n, r_n)$$

where $\bar{\lambda} = \lambda(1 + \eta)$, BS is the classic Black–Scholes call/put prices with proxy volatility $\sigma_n = \sqrt{\sigma^2 + \frac{n\sigma_J^2}{\tau}}$ and proxy risk-free rate $r_n = r - \lambda\eta + n \frac{\log(1+\eta)}{\tau}$.

Chapter 27

Interest Rate Term Structure Modelling*



In Part II Chap. 17, we have presented the pricing for vanilla interest rate products that do not require the modelling of interest rate term structure. However, structures with features such as path-dependency or callability usually require the modelling of term structure for pricing and risk management. In this chapter, we first present the concepts and the relationships between zero coupon bond, short rate and forward rate that are essential for interest rate term structure modelling. Then we introduce the Heath–Jarrow–Morton framework with the no-arbitrage condition, which will give birth to different short rate models. Lastly, we study the BGM model that deals with the forward rates directly.

As the interest rate term structure modelling is fairly complicated, we limit the presentation to the basic concepts and important results. For a comprehensive study and practical implementation, we refer to [4–6].

27.1 Continuous-Time Modelling of Interest Rate

There are different models to describe the dynamics of the term structure of interest rates. The definition of interest rates used in a model (known as “state-variables”) is often different from the interest rates implied from the contracts of financial instruments. Actually, the interest rates related measures can be constructed from the state variables. The principal component analysis (PCA) on historical data of interest rates confirmed that the dynamics of the term structure are mainly explained by 3 factors: the first factor for parallel shift, the second for yield curve pivoting and the third for curvature change. To describe the dynamics of the term structure, models with more than 2 factors would give satisfaction if efficient computation algorithms can be implemented.

27.1.1 Zero Coupon Bond

In continuous-time modelling, the zero coupon bonds (also known as discount bond), denoted by $P_t(T)$, are considered as the base tradable assets. A zero coupon bond $P_t(T)$ represents the time- t value of an instrument paying out \$1 value at maturity T . The probability $\tilde{\mathbb{P}}$ under which the discounted zero coupon bond prices are martingales is known as risk-neutral probability measure.

The zero coupon bonds for different maturities can be extracted from market traded instruments as explained in Sect. 4.4. Conversely, any interest rate instrument in the market can be represented by zero coupon bonds. To price interest rate derivatives, we usually need zero coupon bonds and stochastic discount factors (e.g., $e^{-\int_0^t r_u du}$) under $\tilde{\mathbb{P}}$ with term structure models.

Example 27.1 At time t , the 3M SOFR forward rate for the period $(T, T + 3M)$ can be determined by

$$\left[\frac{P_t(T)}{P_t(T + 3M)} - 1 \right] / \tau ,$$

where the day count fraction τ for the period $(T, T + 3M)$ is calculated with the relevant day count convention. $\square$

27.1.2 Short Rate

The short rate r_t is the stochastic instantaneous interest rate previously used to define the money market account numéraire

$$M_t = e^{\int_0^t r_u du},$$

and M_t represents the accrued value at time t for \$1 at time 0.

From the risk-neutral pricing formula $P_t(T) = M_t \tilde{\mathbb{E}}[\frac{P_T(T)}{M_T} | \mathcal{F}_t]$ and $P_T(T) = 1$, we obtain the relationship between the market price of zero coupon bond $P_t(T)$ and the short rate r_t :

$$P_t(T) = \tilde{\mathbb{E}}[e^{-\int_t^T r_u du} | \mathcal{F}_t]. \quad (27.1)$$

27.1.3 Forward Rate

The *time- t discrete forward rate* for the time interval $(T, T + \Delta T)$ is defined by the following $\mathcal{F}_t$ -measurable stochastic process $f_t(T, T + \Delta T)$:

$$f_t(T, T + \Delta T) = -\frac{\ln P_t(T + \Delta T) - \ln P_t(T)}{\Delta T},$$

which satisfies

$$\exp(-f_t(T, T + \Delta T)\Delta T) = \frac{P_t(T + \Delta T)}{P_t(T)},$$

i.e., the interest rate at time t to borrow between T and $T + \Delta T$. Then, the *forward rate* is defined as

$$f_t(T) = \lim_{\Delta T \rightarrow 0} -\frac{\ln P_t(T + \Delta T) - \ln P_t(T)}{\Delta T} = -\frac{\partial \ln P_t(T)}{\partial T}.$$

Relationship between the market price of zero coupon bond $P_t(T)$ and the forward rate is

$$P_t(T) = e^{-\int_t^T f_t(u)du}.$$

Note that when $T \downarrow t$, the forward rate is reduced as the short rate $r_t = f_t(t)$.

27.2 Heath–Jarrow–Morton Framework

The Heath–Jarrow–Morton (HJM) framework analyzes the properties of no-arbitrage interest rate term structure models. It models the dynamics of forward rate as

$$f_t(T) = f_0(T) + \int_0^t \alpha_u(T)du + \int_0^t \sigma_u(T) \cdot d\mathbf{W}_u,$$

where $\sigma_t(T)$ and the Brownian motion $\mathbf{W}_t$ are n -dimensional processes and an inner product is carried out for the last term. Specifications of $\alpha_t(T)$ and $\sigma_t(T)$ lead to different modelling choices for $f_t(T)$. An assumption of no-arbitrage will relate $\alpha_t(T)$ and $\sigma_t(T)$ and give the HJM no-arbitrage model for $f_t(T)$.

Proposition 27.1 (HJM Non-Arbitrage Model)

$$\begin{cases} f_t(T) = f_0(T) + \int_0^t \tilde{\alpha}_u(T)du + \int_0^t \sigma_u(T) \cdot d\tilde{\mathbf{W}}_u, \\ \tilde{\alpha}_t(T) = \sigma_t(T) \cdot \int_t^T \sigma_t(u)du \quad (\text{known as “HJM no-arbitrage condition”}), \end{cases}$$

where $\tilde{\mathbf{W}}_t$ is an n -dimensional vector of Brownians under $\tilde{\mathbb{P}}$, the money market account risk-neutral probability measure.

Proof Under interest rate modelling, zero coupon bonds are tradable assets with strictly positive values. Hence, under $\tilde{\mathbb{P}}$, any zero coupon bond deflated by the numéraire $M_t = e^{\int_0^t r_u du}$ is a martingale, which implies

$$\frac{dP_t(T)}{P_t(T)} = r_t dt - \sigma_t^P(T) \cdot d\tilde{\mathbf{W}}_t.$$

Here, the minus sign before σ^P is for notational convenience because it can be absorbed into the Brownian motion. Then by Itô's lemma,

$$d \ln P_t(T) = \left[r_t - \frac{1}{2} \sigma_t^P(T) \cdot \sigma_t^P(T) \right] dt - \sigma_t^P(T) \cdot d\tilde{\mathbf{W}}_t$$

or

$$\ln P_t(T) = \ln P_0(T) + \int_0^t \left[r_u - \frac{1}{2} \sigma_u^P(T) \cdot \sigma_u^P(T) \right] du - \int_0^t \sigma_u^P(T) \cdot d\tilde{\mathbf{W}}_u.$$

From $f_t(T) = -\frac{\partial \ln P_t(T)}{\partial T}$, we obtain

$$df_t(T) = \frac{\partial \sigma_t^P(T)}{\partial T} \cdot \sigma_t^P(T) dt + \frac{\partial \sigma_t^P(T)}{\partial T} \cdot d\tilde{\mathbf{W}}_t.$$

Defining $\sigma_t(T) = \frac{\partial \sigma_t^P(T)}{\partial T}$, we have

$$df_t(T) = \left(\sigma_t(T) \cdot \int_t^T \sigma_t(u) du \right) dt + \sigma_t(T) \cdot d\tilde{\mathbf{W}}_t$$

since $\int_t^T \sigma_t(u) du = \sigma_t^P(u)|_t^T$ and $\sigma_t^P(t) = 0$. □

We note that $f_0(T)$ can be obtained from the market quoted zero coupon prices $P_0(T)$ with $f_0(T) = -\frac{\partial \ln P_0(T)}{\partial T}$. It means that at time-0, the HJM model is consistent with the market price for the interest rate instruments that can be composed of zero coupon bonds. Therefore, the HJM model is arbitrage-free by construction.

From the no-arbitrage condition, we see that $\sigma_t(T)$ specifies the entire model. A particular implementation of the HJM model consists of a specification of $\sigma_t(T)$. In particular, the risk-neutral SDE for $P_t(T)$ is also specified by $\sigma_t(T)$ as

$$\frac{dP_t(T)}{P_t(T)} = r_t dt - \int_t^T \sigma_t(u) du \cdot d\tilde{\mathbf{W}}_t. \quad (27.2)$$

The problem of the general form of the HJM model is that, to obtain the zero coupon bonds between (t_1, t_2) , a continuum of $f_t(s)$, $(t_1 \leq s \leq t_2)$ is required. In practical implementation, dimension reduction is achieved through the following approaches:

- applying particular specification of $\sigma_t(T)$ to obtain lower dimensional Markovian models (see short rate model);
- modelling the discrete forward rate $f_t(t_1, t_2)$ directly (see the BGM model in Sect. 27.4).

27.2.1 No-Arbitrage Short Rate Model

Under $\widetilde{\mathbb{P}}$, the modelling of short rate r_t often requires a few factors to describe the dynamics of the whole term structure. Any no-arbitrage short rate model should satisfy the HJM model as

$$r_t = f_t(t) = f_0(t) + \int_0^t \widetilde{\alpha}_u(t) du + \int_0^t \sigma_u(t) \cdot d\widetilde{\mathbf{W}}_u. \quad (27.3)$$

It is straightforward to get the mean and variance of r_t , when σ is deterministic:

$$\widetilde{\mathbb{E}}[r_t] = f_0(t) + \int_0^t \widetilde{\alpha}_u(t) du \quad \text{and} \quad \widetilde{\text{Var}}(r_t) = \int_0^t \sigma_u(t) \cdot \sigma_u(t) du.$$

27.2.2 Markovian Characterization for Short Rate Models

Performing differential¹ on (27.3), we obtain the SDE for r_t ,

$$dr_t = df_t(t) = f'_0(t)dt + \widetilde{\alpha}_t(t)dt + \int_0^t \frac{\partial \widetilde{\alpha}_u(t)}{\partial t} du dt + \sigma_t(t) \cdot d\widetilde{\mathbf{W}}_t + \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\widetilde{\mathbf{W}}_u dt$$

¹ We know that $d\left(\int_0^t f(t, u) du\right) = f(t, t)dt + \int_0^t df(t, u)du$ for any smooth function $f(t, u)$.

Actually, $d\left(\int_0^t f(t, u) du\right) = \lim_{\Delta t \rightarrow 0} \left[\int_0^{t+\Delta t} f(t+\Delta t, u) du - \int_0^t f(t, u) du \right] = \lim_{\Delta t \rightarrow 0} \left[\int_t^{t+\Delta t} f(t+\Delta t, u) du + \int_0^t f(t+\Delta t, u) du - \int_0^t f(t, u) du \right] = f(t, t)dt + \int_0^t df(t, u)du.$

Similarly, we have $d\left(\int_0^t f(t, u) dW_u\right) = f(t, t)dW_t + \int_0^t df(t, u)dW_u.$

or

$$dr_t = \left[f'_0(t) + \int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du + \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u \right] dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t \quad (27.4)$$

since $\tilde{\alpha}_t(t) = 0$. The drift contains a stochastic term $\int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u$, which is path dependent. Therefore, r_t is not Markovian in general.

The non-Markovian feature makes the implementation difficult. Taking the example of tree method, a non-Markovian model may lead to non-recombining branches, and an explosive number of nodes. Finite difference method is only applicable under Markovian models. However, by choosing some special function forms for $\sigma_u(t)$, it is possible to obtain Markovian models for r_t .

Definition 27.1 (Volatility Separation Condition) The volatility $\sigma_t(T)$ is said to satisfy the separation condition if it takes the form

$$\sigma_t(T) = \mathbf{g}(t)h(T), \quad \text{where } \mathbf{g}(t) \text{ is a vector-valued function and } h(T) > 0.$$

Proposition 27.2 If $\sigma_t(T)$ satisfies the volatility separation condition, then

$$dr_t = \kappa(t)(\theta(t) - r_t)dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t \quad \text{i.e., } r_t \text{ is Markovian,}$$

where

$$\kappa(t) = -\frac{h'(t)}{h(t)}, \quad \theta(t) = \frac{f'_0(t)}{\kappa(t)} + \frac{h^2(t)}{\kappa(t)} \int_0^t \mathbf{g}^\top(u)\mathbf{g}(u)du + f_0(t).$$

This result can be extended to the case where h is an n -dimensional vector $\mathbf{h}$, and $\mathbf{g}$ an $n \times n$ matrix-valued function.

Proof It suffices to calculate the drift term of dr_t in (27.4), i.e.,

$$\int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du \quad \text{and} \quad \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u.$$

Firstly, from the definition of $\tilde{\alpha}$ in the no-arbitrage condition,

$$\tilde{\alpha}_u(t) = \sigma_u(t) \cdot \int_u^t \sigma_u(s)ds = h(t)\mathbf{g}^\top(u) \int_u^t h(s)\mathbf{g}(u)ds.$$

Direct operations give

$$\begin{aligned} \int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du &= \int_0^t \left[h'(t)\mathbf{g}^\top(u) \int_u^t h(s)\mathbf{g}(u)ds + h^2(t)\mathbf{g}^\top(u)\mathbf{g}(u) \right] du \\ &= -\kappa(t) \int_0^t \tilde{\alpha}_u(t)du + \int_0^t h^2(t)\mathbf{g}^\top(u)\mathbf{g}(u)du. \end{aligned}$$

Similarly,

$$\begin{aligned} \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u &= h'(t) \int_0^t \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u = -\kappa(t) \left(\int_0^t \sigma_u(t) \cdot d\tilde{\mathbf{W}}_u \right) \\ &= -\kappa(t) \left(r_t - f_0(t) - \int_0^t \tilde{\alpha}_u(t) du \right). \end{aligned}$$

Hence,

$$\begin{aligned} dr_t &= \left(f_0'(t) + \int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du + \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u \right) dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t \\ &= \kappa(t)(\theta(t) - r_t)dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t. \end{aligned} \quad \square$$

Proposition 27.3 *If $\sigma_t(T)$ satisfies the volatility separation condition and $\mathbf{g}, h$ are deterministic, then*

$$P_t(T) = e^{C_t(T) - D_t(T)r_t},$$

where

$$\begin{aligned} C_t(T) &= \ln \frac{P_0(T)}{P_0(t)} - \int_t^T \left[\int_0^t \left(h(z) \mathbf{g}^\top(u) \int_t^z h(s) \mathbf{g}(u) ds \right) du \right] dz + D_t(T) f_0(t), \\ D_t(T) &= \frac{\int_t^T h(u) du}{h(t)}. \end{aligned}$$

Proof Firstly, using the volatility separation condition in (27.3) gives

$$r_t = f_t(t) = f_0(t) + \int_0^t \left(h(t) \mathbf{g}^\top(u) \int_u^t h(s) \mathbf{g}(u) ds \right) du + \int_0^t h(t) \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u.$$

Then,

$$\begin{aligned} f_t(T) &= f_0(T) + \int_0^t \left(h(T) \mathbf{g}^\top(u) \int_u^T h(s) \mathbf{g}(u) ds \right) du + \int_0^t h(T) \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u \\ &= f_0(T) + \frac{h(T)}{h(t)} \left[\int_0^t \left(h(t) \mathbf{g}^\top(u) \int_u^t h(s) \mathbf{g}(u) ds \right) du \right. \\ &\quad \left. + \int_0^t h(t) \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u \right] + \int_0^t \left(h(T) \mathbf{g}^\top(u) \int_t^T h(s) \mathbf{g}(u) ds \right) du \\ &= f_0(T) + \frac{h(T)}{h(t)} (r_t - f_0(t)) + \int_0^t \left(h(T) \mathbf{g}^\top(u) \int_t^T h(s) \mathbf{g}(u) ds \right) du \end{aligned}$$

and

$$\begin{aligned}
 P_t(T) &= e^{-\int_t^T f_t(u)du} \\
 &= \exp \left\{ -\int_t^T f_0(u)du \right\} \exp \left\{ -\frac{\int_t^T h(u)du}{h(t)}(r_t - f_0(t)) \right. \\
 &\quad \left. - \int_t^T \left[\int_0^t \left(h(z)\mathbf{g}^\top(u) \int_t^z h(s)\mathbf{g}(u)ds \right) du \right] dz \right\} \\
 &= e^{C_t(T) - D_t(T)r_t}.
 \end{aligned}$$

Note that

$$\exp \left\{ -\int_t^T f_0(u)du \right\} = \frac{\exp\{-\int_0^T f_0(u)du\}}{\exp\{-\int_0^t f_0(u)du\}} = \frac{P_0(T)}{P_0(t)}. \quad \square$$

27.3 Short Rate Models

One-factor short rate models are the easiest to handle. However, because there is only one Brownian for the whole term structure, the bond prices for all the maturities are perfectly correlated. So, the one-factor models are only suitable for structures that are mainly sensitive to parallel shifts. Two-factor models capture the yield curve pivoting, which gives satisfaction for the pricing of many interest rate products.

27.3.1 Hull–White One-Factor Model

In the Hull–White (HW) model [51], r_t follows an OU process:

$$dr_t = \kappa(t)(\theta(t) - r_t)dt + \sigma d\tilde{W}_t.$$

We consider the widely used case with deterministic $\theta(t)$, and constants κ and σ . It is a no-arbitrage model because it can be derived from the HJM no-arbitrage model by setting $\sigma_t(T) = \sigma e^{-\kappa(T-t)}$.

r_t follows an OU process, which is a Gaussian process. Its dynamics are known to be *mean reverting*, which means that, with $\kappa > 0$, r_t tends to rise when $r_t < \theta(t)$ and tends to fall when $r_t > \theta(t)$. The parameter κ represents the reverting speed. As a Gaussian process, the model is fully specified by its mean and variance that can be easily obtained. Zero coupon bonds can be calculated with closed-form formula. The HW model belongs to the so-called *arbitrage-free* models, which can calibrate the initial term structure of the yield curve of the market thanks to the time-dependent parameters. The HW model is the most applied model in practice for structures that are mainly sensitive to parallel shifts of the yield curve.

Applying $h(T) = \sigma e^{-\kappa T}$ and $g(t) = e^{\kappa t}$ to the results in Proposition 27.2, we obtain

Corollary 27.1 *For the Hull–White model,*

$$\begin{aligned}\theta(t) &= f_0(t) + \frac{f'_0(t)}{\kappa} + \frac{\sigma^2}{2\kappa^2}(1 - e^{-2\kappa t}), \\ \widetilde{\mathbb{E}}[r_t] &= f_0(t) + \frac{\sigma^2}{2\kappa^2}(1 - e^{-\kappa t})^2, \\ \widetilde{\text{Var}}(r_t) &= \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}), \\ P_t(T) &= e^{C_t(T) + D_t(T)r_t},\end{aligned}$$

where

$$\begin{aligned}C_t(T) &= \ln \frac{P_0(T)}{P_0(t)} - D_t(T) \frac{\partial \ln P_0(t)}{\partial t} - \frac{\sigma^2}{4\kappa^3}(e^{-\kappa T} - e^{-\kappa t})^2(e^{2\kappa t} - 1), \\ D_t(T) &= \frac{1}{\kappa}[1 - e^{-\kappa(T-t)}].\end{aligned}$$

The yield for zero coupon bond is $C_t(T) + D_t(T)r_t$. This implies perfect correlation of yield levels over all maturities. As a consequence, the model is not suitable for structures having significant sensitivities to changes other than parallel shift of the yield curve.

The HW model is equivalent to the one-factor case of the LGM (Linear Gaussian Markov) model, which takes the form $r_t = f_0(t) + x_t$ where x_t is the random state variable following an OU process. Working with $x_t = r_t - f_0(t)$ has the advantage of avoiding the non smoothness issues of $f_0(t)$ from bootstrapped prices of the market zero coupon bonds.

Remark 27.1 By setting $\theta(t)$ to a constant b , we get the Vasicek model $dr_t = \kappa(b - r_t)dt + \sigma d\widetilde{W}_t$. Unlike the Hull–White model, the Vasicek model is an equilibrium model that is not arbitrage free in the sense that the initial term structure of the market may not be satisfied. $\square$

27.3.2 Two-Factor LGM (Linear Gaussian Markov)

In one-factor models, the yields of different maturities are perfectly correlated. For products that are sensitive to the yield curve changes other than parallel shift, more factors need to be introduced. For most of the interest rate structures, two factors give satisfaction to capture the variability of the term structure.

The two-factor LGM model is among the most popular ones used in practice. It is a no-arbitrage model, and it allows analytical solution for the zero coupon bonds. A detailed analysis can be found in [5]. The standard form of two-factor LGM may be written as follows:

$$r_t = f_0(t) + x_t^1 + x_t^2,$$

where x_t^1, x_t^2 are the factors following the mean reverting process

$$d\mathbf{x}_t = [\mathbf{y}_t \mathbf{1} - \kappa(t)\mathbf{x}_t]dt + \boldsymbol{\sigma}(t)^\top d\tilde{\mathbf{W}}_t,$$

with

$$\begin{aligned} \mathbf{x}_t &= [x_t^1, x_t^2]^\top, \quad \mathbf{1} = [1, 1]^\top, \\ \tilde{\mathbf{W}}_t &= [\tilde{W}_t^1, \tilde{W}_t^2]^\top, \text{ where } \tilde{W}_t^1, \tilde{W}_t^2 \text{ are independent Brownians,} \\ \boldsymbol{\sigma}(t) &= \begin{bmatrix} \sigma_{11}(t) & 0 \\ \sigma_{21}(t) & \sigma_{22}(t) \end{bmatrix}. \end{aligned}$$

The other parameters are as follows, which are derived from the HJM model with separability condition on $\boldsymbol{\sigma}$ so that the model is Markovian:

$$\begin{aligned} \kappa(t) &= \text{diag}(\kappa_1(t), \kappa_2(t)), \\ \mathbf{h}(t) &= \begin{bmatrix} e^{-\int_0^t \kappa_1(u)du} \\ e^{-\int_0^t \kappa_2(u)du} \end{bmatrix}, \\ \mathbf{g}(t) &= \begin{bmatrix} \sigma_{11}(t)e^{\int_0^t \kappa_1(u)du} & 0 \\ \sigma_{21}(t)e^{\int_0^t \kappa_1(u)du} & \sigma_{22}(t)e^{\int_0^t \kappa_2(u)du} \end{bmatrix}, \\ \mathbf{y}_t &= \mathbf{H}(t) \left(\int_0^t \mathbf{g}(u)^\top \mathbf{g}(u)du \right) \mathbf{H}(t), \quad \mathbf{H}(t) = \text{diag}(\mathbf{h}(t)). \end{aligned}$$

The model is fully specified by $\kappa_1, \kappa_2, \sigma_{11}, \sigma_{21}$ and σ_{22} .

Extension for Volatility Skew

The LGM model (one and two factors) can be extended to take into account the volatility skew observed in vanilla instruments for pricing certain structures that are sensitive to volatility dynamics. One extension applies local volatility to $g(t)$ in the LGM model, which assumes the volatility separability:

$$\sigma_t(T) = g(t)h(T) \quad \text{with} \quad g(t) = g(t, x_t, y_t).$$

A further extension consists of applying a stochastic term to $g(t, x_t, y_t)$:

$$\sqrt{z(t)}g(t, x_t, y_t)$$

where z follows a CIR process that is uncorrelated to x_t . We refer to [5] for detailed discussions.

27.3.3 CIR (Cox-Ingersoll-Ross) One-Factor Model

The original CIR model is given by

$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}d\tilde{W}_t,$$

where κ, θ, σ are all constants.

Similar to the Vasicek model, it is an equilibrium model that is not arbitrage-free. The particular property of the model is that it guarantees the positivity of the short rate, which used to be a much-desired feature before negative interest rates became a reality for several major currencies in the 2010s.

Like the Hull-White model, the CIR model can be derived from the HJM no-arbitrage short rate model by setting $\sigma_t(T) = \sigma\sqrt{r_t}e^{-\kappa(T-t)}$. Under the CIR model, zero coupon bonds admit an analytical solution as follows [33]:

$$P_t(T) = e^{A_t(T) - B_t(T)r_t}.$$

27.3.4 Affine-Yield Models*

A model satisfying the below features is called an affine-yield model:

- the short rate r_t is an affine function (linear plus translation) of the factors $\mathbf{Y}_t$
- zero coupon bond prices are of the form $P_t(T) = e^{C_t(T) - D_t(T) \cdot \mathbf{Y}_t}$ where $C_t(T)$ and $D_t(T)$ are non-random functions ($D_t(T)$ in vector form).

It can be proved that an affine-yield model takes the below vector form [36]:

$$d\mathbf{Y}_t = (\lambda_0 + \phi\mathbf{Y}_t)dt + \Sigma \begin{bmatrix} \sqrt{\alpha_1 + \beta_1^\top \mathbf{Y}_t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sqrt{\alpha_n + \beta_n^\top \mathbf{Y}_t} \end{bmatrix} d\mathbf{W}_t,$$

where

$$\begin{array}{ll} \lambda_0, \beta_i : \text{vectors} & \alpha_i : \text{scalars} \\ \phi, \Sigma : \text{matrices} & \mathbf{Y}_t : \text{vector of factors (state variables)} \\ \mathbf{W}_t : \text{Brownian motion vector} & \end{array}$$

The Vasicek and CIR models are affine-yield models.

For a one-factor model, we have $r_t = Y_t$ (see the result of the HW model for example), which implies that the yields of zero coupon bonds for different maturities are all linearly dependent on r_t , i.e., 100% correlated.

Interest Rate Derivatives Pricing with Short Rate Models

For pricing interest rate derivatives, short rate models can be applied. As usual, the model parameters are calibrated to match the market price of liquid instruments. Closed-form solutions for zero coupon bonds exist for the popular one-factor models. PDE and tree methods are applicable for low dimension models. In Monte Carlo simulation, the increments of the short rate and the stochastic discount factor can be generated for each time-step. Zero coupon bond prices, if required, can also be simulated by using the solution of (27.2) (see the Monte Carlo method for the BSM model). We recall that the price of a zero coupon bond is the expected value of the stochastic discount factor, as shown by (27.1). To price a bond option with Monte Carlo, the random zero coupon bond prices need to be generated.

Here we take the example of the pricing of a caplet with Monte Carlo method. Assume that it is a call option on 3M SOFR.

- (1) generate one random path under $\tilde{\mathbb{P}}$ of the applied short rate model;
- (2) calculate the 3M SOFR rate with the stochastic discount factor values for this path;
- (3) calculate the present value of the option payoff for this path;
- (4) repeat the above steps and average the present value of the simulated payoffs to obtain the option price.

27.4 BGM Model

The BGM model (Brace–Gatarek–Musiela model, also known as the LMM) describes the forward interest rates that are observable through FRA and swap contracts that are replicable by zero coupon bonds. We have seen that each forward rate R_t^i is a martingale under its natural risk-neutral probability measure $\tilde{\mathbb{P}}^{T_{i+1}}$. The BGM model consists of applying the same numéraire for all the forward rates when pricing with the term structure. If we use $\tilde{\mathbb{P}}^{T_{n+1}}$ (the last one is a common choice known as the terminal measure, but we may choose any other $\tilde{\mathbb{P}}^{T_i}$) for all

R_t^k ($k = 0, 1, \dots, n$), we will need to adjust the drifts accordingly. Assume the following SDE

$$dR_t^k = \sigma_k \cdot d\tilde{\mathbf{W}}_t^{T_{k+1}}$$

where the volatility is deterministic and modelled either as

$$\sigma_k = \begin{cases} \sigma_k(t) R_t^k, & \text{for log normal model,} \\ \sigma_k(t), & \text{for normal model,} \end{cases}$$

and $\tilde{\mathbf{W}}_t^{T_{k+1}}$ is a multi-dimensional Brownian motion. By definition, the components of a multi-dimensional Brownian motion are independent one-dimensional Brownian motions.

Note that each R_t^k can be equivalently expressed with a one-dimensional Brownian motion

$$dR_t^k = \|\sigma_k\| d\tilde{X}_t^{T_{k+1}}$$

where $d\tilde{X}_t^{T_{k+1}} = \frac{1}{\|\sigma_k\|} \sigma_k \cdot d\tilde{\mathbf{W}}_t^{T_{k+1}}$. However, one-dimensional Brownian terms $\tilde{X}_t^{T_{i+1}}$ and $\tilde{X}_t^{T_{k+1}}$ ($\forall i, k$) are correlated by

$$d\tilde{X}_t^{T_{i+1}} d\tilde{X}_t^{T_{k+1}} = \frac{\sigma_i \cdot \sigma_k}{\|\sigma_i\| \|\sigma_k\|} dt \equiv \rho_{i,k} dt.$$

According to Proposition 17.1, the Radon–Nikodym derivative process for the numéraire change from $\tilde{\mathbb{P}}^{T_{i+1}}$ to $\tilde{\mathbb{P}}^{T_i}$ is

$$\left. \frac{d\tilde{\mathbb{P}}^{T_i}}{d\tilde{\mathbb{P}}^{T_{i+1}}} \right|_{\mathcal{F}_t} = Z_t(T_i) = \frac{P_t(T_i)/P_0(T_i)}{P_t(T_{i+1})/P_0(T_{i+1})} = \frac{1 + \tau_i R_t^i}{1 + \tau_i R_0^i},$$

where we have used $R_t^i = \frac{P_t(T_i) - P_t(T_{i+1})}{\tau_i P_t(T_{i+1})}$ and τ_i is the time fraction $\tau_i = T_{i+1} - T_i$. The Radon–Nikodym derivative admits to

$$dZ_t(T_i) = \frac{\tau_i dR_t^i}{1 + \tau_i R_0^i} = \frac{1 + \tau_i R_t^i}{1 + \tau_i R_0^i} \frac{\tau_i dR_t^i}{1 + \tau_i R_t^i} = Z_t(T_i) \frac{\tau_i dR_t^i}{1 + \tau_i R_t^i}.$$

Hence, applying Girsanov theorem in a vectorized fashion gives

$$\begin{aligned}
 d\tilde{\mathbf{W}}_t^{T_n} &= -\frac{d\langle Z_t(T_n), \tilde{\mathbf{W}}_t^{T_{n+1}} \rangle}{Z_t(T_n)} + d\tilde{\mathbf{W}}_t^{T_{n+1}} = -\frac{\tau_n \sigma_n}{1 + \tau_n R_t^n} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}}, \\
 d\tilde{\mathbf{W}}_t^{T_{n-1}} &= -\frac{d\langle Z_t(T_{n-1}), \tilde{\mathbf{W}}_t^{T_n} \rangle}{Z_t(T_{n-1})} + d\tilde{\mathbf{W}}_t^{T_n} = -\sum_{i=n-1}^n \frac{\tau_i \sigma_i}{1 + \tau_i R_t^i} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}}, \\
 &\vdots \\
 d\tilde{\mathbf{W}}_t^{T_{k+1}} &= -\sum_{i=k+1}^n \frac{\tau_i \sigma_i}{1 + \tau_i R_t^i} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}}, \quad k < n.
 \end{aligned}$$

We obtain the SDE for R_t^k under the risk-neutral probability measure $\mathbb{P}^{T_{n+1}}$

$$dR_t^k = \sigma_k \cdot \left[-\sum_{i=k+1}^n \frac{\tau_i \sigma_i}{1 + \tau_i R_t^i} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}} \right], \quad 0 \leq t \leq T_k,$$

or

$$dR_t^k = \left[-\sum_{i=k+1}^n \frac{\tau_i \rho_{k,i} \|\sigma_k\| \|\sigma_i\|}{1 + \tau_i R_t^i} dt + \sigma_k \cdot d\tilde{\mathbf{W}}_t^{T_{n+1}} \right], \quad 0 \leq t \leq T_k.$$

27.4.1 Factor Reduction

To avoid too many factors (Brownians) for long-term products based on forward rates, we can apply the factor reduction technique. Let $\tilde{B}_t^j$ ($j = 1, \dots, d$, typically $d < 10$) be independent Brownians and replace $d\tilde{X}_t^{T_{i+1}} = \frac{1}{\|\sigma_i\|} \sigma_i \cdot d\tilde{\mathbf{W}}_t^{T_{i+1}}$ by $\sum_{j=1}^d \theta_j^{T_{i+1}} d\tilde{B}_t^j$. Here, $\{\theta_j^{T_{i+1}}\}_{j=1, \dots, d}^{i=0, \dots, n}$ are calibrated to produce $d\tilde{X}_t^{T_{i+1}} d\tilde{X}_t^{T_{k+1}} = \rho_{i,k} dt, \forall i, k$.

Readers may refer to [5, 117] for further analysis on calibration and rank reduction for the BGM model.

27.4.2 Stochastic Volatility

The basic version of the BGM model does not handle volatility skew. It is possible to extend the model with a stochastic component in the volatility term like $\sqrt{v_t} \sigma_i$ where v_t is a positive process. The stochastic process on the volatility is often common to all maturities in the implementation.

Exercises

1. Show that the forward rate is a martingale under the T -forward measure, i.e., for $t \leq T$,

$$\tilde{\mathbb{E}}^T[f_T(T)|\mathcal{F}_t] = f_t(T).$$

2. Assume a one-factor short rate model under the HJM framework with constant volatility, i.e., $\sigma_t(T) = \sigma$. Show that

$$f_t(T) = f_0(T) + \sigma^2 t \left(T - \frac{t}{2} \right) + \sigma \tilde{W}_t$$

and

$$dr_t = \left(\frac{\partial f_0(t)}{\partial t} + \sigma^2 t \right) dt + \sigma d\tilde{W}_t,$$

which is a special case of the *Ho–Lee model* related to the slope of the initial yield curve.

3. Given the following two models under the risk-neutral measure for asset price and bond price respectively:

$$\begin{aligned} dS_t &= r_t S_t dt + \sigma_S S_t d\tilde{W}_t, \\ dP_t(T) &= r_t P_t(T) dt - \sigma_P P_t(T) d\tilde{W}_t, \end{aligned}$$

where r_t is a stochastic process of interest rate, σ_S and σ_P are positive constants.

- (1) Let $\text{For}_t(T) = S_t/P_t(T)$ be the forward price at time t and denote $\sigma = \sigma_S + \sigma_P$. Show that $d\text{For}_t(T) = \sigma \text{For}_t(T) d\tilde{W}_t^T$, where $\tilde{W}_t^T$ is the Brownian motion under the T -forward measure.
- (2) Consider a European cash-or-nothing binary call on the asset S_t , expiring at time T with strike price K . Show that its time- t value is given by

$$P_t(T) \tilde{\mathbb{E}}^T[\mathbf{1}_{\{S_T > K\}}|\mathcal{F}_t] = P_t(T) \tilde{\mathbb{E}}^T[\mathbf{1}_{\{\text{For}_T(T) > K\}}|\mathcal{F}_t] = P_t(T) N(d_2),$$

$$\text{where } d_2 = \left(\log(\text{For}_t(T)/K) - \frac{1}{2}\sigma^2(T-t) \right) / (\sigma\sqrt{T-t}).$$

Chapter 28

Credit Modelling*



In this chapter, we study the case of default in credit modelling. Structural models and intensity models are first covered. We then study the pricing of some multi-name credit products, where default correlations and recovery rates are involved. Interested readers may refer to [20], [49] and [84]. Counterparty risk measures are introduced at the end of the chapter.

28.1 Credit Modelling

28.1.1 Structural Model

Structural models link the default occurrence to the value of the firm. Merton [79] pioneered this approach in 1974. Denote A_t as the asset value at time t of the considered firm. Assume that the total debt of the firm is D for the considered time horizon T . At time T , the debt holder (creditor) will receive the full amount D or what is left of the firm, i.e., $\min\{D, A_T\} = D - (D - A_T)^+$. The value of the debt at time t ($t < T$) is then

$$D_t(T) = \tilde{\mathbb{E}}[e^{-r(T-t)}(D - (D - A_T)^+)|\mathcal{F}_t] = De^{-r(T-t)} - p_t(A_t, D, T).$$

Here $\mathcal{F}_t$ is the natural filtration of A_t . The term $p_t(A_t, D, T)$ is the value of a put option on A_t with D as strike and T as maturity. So, the valuation relies on a European put option pricing.

Let S_t be the equity value of the firm at time t . By the definition of equity, we have

$$S_T = A_T - \min\{D, A_T\} = (A_T - D)^+,$$

Therefore, the equity is valued as a European call option. The model can be further improved by letting the default time be the first time that $A_t < D$.

As the asset value is not observable on a daily basis, practitioners favor intensity models for credit derivatives modelling, where the model parameters can be implied from the markets quotes of liquid credit instruments. As the structural model links the debt and equity value, it has found its application in the strategies for capital structure arbitrage. The model can be used to calculate the stock delta for hedging the position in the debt of the company.

In Moody's KMV¹ and CreditMetrics² models, the asset value of the i -th firm is assumed to follow the SDE

$$dA_t^i/A_t^i = \mu^i dt + \sigma_i dW_t^i.$$

The probability of default for the period $(0, T)$ is calculated from the *distance-to-default* that is defined as

$$d^i(T) = \frac{\mathbb{E}[\ln A_T^i] - \ln D^i}{\sigma_i \sqrt{T}} = \frac{\ln(A_0^i/D^i) + (\mu^i - \frac{1}{2}\sigma_i^2)T}{\sigma_i \sqrt{T}},$$

where D^i is the debt of the i -th firm.

28.1.2 Intensity Model

In intensity models (also known as reduced-form models), the *risk-neutral* distribution of the default time τ is

$$\tilde{\mathbb{P}}\{\tau > t\} = 1 - F(t) = e^{-\int_0^t \lambda_u du}, \quad t > 0,$$

where $\tilde{\mathbb{P}}\{\tau > t\}$ is the *survival probability* and $F(t)$ represents the default probability. When the hazard rate λ is a constant, τ follows exponential distribution. It is the first jump time of a Poisson process. Note that λ can be calculated from the

¹ Founded by Stephen Kealhofer, John McQuown and Oldřich Vašíček and later acquired by Moody's.

² RiskMetrics was established in 1989 by J.P. Morgan and acquired by MSCI in 2010.

market quote of CDS and be used to price other credit derivatives. $\tilde{\mathbb{P}}\{\tau \leq t\}$ is the probability of default before t , and

$$\tilde{\mathbb{P}}\{t_1 < \tau \leq t_2\} = \tilde{\mathbb{P}}\{\tau > t_1\} - \tilde{\mathbb{P}}\{\tau > t_2\}$$

is the probability of default between time t_1 and $t_2 (> t_1)$. From the result in Chap. 13 Sect. 13.8, $\lambda_t dt$ is the probability to have one default in $(t, t + dt)$, conditioned by no default before t .

In this chapter, $P_t(T)$ will denote the discount factor, $P_t(T) = e^{-\int_t^T r_u du}$. Let $\mathcal{F}_t$ be the filtration for the information on default free market until t , which includes $\{\tau > t\}$. In intensity models, the default time τ is not a stopping time of $\mathcal{F}_t$. However, it is a stopping time of the enlarged filtration $\mathcal{G}_t = \mathcal{F}_t \cup \sigma(\{\tau \leq s\}, s < t)$. For financial modelling, we need to assume the so-called $\mathbb{H}$ -hypothesis, which is the conditional independence between $\sigma(\{\tau \leq s\}, s < t)$ and $\mathcal{F}_\infty$ given $\mathcal{F}_t$. Under this technical hypothesis, every $\mathcal{F}_t$ -martingale is a $\mathcal{G}_t$ -martingale and filtration $\mathcal{F}$ is said to be *immersed* in filtration $\mathcal{G}$. For the proof and further details, we refer to [60].

Proposition 28.1 (Filtration Switching Formula) *Let X be an integrable random variable, then*

$$\mathbf{1}_{\{\tau > t\}} \mathbb{E}[X | \mathcal{G}_t] = \mathbf{1}_{\{\tau > t\}} \frac{\mathbb{E}[X \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]}{\mathbb{E}[\mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]} \quad a.s.$$

Sketch of Proof Note that $\forall A \in \mathcal{G}_t$, there exists some event $B \in \mathcal{F}_t$ such that $A \cap \{\tau > t\} = B \cap \{\tau > t\}$. So, there should exist $\mathcal{F}_t$ -measurable r.v. Y_t , s.t.

$$\mathbf{1}_{\{\tau > t\}} \mathbb{E}[X | \mathcal{G}_t] = \mathbf{1}_{\{\tau > t\}} Y_t.$$

Taking conditional expectation $\mathbb{E}[\mathbf{1}_{\{\tau > t\}} \mathbb{E}[X | \mathcal{G}_t] | \mathcal{F}_t] = \mathbb{E}[\mathbf{1}_{\{\tau > t\}} Y_t | \mathcal{F}_t]$, we obtain

$$Y_t = \frac{\mathbb{E}[X \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]}{\mathbb{E}[\mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]}. \quad \square$$

Example 28.1 (Zero Coupon Bond Modelling) Denote

R : recovery rate upon default (usually, $R \approx 40\%$ for senior unsecured bonds);

DP: Default Probability ($DP = \tilde{\mathbb{E}}[\mathbf{1}_{\{\tau \leq T\}} | \mathcal{G}_t]$).

LGD: Loss Given Default ($LGD = 1 - R$);

The value of a zero coupon bond before default is

$$D_t(T) = \tilde{\mathbb{E}}[P_t(T)(\mathbf{1}_{\{\tau > T\}} + R \cdot \mathbf{1}_{\{\tau \leq T\}}) | \mathcal{G}_t] = \tilde{\mathbb{E}}[P_t(T)(1 - LGD \cdot \mathbf{1}_{\{\tau \leq T\}}) | \mathcal{G}_t].$$

If the short rate r_t is deterministic, we have

$$D_t(T) = P_t(T)(1 - LGD \times DP).$$

If $R = 0$, from filtration switching formula in Proposition 28.1, we obtain the value of a zero coupon bond before default

$$\begin{aligned}
 D_t(T) &= \tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}|\mathcal{G}_t] = \frac{\tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]}{\tilde{\mathbb{E}}[\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]} = \frac{\tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}|\mathcal{F}_t]}{\tilde{\mathbb{E}}[\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]} \\
 &= \frac{\tilde{\mathbb{E}}[\tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}|\mathcal{F}_t]|\mathcal{F}_t]}{\tilde{\mathbb{E}}[\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]} = \frac{\tilde{\mathbb{E}}[P_t(T)e^{-\int_0^T \lambda_u du}|\mathcal{F}_t]}{e^{-\int_0^t \lambda_u du}} \\
 &= \tilde{\mathbb{E}}[e^{\int_t^T -(r_u + \lambda_u) du}|\mathcal{F}_t].
 \end{aligned}$$

□

28.2 CDS (Credit Default Swap)

As presented in Part I, Chap. 8, CDS (Credit Default Swap) remains the most widely used credit derivative instrument despite the issues in the financial crisis of 2008. Let N be the notional amount and R the recovery rate that is usually assumed to be around 40% for senior unsecured debts. Assume that

$T_i, i = 1, \dots, m$: dates on which the premiums are paid³ ($T_0 = 0$);

$\delta_i, i = 1, \dots, m$: day count fraction for the period $[T_{i-1}, T_i]$;

Premium Leg: $\sum_{i=1}^m sN\delta_i\mathbf{1}_{\{\tau>T_i\}}$, s is the spread;

Protection Leg: $N(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}$.

We have the following, assuming $N = 1$ without loss of generality:

$$\begin{aligned}
 LEG_{\text{premium}} &= s\tilde{\mathbb{E}}\left[\sum_{i=1}^m \delta_i\mathbf{1}_{\{\tau>T_i\}}P(T_i)\right] \\
 LEG_{\text{protection}} &= \tilde{\mathbb{E}}[(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}P(\tau)].
 \end{aligned}$$

Equating the two legs, we obtain the market pricing of CDS spread⁴

$$s = \frac{\tilde{\mathbb{E}}[(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}P(\tau)]}{\tilde{\mathbb{E}}\left[\sum_{i=1}^m \delta_i\mathbf{1}_{\{\tau>T_i\}}P(T_i)\right]}. \quad (28.1)$$

³ We also made the assumption of no accrued payment for the default period, which can be relaxed in practical implementation.

⁴ In practice, it is sometimes necessary to take into account the risk of the collateral as well as the correlation between the collateral and reference entity.

In practice, the CDS spread is quoted in the market. By intensity model, we can imply asset's default probability for other purposes (e.g., FTD, CDO and convertible bond pricing).

28.3 Credit Triangle

Let us assume that the hazard rate λ and risk-free rate r are deterministic and flat. We have the so-called *credit triangle* that reveals the relationship in continuous time between the CDS spread s , the default rate λ and the recovery rate R :

$$s = (1 - R)\lambda.$$

Proof The PV of the protection leg is

$$\begin{aligned} LEG_{\text{protection}} &= \tilde{\mathbb{E}}[(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}P(\tau)] = (1 - R) \int_0^{T_m} e^{-rt} \lambda e^{-\lambda t} dt \\ &= (1 - R)\lambda \int_0^{T_m} e^{-(r+\lambda)t} dt. \end{aligned}$$

The PV of the premium leg is

$$\begin{aligned} LEG_{\text{premium}} &= s \tilde{\mathbb{E}} \left[\sum_{i=1}^m \delta_i \mathbf{1}_{\{\tau > T_i\}} P(T_i) \right] = s \tilde{\mathbb{E}} \left[\sum_{i=1}^m \tilde{\mathbb{E}} \left[\delta_i \mathbf{1}_{\{\tau > T_i\}} e^{-rT_i} | \mathcal{G}_{T_i} \right] \right] \\ &= s \tilde{\mathbb{E}} \left[\sum_{i=1}^m \delta_i e^{-\lambda T_i} e^{-rT_i} \right] = s \sum_{i=1}^m \delta_i e^{-(r+\lambda)T_i}. \end{aligned}$$

In continuous time, we have

$$LEG_{\text{premium}} \rightarrow s \int_0^{T_m} e^{-(r+\lambda)t} dt.$$

Equating the protection and premium legs, we obtain the result to be proved. $\square$

28.4 Pricing and Hedging of Multi-Name Credit Derivatives*

28.4.1 Copula

Copula (*Latin: link*) describes the dependence structure of random variables in a general way. It combines univariate distribution functions to form multivariate distribution functions.

Definition 28.1 Let C be a function $[0, 1]^n \rightarrow [0, 1]$. It is called “copula” if there exists a vector $(U_1, U_2, \dots, U_n)$ in which each U_i , $i = 1, \dots, n$ is uniformly distributed, s.t.,

$$C(u_1, \dots, u_n) = \mathbb{P}\{U_1 \leq u_1, \dots, U_n \leq u_n\}, \forall (u_1, \dots, u_n) \in [0, 1]^n.$$

We introduced two fundamental theorems on copula. For the proof, we refer to [67].

Theorem 28.1 (Sklar Theorem) Let F be an n -dimensional cumulative distribution function with margins F_i . Then there exists a copula C such that

$$F(x_1, \dots, x_n) = C(F_1(x_1), \dots, F_n(x_n)), \quad \forall x_1, \dots, x_n \in \mathbb{R}.$$

Theorem 28.2 (Fréchet–Hoeffding Bounds) For any copula C , the following Fréchet–Hoeffding bounds hold:

$$\max(1 - n + \sum_{i=1}^n u_i, 0) \leq C(u_1, \dots, u_n) \leq \min(u_1, \dots, u_n).$$

One can construct a copula from any distribution with $F_i(x_i) = u_i$,

$$C(u_1, \dots, u_n) = F(F_1^{-1}(u_1), \dots, F_n^{-1}(u_n)).$$

If $F(\cdot)$ is a multivariate normal distribution, then $C(\cdot)$ is called Gaussian copula that is widely used in pricing portfolio credit structures (CDO, FTD (first-to-default) note, etc.). Student-t copula is sometimes used to induce fatter tails than Gaussian copula.

28.4.2 One-Factor Gaussian Copula Model

In the one-factor Gaussian copula model, the latent variables for reference entities are assumed to be correlated to a common market factor, usually representing the systemic risk of the economy. Let x_i be the latent variable for the i -th name that is a normal random variable. Then,

$$x_i = \beta_i \hat{Z} + \sqrt{1 - \beta_i^2} Z_i,$$

where

$\hat{Z}$: common market factor;

Z_i : idiosyncratic risk of the i -th name;

β_i : correlation of the i -th name to $\hat{Z}$. It may be either estimated with econometric methods (e.g., correlation derived from the prices of the stock and the index

representing $\hat{Z}$), or implied from the market quotes of liquid credit derivative products.

$Z_1, \dots, Z_n, \hat{Z}$: independent standard normal variables.

Note that the latent variable x_i is also a standard normal, and that

- correlation $\rho_{i,j}$ between the names i and j equals to $\beta_i \beta_j$
- conditioned by the common market factor $\hat{Z}$, all latent variables for reference names are independent.

The marginal default distribution of a reference name can be expressed with the cumulative distribution function via a percentile-to-percentile transformation with the latent variable x_i in Gaussian copula: $p_i(t) = \tilde{\mathbb{P}}\{\tau_i \leq t\} = \tilde{\mathbb{P}}\{x_i \leq h_i\} = N(h_i)$. If the default time τ_i for name i follows exponential distribution with time-dependent intensity, then $p_i(t) = 1 - e^{-\int_0^t \lambda_u^i du}$. The marginal default probability conditional on copula factor $\hat{Z}$ can be constructed by the below function (the conditional distribution of x_i is normal with mean $\beta_i \hat{Z}$ and variance $1 - \beta_i^2$):

$$\tilde{\mathbb{P}}\{\tau_i \leq t | \hat{Z}\} = \tilde{\mathbb{P}}\{x_i \leq h_i | \hat{Z}\} = N\left(\frac{N^{-1}(p_i(t)) - \beta_i \hat{Z}}{\sqrt{1 - \beta_i^2}}\right) \quad (28.2)$$

The marginal survival probability conditional on copula factor $\hat{Z}$ is hence

$$\tilde{\mathbb{P}}\{\tau_i > t | \hat{Z}\} = 1 - \tilde{\mathbb{P}}\{\tau_i \leq t | \hat{Z}\} = N\left(\frac{\beta_i \hat{Z} - N^{-1}(p_i(t))}{\sqrt{1 - \beta_i^2}}\right) \quad (28.3)$$

Given the conditional expectation $\tilde{\mathbb{E}}[X | \hat{Z}]$, the unconditional expectation of a variable X is obtained by

$$\tilde{\mathbb{E}}[X] = \int_{-\infty}^{\infty} \tilde{\mathbb{E}}[X | z] \phi(z) dz, \quad \text{where } \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}. \quad (28.4)$$

The integral can be calculated with Gaussian Quadrature, Monte Carlo or any other efficient numerical integration methods.

Under the assumption of a homogeneous pool, the default probabilities and correlations are identical for all names with a common β . Denote $p(t|z) = \tilde{\mathbb{P}}\{\tau \leq t | \hat{Z} = z\} = N\left(\frac{N^{-1}(1 - e^{-\int_0^t \lambda_u^i du}) - \beta z}{\sqrt{1 - \beta^2}}\right)$. The conditional probability of m defaults by time t can be obtained from binomial distribution

$$\binom{n}{m} p(t|z)^m (1 - p(t|z))^{n-m},$$

and the unconditional probability of m defaults by time t is

$$\int_{-\infty}^{\infty} \binom{n}{m} p(t|z)^m (1 - p(t|z))^{n-m} \phi(z) dz.$$

28.4.3 Multi-Name Credit Derivatives Pricing

The typical multi-name credit derivatives include swaps and notes on an FTD event or single tranche CDO. The premium leg and default leg in a multi-name instrument involve the joint default probability distribution and the expected loss computation. The Gaussian copula model can be used to take into account the dependence structure of the defaults, leading to a joint default distribution that can be used to calculate the expected loss at different time horizons. Without loss of generality, we will analyze first-to-default swap and single tranche CDO swap assuming deterministic interest rate and non-homogeneous recovery rates (due to different seniorities of the reference debts or eventual downgrading after trade inception, etc.).

First-To-Default Swap

Assuming deterministic recovery rates and no accrued premium for the default period, the fair value of an FTD swap (protection buyer's view) is

$$\tilde{\mathbb{E}}[(1 - R_{i^*})P(\tau_{i^*})\mathbf{1}_{\{\tau_{i^*} \leq T_m\}}] - s \sum_{k=1}^m \delta_i P(T_k) \tilde{\mathbb{E}}[\mathbf{1}_{\{\tau_{i^*} > T_k\}}],$$

where i^* is the index of the first defaulted name (i.e., $\tau_{i^*} = \min_i \tau_i$) and the other notations are the same as the ones in Eq. (28.1). Under the Gaussian copula model, the first-to-default probability may be obtained by the independence of the defaults conditional to the copula factor $\hat{Z}$ and the integration formula (28.4). The conditional survival probability of first-to-default is

$$\tilde{\mathbb{P}}\{\min_i \tau_i > t | \hat{Z}\} = \prod_i \tilde{\mathbb{P}}\{\tau_i > t | \hat{Z}\}.$$

Therefore, the unconditional survival probability and the default probability are obtained as follows

$$\begin{aligned} \tilde{\mathbb{P}}\{\tau_{i^*} > t\} &= \int_{-\infty}^{\infty} \prod_i \tilde{\mathbb{P}}\{\tau_i > t | z\} \phi(z) dz \\ \tilde{\mathbb{P}}\{\tau_{i^*} \leq t\} &= 1 - \tilde{\mathbb{P}}\{\tau_{i^*} > t\}. \end{aligned} \tag{28.5}$$

The valuation of the premium leg for an FTD swap can be done with formula (28.5):

$$s \sum_{k=1}^m \delta_i P(T_k) \tilde{\mathbb{P}}(\tau_{i^*} > T_k).$$

For a homogeneous multi-name basket with recovery rate R , the expected loss is

$$\tilde{\mathbb{E}}[(1 - R)P(\tau_{i^*})\mathbf{1}_{\{\tau_{i^*} \leq T_m\}}] = (1 - R) \int_0^{T_m} P(t) d\tilde{\mathbb{P}}\{\tau_{i^*} \leq t\}.$$

For a non-homogeneous basket, the knowledge of the defaulted name is needed for the computation of the expected loss. Using the independence of the defaults conditional to the copula factor, the conditional probability for reference i to be the first defaulted name is

$$Q_i(t|\hat{Z}) = \tilde{\mathbb{P}}\{\tau_i \leq t|\hat{Z}\} \prod_{j \neq i} \tilde{\mathbb{P}}\{\tau_j > t|\hat{Z}\}. \quad (28.6)$$

Conditional to the copula factor $\hat{Z}$, the probability density for name i to be the first defaulted name $q_i(t|z)$ is given by the below proposition.

Proposition 28.2

$$q_i(t|z) = \left\{ \frac{d}{dt} \tilde{\mathbb{P}}(\tau_i \leq t \mid \hat{Z} = z) \right\} \prod_{j \neq i} \tilde{\mathbb{P}}(\tau_j > t \mid \hat{Z} = z).$$

Proof Let $h > 0$. We have

$$\begin{aligned} q_i(t|z) &= \lim_{h \rightarrow 0} \tilde{\mathbb{P}} \left\{ \tau_i \in [t, t+h), \tau_j \geq t+h \ (\forall j \neq i) \mid \hat{Z} = z \right\} / h \\ &= \lim_{h \rightarrow 0} \tilde{\mathbb{P}} \left\{ \tau_i \in [t, t+h) \mid \hat{Z} = z \right\} \prod_{j \neq i} \tilde{\mathbb{P}} \left\{ \tau_j \geq t+h \mid \hat{Z} = z \right\} / h \\ &= \frac{d}{dt} \tilde{\mathbb{P}} \left\{ \tau_i \leq t \mid \hat{Z} = z \right\} \prod_{j \neq i} \tilde{\mathbb{P}} \left\{ \tau_j > t \mid \hat{Z} = z \right\}. \end{aligned}$$

□

Hence, the expected loss is

$$\tilde{\mathbb{E}}[(1 - R_{i^*})P(\tau_{i^*})\mathbf{1}_{\{\tau_{i^*} \leq T_m\}}] = \int_{-\infty}^{\infty} \phi(z) \int_0^{T_m} P(t) \sum_{i=1}^m (1 - R_i) q_i(t|z) dt dz.$$

The computation of the probability distribution as well as the expected loss shown in the above equations can be done by numerical integration or by Monte Carlo simulation.

Single Tranche CDO Swap

The valuation of a single tranche CDO with attachment point K_1 and detachment point K_2 is

$$\int_0^{T_m} P(t) d\tilde{\mathbb{E}}[L_t^{K_1, K_2}] - s \sum_{k=1}^m \delta_k P(T_k) (1 - \tilde{\mathbb{E}}[L_{T_k}^{K_1, K_2}]),$$

where $L_t^{K_1, K_2}$ represents the loss for the tranche and the other notations are as usual.

The valuation of CDO payoff (i.e., default leg) is dependent on the distribution of the portfolio loss L_t for $t \in [0, T_m]$. While the expected loss $\tilde{\mathbb{E}}[L_t]$ is independent of the correlation model as it is fully determined by single name CDS market, the variance of L_t is highly dependent on the copula chosen. When using the Gaussian copula, loss variance and all senior tranche prices are increasing functions of default correlation.

In 2004, as a liquid standard tranche market emerged, and implied correlations could be obtained from market information. The deterministic recovery assumption was generally made and the *base tranches* started to be used as basic building blocks for CDO pricing. A base tranche is the one with 0% attachment point, which can be expressed as $\min(L_t, K)$, or equivalently, $L_t - \min(L_t - K)^+$ with detachment point K . Actually, the single tranche loss $L_t^{K_1, K_2}$ can be formulated by the base tranches:

$$\begin{aligned} L_t^{K_1, K_2} &= [\min(L_t, K_2) - \min(L_t, K_1)] / (K_2 - K_1) \\ &= [(L_t - K_1)^+ - (L_t - K_2)^+] / (K_2 - K_1), \end{aligned}$$

where L_t is the portfolio loss. Hence, $L_t^{K_1, K_2}$ can be evaluated as a call spread on the portfolio loss L_t :

$$\tilde{\mathbb{E}}[L_t^{K_1, K_2}] = (\tilde{\mathbb{E}}[(L_t - K_1)^+] - \tilde{\mathbb{E}}[(L_t - K_2)^+]) / (K_2 - K_1).$$

The distribution of L_t and the expected loss for any strike can be done by Monte Carlo or by an iterative method based on convolution (see [7] and [52]) using the independence property conditional to the copula factor $\hat{Z}$. As the market tends to price senior tranches with higher implied correlation, the expected loss of each base tranche is calculated with a strike-dependent copula correlation parameters $\rho(K)$. This results in an upward sloping base tranche correlation skew.

In 2008, as super-senior premium spread rose to such a level that the implied dependence copula went beyond the Fréchet–Hoeffding bound. It became necessary

to relax the deterministic recovery assumption to be able to calibrate the tranche valuation to the observed market prices. On top of the skewed default correlations, the application of recovery variance further contributes to the loss variance and increases the range of the model calibration.

In order to have a material impact on the senior tranche loss distribution, the expected recovery level needs to be correlated to the default rate. This can be done by introducing a functional dependence on the copula factor $\hat{Z}$. Amraoui and Hitier (see [1]) proposed a market model that makes the expected loss distribution similar to that of a portfolio with recoveries R_i marked down to a certain level R_m when the default rate increases. In this model, from the unconditional default probability $p_i(t)$, an implied probability $p_i^m(t)$ is determined by the conservation of expected loss observed in single name CDS market:⁵

$$(1 - R_m)p_i^m(t) = (1 - R_i)p_i(t), \quad \forall i.$$

Using the conditional probability function as defined in Eq. (28.2), the conditional recovery function $R_i(\hat{Z})$ can be derived to make the conditional loss distribution equivalent to that of a marked-down portfolio:

$$(1 - R_i(\hat{Z}))\mathcal{G}_i(p_i^m(t), \hat{Z}) = (1 - R_i)\mathcal{G}_i(p_i(t), \hat{Z}),$$

where $\mathcal{G}_i(q, z) = N\left(\frac{N^{-1}(q) - \beta_i z}{\sqrt{1 - \beta_i^2}}\right)$. This conditional recovery formulation by $R_i(\hat{Z})$

leads to a portfolio loss distribution that is similar to the Gaussian copula loss distribution after the recovery marked down to R_m , and enables reaching higher super-senior tranche prices without having to remark the single name recoveries.

28.4.4 Hedging Multi-Name Credit Derivatives*

The hedging of multi-name credit derivatives is complex and imperfect, especially in case of P/L jumps by credit events. Apart from the CDS spreads of the names, there are other factors impacting the P/L of the hedging party such as correlation, time decay, (cross) gamma, default⁶ and recovery rate. For illustration, we take the example of the risk exposure for X , the issuer of an FTD note of notional N (see [19]).

As the long party of credit protection from the note, X carries out delta-hedging by selling protection through CDS on individual names offsetting the first order exposure of the note. Due to correlation effect, the delta amount on each name is

⁵ In practice, $\min(R_m, R_i)$ is used resulting in $p_i^m \leq 1$. To be more general, we can imagine to relax this constraint but cap the calculated p_i^m by 1.

⁶ By default, we actually mean the *credit event* as defined in the credit linked product.

less than 100% of N , and the higher the correlations, the lower the deltas. However, the sum of the individual deltas is more than N in general, which means that the FTD note is a leverage product providing enhanced yield to investors. The P/L exposure of X is the combined effect of the below:

- short position on default correlations between the names as the correlation increase reduces the probability of default;
- potential negative gamma position on the overall increase in the spreads, which tends to decrease the overall delta amount;
- negative carry from time decay of the note;
- positive individual gamma position because the credit spread increase for an individual name will increase the delta on that name (while the spreads of the other names remain stable), which requires X to sell extra protection on it, implying a sell-high/buy-low position. At the same time, the delta amounts on the other names may slightly drop;
- in case of default, the P/L gain for X from the note is higher than its loss from the hedging CDS on the defaulted name, as the delta is less than 100% of N . However, if the default triggers an overall credit spread increase, X may have potential loss from the unwind of the CDS position on other names.

28.5 Counterparty Risk Measures

In this section, we will introduce the most applied concepts and measures that are used for the management of counterparty risks. Many measures and regulatory requirements related to counterparty risks have been developed and implemented after the financial crisis of 2008. We refer to [24] for a dedicated presentation on counterparty risk management.

In the following, V_t stands for the mark-to-market value of the considered trade or portfolio. Monte Carlo simulation based on the physical probability measure is often necessary to calculate the risk measures, while option pricing should always be done under risk-neutral measure.

28.5.1 PFE (Potential Future Exposure)

PFE measures the worst potential loss of a transaction with a given confidence percentile α (usually 97.5%) during its whole life. Mathematically, $\text{PFE}_\alpha = \inf\{X | \mathbb{P}(V_t \leq X) \geq \alpha, t \in [0, T]\}$. PFE is an important measure for counterparty risk management. A bank may define the credit limit for derivatives trades with a client by a total PFE amount. When the aggregated PFE reaches the limit, the bank will stop making new trades with the client. Another example of PFE application is the determination of IA (Independent Amount) for swaps.

28.5.2 *VaR (Value-at-Risk)*

VaR is defined as the maximum loss over the considered period $([0, T])$ L_T given a certain confidence level or percentile (p). Mathematically, $\text{VaR}(p, T) = \inf\{v : \mathbb{P}\{L_T \leq v\} \geq p\}$. For instance, the 10-day 90% VaR of USD100m means that there is 90% probability that the loss will be less than USD100m for the period of 10 (business) days (in other terms, there is 10% probability that the loss will be more than USD100m for the period of 10 business days). For more details on VaR, we refer to [62].

VaR is a measure for extreme loss, which is widely used at entity level or portfolio level by financial institutions as required by regulators. It can be used as an important metric for consolidated risks including market risk, credit risk, etc. It can also be applied to single trades, similarly to PFE.

28.5.3 *Close-Out Risk*

It is the potential loss during the period between the time the counterparty is deemed to default and the time the transaction is unwound. Usually, the considered period is 10 business days for OTC trades and 2 business days for exchange trades. The Close-Out Risk is measured with a similar percentile for PFE (e.g., 97.5%).

28.5.4 *EPE (Expected Positive Exposure)*

$\text{EPE}_t = \mathbb{E}[V_t^+]$. It measures the exposure of counterparty risk, as the negative mark-to-market value does not constitute a risk.

28.5.5 *CVA (Credit Value Adjustment)*

CVA is the market price of the counterparty credit risk for a considered transaction or portfolio. It is represented mathematically as $\text{CVA} = \text{LGD} \times \sum_{t=0}^T e^{-rt} \mathbb{E}[V_t^+] =$

$(1 - R) \int_0^T e^{-rt} V_t^+ \lambda_c e^{-\lambda_c t} dt$, where λ_c is the default intensity of the counterparty and R the recovery rate. Actually, the CVA represents only the theoretical hedging cost due to the lack of liquidity for the CDS to be used. In practice, the counterparty risk is often proxy hedged with the CDS on liquid indices or reference entities correlated to the target one. The CVA needs to be accounted for the determination of the profit of a trade. To give the background, CVA was largely ignored by most of the financial institutions before the financial crisis of 2008. It started to play an

important role after the crisis. Now, the widely practiced CCP clearing has led to a significant reduction of counterparty risk and hence the CVA amount in the market.

DVA (Debt Value Adjustment) is the CVA seen by the counterparty. $DVA = (1 - R) \int_0^T e^{-rt} V_t^- \lambda_d e^{-\lambda_d t} dt$, λ_d is dealer's own default intensity. It is also known as Debit Value Adjustment.

28.5.6 Wrong-Way Risk

Wrong-Way Risk occurs when the exposure to a counterparty is adversely correlated with the credit quality of that counterparty. For example, buying a put option on gold from a gold producer is a wrong-way risk. Under the current practice, the market risk of a transaction and the counterparty risk are modelled independently. Hence, an exogenous consideration for wrong-way risk is necessary. While trying to avoid wrong-way risks, one should not forget to encourage the transactions with “Right-Way” exposure.

Example 28.2 Consider a long position of an equity call option (assuming the BSM model with zero div, repo) with initial price V_0 . Recall that the 97.5-th percentile of a standard normal variate is 1.96. Then,

$$PFE_T \approx S_0 e^{(r - \frac{1}{2}\sigma^2)T + 1.96\sigma\sqrt{T}} - K$$

$$\text{Close-Out Risk}_t \approx \Delta_t (S_t e^{(r - \frac{1}{2}\sigma^2)\delta + 1.96\sigma\sqrt{\delta}} - S_t) e^{-r\delta} - V_t, \delta = 14/365$$

$$EPE_t = \mathbb{E}[V_t^+] = \mathbb{E}[\mathbb{E}[e^{-r(T-t)}(S_T - K)^+ | \mathcal{F}_t]] \approx e^{rt} V_0$$

$$CVA \approx V_0 s_c T, s_c \text{ is the credit spread of the counterparty.}$$

□

28.5.7 Basel Risk Weight Function*

Banks are required to hold a certain amount of *regulatory risk capital* to cushion the risks and to prevent excessive leverage. The losses of a bank or a large portfolio can be described with 3 marks: expected loss (EL), unexpected loss (UL) and extreme loss. It is admitted that the expected loss is a business cost that is well managed by the appropriate transaction pricing and provisioning. The extreme loss corresponds to the loss beyond VaR threshold, and UL represents the loss between VaR threshold and EL. In 2004, Basel Committee on Banking Supervision (BCBS) proposed to set the risk capital requirement (K) equal to the unexpected loss.

For banks adopting Internal Ratings-Based (IRB) Approach, it is allowed to use their own internal measures for key drivers of credit risk such as probability of default and loss given default, which normally leads to lower risk capital amount compared to the Standardized Approach. To calculate the risk capital for the i -th

performing (non-defaulted) exposure K^i ($K = \sum_i K^i$), BCBS proposed a general formula known as IRB Risk Weight Function (see [13]):

$$K^i = EAD^i \times (LGD^i \times P_{VaR}^i - LGD^i \times PD^i) \times Adj(M^i, PD^i),$$

where

EAD^i : exposure at default;

LGD^i : loss given default (in % of the EAD^i) estimated for stressed situation;

PD^i : probability of default for 1-year horizon and $\sum_j LGD^j \times PD^j = EL$;

P_{VaR}^i : defined as $N\left(\frac{N^{-1}(PD^i) + \beta_i N^{-1}(99.9\%)}{\sqrt{1 - \beta_i^2}}\right)$, which is the adjusted default probability such that $\sum_j LGD^j \times P_{VaR}^j = EL + UL$ (i.e., 1-year, 99.9% VaR threshold);

$Adj(M^i, PD^i)$: adjustment factor⁷ for giving more weight for exposures with longer maturities M and lower PD .

Now, we show that this formula can be interpreted by the one-factor Gaussian copula model⁸. Assume the latent variable for the i -th exposure to be

$$x_i = \sqrt{1 - \beta_i^2} Z_i - \beta_i \hat{Z}, \quad Z_i, \hat{Z} \sim \mathcal{N}(0, 1) \text{ and independent.}$$

where $\hat{Z}$ is the single systematic risk factor, Z_i represents the specific non-systematic risk and β_i is the correlation relating the i -th exposure to $\hat{Z}$.⁹

Under the granular portfolio assumption, the variability of portfolio loss is only driven by the systematic risk factor $\hat{Z}$ (see [44] for the proof).¹⁰ BCBS has chosen the VaR percentile of 99.9%, which implies $\hat{Z} = N^{-1}(99.9\%)$. The default probability conditional on $\hat{Z}$ is obtained by formula (28.2):

$$P_{VaR}^i = N\left(\frac{N^{-1}(PD^i) + \beta_i N^{-1}(99.9\%)}{\sqrt{1 - \beta_i^2}}\right).$$

In BCBS recommendation, banks are required to determine RWA (risk weighted asset) for each exposure and that the risk capital is calculated by $K^i = 8\% \times RWA^i$. Therefore, we can obtain RWA^i from K^i :

$$RWA^i = 12.5 \times EAD^i \times LGD^i \times (P_{VaR}^i - PD^i) \times Adj(M^i, PD^i).$$

⁷ $Adj(M, PD) = \frac{1 + (M - 2.5)b(PD)}{1 - 1.5b(PD)}$, $b(PD) = [0.11852 - 0.05478 \times \ln(PD)]^2$.

⁸ Our simplified analysis may not reflect completely the rationale of BCBS.

⁹ β_i is given by BCBS in [13] according the exposure type: corporate, bank, sovereign, residential mortgage, qualifying revolving retail, other retail, etc.

¹⁰ It could be understood by the fact that conditional to $\hat{Z}$, the non-systematic (idiosyncratic) risk in the portfolio is diversified away.

Exercises

1. Consider the Merton's structural model given by

$$dA_t = rA_t dt + \sigma_A A_t d\tilde{W}_t$$

and let $S_t = A_t - D_t(T)$ where S_t represents the equity value and $D_t(T)$ the debt. Show that the asset volatility σ_A and the equity volatility σ_S are related by

$$\sigma_S = \sigma_A N(d_1) \frac{A_t}{S_t}, \quad d_1 = \frac{\ln \frac{A_t}{L} + (r + \frac{1}{2}\sigma_A^2)(T - t)}{\sigma_A \sqrt{T - t}}.$$

2. Derive the survival probability by time T , i.e., $\tilde{\mathbb{P}}\{\tau > T\}$, when the intensity follows a stochastic CIR process as follows:

$$d\lambda_t = (b - a\lambda_t)dt + \sigma\sqrt{\lambda_t}d\tilde{W}_t,$$

in which it is assumed that the intensity is positive.

3. Calculate the PFE with $\alpha = 97.5\%$ for a long equity put option in which the asset follows the SDE as

$$dS_t = (r - q)S_t dt + \sigma S_t d\tilde{W}_t.$$

Chapter 29

Commodity Modelling*



From the supply-and-demand relation, the time- t commodity futures price F_t for a contract with expiration date T is

$$S_t e^{(r-y)(T-t)},$$

where y is a number reflecting *both* convenience yield and storage costs, so it can be positive or negative. For simplicity, we still call it convenience yield in modelling. One way for commodity modelling is to assume that the spot price follows a geometric Brownian motion under the risk-neutral measure

$$dS_t = (r - y)S_t dt + \sigma S_t d\tilde{W}_t,$$

which leads to

$$dF_t = \sigma F_t d\tilde{W}_t,$$

showing that the futures price F_t is a martingale. It is used for European options that depend mainly on final distributions. However, for path-dependent or some other exotic options, more advanced models may be required to capture the mean-reverting property of some of the commodities as well as the volatility skew.

29.1 Spot Model

The Gabillon model [40], originally proposed for oil futures, is one of the most popular models for commodity derivatives. Define the long-term price of a commodity as the price for delivery at infinite time. Under risk-neutral probability measure, the spot price S_t is modelled along with the long-term price process L_t as

$$\begin{cases} \frac{dS_t}{S_t} = (r_t - y_t)dt + \sigma_t^S d\tilde{W}_t^S, \\ \frac{dL_t}{L_t} = \sigma_t^L d\tilde{W}_t^L, \end{cases}$$

where r_t is the short rate, σ_t^S and σ_t^L are corresponding volatilities, and $d\tilde{W}_t^S d\tilde{W}_t^L = \rho dt$. More importantly, the convenience yield y_t is totally determined by S_t and L_t as

$$y_t = \kappa_t \ln(S_t/L_t) + \delta_t,$$

where κ_t is a positive process representing the speed of mean reversion and δ_t is an extra term.

29.2 Futures Model

Now, we show how to derive the process F_t from the Gabillon spot model for the futures contract with maturity T . Applying the Itô's formula with the functions $\ln S_t$ and $\ln L_t$ on the Gabillon spot model, we obtain the following two SDEs:

$$\begin{cases} d \ln S_t = \left[r_t - y_t - \frac{1}{2}(\sigma_t^S)^2 \right] dt + \sigma_t^S d\tilde{W}_t^S \\ d \ln L_t = -\frac{1}{2}(\sigma_t^L)^2 dt + \sigma_t^L d\tilde{W}_t^L. \end{cases}$$

Denoting the quotient $X_t = S_t/L_t$ (such that $\ln X_t = \ln S_t - \ln L_t$) and plugging in the convenience term y_t , we obtain

$$d(\ln X_t) = (a_t - \kappa_t \ln X_t)dt + \sigma_t^X d\tilde{W}_t^X,$$

where $a_t = r_t + \delta_t + \frac{1}{2}[(\sigma_t^L)^2 - (\sigma_t^S)^2]$ and $\sigma_t^X d\tilde{W}_t^X = \sigma_t^S d\tilde{W}_t^S - \sigma_t^L d\tilde{W}_t^L$. We see that $\ln X_t$ is an OU process that can be solved with the usual method (see Sect. 15.3). Let $K_t = e^{-\int_t^T \kappa_s ds}$. Then,

$$d(K_t \ln X_t) = K_t(a_t dt + \sigma_t^X d\tilde{W}_t^X)$$

$$\ln X_T = K_t \ln X_t + \int_t^T K_u(T)(a_u du + \sigma_u^X d\tilde{W}_u^X).$$

Putting $\ln L_T = \ln L_t - \frac{1}{2} \int_t^T (\sigma_u^L)^2 du + \int_t^T \sigma_u^L d\tilde{W}_u^L$ and the expression of $\tilde{W}_u^X$,

$$\begin{aligned} \ln S_T = & K_t \ln S_t + (1 - K_t) \ln L_t + \int_t^T [K_u(T)a_u - \frac{1}{2}(\sigma_u^L)^2] du \\ & + \int_t^T K_u(T)\sigma_u^S d\tilde{W}_u^S + \int_t^T (1 - K_u(T))\sigma_u^L d\tilde{W}_u^L. \end{aligned}$$

In the following it is assumed that κ_t , σ_t^S and σ_t^L are deterministic. Then, S_T is log-normally distributed. From the property of log-normal distribution, we know that $\mathbb{E}[S_T] = \exp\{\mathbb{E}[\ln S_T] + \frac{1}{2}\text{Var}(\ln S_T)\}$. We have $\mathbb{E}_t[\ln S_T] = A + \int_0^t K_u(T)\sigma_u^S d\tilde{W}_u^S + \int_0^t (1 - K_u(T))\sigma_u^L d\tilde{W}_u^L$ because the expectations of the Brownian integrals from t to T are zero. A is a term that does not contain Brownians and needs not to be detailed here. The variance $\text{Var}_t(\ln S_T) = B$ does not contain any Brownian terms either. The futures price under risk-neutral measure is given by $F_t = \mathbb{E}_t[S_T]$. Hence,

$$\begin{aligned} F_t = \mathbb{E}[S_T | \mathcal{F}_t] = & \exp\left(A + \frac{1}{2}B\right) \\ & \exp\left(\int_0^t K_u(T)\sigma_u^S d\tilde{W}_u^S + \int_0^t (1 - K_u(T))\sigma_u^L d\tilde{W}_u^L\right). \end{aligned}$$

Recall that futures price under the risk-neutral measure is a martingale (i.e., zero drift). Therefore, we obtain

$$dF_t/F_t = K_t \sigma_t^S d\tilde{W}_t^S + (1 - K_t) \sigma_t^L d\tilde{W}_t^L,$$

which is the Gabillon model for futures.

To model the skewed distribution for different maturities, we can add extra terms ω^S and ω^L , which enhance and generalize the original Gabillon model:

$$dF_t/F_t = \omega^S K_t \sigma_t^S d\tilde{W}_t^S + \omega^L (1 - K_t) \sigma_t^L d\tilde{W}_t^L.$$

Here ω^S and ω^L may be dependent on t , T and F_t , which can be calibrated from option prices.

29.3 Multi-Factor Models*

The Gabillon spot and futures models can be further generalized to multi-factors ones. The spot price process with n driving correlated factors is given as:

$$d \ln X_t^i = \left(a_t^i - \sum_{k=1}^n \kappa_t^{ik} \ln X_t^k \right) dt + \sigma_t^i d\tilde{W}_t^i, \quad i = 1, \dots, n.$$

For example, a 1-factor model is $d \ln S_t = (a_t - \kappa_t \ln S_t)dt + \sigma_t d\tilde{W}_t$ while a 3-factor spot price model may take the form:

$$d \ln S_t = (a_t - \kappa_t \ln S_t - \beta_t^{(1)} \ln X_t^{(1)} - \beta_t^{(2)} \ln X_t^{(2)})dt + \sigma_t^S d\tilde{W}_t^S$$

$$d(\ln X_t^{(1)}) = (a_t^{(1)} - \kappa_t^{(1)} \ln X_t^{(1)})dt + \sigma_t^{(1)} d\tilde{W}_t^{(1)}$$

$$d(\ln X_t^{(2)}) = (a_t^{(2)} - \kappa_t^{(2)} \ln X_t^{(2)})dt + \sigma_t^{(2)} d\tilde{W}_t^{(2)}.$$

Futures are the main trading instruments for commodities, and the futures curves are the market observable price references. The multi-factor futures model applies the same factors to all the futures, as shown by the process with expiration date T_i :

$$dF_t^i / F_t^i = \sum_{k=1}^n \omega^{ik} e^{-\int_t^{T_i} \kappa_s^{ik} ds} \sigma_t^k d\tilde{W}_t^k.$$

For example, using a constant κ_1 and defining $\kappa_2 = \ln(1 - e^{-\kappa_1(T_i-t)}) / (T_i - t)$, we retrieve the simplified 2-factor Gabillon futures model presented in the precedent section.

Exercises

1. Consider Schwartz one-factor model [102] for commodity spot price

$$dS_t = (b - a \ln S_t)S_t dt + \sigma S_t d\tilde{W}_t, \quad \text{where } a, b > 0.$$

- (1) Show that $\ln S_t$ follows an OU process.
- (2) Show that the futures price satisfies $dF_t = \sigma e^{-a(T-t)} F_t d\tilde{W}_t$, i.e., volatility decreases for longer maturity.

2. Consider the following the Gabillon futures model

$$\frac{dF_t}{F_t} = K_t \sigma^S d\tilde{W}_t^S + (1 - K_t) \sigma^L d\tilde{W}_t^L,$$

where $K_t = e^{-\kappa(T-t)}$, $\sigma^S > 0$ and $\sigma^L > 0$.

- (1) Derive the volatility of F_t .
- (2) Given that $\kappa = 0.5$, $\sigma^S = 0.3$, $\sigma^L = 0.1$ and the short and long term prices are correlated by -0.4 . Plot the volatility term structure.

Part IV
Structured Products and Solutions

Chapter 30
Structured Products: Structuring Topics

In this chapter, the technical topics on structuring are presented. It is highly recommended to read the introduction to structured products in Chap. 11 as well as some of the product examples in Chap. 31 before reading this chapter.

30.1 Building Blocks

The building blocks of a structured product are underlier, payoff and wrapper. The below table provides an overview of the three blocks.

Underlier	- Equities - Interest Rate - Foreign Exchange - Commodities - Credit - Investment Fund - Customized Index/Strategy -		
Payoff	Access Product - Direct Exposure - Leveraged Exposure	Option Based Structure - Principal Protected - Non-Principal Protected	Dynamic Portfolio - Portfolio Insurance - Dynamic Asset Allocation
Wrapper	- Structured Note - Certificate - Structured Deposit - Structured Fund - Insurance Policy - Over-The-Counter (OTC) Option - Warrant - Swap - Islamic Shariah Compliant Product -		

Structured product issuers are mainly the investment banks that possess the necessary resources, including technical skills, systems, risk management capabilities, and a client base. A number of teams are involved in structured product activities. Front-office teams normally refer to

Sales: advising and servicing clients with the suitable products;

Structuring (or *Financial Engineering*): product solution and pricing;

Trading: managing the risk of clients' trades.

Other functions may include

Back-office/Operations: confirmation, settlement, reporting, etc.

Risk: market risk and credit risk monitoring, pre-trade approval;

Quantitative modelling: pricing and hedging model implementation;

IT: systems for front-office, back-office, etc.

30.2 Underliers

A wide range of assets or references can be used as underlier for structured products, which are shown in the below overview:

Equity	Interest Rates	Foreign Exchange	Commodities
- Single stock - Basket of stocks - Equity index -	- Reference Rate - IRS Rate - OIS Rate - Futures Price -	- FX Spot Rate - Basket of Currencies -	- Futures - Index - Spot Price -
Credit	Fund	Customized Index / Strategy	
- Corporate Bond - Corporate Bonds - Index - Credit Index -	- Mutual Fund - ETF - Hedge Fund - Managed Account -	- Smart Beta - Alternative Beta - Factors - Multi-Asset Strategy -	
			- ESG Index - Climate Change Index -

30.3 Wrapper or Instrument

The choice for the legal form/carrier of a structured product depends on the distribution channel, regulations, credit agreement between the counterparties, etc. The most common wrappers are as follows:

Structured Note: a debt security, in Bearer or Registered form, that pays the interests and principal according to the underlier performance;

Certificate: a registered security that entitles the holder to payments linked to the performance of the underlier. For investors, a structured product wrapped as a note or a certificate from the same issuer has no real difference;

Warrant: it is a securitized option (could be vanilla or exotic option); Warrants can be listed on the exchange (accessible for retail investors) or issued under private placement (known as OTC warrant). We note that a listed warrant with stock underlier does not create dilution to the underlying stock because it only involves secondary market shares;

Swap or Total Return Swap (TRS): in a TRS, one party pays the funding on the notional and receives the total return linked to the performance of the underlier. It is a leverage product for this counterparty, as the engaged amount is less than the notional amount. Initial and variation margins are required for risk management (see Chap. 2).

Swaps can also be used for market access or wholesale transactions for distribution. For example, it is used for a structured product whereby the counterparties are typically two institutions:

- one party is the product distributor, who pays funding to the hedging party. It uses the swap as a back-to-back hedge to repack the product as a note or fund to be sold to end investors who cannot directly invest into the underlying asset or product.¹ Actually, the funding paid to the hedging party is from the investment of the note proceeds from the note investors;
- the other party is the swap hedging party, who provides the cash flows contingent on the performance of the underliers as specified in the product.

For counterparties without CSA agreement, *fully funded swaps* (see Remark 5.3) can be used to mitigate the counterparty risk to the party providing the investment return.

OTC Option: it is a flexible instrument for customized features;

Structured Deposit: offered by deposit taking institutions. Deposits are not transferable;

Structured Fund: issued under the form of fund units that are usually not transferable unless listed;

Insurance Policy: structured product can be included in an insurance product. There exist transferable insurance policies.

Notes/Certificates/Warrants are transferable securities that are unsubordinated and unsecured obligations of the issuer and are ranked *pari passu* among themselves. Investment banks usually set up an *issuance programme* for notes/certificates/warrants with the general terms and conditions governing all

¹ For example, offshore notes cannot be sold to South Korea onshore investors by the current regulation while swaps are allowed. So, such a repackaging is necessary to provide structured products to South Korea onshore investors.

the issued securities. The securities may be settled through a clearing house (e.g., Euroclear, Clearstream) and DvP (Delivery versus Payment) is usually applied to minimize counterparty settlement risk.

The concepts related to the life cycle for structured products can be found in Sect. 2.1 whereby Example 2.1 provides the details on the dates for a structured note.²

30.4 Payoff Structuring Techniques

It is recommended to read this section after Chap. 31 in order to better understand the discussion on structuring techniques.

A product payoff determines how the return is calculated from the performance of the underliers. A rich set of payoffs is being offered in the market to meet the targets of investors with specific market views and risk profiles. The goal of payoff design consists of using the market view to offer the target return/risk profile under the price constraint of investors. To minimize the cost to investors, the payoff needs to offer attractive return when the market view is realized, but provides little return or even incurs loss to investors for other market conditions. A product offering a nice return with no payoff risk, such as straddles³ (long call plus long put), will be too expensive to investors, and hence not viable. The real returns after investment cost from such products are not good in general.

For a bullish structure, lower forward of the underlier leads to a lower price. So, the underlier selection will favor the ones with high dividend, high repo rate and low correlation for worst-of basket (if applicable). The currency of the underlier may matter if it is not the currency of the product, as the correlation FX/underlier impacts the forward (see Chap. 18).

According to the investor's view and the investment budget for the product, the following technical features are often employed in the payoff design:

Basket Performance Variations

- *Worst-of* Basket Performance: the most popular multi-underlier structures are based on the worst performance of the underliers due to the low cost. The features of a product based on worst-of basket often look attractive as it may provide a comfortable protection cushion and/or high potential return. On the other hand, the cost of a *Best-of* based bullish product is normally too high for investors;
- *Rainbow* Basket Performance: for multi-underlier structures, the Rainbow performance is measured as the weighted average of the performances according to the performance ranking, e.g., 70% of the worst, 30% of the second worst. Worst-of performance is a special case of Rainbow performance.

² In modelling, the period between issue date and maturity is used for discounting while the period between trade date and final valuation date is used for the underlier's price diffusion.

³ The delta hedged straddle strategy is often used in trading for its volatility exposure.

Cap on the Return

This feature may help reduce the product's cost (e.g., call spread). Binary option is the basic version of products with capped return. It has a clear return target for the investor. For multi-underlier structures, although it is natural to apply the cap on the averaged performance, it is however more effective for cost reduction if the cap is applied to the performance of each underlier (known as *Individual Cap*) or even to the best performance only.

Barrier

The barrier feature allows to achieve different purposes:

- *protection*: for example, short position of down-and-in put is often seen in the structured products where the investor takes downside risk to fund the return on the upside. The investor is protected until the barrier level;
- *price cheapening*: if a barrier event reduces the potential gain or even incurs loss to the investor, the price of the structure will be cheapened consequently. In the product design, such a barrier is usually set at a level that is unlikely to achieve according to the market view of the investor. One simple example is an up-and-out call option;
- *return lock-in*: the barrier applied over a period of time, either continuously or on discrete points, to allow the investor to lock in the return timely, e.g., one-touch options.

The observation for barrier event can be of American style (sometimes noted as AKI—American knock-in) or European style (sometimes noted as EKI—European knock-in, or AXKI—at expiry knock-in). For basket based structures, the barrier is often observed on the worst-of performance. The best-of performance may be applied for range-bound products (for upper bound) or bearish products.

*Memory Touch*⁴ is a special type of barrier event: for the considered period, the barrier event is deemed to occur if the barrier has been hit by every underlier in a basket but not necessarily at the same time. This feature is often used as an additional condition to lock in a benefit for the investor.

Averaging

It consists of using the averaged price as reference for payoff calculation. The volatility of the averaged price is lower than the volatility of the price itself (e.g., daily closing price). Averaging makes the price cheaper for products with positive vega. For investors, averaging mitigates the risk of abnormal price moves.

Autocallable Feature

As suggested by its name, an *autocallable* structure is early terminated if the underlier price meets a certain condition predefined in the contract, e.g., when an up-and-out barrier is hit. It is implemented in many products, e.g., FCN, Snowball, Accumulator. The autocallable feature allows to lock in the gain for the investor

⁴ The second author introduced this concept to the market in ~2004.

and, for principal non-protected structures, to stop the risk taking by the investor. It usually cheapens the cost of the structure because it avoids potential bigger gain that may grow over time if the underlier performs well.

Target Redemption refers to a special barrier event for early termination. It occurs when the accumulated gain to the investor as specified by the product design has achieved a predefined target. A variation consists of using the accumulated number of times that the intrinsic value has been positive instead of the accumulated gain.

Issuer Callable Feature

An *Issuer callable* structure gives the issuer the right to early terminate the product at one of the predefined dates or within the predefined time windows. Technically speaking, it is an American option to the issuer in which the return for the investor is enhanced by the “option value” for the early termination right given (or, more precisely, sold) to the product issuer. The callable structures are usually designed in a way such that the early termination event provides an attractive benefit to investors. The early termination feature is generally perceived by investors as a lock-in mechanism for the gain. The drawback of the issuer callable product to investors is the uncertainty as to when the product will be called (i.e., early terminated).

Puttable Feature

The structure where the early termination call right is given to the investor is known as a *puttable* structure. This type of product is less common in the market because the call right given to the investor makes the product “more expensive” leading to reduced returns. As investors normally determine the early termination/continuation according to their views on the market instead of the optimal option exercise time calculated with a pricing model, the termination/continuation decision is often not optimal as seen by the hedging party, i.e., the product issuer. So, the hedging party is likely to realize hedging profit in their trading book for such products.

30.5 Investment Leveraging

Apart from portfolio based financing (see Sect. 2.5) and listed derivatives (e.g., futures and options), leveraged investment can also be made by a standalone non-recourse trade that offers linear or close to linear return.

A **Leveraged note** offers a leveraged return of an investment with a fixed maturity, net of financing cost for the leveraged investment. By its legal nature, a note cannot have negative value. Hence, special terms and conditions are required for a leveraged note to make it hedgeable. To see this, let's take a 2-times leveraged note as an example: had the issuer simply hedged with a long position of 2-times face value in the underlier, it would suffer a loss should the underlier price drop by more than 50%, even with zero financing cost. In practice, the issuer usually implements one of the two mechanisms:

- *optional payoff*: the note redemption value is defined as

$$N \times \max \{1 + PR \times (S_T/S_0 - 1), 0\}$$

that is equivalent to $PR \times N \times \max \{S_T/S_0 - K, 0\}$, $K = 1 - 1/PR$. Clearly, it is an ITM call option when $PR > 1$. The investment in the underlier will follow the option's delta.

- *stop-loss*: this approach consists of introducing an early termination feature linked to a pre-agreed *stop loss* level. As soon as the note value (or equivalently, the performance of the investment with leverage minus financing cost) drops below *stop loss* level, the note will be early redeemed at loss for the note investor.

ITM (in-the-money) call option provides leveraged exposure to the performance of the underlier without paying too much time-value for the option. The popular CBBC warrant is an example of such a product. For active investors, there may be a need to early terminate the option to lock in the underlier's good performance. In this case, an American option may be a good choice due to its transparency of early termination price, which is the intrinsic value.

Total Return Swap (TRS) is a preferred trade for institutional investors for leveraged exposure because the money engaged by the investor is only the funding, initial margin (e.g., 15% of the notional) and variation margin during the life of the transaction. The ISDA/CSA master agreement is required in general. The initial margin depends on the PFE (cf Sect. 28.5.1) of the TRS. An efficient way to reduce initial margin consists of resetting the P/L regularly, e.g., monthly, in which case, the swap is called a *resettable TRS*.

30.6 Pricing and Hedging

For single underlier products, PDE method is more robust and efficient. For multi-asset products or multi-factor models, Monte Carlo is the usual numerical method. The pricing of multi-underlier callable and puttable structures requires American Monte Carlo algorithms (see Sect. 21.7). The Least-Squares method proposed by Longstaff and Schwartz (see Sect. 21.7.2) is widely used in practice for issuer callable products. It provides a lower bound of the option value, which is conservative to the issuer because it implies that the issuer “buys” the call option at a lower price. For a puttable structure, upper bound methods (see Sect. 21.7.3) should be used as the issuer “sells” a put option to investors.

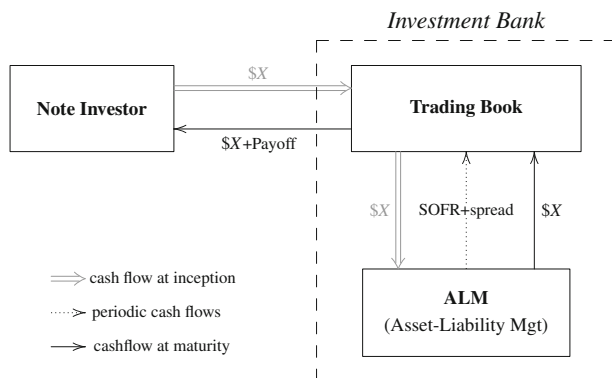
The standard model for equity-linked products is the local volatility model, which is used for the majority of the products. Stochastic volatility models are used for products with forward skew sensitivities or volatility convexity. For interest rate products requiring term structure modelling, the LGM model with one or two factors

and the BGM model are often used in practice, while models with more factors are needed for the products with more complex risks. For foreign exchange rate linked structures, stochastic volatility and local volatility models are frequently used in the market. Model parameters are calibrated with liquid market traded instruments. Hedging costs are considered in pricing by shifting the parameters on the conservative side.

The nature of structured products often introduces gap risk and correlation risk that are difficult or impossible to hedge. The techniques to mitigate these risks are explained in Chap. 20. A trading book for structured products usually has a modest vega limit (e.g., USD1.5m), which leads to frequent mandatory vega-hedging. Some structures, e.g. snowball autocallable structure, have volatility convexity issue that makes trading to buy volatility when it is high and to sell the volatility when it is low during the process of vega-hedging. Unfortunately, there are no liquid instruments for hedging volatility convexity, which limits the sales volume of such products.

30.7 Funding Management

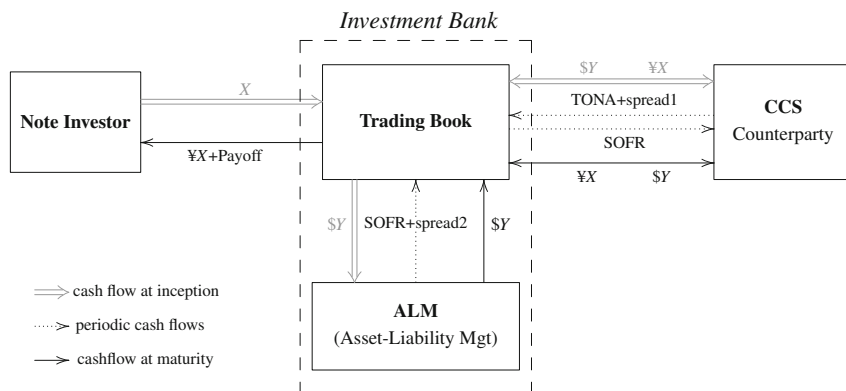
Proceeds of structured notes and certificates are typically managed by the ALM (asset liability management)/Corporate Treasury⁵ of the issuer. Trading book receives the interests of the proceeds from ALM. The following diagram illustrates a typical funding management for structured notes in USD.



ALM manages mainly the domestic currency and eventually some other major currencies (e.g., USD, EUR). If a note denomination currency is not managed by

⁵ The mission of ALM for a financial institution is to manage the financial institution's liquidity, funding, balance sheet and other financial resources/risks. For some banks, Treasury focuses on short term part (up to 1 year) while ALM focuses more on medium and long term financing needs and financial risks.

ALM, trading needs to make a CCS (cross currency swap) to convert the currency into a major currency acceptable by ALM.



30.8 Back-Testing

Back-testing (also known as historical simulation) consists of simulating the return of a structured product by applying the real historical prices of the underlier. It is often included in the product brochure or provided to potential clients at their request as a reference for the product's performance if it had been traded in the past.

The time window for back-testing usually starts from the earliest historical data (or from a meaningful start date if the data history is too long to be relevant). A typical back-testing process is described as follows:

- for the first date of the considered period, a back-test is realized by assuming that the structured product has started at that date. One simulated return of the product is obtained;
- then, the next date will be chosen as the start date for the structured product and another simulated return will be obtained;
- this process is repeated until the expiration date of the product becomes the current date (in which case, the process cannot continue any further).

A distribution is hence obtained from the above “historical” returns of the product. It is customary to show the following back-testing results:

- average IRR (internal rate of return), worst IRR, best IRR;
- return distribution chart illustrating frequencies for return levels.

In the product marketing materials, it should be highlighted that “past performance is not a good indication for future performances”. Unfortunately, this statement proved to be true in real-life cases.

Chapter 31

Popular Option Based Structured Products



As investors' demand continues to grow and hedging capabilities of investment banks continue to improve, new structured products are constantly being introduced to the market. Some of the popular structures are presented in the following sections. Categorizing these products can be challenging, as a structure may be applied to different asset classes and may offer different types of returns depending on the specific parameters. For example, Target Redemption mechanism-based products designed for FX underliers can also be applied to equities.

In this chapter, we organize the structures by underlying asset class rather than by their technical features, return profiles, or suitable market conditions. This categorization follows the typical business organization in the investment banks offering structured products.

The product names are not standardized and are given by the issuers: some suggest technical features (e.g., Fixed Coupon Callable Note) while others are used for marketing purposes (e.g., Himalaya, Athena, etc.). The same structure may also be named differently by different product issuers.

31.1 Equity Structured Products

Equity structured products provide a diverse range of investment solutions and instruments. They are popular among retail investors, who often use them to monetize their views on specific stocks or the stock market as a whole. Moreover, the large pool for underlying assets and the ease of equity derivatives hedging implementation make it relatively easy to create a wide range of product variations with equities.

31.1.1 Equity Linked Note (ELN)

Equity Linked Note (ELN) is one of the most popular short-term wealth management products. It aims to offer enhanced yield to investors who are willing to take on downside risk associated with equities. ELN was previously introduced in Chap. 11 as a typical example of a non-principal protected product (see Sect. 11.3).”

31.1.2 Fixed Coupon Callable Note (FCN)

A fixed coupon callable note (FCN) provides the investor with fixed periodic coupon payments. The underlier in the product is typically the worst-performing one in a basket of two or three equity assets. FCN has an early termination feature: if the underlier’s price is at or above the knock-out barrier on any observation date, the note will be early redeemed at its par value (i.e., denomination or face value). If the note is not early redeemed and reaches its maturity,

- if the underlier has not underperformed beyond strike level, the investor will redeem the note at par;
- otherwise, the investor will receive the worst performing stock at strike price.

Refer to Table 31.1 for an illustrative example of an FCN.

Table 31.1 Illustrative example of FCN (Fixed Coupon Callable Note)

Note tenor	12 Months
Denomination (N)	HKD 1 m
Issue price	$100\% \times N$
Coupon	12% p.a. payable quarterly
Observation dates	The end of each quarterly observation period
Underlying stocks	S^1 and S^2
Underlier performance at time t (P_t)	$P_t = \min \left\{ \frac{S^1_t}{S^1_0}, \frac{S^2_t}{S^2_0} \right\}$, which is the worst of the basket
Initial stock price	S^1_0, S^2_0 , fixed at the inception of the trade
Strike (K)	70%
Knock-out barrier	100%
Early termination	If the stock price has breached Knock-out barrier on any observation date, the Note will be early redeemed at par
Maturity payoff (if not early terminated)	If $P_T > K$, noteholder will receive N; Otherwise, noteholder will receive $N/(K S^*_0)$ stocks of S^* (S^* is the worst performing stock at T) If the currency of S^* is not the currency of the note, the number of stocks will be $X_T N/(K S^*_0)$, X_T is the units of the stock’s currency per note’s currency.

Fixed coupon callable notes are becoming increasingly popular as a yield enhancement product, particularly in low volatility market environments, where it can be challenging for equity linked notes (ELNs) to offer attractive yields. FCNs typically have longer tenors (ranging from 6 months to 2 years) compared to ELNs. In addition to their other benefits, fixed coupon callable notes can also be used by investors who need to generate regular cash flows to pay interest on financing. For example, private banking clients may use credit facilities, collateralized by their assets, to make investments. The credit interest may be covered by the regular coupon payments from an FCN. This gives FCNs an advantage over snowball structures that do not pay coupons before termination.

From a technical perspective, the underliers that are effective to achieve high coupons with low strikes are baskets of stocks with high volatility, low correlation, high dividend yield, and high repo rate. Increasing the number of stocks in a basket can lead to higher coupon levels. If the note and underlying assets are not denominated in the same currency, the correlations between the foreign exchange rate and the assets also impact the conditions of the structure. For more information about the correlation impact on drift or “quanto forward,” readers can refer to Chap. 18, Proposition 18.2.

Further variations may be introduced to meet investors’ needs:

- ATM strike for the put, combined with a knock-in barrier to provide a cushion for downside risk;
- Knock-out barrier that steps down over time to make early termination easier (e.g., 100% for the first month, 99% for the second month. . .)
- coupons based on floating rate (for rising interest rate view), e.g., 3M SOFR + 10% p.a., etc.

31.1.3 Daily Range Accrual Callable Note (DAC)

Daily range accrual callable note (DAC) is a variation of FCN in which, instead of paying a predefined coupon, the coupon payable at the end of each period (e.g., i -th period) is determined according to the below formula

$$Coupon_i = R \times \frac{n_i}{N_i} \times \tau_i$$

where

R : nominal annual interest rate (e.g., 15% p.a.);

n_i : number of days in the period that the underlier performance $P_{t_i} \geq H$; H is the *Coupon barrier* that is often set at strike level for simplification;

N_i : total number of days in the period;

τ_i : day count fraction for the period.

31.1.4 Phoenix Callable Note

A *Phoenix callable note* carries the same downside risk as an FCN or DAC. On the earning side, the investor will receive either a higher or a lower coupon (often set at zero) at the end of each observation period, depending on the performance of the underlier. If the performance is at or above *Coupon Barrier* (e.g., 80%), the investor will receive the higher coupon. Otherwise, they will receive the lower coupon for that period. Compared to an FCN, the coupon level is higher due to the condition to receive it. The pricing and hedging for a Phoenix callable note are similar to those for an FCN, except that payoff smoothing must be applied to the coupon due to its digital feature. Table 31.2 contains an illustrative example of the structure.

Table 31.2 Illustrative example of Phoenix Callable Note

Note tenor	12 Months
Denomination (N)	HKD 1 m
Issue price	$100\% \times N$
Observations	Quarterly
Underlying stocks	S^1 and S^2
Underlier performance at time t (P_t)	$P_t = \min \left\{ \frac{S_t^1}{S_0^1}, \frac{S_t^2}{S_0^2} \right\}$, which is the worst of the basket
Initial stock price	S_0^1, S_0^2 , fixed at the inception of the trade
Strike level (K)	70%
Coupon barrier (H)	80%
Coupon (for the i -th period)	18% (Higher coupon), if $P_{t_i} \geq H$; 2% (Lower coupon), otherwise
Knock-out barrier	100%
Early termination	If the stock price has breached Knock-out barrier at any observation date, the Note will be early redeemed at par
Maturity payoff (if not early terminated)	If $P_T > K$, noteholder will receive N ; Otherwise, noteholder will receive $N/(K S_0^*)$ stocks of S^* (S^* is the worst performing stock at T) In case the currency of S^* is different from the currency of the note, the number of stocks will be $X_T N / (K S_0^*)$

31.1.5 Snowball Structure

Snowball structures are popular among retail investors, particularly in South Korea (where they are known as High-Five or ELS and usually have a basket of two underliers), Japan (where they are mostly based on a single underlier and known as Uridashi), Europe (where they are called Athena), and China. They are income products, sold as notes, certificates, or structured funds, which may early terminate if the underlier price hits a knock-out barrier (i.e., they are *autocallable*). Upon termi-

nation, the investor receives the principal plus a coupon equal to $n \times \text{Coupon Rate}$ where n is the number of observation periods until termination. This type of coupon is known as a *Snowball Coupon*. The investors take on downside risk through a short put option, often with a knock-in barrier (see Table 31.3).

Table 31.3 Illustrative example of an Autocallable Note

Note tenor (T)	3 years
Notional (N)	KRW20bn
Underliers	KOSPI200 Index, HSCEI Index
Performance (P_t)	Worst performer of the underliers at time t measured as $\frac{S_t}{S_0}$
Observation	Every 6 months, including the maturity date
Knock-out barrier	90–90–85–85–80–80, in % of initial index levels (stepping down every 6 months)
Knock-out event	if $P_t \geq$ Knock-out barrier at i -th observation date for the first time, Knock-out event is deemed to have occurred at the i -th period
KO coupon	6.5% p.a., i.e., $i \times \text{KO coupon}/2$ if Knock-out event occurred at the i -th period
Put strike (K)	100% (ATM)
Knock-In barrier	60%, daily observation on closing prices
Knock-In event	if $P_t < \text{Knock-In barrier}$ on any day, Knock-In event is deemed to have occurred
Knock-out termination	if Knock-out event occurred, the note will be (early) terminated immediately at $N(1 + \text{KO coupon})$
Maturity payoff	if Knock-out event has not occurred until maturity, then, – if Knock-In event has not occurred, the note will terminate at $N(1 + \text{Bonus Coupon})$ – if Knock-In event has occurred, the note will terminate at $N(1 - \max\{K - P_T, 0\})$
Bonus coupon	$3 \times 6.5\%$, if not Knocked-In and not Knocked-Out until maturity

“Bonus coupon” is optional. Careful readers may notice that the autocallable with bonus coupon feature offers a similar return to investors as an FCN under most scenarios of underlier performance.

The price discontinuity of the structure due to barriers can be smoothed out using techniques such as barrier shift and call/put spread, as presented in Sect. 20. However, the hedging of snowball structures can still be challenging due to the *volatility convexity* issue (see Remark 14.1 on *vanna* and *volga*), which may cause the product hedger to buy volatility when its level rises and sell it when it falls, e.g. if knock-in event occurred due increased volatility, the hedging issuer who had a flat vega position on the trade risks to buy back volatility at a high level from the market. While the local volatility model is widely used for pricing and hedging autocallable structures, stochastic volatility models better capture the risks from second-order derivatives involving volatility.

31.1.6 Issuer Callable Structure

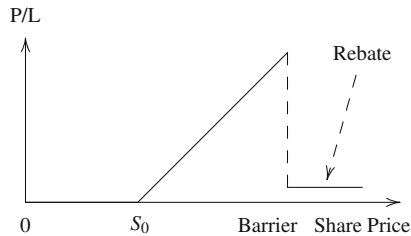
An issuer callable structure gives the issuer the right to early terminate the trade at predefined dates or time windows (see Table 31.4 for illustration). This enhances the return for the investor by adding the “option value” of the early termination right sold to the issuer. Callable structures are designed to provide an attractive benefit to investors upon early termination, which is perceived as a lock-in mechanism for gain. However, there is uncertainty as to when the product will be called. The Least-Squares American Monte Carlo method by Longstaff and Schwartz is commonly used to price issuer callable products with multiple underliers.

Table 31.4 Illustrative example of Issuer Callable Note

Note tenor	2 Years
Notional	TWD800m
Underlier performance	Worst performance of 5 stocks
High coupon	10% p.a.
Low coupon	2% p.a.
Coupon payment	Half yearly
First coupon date	High coupon is guaranteed
From second coupon date onward	→ High coupon: if no stock has ever traded below Barrier in the period → Low coupon: otherwise
Barrier	65%
Final redemption	→ 100%: if Barrier has never been hit (daily monitoring) → $\text{Min}\{100\%, \text{Final Price/Initial Price}\}$: otherwise
Issuer call	Every 6 months, Issuer has the right to early terminate the note at par

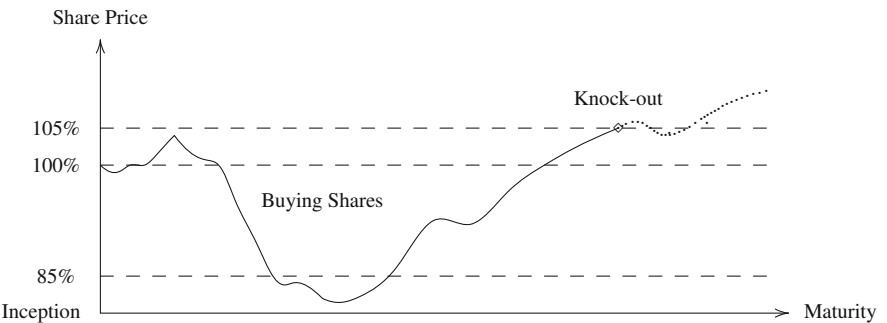
31.1.7 Shark-Fin

A *shark-fin* is a call option with an American or European up-and-out barrier. Due to its low cost, it is often used in principal-protected products, such as structured deposits in Korea with interest equal to the payout of the up-and-out call option on the KOSPI200 index. For equity underliers, the local volatility model is often used to price this structure.



31.1.8 Accumulator

An *accumulator*, also known as a *knock-out forward* or *BBM* (buy below the market), is a popular product for private investors in Asia. It is a contract between the investor and the issuer, where the investor buys a certain number of shares daily at a fixed strike price, at a discount to the initial share price (e.g., 80%), for the contract’s duration. Shares are delivered periodically (e.g., every week), and if the share price exceeds a predefined knock-out barrier level (e.g., 105% of initial share price) on any day, the trade is terminated early. The contract starts with zero value for both parties, like a swap contract. Investors typically post collateral for the transaction (e.g., 35% of the exposure) to cover the issuer’s credit risk exposure. See Table 31.5 for an illustration.



To achieve a lower strike price, investors often invest in a leveraged version of an accumulator, where they buy a leveraged number of shares (typically two times) if the stock price is below the strike price on the considered date. Another common feature is that if early termination occurs before a specified period, the investor is

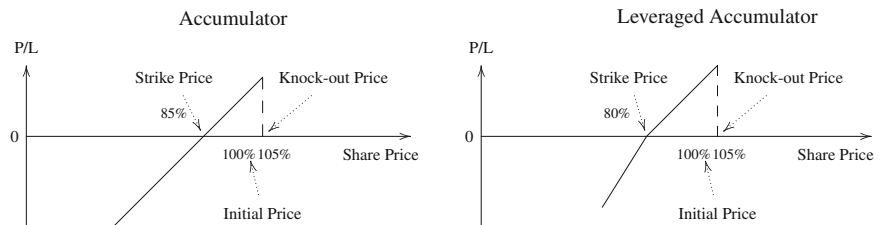
Table 31.5 Illustrative example of (non-leveraged) Accumulator

Maturity	52 weeks
Share name	[●]
Number of shares per day (n)	1000
Knock-out	105% of initial share price, observed daily If the share trades at or above 105% of the Initial Share Price on any day, the trade will be terminated
Strike price	85% of Initial Share Price Client purchases n shares every day at Strike price
Settlement	Settlement at the end of every week
Termination	If no Knock-out, the structure will terminate at the end of the Maturity

still guaranteed to buy the corresponding quantity of shares at the strike price for that full period. For example, the terms may be modified as follows:

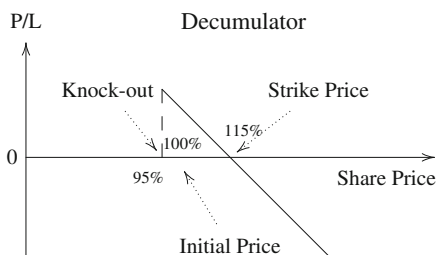
Strike price	82% of Initial Share Price For each business day, if the closing price of the share > strike price, Client purchases n shares at strike price; Otherwise, Client purchases $2 \times n$ shares at strike price
Guaranteed quantity	if early terminated before T_g where T_g is two weeks time after inception, Client will buy $n \times r$ shares at strike price where r is the remaining number of trading days until T_g

Technically, an accumulator is a strip of call and put options with a common up-and-out barrier. The non-leveraged version is a strip of forward contracts with an up-and-out barrier, while the two times leveraged version is a strip of option combinations of long 1 call and short 2 puts with the same up-and-out barrier. The local volatility model is typically used for pricing, and PDE or tree methods are more efficient than Monte Carlo simulation.



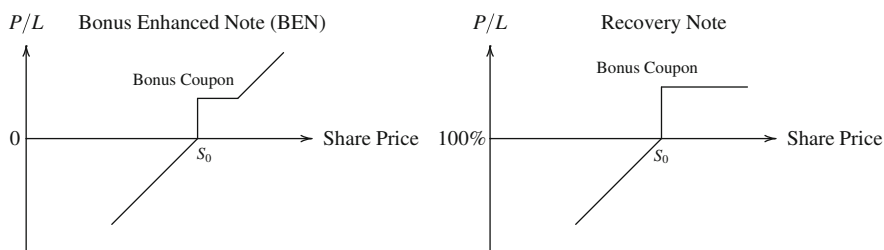
31.1.9 Decumulator

The opposite trade to an accumulator is a *decumulator*, which involves selling shares at a premium to the initial share price (also known as a *reverse knock-out forward* or *SAM* (sell above market)). The investor stops selling shares if the share price falls below a predefined knock-out barrier level.



31.1.10 Bonus Enhanced Note (BEN)

A *bonus enhanced note* (BEN) offers yield enhancement or participation according to the performance of the underlier. It is a bullish product offering unlimited upside on the *price return* of the underlier as shown in the following graph:



Actually, a single stock BEN needs to be structured with stocks showing low forward (high dividend and repo rate) as the investor forgoes the dividend and the security lending fees for the potential benefit shown by the diagonal triangle in the graphic. The tenor of the BEN should cover the ex-dividend date of the underlying stock. See Table 31.6 for illustration.

It is possible to structure a BEN based on the worst-performing stock in a basket of stocks. This allows to include non-dividend paying stocks, as the correlations between the stocks can help to lower the price of the BEN. For a basket BEN, there are several variations that may be offered, including a floor to limit downside risk, or a knock-in barrier for the embedded put.

Another variation, known as *recovery note*, consists of offering a digital payoff as shown in the graphic. As the name suggests, it is often proposed to investors

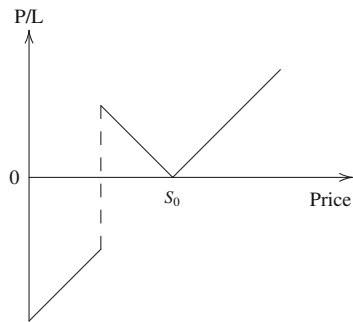
Table 31.6 Illustrative example of Bonus Enhanced Note

Note tenor	3 months
Notional (N)	USD1m
Underlying	A (dividend paying) stock
Coupon	6.5% flat (i.e., 26% p.a.)
Strike (K)	100%
Final redemption	If $\frac{S_T}{S_0} \geq K$, Investor will receive $N \times \max \left\{ 1 + \text{Coupon}, \frac{S_T}{S_0} \right\}$; Otherwise, Investor will receive a number of $\frac{N}{K S_0}$ stocks

who have already experienced losses from holding the underlying stock and wish to recover those losses if the price of the stock finishes above the strike price at maturity. In this case, the investor would sell their stocks and use the proceeds to purchase a recovery note.

31.1.11 *Twin-Win*

The basic version of a *twin-win* is a zero-cost option combination, which consists of a long position in a down-and-out straddle and a short position in a down-and-in put with the same barrier level (see graphics). The product is suitable for investors with a directionless but limited downside market view. The investor enjoys the absolute return if the downside barrier is not hit.



However, the investor may suffer substantial loss if the underlier price dropped beyond the barrier. The structure is cheap when the volatility is high. Hence, it is often proposed to investors when the market has gone through a significant correction. In such a situation, the volatilities are high, but people believe that the market may not suffer another big downturn beyond the barrier.

The same pricing model for standard barrier options may be used to price Twin-Win. The structure is quite sensitive to the volatility and the skew parameters.

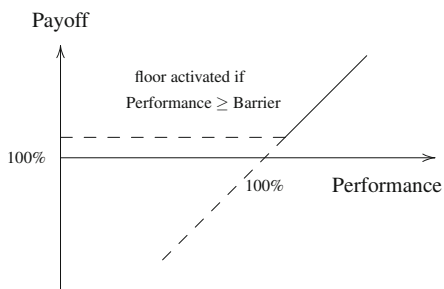
A *twin-win note* is a further tweak that is presented through the example in Table 31.7 with a single stock as underlier. This type of note can also be structured using a worst-of basket as underlier.

Table 31.7 Illustration of Twin-Win Note

Note tenor	6 months
Notional (N)	USD1mio
Underlier	Stock A
Performance (at t)	$P_t = \frac{S_t}{S_0}$, S_t is the closing price for date t
Observation dates	Monthly dates except the last one, i.e., 5 dates in total
KO barrier (H)	100%
KO event	Deemed to have occurred if $P_{t_i} \geq H$ on any observation date t_i
KO coupon	12.5% p.a., i.e., $12.5\% \times i/12$ if knocked out at the i -th observation date
Strike (K)	100%
KI barrier (L)	75%
KI event	Deemed to have occurred if on any date t , $P_t < L$
Early termination	Upon KO Event, the Note will be early redeemed at $N \times (1 + \text{KO Coupon})$
Maturity payoff	At maturity, if not early terminated, Investor will receive: $\rightarrow N(1 + P_T - K)$, if KI Event has never occurred; $\rightarrow N(1 + \max\{0, P_T - K\})$, if KI Event has occurred and $P_T \geq K$; $\rightarrow N/(K S_0)$ units of stocks, if KI Event has occurred and $P_T < K$

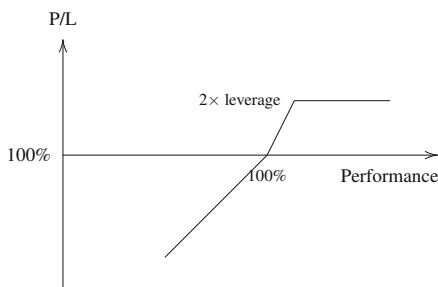
31.1.12 Tracker⁺ Note

This structure offers the underlier's performance with a conditional protection. The protection is activated if the underlier's performance has achieved a pre-defined barrier level during the tenor of the product. The underlier is usually a worst-of basket. A variation consists of offering an additional condition to trigger the protection with the "memory-touch" mechanism: e.g., the barrier event is deemed to happen if each underlier in the basket has breached the barrier at one time or another, but not necessarily at the same time, during the observation period.



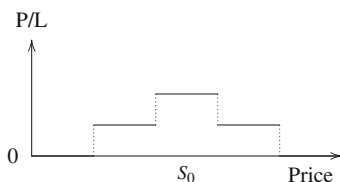
31.1.13 *Booster Note*

Booster note is a bullish product offering leveraged upside until the cap level. This is achieved by giving up the further upside, as shown in the graphic. A floor can be applied by lowering the cap level. If the underlier is a worst-of basket, it is possible to have a floor (e.g., 95%) and a high cap or even no cap.



31.1.14 *Wedding Cake*

The investor will receive a payoff contingent on the scale of the price change for the observation period. The observation may be either at the end of the period or monitored over the whole period.



The investor enjoys the absolute return if the price change is moderate. The risk is the potential low return if the price has moved with a larger scale than expected. This payoff is suitable for investors with a range-bound view. The product may be structured with the observation on the worst and the best performances of a basket of the underliers.

For pricing, we may use the same pricing model for standard European or American binary options. The discontinuity smoothing needs to be applied. Obviously, the structure is sensitive to the volatility of the underlier.

31.1.15 Stellar Note

This is an income structure based on *individually capped basket* of underliers. It pays regular coupons that are the positive basket performances at the observation dates. For investors who are bullish for the long term, but concerned about near-term market volatility, this structure can be an alternative to fixed-income products. See Table 31.8 for an Illustrative example of stellar Note.

Table 31.8 Illustrative example of Stellar Note

Note tenor	10Y
Notional (N)	EUR10mio
Underlying indices	$S^1 = \text{EURSTOXX50}$, $S^2 = \text{HSCEI}$, $S^3 = \text{SPX}$
Coupon payments	Yearly
Coupon(1)	3.5%
Coupon(i), $i = 2, \dots, 10$	$N \times \max\{0, P(t) - K\}$
Performance $P(t)$	$\frac{1}{3} \sum_{i=1}^3 \min \left\{ \frac{S_t^i}{S_0^i}, \text{Cap} \right\}$
Cap	106%
Strike (K)	100%
Final redemption	At par

31.1.16 CBBC: Callable Bull/Bear Contract

CBBC (callable bull/bear contract) is a popular derivative product listed on the HK Stock Exchange. It is a down-and-out call option or up-and-out put option with a deeply in the money strike. The barrier is of American style, which helps to reduce the *time value* of the option, resulting in a delta value close to one. It is a leveraged product for trading that is often sold before the expiry. The drawback of CBBC is the early termination risk that results in a loss to the option owner. There are two types of CBBC:

- N -CBBC: “barrier = strike” and the payout at barrier breach (i.e., rebate) is zero;
- R -CBBC: “barrier” is between “strike” and “spot price” at issuance; The rebate upon the barrier breach is $\max\{S_t^\ell - K, 0\}$ for Bull CBBCs and $\max\{K - S_t^h, 0\}$ for Bear CBBCs, where S_t^ℓ and S_t^h are respectively the minimum and maximum trading price of the underlier between barrier breach time and the next trading day.

The pricing of CBBC follows the same methodology as any other similar barrier options. Table 31.9 provides a product example.

Table 31.9 Illustrative example of CBBC

Product	Type R Bull CBBC
Expiry (T)	182 days
Underlier	Stock XYZ
Spot price (S_0)	\$120
Entitlement (α)	110 : 1; (also known as <i>conversion factor</i>) for scaling the contract size
Strike (K)	\$100
Call price	\$105, which is the down-and-out barrier level
MCE (mandatory call event)	Barrier breach event that leads to the early termination of the contract
Residual Value	Rebate upon MCE, which is $\max\{S_T^\ell - K, 0\}/\alpha$, where S_T^ℓ is the minimum trading price of XYZ between MCE and the next trading day
Final Payoff	If there is no MCE, the final payoff is $\max\{S_T - K, 0\}/\alpha$, where S_T is the closing price at T for the stock
Contract price (P_t)	\$0.1908; market quoted price of the contract
Funding cost (fc)	\$0.009 (1.99% p.a.); fc is implied from P_t by the relation: $P_t = \text{intrinsic}/\alpha + fc$ where $\text{intrinsic} = \max(S_t - K, 0)$; The rate 1.99% is the <i>financial ratio</i> (fr) s.t. $fc = K \times fr \times \frac{T}{365}/\alpha$

31.1.17 Cliquet Option

A *cliquet option* uses the future market prices of the underlier as references (e.g., option strike levels) in the payoff, for example, the call option on the sum of the intrinsic values of forward starting options. The cap and floor are often applied on the intrinsic values in a cliquet option as illustrated by the following example:

$$\max \left\{ 0, \sum_{i=1}^{12} \max \left\{ -4\%, \min \left\{ 2\%, \frac{S_i}{S_{i-1}} - 1 \right\} \right\} \right\},$$

where S_i is the price of the underlier at the end of the i -th month.

A cliquet option's premium can be very low if the cap level is low. It is sometimes used to structure products with tiny premium amount to spend. However, for a cliquet structured product to yield a good return, the underlier's price needs to consistently perform in the right direction during the tenor of the product.

The price of a cliquet option is sensitive to forward skew due to the cap and floor levels applied on forward starting options. The local volatility model is not suitable to price cliquet options because it does not preserve the forward skew of volatilities. Actually, the local volatility surface tends to flatten for the long-term part. More advanced models with volatility dynamics, e.g., stochastic volatility models or jump-diffusion models, are often used in practice to price cliquet options. A cliquet option often implies negative volatility convexity exposure to the issuer, which makes the hedging difficult. Investment banks are cautious about offering this type of product.

31.1.18 *Himalaya*

Initially created by Société Générale around the year 2000, the Himalaya structure is a typical example that demonstrates the flexibility and openness to imagination in financial engineering. The principle of the structure can be described by the following illustrative example of its basic version:

1. Assume 5 yearly observation periods and initially an underlying basket of 5 stocks.
2. At the end of each yearly observation period except the last one:
 - record the performance of the best performing stock in the basket, and then,
 - remove this stock from the basket for the following observation periods
3. For the last observation period, there will be only one stock remaining in the basket. Its performance is then recorded for the last period.
4. The payoff of the structure for the investor is the call option on the arithmetic average of the recorded performances.

In a Himalaya structure, the best performances are locked in at each period. However, if the performance in the last period is very low, it may drag down the final payoff. Since the performances at the observations are always measured against the initial stock prices, the performance for the last period might be significantly negative.

For equity-linked products, the local volatility model with Monte Carlo implementation can be used. Removing a stock may create a discontinuity, but it is usually small and can be neglected in pricing and hedging.

31.2 Fixed-Income Structured Products

Fixed-income derivatives and structured products offer investment and hedging solutions linked to rates and credit for both asset and liability management. Fixed-income derivatives such as interest rate swap, swaption, cap/floor, and CDS are actively used by financial market professionals and money managers. Various structured products have been introduced for financial institutions, corporates as well as retail investors. The tenors of fixed-income products are generally longer than those for equity and FX structured products.

31.2.1 *Range Accrual*

A (daily) range accrual (also called *corridor*) structure has a number of observation periods (e.g., every 6 months) until maturity of the trade. At the end of each period,

the investor receives a coupon determined by the following rate:

$$Coupon_i = R \times \frac{n_i}{N_i} \times \tau_i$$

where

R : nominal interest rate (e.g., 8% p.a.);

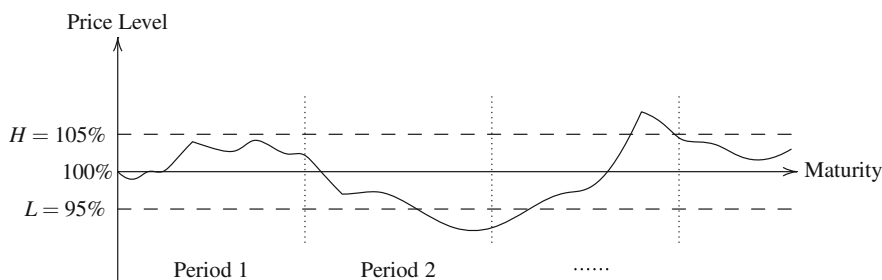
n_i : number of days during the period that the closing price of the underlier has been within the predefined range $[L, H]$;

N_i : total number of days in the considered period;

τ_i : day count fraction for the considered period.

Although the structure is introduced in this section, it is applicable to all asset classes: rates, equities, FX, and commodities. For equity (and some commodities) underliers, it is normally structured with a one-sided barrier (equivalently, the upper barrier is set to $+\infty$). The reason is that people often have bullish views on these underliers due to their natural growth perspective. Daily range accrual is also a popular structure for hybrid products where investors have views on assets from different asset classes (e.g., equity, rate, and FX).

For a basket underlier, range accrual products can be structured with the worst performance for the lower barrier observation and the best performance for the upper barrier observation. This means that the performances of all underliers in the basket must stay within the range for the accrual. With the same option price, the basket version has wider range levels than the single underlier case.



A daily range accrual option can be decomposed into a sum of binary options with maturities on each observation date. The short party of a range accrual option is buying volatility (and selling correlation for the basket version) in general. The pricing model for binary options is applicable for range accrual options.

There are a few other interesting variations of range accrual structures, which can be seen as extensions of barrier options for smoothness:

- full coupon lock-in: as soon as the number of accrued days reaches a certain threshold, the full coupon will be paid to the investor;
- “accrual/decruel” structure: it adds a “decruel” feature, which reduces accrued days if the underlier’s price is outside the range. For instance, we may design a swap paying a coupon of $\max \left\{ -2\%, R \times \frac{n_i - 3 \times (N_i - n_i)}{N_i} \right\} \times T_i$. This decruel feature helps to enhance the level of $R\%$ when a normal range accrual structure is not attractive enough. The payoff is less smooth than a normal daily range accrual, but it is smoother than a binary coupon observed only at the period end.

31.2.2 Inverse Floater

The basic payoff of an *inverse floater* is

$$N \times \max\{0, K - L \times R_T\},$$

where K is the strike, L is the leverage factor and R_T is the floating rate reference (e.g., 3M SOFR) level at maturity date T . It is clearly a European put option on floating rate.

31.2.3 Zero Coupon Callable Note

As indicated by its name, a *zero coupon callable note* is a note that pays no coupon until maturity or early termination. In this structure, the note investor is short a call option (Bermudan option) in order to earn a higher return than the fixed interest rate offered by the same issuer for the same maturity. We illustrate the structure with an example (see Table 31.10).

Table 31.10 Illustrative example of a 10Y (1Y Non-Call) Zero Coupon Callable Note

Note tenor	10Y
Denomination (N)	USD 100,000
Issue price	100% of denomination
IRR	4.2% p.a. (IRR for a non-callable ZC note is 2.8%)
Early termination	At the discretion of the issuer on any Callable date
Callable date	Quarterly, starting from year 2
Early redemption	$N(1 + IRR)^t$, where t is the number of years from inception to early redemption date
Final redemption	$N(1 + IRR)^{10}$

31.2.4 Reverse Convertible Note on Rate

The reverse convertible note for equities (i.e., ELN) has been extended to rate underliers recently. We show it through an example in Table 31.11.

Table 31.11 Illustrative example of Reverse Convertible Note on USD CMS (constant maturity swap) 10Y

Note tenor	1 month
Denomination (N)	USD 100,000
Issue price	100% of denomination
Underlying	USD CMS 10Y
Initial level of CMS 10Y (R_0)	1.95%
Coupon	5% p.a. payable at maturity
Strike (K)	75.5% (or CMS 10Y level 1.4723%)
Final USD CMS 10Y (R_T)	USD CMS 10Y level observed at maturity
Final redemption	If $R_T \geq K \times R_0$, Investor will receive N otherwise, Investor will receive $R_T \times \frac{N}{K \times R_0}$ floored at zero

With a similar analysis for ELN, we can find that the final redemption is $N \times (1 + \text{Coupon}) - N \times \min \left\{ 1, \max \left\{ 0, \frac{1}{K} \left(K - \frac{R_T}{R_0} \right) \right\} \right\}$. Hence, the investor is short a put option on interest rate. Since the underlying interest rate may become negative, the option has a cap to limit the maximum loss to N .

31.2.5 Yield Spread Structure

A *yield spread trade*, also known as a *steepener*, is linked to a CMS spread, the difference between two CMS rates of different maturities (e.g., 20Y CMS rate minus 2Y CMS rate). It provides opportunities to generate return or to hedge duration mismatch by taking exposure to the yield difference between two maturities. In the market, it is common to see spread trades with long tenor and issuer callable feature. See Table 31.12 for an example.

For European CMS spread options, active broker markets exist for some pairs. The modelling can be done on the two underlying CMS rates separately. A copula may be used to take into account the swaption volatility smiles of the two CMS rates (see [6, 16]). For the structure with callable feature, interest rate term structure modelling is required. At least two factors for the rates are necessary to capture the yield spread dynamics and stochastic volatilities.

Table 31.12 Illustrative example of Callable Steepener Note

Note tenor	5 Years, subject to early termination
Denomination (N)	USD 20mio
Issue price	$N \times 100\%$
Final redemption	N (if not called by the issuer before the maturity)
Underlying	USD 10-year CMS rate – 2-year CMS rate
Quarterly coupons	$N \times \min\{6\%, \max\{0, 5.6 \times (10Y \text{ CMS rate} - 2Y \text{ CMS rate})\}\}/4$, if the note is not early terminated. (yield for a normal note would be 2.8%)
Early termination	At the end of each quarter, note issuer has the right to redeem it at par

31.3 Foreign-Exchange Structured Products

Foreign exchange instruments and structured products are actively used by financial institutions and corporates for hedging and investment. Retail investors often use FX products for yield enhancement.

31.3.1 Dual Currency Investment (DCI)

It is a short-term (1 week–12 months) product for yield enhancement that involves a base currency and an alternative currency. The investor of a DCI with initial investment in base currency believes that the alternative currency will not depreciate against the base currency and wants to monetize this view for an enhanced yield. At maturity of the DCI,

- if the FX spot fixing at maturity is equal or higher than *Strike* (expressed as units of alternative currency per base currency): the investor will receive the total amount of principal plus interest in base currency;
- otherwise, the investor will receive the total amount of principal plus interest in alternative currency converted with *Strike*.

See Table 31.13 for an Illustrative example.

In a DCI, the investor's exposure is short a put option on alternative currency at strike, which is shown by the final value measured in the same currency (that is USD in our example):

$$\begin{aligned}
 & N(1 + C)\mathbf{1}_{X_T \geq K} + (X_T N(1 + C)/K)\mathbf{1}_{X_T < K} \\
 &= N(1 + C)\left(1 - \frac{1}{K}(K - X_T)\mathbf{1}_{X_T < K}\right) = N(1 + C)\left(1 - \frac{1}{K} \max\{0, K - X_T\}\right).
 \end{aligned}$$

Table 31.13 Illustrative example Dual Currency Investment (DCI)

Note tenor	2 months
Notional (N)	USD 100,000
Base currency	USD
Alternative currency	JPY
Spot reference (i.e., initial spot)	111.00
Strike (K)	1/113.01 (USD units per JPY)
Coupon (C)	1% (6% p.a.)
Final spot X_T	Expressed as USD units per JPY
Final redemption	If $X_T \geq K$, $N(1 + C) = 101,000$ in USD otherwise, $N(1 + C)/K = 11,414,010$ in JPY

31.3.2 *FX Accumulator*

Accumulators are actively traded on foreign exchange rates, similar to equity accumulators. Active trade tenors include 3M, 6M, and 1 year. See Table 31.14 for an example of a boosted product.

Table 31.14 Illustrative example of Boosted EUR/USD Accumulator—Daily Accrual, Monthly Settlement

Tenor	12M
Daily notional	N euros
Currency pair	EUR/USD
Spot ref	1.1780
Strike (K)	1.1490
Accrual	On each business day, Investor buys EUR/USD at Strike. The quantity is N if the fixing price is above Strike; Otherwise, the quantity will be $2 \times N$
Settlement	Monthly
Knock-out rate	1.2080

31.3.3 *Target Redemption Forward (TARF)*

Target redemption forward (TARF or TRF) is a very popular structure for FX investors. The special feature of a target redemption structure is that when the accumulated profit or intrinsic value for the investor reaches a certain level, the entire structure will be terminated early. See Table 31.15 for an illustration of a boosted TRF.

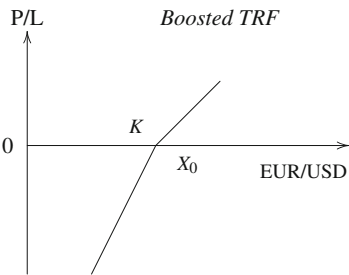


Table 31.15 Illustrative example of a Boosted TRF

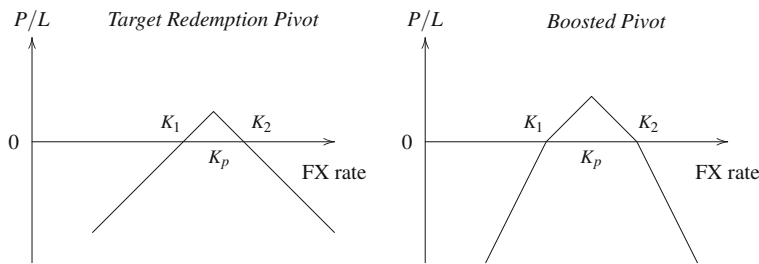
Tenor	12M, monthly settlement
Notional per fixing (N)	EUR200k
Currency pair	EUR/USD
Spot ref	1.1780
Strike (K)	1.1410
Intrinsic value	$\max\{X_t - K, 0\}$
Frequency	Monthly fixing and monthly settlement
Target	10 big figures (i.e., 0.1000), exact (meaning that the total payout at knock-out is exactly 10 big figures)
Knock-out event	If accumulated Intrinsic value $\geq$ Target
Payoff for each fixing (before Knock-out)	If $X_t \geq K$, investor buys N at K ; Otherwise, investor buys $2 \times N$ at K

Where X_t (Expiry Reference Rate) is the fixing at the monthly observation date

A TRF allows the investor to buy a certain currency at a discount (or to sell at a premium) to the initial spot until the early termination, which is triggered by the target redemption mechanism, or its maturity. A variation consists of using the accumulated number of times that the intrinsic value has been positive instead of the accumulated intrinsic value. A target redemption note is known as a *TARN*.

Variations of TRF: Pivot Structures

One popular variation is the *pivot* structure, where the investor buys the foreign currency at a lower strike and sells it at a higher strike. It is suitable for clients who believe that the FX rate will trade within a range for the considered tenor. Target redemption is a natural knock-out mechanism for a pivot structure. Similar to a boosted TRF, a boosted pivot TRF allows the investor to have wider strikes.



We note that in the graphical illustration for pivot structures, we have chosen $K_p - K_1 = K_2 - K_p$ (a butterfly structure), while in general, $K_p - K_1$ may differ from $K_2 - K_p$. Table 31.16 provides an example of pivot TRF.

Table 31.16 Illustrative example of Pivot TRF

Tenor	12M, monthly fixing and settlement		
Notional	AUD100k per Month		
Currency pair	AUD/USD		
Spot ref	$X_0, K_1 \leq X_0 \leq K_2$		
Buy strike	K_1	Sell strike	K_2
Pivot strike	K_p	Target	0.2000 Exact
Intrinsic value	if $X_t < K_p$: $\max\{X_t - K_1, 0\}$; if $X_t \geq K_p$: $\max\{K_2 - X_t, 0\}$		
Knock-out event	if accumulated Intrinsic value $\geq$ Target		
Settlements	if $X_t \leq K_p$: investor buys at K_1 ; if $X_t > K_p$: Investor sells at K_2		

Where X_t (Expiry Reference Rate) is the fixing at the monthly observation date

There are many other variations for pivot target redemption products that are actively traded in the market. For instance, the one-sided pivot shown in Table 31.17.

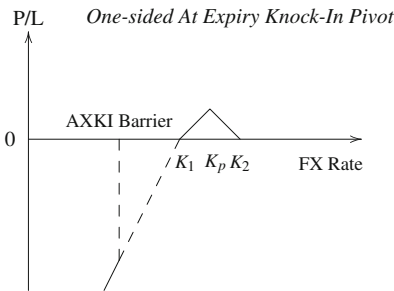


Table 31.17 Illustrative example of one-sided AXKI Pivot TRF

Tenor	12M, monthly settlement		
Currency pair	AUD/USD		
Spot ref	1.0359		
Lower strike (K_1)	0.9900	Upper strike (K_2)	1.0700
Pivot strike (K_p)	1.0300	AXKI barrier	0.9100
Target	0.1500		
Intrinsic value	if $X_t < K_p$: $\max\{X_t - K_1, 0\}$		
	if $X_t \geq K_p$: $\max\{K_2 - X_t, 0\}$		
Knock-out event	if accumulated Intrinsic value $\geq$ Target		
Settlements	if $X_t \geq K_1$ and $X_t \leq K_p$: investor buys AUD1mio at K_1		
	if $X_t > K_p$ and $X_t \leq K_2$: investor sells AUD1mio at K_2		
	if $X_t < \text{AXKI}$: investor buys AUD2mio at K_1		
	Otherwise, no settlement		

Where X_t is the fixing at the monthly observation date (i.e., Expiry Reference Rate)

31.4 Commodities Structured Products

In general, commodities-linked structured products are less traded compared to those linked to other asset classes. Precious metals and oil (Brent or WTI) are more commonly used as underliers than other commodities such as industrial metals and soft commodities. Here are a few examples of structured products linked to commodities:

- reverse convertible linked to the price of oil front-month futures: 6M USD note, put strike 100%, American knock-in barrier at 72% of initial spot, 10% p.a. coupon;
- DAC (daily range accrual callable: 2Y USD note, worst-of basket gold/silver/platinum, nominal Coupon 8%, accrual barrier = put strike = 87%, autocall barrier 100%.

31.5 Hybrid Structured Products

When the underliers belong to different asset classes, the structured product is called a hybrid product. This allows the investor to take advantage of their views on multiple asset classes. The structures for basket underliers presented so far are generally applicable to hybrid underliers. In the following, we will show a few hybrid structures for illustration purpose.

31.5.1 Best-of Profile

A *best-of profile* delivers the best performance of different portfolios. Each portfolio represents a specific investment profile. The profile of a portfolio can be achieved by

- choosing the underliers from different asset classes
- weighting the underliers differently according to the riskiness or geographic exposure.

Table 31.18 Illustrative example of Best-of-Profile

Structure	Option
Notional	USD20mio
Tenor	3 Years
Portfolios	Aggressive Portfolio (P_1): 50% Equity + 30% Commodities + 20% Bond Balanced Portfolio (P_2): 33% Equity + 33% Commodities + 34% Bond Conservative Portfolio (P_3): 20% Equity + 30% Commodities + 50% Bond where Equity = 33% S&P Index + 33% EuroSTOXX + 34% HSCEI Commodities = GSCI Commodities Index Bond = iBoxx Investment Grade Bond Index
Maturity payoff	$\max\{0, \max\{\text{Perf}(1), \text{Perf}(2), \text{Perf}(3)\} - 1\}$ where $\text{Perf}(i) = P_i(\text{final})/P_i(\text{initial}), i = 1, 2, 3$

A best-of profile can be sold either as an option (see Table 31.18) or as a structured note. It is suitable for uncertain market views by offering the best performance among investment portfolios. However, the expected gain is often limited because the portfolios tend to have low volatility.

In general, the product issuer is buying correlation between the portfolios but selling correlation between the underliers within a portfolio. So, the correlation shifts should be applied accordingly in pricing.

31.5.2 Callable Range Accrual

Table 31.19 provides an example of a *callable range accrual* with interest rate and equity as underliers. An interesting feature to observe is that the ranges for both equity and interest rate are set to their respective levels.

Table 31.19 Illustrative example of Callable Daily Range Accrual

Note tenor	10Y, NC1Y (no call for the first year)
Notional (N)	USD20mio
Underliers	SPX Index, SOFR 3M
Equity condition	SPX Index closes at or above 50% of initial level
Rate condition	SOFR 3M fixing is below 5% (spot rate ref 1.34%)
Quarterly coupon	$N \times 3.35\% \text{ p.a.} \times m/M$, where m is the number of days both Equity condition and Rate condition are satisfied, and M is the total number of days in the period
Issuer call	Quarterly after year 1, issuer has the right to early redeem note at par
Final redemption	if not early redeemed, the final redemption is N

31.5.3 Gap Note*

A *gap* refers to a sudden large move of the underlier's price. In brokers market, it is sometimes possible to buy protections on gap risk of equity indices through a product known as a *stability note* or *gap note* (see Table 31.20 for illustration).

Table 31.20 Illustrative example of Gap Note

Note Tenor	6M
Denomination (N)	USD2mio
Issue price	$100\% \times N$
Underlying index (I)	S&P 500, I_t is the Index closing price at date t
Quarterly coupon	3M SOFR + 3.5%p.a., until maturity or early redemption event
Strike (K)	90%
Early redemption event	if $I_t/I_{t-1} < K$ on any business day t ;
Early redemption	$N \times (1 - \min\{1, 10 \times \max(0, K - I_t/I_{t-1})\})$
Final redemption	If not early redeemed: N

This structure cannot be delta-hedged as a normal option. The note investor takes the gap risk for income. The pricing is mostly driven by demand/offer. A jump-diffusion model may be used for the consistency of the prices across different strikes/tenors.

31.6 Fund Linked Structured Products

Structured products are mostly linked to mutual funds rather than hedge funds. Mutual funds offer daily liquidity for subscription and redemption, and can be sold to retail investors. They are highly regulated and have transparent investment objectives and risk management methodologies. To offer fund linked derivatives, the derivatives issuer must have an agreement with the fund manager to transact fund

units at NAV without subscription or redemption fees. It is customary for the fund manager to rebate part of the management fees on transacted fund units for delta hedging to the derivatives issuer as an incentive to increase the fund's AUM (assets under management). This rebate is fairly priced into the structured products, helping to improve the features.

Hedge funds usually do not provide daily liquidity, making it difficult to hedge options. However, it is possible to hedge the option on a managed account set up by the derivatives issuer to replicate the hedge fund. The hedge fund manager acts as the investment advisor, ensuring that the managed account contains the same assets as the hedge fund itself. Since the managed account is transparent with regard to its invested assets, the derivatives issuer can perform delta hedging on a daily basis by adjusting the assets in the managed account.

In general, fund-linked structured products favor those with smooth gamma, non-negative delta, and no discontinuity. To reduce the cost of a fund-linked derivative, a *synthetic dividend*¹ may be introduced to an accumulation share class of the fund. The payoff of the product is then linked to a newly created fund underlier with a synthetic dividend, which has a lower price due to its lower forward. The volatility market for funds is limited, so structured products often require a volatility target on the underlying funds (see 32.1). Derivatives issuers are reluctant to sell structures with a volatility target that may require them to buy volatility (e.g., in a call spread, where the short option at the upper strike requires the issuer to buy volatility for that strike). CPPI/TIPP structures, which do not require volatility hedging, are often used for funds, especially those with low volatility.

The maximum issue size for a fund-linked structured product is limited to a certain percentage of the AUM of the fund. If the delta becomes significant compared to the AUM of the fund, the product issuer (the hedging party) may gain majority control of the fund and may need to consolidate its investment in the fund into its balance sheet. In such cases, the issuer may choose to relinquish majority control by issuing an official notice or request.

In the following, we will present a few popular structured products linked to funds through examples. The most common purposes include principal protection, leveraged return and financing. Total return swap is an efficient derivative to gain exposure to funds' performances with leverage.

31.6.1 Bullish Note

This structure offers a geared upside at maturity with a participation rate (PR) that can be lower or higher than 100%. The underlier can be a single fund or a basket of funds, and a volatility target may be applied. For instance, the product example in Table 31.21 offers a leveraged upside due to its very low volatility. The volatility

¹ The concept of synthetic dividend is also applied to total return indices or systematic strategies.

target is not applied because of the availability of volatility hedging in the market for this particular fund. The embedded payoff of the structure is a leveraged call option.

Table 31.21 Illustrative example of Bullish Note

Note tenor	3 Years
Denomination (N)	USD100,000
Currency	USD
Issue price	100%
Underlying (S)	A fixed-income fund with low volatility
Performance (P_T)	S_T / S_0
PR	150%
Redemption	$N \times (1 + PR \times \max\{0, P_T - 1\})$

For non-principal protected products, an alternative payoff structure may be considered. It provides a geared upside, while the downside exposure is not geared, making it comparable to direct fund exposure. Additionally, a floor is often introduced in this type of structure, represented by the *maximum loss* in the formula below. This helps to limit the potential loss for the investor:

$$PR \times \max\{0, P_T - 1\} - \min(\text{maximum loss}, \max\{0, 1 - P_T\}).$$

31.6.2 Bullish Coupon Note

For investors requiring periodic payoffs, we present an example of bullish coupon note linked to the performance of a fund, as shown in Table 31.22. In this structure, the annual payoff is the performance of the fund, measured from the beginning and divided by the number of years that have passed.

Table 31.22 Illustrative example of Bullish Coupon Note

Note tenor	3 Years
Denomination (N)	USD100,000
Currency	USD
Issue price	100%
Underlying (S)	A fund with volatility target
Coupon for year i	$PR \times \max\left\{0, \frac{S_{t_i}}{S_0} - 1\right\} / i$
PR	100%
Redemption	At par

A popular variation to this structure consists of offering an annual coupon, which is the fund's performance at the year-end reduced by the coupons received

previously: $Coupon(i) = \max\{0, \frac{S_i}{S_0} - 1 - \sum_{k=1}^{i-1} Coupon(k)\}$ for $i \geq 2$. In essence, this payout is equivalent to providing the investor with the highest yearly performance, settled at different times.

31.7 Credit Linked Structured Products

31.7.1 Credit Linked Note (CLN)

A CLN linked to a single reference entity is a popular credit structured product sold to investors (see Chap. 8). CLN issuers often offer structures that are tailored to satisfy the client's need, and it is common to see zero recovery in the credit linked notes sold to individual investors, where the implied recovery rate is priced in the structure for yield enhancement.

In the years shortly before the financial crisis of 2008, CLNs linked to FTD (first-to-default) basket of reference entities were actively traded. The default event for FTD basket is the first credit event among all the reference entities in the basket, making default correlation an important factor for pricing such CLNs. However, after the financial crisis of 2008, implied default correlations increased significantly, making product features with first-to-default baskets less attractive to investors.

31.7.2 CLN on Credit Index

A CLN linked to a credit index is less risky compared to a single name CLN. This is because the credit index helps to diversify the risk across the constituent reference entities, reducing the default impact of any single entity. Additionally, the liquidity on credit index is generally better than the liquidity for a single reference entity. An example of a credit index linked note can be found in Table 31.23. For sophisticated investors, index single tranche linked products (similar to CDOs) can also be structured.

31.8 Equity Derivatives for Corporates

There are several popular strategic equity solutions using derivatives for the purpose of

Financing: equity linked financing structures can be used by corporates or their major shareholders. The margin loan and collar financing to be presented later belong to the type of financing known as *non-recourse* loan, meaning that the

Table 31.23 Illustrative example of Credit Index Linked Note

Note tenor	5 Years
Outstanding principal amount	Initial value is USD10,000,000
Underlier	iTraxx Asia ex Japan IG Index: 40 constituents, each with 2.5% weighting
Coupon rate	SOFR 3M + 1.5%
Quarterly coupon	Outstanding principal amount $\times$ Coupon rate $\times$ day count fraction
Credit event	Outstanding principal amount will be written off by 2.5% for each credit event (i.e., zero recovery). The coupon will continue to accrue based on the reduced Outstanding principal amount
Redemption	Outstanding principal amount

lender cannot go after borrower's other assets except those posted as collateral for the loan transaction.

Shares Buy-Back: there are different reasons for a corporate to buy back its own shares from the market. For example, corporates may need shares to pay the executive employees as part of their incentive compensation. New issuance of shares will dilute the value of existing shares, so corporates often choose to buy back shares from the market.

Share Disposal: corporates may need to sell large quantity of shares that they hold for strategic reasons.

31.8.1 Financing

Convertible Bonds (CB)

A company may raise financing through different channels such as direct capital increase, rights issue, preferred shares, bonds, and others. A *convertible bond* is an alternative to a cash bond, which gives bondholders the right to convert the CB into a fixed number of common shares during the conversion period. It is a hybrid structure of (credit bond + call), which often has the early termination feature with the bondholder's put option and the issuer's (soft) call option.

For investors of convertible bonds, there are several attractive features

- upside potential of the underlying stock
- downside protection when the underlying stock is performing poorly
- investment or arbitrage opportunity for equity option investors or credit investors

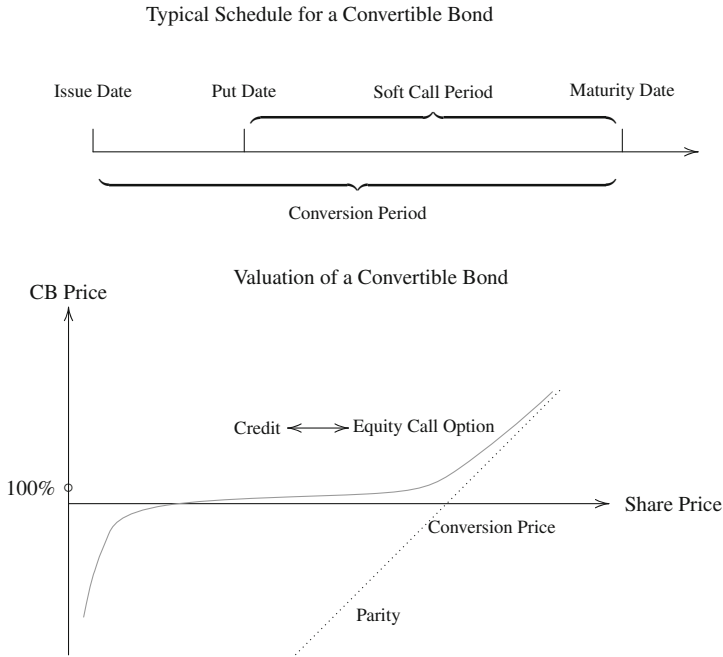
Hedge funds hold an important part of issued convertible bonds because this hybrid credit/equity instrument allows the implementation of various trading strategies. Convertible bonds offer exposure to the implied volatility of the underlying stocks, which are often not blue-chip stocks. For fast-growing companies, convertible

bonds are a preferred instrument for financing. The advantages for the issuer of a convertible bond include

- lower financing cost because the coupon offered by a convertible bond is lower compared to a normal bond
- delayed dilution for common stock and the earnings-per-share (EPS)
- issuer is selling at a premium to the prevailing stock price

A convertible bond is a hybrid structure involving equity and credit modelling. The local volatility model for equity part combined with a jump component for credit modelling is a natural approach. For interested readers, we refer to [3, 21].

In Table 31.24, we summarize the common features of convertible bonds, along with an example. The graphics provide further illustration of the schedule and valuation of convertible bonds.



To avoid dividend risk, it is common to assume zero dividend for convertible bonds in pricing and to apply an adjustment to the conversion price at each dividend payment using the method explained in Sect. 16.7.4 and Sect. 5.2.6. Following a corporate action, the conversion price will be adjusted from K to $\hat{K} = K(1 - d)$ and the share notional is adjusted from N/K to $N/[K(1 - d)] = N/\hat{K}$ while the cash notional N remains unchanged. Actually, the adjustment formulas as presented in Sect. 16.7.4 and Sect. 5.2.6 are often found in the prospectus of convertible bonds.

Table 31.24 Main features of a Convertible Bond

Typical features	Explanation	Example
Issuer		XYZ company
Underlying share		XYZ common share
Maturity	Tenor of the CB	5Y
Denomination	Face value of the CB	USD10,000
Coupon rate	Interest rate offered by the CB	2.75% p.a.
Conversion price	Price at which the convertible bond is converted into common stocks	10.76
Conversion ratio	Number of shares of a common stock received by the CB holder when the bond is converted; It is equal to Denomination/Conversion Price	929.368
Put	Bondholder's right to redeem the bond at a pre-determined price on a pre-determined date prior to maturity	On April 24, 2011, the bondholder will have the right to redeem the bond at 111.54% of the principal amount
Soft call	Issuer's right to call the CB if the underlying share price breaches a pre-specified upper limit for at least a pre-defined number of days during the soft call period	After April 24, 2011, the issuer has the right to early redeem the bond at 130% of denomination, if the share has been traded above 130% of conversion price for at least 20 during 30 consecutive trading days
Early redemption		100% of denomination
Final redemption		100% of denomination if no early redemption occurred

Margin Loan

A *stock margin loan* is a single trade based solution to obtain financing with the shares of a corporate or any other tradable shares as collateral. The ratio of the borrowed amount to the collateral value is called the loan-to-value (LTV). The LTV tends to be high when the collateral has low volatility and high liquidity. For stocks, the LTV often ranges from 30% to 60%. The size of the loan is conditioned by both the averaged daily trading turn-over volume (ADTV) and the free-float of the stock.

In a stock margin loan transaction, the margin call mechanism is defined in the contract. As soon as the collateral value drops by a certain threshold (e.g., 15%),² the margin call (mostly in cash) will be triggered. In case of non-payment for margin call, forced termination is applicable. The shares used as collateral are kept in a special account (e.g., escrow account). Hence, the legal ownership of the shares still

² The threshold is sometime expressed in terms of LTV, e.g., Initial LTV = 35%, Warning LTV = 45%, Top-Up LTV = 50%. Should the LTV exceed the Top-Up LTV, the borrower shall post cash or other acceptable assets with 5 days to reduce the LTV below Top-Up LTV.

belongs to the borrower. This leads to the following advantages of a stock margin loan to the borrowers:

- the voting right is retained by the borrower for the shares used as collateral
- there may not be regulatory disclosure requirement for the substantial share holders.

The drawbacks of a stock margin loan include:

- low LTV (e.g., below 50%).
- margin call that needs to be handled over the whole life of the loan.

Collar Financing

A *collar financing* is a financing trade that is applicable if the underlying share has sufficient liquidity to allow hedging of the embedded derivatives. The word collar is used due to the long put and short call options employed in the typical structure as explained below. Actually, the call option is not mandatory. It is introduced to reduce the cost and can take other forms (e.g., 50% geared call). However, the put option is mandatory as it “transforms” the credit risk into a market risk that can be hedged by the lender. In a typical collar financing trade, the borrower

- purchases a put option with a lower strike (e.g., 80%) from the lender
- optionally, sells a call option with a higher strike (e.g., 115%) to the lender to lower the total cost
- delivers shares to the lender as collateral
- obtains a loan (e.g., at a rate of $\text{SOFR} + 2.5\%$).

The lender can lend an amount up to the PV of the put strike. At maturity, the borrower pays back the loan and the lender returns the collateral that may be adjusted by the difference between the strike price and the prevailing share price (collar payoff).

A collar financing structure has several advantages over the margin loan structure for the borrower, including

- no margin call: this is because the value decrease of the shares in collateral is compensated by the valuation increase of the put option;
- high LTV (loan-to-value) : close to the put strike
- low financing cost: the call option lowers the financing cost and enables financing at a reasonable interest rate level;
- downside protection: the put option hedges the downside risk of the share price for the borrower.

Compared to a stock margin loan structure, the drawbacks of a collar financing are:

- upside share performance is capped at call strike
- voting right lost during the tenor of the trade

- potential disclosure requirement for substantial shareholders (according to the applicable regulation).

Actually, a collar financing may be structured and transacted as an *in-the-money call spread* (e.g., 80–115%). In this case, the “borrower” needs to sell the stocks and buy the in-the-money call spread with a fraction of the selling proceeds. The choice of the structure will depend on the context of the borrower and the lender. A *prepaid variable forward* is another contract that can be used for the trade implementation.

31.8.2 Shares Buy-Back and Disposal

Structures for Fixed Price but Uncertain Quantity

A corporate may use derivatives to buy back shares at a discount to the initial price. In the structures below, the price is fixed at a discount to the initial price at inception of the trade. However, the quantity that the corporate will purchase depends on the future market price of the underlying stock.

- put writing: the corporate sells an OTM put option, i.e., the obligation to buy shares at a strike price set at a discount to the initial price if the share price drops below the strike. The corporate gains the option premium. If the stock price has not increased more than the option premium in percentage, the corporate effectively lowered the buy-back cost compared to a straight purchase at the initial price.
- accumulator: the corporate buys a number of shares at a fixed price at a discount to the initial stock price every day. The trade will be early terminated if the share price trades above a pre-defined knock-out barrier on the upside.

Structures for Fixed Quantity but Uncertain Price

An *accelerated shares buy-back (ASB)* is a popular trade used by corporates, especially in North America, to buy back a fixed quantity of shares within a time window. The purchase price is uncertain but at a discount to the averaged value of the share for the purchase period. The trade can be described as follows:

- The corporate enters into a contract, at zero cost, with an investment bank to buy a fixed number of shares based on the average price over the accumulation period (equally weighted) with a discount (e.g., 1%). The reference price is usually VWAP (volume weighted average price).
- The investment bank buys shares gradually from the market, but has the discretion for the effective daily quantities and termination date. Actually, the investment bank tends to buy more when the price is low, and buy less (or even sell shares) when the share price is high. As a result, the investment bank

guarantees a final purchase price at a discount to the average price over the accumulation period (equally weighted).

This structure is a kind of American option. The quantities that the bank buys over the period are determined by the delta of the option. The vega hedging will also be necessary because otherwise, the bank would not be able to hedge the trade should the share price keep rising (implying very low realized volatility). A PDE pricer under the local volatility model may be used in practice.

Apart from the ASB trade, there is a systematic execution strategy that tends to outperform the averaged price during the buy-back period with equal weight. It consists of buying back the quantity that is determined by the running average of the share price:

- if the current price is higher than the running average, smaller quantity will be bought;
- if the current price is lower than the running average, greater quantity will be bought.

The solutions for shares disposal are the opposite ones compared to shares buy-back, i.e.,

- writing call instead of writing put
- decumulator instead of accumulator
- *accelerated shares disposal*, which allows selling a fixed number of shares at a premium of the average price over the disposal period (equally weighted).

31.8.3 Greenshoe (Over-Allotment) Option*

In the context of an initial public offering (IPO), a Greenshoe or “over-allotment” option is a physically settled call option granted to the underwriters by the issuer, giving them the right to purchase, after the offering date, up to the option notional (typically 15% of the total issuance) additional shares at the offering price. It enables the underwriters to support/stabilize the share price by overselling (shorting) the shares at the offering price and buying back from the market in large blocks if the share price falls, or from the issuer otherwise.

Taking the example of an IPO with an offering price at \$100, if the share price falls below \$100, the underwriters can buy shares at lower prices, e.g., \$90 on average, in the market to cover their short position and make a profit of \$10 per share. However, if the share price rises, the underwriters can exercise the option to buy shares from the issuer at \$100 for closing their short position.

The term Greenshoe comes from the name of the first company (Green Shoe Manufacturing) to use this mechanism as a commercial agreement for price stabilization. Although it is technically a type of call option, the risk-neutral hedging/pricing does not apply and the issuer does not normally hedge the option.

Exercises

1. For a structured note, there are time lags between the trade and issue date, final valuation date and maturity date (see Example 2.1). Now, consider an FCN product described in the table.

FCN Tenor	6 Months + 2 Weeks		
Trade date	03 Oct 2017 (Tue)	Issue date	17 Oct 2017 (Tue)
Final valuation date (T)	17 Apr 2018 (Tue)	Maturity date	24 Apr 2018 (Tue)
Denomination (N)	USD 100k	Issue price	100% of N
Coupon	1% flat paid at each monthly observation (i.e., ~12% p.a.)		
Put Strike (K)	To be calculated		
Underlying basket	Stock A (US stock) and Stock B (Japanese stock)		
Performance (P)	Worst-of basket		
Observation	17 Nov 2017, 19 Dec 2017, 17 Jan 2018, 17 Feb 2018, 17 Mar 2018, 17 Apr 2018		
Knock-out barrier	100% at each monthly observation date		
Spot reference— S_0	USD 50.00 for Stock A, JPY 20000.00 for stock B		
Final redemption	If $P_T \geq K$ then 100% of N , Otherwise, $X_T \times N / (K \times S_0^*)$ shares of S^* , where S_0^* is the initial price of the worst performing stock at maturity; $X_T = 1$ if S^* is US stock, otherwise, X_T is the prevailing USD/JPY exchange rate at T		
Issuer/Distributor margin	0.60%		

Use the BSM model and the below parameters to make Excel-VBA coding to

- (1) estimate Strike K with Monte Carlo simulation (at least 10,000 paths);
- (2) calculate the 95% confidence interval for issue price with strike fixed at 90%.

	Stock A	Stock B	USD/JPY	USD	JPY
Volatility	38%	40%	9.50%		
Dividend rate	0	0			
Repo (continuous)	2%	4%			
Interest rate (continuous)				1.6%	0
(correl.) Stock A	1				
(correl.) Stock B	0.5	1			
(correl.) USD/JPY	0.25	0.20	1		

2. An investment bank (IB) wants to apply the Black–Scholes–Merton model to price a call option on the excess return of a mutual fund. The IB will receive a

retrocession of 0.30% from the management fees of the fund. Please write down the SDE (Stochastic Differential Equation) to be used for pricing.

3. A Chinese bank issues a 2Y, 100% principal protected note to their clients:

Tenor	2 years	Note notional	CNY 1000
Underlying stocks	1 JP stock/1 US stock	Initial spot	8975/443
Observation	Quarterly, on the worst performance		
Knock-out level	100% of initial spot levels		
Snowball coupon rate	?% per annum		
KO coupon	$(\text{Snowball coupon rate}/4) \times i$, if KO at i -th quarter		

The issuer bank (A) receives the proceeds of 100% notional from the clients and makes a back-to-back hedge for the performance of the product via a swap with an investment bank (B):

- A pays a CNY funding of 3.5% p.a. (continuous rate, quarterly payable,) to B until the product's termination;
 - B pays the performance payoff of the product to A ;
 - A pays back 100% notional + performance payoff to the clients at maturity.
- Implement a Monte Carlo pricer using the Black–Scholes–Merton model with the below market data to solve the “Snowball Coupon Rate”.

	US stock	JP stock	USD/CNY	JPY/CNY	USD	JPY	CNY
Volatility	30%	28%	8%	11%			
(correl.) US stock	1	60%	10%				
(correl.) JP stock	60%	1		–40%			
Continuous rate					3.6%	3.3%	3.5%

4. Consider an accumulator whereby the client can buy, every month, from an investment bank 1/12-th of the total number of shares at a discounted price of the initial spot price of the share:

Total number of shares	1 million	Tenor	12 Months
Settlement frequency	Monthly	Strike	?% of Initial Spot
Knock-out level	105% of Initial Spot	Initial spot	HKD 8.8

Solve the Strike (in %) for 0 premium with the Black–Scholes–Merton model with interest rate = 2.5% p.a.(continuous) and volatility = 27.5%.

5. What is the difference between “Issuer Callable” and “Autocallable” structures?

Chapter 32

Structured Products with Dynamic Asset Allocation and Systematic Strategies



The structured solutions presented so far are mostly option-based payoffs on traditional asset classes. In this chapter, we introduce the structured solutions involving non-option based techniques to provide protection and systematic investment strategies as underlier.

32.1 Principal Protection with Dynamic Asset Allocation

Dynamic asset allocation is the mechanism to control the investment timing and quantity into tradable assets such as equities, bonds, currencies, and commodities. It can be used to achieve one or more of the following targets:

- optimizing risk-adjusted return;
- dynamic asset mix according to the performance cycles;
- providing protection.

Consider an investment portfolio with n assets, risky or not. At time- t , the portfolio's NPV (net asset value) is V_t , the price of the i -th asset is $P_{i,t}$ and the exposure or weight to i -th asset is $w_{i,t}$ ($\sum_{i=1}^n w_{i,t} = 1$) that may take negative value for short position. Let q_t be the portfolio management cost and fees for the period $[t-1, t]$. Denoting $V_{t-1}^* = V_{t-1} - q_t$, we have

$$V_t = V_{t-1}^* \left[1 + \sum_{i=1}^n w_{i,t-1} \left(\frac{P_{i,t}}{P_{i,t-1}} - 1 \right) \right].$$

In particular, for a portfolio with two assets, a risky asset P_t and a safe asset M_t (i.e., a risk-free asset), we have the following relation

$$V_t = V_{t-1}^* \left[1 + w_{t-1} \left(\frac{P_t}{P_{t-1}} - 1 \right) + (1 - w_{t-1}) \left(\frac{M_t}{M_{t-1}} - 1 \right) \right]. \quad (32.1)$$

We note that the portfolio investment is self-financing and the financing cost is implicitly reflected by the performance of the monetary asset (i.e., cash position).

32.1.1 Volatility Target

Volatility target mechanism aims at controlling the volatility of the exposure to a risky asset (or a basket of assets) by dynamically rebalancing the exposures between the asset and cash (or cash equivalent) as the volatility of cash instrument is negligible:

$$w_t = \min \left\{ cap, \frac{\sigma_{\text{target}}}{\sigma_{\text{realized}}(t)} \right\},$$

where

σ_{target} : target level for the volatility of the strategy (e.g., 5%);

$\sigma_{\text{realized}}(t)$: realized volatility of the asset at t over an observation window (e.g., calculated from the daily returns in the past 3 months);

cap : maximum exposure (e.g., 150%) to limit the leverage.

The value of a volatility target strategy V_t is based on the performance of the asset and the return from the investment in cash as shown in formula (32.1). A rebalancing threshold for the exposure changes can be introduced to avoid frequent small amount transactions. Due to cap , the effective volatility of the strategy, i.e., the portfolio of asset and cash, can be well below the target level when the realized volatility of the asset is low.

Volatility target mechanism is commonly applied to investment strategies for

(a) controlling the risk from the exposure to a risky asset, as it tends to

- deleverage when σ_{realized} increases, leading to decreased exposure to the asset and increased investment in cash;
- leverage when σ_{realized} decreases, leading to increased exposure to the asset and decreased investment in cash.

In stressed market condition, the volatilities tend to increase.

(b) offering options with exposure to the assets without liquid volatility market (e.g., mutual funds). The underliers for the options are investment portfolios with volatility target so that the volatility hedging will not be necessary.

However, the option writer is only comfortable to offer the option payoffs with positive vega (i.e., option writer sells volatility) such as call option, Asian option, etc. This is because the option writer has limited risk to sell volatility at a level slightly above the volatility target (e.g., 1.5% above) but it is difficult to determine the volatility level to buy. That's why the option traders are reluctant to offer call spread (short volatility on the second leg) on volatility target strategies.

32.1.2 *Actively Managed Portfolio with Protection*

A risk-averse investor may want to keep an active role in asset allocation, but needs protection on the investment. This can be achieved by a call option on the investor managed investment portfolio with volatility target. In such a product, the asset universe, restrictions as well as the volatility target are predefined and agreed by both parties. The investor, i.e., the option holder, has the right to change the asset allocation within the pre-agreed asset universe at his/her sole discretion through orders to the option writer. The frequency and operating procedures should be well documented in the term sheet, which may impact the option price as well as transaction costs for the asset allocation orders. The investor's asset allocation decision could be further assisted by systematic/algorithmic investment strategies provided by the option writer or other vendors.

For risk management, the option writer makes the delta rebalancing and adjustment due to the change of asset allocation, the gamma, and the volatility target.

32.1.3 *Gap Risk Based Asset Allocation: CPPI and TIPP*

Consider a portfolio investing in one or more risky assets that we call *underlier* for simplification. We use the term *gap* for a significant underlier price change from one level to another with no trading possibility in between, for instance, a sudden and large price move due to an unpredictable event/news or the difference between closing and opening prices.

A gap risk based asset allocation strategy dynamically adjusts the exposure to the underlier to provide capital protection without using options. It is particularly useful when the underlier has no liquid volatility market. The issuer of a product based on such strategy guarantees a protection floor via a fee paid by the investor. Denote

g : *gap* size in % of the underlier price for the period $[t - 1, t]$ (e.g., 20% for 1-day¹ drop of an equity index) that is acceptable to the product issuer to provide the protection;

¹ It could be more than one calendar day in case of limit-down/circuit-breaker events.

m : multiplier, which is defined by $m = 1/g$, e.g., $m = 1/20\% = 5$;

E_t : investment to the underlier for the period $[t-1, t]$. We note that E_t is equivalent to $w_{t-1} V_{t-1}^*$ in formula (32.1);

F_t : protection floor to the investor at time t ;

C_t : known as *cushion*, which is defined as

$$C_t = V_{t-1}^* - F_t. \quad (32.2)$$

A gap risk based asset allocation strategy consists of dynamically adjusting the exposure to the underlier according to the below formula over time:

$$E_t = m \times C_t. \quad (32.3)$$

Should the underlier price drop by g , the loss for the investor will be exactly C_t ($= E_t \times g$). If the underlier price drops by more than g , the product issuer/guarantor will take the loss beyond the cushion C_t but not the investor. This guarantee is known as a *gap put option*, *crash put option* or *CPPI put option* and its risk can be partially hedged with a stability note (see Sect. 31.5.3) with resetting frequency matching that of the product.

Example 32.1 Assume $g = 25\%$ and $q_t = 0$. Then, for $V_0 = \$100$ and $F_1 = \$85$, we have $m = 1/g = 4$ and $E_1 = m \times C_1 = 4 \times (\$100 - \$85) = \60 . $\square$

In practice, a *threshold* is often applied to the exposure adjustment in the gap risk based approach to avoid small transactions. The rebalancing schedule may be longer than daily (e.g., weekly), which saves transaction cost at the expense of higher gap risk.

The drawback of this allocation mechanism is that, if the underlier is volatile, it risks making the investor “buy high/sell low”. So, low volatility underliers (e.g., fixed-income funds) are more suitable for such strategies.

Two popular gap risk based structures, CPPI and TIPP, are presented in the sequel.

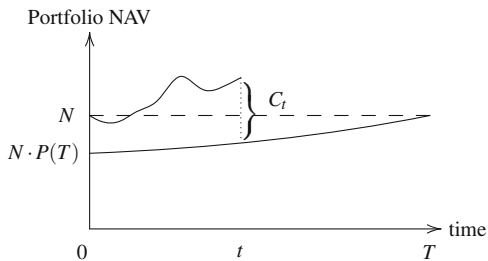
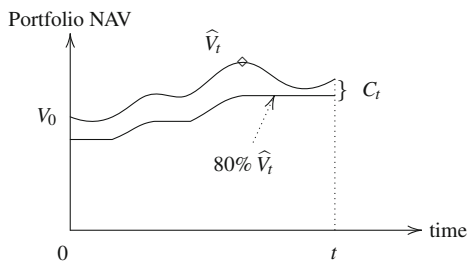
CPPI (Constant Proportion Portfolio Insurance)

In its basic version, a CPPI (see [89]) offers 100% principal protection of notional N ($= V_0$) at product maturity T . Denoting $P_t(T)$ for the zero coupon bond maturing at T , the floor at t is hence $F_t = N \times P_t(T)$ (see Fig. 32.1).

The implementation of a CPPI strategy can be as follows:

- invest $E_t = m \times C_t$ in the risky asset with C_t as defined in formula (32.2);
- invest the amount F_t into $P_t(T)$.

Should V_t drop by C_t or more, there will be no more investment in the risky asset until the product maturity T and the strategy is said to be *cash-locked*.

Fig. 32.1 Illustration for CPPI**Fig. 32.2** Illustration for TIPP

There are numerous CPPI variations including the fixed floor level to avoid interest rate impact on the cushion and floor, guaranteed minimum exposure to the risky asset to prevent cash-lock, variable multiplier, or even a volatility target for controlling the exposure. Some variations are hybrid options with risks linked to interest rate and the risky asset. For CPPI modelling, readers may refer to [31], which uses a jump-diffusion model in continuous-time framework.

TIPP (Time Invariant Portfolio Protection)

TIPP (see [38]) is a structure without the need of a fixed maturity. It provides a floor that is typically a percentage p (e.g., $p = 80\%$) of the running highest NAV (denoted as $\hat{V}_t$) since inception until t (see Fig. 32.2 with $p = 80\%$).

The floor level is thus

$$F_t = p \times \hat{V}_{t-1} = p \times \max_{0 \leq s < t-1} V_s$$

The idea resembles the trailing stop method to secure profit in trading strategies. A TIPP may be implemented as follows:

- invest $E_t = m \times C_t$ in the risky asset with C_t as defined in (32.2);
- carry $(V_{t-1}^* - E_t)$ in money market at overnight rate.

One may argue that if the interest rate is positive, there will never be a cash-lock situation because the floored amount will accrue the cushion every day. But it will take a long time to accumulate a meaningful cushion size.

32.2 Structured Products with Systematic Strategies

Structured business has experienced a fantastic development since the late 1990s. In the early stages, much effort was spent on payoff engineering. Within a few years, payoff engineering had reached its maturity stage, with a variety of product offerings and the capability to take advantage of all sorts of market views across asset classes.

Guided by research and the experience of asset managers, financial engineers in investment banks began to focus on rule-based investment strategies known as *systematic strategies* or *quantitative investment strategies (QIS)*.² The investment allocation in such strategies is rebalanced periodically, e.g., monthly, with an algorithm following a set of pre-defined rules that are disclosed to investors in general. Systematic strategies may be categorized as follows

Beta strategy: a strategy with its returns correlated to the performance of market traded assets (e.g., bonds, stocks) or benchmark (indices);

Smart beta strategy: a strategy aiming to provide higher Sharpe ratio than the market traded assets; Its return is related to the performance of factors other than just the market traded assets through, typically, long-only exposure.

Alternative beta strategy: a strategy designed to achieve uncorrelated returns to the market traded assets with full exposure to factors; The strategy is generally implemented with diversified and/or long-short positions.

The advantages for structured products linked to systematic strategies include

- low cost: the management cost is low compared to an actively managed fund;
- transparency: the strategy rules are made transparent to investors;
- consistency: the investment decisions are algorithm-driven, which avoids the impact of human behaviours;
- protection: with the techniques of “volatility target + option” or CPPI, principal protected products may be offered for systematic strategies. A traditional actively managed funds do not offer principal protection.

On the other hand, there are several well-known drawbacks to such products:

- non-flexibility: the rules are never perfect. Amending the rules is difficult if there are existing products linked to the strategy. To change the rules, a new strategy needs to be set up and the investor should unwind the existing trade to invest into the new strategy. In comparison, fund managers have the flexibility to make investment decisions.
- time lag: most of the strategies use historical data to obtain signals for the algorithms. The asset price regime change is detectable only after a time lag.
- back-testing optimization: the strategy parameters as well as the rebalancing schedules may be optimized to achieve a nice back-testing result. Were the

² A bank cannot directly issue products with discretionary asset management, which requires the asset management license.

schedules shifted to other days, the results might be quite different with the same parameters. Actually, a systematic strategy may show excellent back-tested performance but deliver poor real performance after its launch.

- pricing risk: an implemented strategy is proprietary to the provider (e.g., an investment bank) in general. Very often, only the provider can offer products linked to the strategy. Investors may not be able to use competition to achieve pricing efficiency.

Investment banks often brand their systematic strategies as **customized indices** and publish the index values on a market data system, e.g., Bloomberg. A such customized index often has a calculation agent by a well-know financial service provider such as S&P for the index values. The index rules together with the risk disclosures are made available to investors. Index rules provide all the details including the investment strategy, fees, investment restrictions, exposure limits, treatment of corporate actions treatment, tax, FX hedging, costs, . . .

Factor models and portfolio theory are widely used in systematic strategies. In the following, we will introduce the basic concepts of these topics.

32.2.1 *Introduction to Factor Models*

Many systematic strategies are based on *factors* that are quantifiable features that can explain the performance differences of the assets (e.g., momentum, value). The factors may be identified with the techniques of econometrics or data analysis. Significant advances have been made over the past two decades in understanding the *risk premia*, i.e., the returns in excess of the risk-free rate through the exposure to the (risk) factors. Nowadays, many factors are made into indices by index providers such as MSCI. Quantitative traders use extensively and dynamically a large variety of factors for investing in the market.

Single-Factor Model: Capital Asset Pricing Model (CAPM)

The original CAPM is a single-period and single-factor model proposed by Treynor [114], Sharpe [104], Lintner [71], and Mossin [82] independently, which has influenced significantly the financial industry. The quantitative concepts developed in the model serve as standard measures for investment risk and return analysis.

The CAPM can be derived from portfolio theory (see Proposition D.3 in the annex). It assumes that in equilibrium, the aggregated demand equals the aggregated supply making the whole market capitalization. The weighting of the **market portfolio** (see Proposition D.2 in the annex) should reflect the proportions of the capitalization of all the assets. For equity assets, a main stock market index can be

a good proxy for the market portfolio. Under the CAPM, the investment return into a single asset i is described by the mathematical form of a linear regression:

$$r_i - r_f = \beta_i(r_m - r_f) \text{ or } \mu_i - r_f = \beta_i(\mu_m - r_f) + \epsilon_i$$

where

μ_i : return of asset i ;

μ_m : return of the market portfolio

r_f : risk-free rate;

r_i : ($= \mathbb{E}[\mu_i]$) **expected return** or **required rate of return** of asset i . ($r_i - r_f$) is known as **risk premium** for asset i ;

r_m : ($= \mathbb{E}[\mu_m]$) expected return of the market portfolio. ($r_m - r_f$) is the **market factor**, which reflects the systematic risk that cannot be diversified away. Systematic risk is attributable to macroeconomic, geopolitical, or other general factors that impact the economy and the prices of all securities in the market.

β_i : measures the risk exposure to the market factor for asset i . Mathematically, $\beta_i = \text{Cov}[\mu_i, \mu_m] / \sigma_m^2$ where σ_m^2 the variance of the market portfolio;

ϵ_i : asset-specific idiosyncratic risk or non-systematic risk that can be diversified away by increasing the number of assets in the investment portfolio. It is a white noise (i.e., $\mathbb{E}[\epsilon_i] = 0$) and $\text{Cov}[\epsilon_i, \mu_m] = 0$.

The **Index Model** is an extension to the CAPM by adding a non-systematic return α :

$$r - r_f = \alpha + \beta(r_m - r_f),$$

where α is the unexplained return that is uncorrelated to r_m . For an actively managed fund, its α is an indicator of the fund manager's skills. However, some people believe that the observed α is mostly generated by the β of some other (hidden) factors and that the true α may not exist.

There are two widely used indicators to measure investment performance R , which are based on the Index Model and the CAPM:

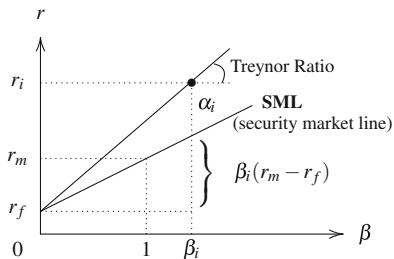
Jensen's Alpha $= R - [r_f + \beta(r_m - r_f)]$, measures the absolute return of the investment in extra of the theoretical expected return derived from the market performance.

Treynor ratio $= (R - r_m) / \beta$ (if $\beta > 0$), is a systematic risk-adjusted measurement for the investment return.

Obviously, the Sharpe ratio and Treynor ratio are not suitable measures for negative excess returns.

The couple (β_i, r_i) can be represented by a point on the β - r plane. In particular, the point for the market portfolio $(1, r_m)$ and the point for risk-free asset $(0, r_f)$ form a line known as the **security market line** (SML). An asset is over-priced if it is below this level, and under-priced if it is above this level (Fig. 32.3).

Fig. 32.3 Illustration of different concepts related to the CAPM



Multi-Factor Asset Pricing Models

The **arbitrage pricing theory (APT)**, proposed by Stephen Ross [98], extends the CAPM by allowing multiple risk factors, resulting in a multiple linear regression formulation,

$$\mu_i - r_f = \sum_{k=1}^K \beta_{ik}(F_k - r_f) + \epsilon_i, \quad (32.4)$$

where μ_i, r_f, ϵ_i are defined as before, and F_k is the return of the k -th systematic risk factor that is common to all assets. The i -th asset's return is now described by a set of betas β_{ik} (also known as *factor loadings*), generalizing the single market factor in the CAPM. The APT uses fewer assumptions than the CAPM. Statistical methods, such as regression or principal component analysis (PCA), may help to identify the factors in (32.4) from the given asset returns $\mu_i, i = 1, 2, \dots, n$.

Some Well-Known Factors

In the following, we introduce some risk factors that are well studied in academic research and believed to offer risk premia from empirical tests. They can be materialized with market transactions. Evidently, the list of factors has no limit.

Momentum strategy is based on the belief that increases in the price of an asset (or instrument) will be likely followed by additional gains and vice versa for declining values. This strategy consists of taking long position in the asset with upward trends (and eventually short sell securities with downward trends). For the implementation, practitioners need to make the choices regarding:

- observation window: for calculating price momentum, e.g., 3 months;

- return calculation: the return type may be total return, z -score³ or some other return types. The historical data may be weighted equally or exponentially to give higher importance to recent data;
- other signals: e.g., the crossing of short-term moving average over long-term moving average.

Value factor investing has long existed and is designed to hold a long position on some under-valued assets and eventually a short position of some over-valued assets. Prices are not directly used to represent the factor but instead financial ratios are used, e.g., price-to-book (P/B) ratio, forward price/earnings (P/E) ratio, dividend yield, cash-flow to price ratio. Equity valuation, bond rating and other financial analysis may be found on public or private sources, e.g., Bloomberg (“ANR” command shows buy/sell/hold ratings from analysts), conviction list from research teams of investment banks, financial analytics systems such as STARMINE.

Size factor is generally defined by the market capitalization such as small caps, mid caps and big caps. A common size strategy is set up by buying small caps and short-selling big caps. Practical implementation might take into account the liquidity and transaction cost when trading small caps. The borrow availability and cost also need to be considered in building a short position.

Carry trade aims to create value by carrying a certain position over time and earning on the accrual, among other things. For example, an FX carry trade consists of borrowing in a low-interest-rate currency (e.g., JPY) to invest in a high-interest-rate currency (e.g., AUD). This strategy will be a winning trade that earns approximately the interest rate difference if the FX rate has not changed significantly by the end of the trade tenor.

Volatility strategies capture the opportunities from volatility exposures, e.g.,

- receiving implied volatility and paying realized volatility: it is based on the observation that implied volatility is often higher than realized volatility.
- *tail-risk hedging* strategy based on implied volatility (through trades on VIX or forward-starting variance swaps): it jumps up during extreme market events.
- *call overwriting* (or *buy/write*) strategy: it consists of holding a long position of certain securities and selling (either regularly or opportunistically) short-term call options on (some of) these securities.

Geography factor targets to take advantage of either the high growth potential of certain geographical regions or the growth differential between geographical regions. An example is taking a long position in emerging market and a short position in developed market.

Rolling futures strategy consists of keeping an exposure (long or short) in futures continually. The simplest implementation consists of keeping the exposure in the

³ z -score is defined as $\frac{\text{return} - \text{averaged return}}{\text{standard deviation of the return}}$.

nearest (front-month) futures because of its high liquidity, and switching (rolling) to the next nearest futures a few days before the expiry date. A downward sloping futures price curve (backwardation) tends to make profit on rolling to the long exposure if the price curve remains similar over time. Conversely, in an upward sloping (contango) situation, the long exposure tends to suffer loss on rolling. The return of futures is close to excess return due to the low capital cost.

32.2.2 Portfolio Approach

Notations and Basic Properties for Portfolios

In the following, we focus on the allocation to multiple risky assets. The general notations will be used throughout the section:

n	number of assets in the portfolio;
w_i	weight for asset i that may take negative value for short position;
μ_i	return of asset i ;
r_i	expected return of asset i ($r_i = \mathbb{E}[\mu_i]$);
σ_i	standard deviation (volatility) of asset i ($\sigma_i^2 = \text{Var}(\mu_i)$);
$\mathbf{w}, \boldsymbol{\mu}, \mathbf{r}$	vectors of w_i, μ_i, r_i ;
ρ_{ij}	correlation between assets i and j , $\rho_{ii} \equiv 1, \forall i$;
$\boldsymbol{\Sigma}$	covariance matrix for asset returns ($\boldsymbol{\Sigma} = [\rho_{ij}\sigma_i\sigma_j]_{n \times n}$);
μ	portfolio return ($\mu = \sum_{i=1}^n w_i \mu_i$);
r	expected portfolio return ($r = \sum_{i=1}^n w_i r_i$);
σ^2	portfolio return variance;
$\mathbf{1}$	$= [1, \dots, 1]^\top$.

The quantities r_i, σ_i , and ρ_{ij} are assumed to be known. For instance, historical data from a time window can be used to sample μ_i while r_i, σ_i , and ρ_{ij} can be determined with the time series of μ_i .

Some Properties for Portfolios:

- portfolio variance: $\sigma^2 = \sum_{i,j=1}^n w_i w_j \rho_{ij} \sigma_i \sigma_j = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$
- covariance of two portfolios $(\boldsymbol{\mu}, \mathbf{w})$ and $(\hat{\boldsymbol{\mu}}, \hat{\mathbf{w}})$: $\sum_{i,j=1}^n w_i \hat{w}_j \rho_{ij} \sigma_i \sigma_j = \mathbf{w}^\top \boldsymbol{\Sigma} \hat{\mathbf{w}}$
- for invertible $\boldsymbol{\Sigma}$: $(\boldsymbol{\Sigma}^{-1})^\top = \boldsymbol{\Sigma}^{-1}$

Proof By definition, the covariance of the returns of portfolios $(\boldsymbol{\mu}, \mathbf{w})$ and $(\hat{\boldsymbol{\mu}}, \hat{\mathbf{w}})$ is

$$\begin{aligned} & \mathbb{E}\left[\left(\sum_{i=1}^n w_i \mu_i - \mathbb{E}\left[\sum_{i=1}^n w_i \mu_i\right]\right)\left(\sum_{j=1}^n \hat{w}_j \mu_j - \mathbb{E}\left[\sum_{j=1}^n \hat{w}_j \mu_j\right]\right)\right] \\ &= \mathbb{E}\left[\sum_{i=1}^n w_i (\mu_i - r_i) \sum_{j=1}^n \hat{w}_j (\mu_j - r_j)\right] = \sum_{i,j=1}^n w_i \hat{w}_j \rho_{ij} \sigma_i \sigma_j. \end{aligned}$$

With two identical portfolios, we obtain the portfolio return variance σ^2 .

From $(\boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma})^\top = \mathbf{I}$, we obtain the symmetry property $(\boldsymbol{\Sigma}^{-1})^\top = \boldsymbol{\Sigma}^{-1}$. □

Mean-Variance Optimization (MVO)

The celebrated **modern portfolio theory** (MPT) pioneered by Nobel laureate Harry Markowitz [75] makes use of the expected returns and covariance of the components of a portfolio to solve the optimal asset allocation $\mathbf{w}$ with either of the following equivalent problems known as **mean-variance optimization** (MVO):⁴

- highest expected return given the portfolio variance;
- lowest variance given the portfolio expected return.

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}, \quad \text{subject to the constraints} \quad \begin{cases} \mathbf{r}^\top \mathbf{w} = r, \\ \mathbf{1}^\top \mathbf{w} = 1, \end{cases}$$

Actually, any portfolio can be simply characterized by a pair (r, σ) and therefore it is convenient to visualize portfolios as a point on the r - σ plane (see Fig. 32.4). The set of MVO solutions (see Proposition D.1 in the annex) for different r constitutes a hyperbola on the r - σ plane, and the upward sloping part is called the **efficient frontier**. To gain higher return, risk-averse investors have to move up the efficient frontier and endure more risk.

In practice, it is often necessary to set other (linear) **constraints** on $\mathbf{w}$, such as a long-only scenario (i.e., $w_i \geq 0$), or certain weight limits on some of the assets (e.g., $w_2 + w_3 \leq 30\%$, $0 \leq w_1 + w_4 + w_5 \leq 100\%$, etc.). The constrained optimization problem can be solved using numerical methods, and solvers are available.

If a risk-free asset is added to the portfolio represented by $\mathbf{w}$ (see [113]), the MVO problem becomes

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}, \quad \text{subject to constraints } \mathbf{r}^\top \mathbf{w} + (1 - \mathbf{1}^\top \mathbf{w}) r_f = r.$$

⁴ Normally, $\mathbf{w}$ is mostly impacted by the expected returns of the components.

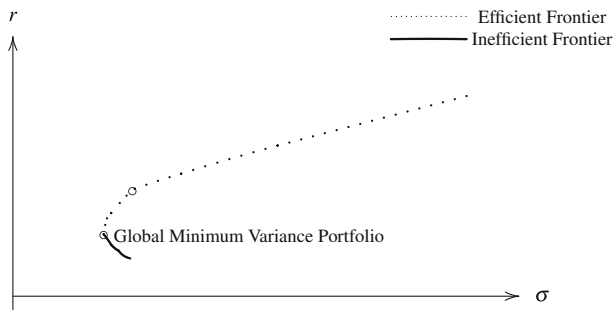


Fig. 32.4 Illustration of efficient frontier on the risk-return plane

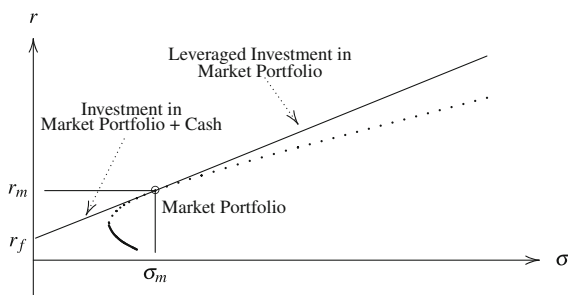


Fig. 32.5 The terms r_m and σ_m represent respectively the expected return and volatility of the market portfolio. Sharpe ratio of market portfolio is the slope of the line: $\frac{r_m - r_f}{\sigma_m}$

We note that the weights for risky assets portfolio together with the risk-free asset sum up to 1 and that $\mathbf{w}^\top \mathbf{1} > 1$ implies borrowing risk-free asset. The solution is provided by Proposition D.2 in the annex, which shows that the efficient frontier of the whole portfolio (risky + risk-free assets) is a straight line known as **capital allocation line** (CAL) with intercept at r_f and slope θ (see Fig. 32.5) :

$$\theta = \frac{r - r_f}{\sigma} \text{ known as } \mathbf{Sharpe\ ratio}, \text{ which is the excess return over volatility.}$$

CAL happens to be tangential to the efficient frontier of the risky assets portfolio (i.e., excluding the risk-free asset) at (m), which is known as **market portfolio**. The half line above (m) involves borrowing at r_f to leverage the investment in (m), while the segment below (m) implies investment in both the risk-free asset and (m).

Apart from the dynamic portfolios with MVO, there are a few other popular but simpler strategies that are worth introducing:

Risk Parity is a strategy consists of allocating the same risk quantity (i.e., volatility) across all the component assets of a portfolio. It can be achieved by periodically resetting the weights so that $w_i \sigma_i = w_j \sigma_j, \forall i, j = 1, \dots, n$.

Risk parity is easy to implement⁵ and particularly convenient for long-only asset allocation. For each asset, its weight reduces if the volatility increases.

Risk Budgeting consists of allocating a risk budget in terms of volatility or variance for each asset in a portfolio. The investor may achieve the desired return profile through his risk appetite/tolerance. For example, one may apply 30% on equities, 20% on emerging market bonds, 40% on investment grade bonds and 10% on commodities.

Structured Products with Rebalancing Portfolios

Many investment banks have offered structured products with systematic strategies with portfolio approach, especially with MVO (e.g., on a basket of 10 underliers of different asset classes). The implementation is described as follows:

- the strategy typically invests into an asset universe across different asset classes covering equity, bond, and commodities market (through indices, ETFs, or large mutual funds);
- it often applies a fixed target portfolio volatility and provides excess return;⁶
- the portfolio is rebalanced periodically, e.g., every month, quarter, or semester;
- between the rebalancing dates, the volatility target mechanism may be applied to control the portfolio volatility;
- it is possible to write options on the strategy if the volatility target is applied.

Exercises

1. Show that the equal weighting asset allocation scheme is essentially a “buy low/sell high” strategy.
2. When solving the MVO problem, one can remove the constraint $\mathbf{r}^T \mathbf{w} = r$ and obtain the *Global Minimum Variance* portfolio. Following the Proposition D.1, prove that the weights take the form of $\mathbf{w} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}$.
3. Analyze the general pros and cons of systematic strategies and the structured products on systematic strategies.

⁵ To determine the weighting w_i , we use the relation $1 = \sum_{j=1}^n w_j = \sum_{j=1}^n \frac{w_i \sigma_i}{\sigma_j}$. Pulling w_i out of the summation gives $w_i = \frac{1/\sigma_i}{\sum_{j=1}^n (1/\sigma_j)}$.

⁶ Excess return allows low option price due to its low forward.

Appendix A

Some Elements in Probability Theory, Linear Algebra and Analysis

A.1 Multivariate Random Variable

Let $\mathbf{X}$ be a column vector of p random variables $[X_1, X_2, \dots, X_p]^\top$ on some probability space. The expectation $\mathbb{E}[\mathbf{X}]$ is also a column vector, containing the expectations in each dimension. The variance $\text{Var}(\mathbf{X})$ is a $p \times p$ matrix defined as

$$\text{Var}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top],$$

which is also known as the variance-covariance matrix. Let $\mathbf{Y}$ be another q -dimensional multivariate random variable. The $p \times q$ *cross-covariance* matrix between $\mathbf{X}$ and $\mathbf{Y}$ is defined as

$$\text{Cov}(\mathbf{X}, \mathbf{Y}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])^\top] = \mathbb{E}[\mathbf{XY}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{Y}]^\top.$$

Let $\mathbf{A}$ and $\mathbf{B}$ be non-random $n \times p$ and $m \times q$ matrices respectively. We have the following proposition.

Proposition A.1 *For linear transformations $\mathbf{AX}$ and $\mathbf{BY}$, we have*

$$\begin{aligned}\mathbb{E}[\mathbf{AX}] &= \mathbf{A}\mathbb{E}[\mathbf{X}], \\ \text{Var}(\mathbf{AX}) &= \mathbf{A}\text{Var}(\mathbf{X})\mathbf{A}^\top, \\ \text{Cov}(\mathbf{AX}, \mathbf{BY}) &= \mathbf{A}\text{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{B}^\top.\end{aligned}$$

Proof The expectation follows directly from linearity. Then, we only need to show the covariance case

$$\begin{aligned}\text{Cov}(\mathbf{AX}, \mathbf{BY}) &= \mathbb{E}[\mathbf{AX}(\mathbf{BY})^\top] - \mathbb{E}[\mathbf{AX}]\mathbb{E}[\mathbf{BY}]^\top \\ &= \mathbf{A}\mathbb{E}[\mathbf{XY}^\top]\mathbf{B}^\top - \mathbf{A}\mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{Y}]^\top\mathbf{B}^\top \\ &= \mathbf{A}(\mathbb{E}[\mathbf{XY}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{Y}]^\top)\mathbf{B}^\top = \mathbf{A}\text{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{B}^\top.\end{aligned}$$

□

A.2 Multivariate Normal Distribution

Assume that the multivariate random variable $\mathbf{X}$ follows the multivariate normal distribution with $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Proposition A.2 (Linear Transformation Property of Multivariate Normal Vector) *Let $\mathbf{A}$ be an $n \times n$ non-random matrix. The linear transformation $\mathbf{AX}$ also follows multivariate normal distribution, with*

$$\mathbf{AX} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Proof Denote $\mathbf{X}$ as $[X_1, X_2, \dots, X_n]^\top$. By definition of the multivariate normal distribution, any linear combination of $X_i, i = 1, \dots, n$ is normally distributed. As each entry of the vector $\mathbf{AX}$ is a linear combination of $X_i, i = 1, \dots, n$, any linear combination of the entries must also be normally distributed. Hence, $\mathbf{AX}$ follows the multivariate normal distribution characterized by the expectation and covariance. From Proposition A.1, we obtain

$$\mathbb{E}[\mathbf{AX}] = \mathbf{A}\mathbb{E}[\mathbf{X}] = \mathbf{A}\boldsymbol{\mu} \quad \text{and} \quad \text{Cov}(\mathbf{AX}, \mathbf{AX}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top.$$

□

Proposition A.3 (Some Properties of Covariance and Correlation)

(1) *The correlation between any two random variables X and Y , ρ_{XY} , satisfies*

$$-1 \leq \rho_{XY} \leq 1.$$

(2) *Let $\mathbf{X} \in \mathbb{R}^n$ be a random vector with mean $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^\top$ and standard deviation $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n)^\top$, $\forall \sigma_i > 0$. The covariance matrix $\boldsymbol{\Sigma}$ and correlation matrix $\mathbf{C}$ are positive semi-definite, i.e., $\forall \mathbf{a} \in \mathbb{R}^n$,*

$$\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} \geq 0, \quad \mathbf{a}^\top \mathbf{C} \mathbf{a} \geq 0.$$

(3) Any correlation matrix $\mathbf{C}$ and $\mathbf{\Sigma}$ can be diagonalized with non-negative eigenvalues:

$$\mathbf{C} = \mathbf{U}^\top \mathbf{\Lambda} \mathbf{U}, \quad \text{where } \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n), \lambda_i \in \mathbb{R}, \text{ and } \lambda_i \geq 0, \forall i.$$

Proof

(1) Denote $\hat{X} = X - \mathbb{E}[X]$ and $\hat{Y} = Y - \mathbb{E}[Y]$. $\forall t \in \mathbb{R}$, from $\mathbb{E}[(t\hat{X} + \hat{Y})^2] \geq 0$, we obtain

$$\mathbb{E}[\hat{X}^2]t^2 + 2\mathbb{E}[\hat{X}\hat{Y}]t + \mathbb{E}[\hat{Y}^2] \geq 0.$$

The discriminant of the above quadratic equation in t must satisfy

$$(2\mathbb{E}[\hat{X}\hat{Y}])^2 - 4\mathbb{E}[\hat{X}^2]\mathbb{E}[\hat{Y}^2] \leq 0.$$

Hence, $(\mathbb{E}[\hat{X}\hat{Y}])^2 \leq \mathbb{E}[\hat{X}^2]\mathbb{E}[\hat{Y}^2]$ and $|\rho| = \frac{|\mathbb{E}[\hat{X}\hat{Y}]|}{\sqrt{\mathbb{E}[\hat{X}^2]\mathbb{E}[\hat{Y}^2]}} \leq 1$.

(2) Denoting $\hat{\mathbf{X}} = \mathbf{X} - \boldsymbol{\mu}$ and noticing that $\mathbf{a}^\top \hat{\mathbf{X}}$ is a scalar, we have

$$\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a} = \mathbf{a}^\top \mathbb{E}[\hat{\mathbf{X}}\hat{\mathbf{X}}^\top] \mathbf{a} = \mathbb{E}[\mathbf{a}^\top \hat{\mathbf{X}}\hat{\mathbf{X}}^\top \mathbf{a}] = \mathbb{E}[(\mathbf{a}^\top \hat{\mathbf{X}})^2] \geq 0.$$

Let $\mathbf{D} = \text{diag}(\sigma_i)_{i=1:n}$ ($\mathbf{D}^\top = \mathbf{D}$). From the relation $\mathbf{\Sigma} = \mathbf{D}\mathbf{C}\mathbf{D}^\top$, we have

$$\mathbf{a}^\top \mathbf{C} \mathbf{a} = \mathbf{a}^\top \mathbf{D}^{-1} \mathbf{\Sigma} (\mathbf{D}^\top)^{-1} \mathbf{a} = \mathbf{b}^\top \mathbf{\Sigma} \mathbf{b} \geq 0 \quad (\mathbf{b} = (\mathbf{D}^\top)^{-1} \mathbf{a}).$$

(3) According to Spectral Theorem, an important result of linear algebra [61, 108], all real-valued symmetric $n \times n$ matrices $\mathbf{M}$ can be decomposed

$$\mathbf{M} = \mathbf{U}^\top \mathbf{\Lambda} \mathbf{U}$$

where $\mathbf{U}$ is a matrix composed of orthonormal eigenvectors and $\lambda \in \mathbb{R}$ are the eigenvalues (spectrum of a matrix is the set of its eigenvalues). Also, $\mathbf{M}$ is positive semi-definite (positive definite) if and only if all its eigenvalues $\lambda_i \geq 0$ (> 0).

As $\mathbf{C}$ and $\mathbf{\Sigma}$ are real-valued symmetric $n \times n$ positive semi-definite matrices, we get the expected conclusion.

□

A.3 Unbiased Estimators

Proposition A.4 (Unbiased Estimators) *Let X be a random variable with mean μ and variance σ^2 . Assume that $X_1, \dots, X_n$ are i.i.d. samples of X . Define the sample $\bar{X}$ by $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Then,*

- (1) $\bar{X}$ is an unbiased estimator of μ ,
- (2) $\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of σ^2 .

Proof (1) is proved by below equation:

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} n\mu = \mu.$$

Now, let us denote $x_i = X_i - \mu$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. From independence of X_i 's, we have $\mathbb{E}[x_i x_j] = 0, \forall i \neq j$ and $\mathbb{E}[\bar{x}^2] = \frac{1}{n} \sigma^2$. We note that $X_i - \bar{X} = x_i - \bar{x}$. The following check proves (2),

$$\begin{aligned} \mathbb{E}\left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2\right] &= \frac{1}{n-1} \mathbb{E}\left[\sum_{i=1}^n (x_i^2 + \bar{x}^2 - 2x_i \bar{x})\right] \\ &= \frac{1}{n-1} (n\sigma^2 + \mathbb{E}[n\bar{x}^2 - 2n\bar{x}^2]) = \frac{1}{n-1} (n\sigma^2 - \sigma^2) = \sigma^2. \end{aligned}$$

□

A.4 Multiple Least-Squares Regression

Let $\mathbf{x} = [x_1, \dots, x_m]^T$ be a vector of centered random variables and $\mathbf{a} = [a_1, \dots, a_m]^T$ be a real vector. Suppose that

$$y = \sum_{i=1}^m a_i x_i + \epsilon$$

where ϵ is a white noise (i.e., a random variable with expected value of zero and finite variance). The vector $\mathbf{a}$ can be estimated by the least-squares method that minimizes $\mathbb{E}\left[(y - \sum_{i=1}^m a_i x_i)^2\right]$. Set the first order derivative w.r.t. a_k ($\forall k$) to zero (i.e.,

first-order condition for optimization problems): $\mathbb{E}[2(y - \sum_{i=1}^m a_i x_i)(-x_k)] = 0$, $k = 1, \dots, m$. It follows that

$$\mathbb{E}[yx_k] = \sum_{i=1}^m \mathbb{E}[x_i x_k] a_i, \forall k; \quad \text{i.e., } \mathbb{E}[y\mathbf{x}] = \mathbb{E}[\mathbf{xx}^\top] \mathbf{a}.$$

Hence, we obtain the regression result

$$\mathbf{a} = \mathbb{E}[\mathbf{xx}^\top]^{-1} \mathbb{E}[y\mathbf{x}].$$

□

A.5 Plackett's Identity

Proposition A.5 Consider an n -dimensional normal variate with zero mean vector and covariance matrix $\Sigma = [\sigma_{ij}]$, and its density function is $f(x_1, x_2, \dots, x_n)$. Then, f satisfies the below equality known as Plackett's identity [92]:

$$\frac{\partial f}{\partial \sigma_{ij}} = w_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j}, \quad w_{ij} = 1 \text{ for } i \neq j \text{ and } w_{ij} = \frac{1}{2} \text{ for } i = j.$$

Proof Firstly, we can write f as the inverse transform of its characteristic function,

$$f = \frac{1}{(2\pi)^n} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} e^{-it^\top \mathbf{x} - \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t}} dt_1 \dots dt_n,$$

where $\mathbf{x}$ and $\mathbf{t}$ are n -dimensional vectors. By direct differentiation, we have

$$\frac{\partial f}{\partial \sigma_{ij}} = -\frac{1}{(2\pi)^n} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} w_{ij} t_i t_j e^{-it^\top \mathbf{x} - \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t}} dt_1 \dots dt_n$$

and

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = -\frac{1}{(2\pi)^n} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} t_i t_j e^{-it^\top \mathbf{x} - \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t}} dt_1 \dots dt_n.$$

By comparing the above two equations, we can obtain the Plackett's identity. □

A.6 Dirac Delta Function

Dirac delta function, denoted as $\delta(x)$, is a generalized function that is always used inside an integration. Instead of giving its rigorous definition and properties, we outline only the basic features used in this book. Let $H(x) = 1_{\{x \geq 0\}}$, which is known as the Heaviside step function. Then, $\delta(x)$ can be considered as $\frac{dH}{dx}$ that satisfies $\delta(x) = 0$ for $x \neq 0$ and $\int_{-\infty}^{+\infty} \delta(x) dx = \int_{-\infty}^{+\infty} dH(x) = 1$. In particular, it has the *sampling property*:

$$\int_{-\infty}^{+\infty} f(x) \delta(x - a) dx = f(a).$$

This property can be heuristically understood with $\int_{-\infty}^{+\infty} f(x) \delta(x - a) dx = \int_{-\infty}^{+\infty} f(a) \delta(x - a) dx = f(a) \int_{-\infty}^{+\infty} \delta(x - a) dx = f(a)$.

A.7 Box–Muller Method for Uniform Variables

If u_1 and u_2 are two independent $[0, 1]$ uniform random variables, then ϵ_1 and ϵ_2 defined below are independent normal random variables:

$$\epsilon_1 = \sqrt{-2 \ln u_1} \cos(2\pi u_2), \quad \epsilon_2 = \sqrt{-2 \ln u_1} \sin(2\pi u_2).$$

Proof We follow [11] for the proof. Obviously, $\alpha = 2\pi u_2$ is a $[0, 2\pi]$ uniform random variable with density $f_\alpha(\theta) = \frac{1}{2\pi}$. Let τ be an exponential random variable with distribution function $\mathbb{P}[\tau \leq t] = 1 - e^{-t/2}$. Then, for $r > 0$,

$$\mathbb{P}[\sqrt{\tau} \leq r] = \mathbb{P}[\tau \leq r^2] = 1 - e^{-r^2/2}$$

and the density for $\sqrt{\tau}$ is $f_\tau(r) = r e^{-r^2/2}$. Hence, the joint distribution function for $(\alpha, \sqrt{\tau})$ is

$$\begin{cases} f(\theta, r) = f_\alpha(\theta) f_\tau(r) = \frac{1}{2\pi} r e^{-r^2/2}, & \text{if } 0 \leq \theta \leq 2\pi \text{ and } r > 0; \\ f(\theta, r) = 0, & \text{otherwise.} \end{cases}$$

Denote $x = \sqrt{\tau} \cos \alpha$, $y = \sqrt{\tau} \sin \alpha$. Let h be an arbitrary integrable function. From the link between polar integral and double integral, we obtain

$$\mathbb{E}[h(x, y)] = \int_0^{2\pi} \int_0^\infty h(r \cos \theta, r \sin \theta) \frac{1}{2\pi} r e^{-r^2/2} dr d\theta$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u, v) \frac{1}{2\pi} e^{-(u^2+v^2)/2} du dv.$$

By the arbitrariness of h , we conclude that x and y are two independent standard normal variables. Applying the inverse sampling method, τ can be simulated by $-2 \ln(1 - u)$ where u is a $[0, 1]$ uniform random. As $1 - u$ and u_1 follow the same distribution, $-2 \ln(1 - u)$ can be expressed as $-2 \ln u_1$. So, for simulating x and y , we can do

$$\epsilon_1 = \sqrt{-2 \ln u_1} \cos(2\pi u_2), \quad \epsilon_2 = \sqrt{-2 \ln u_1} \sin(2\pi u_2).$$

□

A.8 Approximation of Distribution Function for Normal Variables

The calculations of the cumulative distribution function for normal variables $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}u^2} du$ and its inverse function $N^{-1}(p)$ are provided in many software packages (e.g., NORMSDIST and NORMSINV in Excel). But it is still useful to provide them for quick implementation. For more accurate methods, we refer to [67] and [43].

$$N(x) \approx \begin{cases} f(x) & \text{for } x > 0 \\ 1 - f(-x) & \text{for } x \leq 0 \end{cases} \quad \text{by the symmetry property}$$

$$f(u) = 1 - \frac{1}{2}(1 + c_1 u + c_2 u^2 + c_3 u^3 + c_4 u^4)^{-4}$$

where

$$c_1 = 0.196854; c_2 = 0.115194; c_3 = 0.000344; c_4 = 0.019527.$$

The approximation for the inverse function is:

$$N^{-1}(p) \approx \begin{cases} g(p), & \text{for } 0.5 \leq p < 1 \\ -g(1 - p), & \text{for } 0 < p < 0.5 \end{cases} \quad \text{by the symmetry property}$$

$$g(u) = \begin{cases} \sum_{i=0}^3 a_i (u - \frac{1}{2})^{2i+1} / [1 + \sum_{i=0}^3 b_i (u - \frac{1}{2})^{2i}], & \text{for } 0.5 \leq u \leq 0.92 \\ \sqrt{\ln(1 - u)}, & \text{for } u > 0.92 \end{cases}$$

where

$$\begin{aligned} a_0 &= 2.50662823884; & b_0 &= -8.47351093090; \\ a_1 &= -18.61500062529; & b_1 &= 23.08336743743; \\ a_2 &= 41.39119773534; & b_2 &= -21.06224101826; \\ a_3 &= -25.44106049637; & b_3 &= 3.13082909833. \end{aligned}$$

A.9 Inversion of Characteristic Functions

Let $F(x)$ be a distribution function for random variable X . The *characteristic function* is defined as

$$\phi(t) = \mathbb{E}[e^{itX}], \quad t \in \mathbb{R}, \quad i^2 = -1.$$

If $\phi(t)$ is integrable, the inversion from $\phi(t)$ for $F(x)$ can be found and it is unique. Several inversion calculation formulas exist and we cite the following well-known ones:

$$\left\{ \begin{array}{l} F(b) - F(a) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-ita} - e^{-itb}}{it} \phi(t) dt, \text{ for } a < b; \\ F(x) = \frac{1}{2} + \frac{1}{2\pi} \int_0^{\infty} \frac{e^{itx} \phi(-t) - e^{-itx} \phi(t)}{it} dt; \\ F(x) = \frac{1}{2} - \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{e^{-itx} \phi(t)}{it} \right] dt. \end{array} \right.$$

The derivation details for first two formulas can be found in [42]. Here, we derive the third inversion formula from the second one. First, let's recall some properties for complex numbers. The conjugate of any complex number $z = x + iy$ is defined as $\bar{z} = x - iy$. It can be checked that $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$ for any complex numbers z_1 and z_2 . Euler's formula is $e^{iz} = \cos z + i \sin z$. The real part and the imaginary part of z are respectively $\operatorname{Re}[z] = \frac{z + \bar{z}}{2}$, and $\operatorname{Im}[z] = \frac{z - \bar{z}}{2i}$. We can check that $\overline{\phi(t)} = \phi(-t)$: actually, $\overline{\mathbb{E}[e^{itX}]} = \overline{\mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)]} = \mathbb{E}[\cos(tX)] - i\mathbb{E}[\sin(tX)] = \mathbb{E}[\cos(-tX)] + i\mathbb{E}[\sin(-tX)] = \mathbb{E}[e^{-itX}]$. Using these results, we can derive the third inversion formula:

$$\begin{aligned} F(x) &= \frac{1}{2} + \frac{1}{2\pi} \int_0^{\infty} \frac{e^{itx} \phi(-t) - e^{-itx} \phi(t)}{it} dt \\ &= \frac{1}{2} + \frac{1}{2\pi} \int_0^{\infty} \frac{\overline{e^{-itx} \phi(t)} - e^{-itx} \phi(t)}{it} dt \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} + \frac{1}{2\pi} \int_0^\infty \frac{\overline{e^{itx}\phi(-t)} - e^{-itx}\phi(t)}{it} dt \\
&= \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-itx}\phi(t)}{it} \right] dt
\end{aligned}$$

We obtain that $\mathbb{P}[X > x] = 1 - F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-itx}\phi(t)}{it} \right] dt.$

Appendix B

Closed-Form Solutions to Some Options

All the formulas presented in this section are based on the Black–Scholes–Merton model with the below notations:

T	Time to maturity
K	Strike or exercise price
r	Interest rate (continuous, ACT/365)
q	repo (securities lending fee) rate + continuous dividend
σ	Volatility
S_0	Spot price

The vanilla options considered in the section are all of European style.

B.1 Vanilla Options on Stocks

Let c and p be respectively vanilla call and put option prices.

$$c = S_0 e^{-qT} N(d_1) - e^{-rT} K N(d_2),$$

$$p = e^{-rT} K N(-d_2) - S_0 e^{-qT} N(-d_1)$$

where $d_1 = d_2 + \sigma\sqrt{T}$, $d_2 = [\ln \frac{S_0}{K} + (r - q - \frac{1}{2}\sigma^2)T]/(\sigma\sqrt{T})$.

While all the Greeks can be obtained by finite difference, we nevertheless provide the closed-form formulas for vanilla options on a non-dividend paying stock, which may be useful for making quick analysis. The derivations are straight forward using formula (14.23).

$$\begin{aligned}
 \Delta: \frac{\partial c}{\partial S_0} &= N(d_1); & \frac{\partial p}{\partial S_0} &= N(d_1) - 1; \\
 \Gamma: \frac{\partial^2 c}{\partial S_0^2} &= N'(d_1)/(S_0 \sigma \sqrt{T}); & \frac{\partial^2 p}{\partial S_0^2} &= N'(d_1)/(S_0 \sigma \sqrt{T}); \\
 \Theta: \frac{\partial c}{\partial t} &= -\frac{S_0 N'(d_1) \sigma}{2\sqrt{T}} - r K e^{-rT} N(d_2); & \frac{\partial p}{\partial t} &= r K e^{-rT} N(-d_2) - \frac{S_0 N'(d_1) \sigma}{2\sqrt{T}}; \\
 \mathcal{V}: \frac{\partial c}{\partial \sigma} &= S_0 \sqrt{T} N'(d_1); & \frac{\partial c}{\partial \sigma} &= S_0 \sqrt{T} N'(d_1); \\
 \rho: \frac{\partial c}{\partial r} &= K T e^{-rT} N(d_2); & \frac{\partial p}{\partial r} &= -K T e^{-rT} N(-d_2);
 \end{aligned}$$

B.2 Vanilla Options on Foreign Exchange Rate

Let X_t , r_f , and r_d be respectively the foreign exchange rate (units of domestic currency per foreign currency) at time t , the continuous interest rates of the foreign currency, and the domestic currency. Denoting c_X and p_X for FX call and FX put option prices, we have

$$\begin{aligned}
 c_X &= X_0 e^{-r_f T} N(d_1) - e^{-r_d T} K N(d_2), \\
 p_X &= e^{-r_d T} K N(-d_2) - X_0 e^{-r_f T} N(-d_1)
 \end{aligned}$$

where $d_1 = d_2 + \sigma \sqrt{T}$, $d_2 = [\ln \frac{S_0}{K} + (r_d - r_f - \frac{1}{2} \sigma^2) T] / (\sigma \sqrt{T})$.

B.3 Vanilla Options on Futures

Let F_t be the price of the futures contract at time t , c_F and p_F be respectively futures call and put option prices.

$$\begin{aligned}
 c_F &= e^{-rT} [F_0 N(d_1) - K N(d_2)], \\
 p_F &= e^{-rT} [K N(-d_2) - S_0 N(-d_1)]
 \end{aligned}$$

where $d_1 = d_2 + \sigma \sqrt{T}$, $d_2 = [\ln \frac{F_0}{K} - \frac{1}{2} \sigma^2 T] / (\sigma \sqrt{T})$.

B.4 Barrier Options

We provide the formulas for American barrier call and put options with barrier H that is continuously monitored (i.e., American barrier). The notations in [49] are used for the option prices:

c_{ui}	up-and-in call
c_{uo}	up-and-out call, $c_{uo} = c - c_{ui}$ by <i>in-out parity</i>
c_{di}	down-and-in call
c_{do}	down-and-out call, $c_{do} = c - c_{di}$ by <i>in-out parity</i>
p_{ui}	up-and-in put
p_{uo}	up-and-out put, $p_{uo} = p - p_{ui}$ by <i>in-out parity</i>
p_{di}	down-and-in put
p_{do}	down-and-out put, $p_{do} = p - p_{di}$ by <i>in-out parity</i>

All the above option prices can be obtained by the following closed-form solutions:

$$\begin{aligned}
c_{ui} &= S_0 e^{-qT} N(x_1) - K e^{-rT} N(x_1 - \sigma \sqrt{T}) \\
&\quad - S_0 e^{-qT} (H/S_0)^{2\lambda} [N(-y) - N(-y_1)] \\
&\quad + K e^{-rT} (H/S_0)^{2\lambda-2} [N(-y + \sigma \sqrt{T}) - N(-y_1 + \sigma \sqrt{T})] \\
c_{di} &= S_0 e^{-qT} (H/S_0)^{2\lambda} N(y) - K e^{-rT} (H/S_0)^{2\lambda-2} N(y - \sigma \sqrt{T}) \\
p_{ui} &= -S_0 e^{-qT} (H/S_0)^{2\lambda} N(-y) + K e^{-rT} (H/S_0)^{2\lambda-2} N(-y + \sigma \sqrt{T}) \\
p_{di} &= -S_0 e^{-qT} N(-x_1) + K e^{-rT} N(-x_1 + \sigma \sqrt{T}) \\
&\quad + S_0 e^{-qT} (H/S_0)^{2\lambda} [N(y) - N(y_1)] \\
&\quad - K e^{-rT} (H/S_0)^{2\lambda-2} [N(y - \sigma \sqrt{T}) - N(y_1 - \sigma \sqrt{T})]
\end{aligned}$$

where

$$\begin{aligned}
\lambda &= (r - q + \frac{1}{2}\sigma^2)/\sigma^2 \\
y &= [(\ln \frac{H^2}{S_0 K})/(\sigma \sqrt{T})] + \lambda \sigma \sqrt{T} \\
x_1 &= [(\ln \frac{S_0}{H})/(\sigma \sqrt{T})] + \lambda \sigma \sqrt{T} \\
y_1 &= [(\ln \frac{H}{S_0})/(\sigma \sqrt{T})] + \lambda \sigma \sqrt{T} \\
c, p &\text{ closed-form solutions for vanilla call and put options (presented previously).}
\end{aligned}$$

Appendix C

Sobol Method for Low-Discrepancy Sequence

C.1 Sobol Sequence Generation

The base-2 van der Corput sequence introduced in Sect. 21.6.1 provides $[0, 1)$ uniform points $0.a_0(\mathbf{i})a_1(\mathbf{i})\dots a_n(\mathbf{i})$, where $\mathbf{i}$ is a whole number. In Sobol method, to generate multi-dimensional uniform points, each dimension $(1 < k \leq d)$ uses a difference type of generator matrix $V^k(\mathbf{i})$, that we denote by V^k , to lighten the notation to permute the base-2 van der Corput representations. The formula (21.2) can be written as

$$[s_0, s_1, \dots, s_n]^T = V^k[a_0, a_1, \dots, a_n]^T \pmod{2}, \quad (\text{C.1})$$

with $V^k = [\mathbf{v}_1^k, \mathbf{v}_2^k, \dots, \mathbf{v}_n^k]$ where $\mathbf{v}_i^k = [v_{i1}^k, \dots, v_{in}^k]^T$ is the vector representation of the binary-rational number called *direction number* $v_i^k = 0.v_{i1}^k \dots v_{in}^k, 0 < v_i^k < 1, v_{ij}^k \in \{0, 1\}$, i.e.,

$$V^k = \begin{pmatrix} v_{11}^k & v_{21}^k & \dots & v_{n1}^k \\ v_{12}^k & v_{22}^k & \dots & v_{n2}^k \\ \vdots & & \dots & \\ v_{1n}^k & v_{2n}^k & \dots & v_{nn}^k \end{pmatrix}, \quad \text{e.g., } V^k = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{C.2})$$

Equation (C.1) can be rewritten as

$$[s_0^k, s_1^k, \dots, s_n^k]^T = a_0 \mathbf{v}_1^k \oplus a_1 \mathbf{v}_2^k \oplus \dots \oplus a_{n-1} \mathbf{v}_n^k \quad (\text{C.3})$$

where $\oplus$ is the XOR (exclusive OR) operation, which is a bit-wise operator with the rules: $0 \oplus 0 = 0, 0 \oplus 1 = 1, 1 \oplus 0 = 1, 1 \oplus 1 = 0$. For example, $101 \oplus 011 = 110$. It is equivalent to a mod 2 addition operation.

In the sequel, if there is no specification, the symbols $i, j, k, d, h, n, z, p, q, t, \mathbf{i}$ all represent non-negative integers and $x \in \mathbb{R}$. To lighten the notation, we will drop off the dimension index k when it does not create confusion.

C.2 Simplification of Sobol Sequence Generation with Gray Code

Sobol sequence generation can be further simplified by using Gray codes¹ that are different representations of binary numbers with the particular feature that the Gray codes for any two successive numbers $\mathbf{i}$ and $\mathbf{i} + 1$ have exactly one different digit with index $h(\mathbf{i})$ (also denoted as h to lighten the notation). The use of Gray code $g_0(\mathbf{i}) \dots g_{n-1}(\mathbf{i})$ instead of $a_0(\mathbf{i}) \dots a_{n-1}(\mathbf{i})$ only changes the order of the binary digits but it does not impact the outcome of Sobol method (see [8]). Hence, for one step from $\mathbf{i}$ to $\mathbf{i} + 1$, the calculation is

$$\begin{aligned} & a_0(\mathbf{i} + 1)v_1 \oplus \dots \oplus a_h(\mathbf{i} + 1)v_h \dots \oplus a_{n-1}(\mathbf{i} + 1)v_n \\ \Leftrightarrow & g_0(\mathbf{i} + 1)v_1 \oplus \dots \oplus g_h(\mathbf{i} + 1)v_h \dots \oplus g_{n-1}(\mathbf{i} + 1)v_n \\ = & g_0(\mathbf{i})v_1 \oplus \dots \oplus (g_h(\mathbf{i}) \oplus 1)v_h \dots \oplus g_{n-1}(\mathbf{i})v_n \end{aligned}$$

This leads to the simplified recursive process for Sobol sequence generation:

$$s(\mathbf{i} + 1) = s(\mathbf{i}) \oplus v_h, \quad \text{where } g_h(\mathbf{i} + 1) \neq g_h(\mathbf{i}), \quad \mathbf{i} = 0, 1, 2, \dots \quad (\text{C.4})$$

C.3 Direction Numbers

The direction numbers are linked to a special type of polynomial of the form

$$x^q + c_1 x^{q-1} + \dots + c_{q-1} x + c_q, \quad (c_q \equiv 1) \quad \text{with } c_i \in \{0, 1\}, \quad \forall i$$

¹ Gray code of binary number n is $n \oplus \lfloor n/2 \rfloor$ where $\lfloor n/2 \rfloor$ is the integer part of $n/2$, e.g.,:

Decimal number	0	1	2	3	4	5	6	7	8
Binary number	0	1	10	11	100	101	110	111	1000
Gray code	0	1	11	10	110	111	101	100	1100

where the addition of coefficients of the polynomial is XOR.², e.g., $ax^2 + bx + cx^2 = (a \oplus c)x^2 + bx$. A polynomial satisfying the below two conditions is known as a *primitive polynomial*:

- it cannot be factored by another polynomial;
- the smallest p for which the polynomial divides $x^p + 1$ is $p = 2^q - 1$.

Here are some examples for primitive polynomials³: 1 for $q = 0$, $x + 1$ for $q = 1$, $x^2 + x + 1$ for $q = 2$, $x^3 + x + 1$ and $x^3 + x^2 + 1$ for $q = 3$. Many references, e.g., [58], [43], indicate the primitive polynomials already found.

For each dimension, a specific primitive polynomial is applied to generate the direction numbers. Let q ($q > 0$), $c_1, \dots, c_{q-1}$ be the degree and coefficients of the primitive polynomial for that dimension, the direction numbers v_i are generated as follows⁴:

$$\begin{cases} \text{for } 1 \leq i \leq q, & v_i = m_i/2^i \text{ where } m_i \text{ can be any odd integer } < 2^i; \\ \text{for } q < i \leq n, & v_i = c_1 v_{i-1} \oplus \dots \oplus c_{q-1} v_{i-q+1} \oplus v_{i-q} \oplus \frac{v_{i-q}}{2^q}. \end{cases} \quad (\text{C.5})$$

We note that the produced matrix V^k is an upper triangle with 1 on the diagonal which implies that $\det(V^k) = 1$ ($\det(\cdot)$ is the determinant operator) and it is invertible. For $q = 0$, we can apply $v_i = \frac{1}{2^i}$, $\forall i > 0$, which produces the van der Corput sequence as V^k will be an identity matrix.

Example C.1 We illustrate the direction number generation with $x^3 + x^2 + 1$ ($q = 3$, $c_1 = 1, c_2 = 0, c_3 = 1$). Let $v_1 = 0.1, v_2 = 0.11, v_3 = 0.011$. Then,
 $v_4 = c_1 v_3 \oplus c_2 v_2 \oplus v_1 \oplus v_1/2^3 = 0.011 \oplus 0 \oplus 0.1 \oplus 0.1/2^3 = 0.1111$,
 $v_5 = c_1 v_4 \oplus c_2 v_3 \oplus v_2 \oplus v_2/2^3 = 0.1111 \oplus 0 \oplus 0.11 \oplus 0.11/2^3 = 0.00101$.
 The generator matrix V is actually the one shown in (C.2).

The choice of initial direction numbers v_i^k ($1 \leq i \leq n$) is important to avoid correlations between sequences projected to the different dimensions. For example, if $v_i^1 = v_i^2$, the first q columns in V^1 and V^2 will be identical, resulting in the same first $2^q - 1$ sequence numbers for these dimensions. Property-A and Property-A' to be introduced below are helpful for selecting the initial direction numbers.

² The c_i are elements of GF(2), i.e., Galois Field of order 2, where the addition is $\oplus$ and the multiplication is as usual: $c_i \oplus c_i = 0$, $c_i c_i = c_i^2 = c_i$ and $-1 = 1$ (because $1 + 1 = 0$).

³ For example, $x^2 + 1$ is not a primitive polynomial because it can be factored as $(x + 1)(x + 1) = x^2 + x + x + 1 = x^2 + 1$, where we have used $cx + cx = (c + c)x = 0$ with $c = 1$.

⁴ In some references (e.g., [93][43]) the equivalent recurrence with $m_i = 2^i v_i$ is used:

$m_i = 2c_1 m_{i-1} \oplus \dots \oplus 2^{q-1} c_{q-1} m_{i-q+1} \oplus 2^q m_{i-q} \oplus m_{i-q}, \quad \forall i > q.$

C.4 Comments on Uniformity

A cube formed by $\prod_{k=1}^d [j_k/2^{z_k}, (j_k + 1)/2^{z_k})$ with $z_k > 0$ and $0 \leq j_k < 2^{z_k}$ is a subset of $[0, 1)^d$. Its volume is $\frac{1}{2^z}$, $z = \sum_{k=1}^d z_k$ and $[0, 1)^d$ is composed by 2^z such disjoint cubes. For a d -dimensional sequence of points x_j ($j > 0$) in $[0, 1)^d$, a *binary segment of length* 2^z refers to the points $\{x_i : j2^z < i \leq (j + 1)2^z\}$ with $j \geq 0$ and $z \geq t + 1$. The sequence is called a (t, d) -sequence ($t \geq 0$) if each binary segment of length 2^z has 2^t points in each cube with the volume of $2^t \frac{1}{2^z}$. A $(0, d)$ -sequence has the best uniformity as it allocates one point to each cube with the volume of $1/2^z$. Actually, van der Corput sequence is a $(0, 1)$ -sequence.

The d -dimensional Sobol sequence has the below uniformity features:

- (1) if the sequence has $2^n - 1$ points, $\forall n$, its projected sequence on a single dimension is a permutation of the van der Corput points of $2^n - 1$ points;
- (2) it is a (t, d) -sequence with $t \leq \sum_{k=1}^d q_k - d$ where q_k is the polynomial degree for the k -th dimension. Hence, lower the sum of polynomial degrees, better the uniformity.

A d -dimensional Sobol sequence is said to possess

- Property-A: if for any binary segment of length 2^d , there is only one point in each of the cube $\prod_{k=1}^d [\frac{b_k}{2}, \frac{b_k+1}{2})$, $b_k \in \{0, 1\}$. It is satisfied if and only if $\det(V_d) = 1 \pmod{2}$,
- Property-A': if for any binary segment of length 2^{2d} , there is only one point in each of the cube $\prod_{k=1}^d [\frac{b_k}{4}, \frac{b_k+1}{4})$, $b_k \in \{0, 1, 2, 3\}$. It is satisfied if and only if $\det(V_{2d}) = 1 \pmod{2}$,

$$\text{where } V_d = \begin{pmatrix} v_{11}^1 & v_{12}^1 & \cdots & v_{1d}^1 \\ v_{11}^2 & v_{12}^2 & \cdots & v_{1d}^2 \\ \vdots & \vdots & & \vdots \\ v_{11}^d & v_{12}^d & \cdots & v_{1d}^d \end{pmatrix}, \quad V_{2d} = \begin{pmatrix} v_{11}^1 & v_{12}^1 & \cdots & v_{1,2d}^1 \\ \vdots & \vdots & & \vdots \\ v_{11}^d & v_{12}^d & \cdots & v_{1,2d}^d \\ v_{21}^1 & v_{22}^1 & \cdots & v_{2,2d}^1 \\ \vdots & \vdots & & \vdots \\ v_{21}^d & v_{22}^d & \cdots & v_{2,2d}^d \end{pmatrix} \quad (\text{see [107]}).$$

C.5 Comments on the Rationale of the Method

Readers should have noticed that Sobol method is far from intuitive. It is rather a quite involved mathematical construct using the recurrence operator:

$$L(u_i) = u_{i+q} \oplus c_1 u_{i+q-1} \oplus \cdots \oplus c_{q-1} u_{i+1} \oplus c_q u_i, \quad \text{where } c_i, u_i \in \{0, 1\}, \quad c_q \equiv 1$$

which becomes a so-called *monocyclic operator* if the solutions to $L(u_i) = 0$ have a period of $2^q - 1$. It is known that the coefficients of a primitive polynomial can be

used to obtain a monocyclic operator with the same coefficients c_i , and vice versa (see [76], [70]). The recurrence relation for the construction of direction numbers in equation (C.5) comes from the application of the below equation (by XOR-ing the term $v_i \oplus v_{i-q}/2^q$ on both sides):

$$L(v_{i-q}) = v_{i-q}/2^q, \text{ for } i > q \text{ where } L(v_{i-q}) \text{ is a monocyclic operator.}$$

As the matrix V for direction numbers hence constructed is invertible, we have $V\mathbf{e}_1 \neq V\mathbf{e}_2 \pmod{2}$ if $\mathbf{e}_1 \neq \mathbf{e}_2$. Hence, the first $2^n - 1$ non-zero van der Corput points transformed by V will be the same points in different order, as all possible points with digit length n are included, which proves the uniformity feature (1). The uniformity feature (2) is proved in [106] (Theorem 3.3) by the fact that the linear equation system for Sobol digits in (C.3) with direction numbers as coefficients has 2^t solutions for each volume of $2^{t \frac{1}{2z}}$. In the proof, the monocyclic operators $L(v_{i-q})$ are used as elementary transformation to prove that the rank of the linear equation system with z unknowns is $z - t$, leading to $z - (z - t) = t$ arbitrary binary digits which implies 2^t solutions.

Appendix D

Some Results of Portfolio Theory

In this part, we use the notations in Sect. 32.2.2.

Proposition D.1 (Mean-Variance Optimization (MVO)) *Let $\mathbf{r} = [r_1, \dots, r_n]^\top$ be the expected returns of n assets. Given an expected portfolio return r , the optimal asset allocation $\mathbf{w}$ is determined by the following quadratic programming problem known as mean-variance optimization (MVO):*

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w}, \quad \text{subject to the constraints} \quad \begin{cases} \mathbf{r}^\top \mathbf{w} = r, \\ \mathbf{1}^\top \mathbf{w} = 1, \end{cases}$$

The solution is $\mathbf{w} = \mathbf{h} + r\mathbf{g}$ with the parameters defined as $\mathbf{h} = \Sigma^{-1} \frac{\mathbf{b}\mathbf{1} - \mathbf{a}\mathbf{r}}{\mathbf{bc} - \mathbf{a}^2}$, $\mathbf{g} = \Sigma^{-1} \frac{\mathbf{c}\mathbf{r} - \mathbf{a}\mathbf{1}}{\mathbf{bc} - \mathbf{a}^2}$, $\mathbf{a} = \mathbf{1}^\top \Sigma^{-1} \mathbf{r}$, $\mathbf{b} = \mathbf{r}^\top \Sigma^{-1} \mathbf{r}$, $\mathbf{c} = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$.

The variance of the optimal portfolio is $\sigma^2 = \mathbf{w}^\top \Sigma \mathbf{w} = \frac{\mathbf{b} - 2\mathbf{a}r + \mathbf{c}r^2}{\mathbf{bc} - \mathbf{a}^2}$.

Proof The proof is extracted from [77]. First-order condition on the Lagrangian $\mathcal{L}(\mathbf{w}, \lambda_1, \lambda_2) = \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} + \lambda_1(r - \mathbf{r}^\top \mathbf{w}) + \lambda_2(1 - \mathbf{1}^\top \mathbf{w})$ implies $\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \Sigma \mathbf{w} - \lambda_1 \mathbf{r} - \lambda_2 \mathbf{1} = 0$, $\frac{\partial \mathcal{L}}{\partial \lambda_1} = r - \mathbf{r}^\top \mathbf{w} = 0$, and $\frac{\partial \mathcal{L}}{\partial \lambda_2} = 1 - \mathbf{1}^\top \mathbf{w} = 0$. From the first equation, we obtain $\mathbf{w} = \Sigma^{-1}(\lambda_1 \mathbf{r} + \lambda_2 \mathbf{1})$. Replacing $\mathbf{w}$ in the two constraint equations, we have

$$\begin{aligned} \mathbf{r}^\top \mathbf{w} &= \mathbf{r}^\top \Sigma^{-1}(\lambda_1 \mathbf{r} + \lambda_2 \mathbf{1}) = r \\ \mathbf{1}^\top \mathbf{w} &= \mathbf{1}^\top \Sigma^{-1}(\lambda_1 \mathbf{r} + \lambda_2 \mathbf{1}) = 1, \end{aligned}$$

which can be expressed as $\begin{bmatrix} \mathbf{b} & \mathbf{a} \\ \mathbf{a} & \mathbf{c} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} r \\ 1 \end{bmatrix}$. Hence, $\lambda_1 = \frac{\mathbf{c}r - \mathbf{a}}{\mathbf{bc} - \mathbf{a}^2}$, $\lambda_2 = \frac{\mathbf{b} - \mathbf{a}r}{\mathbf{bc} - \mathbf{a}^2}$, and $\mathbf{w} = \Sigma^{-1} \frac{\mathbf{r}(\mathbf{c}r - \mathbf{a}) + \mathbf{1}(\mathbf{b} - \mathbf{a}r)}{\mathbf{bc} - \mathbf{a}^2} = \mathbf{h} + r\mathbf{g}$.

Lastly, we have $\sigma^2 = \mathbf{w}^\top \Sigma \mathbf{w} = (\mathbf{h} + r\mathbf{g})^\top \Sigma (\mathbf{h} + r\mathbf{g}) = \frac{\mathbf{b} - 2\mathbf{a}r + \mathbf{c}r^2}{\mathbf{bc} - \mathbf{a}^2}$. □

Proposition D.2 (Market Portfolio and Sharpe Ratio) *If the risk-free asset yielding r_f is added to the portfolio with risky assets (see [113]), the MVO problem becomes*

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w}, \text{ subject to constraints } \mathbf{r}^\top \mathbf{w} + (1 - \mathbf{1}^\top \mathbf{w}) r_f = r.$$

where $\mathbf{w}$ represents the weights of the risky assets portfolio ($\mathbf{w}^\top \mathbf{1} > 1$ means borrowing from the money market). We have

$$r = r_f + \theta \sigma \text{ i.e., } \theta = \frac{r - r_f}{\sigma}, \text{ which is known as } \textit{Sharpe ratio}.$$

The optimal risky assets portfolio is called **market portfolio**.

Proof The first order condition on the Lagrange implies $\Sigma \mathbf{w} + \lambda(\mathbf{r} - r_f \mathbf{1}) = 0$, $r_f - r + (r^\top - r_f \mathbf{1}^\top) \mathbf{w} = 0$. The solution to the above equations is $\mathbf{w} = (r - r_f) \Sigma^{-1} (\mathbf{r} - r_f \mathbf{1}) / \theta^2$ where $\theta^2 = (\mathbf{r} - r_f \mathbf{1})^\top \Sigma^{-1} (\mathbf{r} - r_f \mathbf{1})$. By $\sigma^2 = \mathbf{w}^\top \Sigma \mathbf{w}$, we get

$$r = r_f + \theta \sigma.$$

□

Proposition D.3 (CAPM from the Portfolio Theory) *Let $\mathbf{w}_m$, r_m , σ_m^2 be respectively the weights, expected return, variance of the market portfolio and Cov_m be the vector of covariances of every asset with the market portfolio, then*

$$\mathbf{r} - r_f \mathbf{1} = \beta (r_m - r_f) \text{ where } \beta = \frac{\text{Cov}_m}{\sigma_m^2}.$$

The entry for the i -th asset reads $r_i - r_f = \beta_i (r_m - r_f)$.

Proof From the proof of Proposition D.2, the market portfolio weights are as follows:

$$\mathbf{w}_m = \Sigma^{-1} (\mathbf{r} - r_f \mathbf{1}) h \text{ where } h = \frac{r_m - r_f}{(\mathbf{r} - r_f \mathbf{1})^\top \Sigma^{-1} (\mathbf{r} - r_f \mathbf{1})}.$$

We have

$$\text{Cov}_m = \text{diag}(1, \dots, 1) \Sigma \mathbf{w}_m = (\mathbf{r} - r_f \mathbf{1}) h$$

$$\sigma_m^2 = \mathbf{w}_m^\top \Sigma \mathbf{w}_m = (r_m - r_f) h$$

$$\text{Hence, } \mathbf{r} - r_f \mathbf{1} = \frac{\text{Cov}_m}{\sigma_m^2} (r_m - r_f)$$

□

Appendix E

Representation of CIR Process as a Squared Bessel Process

Some of the important stochastic processes used in financial modelling such as CEV process (used in the SABR model) and CIR process (used in the CIR short rate model and the Heston model) can be represented by a time-changed squared Bessel process. This helps to understand the properties of these processes because squared Bessel process is well studied and many properties are well known. In this appendix, we will only illustrate the representation of CIR process. We refer to [60] for a rigorous and detailed presentation.

E.1 Squared Bessel Process

Squared Bessel process is one of the most important one-dimensional stochastic processes. It is defined as a generalization of the process followed by the squared norm of a multi-dimensional Brownian motion. Denote $Z_t = \|\mathbf{B}_t\|$ where $\mathbf{B}_t = (B_t^1, \dots, B_t^k)$ is a k -dimensional Brownian motion, and $Z_t^2 = \sum_{i=1}^k (B_t^i)^2$. By Itô–Doebelin formula, we obtain

$$dZ_t^2 = \sum_{i=1}^k 2B_t^i dB_t^i + kdt = 2Z_t \sum_{i=1}^k \frac{B_t^i}{\|\mathbf{B}_t\|} dB_t^i + kdt = 2Z_t dW_t + kdt,$$

where we use the fact that $W_t = \sum_{i=1}^k \int_0^t \frac{B_s^i}{\|\mathbf{B}_s\|} dB_s^i$ is a one-dimensional Brownian motion that can be verified with Levy's Theorem.

Squared Bessel process is defined as the generalization of the process followed by Z_t^2 with k extended to non-integer values, i.e., $k \in \mathbb{R}$:

$$dX_t = 2\sqrt{X_t}dW_t + kdt.$$

In our context, we only consider $k \geq 0$ and strictly positive initial value $X_0 > 0$.

The process X_t follows *Chi-squared* distribution extended to non-integer degrees of freedom (cf. [60]). Regarding the squared Bessel process, an important focus is the condition for X_t reaching zero. The following features are useful for the applications in financial modelling

- (i) $k > 2$: X_t will never reach 0 and $X_t \rightarrow +\infty$ as $t \rightarrow +\infty$.
- (ii) $k = 2$: X_t will never reach 0 and X_t is *recurrent*, meaning that the probability for X_t to get infinitely close to any non-negative value in any finite time is strictly positive.
- (iii) $0 < k < 2$: X_t reaches 0 in finite time. However, it leaves 0 instantaneously.
- (iv) $k = 0$: X_t reaches 0 in finite time and will stay at 0 afterward.

For the proofs, we refer to [64] for $k = 2$. For a comprehensive analysis, we refer to [60, 96] and [73]. For helping to understand the properties, we mention some intuitions behind. In the case of one dimension (i.e., $k = 1$), $\sqrt{X_t}$ is a standard Brownian motion before hitting 0. So, it reaches 0 in finite time as it is the case for standard Brownian motion. For $0 \leq k < 1$, the process can be written as the sum of a squared Bessel process with dimension 1 and a strictly decreasing term: $dX_t = (2\sqrt{X_t}dW_t + 1dt) + (k - 1)dt$. So, X_t reaches 0 for $0 \leq k < 1$. Similarly, for $k > 2$ with $dX_t = (2\sqrt{X_t}dW_t + 2dt) + (k - 2)dt$, X_t will never reach 0 if it never reaches zero for $k = 2$. Whenever the point 0 is reached, the staying time should be zero (i.e., it leaves 0 instantaneously). Otherwise, the SDE can not hold because in such a case, the kdt term is strictly positive but all the other terms are zero for the time period that X_t stayed at zero.

E.2 Representation of CIR Process

Consider the CIR process, which follows the below SDE under the assumption that $b > 0$, $r_0 > 0$, $\sigma > 0$ and $a \in \mathbb{R}$,

$$dr_t = (a - br_t)dt + \sigma\sqrt{r_t}dW_t.$$

First, we try to remove r_t in the drift. Applying Itô's Product Rule to $Y_t = e^{bt}r_t$, we have

$$dY_t = e^{bt}dr_t + br_te^{bt}dt = e^{bt}(adt + \sigma\sqrt{r_t}dW_t) = ae^{bt}dt + \sigma\sqrt{e^{bt}}\sqrt{Y_t}dW_t.$$

Denote $\tau(t)$ such that $t = \frac{\sigma^2}{4} \int_0^{\tau(t)} e^{bs} ds$ for any given t . The function $\tau(t)$ exists because $f(x) = \frac{\sigma^2}{4} \int_0^x e^{bs} ds$ is an increasing continuous function with $f(0) = 0$ and $f(+\infty) = +\infty$. Hence, integration from 0 to $\tau(t)$ gives

$$Y_{\tau(t)} = r_0 + a \int_0^{\tau(t)} e^{bs} ds + 2 \int_0^{\tau(t)} \frac{\sigma}{2} \sqrt{e^{bs}} \sqrt{Y_s} dW_s.$$

The first integral is equivalent to the following, which is straightforward from the definition of $\tau(t)$,

$$a \int_0^{\tau(t)} e^{bs} ds = \frac{4a}{\sigma^2} t.$$

For the second integral, we will apply the so-called *time-change*. Actually, for any continuous martingale M_t , it is known that $M_t = W_{\langle M_t, M_t \rangle}$. This helps to understand time-change, which implies:

$$\int_0^T \beta_t dM_t = \int_0^{\langle M_T, M_T \rangle} \beta_{u(t)} dM_{u(t)} = \int_0^{\langle M_T, M_T \rangle} \beta_{u(t)} dW_t,$$

where $u(t)$ is the value such that $\langle M_{u(t)}, M_{u(t)} \rangle = t$.

Now, for our case, denote $M_t = \int_0^t \frac{\sigma}{2} \sqrt{e^{bs}} dW_s$. M_t is a continuous martingale and $\langle M_{\tau(t)}, M_{\tau(t)} \rangle = \frac{\sigma^2}{4} \int_0^{\tau(t)} e^{bs} ds = t$. Applying time-change, we have for the second integral,

$$\int_0^{\tau(t)} \frac{\sigma}{2} \sqrt{e^{bs}} \sqrt{Y_s} dW_s = \int_0^{\tau(t)} \sqrt{Y_s} dM_s = \int_0^t \sqrt{Y_{\tau(s)}} dW_s.$$

And all together,

$$Y_{\tau(t)} = r_0 + \frac{4a}{\sigma^2} t + 2 \int_0^t \sqrt{Y_{\tau(s)}} dW_s.$$

Setting $X_t = Y_{\tau(t)}$, $X_0 = r_0$ and $k = \frac{4a}{\sigma^2}$, we obtain the representation of CIR process as a squared Bessel process

$$X_t = r_0 + kt + 2 \int_0^t \sqrt{X_s} dW_s.$$

From the properties of a squared Bessel process, we obtain the following:

- if $a \geq \frac{\sigma^2}{2}$ (i.e. $k \geq 2$), then r_t will never reach zero.
- if $0 < a < \frac{\sigma^2}{2}$ (i.e. $0 < k < 2$), then r_t has a non-zero probability to touch 0 in any time interval but r_t will leave 0 instantaneously if it does.

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